Environmental Science: Theory into Practice

A Textbook for the Ability Enhancement Course (AEC I & II)

University of Delhi · Undergraduate Curriculum Framework 2022

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Revised Edition

Contents

Preface vii
How to Use This Book viii
Syllabus Map ix
Part I: Year 1 1
Chapter 1: Introduction to Environmental Studies 2
1.1 Environment and Environmental Studies 2
1.2 Multidisciplinary Nature of Environmental Studies 3
1.3 Components of Environment 6
1.4 Scope and Importance of Environmental Studies 13
1.5 Sustainability and Sustainable Development 15
1.6 Brief History of Environmentalism: The Global Story 23
1.7 Brief History of Environmentalism in India 26
Chapter 1 Review 32
Chapter 2: Ecosystems 39
2.1 Definition and Concept of Ecosystem 39
2.2 Structure of an Ecosystem 42
2.3 Functions of an Ecosystem 46
2.4 Energy Flow in Ecosystems 46
2.5 Food Chains, Food Webs and Trophic Levels 51
2.6 Ecological Succession 56
2.7 Biogeochemical Cycles (Nutrient Cycling) 61
2.8 Productivity 68
2.9 Ecological Pyramids 71
2.10 Ecosystem Homeostasis 73
2.11 Types of Ecosystems 77
2.12 Tundra Ecosystem 78
2.13 Forest Ecosystem 82
2.14 Grassland Ecosystem 86
2.15 Desert Ecosystem 89
2.16 Aquatic Ecosystems 92
2.17 Ponds and Lakes 93
2.18 Streams and Rivers 97
2.19 Oceans 100

2.20 Estuaries, Mangroves and Lagoons 103
2.21 Importance and Threats of Ecosystems in India: A Comparison 107
2.22 Ecosystem Services 107
2.23 Ecosystem Preservation and Conservation Strategies 112
2.24 Basics of Ecosystem Restoration 115
Chapter 2 Review 121
Chapter 3: Natural Resources 134
3.1 Mineral Resources 135
3.2 Soil 139
3.3 Agricultural Crops 142
3.4 Natural Forest Products 145
3.5 Medicinal Plants 147
3.6 Forest-Based Industries and Livelihoods 149
3.7 Land Cover and Land-Use Change 151
3.8 Land Degradation 153
3.9 Soil Erosion and Desertification 155
3.10 Causes of Deforestation 158
3.11 Impacts of Mining and Dam Building 161
3.12 Water Resources: Natural and Man-made Sources 167
3.13 Uses of Water 173
3.14 Over-exploitation of Surface and Groundwater 175
3.15 Floods and Droughts 180
3.16 International and Inter-state Conflicts over Water 183
3.17 Energy Resources: Renewable and Non-renewable 186
3.18 Alternate Energy Sources and Growing Energy Needs 188
3.19 Energy Contents of Coal, Petroleum, Natural Gas and Biogas 191
3.20 Agro-residues as Biomass Energy 195
3.21 Case Studies: Natural Resources in India 198
Chapter 3 Review 206
Chapter 4: Environmental Pollution and Control 219
4.1 Pollution and Pollutants 219
4.2 Air Pollution 221
4.3 Primary and Secondary Air Pollutants 229
4.4 Water Pollution 232
4.5 Soil Pollution 237

4.6 Thermal Pollution 239
4.7 Noise Pollution 241
4.8 Air and Water Quality Standards 244
4.9 Nuclear Hazards and Human Health Risks 248
4.10 Solid Waste: Types, Sources and Generation in India 254
4.11 The Waste Hierarchy and Integrated Waste Management 256
4.12 Segregation at Source and Collection 258
4.13 Processing and Disposal of Municipal Solid Waste 259
4.14 Control Measures for Urban Waste 263
4.15 Industrial Waste: Types and Control Measures 265
4.16 Hazardous Waste and Biomedical Waste 267
4.17 E-Waste: Composition, Hazards and Management 270
4.18 Pollution Control Measures: Legal Methods 272
4.19 Pollution Control Measures: Biological Methods 275
4.20 Pollution Control Measures: Physico-Chemical Methods 277
4.21 Role of Pollution Control in Sustainability 281
4.22 Case Study: The Ganga Action Plan and Namami Gange 282
4.23 Case Study: Delhi Air Pollution and Public Health 285
4.24 Case Study: Plastic Waste Management Rules 289
4.25 Case Study: The Bhopal Gas Tragedy 291
Chapter 4 Review 295
Part II: Year 2 308
Chapter 5: Global Environmental Issues and Policies 309
5.1 Earth's Climate System and the Causes of Climate Change 309
5.2 Global Warming 314
5.3 Ozone Layer Depletion 317
5.4 Acid Rain 322
5.5 Impacts of Climate Change 325
5.6 International Agreements and Programmes 329
5.7 The Earth Summit, 1992 330
5.8 The United Nations Framework Convention on Climate Change (UNFCCC) 331
5.9 The Kyoto Protocol 332
5.10 The Paris Agreement and Recent Climate Conferences 333
5.11 The Montreal Protocol 335
5.12 The Convention on Biological Diversity 336

5.13 The Ramsar Convention on Wetlands 339
5.14 The Chemical Weapons Convention 341
5.15 CITES: Regulating Trade in Endangered Species 342
5.16 The United Nations Environment Programme (UNEP) 343
5.17 India and the International Agreements: An Overview 344
5.18 Sustainable Development Goals and India 345
5.19 India's National Action Plan on Climate Change 346
5.20 Environment Legislation in India 350
5.21 The Wild Life (Protection) Act, 1972 351
5.22 The Water (Prevention and Control of Pollution) Act, 1974 352
5.23 The Forest (Conservation) Act, 1980 354
5.24 The Air (Prevention and Control of Pollution) Act, 1981 355
5.25 The Environment (Protection) Act, 1986 356
5.26 The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006 358
Chapter 5 Review 361
Chapter 6: Biodiversity and Conservation 368
6.1 What Is Biodiversity? 368
6.2 Levels of Biological Diversity 369
6.3 India as a Mega-Biodiversity Nation 373
6.4 Biogeographic Zones of India 375
6.5 Biodiversity Hotspots 377
6.6 Endemic and Endangered Species of India 379
6.7 IUCN Red List Criteria and Categories 380
6.8 Value of Biodiversity 383
6.9 Sacred Groves 387
6.10 Threats to Biodiversity 390
6.11 Biodiversity Conservation Strategies: In-situ and Ex-situ 400
6.12 National Parks, Wildlife Sanctuaries and Biosphere Reserves 403
6.13 Keystone, Flagship, Umbrella and Indicator Species 406
6.14 Species Reintroduction and Translocation 407
6.15 Conservation in Practice: Indian Case Studies 410
6.16 Project Tiger 412
6.17 Project Elephant 416
6.18 Vulture Conservation Breeding Programme 420

6.19 Project Great Indian Bustard 425
6.20 Crocodile Conservation Project 428
6.21 Silent Valley Movement 431
6.22 Save Western Ghats Movement 433
6.23 Lessons Across the Case Studies 437
Chapter 6 Review 439
Chapter 7: Human Communities and the Environment 452
7.1 Human Population Growth: Impacts on Environment, Health and Welfare 452
7.2 Carbon Footprint 459
7.3 Resettlement and Rehabilitation of Project-Affected Persons and Communities 464
7.4 Environmental Movements in India 470
7.5 Environmental Justice: The National Green Tribunal 478
7.6 Environmental Philosophy: Ethics, Religion and Culture 482
7.7 Environmental Communication and Public Awareness 486
Chapter 7 Review 493
Appendix A: Practical and Outreach Activities 500
A.1 How Practical Work Is Organised and Assessed 500
A.2 Unit 1: Introduction to Environmental Studies 502
A.3 Unit 2: Ecosystems 506
A.4 Unit 3: Natural Resources 514
A.5 Unit 4: Environmental Pollution and Control 522
A.6 Unit 5: Global Environmental Issues and Policies 537
A.7 Unit 6: Biodiversity and Conservation 548
A.8 Unit 7: Human Communities and the Environment 558
A.9 Assessment Rubric for Practical Reports 564
Appendix B: Glossary 566

Preface

This book is written for undergraduate students of the University of Delhi taking the Ability Enhancement Course (AEC) "Environmental Science: Theory into Practice" under the Undergraduate Curriculum Framework (UGCF) 2022. The course is compulsory for students of every discipline, so the book assumes no more than school-level science. Every technical term is explained when it first appears, and the text tries to show why each idea matters for India and for the reader's own life.

This revised edition is a complete rewrite of an earlier, much longer draft. That draft contained useful material but also much repetition, uneven coverage of the syllabus and a heavy reliance on examples from other countries. In this edition the text has been shortened and reorganised to follow the syllabus unit by unit, facts and figures have been checked and updated to 2025, including recent amendments to India's environmental laws and the country's current climate commitments, and original diagrams have been added to explain key processes.

Three principles have guided the revision. First, the book is **India-focused**: every concept is illustrated with Indian places, species, institutions, policies and data, while global examples are used where they teach something important. Second, it is **concise**: each chapter aims to explain the essentials clearly rather than to cover everything. Third, it is **practical**: as the course title suggests, theory should lead to practice. Activity boxes in every chapter and the practical exercises in Appendix A ask students to observe, measure, survey and act on the environment around them, on campus, at home and in the city.

Environmental problems are among the most important challenges facing India and the world. This book is meant to help students understand those problems, think critically about their causes and solutions, and adopt sustainability as a practice in their own lives.

Deepak Gusain

How to Use This Book

The book follows the syllabus in seven chapters. Chapters 1 to 4 form Part I, for the first year of the course; Chapters 5 to 7 form Part II, for the second year. Each chapter uses the same set of features.

- **Learning Objectives** at the start of each chapter list what you should be able to do after studying it. Use them to plan your reading and to check your understanding afterwards.
- **Numbered sections and subsections** follow the wording of the syllabus as closely as possible, so that you can quickly find each topic.
- **Figures** are original diagrams and charts that explain processes (such as nutrient cycles or the layers of the atmosphere) or present real data with their sources. Study each figure together with the text that refers to it.
- **Tables** summarise classifications, comparisons and data. Figures quoted in the text and tables are given with their year and source.
- **Case Study** boxes describe real Indian (and some global) examples in detail, such as river-water disputes, conservation programmes and environmental movements. They show how concepts work in practice and are useful for long-answer questions.
- **Did You Know?** boxes present striking facts or correct common misconceptions.
- **Think and Do** boxes suggest short activities, discussions or observations that you can carry out in class, on campus or at home.
- **Key Points** boxes summarise long sections in a few bullets for quick revision.
- **End-of-chapter material** includes:
  - a **Summary** of the main ideas;
  - **Key Terms** with one-sentence definitions;
  - **Multiple-Choice Questions**, with an **answer key** at the end of the book;
  - **Short-Answer Questions** of the 2–5 mark type and **Long-Answer Questions** of the 10–15 mark type;
  - **Further Reading**, listing standard textbooks, official reports and websites.
- **Appendix A: Practical and Outreach Activities** describes the practical exercises prescribed in the syllabus for all seven units, with aims, materials, methods and suggestions for reporting. Your college will choose which practicals you perform.
- The **Glossary** at the end of the book brings together the key terms from all chapters in alphabetical order.

A suggested way to study each chapter is to read the learning objectives, read the text section by section with its figures and boxes, attempt the Think and Do activities, and then use the summary, key terms and questions for revision. Cross-references such as "(see Chapter 5)" point to the chapter where a topic is treated in full.

Syllabus Map

The table below shows where each unit of the University of Delhi AEC syllabus "Environmental Science: Theory into Practice" (UGCF 2022) is covered in this book. Units 1 to 4 are studied in Year 1 and form Part I of the book; Units 5 to 7 are studied in Year 2 and form Part II.

Mapping of syllabus units to chapters

Part	Syllabus unit	Main topics	Chapter
Part I (Year 1)	Unit 1: Introduction to Environmental Studies	Multidisciplinary nature; atmosphere, hydrosphere, lithosphere, biosphere; scope and importance; sustainability and sustainable development; history of environmentalism	Chapter 1
Part I (Year 1)	Unit 2: Ecosystems	Concept, structure and functions; energy flow, food webs, succession, nutrient cycles; productivity, pyramids, homeostasis; ecosystem types in India; ecosystem services, conservation and restoration	Chapter 2
Part I (Year 1)	Unit 3: Natural Resources	Land, minerals, soil, forests; land degradation and deforestation; mining and dams; water resources, floods, droughts and conflicts; energy resources; case studies	Chapter 3
Part I (Year 1)	Unit 4: Environmental Pollution and Control	Air, water, soil, thermal and noise pollution; nuclear hazards; solid, hazardous and e-waste; pollution control measures; case studies	Chapter 4
Part II (Year 2)	Unit 5: Global Environmental Issues and Policies	Climate change, ozone depletion, acid rain; international agreements; SDGs in India and NAPCC; Indian environmental legislation	Chapter 5
Part II (Year 2)	Unit 6: Biodiversity and Conservation	Levels of biodiversity; India as a megadiverse nation; hotspots; IUCN categories; values and threats; in-situ and ex-situ conservation; case studies	Chapter 6
Part II (Year 2)	Unit 7: Human Communities and the Environment	Population and carbon footprint; resettlement and rehabilitation; environmental movements; National Green Tribunal; ethics and religion; communication and awareness	Chapter 7
Parts I and II	Practicals for Units 1 to 7	Field visits, surveys, monitoring with kits, worksheets, models, data analysis and reports	Appendix A

Some topics appear in more than one unit of the syllabus. To avoid repetition, each such topic is treated in full in one chapter and briefly elsewhere with a cross-reference. For example, the concept of the Sustainable Development Goals is explained in Chapter 1, while India's progress on them is discussed in Chapter 5; the Chipko and Appiko movements are treated in full in Chapter 7 and summarised in Chapter 3.

Part I: Year 1

Units 1–4 · Semester I/II

Chapter 1: Introduction to Environmental Studies

Every breath you take, every glass of water you drink and every meal you eat depends on the environment. So does every job, factory, farm and city. Yet human activity is now changing the air, water, land and living world faster than at any time in history. Delhi's winter smog, falling water tables in Punjab, retreating Himalayan glaciers and record heatwaves are all signs of this change. Environmental studies is the field that tries to understand these changes as a whole: how nature works, how people alter it, and how societies can meet their needs without destroying the systems that keep them alive. This chapter introduces the field, the four major components of the environment, the scope and importance of the subject, the idea of sustainable development, and the history of environmental thought and action in the world and in India.

Learning Objectives

- Define environment, environmental science and environmental studies, and distinguish between them.
- Explain the multidisciplinary nature of environmental studies with examples of contributing disciplines.
- Describe the atmosphere, hydrosphere, lithosphere and biosphere, and explain how they interact.
- Discuss the scope and importance of environmental studies for careers, citizenship and India's development.
- Explain sustainability and sustainable development, including the Brundtland definition, the three-pillar and nested models, strong and weak sustainability, carrying capacity and the ecological footprint.
- List the 17 Sustainable Development Goals and relate them to environmental problems in India.
- Trace the major milestones in the history of environmentalism, globally and in India.

1.1 Environment and Environmental Studies

1.1.1 Meaning of Environment

The word **environment** comes from the French *environner*, "to surround". In the simplest sense, the environment of an organism is everything that surrounds it and affects its life: air, water, soil, light, temperature, other organisms and, for human beings, the social and economic conditions in which they live.

Indian law gives a working definition. The Environment (Protection) Act, 1986 states that environment "includes water, air and land and the inter-relationship which exists among and between water, air and land, and human beings, other living creatures, plants, micro-organism and property". It includes living and non-living parts together and stresses the **inter-relationships** between them.

The environment has two broad kinds of components:

- **Abiotic components** are the non-living physical and chemical factors: sunlight, temperature, air, water, soil, minerals and nutrients.
- **Biotic components** are the living organisms: plants, animals, fungi and micro-organisms, including human beings.

It is also useful to separate the **natural environment** (forests, rivers, mountains, oceans, wildlife) from the **built environment** made by people (cities, roads, dams, farms, factories). In India very little of the natural environment is untouched; most landscapes are a mix of the two, such as a village with its fields, ponds, grazing land and sacred grove.

1.1.2 Environmental Science and Environmental Studies

The two terms are often used interchangeably, but they have different emphases.

Environmental science is the scientific study of the natural world and of how human activity affects it. It uses the methods of physics, chemistry, biology and earth science: observation, measurement, experiment and modelling. An environmental scientist might measure the concentration of $PM_{2.5}$ in Delhi's air, test arsenic in groundwater in Bihar, or count birds in a wetland.

Environmental studies is broader. It combines environmental science with the social sciences and humanities (economics, sociology, political science, law, ethics, history) to understand why environmental problems arise and how societies can solve them. An environmental studies scholar asks not only *what* is polluting the Yamuna, but also *why* sewage still enters it, *who* bears the cost, and *which* laws, institutions and behaviours could change the situation.

Table 1.1: Environmental science and environmental studies compared

Feature	Environmental science	Environmental studies
Main focus	Natural processes and human impacts on them	Human–environment relationships as a whole
Main disciplines	Physics, chemistry, biology, geology	Natural sciences plus economics, law, sociology, ethics
Typical methods	Measurement, experiment, modelling	Data plus policy analysis, surveys, case studies
Typical question	How much fluoride is in this groundwater?	Why do villagers still drink it, and what policy would help?
Example careers	Hydrologist, ecologist, air-quality analyst	Policy analyst, sustainability manager, environmental lawyer

The two are complementary. Science tells us how much of a pollutant is harmful; studies of law, economics and society tell us how to keep exposure below that level. **Ecology**, the study of the relationships between organisms and their environment, is a core science within both fields (see Chapter 2).

1.2 Multidisciplinary Nature of Environmental Studies

Environmental problems do not respect the boundaries between university departments. Climate change is a problem of physics (how greenhouse gases trap heat), chemistry (how carbon moves

between air, water and rocks), biology (how species respond), economics (who pays to cut emissions), law (international agreements) and ethics (what the present generation owes to future ones). No single discipline can understand it fully. This is why environmental studies is called a **multidisciplinary** field: it draws on knowledge from many disciplines at once.

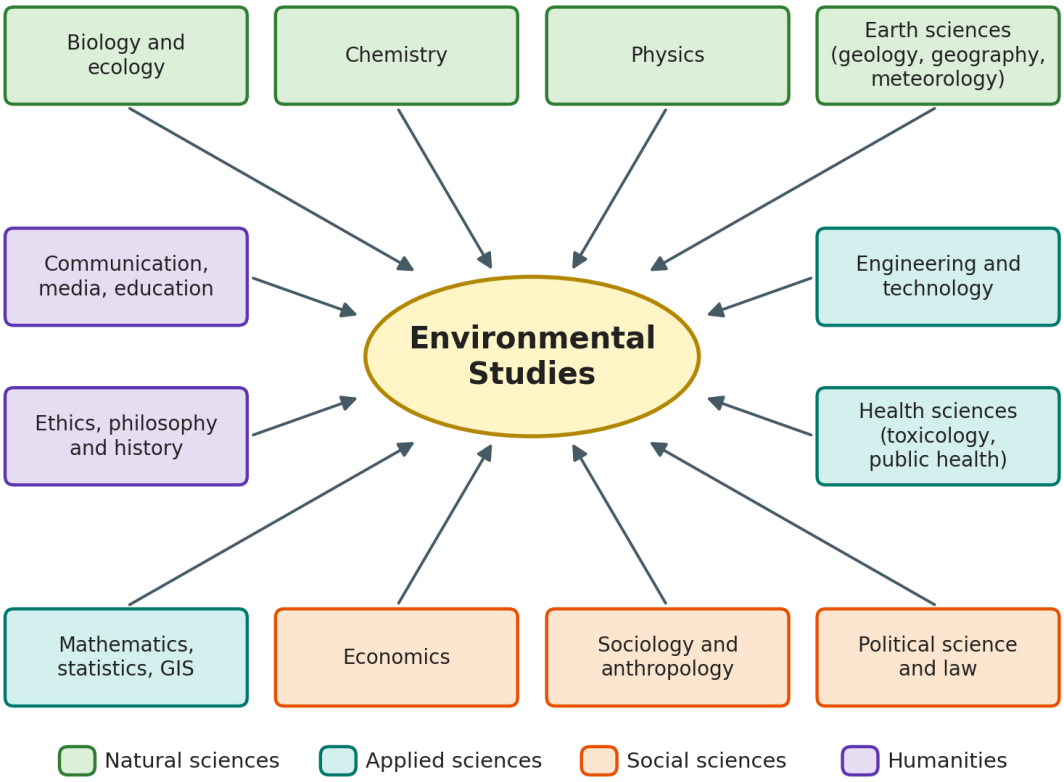


Figure 1.1: *Disciplines that contribute to environmental studies, grouped as natural sciences, applied sciences, social sciences and humanities. Source: Author.*

1.2.1 Contributing Disciplines

The figure above groups the main contributing disciplines by the kind of question each helps to answer.

- **Natural sciences** explain how the environment works. **Biology and ecology** study organisms, food chains and ecosystems; **chemistry** studies the composition of air, water and soil and the behaviour of pollutants; **physics** explains energy flow, radiation and the Earth's heat balance; and the **earth sciences** (geology, geography, meteorology and hydrology) study rocks, landforms, weather, climate and water movement.
- **Applied sciences** turn this understanding into solutions. **Engineering** designs sewage treatment plants, pollution-control devices and renewable energy systems; **health sciences** such as toxicology and epidemiology study how pollutants and diseases affect people; and **mathematics, statistics and geoinformatics** provide data analysis, models, remote sensing and **GIS** (Geographic Information Systems, computer tools that store and map spatial data).
- **Social sciences** explain human causes and consequences. **Economics** studies costs, benefits, market failures and green taxes; **sociology and anthropology** study population, culture, livelihoods and traditional knowledge; and **political science and law** study governance, environmental laws, courts and treaties.

- **Humanities** shape values and communication. **Ethics and philosophy** ask what people owe to nature and future generations; **history** shows how past societies changed their environments; and **communication and education** help change awareness and behaviour.

1.2.2 Multidisciplinary and Interdisciplinary Approaches

A **multidisciplinary approach** places several disciplines side by side, each studying the same problem in its own way. A team studying a polluted lake might include a chemist, a biologist and an economist, each writing a separate report.

An **interdisciplinary approach** goes further and integrates the disciplines, so that the methods and findings of one shape the questions of another. The team might jointly build a model that links the chemistry of phosphorus in the lake to fish catches, fishers' incomes and the rules for discharging sewage. Environmental studies aims for this integrated, interdisciplinary understanding, because real problems have connected causes and need connected solutions.

Case Study: One Problem, Many Disciplines: Crop-Residue Burning in Punjab and Haryana

Every October and November, farmers in Punjab and Haryana burn the straw left after the paddy harvest so that they can sow wheat quickly. The smoke adds to the dense smog over the Indo-Gangetic Plain, including Delhi.

Understanding this problem needs many disciplines. **Chemists** identify the pollutants released ($PM_{2.5}$, carbon monoxide, black carbon). **Meteorologists** explain why calm winds and low winter temperatures trap the smoke near the ground. **Agronomists** note that the window between paddy harvest and wheat sowing is only two to three weeks, partly because a 2009 Punjab law delayed paddy transplanting to save groundwater, pushing the harvest later. **Economists** compare the cost of burning (almost nothing for the farmer) with the cost of machines such as the Happy Seeder or of collecting straw for biomass power. **Sociologists** study why small farmers hesitate to adopt new practices, **lawyers** examine why bans on burning are hard to enforce, and **doctors** measure the rise in breathing problems.

Durable solutions, such as subsidised machines, markets for straw and crop diversification away from water-hungry paddy, come only when these insights are combined. The case also shows how a policy meant to solve one environmental problem (groundwater depletion) can worsen another (air pollution). The Delhi air pollution case is discussed in detail in Chapter 4.

1.2.3 Environmental Studies in Indian Education

Environmental studies is taught to every Indian undergraduate partly because of a court case. In 1991, in a public interest petition filed by advocate M.C. Mehta, the Supreme Court directed that environmental education be made compulsory in schools and asked the University Grants Commission (UGC) to consider making it a compulsory subject in college education. Following further orders in 2003, the UGC introduced a compulsory course in environmental studies for all undergraduate programmes. The University of Delhi's Ability Enhancement Course "Environmental Science: Theory into Practice", for which this book is written, continues that tradition under the Undergraduate Curriculum Framework (UGCF) 2022, with a stronger emphasis on practical and field work.

1.3 Components of Environment

Earth scientists divide the planet's surface environment into four interacting **spheres**: the **atmosphere** (air), the **hydrosphere** (water), the **lithosphere** (rock and soil) and the **biosphere** (life). Together they form the **Earth system**. Energy from the Sun drives this system, while matter such as water, carbon and nitrogen moves continuously from one sphere to another. A change in one sphere usually causes changes in the others.

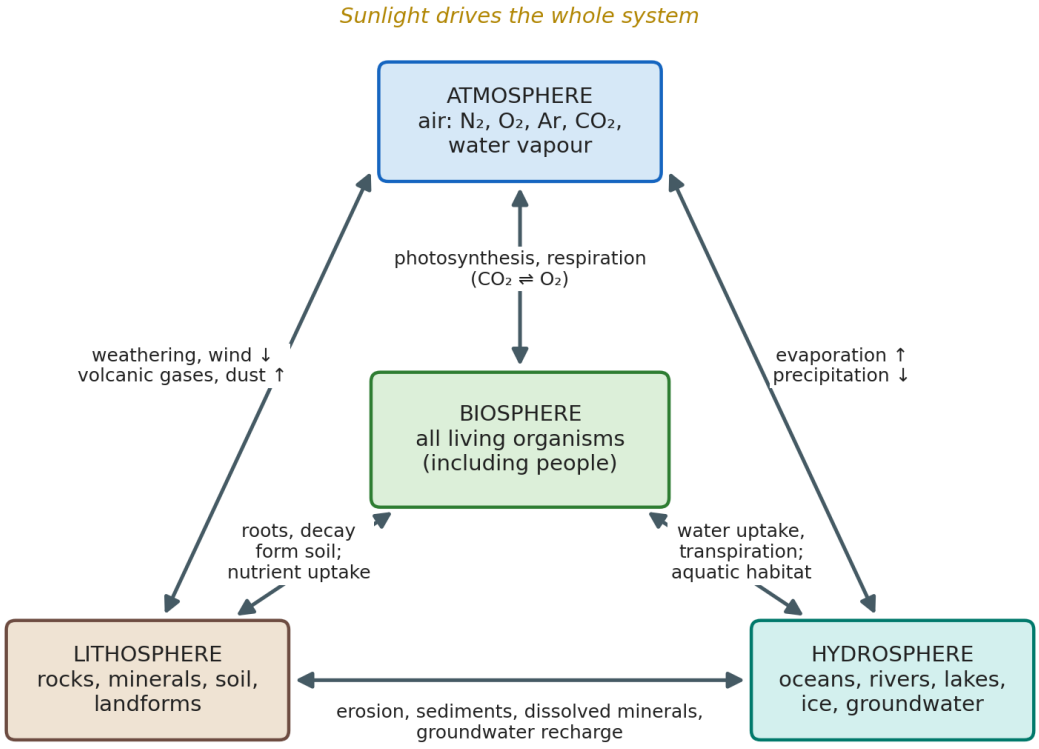


Figure 1.2: *The four components of the environment and some of the exchanges of matter and energy between them. Source: Author.*

1.3.1 Atmosphere

The **atmosphere** is the layer of gases that surrounds the Earth, held in place by gravity. Although it extends hundreds of kilometres upward, it is very thin compared with the size of the planet: about half of its mass lies below 5.5 km, and about 99% lies below 30 km. Without it there would be no breathable oxygen, no rain, no protection from harmful ultraviolet radiation, and an average surface temperature of about −18 °C instead of the present 15 °C.

Composition

Dry air is a mixture of gases whose proportions are nearly constant up to about 80 km. The table below gives the main constituents. In addition, air contains a variable amount of **water vapour** (from almost 0% over deserts to about 4% in the humid tropics) and tiny solid or liquid particles called **aerosols** (dust, sea salt, smoke, pollen).

Table 1.2: Composition of dry air near the Earth's surface (by volume)

Gas	Formula	Share of dry air	Main role
Nitrogen	N_2	78.08%	Dilutes oxygen; source of nitrogen for life after fixation
Oxygen	O_2	20.95%	Respiration and combustion
Argon	Ar	0.93%	Inert gas
Carbon dioxide	CO_2	about 0.042% (423 ppm in 2024)	Photosynthesis; main long-lived greenhouse gas
Neon, helium, krypton	Ne, He, Kr	traces	Inert gases
Methane	CH_4	about 1.9 ppm	Greenhouse gas
Nitrous oxide	N_2O	about 0.34 ppm	Greenhouse gas; harms ozone layer
Ozone	O_3	variable, mostly in stratosphere	Absorbs ultraviolet radiation

The concentration of carbon dioxide has risen from about 280 parts per million (**ppm**) before the Industrial Revolution to a global average of about 423 ppm in 2024 (NOAA), mainly because of burning fossil fuels and clearing forests. Small as this share looks, it controls much of the Earth's heat balance (see Chapter 5).

Layers of the Atmosphere

Scientists divide the atmosphere into layers according to how temperature changes with height, as the figure below shows. The boundaries between layers are called "pauses".

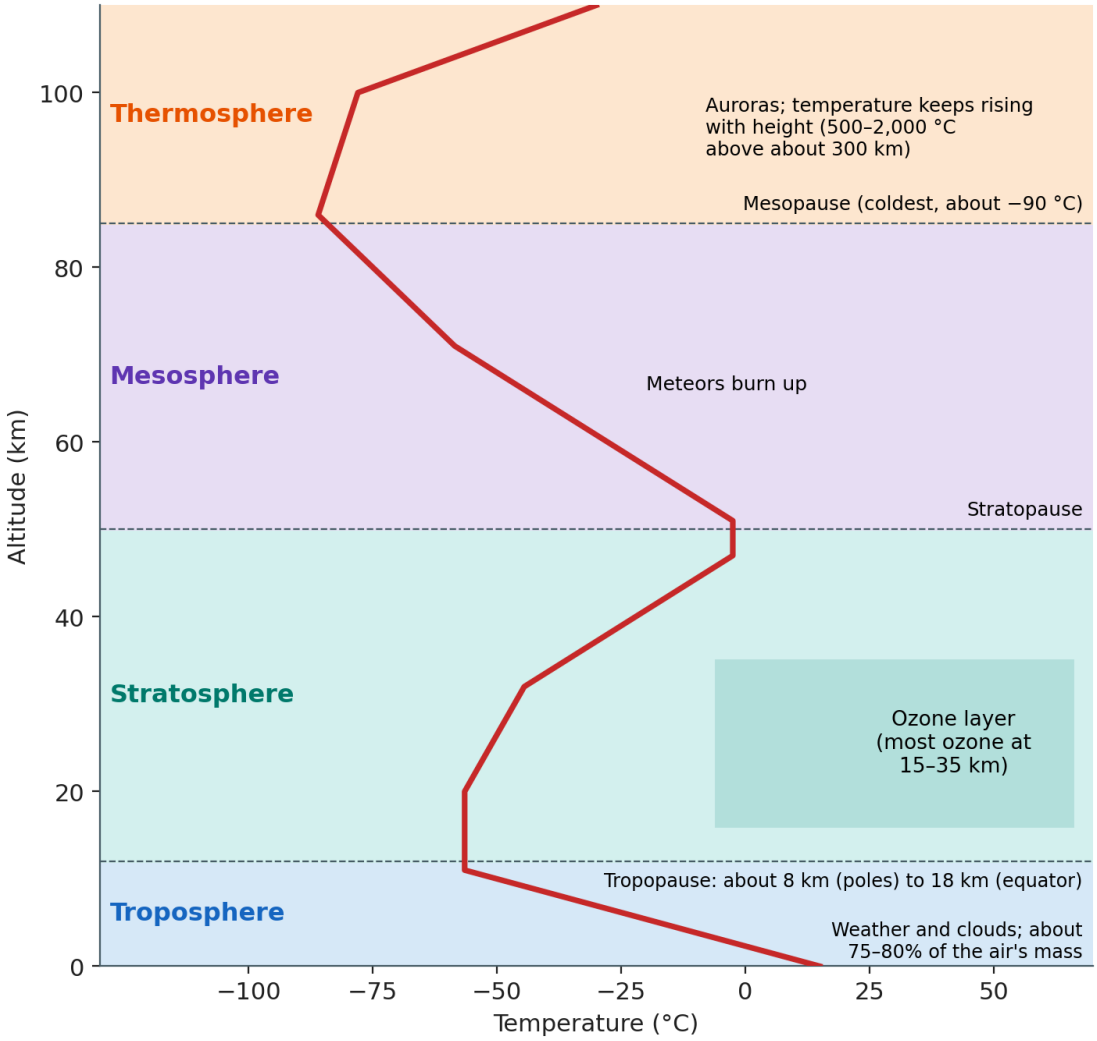


Figure 1.3: *Layers of the atmosphere, showing how air temperature changes with altitude. The temperature line is an approximate mid-latitude average. Source: Author, based on the International Standard Atmosphere.*

1. **Troposphere** (surface to about 8 km at the poles and about 18 km at the equator). Temperature falls with height, by roughly 6.5 °C per km on average. This layer holds about 75–80% of the atmosphere's mass and almost all its water vapour, so all weather (clouds, rain, storms, the monsoon) happens here. Most air pollution is also trapped in the troposphere. Its upper boundary is the **tropopause**.
2. **Stratosphere** (tropopause to about 50 km). Temperature stays steady and then rises with height, because ozone absorbs ultraviolet radiation and warms the air. Most of the atmosphere's ozone lies between about 15 and 35 km, in the **ozone layer**. Because warmer air lies above cooler air, the stratosphere is stable, with little vertical mixing.
3. **Mesosphere** (about 50 to 85 km). Temperature falls again with height, reaching the coldest point in the atmosphere, about −90 °C, at the **mesopause**. Most meteors burn up here.
4. **Thermosphere** (about 85 km to 500–1,000 km). The very thin gas absorbs high-energy solar radiation, so temperature rises sharply, to 500–2,000 °C depending on solar activity, although the air is so thin that it would not feel hot. Auroras occur here, and the International Space Station orbits within it. The electrically charged part of the upper atmosphere (from about 60 km upward), called the **ionosphere**, reflects some radio waves.

5. **Exosphere** (above the thermosphere). The outermost fringe, where gas molecules gradually escape into space.

> **A Thin Shield**
>
> If all the ozone in a column of air were brought down to sea-level pressure and temperature, it would form a layer only about 3 mm thick. This amount, 300 **Dobson units**, is enough to absorb most of the Sun's harmful UV-B radiation. Damage to this thin shield by chlorofluorocarbons (CFCs) is discussed in Chapter 5.

Functions of the Atmosphere

- **Life support**: supplies oxygen for respiration and carbon dioxide for photosynthesis.
- **Temperature regulation**: water vapour, CO_2 and other **greenhouse gases** absorb heat radiated by the Earth's surface and keep the planet warm enough for life (the natural **greenhouse effect**).
- **Protection**: the ozone layer absorbs most ultraviolet radiation; the upper atmosphere burns up meteors.
- **Weather and climate**: the atmosphere carries heat and moisture around the globe; the monsoon brings about three-quarters of India's annual rainfall.
- **Dispersal**: winds spread pollen, seeds and, unfortunately, pollutants.

1.3.2 Hydrosphere

The **hydrosphere** includes all the water on Earth, in oceans, seas, rivers, lakes, glaciers, ice caps, groundwater, soil moisture and water vapour. Water covers about 71% of the Earth's surface.

Distribution of Water

Earth holds about 1.386 billion cubic kilometres of water (USGS), but very little of it is fresh and easily available.

Table 1.3: Distribution of the Earth's water (USGS, based on Shiklomanov)

Store	Share of all water	Share of freshwater
Oceans, seas, saline lakes and saline groundwater	about 97.5%	—
All freshwater	about 2.5%	100%
Glaciers, ice caps and permanent snow	about 1.7%	about 68.7%
Fresh groundwater	about 0.76%	about 30.1%
Surface and other freshwater	about 0.03%	about 1.2%

"Surface and other freshwater" includes lakes, rivers, swamps, soil moisture, water vapour and the water in living things. Thus less than 1% of the Earth's water is liquid freshwater that people can readily use, and most of that is underground. Water moves continuously between these stores through the **hydrological cycle** (evaporation, condensation, precipitation, infiltration and runoff), described in Chapter 2.

Hydrosphere in India

India has about 18% of the world's population but only about 4% of its freshwater resources. Most of its rain falls in four monsoon months, and much of it runs off quickly to the sea. Himalayan glaciers and snow feed the perennial Indus, Ganga and Brahmaputra river systems, while peninsular rivers such as the Godavari, Krishna and Kaveri depend on monsoon rain. India is the world's largest user of groundwater, which supplies most of its irrigation and drinking water. Average annual water availability per person has fallen below 1,700 m^3, the level commonly used to define **water stress**. Water resources, floods, droughts and water conflicts are discussed in Chapter 3.

Functions of the Hydrosphere

The hydrosphere provides **habitat** for aquatic life, acts as a **solvent** that carries nutrients through soils and organisms, and **regulates climate**: the oceans store vast amounts of heat, absorb roughly a quarter of the CO_2 released by human activity, and redistribute heat through currents. Rivers, waves and glaciers **shape the land**, building plains such as the Indo-Gangetic Plain and deltas such as the Sundarbans. Water is also a **resource** for drinking, irrigation, fisheries, hydropower, transport and industry.

1.3.3 Lithosphere

The **lithosphere** (from Greek *lithos*, stone) is the rigid outer shell of the Earth. It consists of the **crust** and the uppermost part of the **mantle**, and is on average about 100 km thick.

Structure

- **Continental crust** forms the continents. It is 30–50 km thick on average and much thicker, up to about 70 km, under the Himalaya and Tibet. It is made largely of lighter, granite-type rocks.
- **Oceanic crust** lies beneath the oceans. It is thinner (about 5–10 km) and made of denser basalt.
- **Uppermost mantle**: the solid rock just below the crust that forms the base of the lithosphere. Beneath it lies a hotter, slowly flowing layer called the **asthenosphere**.

The crust is made mostly of oxygen (about 46% by mass) and silicon (about 28%), with aluminium, iron, calcium, sodium, potassium and magnesium, combined as **minerals**. Minerals combine to form **rocks**, which are classified as igneous (formed from cooled molten rock, like the Deccan basalts), sedimentary (formed from deposited sediments, like sandstone and limestone) and metamorphic (changed by heat and pressure, like marble and slate). Over millions of years rocks change from one type to another through the **rock cycle**.

The lithosphere is broken into large **tectonic plates** that move a few centimetres a year over the asthenosphere. The Indian Plate is moving north at about 5 cm per year and pushing into the Eurasian Plate. This collision, which began about 50 million years ago, raised the Himalaya and still causes earthquakes across northern and north-eastern India.

Soil

Soil is the thin, loose upper layer of the lithosphere in which plants grow. It forms when rocks are broken down by **weathering** (physical, chemical and biological) and mixed with organic matter from dead plants and animals, air and water. Soil is a meeting point of all four spheres. Because a few centimetres of fertile topsoil can take several hundred years to form, soil lost to erosion is, in practical

terms, a non-renewable resource. India's major soil groups include alluvial, black (regur), red, laterite, desert and mountain soils.

Functions of the Lithosphere

The lithosphere is the **foundation** for land ecosystems, farms and cities; the **resource base** for minerals, building materials, fossil fuels and groundwater (see Chapter 3); the source of **soil**; and the site of slow **geological recycling** of materials, including part of the carbon cycle. Earthquakes, volcanoes and landslides also originate in it.

1.3.4 Biosphere

The **biosphere** is the part of the Earth where life exists, together with all the living organisms in it. It reaches from the deepest ocean trenches (about 11 km below sea level) to several kilometres up in the atmosphere, where bacteria and spores drift. Most life, however, is crowded into a narrow zone: the upper layers of soil, the land surface and its vegetation, and the sunlit upper waters of seas and lakes. The biosphere is not separate from the other spheres; it exists where they overlap.

Living things are organised in levels of increasing complexity:

$$\text{Organism} \rightarrow \text{Population} \rightarrow \text{Community} \rightarrow \text{Ecosystem} \rightarrow \text{Biome} \rightarrow \text{Biosphere}$$

A **population** is a group of individuals of one species living in an area; a **community** is all the populations of different species living together; an **ecosystem** is a community together with its physical environment; and a **biome** is a large region with a characteristic climate and vegetation, such as tropical rain forest or desert (see Chapter 2).

Functions of the Biosphere

- **Energy capture**: green plants, algae and some bacteria convert sunlight into chemical energy by **photosynthesis**, feeding almost all other life.

$$6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

- **Biogeochemical cycling**: living things move carbon, nitrogen, phosphorus and sulfur between air, water and rock.
- **Regulating the other spheres**: the oxygen in today's atmosphere was produced by photosynthesis over billions of years; forests influence rainfall; corals and shellfish build limestone.
- **Biodiversity and ecosystem services**: food, timber, medicines, pollination, clean water and soil fertility (see Chapters 2 and 6).

India is one of the world's 17 **megadiverse** countries: with about 2.4% of the world's land area, it holds roughly 7–8% of recorded species and parts of four global biodiversity hotspots (see Chapter 6).

1.3.5 Interactions Among the Components

The four spheres constantly exchange matter and energy, as the figure at the start of this section shows. Some examples:

- **Atmosphere–hydrosphere**: evaporation from oceans supplies water vapour; condensation and precipitation return it. The ocean absorbs CO_2 from the air, which is making seawater more acidic.
- **Atmosphere–lithosphere**: wind and rain weather rocks; volcanoes release gases and ash; dust from the Thar Desert and West Asia is lifted into the air over north India before the monsoon.
- **Hydrosphere–lithosphere**: rivers erode mountains and deposit silt on plains and deltas; rainwater seeps into rock to recharge **aquifers** (underground water-bearing layers of rock or sediment).
- **Biosphere–all spheres**: plants take CO_2 from the air and release oxygen and water vapour (**transpiration**); roots break rocks and hold soil; decomposers return nutrients to soil and water. Human beings, the most powerful part of the biosphere, now alter all three other spheres.

Case Study: The Indian Monsoon: Four Spheres Working Together

The south-west monsoon shows how closely the spheres are linked. In summer the land of the Indian subcontinent heats up faster than the surrounding ocean (**lithosphere–atmosphere** interaction). Hot air rises over land, and moist winds blow in from the Arabian Sea and the Bay of Bengal, carrying water evaporated from the ocean (**hydrosphere–atmosphere**). The Western Ghats and the hills of the north-east force these winds to rise and cool, producing heavy rain on the windward slopes; Mawsynram and Cherrapunji in Meghalaya are among the wettest places on Earth. The rain fills rivers, tanks and aquifers (**hydrosphere–lithosphere**) and triggers the sowing of *kharif* crops on which hundreds of millions of people depend (**biosphere**). Forests return much of this moisture to the air through transpiration, recycling rainfall inland.

Human activity now affects every link in this chain. Aerosol pollution can weaken the monsoon circulation, deforestation reduces moisture recycling, and a warmer atmosphere holds more water vapour, making extreme rainfall more frequent. Because a weak monsoon reduces crop output and rural incomes, the monsoon is an Earth-system process with direct economic and social consequences.

Think and Do: Tracing a Glass of Water

Take a glass of tap water at home or college. Working backwards, trace where it came from (a tap, a tank, a borewell or a river), how it got there, and which spheres it passed through. Then trace where it will go after you use it. Draw your answer as a flow diagram linking the atmosphere, hydrosphere, lithosphere and biosphere. This is a simple version of the interdependence worksheets in Appendix A.

Key Points

- The environment has four interacting components: atmosphere, hydrosphere, lithosphere and biosphere.
- Dry air is about 78% nitrogen, 21% oxygen and 0.93% argon; CO_2 is only about 0.04%, but its concentration has risen by about half since pre-industrial times.
- The atmosphere is layered by temperature: troposphere, stratosphere (with the ozone layer), mesosphere, thermosphere and exosphere.
- Only about 2.5% of the Earth's water is fresh, and most of that is frozen or underground.
- The lithosphere supplies soil, minerals and fuels; soil forms very slowly.
- The biosphere lies where the other three spheres overlap and actively shapes them.

1.4 Scope and Importance of Environmental Studies

1.4.1 Scope

The **scope** of a subject is the range of topics and activities it covers. Because every human activity uses resources and produces waste, the scope of environmental studies is very wide. Its main areas, and the chapters of this book that cover them, are:

1. **Natural resources**: forests, water, soil, minerals and energy, and how to use and share them sustainably (Chapter 3).
2. **Ecosystems**: their structure, functions and services (Chapter 2).
3. **Biodiversity and conservation** (Chapter 6).
4. **Pollution and waste**: sources, effects and control (Chapter 4).
5. **Global environmental change and policy**: climate change, ozone depletion, international agreements and Indian environmental laws (Chapter 5).
6. **Population, society, ethics and movements**: how people affect and are affected by the environment, environmental justice and communication (Chapter 7).
7. **Tools and solutions**: **Environmental Impact Assessment (EIA)**, the study of a project's likely environmental effects before it is approved; clean technologies; and the **circular economy**, an economic model in which products and materials are kept in use for as long as possible instead of being discarded after one use.

1.4.2 Importance

Environmental studies matters at several levels.

For health and well-being. The environment directly shapes human health. According to the *State of Global Air 2024* report, air pollution was linked to about 2.1 million deaths in India in 2021. Unsafe water and poor sanitation still cause diarrhoeal disease, and heatwaves are becoming more frequent and intense. Knowing how pollutants and diseases spread helps people protect themselves and demand better services.

For the economy and livelihoods. A large share of India's workforce depends directly on agriculture, fisheries, forestry and livestock, all sensitive to climate and resource degradation. Environmental damage has real economic costs, while green sectors such as solar energy and recycling create new jobs.

For informed citizenship. Article 51A(g) of the Constitution of India makes it a fundamental duty of every citizen "to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures". Citizens who understand environmental issues can make better choices as consumers and voters, take part in public hearings on projects, use the Right to Information, file complaints with pollution control boards, and participate in local bodies that manage water, forests and waste.

For policy and decision-making. Governments, companies and courts need people who can read scientific evidence and weigh environmental, social and economic trade-offs.

For future generations and other species. The choices made in the coming decades about energy, land and consumption will shape the planet for centuries. Environmental studies encourages a sense of responsibility towards people not yet born and towards non-human life.

1.4.3 Environmental Challenges Facing India

India's environmental challenges are shaped by its large and growing population, rapid economic growth, widespread poverty, high dependence on natural resources and high vulnerability to climate change. The table below summarises some of the most important.

Table 1.4: Selected environmental challenges in India

Challenge	Indicator	Covered in
Population pressure	About 1.46 billion (146 crore) people in 2025 on 2.4% of world land area (UN WPP 2024)	Chapter 7
Air pollution	About 2.1 million deaths linked to air pollution in 2021 (State of Global Air 2024)	Chapter 4
Water stress	About 18% of world population, about 4% of freshwater; falling water tables in the north-west	Chapter 3
Land degradation	About 97.8 million hectares (nearly 30% of land) degraded in 2018–19 (ISRO Desertification Atlas 2021)	Chapter 3
Forest cover	Forest and tree cover 25.17% of geographical area (ISFR 2023), below the 33% goal of the National Forest Policy 1988	Chapter 3
Waste	Over 1.6 lakh tonnes of municipal solid waste a day (CPCB 2020–21)	Chapter 4
Biodiversity loss	Habitat loss, poaching, invasive species and human–wildlife conflict	Chapter 6
Climate change	Heatwaves, erratic monsoon, glacier retreat, sea-level rise along a long coastline	Chapter 5

At the same time, India's per-person environmental impact is low by world standards. Its carbon dioxide emissions per person are less than half the world average, even though its total emissions are the third largest of any country. This tension, between the need to lift millions out of poverty and the

need to protect the environment, runs through all of Indian environmental policy and is a central theme of this book.

1.4.4 Careers in the Environmental Field

Environmental knowledge opens careers for students of every discipline.

- **Science and technology**: analyst in pollution control boards and laboratories; hydrologist; ecologist; wildlife biologist; GIS specialist.
- **Engineering and industry**: environmental engineer; renewable energy planner; energy auditor; environment, health and safety (EHS) manager.
- **Business and finance**: sustainability or ESG (environmental, social and governance) analyst; carbon-market specialist; corporate social responsibility (CSR) manager. Since 2022–23, India's top 1,000 listed companies by market value have had to publish a **Business Responsibility and Sustainability Report** (BRSR) under SEBI rules, creating demand for people who can measure and report environmental performance.
- **Government and law**: Indian Forest Service; officers in the Ministry of Environment, Forest and Climate Change and state departments; environmental lawyers; urban and regional planners.
- **Social sector and communication**: NGOs and community organisations; environmental educators; journalists, writers, documentary makers and designers.
- **Research and teaching**: universities and institutes such as the Wildlife Institute of India and the Botanical and Zoological Surveys of India.

1.5 Sustainability and Sustainable Development

1.5.1 The Concept of Sustainability

To **sustain** something is to keep it going. **Sustainability** is the ability of a system, whether a forest, a fishery, a farm, a city or the whole Earth, to continue functioning and providing for its users over a long period without running down the resources and conditions it depends on.

The idea is old. Indian villages that harvested only the dead wood from a common forest, or fishers who stopped fishing during the breeding season, were practising sustainability long before the word existed. In Europe, the German forester Hans Carl von Carlowitz argued in 1713 that forests should be cut no faster than they could regrow, so that timber supply would be "sustained" indefinitely. The same principle applies to any **renewable resource**: harvest should not exceed the rate of regeneration.

A simple way to see sustainability is to think of nature as a bank account. **Natural capital** is the stock of natural resources and ecosystems (forests, soils, fish stocks, aquifers, a stable climate). The **interest** is the flow of goods and services it produces each year (timber growth, new fish, groundwater recharge). Living sustainably means living on the interest without spending the capital. When Punjab pumps more groundwater each year than the rain replaces, it is spending its capital, and the water table falls.

1.5.2 Sustainable Development

The Brundtland Definition

In 1983 the United Nations set up the **World Commission on Environment and Development (WCED)**, chaired by Gro Harlem Brundtland, former Prime Minister of Norway. Its 1987 report, *Our Common Future*, popularly called the **Brundtland Report**, gave the most widely used definition:

Sustainable development is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

The report explained that the definition contains two key ideas:

- **Needs**, in particular the essential needs of the world's poor, "to which overriding priority should be given". Sustainable development is therefore not just about protecting nature; it is also about ending poverty.
- **Limitations** imposed by the state of technology and social organisation on the environment's ability to meet present and future needs.

Sustainability and sustainable development are closely related. Sustainability describes the goal, a state that can last; sustainable development describes the process of improving human well-being while moving towards that goal.

Principles of Sustainable Development

Several principles follow from the Brundtland idea and have been adopted in international law and in Indian court judgments:

- **Intergenerational equity**: the present generation must pass on to future generations an environment at least as healthy and productive as the one it received.
- **Intragenerational equity**: fairness among people alive today, between rich and poor, and between developed and developing countries.
- **Precautionary principle**: where there is a threat of serious or irreversible damage, lack of full scientific certainty should not be used as a reason to postpone measures to prevent environmental harm.
- **Polluter pays principle**: those who cause pollution should bear the costs of preventing it and repairing the damage.
- **Public participation**: people affected by decisions should have access to information and a voice in decision-making.

In *Vellore Citizens' Welfare Forum v. Union of India* (1996), the Supreme Court held that the precautionary principle and the polluter pays principle are part of the environmental law of India.

1.5.3 Three Pillars and the Nested Model

The most common way of picturing sustainable development is as three **pillars** or dimensions that must be balanced:

- **Environmental sustainability**: use renewable resources no faster than they regenerate, use non-renewable resources efficiently, keep wastes within the capacity of nature to absorb them, and conserve biodiversity.

- **Social sustainability**: meet basic needs (food, water, health, education, housing), reduce inequality, protect cultures and human rights, and ensure participation.
- **Economic sustainability**: maintain productive economies that provide decent livelihoods and use resources efficiently over the long term, not just maximise short-term growth.

The idea of a **triple bottom line** (people, planet, profit), proposed by John Elkington in 1994, applies the same three dimensions to businesses, which are asked to report social and environmental performance alongside profits.

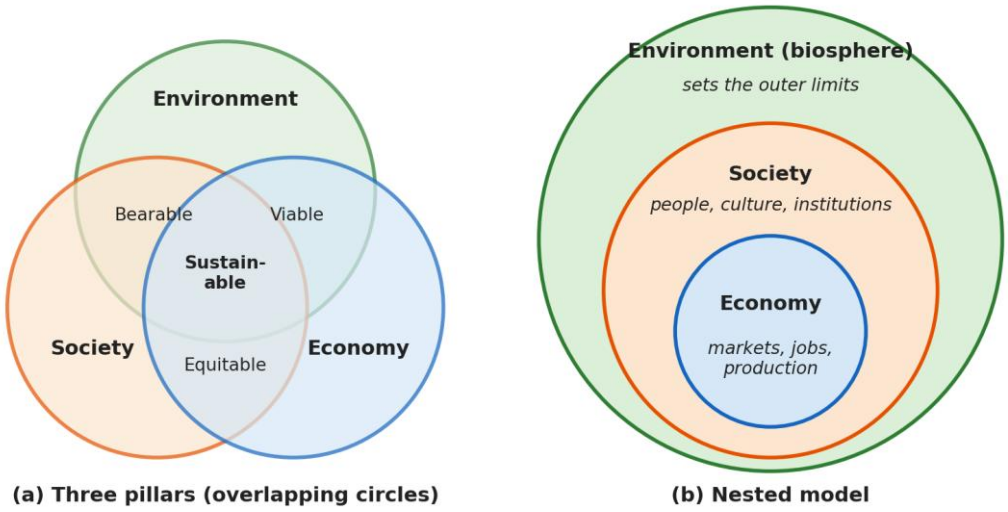


Figure 1.4: *Two ways of picturing sustainable development: (a) three overlapping pillars, where development is sustainable only where all three overlap; (b) the nested model, in which the economy exists within society and both depend on the environment. Source: Author.*

In panel (a), development that satisfies only two dimensions is merely "bearable" (environment and society), "viable" (environment and economy) or "equitable" (society and economy); only the central overlap is sustainable.

Critics point out that the three-pillar picture can suggest that the three dimensions are equal and can be traded against one another, for example, that a large economic gain can justify a large environmental loss. The **nested model** (panel b) corrects this. The economy is part of society, and society exists within, and depends entirely on, the biosphere. An economy cannot exist without a society, and no society can survive without clean air, water, fertile soil and a stable climate. The environment therefore sets the outer limits within which social and economic goals must be pursued.

1.5.4 Weak and Strong Sustainability

The question of whether natural capital can be replaced by human-made capital divides two schools of thought.

Weak sustainability holds that what matters is the total stock of capital (natural, human-made and human) passed on to future generations. If a forest is cut down and the income is invested in schools or factories of equal value, the total has not fallen; natural and human-made capital are treated as **substitutes**.

Strong sustainability holds that much natural capital cannot be replaced. No investment can substitute for a stable climate, the ozone layer or an extinct species. This **critical natural capital** must

be maintained in its own right. The ecological economist Herman Daly suggested three simple rules for strong sustainability:

1. Harvest renewable resources no faster than they regenerate.
2. Release wastes no faster than the environment can safely absorb them.
3. Use non-renewable resources no faster than renewable substitutes are developed.

Table 1.5: Weak and strong sustainability compared

Feature	Weak sustainability	Strong sustainability
Key assumption	Natural and human-made capital are interchangeable	Critical natural capital cannot be replaced
What must be maintained	Total capital stock	Natural capital itself, plus total capital
Example judgement	Mining a forest is acceptable if revenues are reinvested well	Some forests, rivers and species must be protected regardless of revenue
Associated discipline	Mainstream (neoclassical) economics	Ecological economics

In practice, most environmental policy lies between the two. Indian law, for example, allows forest land to be diverted for projects but requires **compensatory afforestation** and payments into a fund (an approach closer to weak sustainability), while banning all hunting of species listed in Schedule I of the Wildlife (Protection) Act (closer to strong sustainability).

1.5.5 Carrying Capacity and Limits

Carrying capacity is the maximum population of a species that an environment can support indefinitely without being degraded. When a population of deer, for example, grows beyond the carrying capacity of a grassland, overgrazing damages the vegetation and the population eventually crashes (see Chapter 2).

For human beings, carrying capacity is harder to define because technology and trade can increase it (through irrigation, fertilisers or imported food) and consumption can reduce it. A widely used way of expressing human impact is the **IPAT equation**, proposed by Paul Ehrlich and John Holdren in 1971:

$$I = P \times A \times T$$

Here **I** is environmental impact, **P** is population, **A** is affluence (consumption per person) and **T** is technology (impact per unit of consumption). The equation shows that a small population with very high consumption can have as large an impact as a large population with low consumption, and that cleaner technology can reduce impact even as population and incomes grow.

Two influential ideas have applied the concept of limits to the whole planet:

- ***The Limits to Growth*** (1972), a report for the Club of Rome by Donella Meadows and colleagues, used computer models to show that if growth in population, industry, pollution and resource use continued unchecked, the world would reach its limits within about a century, followed by decline.

- **Planetary boundaries**, proposed in 2009 by Johan Rockström and colleagues at the Stockholm Resilience Centre, identify nine Earth-system processes (including climate change, biodiversity loss, nitrogen and phosphorus flows, freshwater use, land-system change and ocean acidification) that define a "safe operating space for humanity". The *Planetary Health Check 2025* concluded that seven of the nine boundaries have now been crossed, with only stratospheric ozone and atmospheric aerosol loading still within safe limits at the global scale.

1.5.6 Ecological Footprint

The **ecological footprint**, developed by Mathis Wackernagel and William Rees in the 1990s, measures how much biologically productive land and water area a person, city or country needs to produce the resources it consumes and to absorb its wastes, especially carbon dioxide. It is expressed in **global hectares (gha)**, hectares of land with world-average productivity. The footprint is compared with **biocapacity**, the productive area actually available.

If the footprint is smaller than biocapacity, there is an **ecological reserve**; if it is larger, there is an **ecological deficit**, met by importing resources, depleting stocks such as groundwater, or accumulating wastes such as CO_2 in the atmosphere.

According to the Global Footprint Network, humanity as a whole currently uses resources about 1.8 times as fast as the Earth's ecosystems can regenerate them, as if it had 1.8 Earths. **Earth Overshoot Day**, the date each year by which humanity has used up nature's budget for the year, fell on 24 July in 2025.

India's footprint per person is low, roughly 1.2 gha, less than half the world average. But India's biocapacity per person is even lower, about 0.4 gha, because so many people share limited land. India therefore runs an ecological deficit: its total footprint is roughly two to three times its biocapacity. A typical resident of the United States, by contrast, has a footprint several times larger than an Indian's. The ecological footprint thus highlights both the limits of the planet and the large inequalities in how its resources are used. The related concept of the **carbon footprint**, which counts only greenhouse gas emissions, is discussed in Chapter 7.

1.5.7 The Sustainable Development Goals

From the MDGs to the SDGs

International efforts to put sustainable development into practice include *Agenda 21*, the action plan adopted at the Rio Earth Summit in 1992, and the eight **Millennium Development Goals (MDGs)**, adopted in 2000 for the period up to 2015. The MDGs focused mainly on poverty, hunger, health and education in developing countries, with only one goal on environmental sustainability. They made real progress but were criticised for neglecting the environment, inequality and the responsibilities of rich countries.

At the Rio+20 conference in 2012, countries agreed to create a broader set of goals. On 25 September 2015, all 193 member states of the United Nations adopted ***Transforming Our World: the 2030 Agenda for Sustainable Development***. At its heart are the **Sustainable Development Goals (SDGs)**: 17 goals with 169 targets, measured by more than 230 indicators, to be achieved by 2030.

The SDGs differ from the MDGs in three important ways:

- They are **universal**: they apply to all countries, rich and poor, not just to developing countries.

- They are **integrated**: economic, social and environmental goals are treated as interdependent. Progress on clean energy (SDG 7), for example, supports health (SDG 3) and climate action (SDG 13).
- They pledge to **leave no one behind**, giving priority to the poorest and most marginalised groups.

The 2030 Agenda groups its goals under five themes, often called the **five Ps**: People, Planet, Prosperity, Peace and Partnership.

Table 1.6: The 17 Sustainable Development Goals

Goal	Short title	Aim (summary)	An Indian link
1	No Poverty	End poverty in all its forms everywhere	National Multidimensional Poverty Index
2	Zero Hunger	End hunger, improve nutrition, promote sustainable agriculture	National Food Security Act 2013
3	Good Health and Well-being	Healthy lives and well-being for all ages	Ayushman Bharat; air-quality health risks
4	Quality Education	Inclusive, equitable quality education	Environmental education in schools and colleges
5	Gender Equality	Equality and empowerment of women and girls	Ujjwala clean cooking gas for women
6	Clean Water and Sanitation	Water and sanitation for all, managed sustainably	Jal Jeevan Mission; Swachh Bharat Mission
7	Affordable and Clean Energy	Affordable, reliable, sustainable, modern energy	500 GW non-fossil capacity target for 2030
8	Decent Work and Economic Growth	Inclusive growth and decent jobs	Green jobs in solar and recycling
9	Industry, Innovation and Infrastructure	Resilient infrastructure, sustainable industry	Cleaner technology in MSMEs
10	Reduced Inequalities	Reduce inequality within and among countries	Forest Rights Act 2006
11	Sustainable Cities and Communities	Safe, resilient, sustainable cities	National Clean Air Programme
12	Responsible Consumption and Production	Sustainable patterns of consumption and production	Mission LiFE; Extended Producer Responsibility
13	Climate Action	Urgent action on climate change and its impacts	National Action Plan on Climate Change
14	Life Below Water	Conserve oceans, seas and marine resources	Coastal Regulation Zone rules; mangrove protection
15	Life on Land	Protect forests and land ecosystems, halt biodiversity loss	Project Tiger; restoring degraded land
16	Peace, Justice and Strong Institutions	Just, peaceful, inclusive societies and institutions	National Green Tribunal; Right to Information

Goal	Short title	Aim (summary)	An Indian link
17	Partnerships for the Goals	Strengthen global partnership and means of implementation	International Solar Alliance

At least six goals (6, 7, 12, 13, 14 and 15) deal directly with the environment, and most of the others depend on it. No country can end hunger (SDG 2) if its soils are degraded and its monsoon unreliable.

Progress

Global progress is slow. The *Sustainable Development Report 2025* (UN Sustainable Development Solutions Network) estimated that only about 17% of SDG targets worldwide were on track to be met by 2030. India ranked 99th of 167 countries in that report, entering the top 100 for the first time; it had made strong progress on goals such as poverty, energy access and sanitation but lagged on several environmental goals. Within India, NITI Aayog tracks the performance of states and union territories through the **SDG India Index**. India's SDG framework, the SDG India Index and the National Action Plan on Climate Change are discussed in Chapter 5.

> **Think and Do: Analysing a Country's SDG Performance**
>
> Choose any country (for example, India, Bangladesh, Sweden or Brazil). Using the free SDG dashboards of the Sustainable Development Report or the UN SDG database, record its status (on track, moderately improving, stagnating or decreasing) for SDGs 6, 7, 12, 13, 14 and 15. Identify the goal on which it performs best and worst, and suggest one reason for each. This is one of the Unit 1 practicals described in Appendix A.

1.5.8 Criticisms of Sustainable Development

- **Vagueness**: the term is so broad that almost anyone can claim to support it. Companies sometimes use it for **greenwashing**, giving a misleading impression that their products or activities are environmentally friendly.
- **The growth question**: some scholars argue that endless economic growth on a finite planet is impossible, and that rich countries need to reduce material consumption, not just make it more efficient. Others reply that growth is still essential to end poverty in countries like India and that technology can **decouple** growth from environmental damage, so that the economy grows while resource use and pollution fall.
- **Equity between countries**: developing countries argue that rich countries, which caused most past pollution and greenhouse gas emissions, should bear a larger share of the costs. This is reflected in the principle of **common but differentiated responsibilities**, discussed in Chapter 5.
- **Implementation gap**: commitments are often not matched by funding, laws or enforcement.

Despite these criticisms, sustainable development remains the main framework for discussing environment and development together.

1.5.9 Sustainability in Practice

Sustainability can be applied at every scale. For individuals and households, India's **Mission LiFE** (Lifestyle for Environment), launched in October 2022, promotes simple actions such as saving water and energy, avoiding single-use plastic and cutting food waste. For industries, tools include **life-cycle assessment** (measuring a product's impact from raw-material extraction to disposal), environmental management systems such as ISO 14001, cleaner production, and **Extended Producer Responsibility (EPR)**, which makes producers responsible for collecting and recycling their packaging, batteries and electronic goods. Cities can promote public transport, rainwater harvesting and waste segregation; farmers can diversify crops, irrigate efficiently and adopt organic or natural farming.

Case Study: Sikkim: India's First Fully Organic State

In 2003 the Sikkim Legislative Assembly resolved to convert the state's agriculture to organic farming. Over the following years the state phased out chemical fertilisers and pesticides, banned their sale, trained farmers in composting and biological pest control, and certified farmland as organic. In January 2016 Sikkim was declared India's first fully organic state, with about 75,000 hectares of farmland under certified organic management.

The programme illustrates all three pillars of sustainability. **Environmentally**, it reduced chemical pollution of soils and the state's rivers and protected its rich biodiversity, including many orchids and medicinal plants. **Socially**, it promoted healthier food and drew on traditional farming knowledge. **Economically**, it aimed to earn premium prices for organic products such as large cardamom and ginger and to support organic and eco-tourism; Sikkim received the Future Policy Award from the UN Food and Agriculture Organization and partner organisations in 2018.

The case also shows the difficulties. Yields of some crops fell in the early years, farmers needed markets and fair prices for certified produce, and much of the state's food is still imported from other states. Sustainability transitions need support, patience and trade-offs, not just a single decision.

Think and Do: A Sustainability Framework for an Industry

Choose an industry common near you, such as a brick kiln, a dairy, a textile dyeing unit or a hotel. Using a worksheet with three columns (Environmental, Social, Economic), list its main inputs (raw materials, water, energy, labour) and outputs (products, wastes, emissions). For each column, identify one problem and one practical improvement, for example switching a brick kiln to zig-zag firing to save coal and reduce smoke. Appendix A gives a full worksheet.

Key Points

- Sustainable development is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (Brundtland Report, 1987).
- It rests on environmental, social and economic pillars; the nested model shows the economy and society depending on the environment.
- Weak sustainability allows natural capital to be replaced by human-made capital; strong sustainability insists that critical natural capital be preserved.
- Carrying capacity, planetary boundaries and the ecological footprint all express the idea that the Earth has limits.
- The 17 SDGs (2015–2030) are universal, integrated goals for people, planet, prosperity, peace and partnership.

1.6 Brief History of Environmentalism: The Global Story

Environmentalism is the set of ideas and social movements that seek to protect the natural environment and human health from harmful human activity. Its concerns have shifted over time: from protecting forests and wild places, to fighting pollution, to the global problems of climate change and biodiversity loss, and to questions of justice about who bears environmental harm. The timeline below marks the main milestones described in this section.

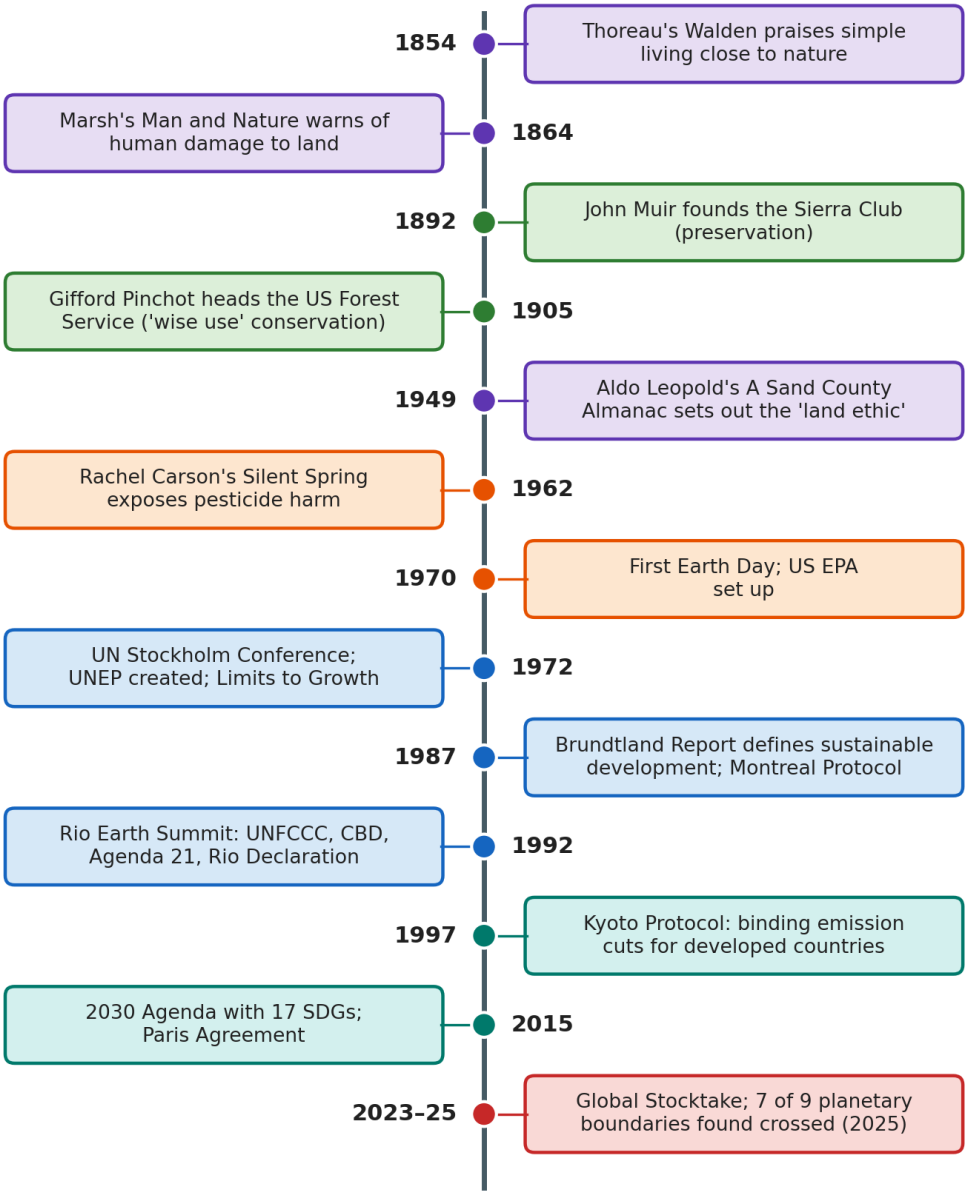


Figure 1.5: *Major milestones in the history of environmentalism worldwide. Source: Author.*

1.6.1 Early Roots

Concern for nature is ancient. Many traditional societies, including the forest-dwelling communities of India, regulated hunting, fishing and tree-cutting through custom and religion. Modern environmentalism, however, grew mainly as a response to the **Industrial Revolution** of the eighteenth and nineteenth centuries, which brought coal smoke, polluted rivers, crowded cities and rapid clearing of forests.

1.6.2 Preservation and Conservation

In the nineteenth-century United States, two different attitudes to nature developed.

- The **preservationist** view held that wild nature has value in itself and should be protected from use. The writer **Henry David Thoreau** described a simple life in the woods in *Walden* (1854) and later wrote, in his essay "Walking", that "in Wildness is the preservation of the World". The

Scottish-American naturalist **John Muir** campaigned for Yosemite to become a national park (1890) and founded the Sierra Club in 1892.

- The **conservationist** view, associated with **Gifford Pinchot**, the first chief of the US Forest Service (1905), held that natural resources should be managed scientifically for "the greatest good of the greatest number for the longest time". Conservation meant wise use, not non-use.

Earlier, in *Man and Nature* (1864), the American diplomat **George Perkins Marsh** had argued that human beings were transforming the Earth, often destructively, and cited deforestation around the Mediterranean as a warning. The two views clashed over the Hetch Hetchy valley in Yosemite, which Muir wanted preserved and Pinchot supported damming for San Francisco's water supply; the dam was approved in 1913. The tension between preservation and use still runs through debates over national parks, dams and mining in India and elsewhere.

In *A Sand County Almanac* (1949), the American forester and ecologist **Aldo Leopold** proposed a **land ethic**: a duty of care that extends from people to "soils, waters, plants, and animals, or collectively: the land". His idea that humans are plain members, not conquerors, of the land community became a foundation of environmental ethics (see Chapter 7).

1.6.3 The Rise of Modern Environmentalism

After the Second World War, rapid industrial and agricultural growth brought new problems: synthetic pesticides, toxic chemicals, nuclear fallout and severe urban smog, such as the London smog of December 1952, which is estimated to have caused around 4,000 deaths within days and several thousand more in the following months.

In 1962 the American biologist **Rachel Carson** published ***Silent Spring***. She showed how DDT and other persistent pesticides accumulated in food chains, killing birds and fish and threatening human health. Despite attacks from the chemical industry, the book changed public opinion and led to restrictions on DDT in many countries, including a ban on its agricultural use in the United States in 1972. By linking environmental damage to human health, *Silent Spring* is often said to have started the modern environmental movement.

The movement grew quickly. On 22 April 1970 about 20 million Americans took part in the first **Earth Day**, and the US Environmental Protection Agency was set up the same year. Other industrialised countries passed their first major pollution laws in the same period.

1.6.4 Stockholm 1972 and Global Environmental Governance

The **United Nations Conference on the Human Environment**, held in Stockholm from 5 to 16 June 1972, was the first major international meeting on the environment. It adopted the Stockholm Declaration, whose principles affirmed both the right to a healthy environment and the responsibility to protect it, and it led to the creation of the **United Nations Environment Programme (UNEP)**, headquartered in Nairobi. The opening day, 5 June, is now celebrated as **World Environment Day**. In the same year, *The Limits to Growth* warned of the physical limits to growth on a finite planet.

Over the next two decades, environmental problems were increasingly seen as global:

- **1985–1987**: scientists reported a seasonal "hole" in the ozone layer over Antarctica (1985). The **Montreal Protocol** (1987) on substances that deplete the ozone layer followed and is widely regarded as the most successful environmental treaty (see Chapter 5).

- **1987**: the **Brundtland Report** introduced sustainable development as the bridge between environmental protection and economic development.
- **1988**: the World Meteorological Organization and UNEP set up the **Intergovernmental Panel on Climate Change (IPCC)** to assess climate science.
- **1992**: the **UN Conference on Environment and Development**, or **Earth Summit**, in Rio de Janeiro adopted the Rio Declaration, *Agenda 21* and the Forest Principles, and opened for signature the UN Framework Convention on Climate Change (UNFCCC) and the Convention on Biological Diversity (CBD).
- **1997**: the **Kyoto Protocol** set binding emission-reduction targets for developed countries; it entered into force in 2005.
- **2015**: the 2030 Agenda and SDGs were adopted, and the **Paris Agreement** committed almost all countries to holding global warming "well below 2 °C" above pre-industrial levels and pursuing efforts to limit it to 1.5 °C, through nationally determined contributions (NDCs) updated every five years.

These agreements are examined in detail in Chapter 5.

1.6.5 Environmental Justice and New Movements

From the 1980s, the **environmental justice** movement highlighted that pollution and environmental hazards fall most heavily on poor and marginalised communities, and that these communities often have the least say in decisions. Movements in the Global South, including many in India, showed that environmentalism is often a struggle by poor people to defend the forests, rivers and land on which their livelihoods depend, an idea that historians Ramachandra Guha and Joan Martinez-Alier called the "environmentalism of the poor". Since 2018, youth movements such as Fridays for Future have brought a new generation into climate activism.

1.6.6 The Anthropocene

In 2000 the atmospheric chemist Paul Crutzen, who shared the 1995 Nobel Prize in Chemistry for work on the ozone layer, and the biologist Eugene Stoermer proposed that the Earth had entered a new geological epoch, the **Anthropocene** (from Greek *anthropos*, human), in which human activity has become the dominant force shaping the planet. Evidence includes the rise in atmospheric CO_2, global spread of plastics and concrete, radioactive fallout from nuclear tests, extinction rates far above the natural background, and the doubling of reactive nitrogen in the environment. The sharp rise in almost every measure of human activity and environmental change after about 1950 is called the **Great Acceleration**.

A working group of geologists proposed formally recognising the Anthropocene as an epoch beginning in the early 1950s. In March 2024, however, the International Union of Geological Sciences confirmed a vote of its Subcommission on Quaternary Stratigraphy rejecting the proposal, so the Anthropocene is not part of the official geological time scale. The term remains widely used, informally, to describe the present era of human-dominated environmental change.

1.7 Brief History of Environmentalism in India

India's environmental history differs from that of Western countries in an important way. In the West, early environmentalism was largely about protecting wilderness for recreation and its own sake. In

India, most environmental conflicts have involved people whose daily survival depends on forests, water, fish and land: peasants, tribal communities, pastoralists and fishers. Indian environmentalism has therefore always been closely linked to questions of livelihood, rights and social justice. The timeline below summarises the milestones discussed in this section.

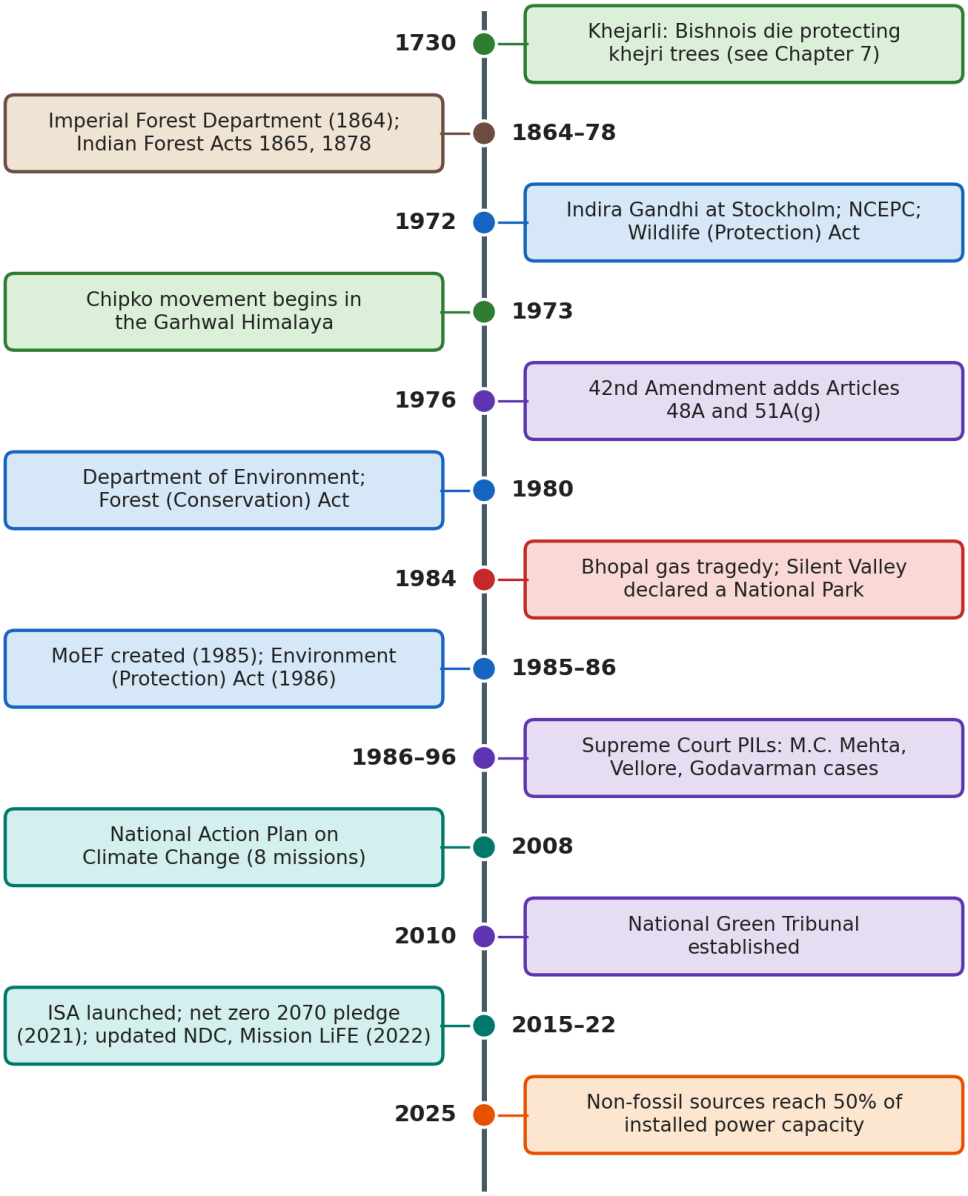


Figure 1.6: *Major milestones in the history of environmentalism in India. Source: Author.*

1.7.1 Traditions of Conservation

Indian religious and cultural traditions contain many ideas of respect for nature. The Prithvi Sukta of the Atharva Veda describes the Earth as a mother: "*Mata bhumih putro aham prithivyah*" (the Earth is my mother and I am her son). Jainism and Buddhism stress **ahimsa**, non-violence towards all living beings. Around the third century BCE, the Mauryan emperor **Ashoka** listed animals that were not to be killed and forbade the burning of forests without reason in his fifth pillar edict. Kautilya's *Arthashastra* describes officials responsible for forests and protected forest areas.

Communities across India protected **sacred groves**, patches of forest dedicated to local deities, such as the *orans* of Rajasthan, the *devrais* of Maharashtra, the *kavus* of Kerala and the *law kyntang* of Meghalaya (see Chapter 6). They built water-harvesting systems suited to local conditions: *johads* and *kunds* in Rajasthan, stepwells in Gujarat, *eris* (tanks) in Tamil Nadu, *kuhls* (diversion channels) in Himachal Pradesh and bamboo drip irrigation in Meghalaya. In 1730 at Khejarli village near Jodhpur, 363 members of the **Bishnoi** community, led by Amrita Devi, gave their lives to stop the felling of khejri trees (see Chapter 7).

1.7.2 Colonial Forestry and Resistance

British rule changed the relationship between people and forests. The expansion of railways from the 1850s created huge demand for timber for sleepers and fuel, and the colonial state wanted to control forests as a source of revenue. The **Imperial Forest Department** was set up in 1864 under the German forester Dietrich Brandis. The **Indian Forest Acts** of 1865 and 1878, and later the Indian Forest Act of 1927 (still in force), allowed the state to declare large areas as **reserved forests**, where local people's customary rights to collect fuelwood, graze cattle, hunt and practise shifting cultivation were curtailed or extinguished.

This "scientific forestry" favoured commercially valuable trees such as teak, sal and deodar, and later chir pine, often at the expense of the mixed forests that met villagers' needs. It provoked widespread resistance: the Bastar rebellion of 1910, forest protests in Kumaon in the 1910s and 1920s, and forest satyagrahas during the freedom struggle in the 1920s and 1930s, such as the protest at Tilari in Tehri Garhwal in 1930, where state forces fired on villagers. Historians Madhav Gadgil and Ramachandra Guha, in *This Fissured Land* (1992), argued that colonial forestry broke the link between communities and the forests they had long managed.

The colonial period also saw the first wildlife laws, such as the Madras Wild Elephant Preservation Act of 1873, and India's first national park, Hailey National Park, established in 1936 and now called Jim Corbett National Park.

1.7.3 People's Movements after Independence

After independence, India's planners emphasised rapid industrialisation, large dams and the expansion of agriculture. The National Forest Policy of 1952 continued the colonial emphasis on commercial forestry. From the 1970s, a series of people's movements challenged this model of development:

- The **Chipko movement** began in 1973 in the Garhwal Himalaya, when villagers, many of them women, hugged trees to stop commercial felling (see Chapter 7).
- The **Silent Valley** campaign in Kerala (1970s–1984) stopped a hydroelectric dam in a rain forest and led to the area being declared a national park (see Chapter 6).
- The **Appiko movement** (1983) in the Western Ghats of Karnataka and the **Narmada Bachao Andolan** (from the late 1980s) against large dams on the Narmada are discussed in Chapters 3 and 7.

1.7.4 Stockholm 1972 and the Building of Institutions

The Stockholm Conference of 1972 had a strong influence on Indian policy. Prime Minister **Indira Gandhi** was the only head of government, apart from the host, to attend. In her address she argued

that environmental protection could not be separated from the fight against poverty, asking: "Are not poverty and need the greatest polluters?" This statement became a reference point for the developing world's position that environment and development must be addressed together.

In the same year, India set up a **National Committee on Environmental Planning and Coordination (NCEPC)** and passed the **Wildlife (Protection) Act, 1972**. Project Tiger followed in 1973 and the **Water (Prevention and Control of Pollution) Act** in 1974, which created the central and state pollution control boards.

In 1976 the **42nd Amendment** to the Constitution made environmental protection a constitutional responsibility. It added:

- **Article 48A** (a Directive Principle of State Policy): "The State shall endeavour to protect and improve the environment and to safeguard the forests and wild life of the country."
- **Article 51A(g)** (a Fundamental Duty of citizens), quoted earlier in this chapter.

The amendment also moved "forests" and "protection of wild animals and birds" from the State List to the Concurrent List, so that both the Centre and the states could legislate on them.

On the recommendation of a committee chaired by N.D. Tiwari, the Government of India created a separate **Department of Environment** in November 1980. The Forest (Conservation) Act, 1980, required central approval before forest land could be used for non-forest purposes. In 1985 the department became the **Ministry of Environment and Forests (MoEF)**, which was renamed the **Ministry of Environment, Forest and Climate Change (MoEFCC)** in 2014. The Air (Prevention and Control of Pollution) Act followed in 1981. The details of these laws and their recent amendments are covered in Chapter 5.

1.7.5 Bhopal and the Legislative Response

On the night of 2–3 December 1984, about 40 tonnes of the toxic gas methyl isocyanate leaked from the Union Carbide pesticide factory in **Bhopal**. Thousands died within days and hundreds of thousands suffered lasting injuries, making it the world's worst industrial disaster (see Chapter 4). Bhopal exposed the weakness of existing laws on industrial hazards and led to a wave of legislation:

- The **Environment (Protection) Act, 1986**, an "umbrella" law giving the central government broad powers to set standards, regulate industries and hazardous substances, and take action against polluters.
- The Factories (Amendment) Act, 1987, which added special provisions for hazardous processes.
- Rules for the manufacture, storage and import of hazardous chemicals (1989) and for hazardous wastes.
- The **Public Liability Insurance Act, 1991**, requiring industries handling hazardous substances to carry insurance to provide immediate relief to victims of accidents.

1.7.6 Judicial Activism

From the 1980s, the Supreme Court and High Courts became major actors in environmental protection. **Public interest litigation (PIL)** allowed any citizen or organisation to approach the courts on behalf of the public or of people unable to do so themselves. The courts interpreted the **right to life** under Article 21 of the Constitution to include the right to a pollution-free environment and clean water, for example in *Subhash Kumar v. State of Bihar* (1991).

The Delhi lawyer **M.C. Mehta**, who received the Goldman Environmental Prize in 1996, filed dozens of landmark PILs, beginning with a 1984 petition on air pollution damaging the Taj Mahal:

- **Oleum gas leak case** (1985–86): after a leak of oleum gas from a fertiliser plant in Delhi, the Supreme Court laid down the principle of **absolute liability**: an enterprise engaged in a hazardous activity is fully liable for any harm caused, with no exceptions.
- **Ganga pollution cases** (from 1987): the Court ordered tanneries in Kanpur that had not installed treatment plants to close, and directed municipalities to treat sewage.
- **Taj Trapezium case** (1996): industries around the Taj Mahal in Agra were ordered to switch to cleaner fuel or relocate to protect the monument from air pollution.
- **Delhi vehicular pollution case** (1998 onwards): the Court ordered Delhi's buses, autorickshaws and taxis to switch to compressed natural gas (CNG) (see Chapters 4 and 7).

Other key cases include *Rural Litigation and Entitlement Kendra v. State of Uttar Pradesh* (1985), which halted limestone quarrying in the Doon Valley; *Vellore Citizens' Welfare Forum* (1996), which adopted the precautionary and polluter pays principles; and *T.N. Godavarman Thirumulpad v. Union of India* (from 1995), in which the Court extended forest protection to all areas that are forests in the ordinary dictionary sense, regardless of ownership, and has continued to supervise forest matters for decades.

Judicial activism has limits. Courts sometimes take on executive functions; many orders take years to implement and need continuous supervision; closures of polluting units have sometimes cost workers their jobs without giving them a voice; and the Ganga and Yamuna remain polluted decades after the first orders. Nevertheless, it made Indian environmental law one of the most active in the world and gave citizens a powerful tool.

1.7.7 The National Green Tribunal

To reduce the burden on ordinary courts and provide expert, speedy justice in environmental cases, Parliament passed the **National Green Tribunal Act, 2010**. The **National Green Tribunal (NGT)**, set up in October 2010 with its principal bench in New Delhi, has both judicial and expert members and hears cases relating to environmental protection, forest conservation and compensation for environmental damage (see Chapter 7).

1.7.8 Environmental Policy in the Climate Era

Since the 2000s, climate change has become central to Indian environmental policy. India's position has combined two elements: insistence on equity and **common but differentiated responsibilities**, since developed countries caused most historical emissions, and growing domestic action on clean energy.

- **2008**: the **National Action Plan on Climate Change (NAPCC)** set out eight national missions, including the National Solar Mission (see Chapter 5).
- **2015**: at the Paris climate conference, India and France launched the **International Solar Alliance**, headquartered in Gurugram.
- **2021**: at the Glasgow conference, India announced a goal of **net-zero emissions by 2070** and of 500 GW of non-fossil electricity capacity by 2030.
- **2022**: India's **updated NDC** committed to reduce the emissions intensity of its GDP by 45% from 2005 levels by 2030, to obtain about 50% of its installed electricity capacity from non-

fossil sources by 2030, and to create an additional carbon sink of 2.5–3 billion tonnes of CO_2-equivalent through forests and tree cover. India also launched **Mission LiFE**.

- **2025**: in June 2025, non-fossil sources (renewables, large hydro and nuclear) reached about 50% of India's installed electricity capacity, five years ahead of the NDC target, although coal still generates most of the electricity actually consumed.

Key Points

- Modern environmentalism grew from nineteenth-century preservation (Thoreau, Muir) and conservation (Pinchot), Leopold's land ethic and Carson's *Silent Spring*.
- Stockholm (1972), Brundtland (1987), Rio (1992), Kyoto (1997) and Paris (2015) are the main milestones of global environmental governance.
- The Anthropocene is a widely used informal term for the era of human-dominated change, though it was not adopted as a formal geological epoch in 2024.
- Indian environmentalism is rooted in traditions of conservation and in struggles over livelihoods, from colonial forest protests to Chipko.
- India built its environmental institutions after Stockholm 1972: NCEPC (1972), the 42nd Amendment (1976), the Department of Environment (1980) and MoEF (1985).
- Bhopal (1984) led to the Environment (Protection) Act 1986; courts, led by PILs such as those of M.C. Mehta, and later the NGT (2010) became key enforcers; climate policy now drives much of India's environmental agenda.

Chapter 1 Review

Summary

- The environment includes all the living (biotic) and non-living (abiotic) things around an organism and the relationships among them; the Environment (Protection) Act, 1986 defines it in terms of water, air, land, living things and property and their inter-relationships.
- Environmental science studies natural processes and human impacts on them; environmental studies is broader, combining the natural sciences with the social sciences and humanities.
- Environmental studies is multidisciplinary because real problems, such as crop-residue burning or climate change, have physical, biological, economic, social, legal and ethical dimensions; it aims for an integrated, interdisciplinary understanding.
- The environment has four interacting components: atmosphere, hydrosphere, lithosphere and biosphere. Sunlight drives the exchange of energy and matter among them.
- Dry air is about 78% nitrogen, 21% oxygen and 0.93% argon, with about 0.04% CO_2 (about 423 ppm in 2024). The atmosphere is layered into the troposphere, stratosphere (containing the ozone layer), mesosphere, thermosphere and exosphere.
- Only about 2.5% of the Earth's water is fresh, and most of it is locked in ice or underground; India has about 18% of the world's people but only about 4% of its freshwater.
- The lithosphere supplies soil, minerals and fuels; soil forms very slowly. The biosphere is the zone where the other three spheres overlap, and it actively shapes them.
- The scope of environmental studies covers resources, ecosystems, biodiversity, pollution, global change, society, law and sustainable technology; it matters for health, livelihoods, citizenship and policy.
- Sustainable development is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (Brundtland Report, 1987). It gives priority to the needs of the poor and recognises environmental limits.
- The three-pillar model balances environmental, social and economic goals; the nested model places the economy within society and both within the environment.
- Weak sustainability accepts substitution of natural capital by human-made capital; strong sustainability requires critical natural capital to be preserved.
- Carrying capacity, the IPAT equation, planetary boundaries and the ecological footprint all express the limits of the Earth; humanity currently uses resources about 1.8 times as fast as ecosystems can regenerate them.
- The 17 Sustainable Development Goals, adopted in 2015 for 2030, are universal and integrated; at least six deal directly with the environment.
- Global environmentalism developed from preservation and conservation in the nineteenth century through *Silent Spring* (1962), Stockholm (1972), Brundtland (1987), Rio (1992), Kyoto (1997) and Paris (2015).
- In India, environmentalism grew from cultural traditions and struggles over forests and livelihoods; key milestones include the 42nd Amendment (1976), the Department of Environment (1980), MoEF (1985), the Environment (Protection) Act after Bhopal (1986), judicial activism through PILs, the NGT (2010) and climate commitments such as net zero by 2070.

Key Terms

Environment: The sum of all living and non-living things surrounding an organism, together with the relationships among them.

Abiotic component: A non-living physical or chemical factor of the environment, such as sunlight, temperature, water or soil.

Biotic component: A living part of the environment, such as a plant, animal or micro-organism.

Environmental science: The scientific study of natural processes and of how human activities affect them.

Environmental studies: A multidisciplinary field that combines the natural sciences, social sciences and humanities to understand and solve environmental problems.

Multidisciplinary approach: Study of a problem by several disciplines working side by side, each using its own methods.

Interdisciplinary approach: Study of a problem in which knowledge and methods from different disciplines are integrated into a combined understanding.

Atmosphere: The layer of gases surrounding the Earth, held by gravity.

Troposphere: The lowest layer of the atmosphere, up to about 8–18 km, where weather occurs and temperature falls with height.

Stratosphere: The atmospheric layer from the tropopause to about 50 km, containing the ozone layer, where temperature rises with height.

Ozone layer: The region of the stratosphere, mainly between 15 and 35 km, where ozone is concentrated and absorbs harmful ultraviolet radiation.

Hydrosphere: All the water on Earth in liquid, solid and gaseous forms.

Lithosphere: The rigid outer shell of the Earth, made up of the crust and the uppermost mantle.

Biosphere: The part of the Earth where life exists, together with all living organisms.

Sustainability: The ability of a system to continue functioning over the long term without exhausting the resources and conditions it depends on.

Sustainable development: Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Natural capital: The stock of natural resources and ecosystems that provides goods and services to people.

Intergenerational equity: The principle that present generations must pass on a healthy and productive environment to future generations.

Precautionary principle: The principle that lack of full scientific certainty should not delay action to prevent serious or irreversible environmental harm.

Polluter pays principle: The principle that those who cause pollution should bear the cost of preventing and repairing it.

Weak sustainability: The view that natural capital can be replaced by human-made capital as long as the total capital stock does not fall.

Strong sustainability: The view that critical natural capital cannot be replaced and must be maintained in its own right.

Carrying capacity: The maximum population of a species that an environment can support indefinitely without being degraded.

Planetary boundaries: Limits for nine Earth-system processes that together define a safe operating space for humanity.

Ecological footprint: The area of biologically productive land and water needed to supply the resources a population consumes and absorb its wastes, measured in global hectares.

Biocapacity: The capacity of an area's ecosystems to produce useful biological materials and absorb wastes.

Sustainable Development Goals (SDGs): The 17 global goals with 169 targets adopted by the United Nations in 2015 for achievement by 2030.

Greenwashing: Giving a misleading impression that a product, company or policy is environmentally friendly.

Anthropocene: An informal name for the present era in which human activity has become the dominant influence on the Earth's environment.

Public interest litigation (PIL): A legal action filed in court on behalf of the public or of people unable to approach the court themselves.

Multiple-Choice Questions

1. Which law defines environment as including water, air and land and the inter-relationship among them and living creatures and property?

(a) Water (Prevention and Control of Pollution) Act, 1974
(b) Wildlife (Protection) Act, 1972
(c) Environment (Protection) Act, 1986
(d) National Green Tribunal Act, 2010

2. A team in which a chemist, an economist and a sociologist jointly build a single model linking lake pollution, fish catch and fishers' incomes is using which approach?

(a) Interdisciplinary
(b) Single-disciplinary
(c) Multidisciplinary only
(d) Reductionist

3. After nitrogen and oxygen, which is the most abundant gas in dry air?

(a) Carbon dioxide
(b) Argon
(c) Water vapour
(d) Neon

4. In which layer of the atmosphere does almost all weather occur?

(a) Stratosphere
(b) Mesosphere
(c) Thermosphere
(d) Troposphere

5. Temperature increases with height in the stratosphere mainly because:

(a) Ozone absorbs ultraviolet radiation
(b) It is closer to the Sun
(c) Clouds trap heat there
(d) Greenhouse gases are concentrated there

6. The coldest region of the atmosphere, at about −90 °C, is found at the:

(a) Tropopause
(b) Stratopause
(c) Mesopause
(d) Earth's surface at the poles

7. The largest store of the Earth's freshwater is:

(a) Rivers and lakes
(b) Groundwater
(c) Atmospheric water vapour
(d) Glaciers, ice caps and permanent snow

8. The lithosphere consists of:

(a) The crust and the uppermost mantle
(b) The crust only
(c) The mantle and the core
(d) Soil and groundwater only

9. The Brundtland Report, which gave the most widely used definition of sustainable development, was titled:

(a) The Limits to Growth
(b) Our Common Future
(c) Agenda 21
(d) Transforming Our World

10. According to the Brundtland Report, which concept within sustainable development should receive "overriding priority"?

(a) Economic growth of industrialised countries
(b) Protection of wilderness for recreation
(c) The essential needs of the world's poor
(d) Technological innovation

11. The view that a forest may be cleared if the income is invested in schools and factories of equal value reflects:

(a) Strong sustainability
(b) The precautionary principle
(c) Deep ecology
(d) Weak sustainability

12. In the IPAT equation, the letter A stands for:

(a) Affluence, or consumption per person
(b) Area of land available
(c) Agriculture
(d) Atmospheric pollution

13. India's ecological footprint per person is low, yet India runs an ecological deficit because:

(a) India exports most of its biocapacity
(b) Its biocapacity per person is even lower than its footprint per person
(c) Its footprint is higher than the world average
(d) Ecological footprint does not include carbon emissions

14. How many Sustainable Development Goals and targets were adopted in the 2030 Agenda?

(a) 8 goals and 21 targets
(b) 12 goals and 100 targets
(c) 17 goals and 169 targets
(d) 17 goals and 231 targets

15. SDG 12 is concerned with:

(a) Climate action
(b) Responsible consumption and production
(c) Life on land
(d) Clean water and sanitation

16. Whose 1962 book on the effects of DDT and other pesticides is credited with launching the modern environmental movement?

(a) Aldo Leopold
(b) John Muir
(c) Gifford Pinchot
(d) Rachel Carson

17. The 1972 Stockholm Conference led directly to the creation of:

(a) The United Nations Environment Programme
(b) The Intergovernmental Panel on Climate Change
(c) The Convention on Biological Diversity
(d) The International Solar Alliance

18. Articles 48A and 51A(g) were added to the Constitution of India by the:

(a) 44th Amendment, 1978
(b) 73rd Amendment, 1992
(c) 42nd Amendment, 1976
(d) 86th Amendment, 2002

19. Which of the following was enacted as an "umbrella" environmental law in the aftermath of the Bhopal gas tragedy?

(a) Forest (Conservation) Act, 1980
(b) Environment (Protection) Act, 1986
(c) Air (Prevention and Control of Pollution) Act, 1981
(d) Biological Diversity Act, 2002

20. The principle of "absolute liability" for hazardous industries was laid down by the Supreme Court in the:

(a) Oleum gas leak case
(b) Taj Trapezium case
(c) Godavarman forest case
(d) Vellore Citizens' Welfare Forum case

Short-Answer Questions

1. Distinguish between environmental science and environmental studies, with one example of a question each would ask.
2. Why is environmental studies called a multidisciplinary subject? Illustrate with the example of crop-residue burning.
3. Describe the composition of dry air and explain why carbon dioxide matters despite its small share.
4. Name the layers of the atmosphere and state one distinctive feature of each.
5. Explain, with figures, why freshwater is a scarce resource although water covers most of the Earth's surface.
6. Give three examples of interactions between the biosphere and the other components of the environment.
7. State the Brundtland definition of sustainable development and explain its two key concepts.
8. Differentiate between weak and strong sustainability with one Indian example.
9. What is the ecological footprint? Why does India have an ecological deficit despite a low footprint per person?
10. How do the Sustainable Development Goals differ from the Millennium Development Goals?
11. What changes did the 42nd Amendment (1976) bring to the Constitution with respect to the environment?
12. Distinguish between the preservation and conservation approaches to nature, naming one person associated with each.

Long-Answer Questions

1. Describe the four components of the environment. Using the Indian monsoon as an example, explain how they interact and how human activities are altering these interactions.
2. Discuss the scope and importance of environmental studies with reference to India's major environmental challenges and the role of informed citizens.
3. Explain the concept of sustainable development. Compare the three-pillar and nested models, and discuss whether India's development path reflects weak or strong sustainability.

4. Explain carrying capacity, planetary boundaries and the ecological footprint. How do these concepts help us understand the limits to growth, and what are their implications for India?
5. List the 17 Sustainable Development Goals. Choose any three closely related to the environment and analyse India's challenges in achieving them.
6. Trace the history of environmentalism in India from traditional conservation practices to the climate era, highlighting the roles of the Stockholm Conference, the Bhopal tragedy and the judiciary.

Further Reading

- World Commission on Environment and Development (1987). *Our Common Future*. Oxford University Press.
- Carson, R. (1962). *Silent Spring*. Houghton Mifflin.
- Gadgil, M. & Guha, R. (1992). *This Fissured Land: An Ecological History of India*. Oxford University Press.
- Raven, P. H., Hassenzahl, D. M., Hager, M. C., Gift, N. Y. & Berg, L. R. (2015). *Environment* (9th ed.). Wiley.
- Bharucha, E. (2013). *Textbook of Environmental Studies for Undergraduate Courses* (2nd ed.). Universities Press for the University Grants Commission.
- Guha, R. (2000). *Environmentalism: A Global History*. Longman.
- United Nations (2015). *Transforming Our World: The 2030 Agenda for Sustainable Development*. https://sdgs.un.org/2030agenda
- Ministry of Environment, Forest and Climate Change, Government of India. https://moef.gov.in

Answers to Multiple-Choice Questions

1. (c)	2. (a)	3. (b)	4. (d)	5. (a)	6. (c)	7. (d)	8. (a)	9. (b)	10. (c)
11. (d)	12. (a)	13. (b)	14. (c)	15. (b)	16. (d)	17. (a)	18. (c)	19. (b)	20. (a)

Chapter 2: Ecosystems

A teak forest in Madhya Pradesh, a village pond in Haryana, the mangroves of the Sundarbans and the lawn outside your classroom all have something in common. In each, living things such as plants, animals and microbes are tied to one another and to the sunlight, air, water and soil around them. Energy passes from one organism to the next, nutrients are used again and again, and the whole system changes over time and recovers from shocks. Ecologists call such a unit an **ecosystem**. This chapter explains what an ecosystem is, what it is made of and how it works: how energy flows through it, how food chains and food webs link its members, how communities change through ecological succession, and how carbon, nitrogen, phosphorus, sulfur and water cycle through it. It then looks at productivity, ecological pyramids and the self-regulation, or homeostasis, that keeps ecosystems stable. The second half of the chapter surveys the major types of ecosystems found in India and the world, the services they give to people, and how they can be conserved and restored.

Learning Objectives

- Define an ecosystem and explain why it is treated as an open, self-regulating system.
- Describe the abiotic and biotic components of an ecosystem and the role of limiting factors.
- Explain energy flow, food chains, food webs, ecological pyramids and biomagnification using Indian examples.
- Describe the stages of ecological succession and the carbon, nitrogen, phosphorus, sulfur and water cycles, and show how human activity alters them.
- Compare the productivity of different ecosystems and explain how feedback, resilience and tipping points govern ecosystem stability.
- Identify the main types of terrestrial and aquatic ecosystems (tundra, forest, grassland, desert, ponds, lakes, rivers, oceans, estuaries) with Indian examples, their importance and the threats they face.
- Classify ecosystem services into provisioning, regulating, cultural and supporting services.
- Outline strategies for ecosystem preservation and conservation and the basic steps of ecosystem restoration.

2.1 Definition and Concept of Ecosystem

An **ecosystem** is a community of living organisms interacting with one another and with their non-living physical and chemical environment, forming a unit through which energy flows and materials cycle. The word joins "eco" (from the Greek *oikos*, home) and "system" (an organised whole whose parts depend on one another). The idea is simple but powerful: an organism cannot be understood apart from its surroundings, and the surroundings themselves are shaped by the organisms that live in them.

The ecosystem is the basic working unit of ecology, the science that studies the relationships of organisms with one another and with their environment. It is the level at which ecologists can measure how much sunlight a forest captures, how much fish a lake can yield, or how fast a field recovers after it is abandoned.

2.1.1 Origin of the ecosystem concept

People have long noticed that living things depend on their surroundings. As a science, however, the idea developed over about a hundred years.

The German biologist **Ernst Haeckel** coined the word "ecology" in 1866. In 1877 another German, **Karl Möbius**, used the term **biocoenosis** for the community of organisms living together on an oyster bank. In 1887 the American naturalist **Stephen A. Forbes** described a lake as a "microcosm", a small world in which plants, animals and water form one interacting whole. These ideas still treated the living community and its environment as separate.

The word **ecosystem** was introduced by the British ecologist **Arthur G. Tansley** in 1935. Tansley argued that organisms and their physical environment form a single system, and that the non-living "factors" are as much a part of that system as the plants and animals. He also stressed that an ecosystem is a unit chosen by the ecologist for study; it can be as small as a pond or as large as an ocean.

The concept became quantitative in 1942, when the American ecologist **Raymond Lindeman** published a study of energy flow in Cedar Bog Lake, Minnesota. Lindeman measured how much energy passed from plants to herbivores to carnivores and showed that much of it was lost at each step. In the 1950s and 1960s the brothers **Eugene P. Odum** and **Howard T. Odum** built on this work and made ecosystem ecology a central part of the discipline. In India, ecologists such as **R. Misra**, often called the father of Indian ecology, and later J. S. Singh and S. P. Singh, carried out pioneering studies of the structure and productivity of Indian grasslands and forests.

Table 2.1: Milestones in the development of the ecosystem concept

Year	Scientist	Contribution
1866	Ernst Haeckel	Coined the term "ecology"
1877	Karl Möbius	Used "biocoenosis" for a community of organisms
1887	Stephen A. Forbes	Described a lake as a "microcosm"
1935	Arthur G. Tansley	Introduced the term "ecosystem"
1942	Raymond Lindeman	First energy budget of an ecosystem (Cedar Bog Lake)
1953	Eugene P. Odum	*Fundamentals of Ecology* popularised the ecosystem approach

2.1.2 Levels of ecological organisation

Living matter is organised in a hierarchy. Biology begins with molecules, cells, tissues and organs. Ecology starts at the level of the whole organism and works upward:

1. **Organism (individual)**: a single living being, such as one neem tree or one tiger.
2. **Population**: a group of individuals of the same species living in an area at the same time, such as all the chital in Kanha National Park.
3. **Community**: all the populations of different species living and interacting in an area, such as the trees, grasses, deer, tigers, birds, insects and microbes of Kanha.
4. **Ecosystem**: the community together with its non-living environment of soil, water, air, light and climate.

5. **Landscape**: a mosaic of interacting ecosystems, such as forest patches, grasslands, streams and villages in a valley.
6. **Biome**: a large region with a characteristic climate and type of vegetation, such as tropical rainforest, desert or tundra.
7. **Biosphere**: the thin layer of the Earth, including air, land and water, where life exists. It is the largest ecosystem.

Each level has **emergent properties** that cannot be predicted from the level below: a single fish has no birth rate, but a population does; an ecosystem shows productivity and nutrient cycling, which depend on living and non-living parts acting together.

2.1.3 Ecosystems as open systems

In physics, a **closed system** exchanges energy but not matter with its surroundings, while an **open system** exchanges both. Ecosystems are **open systems**. Sunlight enters and heat leaves. Water, nutrients, sediment, seeds and animals move in and out across their boundaries. A river carries soil and leaf litter from the Himalaya to the Sundarbans; migratory birds such as bar-headed geese carry nutrients from Central Asia to wetlands in India each winter; and monsoon winds bring moisture evaporated from the Arabian Sea and the Bay of Bengal to forests far inland.

Two flows define every ecosystem:

- **Energy flows one way.** It enters mostly as sunlight, is stored in food, and leaves as heat. It cannot be recycled, so every ecosystem needs a constant supply.
- **Matter cycles.** Carbon, nitrogen, phosphorus, water and other materials pass from the environment into organisms and back again, and are reused many times.

Ecosystems are also **dynamic**, changing through succession, and within limits **self-regulating**, a property discussed later under homeostasis.

2.1.4 Size, boundaries and kinds of ecosystems

Ecosystems come in every size. A drop of pond water, a rotting log, a temple tank, the Chilika lagoon, the Western Ghats and the Indian Ocean can each be studied as an ecosystem. The boundary is set by the question being asked. Ecologists often describe three broad scales: **micro-ecosystems** (a puddle, the underside of a stone), **meso-ecosystems** (a forest, a lake) and **macro-ecosystems** or biomes (a tropical rainforest region or an ocean).

Natural boundaries are rarely sharp. Where two ecosystems meet, there is usually a zone of transition called an **ecotone**, such as the mangrove belt between land and sea, the reed fringe of a lake, or the Terai belt between the Himalayan foothills and the Gangetic plain. Ecotones often contain species from both neighbouring ecosystems as well as some found only there, so they can be unusually rich in life. This tendency is called the **edge effect**.

Ecosystems can also be classified by origin:

- **Natural ecosystems** develop and maintain themselves without human management. Examples are the evergreen forests of the Western Ghats, the alpine meadows of Ladakh, and the open ocean.

- **Managed or artificial (human-made) ecosystems** are created and maintained by people, who add energy and materials such as fertiliser, irrigation water, fuel and labour. Examples are croplands, plantations, aquaculture ponds, gardens, reservoirs and cities.

Natural ecosystems are further divided into **terrestrial** (forest, grassland, desert, tundra) and **aquatic** (freshwater and marine) types. These are described in detail in the second half of this chapter.

> **Think and Do: Natural or managed?**
>
> Open Google Earth or Google Maps and look at your college and its surroundings. Mark three natural or semi-natural patches (for example a ridge forest, a wetland or a grove) and three managed ecosystems (a lawn, a farm, a park). For each, list one sign of human management, such as straight edges, irrigation channels or mown grass.

2.2 Structure of an Ecosystem

The **structure** of an ecosystem is the composition and arrangement of its parts: which species are present and in what numbers, how much living and dead material there is, how organisms are arranged in space, and what the physical and chemical conditions are. Every ecosystem, whatever its size, has two kinds of components: **abiotic** (non-living) and **biotic** (living). The two are so closely linked that neither can be understood alone.

2.2.1 Abiotic components

The **abiotic components** are the non-living physical and chemical factors of the environment. They supply the energy and raw materials that organisms need, and they set the conditions under which organisms must live. They can be grouped as follows.

Table 2.2: Abiotic components of an ecosystem

Group	Factors	Example of effect
Climatic	Light, temperature, rainfall, humidity, wind	Rainfall decides whether a region supports forest, grassland or desert
Edaphic (soil)	Soil texture, structure, water content, pH, minerals	Saline soils of the Rann of Kutch support only salt-tolerant plants
Topographic	Altitude, slope, aspect (direction a slope faces)	South-facing Himalayan slopes are warmer and drier than north-facing ones
Inorganic substances	Water, CO_2, O_2, N_2, nitrates, phosphates, sulfates	Phosphate supply limits algal growth in many lakes
Organic substances	Proteins, carbohydrates, lipids, humus in dead matter	Leaf litter and humus link living and non-living parts

Light drives photosynthesis and many animal rhythms; **temperature** controls the rate of chemical reactions in organisms; and **water** availability shapes vegetation from the rainforests of the Andaman Islands, which receive about 3,000 mm of rain a year, to the western Thar desert, where parts receive less than 150 mm.

The **organic substances** deserve special mention. Dead leaves, wood, animal remains and wastes form **detritus**, and partly decomposed organic matter in soil forms **humus**. These materials are not alive, but they come from living things and are broken down by living things, so they form a bridge between the biotic and abiotic worlds.

2.2.2 Biotic components

The **biotic components** are all the living organisms in an ecosystem. They are classified by how they obtain food, that is, by their **trophic** (feeding) role.

Producers (autotrophs)

Producers or **autotrophs** ("self-feeders") make their own food from simple inorganic substances. Most are **photoautotrophs**: green plants, algae and cyanobacteria that use sunlight to carry out **photosynthesis**:

$$6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$$

The sugar made in photosynthesis stores chemical energy and supplies the carbon skeletons for all other organic molecules. In a forest the main producers are trees, shrubs and herbs; in a grassland, grasses; in a pond or the sea, tiny floating algae called **phytoplankton**, together with rooted and floating plants.

A few producers are **chemoautotrophs**. They obtain energy by oxidising inorganic chemicals such as hydrogen sulfide, ammonia or iron compounds instead of using light. Chemosynthetic bacteria support whole communities of worms, clams and crabs around deep-sea **hydrothermal vents**, where no sunlight reaches. Nitrifying bacteria in soil, which are important in the nitrogen cycle, are also chemoautotrophs.

Consumers (heterotrophs)

Consumers or **heterotrophs** ("other-feeders") cannot make their own food and obtain energy by eating other organisms. They are grouped by what they eat:

- **Primary consumers** or **herbivores** feed on producers. Examples: grasshoppers, cattle, chital, nilgai, elephants, and in water, zooplankton that graze on phytoplankton.
- **Secondary consumers** feed on herbivores. These are **carnivores** (frogs, insect-eating birds, small fish) or **omnivores** that eat both plants and animals (bears, crows, many humans).
- **Tertiary consumers** feed on secondary consumers. Examples: snakes that eat frogs, large fish that eat smaller fish.
- **Top carnivores** or **apex predators** sit at the end of food chains and are not normally preyed upon as adults. Examples: the tiger, the Asiatic lion, the snow leopard, the mugger crocodile, the saltwater crocodile.

Other consumer roles include **parasites**, which live on or in a host and feed on it without usually killing it quickly (lice, ticks, tapeworms, the parasitic plant *Cuscuta*), and **scavengers**, which feed on dead animals they did not kill (vultures, jackals, hyenas).

Decomposers and detritivores

When organisms die, or shed leaves, hair and dung, their remains still contain energy and nutrients. Two groups of organisms recover them.

Detritivores are animals that eat detritus, breaking it into smaller pieces and digesting part of it. Earthworms, termites, millipedes, woodlice, dung beetles, many crabs and some snails are detritivores. Charles Darwin estimated that earthworms pass huge quantities of soil through their bodies; in India, termites are major processors of dead wood and leaf litter in dry deciduous forests and savannas.

Decomposers or **saprotrophs** are mainly **bacteria** and **fungi**. They secrete digestive enzymes onto dead matter and absorb the dissolved products. In doing so they convert complex organic compounds into simple inorganic substances such as CO_2, water, ammonium, phosphate and sulfate. This process, called **mineralisation**, returns nutrients to the soil and water where producers can absorb them again.

The difference is important. Detritivores fragment dead matter and increase its surface area; decomposers complete its chemical breakdown. Without decomposers, dead material would pile up and nutrients would stay locked in it; producers would starve for nitrogen and phosphorus and the ecosystem would run down.

Table 2.3: Biotic components in three Indian ecosystems

Role	Village pond	Tropical deciduous forest	Grassland
Producers	Phytoplankton, *Hydrilla*, lotus, water hyacinth	Sal, teak, bamboo, shrubs, grasses	Grasses such as *Dichanthium*, herbs, legumes
Primary consumers	Zooplankton, snails, tadpoles	Chital, sambar, langur, insects	Grasshoppers, blackbuck, cattle, rodents
Secondary consumers	Small fish, frogs, water beetles	Wild dog (dhole), jungle cat, insect-eating birds	Frogs, lizards, insect-eating birds
Top consumers	Large fish, kingfisher, heron	Tiger, leopard	Wolf, raptors such as the tawny eagle
Detritivores and decomposers	Bacteria, fungi, worms in sediment	Termites, earthworms, fungi, bacteria	Dung beetles, termites, fungi, bacteria

2.2.3 Interactions between biotic and abiotic components

The abiotic environment shapes the community, but the community also changes its environment. Plants take up water and minerals from the soil, and when they die their remains build humus that improves the soil. A forest canopy cools the ground beneath it, raises humidity and, through transpiration, returns water vapour to the air that later falls as rain. Corals build entire reefs of calcium carbonate. Mangrove roots trap silt and slowly build new land along the coasts of the Sundarbans and the Gulf of Kutch. Because of these two-way interactions, ecologists treat the ecosystem, not the community alone, as the unit of study.

Species with an effect far larger than their numbers suggest, such as fig trees that feed many animals when other fruit is scarce, are called **keystone species**; organisms that build or modify habitats, such as termites and corals, are called **ecosystem engineers** (see Chapter 6).

2.2.4 Spatial structure: stratification and zonation

Ecosystems also have a physical arrangement. In a forest, plants are arranged in vertical layers, a pattern called **stratification**: emergent trees, the main canopy, an understorey of smaller trees, a shrub layer, a herb layer, and the forest floor with its litter. Tropical evergreen forests of the Western Ghats and the North-East may have four or five distinct layers, each with its own animals. Soil also has layers, called **horizons**.

Lakes and seas show a similar **zonation** from the sunlit surface to the dark bottom and from the shallow margin to open water (see the section on aquatic ecosystems). Such layering creates many small habitats, or **niches**, and so allows many species to coexist.

2.2.5 Limiting factors and the law of tolerance

An organism's growth or survival often depends less on the resources it has in plenty than on the one that is in shortest supply. A **limiting factor** is any abiotic or biotic factor that restricts the growth, abundance or distribution of an organism or population.

The German chemist **Justus von Liebig** observed in 1840 that crop yield was often limited not by the nutrient needed in the largest amount but by the one that was scarcest. This is **Liebig's law of the minimum**: the growth of an organism is controlled by the essential factor present in the minimum quantity relative to need. A field rich in nitrogen but poor in phosphorus will not grow a good crop until phosphorus is added. In 1905 the British plant physiologist **F. F. Blackman** extended the idea to rates of processes such as photosynthesis: when a process depends on several factors, its rate is limited by the factor that is in shortest supply (**Blackman's law of limiting factors**).

Too much of a factor can also be harmful. In 1913 the American zoologist **Victor Shelford** proposed the **law of tolerance**: each organism has a minimum and a maximum for each environmental factor, and it can survive only between these limits. The range between them is its **range of tolerance**. Within this range lies an **optimum** where the organism does best; towards either limit it experiences stress, and beyond them it cannot survive.

Organisms differ greatly in their tolerance. Species with a wide range are described with the prefix **eury-** and those with a narrow range with **steno-**:

- **Eurythermal** organisms tolerate a wide range of temperature (many insects, the house crow); **stenothermal** ones need a narrow range (reef-building corals, which bleach when sea temperatures stay about 1 °C above the normal summer maximum for several weeks).
- **Euryhaline** organisms tolerate a wide range of salinity (mangrove trees, mud crabs, the milkfish of estuaries); **stenohaline** ones do not (most freshwater fish, which die in seawater).
- **Euryphagic** animals eat many kinds of food (crows, jackals); **stenophagic** ones are specialists (the koala in Australia, or the larvae of many butterflies that feed on a single plant family).

Species with wide tolerance for many factors tend to be widely distributed; specialists with narrow tolerance are often restricted and more vulnerable to change. This is why stenotopic species such as certain stream insects and lichens are used as **indicator species** of environmental quality: their presence or absence reveals conditions such as pollution.

Limiting factors in Indian ecosystems

- In the Thar desert, **water** limits plant growth for most of the year.
- In the high Himalaya and Ladakh, **low temperature** and a short growing season limit plants.
- In many Indian lakes and reservoirs, **phosphorus** is the limiting nutrient; sewage that adds phosphorus causes algal blooms.
- In the deep sea and inside dense forests, **light** limits photosynthesis.
- In the Arabian Sea, productivity rises sharply in the monsoon months when winds bring nutrient-rich deep water to the surface.

Key Points

- An ecosystem is a community of organisms plus its non-living environment, linked by energy flow and nutrient cycling (Tansley, 1935).
- Ecosystems are open, dynamic and self-regulating systems of every size, from a puddle to the biosphere.
- Abiotic components include climatic, edaphic, topographic, inorganic and organic factors.
- Biotic components are producers, consumers, and detritivores and decomposers that recycle nutrients.
- Limiting factors (Liebig, Blackman) and ranges of tolerance (Shelford) govern where and how well organisms live.

2.3 Functions of an Ecosystem

The **function** of an ecosystem is what it does: the processes by which it captures energy, produces organic matter, passes food from one organism to another, recycles materials and changes over time. The syllabus groups these functions into three kinds, and the rest of this chapter follows the same plan:

- **Physical function**: the flow of energy through the ecosystem.
- **Biological functions**: food chains and food webs, which carry energy and materials between organisms, and ecological succession, by which communities develop and change.
- **Biogeochemical function**: the cycling of nutrients and water between the living and non-living parts of the ecosystem.

Three further ideas describe how well an ecosystem performs these functions: **productivity** (the rate at which organic matter is made), **ecological pyramids** (the quantitative relationship between feeding levels) and **homeostasis** (the ability to stay stable).

2.4 Energy Flow in Ecosystems

Every activity of life needs energy: growing, moving, keeping warm, reproducing, and even repairing cells. The movement of energy through an ecosystem, from the Sun to producers, to consumers and

decomposers, and finally out as heat, is called **energy flow**. It is the most basic **physical function** of an ecosystem.

2.4.1 The Sun as the source of energy

Almost all energy in ecosystems comes from the Sun. At the top of the atmosphere the Earth receives about 1,361 watts per square metre of solar energy, a value called the **solar constant**. About 30% of this is reflected back to space by clouds, ice and land, and much of the rest is absorbed by the atmosphere, land and oceans and heats them.

Only a small fraction of the sunlight that reaches plants is captured in photosynthesis. Roughly half of incoming solar radiation lies in the wavelengths that plants can use (400 to 700 nanometres), called **photosynthetically active radiation (PAR)**. Some light is reflected by leaves, some passes through, and much of the absorbed energy is lost as heat. As a result, natural ecosystems typically fix only about 1% of the sunlight falling on them as chemical energy, and even very productive crops such as sugarcane rarely exceed 2 to 3%. This tiny fraction supports almost all life on Earth. The main exception is chemosynthetic communities, such as those at deep-sea hydrothermal vents.

2.4.2 The laws of thermodynamics

Energy flow in ecosystems obeys two laws of physics.

The **first law of thermodynamics** states that energy can be neither created nor destroyed; it can only change from one form to another. In an ecosystem, light energy is changed into chemical energy in food, and chemical energy is changed into the mechanical energy of muscles, the electrical energy of nerves, and heat. At every stage the energy can be accounted for.

The **second law of thermodynamics** states that whenever energy is transformed, some of it is degraded into a less useful, more dispersed form, usually heat. No transformation is 100% efficient. Put another way, the **entropy** (disorder) of the universe always increases. For ecosystems this has three consequences:

1. When one organism eats another, most of the food energy is not converted into the eater's body. It is lost as heat in respiration or passed out in faeces.
2. Energy cannot be recycled. Once it has been degraded to heat and radiated into space, it cannot be used again by organisms.
3. An ecosystem therefore needs a continuous input of energy from the Sun to keep running. Organisms maintain their highly ordered bodies only by constantly using energy.

Because of the second law, energy flow is **unidirectional** (one-way), from the Sun through producers and consumers and out as heat. This contrasts with materials such as carbon and nitrogen, which **cycle**.

2.4.3 Trophic levels

The position an organism occupies in the sequence of energy transfer is its **trophic level** (from Greek *trophe*, nourishment). Producers form the **first trophic level (T_1)**, herbivores the **second (T_2)**, carnivores that eat herbivores the **third (T_3)**, and so on. Decomposers act on dead matter from every level.

An organism's trophic level is defined by its function, not its species. Many animals feed at more than one level. A sloth bear eating fruit is at T_2; the same bear eating termites that fed on dead wood is feeding at a higher level. Humans are herbivores when eating rice and dal, and secondary or higher consumers when eating fish or chicken.

2.4.4 The ten per cent law

As energy passes from one trophic level to the next, a large part is lost at each step. Consider what happens to the energy in the grass eaten by a chital:

- Part of the grass is not eaten at all; it dies and passes to decomposers.
- Of what is eaten, part is not digested and leaves as dung.
- Of what is digested and absorbed (**assimilated**), most is used in **respiration** to power the chital's life activities and is lost as heat.
- Only the remainder is converted into new chital tissue (growth and reproduction), which is available to a tiger or leopard that eats it.

Based on his study of Cedar Bog Lake, Lindeman (1942) concluded that only a small fraction of the energy at one trophic level becomes available to the next. This idea is commonly summarised as **Lindeman's ten per cent law**: on average, only about 10% of the energy stored at one trophic level is transferred to and stored at the next higher level; about 90% is lost, mainly as heat.

If producers in an area fix 10,000 kilojoules (kJ) of energy, herbivores will store about 1,000 kJ, primary carnivores about 100 kJ, and secondary carnivores about 10 kJ. The 10% figure is a convenient average, not a fixed rule. Measured transfer efficiencies usually range from about 5% to 20%. They tend to be lower for warm-blooded animals such as birds and mammals, which spend much energy keeping their bodies warm, and higher for cold-blooded animals such as fish and insects. Lindeman's own data show this variation.

Table 2.4: Energy flow in Cedar Bog Lake, Minnesota (Lindeman, 1942)

Trophic level	Energy fixed or taken in ($cal/cm^2/year$)	Lost in respiration ($cal/cm^2/year$)	Efficiency of transfer
Producers	111.3	23.4	about 0.1% of sunlight
Herbivores	14.8	4.4	13.3%
Carnivores	3.1	1.8	22.3%

Lindeman's figures show that the producers captured only about 0.1% of the sunlight reaching the lake, and that the efficiency of transfer was about 13% from plants to herbivores and about 22% from herbivores to carnivores. The exact percentages differ from 10%, but the general lesson stands: energy decreases sharply at each step.

Why there are so few tigers

India had an estimated 3,682 wild tigers in the 2022 All India Tiger Estimation, compared with lakhs of chital, sambar and wild pigs and vast quantities of grass. Field studies by K. Ullas Karanth and colleagues suggest that an adult tiger kills roughly 50 deer-sized animals a year, which only a prey population of several hundred animals can sustain. Because energy is lost at every step, top predators are always few, need large areas, and are often the first to disappear when habitats shrink.

2.4.5 Why food chains are short

Since about 90% of energy is lost at each transfer, very little remains after four or five steps. A fifth-level consumer would receive only about 0.01% of the energy fixed by producers and would need an enormous area to find enough food. This is why most food chains have only three to five trophic levels. Food chains tend to be somewhat longer in productive, stable ecosystems such as large lakes and coastal seas, and shorter in unproductive ones such as deserts and the tundra. The same principle explains why a plant-based diet can feed more people from the same land.

2.4.6 Single-channel energy flow model

Ecologists use simple diagrams, called **energy flow models**, to show how energy enters, moves through and leaves an ecosystem. The simplest is the **single-channel** (or universal) energy flow model associated with E. P. Odum.

In this model, energy moves in a single line from producers to herbivores to carnivores. At each trophic level, a box represents the energy present, and arrows show what happens to it:

- Energy enters the producer level as sunlight; only a small part is fixed as **gross primary production**.
- At each level, part of the energy is used in **respiration** and lost as **heat**.
- Part of the energy at each level is not consumed but passes, as dead matter and wastes, to **decomposers**.
- The remainder is passed to the next trophic level, which therefore receives much less.

The upper part of the figure at the end of this section shows this model. It makes two points clear. Energy flow is one-way, with no arrow returning from heat to the Sun or from carnivores to plants. And the quantity of energy shrinks steadily along the channel.

2.4.7 Y-shaped energy flow model

In real ecosystems, energy fixed by producers takes two main routes. Some is eaten by herbivores while the plants are alive, but much of it, often most, is never eaten. Leaves fall, plants die, and animals produce dung and carcasses. This dead organic matter supports a separate set of organisms: detritivores and decomposers, and the animals that feed on them.

The **Y-shaped** (or **two-channel**) energy flow model, also proposed by E. P. Odum, shows this division. The stem of the Y represents the producers. One arm is the **grazing channel**, in which energy flows from living plants to herbivores and then to carnivores. The other arm is the **detritus channel**, in which energy flows from dead organic matter to detritivores, decomposers and their predators. The

two arms are linked: dead herbivores and carnivores and their wastes enter the detritus channel, and animals such as birds and fish that eat detritivores (earthworms, insect larvae, crabs) carry energy back into the grazing channel.

The relative size of the two arms differs between ecosystems:

- In **forests**, herbivores usually eat less than a tenth of the plant production. Most leaves, wood and roots die uneaten, so the **detritus channel carries most of the energy**. The same is true of mangroves and many wetlands.
- In **heavily grazed grasslands** and in **open seas** and lakes, where herbivores eat a large share of fast-growing grasses or phytoplankton, the **grazing channel is relatively larger**.

The lower part of the same figure shows the Y-shaped model. It is more realistic than the single-channel model because it recognises the role of decomposers in energy flow.

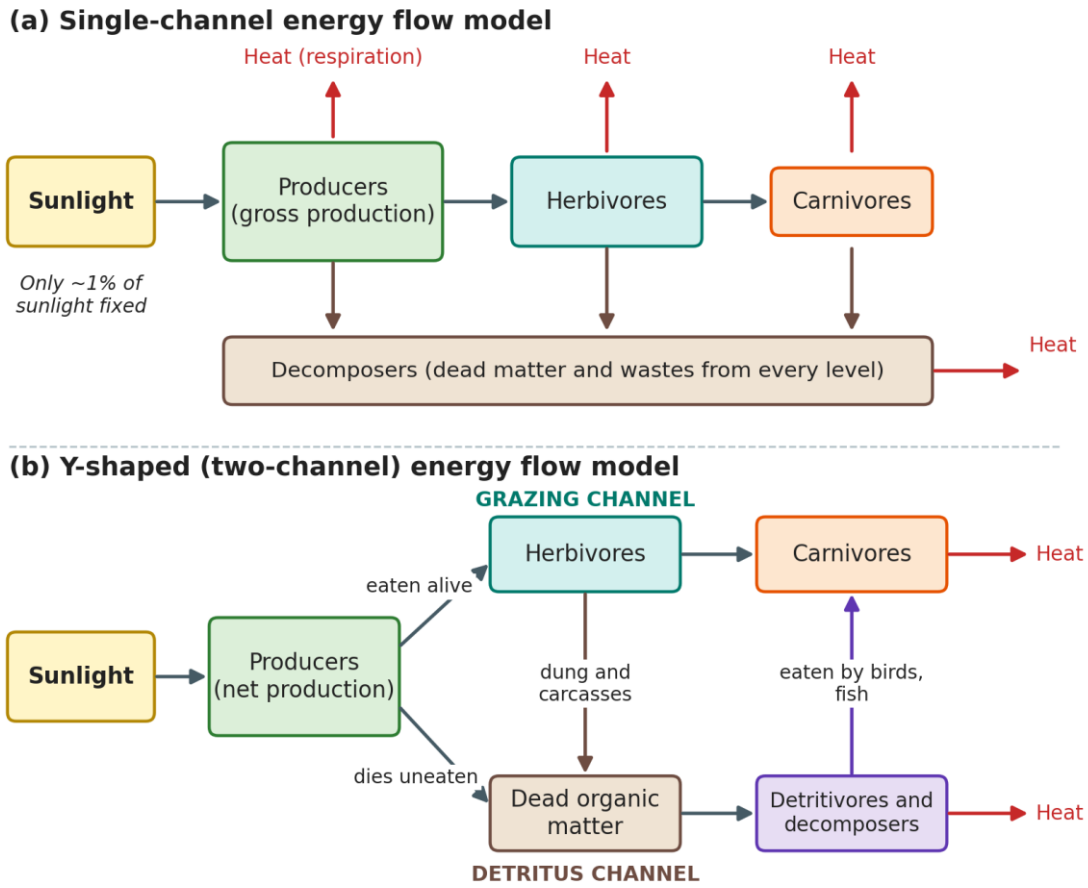


Figure 2.1: *Energy flow models. Top: single-channel model, in which energy passes in a line from producers to herbivores to carnivores and at each level part is lost as respiratory heat or passed to decomposers. Bottom: Y-shaped (two-channel) model, in which production is divided between a grazing channel and a detritus channel. Source: Author, after E. P. Odum.*

Key Points

- Energy enters ecosystems as sunlight; only about 1% of it is fixed by producers.
- The first law of thermodynamics says energy is conserved; the second law says each transformation loses some energy as heat.
- Energy flow is one-way and cannot be recycled; matter cycles.
- On average about 10% of energy passes from one trophic level to the next (Lindeman's ten per cent law); real values range from about 5% to 20%.
- The single-channel model shows a linear flow; the Y-shaped model separates grazing and detritus channels.

2.5 Food Chains, Food Webs and Trophic Levels

Energy and nutrients move through an ecosystem when one organism eats another. These feeding relationships are the main **biological function** that links the members of a community. They were first systematically described by the British ecologist **Charles Elton** in his book *Animal Ecology* (1927), which introduced the ideas of food chains, food webs and pyramids of numbers.

2.5.1 Food chains

A **food chain** is a sequence of organisms in which each one eats the one before it and is eaten by the one after it, so that energy and nutrients pass along the chain. Each link in the chain is a trophic level. A food chain is written with arrows that point in the direction of energy flow, from the eaten to the eater:

Grass → Grasshopper → Frog → Snake → Peacock

Food chains have several general features. Energy flow along them is one-way. Each step loses most of the energy, so chains are usually short. The organisms at higher levels are usually larger and fewer than those below, though there are exceptions. And because many animals eat more than one kind of food, a single food chain is always a simplification. Ecologists recognise three main types.

Grazing food chain

A **grazing food chain** begins with living green plants, which are eaten by herbivores, which are eaten by carnivores. It is the type most people think of. Examples from India include:

- **Grassland**: Grass → Grasshopper → Frog → Snake → Hawk
- **Tropical forest (Kanha or Bandhavgarh)**: Grass and shrubs → Chital → Tiger
- **Pond or lake**: Phytoplankton → Zooplankton → Small fish (such as *Puntius*) → Large fish (such as murrel) → Kingfisher or heron
- **Sea off the Kerala coast**: Diatoms (phytoplankton) → Zooplankton → Oil sardine → Tuna → Shark
- **Farmland**: Wheat → Rat → Snake → Eagle

Grazing chains depend directly on the current capture of sunlight. They dominate in ecosystems where herbivores eat a large share of plant production, such as the open sea and heavily grazed grasslands.

Detritus food chain

A **detritus food chain** begins with dead organic matter (detritus), such as fallen leaves, dead wood, dead animals and dung. The detritus is eaten by detritivores and broken down by decomposers, which in turn are eaten by predators. Examples:

- **Mangrove forest (Sundarbans or Bhitarkanika)**: Fallen mangrove leaves → Fungi and bacteria → Crabs, shrimps and detritus-feeding fish → Larger fish → Fishing cat, birds or humans
- **Forest floor**: Leaf litter → Earthworms and millipedes → Ground-feeding birds (such as babblers) → Shikra
- **Grassland**: Cattle dung → Dung beetles → Lizards and birds

The detritus chain is less visible but often carries more energy than the grazing chain. In most forests, wetlands and mangroves, the majority of plant production is never eaten alive and enters the detritus chain instead. It is also the route by which nutrients are released from dead matter and returned to producers. The Sundarbans fisheries, which support lakhs of people, depend heavily on the detritus produced by mangrove leaves.

Table 2.5: Grazing and detritus food chains compared

Feature	Grazing food chain	Detritus food chain
Starts with	Living green plants	Dead organic matter
First consumers	Herbivores	Detritivores and decomposers
Energy source	Direct, current solar energy fixed by plants	Energy stored in dead matter
Main ecosystems	Open sea, lakes, grazed grasslands	Forests, mangroves, wetlands, soil
Role in nutrient cycling	Small	Large, since it mineralises nutrients

Parasitic food chain

A **parasitic food chain** passes energy from a larger host to smaller parasites, and sometimes to still smaller parasites of those parasites (**hyperparasites**). For example: Tree → Birds living on the tree → Lice on the birds → Microbes on the lice. Here the organisms become smaller and more numerous at each step, which affects the shape of the pyramid of numbers, as discussed later.

2.5.2 Food webs

In nature, food chains rarely stay separate. A frog may eat grasshoppers, beetles and spiders; it may be eaten by snakes, herons or mongooses. A snake may eat frogs, rats and birds. When all the feeding links in a community are drawn together, they form a network called a **food web**. A food web is a set of interconnected food chains that shows all the possible paths by which energy and nutrients move in an ecosystem.

The figure below shows a simplified food web from a village pond and its surrounding grassland in north India. Even with only a dozen kinds of organisms, the links quickly become complex. Real food webs include hundreds of species and thousands of links, most of them involving insects, worms and microbes that are easily overlooked.

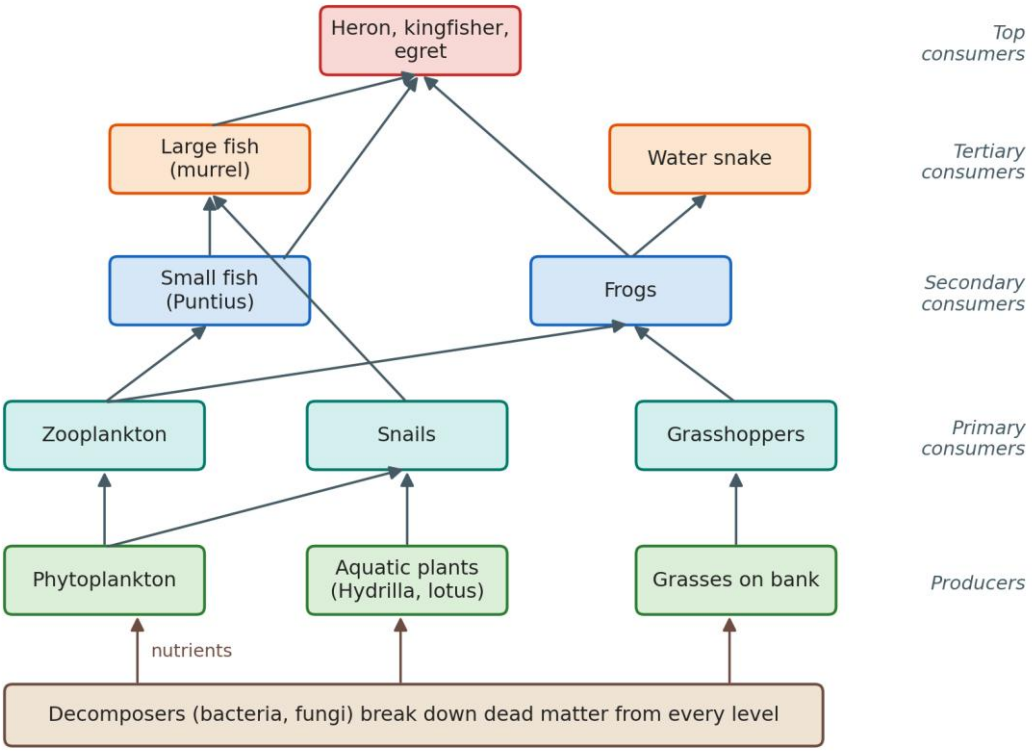


Figure 2.2: *A simplified food web of a village pond and its grassy margin in north India. Arrows point from food to feeder. Many animals feed at more than one trophic level, and all dead matter passes to decomposers. Source: Author.*

Importance of food webs

Food webs matter for several reasons.

1. **Stability.** In a food web, most animals have alternative foods. If one prey species declines, a predator can switch to another, and the community is less likely to collapse. A system with many links and many species performing similar roles tends to be more stable than one built on a single chain.
2. **Indirect effects.** Because species are linked, a change at one point can spread through the web. A **trophic cascade** occurs when a change in the numbers of a predator alters the numbers of its prey and, through them, the plants or animals at lower levels.
3. **Movement of pollutants.** Food webs also carry harmful substances that organisms cannot break down, as explained under biomagnification.

Case Study: When India lost its vultures

In the early 1990s India had tens of millions of vultures, mainly the white-rumped, Indian and slender-billed vultures (*Gyps* species). They were the country's most efficient scavengers, stripping a dead cow to bones within hours. Within about a decade, their populations fell by more than 95%. Scientists traced the collapse to **diclofenac**, a cheap painkiller given to cattle. Vultures that ate the carcasses of treated cattle died of kidney failure. India banned diclofenac for veterinary use in 2006.

The loss of a single group of scavengers rippled through the food web. Carcasses rotted for longer and were eaten instead by feral dogs and rats, whose numbers grew. Dog bites and rabies increased, and pathogens from carcasses spread into water. An economic study by Eyal Frank and Anant Sudarshan, published in 2024, estimated that the decline of vultures was associated with about 1,00,000 additional human deaths a year in India between 2000 and 2005, roughly half a million in total.

The vulture story shows how a link in the detritus pathway that most people never notice can protect human health. India's vulture conservation and breeding programme is discussed in Chapter 6.

Because many animals feed at several levels in a food web, ecologists sometimes give them a fractional trophic level that averages their diet; a fish whose diet is half plants and half herbivores has a trophic level of 2.5.

2.5.3 Biomagnification

Food chains carry not only energy and nutrients but also some pollutants. A substance that is **persistent** (not easily broken down by organisms or the environment) and **fat-soluble** (stored in fatty tissues rather than excreted in urine) can build up in living things. Two related processes are involved.

Bioaccumulation is the build-up of a substance in an individual organism over its lifetime, because the organism takes it in faster than it can break it down or excrete it. An old fish usually contains more mercury than a young fish of the same species.

Biomagnification (also called **biological magnification**) is the increase in concentration of a substance at each successive trophic level of a food chain. It happens because each consumer eats many organisms from the level below and retains the persistent substance from all of them, while most of the food's mass is burned in respiration. As a result, top predators may carry concentrations thousands or even millions of times higher than in the surrounding water or soil.

Substances that biomagnify include organochlorine pesticides such as **DDT**, aldrin, dieldrin, endosulfan and hexachlorocyclohexane (HCH, also called BHC); industrial chemicals such as **polychlorinated biphenyls (PCBs)**; and **methylmercury**, the organic form of mercury. Many of these are listed as **persistent organic pollutants** under the Stockholm Convention (2001).

DDT

DDT (dichlorodiphenyltrichloroethane) was used widely after the Second World War to kill mosquitoes and crop pests. It is very persistent and fat-soluble. Classic studies of an estuary on Long

Island, New York, in the 1960s, by George Woodwell and colleagues, traced DDT through an aquatic food chain. The figures below, widely used in textbooks, summarise the pattern.

Table 2.6: Biomagnification of DDT in an aquatic food chain

Level in food chain	DDT concentration (parts per million)	Approximate magnification over water
Water	0.000003 (3 parts per trillion)	1
Zooplankton	0.04	about 13,000 times
Small fish	0.5	about 1,70,000 times
Large fish	2	about 6,70,000 times
Fish-eating birds	25	about 80,00,000 (8 million) times

High DDT levels interfere with calcium metabolism in birds, so they lay eggs with thin shells that break during incubation. This caused steep declines of the bald eagle, the peregrine falcon, ospreys and pelicans in North America and Europe in the 1950s and 1960s. **Rachel Carson** drew public attention to these effects in her book *Silent Spring* (1962), which helped launch the modern environmental movement. After DDT was banned in the United States (1972) and elsewhere, many of these birds recovered.

In India, the use of DDT in agriculture was withdrawn in 1989, but it is still permitted in restricted quantities for controlling malaria mosquitoes, as allowed by the Stockholm Convention. Studies over several decades have detected residues of DDT and HCH in Indian food, fish, birds and human breast milk, although levels have generally declined since the agricultural ban. In 2011 the Supreme Court banned the manufacture and use of endosulfan, another persistent organochlorine, after decades of aerial spraying on cashew plantations in Kasaragod, Kerala, were linked to serious health problems in local people.

Mercury

Mercury released from coal burning, mining, chlor-alkali plants and other industries settles in water, where bacteria convert it into **methylmercury**. Methylmercury is absorbed easily, binds to proteins in muscle, and biomagnifies so strongly that large predatory fish such as tuna, shark and swordfish may contain a million times the concentration found in the water.

The world's best-known case of mercury poisoning occurred at **Minamata**, Japan. From the 1930s to 1968 the Chisso Corporation discharged wastewater containing methylmercury into Minamata Bay. It accumulated in fish and shellfish, which were the staple food of local fishing families. From 1956 doctors recognised a disease, later called **Minamata disease**, that caused numbness, loss of coordination, blindness, birth defects and death. Cats that ate fish died first. Thousands of people were eventually recognised as victims. The tragedy led to the **Minamata Convention on Mercury** (2013), which India ratified in 2018.

In India, mercury pollution has been documented in the coal and power belt of **Singrauli** (on the Madhya Pradesh–Uttar Pradesh border), where studies have reported elevated mercury in fish, soil and the hair and blood of residents, and at **Kodaikanal** in Tamil Nadu, where a mercury thermometer factory was shut down by regulators in 2001 after mercury waste was found dumped near the town.

Why top predators and people are at risk

Biomagnification means that the organisms most at risk from persistent pollutants are the top carnivores: eagles, otters, dolphins, tigers and humans who eat large fish. It also means that a pollutant present in water at levels too low to measure easily can still poison a fish-eating bird. This is why many countries advise pregnant women and children to limit their consumption of large predatory fish.

Substances that are water-soluble or quickly broken down, such as most modern organophosphate pesticides, do not usually biomagnify, though they may still be highly toxic.

Key Points

- A food chain is a linear series of feeding steps; a food web is a network of interconnected food chains.
- Grazing food chains begin with living plants; detritus food chains begin with dead organic matter and often carry most of the energy in forests and mangroves.
- Food webs give stability and allow trophic cascades; keystone species and scavengers such as vultures can have large effects.
- Persistent, fat-soluble substances such as DDT and methylmercury biomagnify, reaching their highest concentrations in top predators.

2.6 Ecological Succession

Ecosystems are not fixed. A pond left alone slowly fills with silt and plants and becomes a marsh, then a meadow, and perhaps eventually a forest. A field abandoned by a farmer is first covered by weeds, then grasses and shrubs, and then trees. This orderly, directional change in the species composition of a community over time is called **ecological succession**. It is the second major **biological function** of ecosystems.

The idea was developed by the American botanist **Henry C. Cowles**, who in 1899 described how vegetation changed from bare sand to forest across the sand dunes of Lake Michigan, and by **Frederic Clements**, whose book *Plant Succession* (1916) set out the stages and terms still used today.

2.6.1 Key terms

- **Sere**: the whole sequence of communities that replace one another in an area, from the first colonists to the final stable community.
- **Seral stage** (or seral community): each intermediate community in a sere.
- **Pioneer species**: the first organisms to colonise a bare or disturbed area. They are usually hardy, fast-growing, short-lived and good at dispersing, such as lichens on rock or weeds on a cleared field.
- **Climax community**: the relatively stable, self-maintaining community at the end of succession, in balance with the local climate and soil. It changes little as long as conditions remain the same.

2.6.2 Process of succession

Clements described succession as a series of steps that occur in every sere:

1. **Nudation**: a bare area is created, by a volcanic eruption, landslide, flood, fire, glacier retreat or human clearing.
2. **Invasion**: seeds, spores and animals arrive (**migration**), establish themselves (**ecesis**), and multiply and spread (**aggregation**).
3. **Competition and coaction**: as numbers increase, species compete for space, light, water and nutrients, and affect one another as predators, parasites or partners.
4. **Reaction**: the organisms change their own environment, for example by building soil, adding organic matter, providing shade and trapping moisture. These changes often make the site less suitable for the existing species and more suitable for new ones, which then replace them.
5. **Stabilisation**: a climax community develops that is in balance with the environment and can maintain itself.

The idea that each stage prepares the way for the next is called **facilitation**. Later ecologists (J. H. Connell and R. O. Slatyer, 1977) pointed out that succession can also proceed by **tolerance** (later species simply tolerate lower resource levels and outlast earlier ones) or by **inhibition** (early species resist invasion until they die or are disturbed). All three mechanisms operate in real seres.

2.6.3 Primary and secondary succession

Primary succession begins on a surface that has never had life or soil, such as bare rock, fresh lava, new sand dunes, or ground left by a retreating glacier. Because soil must first be built from rock, primary succession is very slow and may take hundreds to thousands of years to reach a climax.

Secondary succession begins on a site where an existing community has been removed but the soil remains, such as an abandoned field, a burnt forest, a clear-felled area or a site of flood damage. Since soil, seeds, roots and soil organisms are already present, secondary succession is much faster and can produce a forest in decades.

Table 2.7: Primary and secondary succession compared

Feature	Primary succession	Secondary succession
Starting surface	Bare, lifeless; no soil	Soil present; previous life removed
Causes	Lava flow, rock exposure, new dune or island, glacier retreat	Fire, logging, abandoned farming, flood, landslide that leaves soil
Pioneers	Lichens, mosses, cyanobacteria	Annual weeds and grasses from the seed bank
Speed	Very slow (centuries to millennia)	Faster (decades to a century or two)
Indian examples	Lava on Barren Island; rock faces in the Aravallis	Fallow land after shifting cultivation in the North-East; abandoned fields

2.6.4 Types of succession by starting habitat

Seres are named after the habitat in which they begin:

- **Hydrosere** (**hydrarch** succession): starts in water, such as a pond or lake.
- **Xerosere** (**xerarch** succession): starts in a dry habitat. A xerosere on bare rock is called a **lithosere**; one on sand is a **psammosere**.
- **Halosere**: starts in saline water or soil, such as a salt marsh or the edge of an estuary.

Both hydrarch and xerarch succession tend towards a **mesic** climax, that is, a community with moderate moisture. Wet sites become drier as they fill with sediment and organic matter; dry sites become moister as soil develops and holds water.

Xerarch succession (lithosere)

Xerarch succession on bare rock passes through the following stages:

1. **Crustose lichen stage**: lichens such as *Rhizocarpon* and *Lecanora* grow as thin crusts on the rock. They secrete acids that slowly dissolve the rock surface, and when they die they add a little organic matter.
2. **Foliose lichen stage**: leaf-like lichens (*Parmelia*, *Dermatocarpon*) cover the crustose forms, trap dust and hold more water.
3. **Moss stage**: mosses such as *Funaria*, *Polytrichum* and *Tortula* grow in the thin soil, accumulating more organic matter and holding moisture.
4. **Herb stage**: annual and then perennial grasses and herbs establish as soil deepens. Their roots break the rock further.
5. **Shrub stage**: shrubs shade out many herbs and deepen the soil.
6. **Climax forest stage**: trees establish and a forest develops whose type depends on climate.

In India such sequences can be seen on exposed rock faces in the Aravallis and on old rock-cut walls, where lichens and mosses gradually give way to grasses and shrubs.

Hydrarch succession (hydrosere)

Hydrarch succession begins in a pond or lake and gradually converts it into land:

1. **Phytoplankton stage**: microscopic algae, cyanobacteria and diatoms grow in the water. When they die, they add organic matter to the bottom.
2. **Submerged plant stage**: rooted plants that grow fully under water, such as *Hydrilla*, *Vallisneria* and *Potamogeton*, colonise the soft mud of the bottom. They trap silt and add organic matter, so the pond becomes shallower.
3. **Floating plant stage**: rooted plants with floating leaves, such as lotus (*Nelumbo*), water lily (*Nymphaea*) and *Trapa* (water chestnut, singhara), cover the surface. Free-floating plants such as *Lemna*, *Pistia* and water hyacinth (*Eichhornia*) may also appear. They shade out submerged plants.
4. **Reed swamp (amphibious) stage**: as the water becomes shallow, emergent plants with roots in water and shoots in the air, such as *Typha* (cattail), *Phragmites* and *Scirpus*, take over.
5. **Sedge meadow (marsh meadow) stage**: sedges (*Carex*, *Cyperus*) and rushes (*Juncus*) grow on the now waterlogged soil; evaporation and transpiration dry it further.
6. **Woodland (shrub) stage**: shrubs and moisture-loving trees such as willows (*Salix*), *Cornus* and *Alnus* appear.

7. **Climax forest stage**: a forest suited to the regional climate develops, for example a tropical deciduous forest in much of peninsular India.

The figure below summarises these stages.

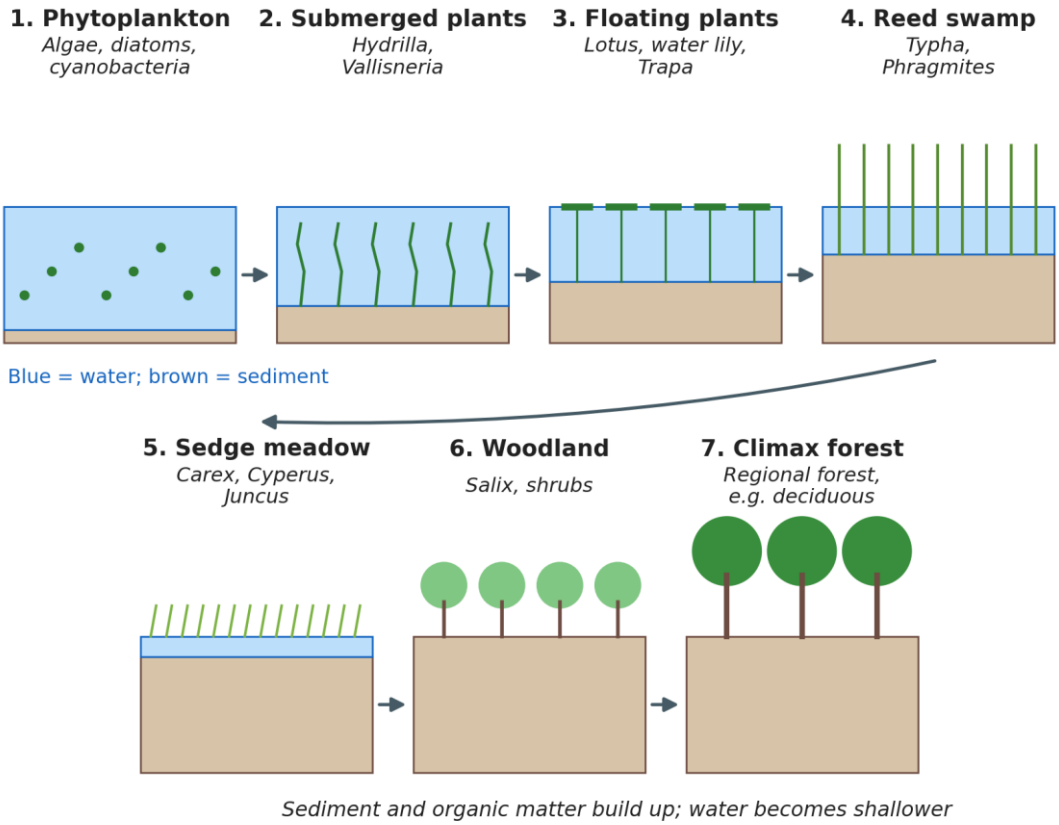


Figure 2.3: *Stages of hydrarch succession, in which a pond is gradually filled by sediment and organic matter and replaced by marsh, meadow and finally forest. Source: Author.*

Many Indian ponds and lakes show stages of hydrarch succession, often speeded up by human activity. Village ponds and urban lakes receiving sewage become choked with water hyacinth and silt and shrink year by year. In **Loktak Lake** in Manipur, floating mats of vegetation and soil called **phumdis** cover much of the surface; one part of the lake forms Keibul Lamjao National Park, the only floating national park in the world and home of the sangai deer. Changes in water levels after the Ithai barrage was built in 1983 have altered how phumdis form and decay, and their management is a major concern.

2.6.5 Autogenic and allogenic succession

Succession driven by the organisms themselves, through changes they make to the site, is called **autogenic succession**. Succession driven by external forces, such as changes in climate, siltation from a river, fire or grazing, is **allogenic succession**. Most seres involve both.

Succession dominated by green plants is **autotrophic**; succession on dead organic matter such as a fallen log or dung pat, dominated by bacteria, fungi and animals, is **heterotrophic**.

2.6.6 The climax community

Clements believed that every region has a single climax determined by its climate, the **monoclimax theory**. In central India, for example, this might be a tropical dry deciduous forest of teak or sal.

Tansley and others argued that local soil, topography, fire or grazing can produce several stable communities in the same climate, the **polyclimax theory**. Robert Whittaker's **climax pattern hypothesis** sees climax vegetation as a continuous pattern that changes gradually along gradients of moisture, temperature and soil.

Today ecologists regard the climax as a **dynamic** state, not a fixed end-point. Even a mature forest contains patches at different stages of recovery after tree falls, fires or storms. Communities held below the climax by repeated disturbance are called **disclimax** (disturbance climax) or **subclimax** communities. Many grasslands in India, such as the grassy blanks of the Western Ghats and parts of the Terai, are maintained by fire, grazing or flooding, and would turn into forest if these were removed.

2.6.7 Changes during succession

As succession proceeds, the community and ecosystem change in predictable ways. Eugene Odum summarised these trends in an influential 1969 paper, *The Strategy of Ecosystem Development*.

Table 2.8: Typical trends during succession (after E. P. Odum, 1969)

Feature	Early (pioneer) stages	Late (climax) stages
Species diversity	Low	High
Size and life span of organisms	Small, short-lived	Large, long-lived
Biomass (standing crop)	Low	High
Net production compared with respiration	Production exceeds respiration	Production roughly equals respiration
Food chains	Short, mainly grazing	Long, web-like, mainly detritus
Nutrient cycles	Open; nutrients easily lost	Closed; nutrients held and recycled
Stability	Low	High

2.6.8 Ecological succession in India

- **Primary succession on Barren Island**: Barren Island in the Andaman Sea is India's only active volcano. It has erupted repeatedly since 1991, burying parts of the island under fresh lava and ash, on which pioneer plants and lichens are beginning primary succession.
- **Landslide scars in the Himalaya**: landslides on steep slopes, such as those after the 2013 Kedarnath disaster, expose rock and rubble on which succession restarts, with grasses, shrubs and alder (*Alnus nepalensis*) as early colonists.
- **New river islands and coastal land**: chars (sandbars) in the Brahmaputra and Ganga, and mudflats in the Sundarbans, are colonised in turn by grasses, then shrubs and, on the coast, mangroves.
- **Secondary succession after shifting cultivation**: in the North-East, fields cleared and burnt for *jhum* cultivation are left fallow after one or two years of cropping. Weeds, grasses and bamboo grow first, followed by secondary forest. Studies at the North-Eastern Hill University, Shillong, by P. S. Ramakrishnan and colleagues showed that soil fertility recovers adequately

only if the fallow period is long, about 10 years or more. Shortened cycles of three to five years, caused by population pressure, prevent the forest from recovering.

- **Deflected succession on the Delhi Ridge**: the invasive mesquite *Prosopis juliflora* (vilayati kikar), planted in the early twentieth century, has spread across much of the Delhi Ridge and suppresses native Aravalli species. Restoration at the Aravalli Biodiversity Park in Delhi has worked by removing *Prosopis* and planting native species such as dhau (*Anogeissus pendula*) and khair (*Senegalia catechu*), guiding succession towards a native community.

Case Study: Shifting cultivation and succession in North-East India

Shifting cultivation, called *jhum* in the North-East, is a traditional system in which a patch of forest is cut and burnt, cropped for a year or two, and then left fallow so that natural secondary succession can restore the soil. With long fallows the system can be sustainable: the forest regrows, roots and leaf litter rebuild soil organic matter, and nitrogen-fixing plants and bamboos restore fertility.

Research led by P. S. Ramakrishnan in Meghalaya in the 1970s and 1980s showed how the length of the jhum cycle controls this process. With fallows of 10 years or more, soil nutrients and crop yields recovered. As population grew and fallows shrank to about 5 years, weeds and bamboos dominated, soils became poorer and erosion increased. The research also showed that traditional practices, such as retaining tree stumps, planting alder in Nagaland and growing mixed crops, speed up recovery. This is an example of how understanding succession can make a traditional land use more sustainable.

Key Points

- Ecological succession is the orderly change of a community over time, passing through seral stages from pioneers to a climax.
- Its steps are nudation, invasion, competition and coaction, reaction and stabilisation.
- Primary succession starts on bare surfaces without soil and is slow; secondary succession starts where soil remains and is faster.
- Xerarch succession starts on dry rock (lichens → mosses → herbs → shrubs → forest); hydrarch succession starts in water (phytoplankton → submerged → floating → reed swamp → sedge meadow → woodland → forest).
- Climax communities are dynamic; fire, grazing and invasive species can hold or deflect succession.

2.7 Biogeochemical Cycles (Nutrient Cycling)

Unlike energy, matter is not lost from the Earth. The carbon in your body may once have been part of a dinosaur, a Himalayan oak or a bubble of CO_2 above the Arabian Sea. The repeated movement of chemical elements between living organisms and the non-living environment of air, water, rocks and soil is called a **biogeochemical cycle** ("bio" for life, "geo" for the Earth, "chemical" for the elements). It is also called **nutrient cycling**, and it is the **biogeochemical function** of ecosystems.

2.7.1 Nutrients and their cycles

Organisms need about 30 to 40 chemical elements. **Macronutrients** are needed in large amounts: carbon, hydrogen, oxygen, nitrogen, phosphorus, sulfur, potassium, calcium and magnesium. **Micronutrients** or trace elements, such as iron, zinc, copper, manganese, boron and molybdenum, are needed in tiny amounts but are still essential.

Each element is held in **pools** or **reservoirs**. A **reservoir pool** is large and slow-moving, usually non-biological, such as carbonate rocks or the nitrogen of the atmosphere. An **exchange** or **cycling pool** is smaller and moves quickly between organisms and their immediate environment, such as the nitrate in soil water. The movement between pools is called **flux**.

Biogeochemical cycles are of two kinds:

- **Gaseous cycles**, in which the main reservoir is the atmosphere or ocean. The carbon, nitrogen and oxygen cycles are gaseous. They tend to be fast and well buffered, because the large atmospheric pool smooths out local changes.
- **Sedimentary cycles**, in which the main reservoir is rocks and sediments of the Earth's crust. The phosphorus, calcium and potassium cycles are sedimentary; the sulfur cycle has both gaseous and sedimentary phases. Sedimentary cycles are slow, and elements lost to deep ocean sediments may be unavailable for millions of years.

Decomposers are central to every cycle. By breaking down dead matter they return nutrients to the exchange pool. In tropical rainforests, decomposition is so fast that most nutrients are held in living plants rather than the soil, which is why cleared rainforest land often loses fertility within a few years.

2.7.2 Water cycle

The **water cycle** or **hydrological cycle** is the continuous movement of water between the oceans, the atmosphere and the land. It is driven by solar energy and gravity, and it carries dissolved nutrients and heat around the planet.

About 97% of the Earth's water is in the oceans. Of the small fraction that is fresh, most is locked in ice caps and glaciers, and nearly all of the rest is groundwater; lakes, rivers, soil moisture and the atmosphere together hold well under 1% of fresh water. Yet this small, fast-moving part sustains land life.

The main processes are:

1. **Evaporation**: the Sun heats oceans, lakes and soil, and water changes into vapour. Most evaporation occurs over the oceans.
2. **Transpiration**: plants absorb water through roots and release vapour through pores (stomata) in their leaves. Together, evaporation from land and transpiration are called **evapotranspiration**. A large tree can transpire hundreds of litres a day.
3. **Condensation**: rising vapour cools and condenses on dust particles to form clouds.
4. **Precipitation**: water returns to the surface as rain, snow or hail.
5. **Infiltration and percolation**: part of the precipitation soaks into the soil and moves down to recharge **groundwater** in aquifers.
6. **Runoff**: water that does not infiltrate flows over the surface into streams and rivers and back to the sea. Groundwater also seeps slowly into rivers, keeping them flowing in the dry season.

Water spends very different lengths of time in each pool. A molecule stays in the atmosphere for only about nine days on average, in a river for weeks, in a lake for years, in deep groundwater for thousands of years and in the Antarctic ice sheet for many thousands of years.

In India, the water cycle is dominated by the **monsoon**. The southwest monsoon (June to September) brings about three-quarters of the country's annual rainfall. According to the Central Water Commission's 2019 reassessment, India's river basins receive an average of about 3,880 billion cubic metres (BCM) of rainfall a year and have average annual water resources of about 1,999 BCM, of which only around 1,100 BCM is considered utilisable. Forests play a key role: the forests of the Western Ghats and the Himalaya slow runoff, recharge springs and keep rivers such as the Kaveri and Krishna flowing after the rains end. Deforestation, paving of cities and over-pumping of groundwater all alter the local water cycle. (India's water resources are discussed in Chapter 3.)

2.7.3 Carbon cycle

Carbon is the backbone of all organic molecules: carbohydrates, proteins, fats and nucleic acids. The **carbon cycle** describes how carbon moves between the atmosphere, living things, soils, oceans and rocks.

The main processes, shown in the figure below, are:

- **Photosynthesis**: producers on land and in water take CO_2 from the air or water and convert it into sugars. This is the main route by which carbon enters living systems.
- **Respiration**: plants, animals and decomposers break down sugars to release energy, returning CO_2 to the air or water.
- **Feeding**: carbon passes along food chains as organisms eat one another.
- **Decomposition**: decomposers break down dead matter, releasing CO_2 (or methane, CH_4, in waterlogged places such as rice paddies and wetlands, where oxygen is scarce).
- **Ocean exchange**: CO_2 dissolves in seawater and forms carbonic acid, bicarbonate and carbonate. Marine organisms use carbonate to build shells and skeletons; when they die, some sink and form limestone and other sediments.
- **Fossil fuel formation**: over millions of years, some dead organic matter was buried before it could decompose, forming coal, oil and natural gas.
- **Combustion**: burning of fossil fuels and of wood and other biomass releases stored carbon as CO_2.
- **Volcanic activity and weathering**: volcanoes release CO_2 from the Earth's interior, while the slow weathering of rocks removes CO_2 from the air over geological time.

The largest stores of carbon are in rocks and ocean sediments. Among the active pools, the deep ocean holds far more carbon than the atmosphere, and soils hold more than living vegetation. Plants and soils on land, and the oceans, each take up roughly a quarter of the CO_2 that humans emit; the remainder accumulates in the atmosphere.

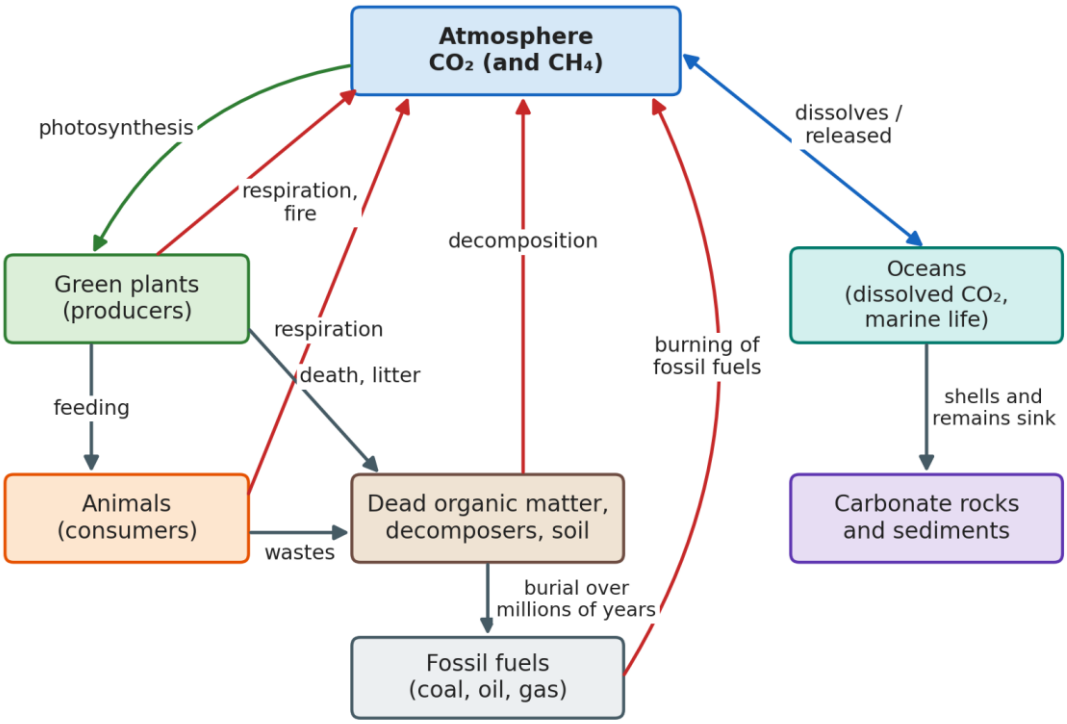


Figure 2.4: *The carbon cycle. Photosynthesis removes CO_2 from the atmosphere; respiration, decomposition and combustion return it. Oceans exchange CO_2 with the air, and buried organic matter forms fossil fuels. Source: Author.*

Human disturbance of the carbon cycle

Since the Industrial Revolution, humans have moved carbon from long-term stores (fossil fuels, forests and soils) into the atmosphere far faster than natural processes can absorb it. The concentration of CO_2 in the atmosphere has risen from about 280 parts per million (ppm) before 1750 to about 419 ppm in 2023, according to the Global Carbon Budget 2024. Fossil fuel use alone released about 37 billion tonnes of CO_2 in 2023. This extra CO_2 is the main cause of global warming and also makes the oceans more acidic, which harms corals and shell-forming organisms. (Climate change is discussed in Chapter 5.)

India's forests are an important carbon store. The *India State of Forest Report 2023* estimated the carbon stock of India's forests at about 7,285 million tonnes, an increase of about 81.5 million tonnes over 2021. Under its updated Nationally Determined Contribution (2022), India has committed to creating an additional carbon sink of 2.5 to 3 billion tonnes of CO_2 equivalent through additional forest and tree cover by 2030. Mangroves, seagrasses and salt marshes, called **blue carbon** ecosystems, store carbon very efficiently in their waterlogged soils, which makes the Sundarbans and other Indian mangroves valuable for climate as well as for biodiversity.

2.7.4 Nitrogen cycle

Nitrogen is an essential part of proteins, nucleic acids (DNA and RNA) and chlorophyll. Although nitrogen gas (N_2) makes up about 78% of the atmosphere, most organisms cannot use it directly, because the two nitrogen atoms are held by a very strong triple bond. Nitrogen must first be **fixed**, that is, converted into reactive forms such as ammonia (NH_3), ammonium (NH_4^+) or nitrate (NO_3^-). For this reason, nitrogen is often the nutrient that limits plant growth on land. The cycle depends heavily on bacteria and has five main steps, summarised in the figure below.

Nitrogen fixation

Nitrogen fixation converts atmospheric N_2 into ammonia or nitrate. It happens in three ways:

- **Biological fixation** by certain bacteria and cyanobacteria that have the enzyme **nitrogenase**. Some are **free-living** in soil or water, such as *Azotobacter* and *Clostridium* and cyanobacteria such as *Anabaena* and *Nostoc*. Others are **symbiotic**, such as *Rhizobium*, which lives in the root nodules of legumes (pulses such as gram, arhar, moong and soybean, and trees such as *Sesbania*). In Indian rice paddies, the small water fern *Azolla*, which carries the cyanobacterium *Anabaena azollae*, is used as a **biofertiliser**.
- **Atmospheric fixation** by lightning, whose energy combines nitrogen and oxygen into nitrogen oxides that are washed down by rain as nitrate.
- **Industrial fixation** by the **Haber–Bosch process**, which combines N_2 and hydrogen at high temperature and pressure to make ammonia for fertilisers such as urea.

Ammonification, nitrification, assimilation and denitrification

- **Assimilation**: plants absorb ammonium and nitrate from the soil and use them to make proteins and nucleic acids. Animals obtain nitrogen by eating plants or other animals.
- **Ammonification**: when organisms die or excrete wastes (urea, uric acid), decomposer bacteria and fungi convert the organic nitrogen back into ammonia or ammonium. This is also called mineralisation.
- **Nitrification**: chemoautotrophic bacteria oxidise ammonium in two steps. *Nitrosomonas* converts ammonium to nitrite (NO_2^-), and *Nitrobacter* converts nitrite to nitrate (NO_3^-), the form most easily taken up by plants.

$$NH_4^+ \rightarrow NO_2^- \text{ (Nitrosomonas)} \rightarrow NO_3^- \text{ (Nitrobacter)}$$

- **Denitrification**: in waterlogged, oxygen-poor soils and sediments, bacteria such as *Pseudomonas* and *Thiobacillus denitrificans* convert nitrate back into nitrogen gas (N_2), with some nitrous oxide (N_2O), which escapes into the atmosphere. This completes the cycle.

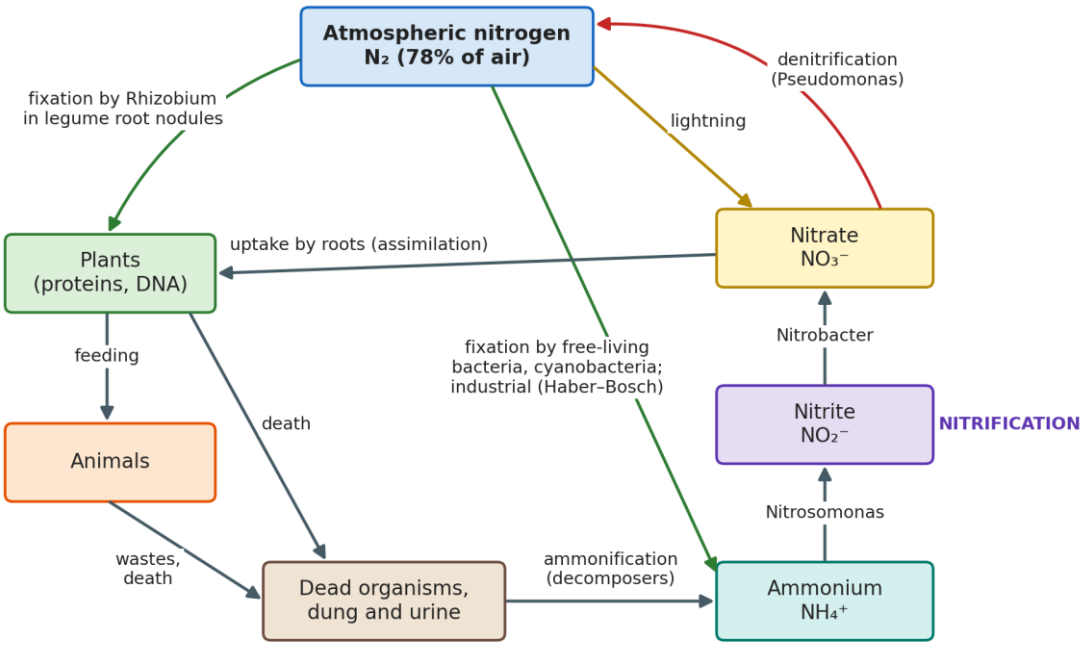


Figure 2.5: *The nitrogen cycle. Atmospheric N_2 is fixed by bacteria, lightning and industry; ammonium and nitrate are taken up by plants and passed along food chains; decomposers return ammonium; nitrifying bacteria convert it to nitrate; and denitrifying bacteria return N_2 to the air. Source: Author.*

Human disturbance of the nitrogen cycle

Through fertiliser manufacture, the growing of legume crops and the burning of fuels, humans now fix more nitrogen each year than all natural processes on land combined. India is one of the world's largest consumers of nitrogen fertiliser, mostly as urea. The *Indian Nitrogen Assessment* (2017) reported that rice and wheat take up only about a third of the fertiliser nitrogen applied to them. The rest is lost: it **leaches** as nitrate into groundwater (nitrate levels above safe limits have been reported in wells in parts of Punjab, Haryana and western Uttar Pradesh); it **runs off** into rivers and lakes, causing **eutrophication** (excessive growth of algae and plants that depletes oxygen); or it escapes as ammonia and as **nitrous oxide**, a powerful greenhouse gas that also damages the ozone layer. Nitrogen oxides from vehicles and power plants contribute to smog and acid rain.

India has responded with measures such as making all urea **neem-coated** from 2015 (the neem oil slows the conversion of urea to nitrate, so less is lost), promoting **soil health cards** that advise farmers on correct fertiliser doses, and encouraging biofertilisers, pulses in crop rotations and natural farming.

2.7.5 Phosphorus cycle

Phosphorus is part of DNA, RNA, ATP (the energy carrier of cells), cell membranes, and bones and teeth. The **phosphorus cycle** is a **sedimentary cycle**. It has no significant gaseous phase, so phosphorus does not circulate through the atmosphere except as dust.

The main reservoir is phosphate rock, such as **apatite**. The cycle runs as follows:

1. **Weathering** of rocks slowly releases phosphate ions (PO_4^{3-}) into soil and water.
2. **Uptake**: plants absorb dissolved phosphate through their roots, often with the help of **mycorrhizal fungi** associated with roots. Animals obtain it by eating plants.

3. **Return**: decomposers release phosphate from dead organisms and wastes back to the soil.
4. **Loss to sediments**: some phosphate is washed into rivers and the sea, where it settles and becomes part of ocean sediments. Over millions of years these may be lifted up by geological movements to form new rock.
5. **Seabird link**: in some coastal areas, fish-eating seabirds deposit phosphorus-rich droppings called **guano**, which returns some phosphorus from sea to land.

Because the cycle is so slow and phosphate binds tightly to soil particles, phosphorus is often the nutrient that **limits** growth, especially in freshwater lakes. For the same reason, even small additions of phosphorus from sewage, detergents and farm runoff can cause **eutrophication**. In Bengaluru, Bellandur Lake, which receives large volumes of untreated sewage, has repeatedly produced toxic foam and has even caught fire, partly because of excess nutrients and detergents.

India has limited deposits of rock phosphate, mainly in Rajasthan (around Udaipur) and Madhya Pradesh, and depends on imports for most of the phosphorus used in its fertilisers such as DAP (diammonium phosphate). Since phosphate rock is a finite resource, recovering phosphorus from sewage and manure is increasingly seen as important.

2.7.6 Sulfur cycle

Sulfur is part of certain amino acids (cysteine and methionine) and therefore of most proteins. The **sulfur cycle** has both sedimentary and gaseous phases.

- **Reservoirs**: most sulfur is stored in rocks and sediments as sulfate minerals such as **gypsum** and sulfide minerals such as **pyrite**, and in ocean water as dissolved sulfate.
- **Uptake**: plants absorb sulfate (SO_4^{2-}) from soil and water and build it into proteins; animals obtain it by eating plants.
- **Decomposition**: when organisms die, decomposers release sulfur as **hydrogen sulfide** (H_2S), the gas that gives rotten eggs and polluted drains their smell.
- **Oxidation and reduction by bacteria**: in the presence of oxygen, sulfur bacteria (such as *Thiobacillus* and *Beggiatoa*) oxidise H_2S and elemental sulfur to sulfate. In oxygen-free mud, sulfate-reducing bacteria (such as *Desulfovibrio*) convert sulfate back to H_2S.
- **Natural gaseous inputs**: volcanoes release sulfur dioxide (SO_2) and H_2S; marine phytoplankton produce **dimethyl sulfide** (DMS), which helps form cloud droplets over the oceans.
- **Return to the surface**: in the atmosphere, SO_2 is oxidised to sulfate and sulfuric acid and falls back in rain or as dry particles.

Human disturbance. Burning of coal and oil, and smelting of sulfide ores, release large amounts of SO_2. Analyses of satellite data published in 2019 identified India as the world's largest emitter of human-made SO_2, mainly from coal-fired power plants. SO_2 forms fine sulfate particles that harm human health and contribute to **acid rain**, which can damage soils, lakes and buildings such as the Taj Mahal (acid rain is discussed in Chapter 5). India's 2015 emission standards required coal power plants to install **flue gas desulfurisation (FGD)** units to remove SO_2, but compliance was slow. In July 2025 the Ministry of Environment, Forest and Climate Change revised the norms, keeping FGD mandatory mainly for plants near large cities and critically polluted areas and exempting most other plants subject to stack-height requirements.

Table 2.9: Summary of major biogeochemical cycles

Cycle	Type	Main reservoir	Key organisms or processes	Main human disturbance
Water	Gaseous (vapour)	Oceans	Evaporation, transpiration, precipitation	Deforestation, over-pumping of groundwater, dams
Carbon	Gaseous	Rocks, oceans, atmosphere	Photosynthesis, respiration	Fossil fuel burning, deforestation
Nitrogen	Gaseous	Atmosphere (N_2)	Fixing, nitrifying and denitrifying bacteria	Fertilisers, vehicle and power-plant emissions
Phosphorus	Sedimentary	Phosphate rocks	Weathering, mycorrhizae, decomposers	Fertilisers, sewage, detergents
Sulfur	Mixed	Rocks, ocean sulfate	Sulfur-oxidising and sulfate-reducing bacteria	Coal burning, ore smelting

Key Points

- Matter cycles between living and non-living pools through biogeochemical cycles; decomposers return nutrients for reuse.
- Gaseous cycles (carbon, nitrogen) have the atmosphere as reservoir; sedimentary cycles (phosphorus) have rocks as reservoir.
- Nitrogen must be fixed by bacteria, lightning or industry before most organisms can use it; nitrifying and denitrifying bacteria complete the cycle.
- Human activities have greatly increased atmospheric CO_2, reactive nitrogen, phosphorus in waters and SO_2 emissions, causing climate change, eutrophication and acid rain.

2.8 Productivity

Productivity is the rate at which organic matter (biomass) is produced in an ecosystem per unit area per unit time. It is expressed as mass, such as grams of dry matter per square metre per year (g/m^2/year), or as energy, such as kilojoules per square metre per year. Productivity is a **rate**. It must not be confused with **standing crop** (or **biomass**), which is the amount of living matter present at a given moment. A grassland may have a small standing crop but high productivity, because the grass is eaten and regrows quickly; an old forest has a huge standing crop but may add little new biomass each year.

2.8.1 Primary productivity

Primary productivity is the rate at which producers convert light energy into chemical energy stored in organic matter.

- **Gross primary productivity (GPP)** is the total rate of photosynthesis, including the organic matter that the plants themselves use up in respiration.

- **Net primary productivity (NPP)** is the rate at which organic matter is stored in plant tissues after the plants' own respiration (R) has been subtracted. It is the food available to herbivores and decomposers.

$$\text{NPP} = \text{GPP} - \text{R}$$

In most plants, respiration uses roughly half of GPP, though the share is higher in hot climates and in old forests with much living wood. Globally, NPP is estimated at about 105 billion tonnes of carbon a year, divided roughly equally between land and oceans (Field and others, 1998).

Primary productivity depends on light, temperature, water and nutrients, and on the length of the growing season. On land, it is highest where it is warm and wet throughout the year and lowest in deserts and the cold tundra. In the oceans, it is highest in shallow coastal waters, estuaries, coral reefs and areas of **upwelling**, where winds bring nutrient-rich deep water to the surface, and lowest in the open ocean, which is like a nutrient-poor "desert" despite plentiful light.

2.8.2 Secondary productivity

Secondary productivity is the rate at which consumers (heterotrophs) form new biomass from the food they assimilate. It is not "gross" or "net" in the same sense, since consumers only convert food that producers have already made. Secondary productivity is always much smaller than primary productivity because of energy losses at each step. The fish catch of a lake, the milk and meat from livestock, and the growth of a deer population are all forms of secondary production.

A further term, **net ecosystem productivity (NEP)**, is GPP minus the respiration of all organisms in the ecosystem (plants, animals and decomposers). A young, growing forest has positive NEP and acts as a **carbon sink**; a mature forest has NEP close to zero; and a degraded or burning forest may have negative NEP and become a **carbon source**.

2.8.3 Productivity of different ecosystems

The table below, based on the classic estimates of R. H. Whittaker and G. E. Likens (1975), compares the NPP of major ecosystems. The chart that follows shows the mean values.

Table 2.10: Net primary productivity of major ecosystems (Whittaker and Likens, 1975)

Ecosystem	Typical range (g dry matter/m^2/year)	Mean (g/m^2/year)	Indian example
Tropical rainforest	1,000–3,500	2,200	Evergreen forests of the Western Ghats, Andamans
Swamps and marshes	800–3,500	2,000	Sundarbans, Keoladeo, Loktak
Algal beds and coral reefs	500–4,000	2,500	Gulf of Mannar, Lakshadweep
Estuaries	200–3,500	1,500	Hooghly estuary, Chilika
Tropical seasonal forest	1,000–2,500	1,600	Sal and teak forests of central India
Savanna	200–2,000	900	Grasslands of the Deccan plateau
Cultivated land	100–3,500	650	Rice–wheat fields of the Indo-Gangetic plain
Temperate grassland	200–1,500	600	Alpine and sub-alpine meadows (bugyals)

Ecosystem	Typical range (g dry matter/m²/year)	Mean (g/m²/year)	Indian example
Tundra and alpine	10–400	140	High-altitude Ladakh and Spiti
Desert scrub	10–250	90	Thar desert
Open ocean	2–400	125	Central Indian Ocean

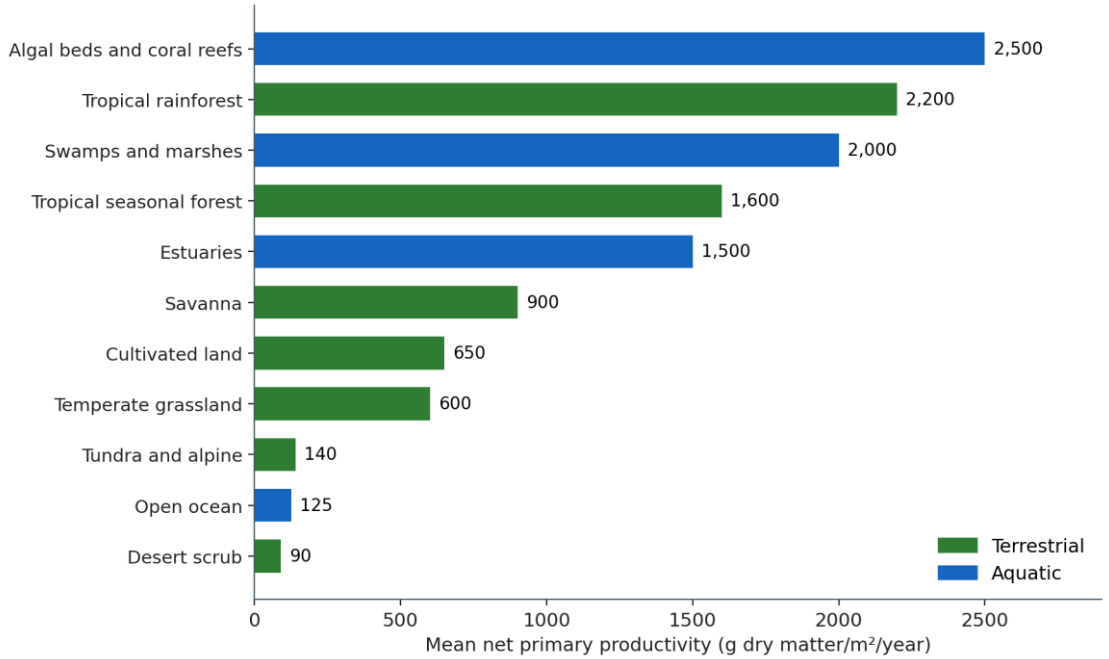


Figure 2.6: *Mean net primary productivity of selected ecosystems, in grams of dry matter per square metre per year. Coral reefs, wetlands and tropical forests are the most productive; deserts and the open ocean the least. Data: Whittaker and Likens (1975).*

Reefs, wetlands, estuaries and tropical rainforests are the most productive natural ecosystems per unit area. Cropland is not necessarily more productive than natural vegetation, though crops channel more of their production into food. The open ocean has low productivity per square metre, but because it covers about two-thirds of the Earth its total production is large.

Productivity in Indian waters

The two seas around peninsular India differ in productivity. In the **Arabian Sea**, strong southwest monsoon winds cause **upwelling** along the coasts of Kerala, Karnataka and Oman, bringing nutrients to the surface. This feeds phytoplankton blooms that support rich fisheries, including the Indian oil sardine (*Sardinella longiceps*) and mackerel of the Kerala coast. The **Bay of Bengal** receives huge volumes of fresh water from the Ganga, Brahmaputra and other rivers, which forms a light surface layer that prevents mixing and keeps surface waters relatively poor in nutrients in the open bay, though its river-fed coasts are productive. Overall, India produced over 18 million tonnes of fish in 2023–24 from marine and inland waters and aquaculture, making it the world's second-largest fish producer (Ministry of Fisheries, 2025).

2.8.4 Measuring productivity

Ecologists measure productivity in several ways. The **harvest method** involves cutting, drying and weighing plants from sample plots (quadrats) at intervals; the increase in dry weight gives NPP. In ponds and lakes, the **light and dark bottle method** is widely used: water samples are sealed in a clear bottle and a blackened bottle and suspended in the water for a few hours. Oxygen rises in the clear bottle (photosynthesis minus respiration) and falls in the dark bottle (respiration only). The difference between the two gives GPP. On a large scale, satellites measure the greenness of vegetation and the colour of the ocean, from which NPP can be estimated for the whole country or planet.

2.9 Ecological Pyramids

An **ecological pyramid** is a graphical representation of the number, biomass or energy at successive trophic levels of an ecosystem. Each trophic level is drawn as a horizontal bar whose width is proportional to the quantity measured, with producers at the base and top carnivores at the apex. The idea was introduced by Charles Elton in 1927, and pyramids are therefore also called **Eltonian pyramids**.

There are three types: pyramids of **number**, **biomass** and **energy**. A pyramid is **upright** when the quantity decreases from the base to the top and **inverted** when it increases.

2.9.1 Pyramid of number

The **pyramid of number** shows the number of individual organisms at each trophic level in a given area.

- **Upright**: in a **grassland**, there are many grass plants, fewer grasshoppers, still fewer frogs and snakes, and very few hawks. A **pond** shows a similar shape, with countless phytoplankton at the base.
- **Partly inverted (spindle-shaped)**: in a **forest**, a few large trees support a great number of insects and birds, which support fewer carnivores. The base is narrower than the second level.
- **Fully inverted**: in a **parasitic** food chain, one tree supports many birds, which support even more lice and other parasites, which support still more hyperparasites.

The pyramid of number is easy to understand and to construct, but it treats a banyan tree and a blade of grass as equal, so its shape depends on the size of organisms rather than their importance.

2.9.2 Pyramid of biomass

The **pyramid of biomass** shows the total dry mass of living organisms (standing crop) at each trophic level, measured as g/m^2 or kg/ha.

- **Upright**: in most **terrestrial** ecosystems, the mass of producers is much greater than that of herbivores, and that of herbivores greater than that of carnivores. In Silver Springs, Florida, H. T. Odum (1957) found 809 g/m^2 of producers, 37 g/m^2 of herbivores, 11 g/m^2 of carnivores and 1.5 g/m^2 of top carnivores.
- **Inverted**: in many **open-water** ecosystems, the standing crop of phytoplankton at a given moment is smaller than that of the zooplankton feeding on it. In the English Channel, measurements reported by Harvey (1950) found about 4 g/m^2 of phytoplankton supporting about 21 g/m^2 of zooplankton and bottom fauna. This is possible because phytoplankton are

tiny and reproduce very fast, replacing themselves within days, while zooplankton live longer and accumulate biomass. A small standing crop with a high rate of turnover can feed a larger standing crop of consumers.

2.9.3 Pyramid of energy

The **pyramid of energy** shows the amount of energy flowing through each trophic level per unit area per unit time, such as kJ/m^2/year. It shows the **rate** of energy flow, not a snapshot.

The pyramid of energy is **always upright**. Because energy is lost as heat at every transfer (the second law of thermodynamics), no trophic level can receive more energy than the level below it. Even where the pyramid of biomass is inverted, as in the English Channel, the energy fixed by phytoplankton over a year is far greater than that passed to zooplankton. For example, in Silver Springs the energy flow was about 20,810 kcal/m^2/year at the producer level, 3,368 at herbivores, 383 at carnivores and 21 at top carnivores (H. T. Odum, 1957).

The pyramid of energy is considered the most fundamental of the three, because it reflects the actual flow of energy, allows different ecosystems to be compared, and takes account of turnover rates. Its drawback is that it is the hardest to measure. The figure below compares the three types of pyramid.

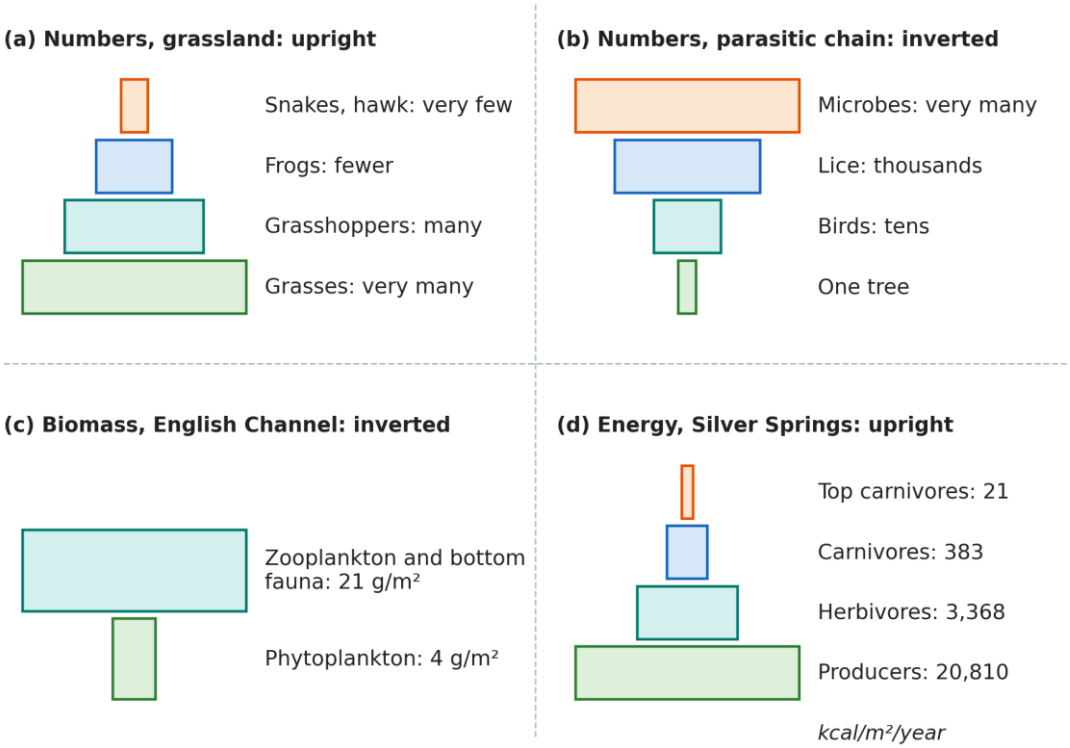


Figure 2.7: *Ecological pyramids. (a) Upright pyramid of number in a grassland. (b) Inverted pyramid of number in a parasitic food chain. (c) Inverted pyramid of biomass in the English Channel (Harvey, 1950), drawn to scale. (d) Upright pyramid of energy in Silver Springs, Florida (H. T. Odum, 1957). Other bar widths are schematic. Source: Author.*

Table 2.11: Shapes of ecological pyramids in different ecosystems

Ecosystem	Pyramid of number	Pyramid of biomass	Pyramid of energy
Grassland	Upright	Upright	Upright
Forest	Partly inverted (spindle)	Upright	Upright

Ecosystem	Pyramid of number	Pyramid of biomass	Pyramid of energy
Pond or open sea	Upright	Often inverted	Upright
Parasitic food chain	Inverted	Usually inverted	Upright

2.9.4 Limitations of ecological pyramids

Pyramids are useful teaching tools, but they simplify nature. They assume that each organism belongs to a single trophic level, although many feed at several. They usually leave out decomposers and detritivores, although these process much of the energy. They do not show the food web connections between species. And the pyramids of number and biomass are snapshots that change with season; a temperate lake has very different pyramids in summer and winter.

> **Think and Do: Pyramid of your college campus**
>
> In groups, mark a 1 m × 1 m quadrat on a campus lawn. Count the grass and herb plants (producers), the insects and other invertebrates feeding on them (herbivores), and the spiders, ants and predatory beetles (carnivores). Count birds seen feeding in the lawn over 30 minutes. Draw a pyramid of number. Then, for one tree on campus, count the leaf-eating insects and insect-eating birds on it. Compare the two shapes and explain the difference.

2.10 Ecosystem Homeostasis

Homeostasis (from the Greek for "staying the same") is the ability of a system to maintain a relatively steady state despite changes in its surroundings. The human body keeps its temperature near 37 °C; in the same way, an undisturbed forest or lake keeps its species composition, biomass and nutrient levels within a fairly narrow range from year to year. Ecosystem homeostasis is not a fixed balance but a **dynamic equilibrium**: populations rise and fall, but they fluctuate around an average because of self-regulating mechanisms.

2.10.1 Feedback mechanisms

Ecosystems regulate themselves through **feedback**, in which the output of a process loops back to influence the process itself. The figure below contrasts the two kinds of feedback.

Negative feedback counteracts change and brings the system back towards its original state. It is the basis of homeostasis. Examples:

- **Predator–prey regulation**: when deer increase, tigers and leopards find food more easily and their numbers rise; more deer are then eaten, so deer numbers fall; with less food, predator numbers decline in turn, and deer recover.
- **Food limitation**: when a herbivore population grows too large, it overgrazes its food plants, so food runs short, births fall and deaths rise, bringing numbers down until vegetation recovers.
- **Carbon uptake**: rising atmospheric CO_2 can stimulate plant growth and dissolve more readily in seawater, removing part of the extra CO_2.

Positive feedback amplifies change and pushes the system further away from its original state. It can drive rapid collapse. Examples:

- **Eutrophication**: nutrients cause algal blooms; when the algae die, decomposers use up oxygen; fish die and decompose, releasing more nutrients and further lowering oxygen.
- **Ice–albedo feedback**: warming melts snow and ice, exposing darker land or sea that absorbs more sunlight, causing more warming and more melting. This is important in the Himalaya and the Arctic.
- **Forest drying**: when a forest is fragmented, the edges dry out and become prone to fire; fire kills trees and makes the remaining forest even drier and more fire-prone.

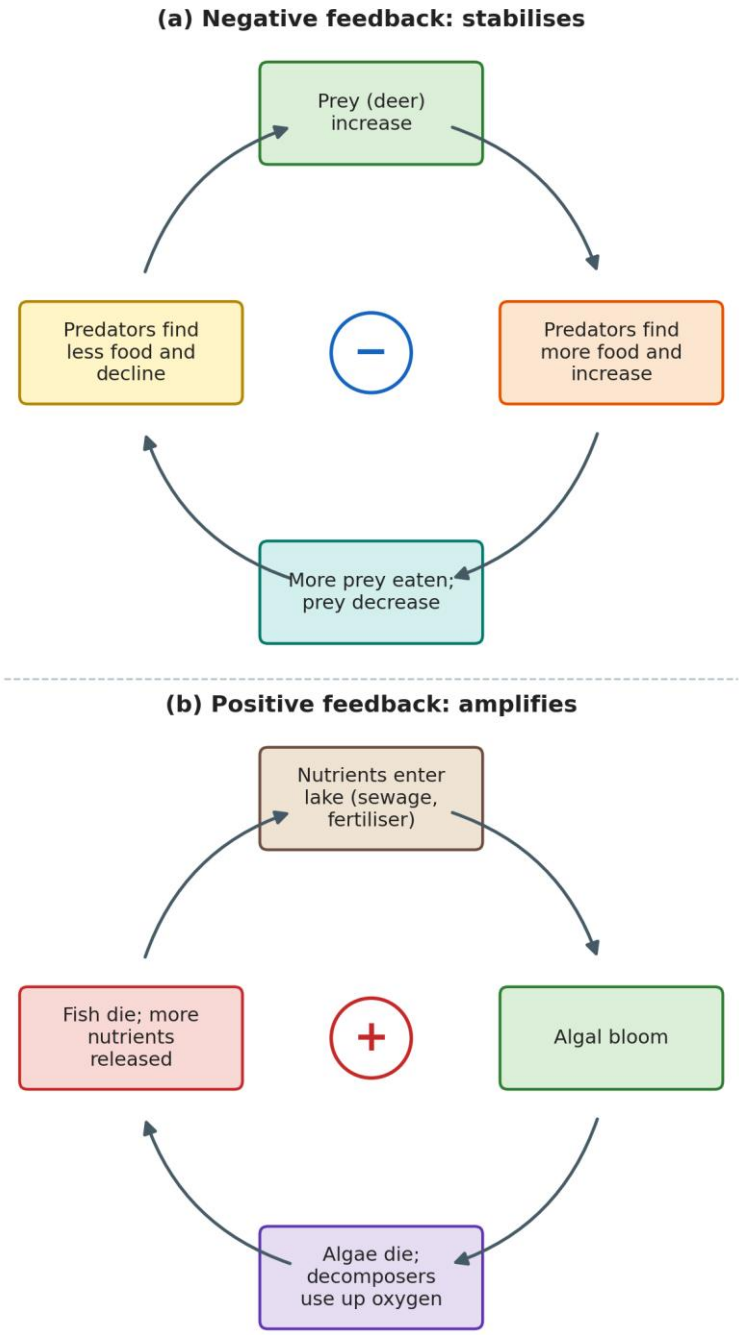


Figure 2.8: *Negative and positive feedback in ecosystems. (a) A negative feedback loop between prey and predator numbers keeps both within limits. (b) A positive feedback loop in a eutrophic lake drives the system further from balance. Source: Author.*

2.10.2 Homeostatic plateau

Negative feedback works only within limits. E. P. Odum described a range of stress, called the **homeostatic plateau**, within which feedback can restore balance. A lake can absorb a little sewage, decomposing it and recovering, because its bacteria and oxygen supply can cope. Beyond the limits of the plateau, positive feedback takes over and the system may change rapidly to a different state. Human activities often push ecosystems beyond these limits.

2.10.3 Resistance and resilience

Ecologists describe ecosystem stability in two ways:

- **Resistance** is the ability of an ecosystem to remain unchanged when disturbed. A mature sal forest with deep roots and thick bark resists a ground fire better than a young plantation.
- **Resilience** is the ability of an ecosystem to recover its structure and function after a disturbance. Grasslands are generally less resistant to fire than forests, as fire burns their above-ground parts, but they are highly resilient, since grasses regrow from underground roots within weeks.

High **biodiversity** usually increases both, because many species perform similar roles. If drought kills one grass species, others take its place; if one prey species declines, a predator switches to another. This "insurance" effect of diversity is one reason why natural forests are more stable than monoculture plantations, and why mixed farming systems are less vulnerable than single crops. The loss of species, as discussed in Chapter 6, reduces this insurance.

2.10.4 Tipping points and regime shifts

When stress exceeds an ecosystem's resilience, it may cross a **threshold** or **tipping point** and shift suddenly into a different, self-reinforcing state. Such a sudden change is called a **regime shift**. Once a tipping point is crossed, simply removing the stress may not be enough to restore the original state.

Examples include a clear lake dominated by rooted plants that suddenly turns into a turbid lake dominated by algae when nutrient loading crosses a threshold; a coral reef that becomes covered in seaweed after repeated bleaching and overfishing of plant-eating fish; and a forest that turns into savanna after repeated fires. Scientists have identified several large **tipping elements** in the Earth system whose collapse would have global effects, including the Amazon rainforest, the Greenland and West Antarctic ice sheets, and the **Indian summer monsoon**, whose stability depends on a balance between atmospheric moisture, land heating and aerosols.

Case Study: Coral bleaching in Lakshadweep

Coral reefs depend on a partnership between coral animals and algae called zooxanthellae, which live in their tissues and supply most of their food through photosynthesis. When sea temperatures stay about 1 °C above the normal summer maximum for several weeks, corals expel their algae and turn white, a process called **bleaching**. If the heat lasts too long, the corals starve and die.

From late 2023 a prolonged marine heatwave affected the Lakshadweep islands as part of the fourth global coral bleaching event. Surveys by the ICAR–Central Marine Fisheries Research Institute in 2024 reported severe bleaching across many reefs, and a CSIR–National Institute of Oceanography study at Kavaratti found about 80% of corals severely bleached. Earlier bleaching events in 1998, 2010 and 2016 had also hit these reefs.

Healthy reefs can recover from a single bleaching event within about a decade if they are not overfished or polluted, an example of resilience. But as bleaching events become more frequent, reefs have less time to recover between them. If recovery fails, reefs may shift to a degraded state dominated by algae and rubble, a regime shift that would threaten the fisheries, tourism and coastal protection on which island communities depend.

2.10.5 Human impacts on homeostasis

People weaken the self-regulating capacity of ecosystems in many ways: by removing top predators and keystone species, simplifying food webs through monocultures, adding nutrients and toxic pollutants, introducing invasive species such as *Lantana camara* and water hyacinth, fragmenting habitats, and changing the climate. Each of these reduces the diversity of species and links that give ecosystems their resilience. Wise management keeps disturbances within the homeostatic plateau and watches for early warning signs, such as slower recovery, that a tipping point is near.

Key Points

- Homeostasis is the dynamic self-regulation that keeps an ecosystem relatively stable.
- Negative feedback restores balance; positive feedback amplifies change.
- Resistance is the ability to withstand disturbance; resilience is the ability to recover; biodiversity strengthens both.
- Beyond the homeostatic plateau, ecosystems can cross tipping points into new states (regime shifts), as seen in eutrophic lakes and bleached coral reefs.

The functions described in this part of the chapter, energy flow, food webs, succession, nutrient cycling, productivity and homeostasis, operate in every ecosystem. The next part of the chapter looks at how they play out in particular kinds of ecosystems, from tundra and forests to grasslands, deserts, ponds, rivers, estuaries and oceans, and at the services these ecosystems provide and how they can be conserved and restored.

2.11 Types of Ecosystems

The structure and functions described so far are common to every ecosystem. Yet a Himalayan meadow, a sal forest in Jharkhand, the sand dunes of Jaisalmer and the mangrove creeks of the Sundarbans look and behave very differently. Climate, mainly temperature and rainfall, decides which plants can grow in a place; the plants in turn decide which animals, fungi and microbes can live there. The result is a set of recognisable ecosystem types that repeat across the world wherever conditions are similar.

A large terrestrial ecosystem type defined by its climate and dominant vegetation is called a **biome**. Tundra, forest, grassland and desert are the four major land biomes covered in this chapter. Aquatic ecosystems are classified instead by the chemistry and movement of their water: **freshwater** (ponds, lakes, streams, rivers, wetlands), **marine** (oceans and seas) and **transitional** systems such as estuaries, where fresh and salt water mix.

Ecosystems can also be **natural** (a forest, a river, a coral reef), running on solar energy and sustaining themselves, or **managed** (a crop field, a tea garden, a fish pond, a city park), created and maintained by people with added energy in the form of fertiliser, irrigation, fuel and labour. Most Indian landscapes are a mosaic of both.

2.11.1 What decides the type of ecosystem

Three groups of abiotic factors shape most of the differences between ecosystem types:

- **Temperature and its seasonality.** Low temperatures slow down photosynthesis, decomposition and every other biological process. Temperature falls with both latitude and altitude, which is why the upper Himalaya has tundra-like conditions despite lying in the subtropics.
- **Water availability.** On land, the balance between rainfall and evaporation decides whether a place supports forest, grassland or desert. Roughly, more than about 1,000 mm of well-distributed rain supports forest in India; 250–1,000 mm with a long dry season favours grassland or open woodland; and less than about 250 mm produces desert.
- **Soil, light and disturbance.** Soil depth and fertility, the amount of light reaching the ground, and recurring disturbances such as fire, grazing, floods and cyclones fine-tune the picture. In water, light, dissolved oxygen, salinity and current take over the role that rainfall plays on land.

The diagram below places the major biomes on a simple grid of mean annual temperature and rainfall, with the approximate positions of some Indian places. Because India spans tropical lowlands, the high Himalaya and the rain shadow beyond it, it contains almost every biome on the grid except true boreal forest. The same factors also set how much new plant matter an ecosystem produces each year, its **net primary productivity**; the comparison of productivity across ecosystem types in the earlier part of this chapter shows that warm, wet systems are the most productive and cold or dry ones the least.

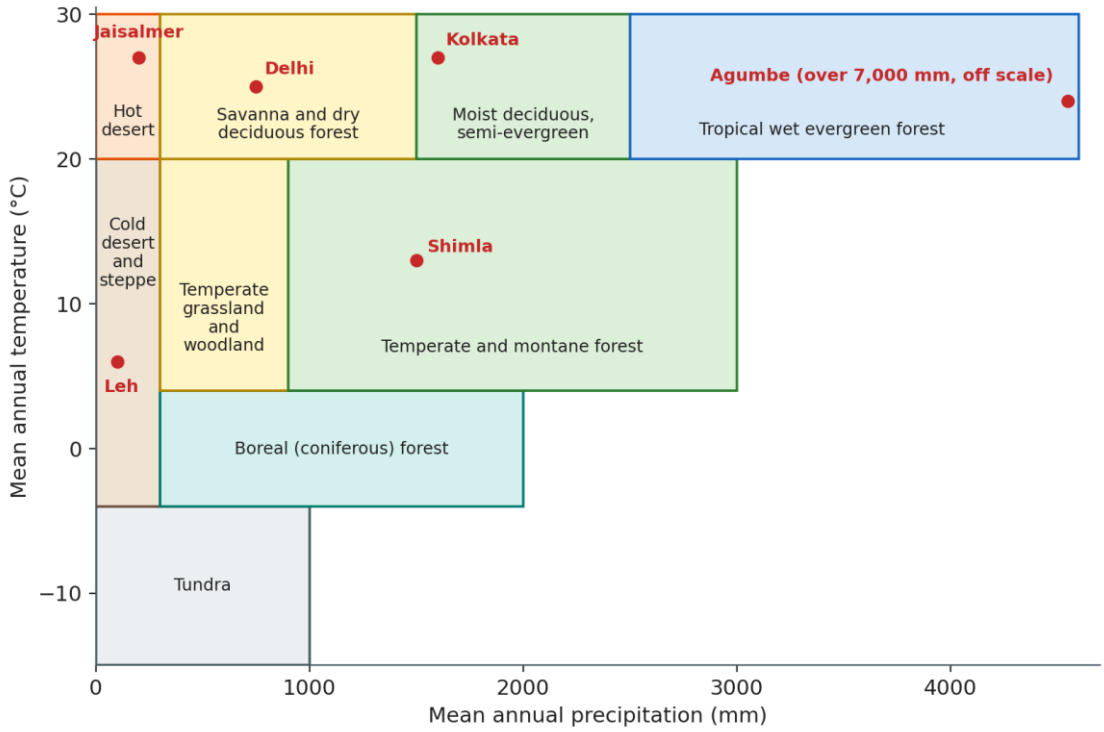


Figure 2.9: *Major biomes in relation to mean annual temperature and precipitation (schematic; real boundaries are gradual and also depend on seasonality and soils). Red dots show approximate positions of Indian places. Source: Simplified after Whittaker (1975), Communities and Ecosystems; Indian climate values approximate (IMD normals).*

2.12 Tundra Ecosystem

Tundra is a treeless biome found where it is too cold, for most of the year, for trees to grow. The word comes from the Finnish *tunturi*, meaning a treeless upland. Tundra is the coldest of all biomes and has the shortest growing season, but it is not lifeless: for a few weeks each summer it bursts into flower and teems with insects and migratory birds.

2.12.1 Arctic and alpine tundra

There are two kinds of tundra. **Arctic tundra** forms a broad belt around the North Pole, north of the coniferous forests (taiga), across northern Alaska, Canada, Greenland, Scandinavia and Russia. **Alpine tundra** occurs on high mountains at any latitude, above the **treeline**, the elevation beyond which trees cannot grow. In the Himalaya the treeline lies at roughly 3,300–4,000 m, depending on rainfall and slope direction.

Table 2.12: Comparison of Arctic and alpine tundra

Feature	Arctic tundra	Alpine tundra
Location	High latitudes (above about 60–70° N)	High altitudes, above the treeline
Day length	24-hour daylight in summer; polar night in winter	Normal day–night cycle
Permafrost	Continuous or widespread	Patchy or absent
Solar radiation	Weak (low sun angle)	Intense, with strong ultraviolet
Soils	Waterlogged, peaty above permafrost	Thin, rocky, well drained
Indian example	None (India's Arctic research station, Himadri, is in Svalbard, Norway)	Ladakh, Lahaul–Spiti, north Sikkim, upper Uttarakhand

2.12.2 Abiotic features

- **Temperature:** Winters are long and severe, often below −30 °C. Summer lasts 6–10 weeks, with mean temperatures of only about 3–12 °C.
- **Precipitation:** Low, generally 150–300 mm a year, much of it as snow. Because evaporation is also low, the ground is often wet in summer.
- **Permafrost:** In the Arctic, the ground below about a metre stays frozen all year. This **permafrost** blocks drainage, so the thawed upper layer (the **active layer**) becomes boggy in summer and dotted with shallow ponds.
- **Soils:** Cold slows decomposition so much that dead plant material piles up as peat. Nutrients, especially nitrogen and phosphorus, are locked in this undecomposed matter and are scarce for plants.
- **Wind:** Strong, drying winds and abrasive ice crystals damage anything that grows tall.

2.12.3 Biotic structure

- **Producers:** Mosses, lichens (including "reindeer moss", actually a lichen), sedges, grasses, cushion-forming herbs and dwarf shrubs such as willow and birch that grow only a few centimetres tall. In the Himalaya, dwarf rhododendrons, junipers, primulas, saxifrages, gentians and the Brahma kamal (*Saussurea obvallata*) are typical.
- **Herbivores:** Lemmings, Arctic hares, caribou (reindeer) and musk oxen in the Arctic; pikas, marmots, Himalayan blue sheep (bharal), ibex and yaks in the Himalaya. Insects such as mosquitoes and midges explode in number in summer.
- **Carnivores:** Arctic fox, wolf, snowy owl and, along the coast, polar bear; in the Himalaya, snow leopard, Tibetan wolf, red fox and golden eagle.

- **Decomposers:** Bacteria and fungi, slowed by the cold. Much of the energy flow is detritus-based, and food chains are short with few trophic levels.

Productivity is very low, about 140 g/m^2 per year on average, and is packed into a brief summer burst.

2.12.4 Adaptations to cold

Table 2.13: Adaptations of tundra organisms

Challenge	Plant adaptations	Animal adaptations
Extreme cold	Low cushion or mat growth near the warmer ground; antifreeze sugars in cells	Thick fur or feathers, fat layers, small ears and tails; counter-current blood flow in legs
Short summer	Perennial habit; flower buds formed the previous year; fast flowering and seed set	Rapid breeding; many birds migrate in only for summer
Wind and snow	Small, leathery or hairy leaves; dense mats that trap heat	Burrowing (pikas, marmots, lemmings) under insulating snow
Scarce nutrients	Nutrient storage in roots; partnerships with mycorrhizal fungi	Wide-ranging foraging; migration (caribou)
Long winter	Dormancy under snow	Hibernation (marmots), food caching (pikas), white winter coats for camouflage

Dark pigments in many alpine flowers and insects absorb more heat and also screen out ultraviolet light, which is intense at high altitude.

2.12.5 Alpine tundra of the Indian Himalaya

India has no Arctic tundra, but large parts of the Greater Himalaya and Trans-Himalaya above the treeline support alpine tundra. Much of this zone is also extremely dry, so it grades into the **cold desert** described later in this chapter.

- **Ladakh:** Above about 4,000 m, the Changthang plateau and Hemis National Park carry sparse alpine meadows and steppe with sedges, cushion plants and dwarf shrubs. Wildlife includes the snow leopard, Tibetan argali, kiang (Tibetan wild ass), Himalayan marmot and the black-necked crane, which breeds at high-altitude lakes such as Tso Kar and Tso Moriri (both Ramsar sites). Changpa herders graze the Changthangi goat, source of pashmina wool.
- **Lahaul–Spiti (Himachal Pradesh):** Pin Valley National Park and Kibber Wildlife Sanctuary protect alpine pastures with ibex, bharal and snow leopard. Spiti and adjoining areas form the Cold Desert Biosphere Reserve (designated by India in 2009). Chandratal, a high-altitude lake, is a Ramsar site.
- **Sikkim:** North Sikkim, around Gurudongmar Lake (above 5,000 m) and the upper reaches of Khangchendzonga National Park, has alpine scrub of rhododendron and juniper and rich alpine meadows. Khangchendzonga National Park was inscribed in 2016 as India's first "mixed" (natural and cultural) World Heritage Site.
- **Uttarakhand and Kashmir:** Alpine meadows locally called **bugyals** (Uttarakhand) and **margs** (Kashmir) lie just above the treeline. The Valley of Flowers and Nanda Devi National Parks (a World Heritage Site) protect some of the best examples.

The Wildlife Institute of India's first national snow leopard survey (2019–2023) estimated 718 snow leopards in India, 477 of them in Ladakh, and found that most live outside protected areas.

Case Study: The snow leopard landscapes of Ladakh

The snow leopard (*Panthera uncia*) is the top predator of India's alpine tundra and cold desert. It hunts bharal, ibex and marmots across steep, rocky terrain at 3,000–5,500 m. Because it needs very large home ranges, it cannot be conserved inside small protected areas alone. Most of its range in Ladakh is shared with pastoral communities whose livestock graze the same pastures and are sometimes killed by snow leopards and wolves.

Conservation in Ladakh has therefore focused on reducing conflict. Communities, supported by organisations such as the Snow Leopard Conservancy India Trust and the Nature Conservation Foundation, have built predator-proof corrals for livestock and set up community-run village "homestays" so that tourism income reaches herders. Livestock insurance schemes compensate losses. India's national Project Snow Leopard (2009) and the SECURE Himalaya project apply a landscape approach that combines protected areas with community lands.

The case shows a general lesson for alpine ecosystems: where people have lived and grazed for centuries, conservation works best when it supports, rather than excludes, local livelihoods.

2.12.6 Importance of tundra ecosystems

- **Carbon store:** Permafrost soils of the northern hemisphere hold roughly 1,400 billion tonnes of carbon, more than one and a half times the carbon now in the atmosphere, locked up because cold prevents decay.
- **Reflecting sunlight:** Snow-covered tundra reflects much of the incoming sunlight (high **albedo**), helping to keep the planet cool.
- **Water towers:** Alpine tundra and the snowfields and glaciers above it feed the headwaters of the Indus, Ganga and Brahmaputra. Alpine meadows act as sponges that release meltwater slowly.
- **Biodiversity and medicine:** Alpine plants such as kutki (*Picrorhiza kurroa*), jatamansi (*Nardostachys jatamansi*) and atis (*Aconitum heterophyllum*) are valued in Ayurvedic and Sowa-Rigpa (Tibetan) medicine.
- **Livelihoods and culture:** Transhumant herders such as the Changpa, Gaddi, Bhotia and Bakarwal depend on summer alpine pastures.

2.12.7 Threats to tundra ecosystems

- **Climate change:** The Arctic has warmed nearly four times as fast as the global average since 1979 (Rantanen and others, 2022). Thawing permafrost releases CO_2 and methane, a feedback that adds to warming. In the Himalaya, rising temperatures are pushing the treeline and shrubs upslope, shrinking alpine meadows and squeezing cold-adapted species towards the summits, where they have nowhere further to go.
- **Glacier retreat:** Most Himalayan glaciers are thinning and retreating, which changes water supply to alpine wetlands and downstream rivers (see Chapter 5).

- **Overgrazing:** Growing herds and the shift to sheep and goats for the market have degraded many bugyals and Ladakhi pastures.
- **Over-harvesting of medicinal plants:** Kutki, jatamansi and the caterpillar fungus (*Ophiocordyceps sinensis*, locally *keeda jadi*) are dug up faster than they can regrow.
- **Tourism and infrastructure:** Roads, trekking, camping and military infrastructure disturb fragile soils and wildlife; recovery is slow because growth is slow. Free-ranging dogs around settlements now kill wildlife and compete with snow leopards and wolves.

Key Points

- Tundra is the cold, treeless biome of high latitudes (Arctic) and high altitudes (alpine).
- Permafrost, a short summer, low productivity and short food chains are its defining features.
- India's alpine tundra lies above the treeline in Ladakh, Spiti, north Sikkim, Uttarakhand and Kashmir.
- Climate warming, overgrazing, medicinal-plant extraction and tourism are the main threats.

2.13 Forest Ecosystem

A **forest** is an ecosystem dominated by trees whose crowns form a more or less continuous canopy. Forests cover about 31 per cent of the world's land (FAO, Global Forest Resources Assessment 2020) and hold most of its land-dwelling species. They are the most complex terrestrial ecosystems, with many layers of life stacked one above the other.

2.13.1 Features and structure

The most visible feature of a forest is its vertical **stratification**, the arrangement of plants in layers:

- **Emergent layer:** scattered giant trees that rise above the main canopy (in tropical evergreen forests only).
- **Canopy:** the closed roof of tree crowns that intercepts most sunlight and rain.
- **Understorey:** young trees and shade-tolerant small trees.
- **Shrub layer and herb layer:** bushes, herbs, ferns and seedlings on the forest floor.
- **Forest floor and soil:** leaf litter, dead wood and the rich community of decomposers beneath.

Climbers (lianas) and **epiphytes** (plants such as orchids and ferns that grow on other plants without harming them) occupy the spaces between layers. Each layer has its own light, humidity and temperature, so a forest creates many **microhabitats** and supports many more species than a grassland of the same area.

Biotic components

- **Producers:** Trees dominate (sal, teak, deodar, oak, pine, dipterocarps), along with shrubs, herbs, climbers, epiphytes, mosses and ferns.
- **Primary consumers:** Leaf-eating insects, termites, squirrels, langurs, deer (chital, sambar, barking deer), gaur and elephants; fruit-eating birds and bats.
- **Secondary and tertiary consumers:** Snakes, birds of prey, civets, leopards, dholes (wild dogs) and tigers.

- **Decomposers:** Fungi, bacteria, termites, earthworms and millipedes, which are especially active in warm, humid forests.

Abiotic components

Forests need relatively high rainfall (generally above about 750–1,000 mm in India) and soils deep enough for tree roots. Under the canopy, light is dim, humidity is high and temperatures vary less than in the open. In tropical forests, most nutrients are held in living plants rather than in the soil, and dead matter decays within weeks. In cold coniferous forests decomposition is slow and thick litter accumulates.

2.13.2 Adaptations of forest organisms

- **Competition for light:** Trees grow tall, straight trunks; climbers use trees for support; epiphytes perch high on branches; forest-floor herbs have broad, thin leaves that capture weak light.
- **Heavy rain:** Leaves of evergreen trees often have pointed "drip tips" that shed water quickly, discouraging fungi and algae.
- **Seasonal drought:** Deciduous trees such as teak and sal shed leaves in the dry season to reduce water loss.
- **Shallow, nutrient-poor tropical soils:** Large trees develop wide buttress roots for support; many plants form mycorrhizal partnerships to capture nutrients quickly.
- **Animals:** Arboreal life (gibbons, langurs, flying squirrels, hornbills) with grasping limbs and gliding membranes; camouflage (leaf insects, the stripes of a tiger in dappled light); loud calls to communicate in dense cover.

2.13.3 Types of Indian forests

The standard classification of Indian forests is that of H. G. Champion and S. K. Seth (*A Revised Survey of the Forest Types of India*, 1968). It recognises **16 type groups** and about 200 individual forest types, based mainly on rainfall, temperature and altitude. For a first course these can be simplified into the classes in the table below.

Table 2.14: Forest types of India (simplified from Champion and Seth, 1968)

Forest type	Climate	Typical trees	Where found
Tropical wet evergreen	Rainfall over 2,500 mm; short dry season	Dipterocarps, *Hopea*, *Mesua*, rattans	Western Ghats, north-east India, Andaman and Nicobar Islands
Tropical semi-evergreen	2,000–2,500 mm	Mix of evergreen and deciduous trees; bamboo	Edges of evergreen belts: Western Ghats, Odisha, Assam
Tropical moist deciduous	1,000–2,000 mm	Sal (*Shorea robusta*), teak, bamboo, mahua	Himalayan foothills, Jharkhand, Odisha, Chhattisgarh
Tropical dry deciduous	700–1,000 mm	Teak, tendu, *Anogeissus*, palas, khair	Peninsular plateau, central India, Gangetic plains

Forest type	Climate	Typical trees	Where found
Tropical thorn	Under 700 mm	*Acacia*, khejri, ber, euphorbias	Rajasthan, Gujarat, rain-shadow Deccan
Tropical dry evergreen	About 1,000 mm, mostly from the north-east monsoon	Small evergreen trees, thorny shrubs	Coromandel coast of Tamil Nadu and Andhra Pradesh
Littoral and swamp	Tidal and freshwater swamps	Mangroves (*Rhizophora*, *Avicennia*, sundari *Heritiera fomes*)	Sundarbans, Bhitarkanika, Godavari–Krishna deltas, Gujarat, Andamans
Subtropical	Hills at about 1,000–1,800 m	Chir pine (*Pinus roxburghii*); broadleaved hill forests	Himalaya, Khasi hills
Montane temperate	About 1,800–3,000 m	Oaks, deodar (*Cedrus deodara*), blue pine, fir, spruce	Himalaya; shola forests of the Nilgiris and Anamalai
Subalpine and alpine	Above about 3,000 m	Birch (*Betula utilis*), silver fir, rhododendron, juniper	Higher Himalaya

By area, the **tropical dry deciduous** forests are India's most extensive type group, accounting for about two-fifths of forest cover, followed by **tropical moist deciduous** forests at about one-fifth (FSI forest type mapping). Tropical wet evergreen forests, though the richest in species, cover only a few per cent.

In the mountains, forest types change with altitude in much the same way they change with latitude across the continents. A climb of a few thousand metres in the Himalaya passes through the equivalent of tropical, temperate and Arctic zones, as the diagram below shows.

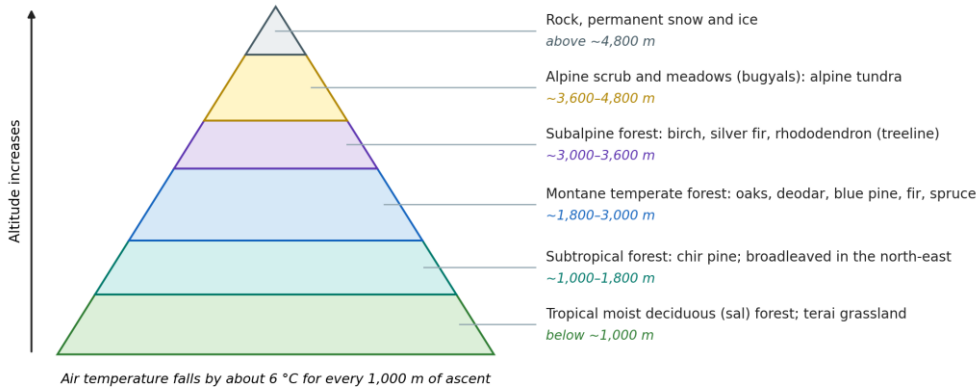


Figure 2.10: *Altitudinal zonation of vegetation on the southern slopes of the Himalaya (schematic; boundaries vary with rainfall, aspect and latitude). Source: Original diagram based on Champion and Seth (1968).*

2.13.4 India's forest cover

The Forest Survey of India (FSI), Dehradun, assesses India's forests every two years using satellite imagery and field inventory, and publishes the results as the *India State of Forest Report* (ISFR). FSI distinguishes two related measures:

- **Forest cover:** all land of more than 1 hectare with tree canopy density above 10 per cent, whatever its ownership or legal status. It includes orchards, bamboo and plantations.

- **Tree cover:** patches of trees smaller than 1 hectare outside recorded forest areas, such as trees on farms and along roads.

FSI further divides forest cover by canopy density into **very dense forest** (canopy density 70 per cent or more), **moderately dense forest** (40–70 per cent) and **open forest** (10–40 per cent). Land with less than 10 per cent canopy and stunted trees is classed as scrub.

Table 2.15: Key figures from the India State of Forest Report 2023

Indicator	Value
Forest cover	7,15,343 km^2 (21.76% of geographical area)
Tree cover	1,12,014 km^2 (3.41%)
Total forest and tree cover	8,27,357 km^2 (25.17%)
Change since ISFR 2021	+1,445 km^2 (forest cover +156 km^2; tree cover +1,289 km^2)
Mangrove cover	About 4,992 km^2 (0.15%)
Forest carbon stock	7,285.5 million tonnes
Largest forest cover by area	Madhya Pradesh, Arunachal Pradesh, Chhattisgarh

Two cautions apply when reading these numbers. First, "forest cover" counts any land with enough tree canopy, including tea gardens, orchards and monoculture plantations, so an increase in cover does not necessarily mean natural forest is recovering. Second, India's National Forest Policy (1988) sets a goal of one-third of the land under forest or tree cover, and the country is still well short of it. India's updated Nationally Determined Contribution (2022) commits to creating an additional carbon sink of 2.5–3 billion tonnes of CO_2 equivalent through additional forest and tree cover by 2030.

2.13.5 Importance of forest ecosystems

- **Provisioning:** Timber, fuelwood, fodder, bamboo, and **non-timber forest products (NTFPs)** such as tendu leaves (for bidis), mahua flowers, sal seeds, lac, honey, gums, resins and medicinal plants. Tens of millions of people, many of them Adivasi, depend on forests for part of their food and income (see Chapter 3).
- **Watershed protection:** Forests on hill slopes absorb rain, recharge groundwater, keep streams flowing after the monsoon and hold soil in place. Many of India's rivers rise in forested catchments.
- **Climate regulation:** Forests store carbon in wood and soil and cool their surroundings through shade and transpiration.
- **Biodiversity:** Forests are home to most of India's terrestrial species, including the tiger, Asian elephant, great hornbill and lion-tailed macaque.
- **Cultural value:** Sacred groves, forest deities and festivals tie many communities to forests (see Chapter 6).

2.13.6 Threats to forest ecosystems

The causes of deforestation and the impacts of mining and dams are discussed in detail in Chapter 3. At the ecosystem level, the main threats are:

- **Diversion of forest land** for mining, dams, roads, railways, transmission lines and industries. Even when compensatory plantations are raised elsewhere, an intact natural forest is replaced by a young, simpler stand.
- **Degradation without loss of cover:** Overgrazing, lopping, heavy fuelwood collection and repeated fires thin out the understorey and stop regeneration, even where satellite images still show canopy.
- **Fragmentation:** Roads and canals cut forests into smaller pieces, isolating animal populations and exposing forest edges to wind, fire and weeds (**edge effects**).
- **Forest fires:** Most fires in India are started by people, often to clear the floor for collecting mahua or tendu or to encourage fresh grass. Chir pine forests of Uttarakhand and dry deciduous forests of central India are especially fire-prone.
- **Invasive alien plants:** *Lantana camara*, *Chromolaena odorata* (Siam weed) and *Mikania micrantha* smother native undergrowth across large areas, reducing fodder for herbivores.
- **Monoculture plantations:** Replacing natural forests with teak, eucalyptus or acacia plantations reduces biodiversity even if tree cover stays the same.
- **Climate change:** Shifts in rainfall and temperature are expected to alter forest types, increase fire risk and push montane species uphill.

Lantana: a garden plant that conquered India's forests

Lantana camara, a shrub from tropical America, was brought to India as an ornamental hedge plant in the early nineteenth century. It now forms dense thickets across a large part of the country's forests, suppressing native seedlings, burning readily and reducing grass for deer and other herbivores. Removing it is labour-intensive. In some places, cut lantana stems are now woven into furniture and crafts, giving local people an income from clearing it.

2.14 Grassland Ecosystem

A **grassland** is an ecosystem dominated by grasses and other non-woody plants, with few or scattered trees. Grasslands occur where rainfall is too low or too seasonal for a closed forest, or where fire, grazing, frost or flooding keep trees out. Grasslands with scattered trees are called **savannas**. Worldwide, grasslands go by many local names: the prairies of North America, pampas of South America, steppes of Central Asia, veld of southern Africa and the savannas of Africa.

2.14.1 Features and structure

- **Climate:** Typically 250–1,000 mm of rain a year with a long dry season. Temperate grasslands have cold winters; tropical savannas are warm all year.
- **Soils:** Grass roots die and regrow every year, adding organic matter deep into the soil. Temperate grassland soils are among the most fertile in the world, which is why so many have been ploughed for farming.
- **Disturbance:** Fire, grazing and drought are not accidents in grasslands but part of how they work. They remove woody seedlings and recycle nutrients, keeping the ecosystem open.

Biotic components

- **Producers:** Grasses (in India, species of *Themeda*, *Dichanthium*, *Bothriochloa*, *Heteropogon*, *Saccharum*, *Cenchrus* and *Lasiurus*), sedges, legumes and other herbs, with scattered trees or shrubs such as *Acacia* and *Ziziphus*.
- **Primary consumers:** Grasshoppers, termites, rodents, hares, and grazing mammals: blackbuck, chinkara, nilgai, wild buffalo, rhinoceros and swamp deer, as well as domestic cattle, sheep and goats.
- **Secondary consumers:** Indian wolf, Indian fox, jackal, striped hyena, snakes, monitor lizards, and raptors such as harriers that migrate to Indian grasslands in winter.
- **Decomposers:** Fungi, bacteria, termites and dung beetles, which bury and recycle dung.

The grazing food chain is more important here than in forests: a large share of plant production is eaten directly by herbivores.

2.14.2 Adaptations of grassland organisms

- **Growing point at ground level:** Grass **meristems** (growing tissue) lie at the base of the plant, not at the tip. When the top is grazed or burnt, the leaf simply grows again from below.
- **Deep, dense fibrous roots and underground stems** (rhizomes) store food, reach deep moisture and survive fire and drought. Often more than half of a grassland's living biomass is underground.
- **C_4 photosynthesis:** Many tropical grasses use this efficient pathway, which works well in hot, bright, dry conditions.
- **Animals:** Speed and keen eyesight to escape predators in the open (blackbuck can run at about 80 km/h); herd living for safety; burrowing (gerbils, porcupines) to escape heat and fire; cryptic colours in ground-nesting birds such as larks and floricans.

2.14.3 Grasslands of India

India's grasslands are diverse but poorly protected and often unrecognised. The main types are:

- **Semi-arid grasslands and savannas of the Deccan and western India:** Open grasslands with scattered thorny trees in Maharashtra, Karnataka, Telangana, Andhra Pradesh, Gujarat and Rajasthan. They are home to the blackbuck, Indian wolf, Great Indian Bustard and lesser florican. Velavadar Blackbuck National Park (Gujarat), Rollapadu Wildlife Sanctuary (Andhra Pradesh) and the grasslands around Solapur (Maharashtra) are examples.
- **Banni grassland (Kachchh, Gujarat):** A saline, arid grassland of about 2,500 km^2 between the Great Rann and mainland Kachchh, often described as one of Asia's largest tropical grasslands. Maldhari pastoralists have bred the Banni buffalo and Kankrej cattle here for centuries.
- **Terai and Brahmaputra floodplain grasslands:** Tall, wet "elephant grass" (*Saccharum*, *Phragmites*, *Arundo*) along the Himalayan foothills and river floodplains in Uttar Pradesh, Uttarakhand, West Bengal and Assam. Kaziranga, Manas, Dudhwa, Jaldapara and Orang protect them. They shelter the greater one-horned rhinoceros, swamp deer (barasingha), wild water buffalo, hispid hare, pygmy hog and Bengal florican.
- **Shola grasslands of the Western Ghats:** On the high hills of the Nilgiris, Anamalai and Palani above about 1,800–2,000 m, rolling montane grasslands alternate with dark patches of stunted evergreen forest (**sholas**) in the sheltered valleys. Eravikulam National Park in Kerala protects

the largest population of the endemic Nilgiri tahr. The grasslands act as sponges that feed rivers such as the Bhavani and Kaveri tributaries.

- **Alpine and subalpine meadows:** The bugyals and margs of the Himalaya, described under tundra, used as summer pastures by transhumant herders.

Grasslands are not "wastelands"

For decades, Indian land records and the *Wastelands Atlas of India* have classed many grasslands, savannas, scrublands and rocky outcrops as "wastelands", meaning unproductive land fit for conversion. This reflects an old forestry view that land without trees is degraded. In reality these are ancient, natural ecosystems with their own specialised species. A 2022 study by M. D. Madhusudan and Abi T. Vanak estimated that India's semi-arid **open natural ecosystems** cover about 10 per cent of the country (over 3,00,000 km^2), and that roughly two-thirds of this area overlaps land labelled as wasteland. Labelling it as waste makes it an easy target for tree plantations, solar parks and industry. Grasslands are protected only where they fall inside protected areas; most lie outside.

2.14.4 Importance of grassland ecosystems

- **Pastoral livelihoods:** Grasslands feed a large part of India's livestock, the largest national herd in the world, and support pastoral communities such as the Maldharis, Rabaris, Dhangars, Gaddis and Gujjars.
- **Soil and water:** Dense grass cover protects soil from wind and water erosion and lets rain soak in. The shola grasslands and alpine meadows are major sources of stream flow.
- **Carbon storage:** Grasslands store most of their carbon underground in roots and soil, where it is safer from fire than carbon held in trees.
- **Biodiversity:** India's grasslands hold some of its most threatened species: the Great Indian Bustard, Bengal florican, lesser florican, pygmy hog, Indian wolf and caracal.
- **Genetic resources:** Wild grasses and indigenous livestock breeds (Banni buffalo, Kankrej and Gir cattle) carry traits for drought and heat tolerance.

2.14.5 Threats to grassland ecosystems

- **Conversion:** Ploughing for agriculture, irrigation projects, industrial estates, and solar and wind parks. Grassland is often allotted for these uses precisely because it is classed as wasteland.
- **Tree planting on grasslands:** Well-meant afforestation drives have planted eucalyptus, acacias and pines on natural grasslands, especially in the shola hills, destroying the habitat they were meant to improve.
- **Invasive species:** *Prosopis juliflora* (vilayati babool) in Banni and other arid grasslands, black wattle (*Acacia mearnsii*) and Scotch broom in the sholas, and *Lantana* in savannas.
- **Overgrazing and loss of common lands:** Shrinking commons concentrate livestock on the remaining patches, preventing grass recovery.

- **Altered fire and flood regimes:** Suppressing all fires lets shrubs invade, while too-frequent fires damage soils. In the terai, embankments and dams change the flood pulse that renews floodplain grasslands.
- **Power lines and infrastructure:** Overhead transmission lines kill large, heavy flying birds such as bustards and floricans.

Case Study: Banni: an invasive tree and a vanishing grassland

In 1961, the Gujarat Forest Department planted *Prosopis juliflora*, a fast-growing thorny tree from Central and South America, along the edge of Banni to stop the Rann's salt from spreading. The tree escaped. Its seeds, eaten and spread by livestock, germinated across the grassland. Studies by the Gujarat Institute of Desert Ecology recorded *Prosopis* on about 6 per cent of Banni in 1997, about one-third by 2009 and more than half by the mid-2010s.

Prosopis shades out grasses, its roots draw down groundwater, and its pods can harm livestock when eaten in excess. Combined with rising soil salinity after dams on rivers that once flooded Banni, grass production has fallen. Many Maldhari families shifted from cattle to the hardier Banni buffalo, and some turned to making charcoal from *Prosopis* wood. The case shows how a single introduction, made with good intentions, can transform an entire ecosystem. Restoration trials now combine uprooting *Prosopis*, sowing native grasses and securing the grazing rights of the Maldharis, who have sought recognition of their community rights over Banni under the Forest Rights Act, 2006.

2.15 Desert Ecosystem

A **desert** is an ecosystem where water loss through evaporation far exceeds rainfall, usually in places receiving less than about 250 mm of rain a year. Deserts are not defined by heat or sand: Antarctica is technically a desert, and most of the world's deserts are rocky or gravelly rather than sandy. Deserts and semi-deserts together cover roughly one-third of Earth's land.

2.15.1 Hot and cold deserts

Table 2.16: Hot and cold deserts compared

Feature	Hot desert	Cold desert
Cause of dryness	Subtropical high-pressure belts; distance from sea	Rain shadow of high mountains; altitude
Temperature	Summer days 40–50 °C; nights much cooler	Short, cool summer; winters below −20 °C
Precipitation	Under 250 mm, mostly in a few monsoon showers	Under about 100–150 mm, much as snow
Typical plants	Thorny shrubs, succulents, ephemeral herbs	Low cushion shrubs, *Artemisia*, seabuckthorn
Indian example	Thar Desert (Rajasthan, Gujarat)	Ladakh, Lahaul–Spiti

2.15.2 Abiotic and biotic features

- **Abiotic:** Very low and unpredictable rainfall, intense sunlight, low humidity, large daily temperature ranges (clear skies let heat escape at night), strong winds, shifting sand or bare rock, and thin, sandy or stony soils that are low in organic matter and often saline.
- **Producers:** Sparse, widely spaced shrubs and trees, grasses, succulents and short-lived herbs.
- **Consumers:** Insects (especially beetles and ants), reptiles, rodents, a few hardy mammals and birds.
- **Decomposers:** Few, because dryness limits microbial activity. Termites and sunlight itself (by breaking down dead matter) help recycle organic material.

Productivity is very low (about 90 g/m^2 per year on average for desert scrub) but bursts briefly after rain.

2.15.3 Adaptations of desert organisms

Desert life follows one of three strategies: **escape** drought (complete life in a short wet spell), **avoid** it (store water or reach groundwater) or **tolerate** it (survive extreme water loss).

Table 2.17: Adaptations of desert plants and animals

Strategy	Plants	Animals
Reduce water loss	Small, waxy or spiny leaves; leaves shed in summer; sunken stomata; CAM photosynthesis (stomata open at night)	Thick skin or scales; concentrated urine and dry dung; no sweat glands in many rodents
Store water	Fleshy succulent stems (*Euphorbia*), water-storing roots	Camels tolerate losing a quarter of body water; fat in hump supplies energy
Find water	Very deep taproots: khejri roots reach tens of metres	Get water from food (seeds, insects) and metabolic water
Avoid heat	Light-coloured, hairy surfaces reflect light	Nocturnal habits, burrowing, summer dormancy (aestivation); long ears and limbs radiate heat
Escape drought	Ephemeral herbs germinate after rain and set seed within weeks	Insects and some frogs remain dormant until rain

2.15.4 The Thar: India's hot desert

The **Thar** or Great Indian Desert covers about 2 lakh km^2 across western Rajasthan, parts of Gujarat, Haryana and Punjab, and neighbouring Pakistan. Annual rainfall ranges from under 150 mm in the west (Jaisalmer) to about 500 mm on its eastern edge, almost all of it during the south-west monsoon. Phalodi recorded 51.0 °C in May 2016, India's highest temperature on record.

The Thar is unusual in two ways. First, it is the most densely populated desert in the world, home to farmers, herders and towns. Second, it is biologically rich for a desert:

- **Plants:** Khejri (*Prosopis cineraria*, the state tree of Rajasthan), rohida (*Tecomella undulata*), ker (*Capparis decidua*), phog (*Calligonum polygonoides*), ber (*Ziziphus*) and sewan grass (*Lasiurus sindicus*), the best natural pasture grass of the desert.

- **Animals:** Chinkara, blackbuck, desert fox, desert cat, Indian spiny-tailed lizard, sandgrouse, and the critically endangered **Great Indian Bustard**, whose last major population survives in and around Desert National Park (Jaisalmer and Barmer).

Desert communities developed ingenious ways of living with scarcity: rainwater-harvesting structures such as **kunds** and **tankas** (covered cisterns), **khadins** (embanked fields that trap runoff) and **johads** (small earthen check dams), together with community-protected grazing lands called **orans** (see Chapter 6).

2.15.5 The cold desert of Ladakh

Ladakh lies in the rain shadow of the Great Himalaya, which blocks the monsoon. Leh receives only about 100 mm of precipitation a year, and winters fall below −20 °C. With altitudes of 3,000 m and above, intense sunlight and thin air, Ladakh combines the dryness of a desert with the cold of the tundra. Lahaul–Spiti in Himachal Pradesh is similar.

- **Plants:** Seabuckthorn (*Hippophae*), wild rose, *Caragana*, *Ephedra*, *Artemisia*, junipers in sheltered places, and willows and poplars planted along irrigation channels.
- **Animals:** Snow leopard, Ladakh urial, Tibetan argali, bharal, kiang, Tibetan wolf, Pallas's cat, Himalayan marmot, and the chiru (Tibetan antelope) and wild yak in the remote Changthang and Chang Chenmo areas.
- **People:** Agriculture depends entirely on meltwater from glaciers and snowfields, channelled to fields by gravity canals.

2.15.6 Importance of desert ecosystems

- **Unique biodiversity:** Species found nowhere else, adapted to extreme conditions; genetic resources for drought tolerance in crops and livestock (Rathi and Tharparkar cattle, Marwari horses, camels).
- **Livelihoods:** Pastoralism, rain-fed farming, desert fruits and gums (ker, sangri pods of khejri), tourism.
- **Knowledge:** Traditional water harvesting and drought-coping practices that are relevant everywhere as climate change makes droughts more common.

2.15.7 Threats to desert ecosystems

- **Desertification:** Land degradation in dry regions caused by overgrazing, over-cultivation, deforestation and poor irrigation, made worse by climate variability. ISRO's *Desertification and Land Degradation Atlas of India* (2021) found that about 97.85 million hectares, or 29.7 per cent of India's land, was undergoing degradation in 2018–19 (see Chapter 3).
- **Canal irrigation:** The Indira Gandhi Canal, which brings Himalayan river water deep into the Thar, has turned desert into farmland but also caused waterlogging and soil salinity and replaced desert habitat with crops.
- **Groundwater depletion:** Tube wells have lowered water tables across Rajasthan and dried up traditional wells and tankas.
- **Renewable energy and power lines:** The Thar and Ladakh are India's prime solar and wind sites. Large parks occupy open habitat, and overhead power lines are the leading cause of death for the Great Indian Bustard. In 2021 the Supreme Court ordered power lines in priority

bustard habitat to be laid underground where feasible; the order was later modified to allow case-by-case assessment.

- **Invasive *Prosopis juliflora*** spreading along roads and canals.
- **In cold deserts:** Rapid growth of tourism, waste and water demand in Leh; glacier retreat threatening irrigation; free-ranging dogs; and large infrastructure projects on fragile pastures.

Think and Do: Desert or grassland?

Using Google Earth, compare a patch of Desert National Park near Jaisalmer with irrigated farmland along the Indira Gandhi Canal about 100 km away. Note the differences in colour, texture and land use. List three ways in which the irrigated area differs from the natural desert as an ecosystem, and one benefit and one cost of the change.

2.16 Aquatic Ecosystems

Water covers about 71 per cent of Earth's surface, and aquatic ecosystems range from a temporary rain-fed pond in a village to the deep ocean. India has a long coastline (11,098.81 km, as re-measured by the Survey of India and the National Hydrographic Office and adopted in 2025), an Exclusive Economic Zone of about 2 million km^2, great river systems, and wetlands covering about 15.98 million hectares, or 4.86 per cent of its area, in 2017–18 (Space Applications Centre, *National Wetland Decadal Change Atlas*, 2022).

Aquatic ecosystems are grouped by salinity and water movement:

- **Freshwater ecosystems** (salinity below about 0.5 parts per thousand, ppt): **lentic** or standing-water systems (ponds, lakes, reservoirs, marshes) and **lotic** or flowing-water systems (springs, streams, rivers).
- **Marine ecosystems** (salinity about 35 ppt): oceans, seas, coral reefs.
- **Transitional or brackish ecosystems** (salinity between the two): estuaries, coastal lagoons, mangroves and salt marshes.

2.16.1 Key abiotic factors in water

- **Light:** Sunlight fades quickly with depth, especially in muddy water. The upper, lit layer where photosynthesis exceeds respiration is the **photic** (or euphotic) **zone**; below it lies the dark **aphotic zone**. The depth at which photosynthesis just balances respiration is the **compensation level**.
- **Temperature:** Water warms and cools slowly, so aquatic temperatures change less than on land, but layering by temperature is important in lakes and seas.
- **Dissolved oxygen (DO):** Water holds far less oxygen than air, and warm water holds less than cold water. Most fish are stressed when DO falls below about 4–5 mg per litre. DO is a key indicator of water quality (see Chapter 4).
- **Salinity:** The concentration of dissolved salts decides which organisms can live in the water, since they must keep the right balance of salt and water in their bodies.
- **Current:** Moving water brings oxygen and food but can sweep organisms away.

- **Nutrients and turbidity:** Nitrogen and phosphorus control plant and algal growth; suspended sediment (turbidity) reduces light.

2.16.2 Life in water

Aquatic organisms are grouped by where and how they live rather than by their taxonomy.

Table 2.18: Ecological groups of aquatic organisms

Group	Description	Examples
Plankton	Small organisms that drift with currents	Phytoplankton (diatoms, green algae, cyanobacteria); zooplankton (rotifers, copepods, water fleas, larvae)
Nekton	Active swimmers that can move against currents	Fish, turtles, dolphins, squid
Benthos	Organisms living on or in the bottom	Snails, mussels, worms, crabs, insect larvae
Periphyton	Algae and microbes attached to submerged surfaces	Films on stones, plant stems
Neuston	Organisms at the water surface	Pond skaters, whirligig beetles, duckweed

2.17 Ponds and Lakes

Lentic ecosystems are bodies of standing water. A **pond** is small and shallow enough for light to reach the bottom everywhere, so rooted plants can grow across it and the water mixes easily. A **lake** is larger and deep enough to have a dark, open-water zone where light does not reach the bottom. The distinction is one of degree; many Indian "tanks" and "sagars" fall in between.

Indian lakes have many origins: **glacial** lakes in the Himalaya (Chandratal), **tectonic** lakes (Wular), **oxbow** lakes cut off by meandering rivers (Kabar Tal in Bihar, the *beels* of Assam), a meteorite **impact crater** lake (Lonar, Maharashtra), **coastal lagoons** (Chilika, Pulicat) and thousands of **human-made** tanks and reservoirs.

2.17.1 Structure of a pond or lake

- **Abiotic components:** Water, dissolved gases (O_2, CO_2), dissolved nutrients (nitrate, phosphate), bottom mud rich in organic matter, light and temperature.
- **Producers:** Phytoplankton (diatoms, *Volvox*, *Chlamydomonas*, cyanobacteria), filamentous algae (*Spirogyra*), and **macrophytes** (large aquatic plants) of three kinds: **submerged** (*Hydrilla*, *Vallisneria*), **floating** (duckweed *Lemna*, water lettuce *Pistia*, water hyacinth *Eichhornia*) and **emergent**, rooted in mud but rising above the water (lotus *Nelumbo*, cattail *Typha*, reeds).
- **Consumers:** Zooplankton, snails, insect larvae (mosquitoes, dragonflies), frogs, fish such as rohu, catla and mrigal in the open water and murrel in the weeds, water snakes, and visiting birds such as herons, kingfishers and ducks.
- **Decomposers:** Bacteria and fungi in the bottom mud, which return nutrients to the water.

2.17.2 Zonation and stratification in lakes

Lakes are divided horizontally and vertically into zones that differ in light, temperature and life, as shown in the diagram below.

- **Littoral zone:** The shallow margin where light reaches the bottom. Rooted plants grow here, and it is usually the richest zone in species: insects, snails, frogs, fish fry and wading birds.
- **Limnetic zone:** The open, sunlit surface water away from the shore, down to the compensation level. Phytoplankton are the main producers; zooplankton and fish feed on them.
- **Profundal zone:** The deep water below the compensation level, where no photosynthesis is possible. Life depends on organic matter sinking from above. It is absent in shallow ponds.
- **Benthic zone:** The bottom sediments at all depths, home to worms, insect larvae, bacteria and fungi that decompose organic matter.

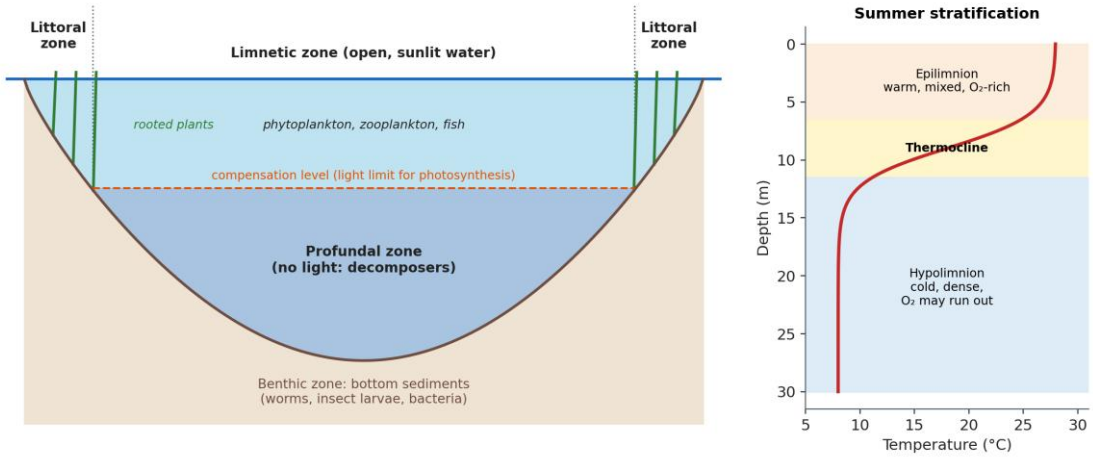


Figure 2.11: *Zones of a lake (left) and summer thermal stratification (right). The littoral zone extends to the depth where light reaches the bottom; below the compensation level lies the dark profundal zone. Source: Original diagram.*

In deep lakes, sunlight warms the surface water in summer. Because warm water is lighter than cold water, the lake separates into layers, a process called **thermal stratification**:

- **Epilimnion:** the warm, well-mixed, oxygen-rich upper layer.
- **Thermocline** (metalimnion): a middle layer in which temperature drops sharply with depth. It acts as a barrier to mixing.
- **Hypolimnion:** the cold, dense bottom layer. Cut off from the air, it can lose its oxygen as decomposers use it up.

When surface waters cool in autumn (or in winter in tropical lakes), the layers reach the same density and wind mixes the whole lake. This **turnover** carries oxygen down and brings nutrients up, often triggering a burst of plankton growth. Many deep Himalayan lakes stratify strongly in summer. In tropical reservoirs, a sudden turnover can bring oxygen-poor bottom water to the surface and cause fish kills.

2.17.3 Nutrient status and eutrophication

Lakes are classified by their nutrient level and productivity:

- **Oligotrophic** lakes are poor in nutrients, clear and deep, with low productivity and plenty of oxygen at all depths. High-altitude lakes such as Tso Moriri are examples.

- **Mesotrophic** lakes are intermediate.
- **Eutrophic** lakes are rich in nutrients, green and turbid with algae, highly productive, and prone to oxygen depletion in deep water.

Eutrophication is the enrichment of a water body with nutrients, mainly nitrogen and phosphorus. It happens naturally over centuries as lakes slowly fill with sediment and organic matter, a part of succession (see the earlier part of this chapter). When human activity speeds it up, it is called **cultural eutrophication**. The sequence is:

1. Sewage, detergents, fertiliser runoff and animal waste add large amounts of phosphate and nitrate.
2. Algae and floating weeds grow explosively, forming **algal blooms** and mats of water hyacinth.
3. The surface becomes green and opaque; submerged plants die for lack of light.
4. When the algae die, bacteria decompose them and use up dissolved oxygen.
5. Fish and other animals suffocate; foul-smelling gases such as hydrogen sulfide and methane are released. Some cyanobacteria produce toxins harmful to people and livestock.
6. The lake becomes shallower and more marshy as dead matter accumulates.

Eutrophication is the most widespread problem of Indian urban lakes. Bellandur Lake in Bengaluru, which receives large volumes of untreated sewage, made national news when toxic foam spilled onto roads and the lake surface caught fire several times between 2015 and 2018. Hussain Sagar in Hyderabad and Dal Lake in Srinagar suffer from sewage inflow and weed growth.

2.17.4 Adaptations of lake and pond organisms

- **Submerged plants** have thin, often finely divided leaves without a waxy cuticle, so they absorb gases and nutrients directly from water; little supporting tissue (water supports them); and air spaces (**aerenchyma**) that keep them buoyant and supply oxygen to roots.
- **Floating plants** such as lotus and water lilies have stomata on the upper leaf surface and waxy, water-repellent leaves. Water hyacinth has swollen, air-filled leaf stalks.
- **Animals** breathe with gills, through moist skin (frogs) or by surfacing for air (diving beetles carry an air bubble). Air-breathing fish such as murrel (*Channa*), magur (*Clarias*) and climbing perch (*Anabas*) survive in warm, oxygen-poor ponds and even in mud during dry spells.

2.17.5 Indian lakes: examples

Table 2.19: Selected lakes of India

Lake	Location and type	Key features	Main problems
Dal	Srinagar, J&K; urban Himalayan lake	Houseboats, floating gardens (*rad*), lotus beds; tourism and fishing	Sewage, siltation, weed growth, encroachment; open water much reduced
Wular	Bandipora, J&K; tectonic floodplain lake on the Jhelum	Traditionally regarded as India's largest freshwater lake; absorbs Jhelum floods; Ramsar site (1990)	Silt from deforested catchment; willow plantations; area has shrunk by more than half in the past century

Lake	Location and type	Key features	Main problems
Loktak	Manipur; floodplain lake	Floating vegetation mats (*phumdis*); Keibul Lamjao, the world's only floating national park; Ramsar site	Ithai barrage altered water levels; thickening phumdis; pollution
Chilika	Odisha; coastal lagoon	Asia's largest brackish-water lagoon; fisheries, Irrawaddy dolphin, wintering birds	Siltation and closing of sea mouth (largely reversed after 2000)
Sambhar	Rajasthan; inland saline lake	India's largest inland salt lake; flamingos	Salt mining, encroachment, mass bird deaths in 2019

Case Study: Loktak and the floating national park

Loktak Lake in Manipur is the largest freshwater lake in north-east India, covering roughly 250–290 km^2 depending on the season. Its most remarkable feature is the **phumdis**: floating mats of vegetation, soil and organic matter, some more than a metre thick. People live on some phumdis in circular fishing huts (*phumshangs*), and enclose patches of water with circular phumdi rings for fish culture (*athaphums*).

The largest continuous phumdi, about 40 km^2 in the south of the lake, forms **Keibul Lamjao National Park**, the only floating national park in the world. It is the last natural home of the sangai or brow-antlered deer (*Rucervus eldii eldii*), the state animal of Manipur, whose population fell to about a dozen animals in the 1970s and has since recovered to a few hundred. The deer's splayed hooves help it walk on the spongy mats.

In 1983 the Ithai barrage was built downstream for the Loktak hydroelectric project. It holds the lake at a high, nearly constant level. Phumdis that once sank to the bottom in the dry season, absorbing nutrients, now float all year, thickening and spreading; the natural flushing of the lake has weakened; and fish migration from the Chindwin–Irrawaddy system has been blocked. Sewage and fertiliser runoff from the catchment add to the nutrient load. Loktak was designated a Ramsar site in 1990 and placed on the Montreux Record (a list of Ramsar sites under threat) in 1993. The Loktak Development Authority now removes excess phumdis and manages the catchment, but balancing hydropower, fisheries and the sangai's habitat remains difficult.

2.17.6 Importance of ponds and lakes

- **Water supply:** Drinking water, irrigation and groundwater recharge. Village ponds and tanks were for centuries the main water source of rural India.
- **Food and livelihoods:** Inland fisheries, water chestnut (*singhara*), lotus stems and seeds (*makhana* from wetlands of Bihar), reeds for mats.
- **Flood control:** Lakes and their marshes store floodwater; Wular absorbs a large part of the Jhelum's floods.
- **Biodiversity:** Breeding and wintering grounds for waterbirds, many of which migrate from Central Asia and Siberia along the Central Asian Flyway.

2.17.7 Threats to ponds and lakes

- **Encroachment and filling** for housing, roads and farms. Cities such as Bengaluru, Hyderabad, Chennai and Delhi have lost a large share of their historic lakes and tanks; their disappearance worsens urban flooding (Chennai 2015).
- **Sewage and cultural eutrophication**, as described above.
- **Siltation** from deforested, eroding catchments, which makes lakes shallower.
- **Invasive weeds and fish:** Water hyacinth, *Salvinia* and *Alternanthera*; exotic fish such as African catfish (*Clarias gariepinus*) and tilapia, which compete with native species.
- **Breaking hydrological links:** Embankments, roads and barrages cut lakes off from the rivers and floodplains that once renewed them.
- **Climate change:** Warmer water holds less oxygen and favours algal blooms; changing rainfall alters water levels.

2.18 Streams and Rivers

Lotic ecosystems are bodies of flowing water: springs, streams and rivers. Unlike a lake, a river is a long, narrow, open system that is closely tied to the land it drains. Everything that happens in a river's **catchment** (or basin, the whole area that drains into it) affects the river, and the river in turn carries water, sediment, nutrients and organisms downstream to the sea.

2.18.1 Features of lotic ecosystems

- **One-way flow:** Water, nutrients and drifting organisms move constantly downstream. Many river animals must hold on, swim against the current or migrate back upstream to breed.
- **High oxygen:** Turbulence mixes air into the water, so fast streams are well oxygenated and rarely stratified.
- **Strong link with land:** In small, shaded streams, most food comes from leaves, twigs and insects falling from the banks (**allochthonous** input, meaning produced outside the stream) rather than from algae growing in the water (**autochthonous** production).
- **Habitat mosaic:** Fast, shallow, stony stretches called **riffles** alternate with deeper, slower **pools**. Riffles are rich in oxygen and insect larvae; pools shelter fish and collect fine sediment.
- **Floodplains:** In their lower reaches, rivers spill over their banks during floods, building fertile floodplains and connecting with oxbow lakes and marshes.
- **Change along the length:** A river changes steadily from its source to its mouth in width, depth, speed, temperature, sediment load and life.

Along its length a typical river has three broad zones:

1. **Headwaters (upper course):** Steep, narrow, cold, fast and clear, with a rocky bed. In the Himalaya these are glacier- and snow-fed torrents.
2. **Middle course:** Gentler slope, wider channel, warmer water, a mix of riffles and pools, and gravel or sand beds.
3. **Lower course and delta:** Flat, wide, deep and slow, carrying heavy loads of silt; the river meanders across a broad floodplain and finally splits into distributaries to form a delta or estuary.

2.18.2 Adaptations of river organisms

- **Holding on:** Hill-stream fishes such as *Garra* and the sucker-catfish *Glyptothorax* have flattened bodies and adhesive pads or sucker-like mouths. Insect larvae (mayflies, stoneflies) are flattened to stay in the thin, slow layer of water on rock surfaces; some have hooks or silk anchors. Caddisfly larvae build heavy cases of sand grains.
- **Strong swimmers:** Streamlined, muscular fish such as the golden mahseer and snow trout swim against strong currents and migrate upstream to spawn.
- **Plants:** Mosses, algae and a few flowering plants with tough, flexible stems and strong anchoring roots grow in fast water; most river plants grow in quiet backwaters.
- **Large lowland rivers:** The Ganges river dolphin is almost blind and finds prey in the muddy water by **echolocation** (sending out clicks and listening for echoes). The gharial has a long, thin snout for snapping fish sideways in water.

2.18.3 The river continuum concept

The **river continuum concept** (Vannote and colleagues, 1980) describes how a river's food base and animal community change in a predictable way from source to mouth.

- In **headwaters**, trees shade the narrow stream, so algae grow poorly. The main food is coarse leaf litter from the banks. Animals called **shredders** (such as some caddisfly and stonefly larvae) chew this material. The stream respires more than it photosynthesises: its ratio of production to respiration (**P/R**) is below 1.
- In the **middle reaches**, the channel widens and sunlight reaches the bed. Algae and water plants flourish, so production exceeds respiration (P/R above 1). **Grazers** that scrape algae and **collectors** that gather fine particles dominate.
- In the **lower river**, the water is deep and muddy, light is again limiting, and the food base is fine organic particles carried down from upstream. **Collectors** that filter particles from the water dominate, and P/R falls below 1 again.

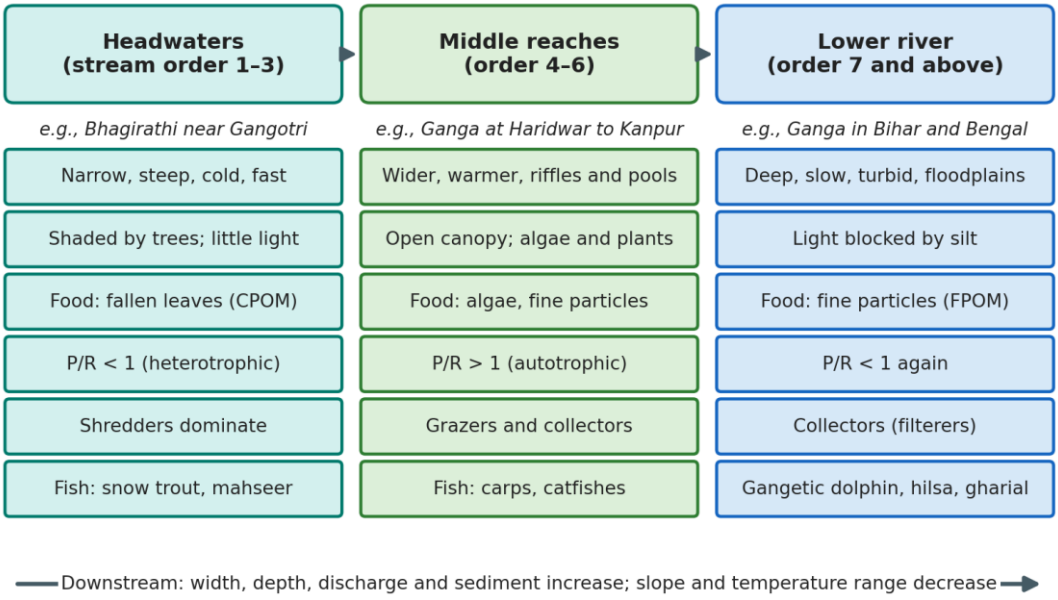


Figure 2.12: *The river continuum concept, simplified, with examples from the Ganga system. P/R is the ratio of photosynthesis to respiration; CPOM and FPOM are coarse and fine particulate organic matter. Source: Original diagram based on Vannote et al. (1980).*

The concept shows that a river is one connected ecosystem: a dam, pollution source or deforested headwater affects everything downstream. A complementary idea, the **flood pulse concept**, emphasises that in large tropical rivers like the Ganga and Brahmaputra, the annual monsoon flood that spreads over the floodplain is the main driver of productivity, fish breeding and nutrient renewal. The amount, timing and quality of flow that a river needs to sustain its ecosystem and the people who depend on it is called its **environmental flow** (e-flow).

2.18.4 Major Indian rivers as ecosystems

Table 2.20: Three Indian rivers as ecosystems

Feature	Ganga	Brahmaputra	Godavari
Source	Gangotri glacier (as the Bhagirathi), Uttarakhand	Chemayungdung glacier in Tibet (as the Yarlung Tsangpo)	Trimbakeshwar, Nashik, Maharashtra
Length in India	About 2,525 km	About 900 km (enters via Arunachal Pradesh)	About 1,465 km
Character	Snow- and rain-fed; vast plains; supports over 40% of India's population in its basin	Braided, very high water and silt load; huge floods	Largest peninsular river, rain-fed; flows east to the Bay of Bengal
Ecological highlights	Ganges river dolphin, gharial, turtles, hilsa; Sundarbans delta	Kaziranga floodplains, Majuli river island, beels (floodplain lakes), dolphins	Delta with Coringa mangroves; Papikonda gorge; Krishna–Godavari delta fisheries
Key threats	Sewage and industrial effluent, barrages, sand mining, reduced flow	Dams in the upper basin, embankments, erosion, floods	Dams and diversions, sand mining, pollution, delta erosion

The **Ganges river dolphin** (*Platanista gangetica*), India's National Aquatic Animal since 2009, is a good indicator of river health because it needs deep, connected, unpolluted channels. The first nationwide survey by the Wildlife Institute of India under Project Dolphin, reported in 2025, estimated about 6,327 river dolphins in India, the great majority in the Ganga and Brahmaputra systems. Barrages split its population into isolated groups, and it is killed accidentally in fishing nets.

2.18.5 Importance of rivers and streams

- **Water** for drinking, irrigation, industry and hydropower; recharge of groundwater in floodplain aquifers.
- **Fertile land:** Floods deposit silt that has made the Indo-Gangetic plains and deltas among the most productive farmland on Earth.
- **Fisheries:** Inland capture fisheries in rivers and floodplain lakes support millions of fishers.
- **Biodiversity corridors:** Rivers and their riparian forests connect habitats across landscapes.
- **Cultural and spiritual value:** Rivers are worshipped across India, and festivals such as the Kumbh Mela are held at their confluences.

2.18.6 Threats to rivers and streams

- **Dams, barrages and diversions** fragment rivers, block fish migration (hilsa catches above Farakka fell sharply after the barrage was built in 1975), trap sediment that should reach deltas, and leave stretches downstream with too little water. Their wider impacts are discussed in Chapter 3.
- **Pollution:** Untreated sewage is the largest source. In 2022 the Central Pollution Control Board identified 311 polluted river stretches on 279 rivers. Industrial effluents (tanneries, distilleries, paper mills) and agricultural runoff add toxins and nutrients (see Chapter 4).
- **Sand and boulder mining** from riverbeds deepens channels, destroys spawning grounds and lowers water tables.
- **Floodplain encroachment and embankments** cut rivers off from their floodplains, destroying wetlands and increasing flood damage downstream.
- **Over-abstraction** of surface and groundwater reduces flow; many rivers that once flowed all year now run dry in summer.
- **Invasive species:** Exotic carps and tilapia in rivers; water hyacinth in slow stretches.
- **Climate change:** Changes in glacier melt and monsoon rainfall alter flow patterns, with more intense floods and longer low-flow periods.

2.19 Oceans

The ocean is the largest ecosystem on Earth, holding about 97 per cent of its water. India is bordered by the Arabian Sea to the west, the Bay of Bengal to the east and the Indian Ocean to the south. The Bay of Bengal is less salty than the Arabian Sea because of the huge freshwater discharge of the Ganga, Brahmaputra and other rivers.

2.19.1 Features and zones of the ocean

The ocean is divided into zones by distance from the shore, by depth and by light, as shown in the diagram below.

- **Intertidal zone:** The strip of shore between high and low tide, alternately exposed and submerged. Its organisms must withstand drying, heat, waves and changing salinity.
- **Neritic zone:** The relatively shallow water (to about 200 m) above the **continental shelf**. Sunlight and nutrients from rivers and the sea floor make it highly productive. Most of the world's fisheries, coral reefs and seagrass beds lie here.
- **Oceanic zone:** The open ocean beyond the shelf, deep and generally poor in nutrients.
- **Pelagic and benthic:** Life in the open water column is **pelagic**; life on the sea floor is **benthic**.
- **Depth zones:** The sunlit **epipelagic** (0–200 m), twilight **mesopelagic** (200–1,000 m), and the permanently dark **bathypelagic**, **abyssopelagic** and **hadal** zones below. Only the epipelagic zone supports photosynthesis.

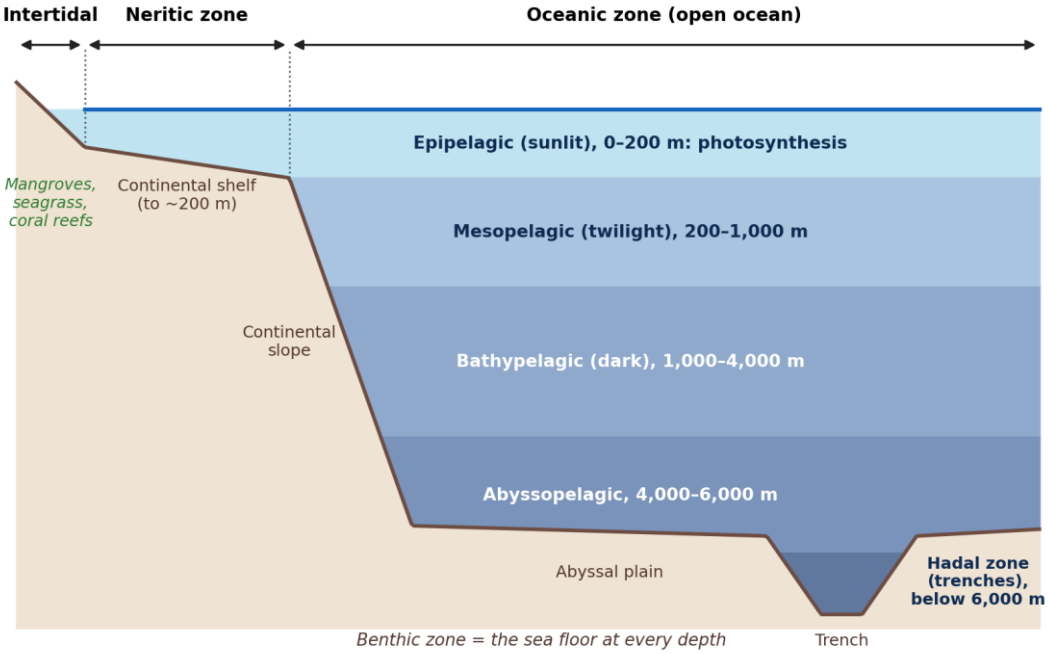


Figure 2.13: *Zones of the ocean (schematic, not to scale). Horizontal zones are defined by distance from the shore and depth of the sea floor; vertical zones by depth and light. Source: Original diagram.*

Productivity in the open ocean is limited by nutrients, which sink below the sunlit layer. It is high where nutrient-rich deep water rises to the surface, a process called **upwelling**. Along the Kerala and Karnataka coasts, upwelling during the south-west monsoon fuels plankton blooms that support India's large oil sardine and mackerel fisheries.

2.19.2 Biotic components

- **Producers:** Phytoplankton (diatoms, dinoflagellates, cyanobacteria), which carry out roughly half of all photosynthesis on Earth; seaweeds; seagrasses in shallow water.
- **Consumers:** Zooplankton (copepods, krill, larvae), fish (sardines, mackerel, tuna, sharks), sea turtles, seabirds and marine mammals (dolphins, whales, dugong).

- **Decomposers:** Bacteria in the water and sediments. In the deep sea, life depends on organic matter drifting down from above; around **hydrothermal vents**, bacteria make food by **chemosynthesis**, using energy from chemicals such as hydrogen sulfide instead of sunlight.

2.19.3 Adaptations of marine organisms

- **Salt balance:** Marine fish drink seawater and excrete salt through their gills; sea turtles and seabirds have salt glands near their eyes or nostrils.
- **Buoyancy:** Swim bladders, oil droplets and spines help plankton and fish stay at the right depth.
- **Streamlined bodies** and powerful tails for fast swimming (tuna, dolphins).
- **Camouflage by counter-shading:** Dark backs and pale bellies hide fish from predators above and below.
- **Deep-sea life:** Bioluminescence (producing light), large eyes or no eyes, slow metabolism, and bodies that withstand enormous pressure.

2.19.4 Coral reefs

Coral reefs are ridges of calcium carbonate built over thousands of years by colonies of tiny animals called coral polyps. Inside the polyps live single-celled algae (**zooxanthellae**) that photosynthesise and supply most of the coral's food; this partnership is why reef-building corals need warm (about 23–29 °C), clear, shallow, sunlit water. Reefs occupy less than 1 per cent of the sea floor yet shelter about a quarter of all known marine species.

Reefs take three main forms: **fringing reefs** along the coast, **barrier reefs** separated from the shore by a lagoon, and **atolls**, rings of coral around a central lagoon. India has reefs in four main areas:

- **Gulf of Mannar** (Tamil Nadu): fringing reefs around 21 islands in the Gulf of Mannar Marine National Park and Biosphere Reserve.
- **Gulf of Kachchh** (Gujarat): India's northernmost reefs, in the Marine National Park, Jamnagar, adapted to high temperature and salinity ranges.
- **Lakshadweep:** a group of atolls in the Arabian Sea.
- **Andaman and Nicobar Islands:** the most extensive and diverse fringing reefs in India.

All reef-building corals are protected under Schedule I of the Wildlife (Protection) Act, 1972.

When water becomes too warm for too long, corals expel their zooxanthellae and turn white, a condition called **coral bleaching**. If the heat persists, the corals starve and die. Major bleaching events struck Indian reefs in 1998, 2010, 2016 and 2024, the last as part of the fourth global bleaching event declared by the US National Oceanic and Atmospheric Administration.

2.19.5 Importance of oceans

- **Climate regulation:** Oceans have absorbed roughly a quarter of the CO_2 released by human activities and more than 90 per cent of the extra heat trapped by greenhouse gases.
- **Oxygen:** Marine phytoplankton produce about half of the oxygen generated by photosynthesis each year.
- **Food and livelihoods:** Marine fisheries supply protein to millions and support a large coastal fishing population in India; seaweed farming is a growing livelihood for women in Tamil Nadu.

- **Weather:** Ocean–atmosphere interactions drive the Indian monsoon.
- **Coastal protection and tourism:** Reefs break waves and attract tourists.

2.19.6 Threats to oceans

- **Overfishing and destructive fishing:** Bottom trawling scrapes the sea floor; bycatch kills turtles, dolphins and sharks. India enforces an annual monsoon fishing ban of 61 days on each coast to protect breeding stocks.
- **Pollution:** Plastics, sewage, industrial waste and oil spills. Nutrient runoff creates low-oxygen "dead zones"; one of the world's largest natural oxygen-minimum zones lies in the Arabian Sea.
- **Warming and acidification:** Warming causes bleaching and shifts fish distribution; dissolving CO_2 makes seawater more acidic, which weakens shells and coral skeletons (see Chapter 5).
- **Coastal development:** Ports, reclamation, sand mining and tourism destroy reefs, beaches and seagrass beds.
- **Sea-level rise** threatening low islands such as Lakshadweep.

> **Dugongs and seagrass**
>
> The dugong (*Dugong dugon*), a plant-eating marine mammal sometimes called the "sea cow", grazes on seagrass meadows. Only about 200–250 dugongs are thought to survive in Indian waters, mostly in the Gulf of Mannar and Palk Bay. In 2022, Tamil Nadu notified India's first **Dugong Conservation Reserve**, covering about 448 km^2 of Palk Bay, to protect the seagrass beds on which they depend.

2.20 Estuaries, Mangroves and Lagoons

An **estuary** is a partly enclosed coastal water body where a river meets the sea and fresh water mixes with salt water. Estuaries, coastal lagoons and the mangrove forests that fringe them are **transitional ecosystems**, or **ecotones**, between land, river and sea.

2.20.1 Features of estuaries

- **Salinity gradient:** Salinity rises from near zero at the river end to near-seawater at the mouth, and shifts twice a day with the tides and seasonally with river flow. Lighter fresh water often flows over a denser wedge of seawater.
- **Nutrient trap:** Rivers bring nutrients and silt, the sea brings more nutrients with each tide, and the mixing traps them. Estuaries are therefore among the most productive ecosystems on Earth (about 1,500 g/m^2 per year on average).
- **Nursery grounds:** Shallow, sheltered, food-rich water makes estuaries the breeding and nursery ground for many fish, prawns and crabs that later live at sea.
- **Organisms:** Most estuarine species are **euryhaline**, able to tolerate wide changes in salinity: mullets, milkfish, mud crabs, prawns, oysters, clams, mudskippers, and fish such as the hilsa that migrate through estuaries between sea and river.

Important Indian estuaries include the **Hooghly** (the main distributary of the Ganga in West Bengal), the Mahanadi, Godavari and Krishna on the east coast, and the Narmada, Mandovi–Zuari (Goa) and the Kochi backwaters on the west coast.

The **Hooghly estuary** shows how human control of flow alters an estuary. To flush silt from the shipping channel to Kolkata port, the Farakka Barrage (1975) diverts Ganga water into the Hooghly. The estuary also receives the sewage and industrial effluent of the Kolkata metropolitan region. It leads into the Indian Sundarbans and supports large fisheries for hilsa and prawn seed.

2.20.2 Mangroves

Mangroves are salt-tolerant trees and shrubs that grow in the intertidal zone of tropical and subtropical coasts, especially on muddy, sheltered shores of estuaries and deltas. India's mangrove cover was about 4,992 km^2 in 2023, roughly 0.15 per cent of its area (ISFR 2023). West Bengal holds about 42 per cent, mainly in the Sundarbans, followed by Gujarat (about 23 per cent) and the Andaman and Nicobar Islands (about 12 per cent). Other important mangroves are Bhitarkanika (Odisha), Coringa (Godavari delta), Pichavaram (Tamil Nadu) and the creeks of Mumbai.

Mangroves face salty, waterlogged, oxygen-poor mud, tides and waves. Their adaptations include:

- **Pneumatophores:** Pencil-like breathing roots that grow upward out of the mud (*Avicennia*, *Sonneratia*), with pores (lenticels) to take in air at low tide.
- **Stilt or prop roots:** Arching roots from the trunk (*Rhizophora*) that anchor the tree in soft mud and resist waves.
- **Salt management:** Some species block salt at the roots; others excrete it through salt glands on their leaves (*Avicennia* leaves are often crusted with salt crystals) or store it in old leaves that are then shed.
- **Succulent, thick, waxy leaves** that conserve fresh water.
- **Vivipary:** Seeds germinate while still attached to the parent tree, producing a long seedling (propagule) that drops into the mud ready to root, or floats until it finds a place to settle.

Case Study: The Sundarbans: the world's largest mangrove forest

The Sundarbans, in the combined delta of the Ganga, Brahmaputra and Meghna, form the largest mangrove forest in the world, about 10,000 km^2, shared between India (about 40 per cent) and Bangladesh. The Indian Sundarbans in West Bengal include about a hundred islands, roughly half of them inhabited by several million people. The forest takes its name from the sundari tree (*Heritiera fomes*).

The Sundarbans are the only mangrove forest with tigers. Sundarbans tigers swim between islands, drink brackish water and hunt fish, crabs, monitor lizards and deer as well as larger prey; the 2022 national tiger estimation recorded roughly a hundred tigers in the Indian Sundarbans. Other species include estuarine crocodiles, Gangetic and Irrawaddy dolphins, fishing cats, river terrapins (*Batagur baska*) and hundreds of bird species. The Sundarbans National Park is a World Heritage Site (1987), the area is a UNESCO biosphere reserve, and in 2019 India designated the Sundarban Wetland as a Ramsar site; at 4,230 km^2 it is the country's largest.

The forest protects Kolkata and the delta villages from cyclones such as Aila (2009) and Amphan (2020), supports fisheries, honey collection and crab catching, and stores large amounts of carbon in its waterlogged soils. But it is under severe stress. Relative sea level is rising faster here than the global average because the delta is also sinking; reduced freshwater flow from upstream and rising seas are making the water saltier, harming the sundari tree; islands such as Lohachara and Ghoramara have been largely or wholly lost to erosion; and earthen embankments built to protect farmland break during cyclones, flooding fields with salt water. Every year people collecting honey, crabs and fish are killed by tigers, and tigers that stray into villages are sometimes killed. Conservation now combines mangrove replanting on mudflats, nylon-net fencing along village edges, community-based eco-development committees and alternative livelihoods to reduce people's dependence on the forest.

2.20.3 Coastal lagoons

A **coastal lagoon** is a shallow water body separated from the sea by a sand bar or barrier island and connected to it by one or more narrow mouths. Its water is brackish, and its salinity depends on the balance between river inflow and seawater entering through the mouths. India's major lagoons are Chilika (Odisha), Pulicat (Andhra Pradesh–Tamil Nadu border) and the backwaters (*kayals*) of Kerala, including Vembanad.

Case Study: Chilika: a lagoon brought back to health

Chilika Lake in Odisha is Asia's largest brackish-water lagoon, about 1,100 km^2 in area. It is fed by distributaries of the Mahanadi and other streams and linked to the Bay of Bengal by a channel through a sand bar. It supports the livelihoods of about two lakh fisherfolk, hosts one of the few lagoon populations of the Irrawaddy dolphin (*Orcaella brevirostris*), and is one of the largest wintering grounds for migratory birds in India, especially at the Nalabana Bird Sanctuary. In 1981 it became one of India's first two Ramsar sites, along with Keoladeo in Rajasthan.

By the 1990s Chilika was dying. Silt from its deforested catchment had made it shallower, and the mouth to the sea was choking and shifting. Less seawater entered, salinity fell, freshwater weeds spread across large areas, and fish, prawn and crab catches collapsed. In 1993 Chilika was placed on the Montreux Record.

The Chilika Development Authority, set up by the Odisha government in 1991, studied the lagoon's hydrology with scientific institutions and, in September 2000, opened a new mouth to the sea near Satapada. Salinity recovered within a year, weeds declined, and fish landings rose several-fold, as did the number of dolphins. The authority also worked on catchment treatment and on settling conflicts between traditional fishers and prawn-farm operators. In 2002 Chilika became the first Asian site to be removed from the Montreux Record and received the Ramsar Wetland Conservation Award. Chilika shows that understanding how an ecosystem works, in this case its water exchange with the sea, is the key to restoring it.

2.20.4 Importance of estuaries, mangroves and lagoons

- **Fisheries:** Nurseries for most commercially important coastal fish and prawns.
- **Coastal protection:** Mangroves reduce wave height and storm-surge damage, as seen during the 1999 Odisha super cyclone, when villages behind mangroves at Kendrapara suffered fewer deaths than those without them (Das and Vincent, 2009).
- **Blue carbon:** Mangroves, salt marshes and seagrasses store carbon in waterlogged sediments for centuries; per hectare, mangroves store several times more carbon than most terrestrial forests.
- **Water purification:** They trap sediment and absorb nutrients and pollutants before they reach the sea.
- **Livelihoods:** Honey, timber, fuelwood, crab and oyster collection, salt pans, tourism.

2.20.5 Threats to estuaries, mangroves and lagoons

- **Conversion** of mangroves and mudflats to shrimp farms, salt pans, ports, industrial estates and housing (for example, along the Andhra Pradesh and Tamil Nadu coasts and in Mumbai).
- **Reduced freshwater and sediment** because of upstream dams and diversions, making estuaries saltier and starving deltas of the silt they need to keep pace with sea-level rise.
- **Pollution** from sewage, industry, ports and oil spills.
- **Siltation** of lagoon mouths from catchment erosion.
- **Overfishing** and the collection of wild prawn seed, which destroys the young of many other species.

- **Climate change:** Sea-level rise, stronger cyclones and warmer water.

2.21 Importance and Threats of Ecosystems in India: A Comparison

The sections above described each ecosystem type separately. The table below brings together their importance and threats in India for quick comparison.

Table 2.21: Importance and threats of India's major ecosystem types

Ecosystem	Indian examples	Key importance	Main threats
Alpine tundra	Ladakh, Spiti, north Sikkim, bugyals of Uttarakhand	Feeds rivers; medicinal plants; pastures; snow leopard habitat	Warming, overgrazing, herb extraction, tourism
Forest	Western Ghats, central Indian sal forests, north-east, Himalaya	Timber and NTFPs; watershed protection; carbon; most biodiversity	Diversion for mining and infrastructure, degradation, fire, invasive weeds
Grassland	Banni, Deccan savannas, terai, sholas	Pastoral livelihoods; soil carbon; bustards, blackbuck, rhinos	Labelled as wasteland; conversion; tree planting; *Prosopis* invasion
Desert	Thar, Ladakh cold desert	Drought-adapted species and breeds; traditional water knowledge	Desertification, canal irrigation, solar parks and power lines, groundwater loss
Ponds and lakes	Dal, Wular, Loktak, urban lakes	Water supply, fisheries, flood storage, waterbirds	Encroachment, sewage and eutrophication, siltation, invasive weeds
Rivers and streams	Ganga, Brahmaputra, Godavari	Water, fertile floodplains, fisheries, culture	Dams, pollution, sand mining, over-abstraction
Oceans and reefs	Gulf of Mannar, Lakshadweep, Andamans	Fisheries, climate regulation, coastal protection	Overfishing, pollution, warming, bleaching
Estuaries and mangroves	Sundarbans, Hooghly, Chilika, Bhitarkanika	Fish nurseries, storm protection, blue carbon	Conversion to aquaculture, reduced river flow, sea-level rise

Across all these ecosystems, the same five direct drivers of decline recur. The global assessment of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES, 2019) ranked them, in order of global impact, as: changes in land and sea use; direct exploitation of organisms; climate change; pollution; and invasive alien species. Behind them lie indirect drivers such as population growth, rising consumption, and policies that do not account for the value of nature, which the next section addresses.

2.22 Ecosystem Services

Ecosystem services are the benefits that people obtain from ecosystems. The idea makes visible something that is easy to forget: clean water, fertile soil, pollinated crops, a stable climate and protection from floods are not free gifts of engineering but products of working ecosystems. When ecosystems are degraded, these services decline, and replacing them with technology is usually costly, partial or impossible.

2.22.1 The Millennium Ecosystem Assessment framework

The most widely used classification comes from the **Millennium Ecosystem Assessment** (MEA), a four-year study by more than 1,300 scientists published in 2005. It grouped ecosystem services into four categories:

1. **Provisioning services:** the products obtained from ecosystems.
2. **Regulating services:** the benefits obtained from the regulation of natural processes.
3. **Cultural services:** the non-material benefits people obtain through spiritual enrichment, recreation, learning and aesthetic experience.
4. **Supporting services:** the underlying processes, such as soil formation and nutrient cycling, that are necessary for all other services.

The MEA linked these services to the components of **human well-being**: security, basic materials for a good life, health, good social relations, and freedom of choice and action. It concluded that about 60 per cent of the ecosystem services it examined (15 of 24) were being degraded or used unsustainably.

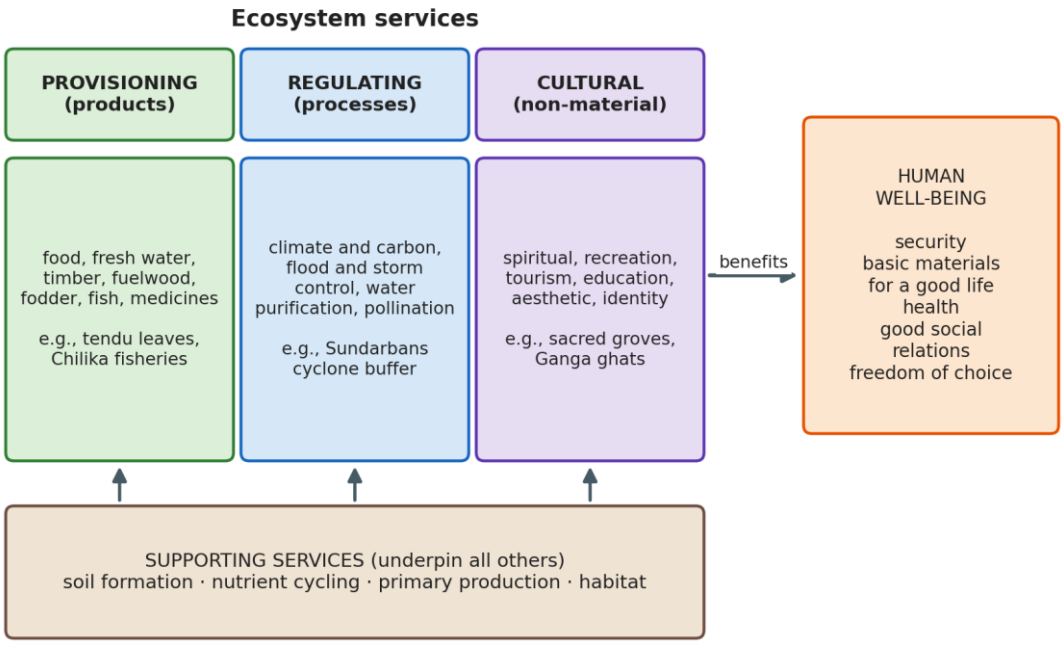


Figure 2.14: *The Millennium Ecosystem Assessment framework. Supporting services underpin provisioning, regulating and cultural services, which together contribute to human well-being. Source: Adapted from Millennium Ecosystem Assessment (2005), with Indian examples added.*

2.22.2 Provisioning services

These are the tangible goods that people harvest or use directly:

- **Food:** crops, livestock, fish, wild fruits, honey, mushrooms. India's marine and inland fisheries, for example, depend entirely on healthy aquatic ecosystems.
- **Fresh water:** for drinking, irrigation and industry, supplied by rivers, lakes, aquifers and forested catchments.
- **Fibre and fuel:** timber, bamboo, cotton, jute, fuelwood, cow dung cake.
- **Fodder:** grasslands, forests and crop residues feed India's livestock.

- **Medicines and biochemicals:** medicinal plants used in Ayurveda, Unani, Siddha and Sowa-Rigpa; plant compounds used in modern drugs.
- **Genetic resources:** wild relatives of crops and traditional livestock breeds that carry genes for disease resistance and drought tolerance.

In central India, the collection and sale of tendu leaves, mahua flowers, sal seeds and other NTFPs provides cash income to millions of forest dwellers during the lean season.

2.22.3 Regulating services

These are the benefits of ecosystems keeping natural processes in balance. They are rarely noticed until they fail:

- **Climate regulation:** Forests, soils, wetlands and oceans absorb and store carbon; vegetation cools the local climate. Urban trees and lakes reduce the urban heat-island effect in cities such as Delhi.
- **Water regulation:** Forests, grasslands and wetlands absorb rainfall, recharge aquifers, reduce flood peaks and sustain dry-season flow.
- **Water purification and waste treatment:** Wetlands and soils filter sediment, nutrients and pathogens.
- **Erosion control:** Roots hold soil on slopes and riverbanks.
- **Hazard protection:** Mangroves, coral reefs and sand dunes reduce damage from cyclones, storm surges and tsunamis.
- **Pollination:** Bees, butterflies, birds and bats pollinate a large share of crops, including most fruits, vegetables, oilseeds and pulses.
- **Pest and disease control:** Birds, bats, spiders and predatory insects keep crop pests in check; vultures once removed livestock carcasses, and their collapse in India since the 1990s has been linked to more feral dogs and a higher risk of rabies.

2.22.4 Cultural services

Cultural services are non-material benefits, closely tied to people's identity and values:

- **Spiritual and religious values:** sacred rivers (Ganga, Narmada, Kaveri), sacred groves, sacred mountains and lakes (Kailash–Mansarovar, Khecheopalri in Sikkim).
- **Recreation and ecotourism:** tiger reserves, hill stations, beaches, birdwatching in Bharatpur and Chilika.
- **Aesthetic and inspirational value:** landscapes and wildlife in art, music and literature, from Kalidasa's *Meghaduta* to Warli and Gond painting.
- **Education and science:** natural ecosystems as outdoor classrooms and field laboratories, such as Delhi's biodiversity parks.
- **Sense of place and cultural heritage:** festivals tied to harvests, rivers and seasons; traditional ecological knowledge.

2.22.5 Supporting services

Supporting services are the basic ecological processes on which all other services depend. They act indirectly or over long periods:

- **Primary production:** capture of solar energy by plants and algae, the base of all food chains.
- **Nutrient cycling:** the carbon, nitrogen, phosphorus and water cycles (described in the earlier part of this chapter).
- **Soil formation:** the slow weathering of rock and build-up of organic matter; a centimetre of fertile topsoil can take hundreds of years to form.
- **Habitat provision:** spaces where species live, breed and shelter, such as mangroves acting as fish nurseries.

Later frameworks, such as The Economics of Ecosystems and Biodiversity (TEEB) and the UN System of Environmental-Economic Accounting, treat supporting processes as part of ecosystem functioning rather than as a separate service, to avoid counting the same benefit twice. The IPBES now uses the broader term **nature's contributions to people**, which gives more weight to indigenous and local knowledge.

Table 2.22: Examples of ecosystem services from Indian ecosystems

Ecosystem	Provisioning	Regulating	Cultural
Himalayan forests	Timber, fuelwood, medicinal herbs	Slope stability, river flow, carbon	Pilgrimage, trekking, sacred groves
Banni grassland	Fodder, milk from Banni buffalo	Soil carbon, dust control	Maldhari pastoral culture
Ganga river	Water, fish, fertile silt	Groundwater recharge, flood dispersal	Ghats, Kumbh Mela, river worship
East Kolkata Wetlands	Fish, vegetables	Natural treatment of city sewage	Livelihood traditions of fishers
Sundarbans	Honey, fish, crabs	Cyclone protection, blue carbon	Bonbibi worship, tourism
Coral reefs (Gulf of Mannar)	Reef fish, seaweed	Wave protection	Snorkelling, research

2.22.6 Trade-offs between services

Ecosystem services are linked, and increasing one often reduces others. Converting a mangrove to a shrimp farm raises one provisioning service (shrimp) but removes storm protection, fish nurseries, carbon storage and the livelihoods of people who collected crabs and fuelwood. Planting eucalyptus on a shola grassland may increase wood production and tree cover but reduces stream flow and wipes out grassland species. Dams increase water supply and hydropower but reduce fisheries and silt delivery downstream.

These trade-offs also have a social side. The people who gain (a shrimp-farm owner, an urban water user) are often not the people who lose (local fishers, downstream farmers). Ecosystem-service thinking helps decision-makers see all the gains and losses, and who bears them, before choosing.

Case Study: East Kolkata Wetlands: a city's free sewage plant

On the eastern edge of Kolkata lies a mosaic of about 12,500 hectares of fishponds (*bheris*), vegetable fields, rice paddies and garbage farms known as the East Kolkata Wetlands. For more than a century, the city's untreated sewage has flowed into these shallow ponds. There, sunlight, algae and bacteria break down the organic matter and kill many pathogens; the algae feed fish; and the nutrient-rich water then irrigates paddy and vegetable fields. The system treats a large share of the city's sewage at almost no cost, supplies Kolkata's markets with fish and vegetables, and employs tens of thousands of people.

The engineer and ecologist Dhrubajyoti Ghosh documented this traditional practice in the 1980s, showing that it was one of the world's largest examples of wastewater-fed aquaculture. The wetlands were designated a Ramsar site in 2002 as a "wise use" wetland. Yet they remain under constant pressure from real-estate development and from the toxic industrial effluent and plastics that now mix with the sewage. The case shows the value of a regulating service (waste treatment) that conventional accounts would record as worthless land.

2.22.7 Valuing ecosystem services

Because most ecosystem services are not bought and sold in markets, they are often treated as having zero value, and so ecosystems are converted even when the losses outweigh the gains. **Economic valuation** tries to estimate the value of these services in money terms, so that they can be counted in decisions.

Economists use the idea of **Total Economic Value**, which has two main parts:

- **Use values: direct use** (timber, fish, tourism), **indirect use** (flood control, pollination, water purification) and **option value** (keeping a resource available for possible future use, such as an undiscovered medicine).
- **Non-use values: existence value** (satisfaction from knowing that tigers or coral reefs exist, even if one never sees them) and **bequest value** (wanting to pass them on to future generations).

Common methods of valuation include:

- **Market prices** for goods that are traded (fish, timber, NTFPs).
- **Cost-based methods:** the **replacement cost** of providing the service artificially (for example, building a sewage treatment plant instead of the East Kolkata Wetlands) or the **avoided damage** cost (cyclone damage prevented by mangroves).
- **Revealed preference methods:** the **travel cost** people spend to visit a park, or the extra price of houses near a lake (**hedonic pricing**).
- **Stated preference methods:** surveys asking people how much they would be willing to pay to protect an ecosystem (**contingent valuation**).

In 1997, Robert Costanza and colleagues estimated the value of the world's ecosystem services at about US$33 trillion a year, more than the world's GDP at the time; an updated study (2014) put it at about US$125 trillion a year. Such global figures are rough and debated, but they make the point that the economy depends on nature, not the other way round.

The Economics of Ecosystems and Biodiversity (TEEB) was an international initiative launched in 2007 by Germany and the European Commission and hosted by UNEP, led by the Indian economist Pavan Sukhdev. Its reports (2008–2010) showed how the economic invisibility of nature drives its loss and recommended ways of building nature's value into policy and business. India launched the **TEEB India Initiative** in 2011, which carried out case studies on forests, inland wetlands and coastal and marine ecosystems.

India already uses valuation in several ways:

- **Net Present Value (NPV) of forests:** Following Supreme Court orders from 2002 onwards, anyone diverting forest land for non-forest use must pay the NPV of the ecosystem services lost, in addition to the cost of compensatory afforestation. These funds are managed under the **Compensatory Afforestation Fund Act, 2016** (CAMPA funds).
- **Environmental-economic accounts:** The Ministry of Statistics and Programme Implementation publishes natural capital accounts for forests, water, soil and wetlands in its *EnviStats India* reports, following the UN System of Environmental-Economic Accounting.
- **Payments for ecosystem services (PES):** Schemes in which those who benefit from a service pay those who maintain it, such as downstream water users compensating upstream communities for protecting a catchment. Sukhomajri, described later in this chapter, is an early Indian example of the principle.

Valuation has limits. Some values, such as the sacredness of a river or the existence of a species, cannot be fairly expressed in money, and putting a price on nature can encourage the idea that anything can be traded away if the price is right. Valuation is best used as one input to decisions alongside ecological limits, rights and ethics.

Key Points

- Ecosystem services are the benefits people obtain from ecosystems: provisioning, regulating, cultural and supporting (MEA, 2005).
- Regulating and supporting services are the least visible and the most often lost.
- Services involve trade-offs, and the winners and losers are often different people.
- Valuation (Total Economic Value, TEEB, NPV of forests) helps bring these services into decisions but cannot capture every value.

2.23 Ecosystem Preservation and Conservation Strategies

Preservation and **conservation** are related but different ideas. **Preservation** means protecting an ecosystem from human use, keeping it as close as possible to its natural state; it rests partly on the belief that nature has value in itself. **Conservation** means managing ecosystems for sustainable use, so that they continue to provide benefits to present and future generations. In practice, India uses both: strict protection for core areas of high value, and wise use across the much larger landscapes where people live and work.

2.23.1 The ecosystem approach

Protecting single species is not enough if their habitat collapses, so modern conservation increasingly works at the level of whole ecosystems and landscapes. The **ecosystem approach**, adopted by the Convention on Biological Diversity in 2000, is a strategy for the integrated management of land, water and living resources that promotes conservation and sustainable use in an equitable way. Its main principles, applied in practice as **ecosystem-based management**, are:

- Manage at the right scale: a river basin, a landscape or a coastline, rather than an isolated patch.
- Maintain ecosystem structure and functioning (water flows, nutrient cycles, food webs), not just individual species.
- Involve all stakeholders, especially local communities, and use both scientific and traditional knowledge.
- Recognise that ecosystems change, and manage adaptively: monitor, learn and adjust.
- Consider the effects of activities on neighbouring ecosystems (for example, catchment deforestation on a downstream lake).
- Balance conservation and use, and share benefits fairly.

2.23.2 Protected areas and landscape connectivity

The most direct preservation strategy is a network of **protected areas**: national parks, wildlife sanctuaries, conservation reserves and community reserves, together with biosphere reserves and marine protected areas. Protected areas cover about 5 per cent of India's land; their types, history and management are discussed in Chapter 6. From an ecosystem point of view, three points matter:

- **Representation:** The network should include examples of every ecosystem type. India's network is strongest for forests and weakest for grasslands, deserts, rivers and the open sea.
- **Connectivity:** Isolated reserves lose species over time. **Wildlife corridors**, such as those linking tiger reserves in central India and the Terai Arc Landscape along the India–Nepal border, allow animals and genes to move between protected areas.
- **Conservation beyond protected areas: Other effective area-based conservation measures (OECMs)** are areas outside the formal protected-area system that still conserve biodiversity effectively, such as community forests, sacred groves, well-managed wetlands and urban biodiversity parks. In 2022, Gurugram's citizen-restored Aravalli Biodiversity Park became the first site in India to be recognised as an OECM. The Kunming–Montreal Global Biodiversity Framework (2022) counts both protected areas and OECMs towards its target of conserving 30 per cent of land and sea by 2030.

2.23.3 Wetland conservation

Wetlands are among India's most threatened ecosystems, and they have a dedicated set of policies:

- **Ramsar Convention:** India has designated more than 90 wetlands as **Ramsar sites** (wetlands of international importance), a number that has risen rapidly since 2019 (see Chapter 5).
- **Wetlands (Conservation and Management) Rules, 2017:** Notified under the Environment (Protection) Act, 1986, these rules replaced the 2010 rules. They set up a **State Wetlands Authority** in each state and union territory to identify, notify and manage wetlands, and a National Wetlands Committee to advise the central government. Within notified wetlands they

prohibit conversion to non-wetland uses, encroachment, setting up or expanding industries, dumping of solid waste and discharge of untreated effluents. Critics note that the rules leave much to state authorities and do not automatically cover wetlands until they are notified.

- **National Plan for Conservation of Aquatic Ecosystems (NPCA):** a centrally sponsored scheme, formed in 2013 by merging the earlier National Lake Conservation Plan and National Wetlands Conservation Programme, that funds integrated management plans for lakes and wetlands: catchment treatment, sewage interception, de-weeding, de-silting, biodiversity conservation and community participation.
- **Amrit Dharohar:** launched on World Environment Day, 5 June 2023, to promote the conservation and wise use of Ramsar sites over three years through community participation (*Jan Bhagidari*), ecotourism and livelihood activities for local people.
- **Save Wetlands Campaign:** a people's campaign launched on World Wetlands Day 2023 to enlist citizens as "wetland mitras" (friends of wetlands) who help map, monitor and protect local wetlands.

2.23.4 Forest, grassland and coastal ecosystems

- **National Mission for a Green India (Green India Mission):** One of the eight missions under India's National Action Plan on Climate Change (see Chapter 5), launched in 2014. Its goals are to increase forest and tree cover on 5 million hectares and improve the quality of forest cover on another 5 million hectares, while improving ecosystem services and the incomes of forest-dependent households. A revised roadmap released in 2025 shifts the focus to restoring vulnerable landscapes: the Aravallis, Western Ghats, Himalaya, mangroves and the arid north-west.
- **Nagar Van Yojana (2020):** creates urban forests on forest land in and around cities, funded from CAMPA; its target was raised to 1,000 Nagar Vans by 2027.
- **Mangrove Initiative for Shoreline Habitats and Tangible Incomes (MISHTI):** launched on 5 June 2023 to restore and plant mangroves on about 540 km^2 across nine coastal states and four union territories over five years (2023–2028), using funds from the compensatory afforestation fund, MGNREGS and other sources.
- **Coastal Regulation Zone (CRZ) Notification, 2019:** regulates construction and industry along the coast and gives the highest protection (CRZ-I) to ecologically sensitive areas such as mangroves, coral reefs, sand dunes and turtle nesting beaches.
- **Grasslands:** India has no dedicated grassland policy. Ecologists have called for grasslands and other open natural ecosystems to be removed from wasteland categories and recognised as ecosystems in their own right, with species recovery programmes such as that for the Great Indian Bustard (see Chapter 6) providing partial protection.

2.23.5 Community-based conservation

Most of India's ecosystems lie outside protected areas and are used daily by people. Their future depends on those people having the rights, incentives and institutions to look after them:

- **Joint Forest Management (JFM):** Since 1990, forest departments and village committees have jointly protected and regenerated degraded forests, with villagers sharing in the produce. Results have been mixed, but JFM brought communities into forest management.

- **Community forest rights:** The Forest Rights Act, 2006 recognises the rights of gram sabhas to protect, regenerate, conserve and manage their **community forest resources** (see Chapter 5). Villages such as Mendha-Lekha in Gadchiroli, Maharashtra, have used these rights to manage and sustainably harvest bamboo.
- **Community-conserved areas:** Sacred groves (*devrai*, *kavu*, *orans*), village forests of Uttarakhand (*van panchayats*), community reserves and the community-protected forests of Nagaland and Meghalaya conserve large areas (see Chapter 6).
- **Biodiversity Management Committees (BMCs):** Under the Biological Diversity Act, 2002, every local body must form a BMC, which documents local biodiversity and knowledge in a **People's Biodiversity Register**.

Community conservation works best when communities have secure rights, gain tangible benefits, and are partners rather than subjects of government programmes.

> **Think and Do: Find an ecosystem service in your neighbourhood**
>
> Choose a lake, park, patch of ridge forest or riverbank near your home or college. List at least two provisioning, two regulating and two cultural services it provides, and the people who benefit from each. Then identify one threat to it and suggest one practical conservation step that your college eco-club could take.

2.24 Basics of Ecosystem Restoration

Protection keeps healthy ecosystems intact, but much of the world's land and water is already degraded. In India alone, about 30 per cent of the land was undergoing degradation in 2018–19 (ISRO, 2021). **Ecological restoration** is the process of assisting the recovery of an ecosystem that has been degraded, damaged or destroyed (Society for Ecological Restoration, SER). Restoration does not replace protection: preventing damage is always cheaper and more certain than repairing it.

2.24.1 Restoration and related terms

Different kinds of repair aim at different goals:

- **Restoration:** returning an ecosystem towards its original native structure, species and functions, using a healthy **reference ecosystem** as the model. Example: re-creating native Aravalli scrub forest on an old mining site.
- **Rehabilitation:** repairing ecosystem functions and services (such as soil cover or water retention) without necessarily restoring the original species mix. Example: planting mixed native and useful trees on eroded village land.
- **Reclamation:** making severely disturbed land, such as a mine dump, stable and usable again, often for a new purpose.
- **Remediation:** removing or neutralising contaminants from soil or water, for example using plants that absorb heavy metals (phytoremediation; see Chapter 4).
- **Afforestation and reforestation:** planting trees where there was no forest, or where forest was recently lost. Tree planting is restoration only if it suits the original ecosystem; planting trees on natural grasslands is not restoration.

2.24.2 Principles of ecological restoration

The SER's *International Principles and Standards for the Practice of Ecological Restoration* (2019) set out eight principles. In simplified form, good restoration:

1. **Engages stakeholders**, especially the local people who depend on the land.
2. **Draws on many kinds of knowledge:** scientific, traditional and local.
3. **Is based on native reference ecosystems**, while allowing for environmental change such as a warming climate.
4. **Supports the ecosystem's own recovery processes** (natural regeneration, seed dispersal, soil formation) rather than replacing them.
5. **Is judged against clear goals and measurable indicators.**
6. **Seeks the highest level of recovery that is achievable.**
7. **Gains value when carried out at large scales**, connecting restored patches across a landscape.
8. **Is part of a continuum** of restorative activities, from reducing pressures to full restoration.

Two practical rules follow: **remove the cause of degradation first** (planting saplings on land that is still overgrazed or mined wastes money), and **use native species suited to the site**.

2.24.3 Passive and active restoration

Table 2.23: Passive and active restoration compared

Feature	Passive restoration	Active restoration
Meaning	Remove the stress and let natural succession repair the ecosystem	Directly intervene to rebuild soil, water, vegetation or animal populations
When suitable	Mild or moderate damage; seeds, roots and soil still present; healthy ecosystems nearby	Severe damage; soil lost; seed sources gone; invasive species dominant; natural recovery stalled
Typical actions	Fencing out livestock, stopping fire or mining, ending pollution, protecting natural regeneration	Earthworks, check dams, soil amendment, removing invasives, planting native species, reintroducing animals, restoring water flows
Cost and speed	Cheap; slow but often produces a more natural result	Costly and labour-intensive; faster where it works
Indian example	Village closures for natural regeneration under JFM; Sukhomajri hills after grazing stopped	Delhi's biodiversity parks; mangrove planting; Chilika's new sea mouth

In practice most projects combine the two: protection first, then active help only where natural recovery does not happen. **Assisted natural regeneration**, which protects and nurtures the seedlings and root stock already on site, is often the most cost-effective middle path.

2.24.4 Steps in a restoration project

The figure below shows the usual sequence of steps in planning and carrying out a restoration project.

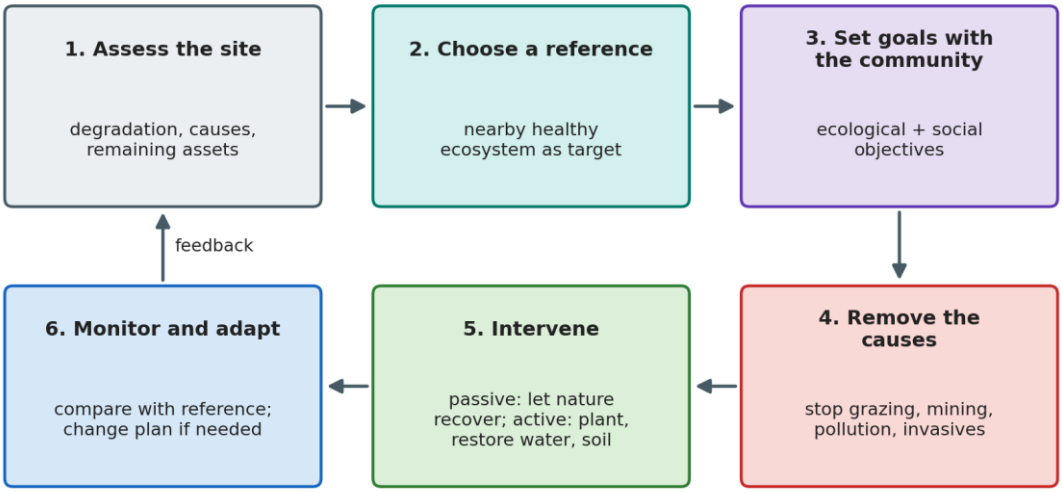


Figure 2.15: *Steps in an ecological restoration project. Monitoring feeds back into assessment and planning, making restoration a cycle of learning (adaptive management). Source: Simplified from SER International Principles and Standards (2019).*

1. **Assess the site:** Map what has been lost, the causes of degradation, and what remains (soil, seed bank, native plants, water sources).
2. **Choose a reference ecosystem:** A nearby, relatively undisturbed example of the same ecosystem type shows what the restored site should eventually resemble.
3. **Set goals with stakeholders:** Agree on ecological goals (native species, water flows, soil cover) and social goals (fodder, water, jobs) with local communities and agencies.
4. **Remove the causes:** Stop the grazing, mining, pollution, drainage or invasion that caused damage.
5. **Intervene:** Allow natural regeneration where possible; otherwise repair soil and water (contour trenches, check dams, removing embankments), control invasive species and plant native species in the right sequence, from hardy pioneers to later-successional species.
6. **Monitor and adapt:** Measure indicators such as plant cover, species richness, soil organic matter, water levels and the return of animals, compare them with the reference site, and change the plan if targets are not being met. This learning cycle is called **adaptive management**.

Restoration applies the ideas of **ecological succession** described earlier in this chapter: practitioners try to kick-start or speed up the natural sequence of communities by supplying what is missing, whether soil, water, seeds or protection.

2.24.5 Global and national commitments

- **UN Decade on Ecosystem Restoration (2021–2030):** Declared by the UN General Assembly in 2019 and led by UNEP and FAO, it calls on governments and citizens to prevent, halt and reverse the degradation of ecosystems on every continent and in every ocean.
- **Bonn Challenge:** A global goal, launched in 2011, to bring 150 million hectares of degraded and deforested land into restoration by 2020 and 350 million hectares by 2030. India joined in 2015 with a pledge of 13 million hectares by 2020 and a further 8 million by 2030. At the 14th Conference of Parties of the UN Convention to Combat Desertification, hosted in New Delhi in

2019, India raised its target to restoring **26 million hectares** of degraded land by 2030, as part of its goal of **land degradation neutrality** (no net loss of healthy land, Sustainable Development Goal target 15.3).

- **Kunming–Montreal Global Biodiversity Framework (2022):** Target 2 aims to have at least 30 per cent of degraded terrestrial, inland water, coastal and marine ecosystems under effective restoration by 2030.
- **National programmes:** the Green India Mission, MISHTI, the Nagar Van Yojana and the **Aravalli Green Wall Project** (launched in 2025), which aims to restore degraded land in a buffer around the Aravalli range across Gujarat, Rajasthan, Haryana and Delhi to check the eastward spread of land degradation.

2.24.6 Restoration in India: examples

Yamuna and Aravalli Biodiversity Parks, Delhi

Delhi's biodiversity parks are a model of urban ecosystem restoration developed by the Delhi Development Authority (DDA) with the Centre for Environmental Management of Degraded Ecosystems (CEMDE), University of Delhi.

- **Yamuna Biodiversity Park:** Begun in 2002 on about 457 acres of degraded, salt-affected farmland on the Yamuna floodplain at Wazirabad in north Delhi, it was redesigned as a set of the river's native ecosystems: floodplain wetlands, grasslands and forest communities typical of the Yamuna basin. Several hundred native plant species were reintroduced in successional stages. The park now attracts large numbers of resident and migratory birds, recharges groundwater and serves as an outdoor laboratory for students. A second phase extends restoration onto the active floodplain.
- **Aravalli Biodiversity Park:** Developed since the mid-2000s on about 692 acres of abandoned stone quarries and *Prosopis juliflora* thickets near Vasant Vihar on the Delhi Ridge. Restoration involved removing invasive *Prosopis*, building soil, harvesting rainwater in old mining pits, and planting native Aravalli species such as dhau (*Anogeissus pendula*), khair, kumtha and salai in community assemblages. Old quarry pits now house a butterfly conservatory, a fernery and an orchid conservatory.

Delhi now has several such parks (including Neela Hauz, Tilpath Valley, Kamla Nehru Ridge, Tughlaqabad and Kalindi), and they are popular field-visit sites for the practicals of this course. Separately, in neighbouring Gurugram, a citizens' group began restoring a roughly 390-acre former mining site in 2010; this Aravalli Biodiversity Park became India's first OECM in 2022.

Sukhomajri: watershed restoration by a village

Sukhna Lake, an artificial lake built for the new city of Chandigarh, was silting up so fast that by the mid-1970s it had lost much of its storage capacity. Scientists of the Central Soil and Water Conservation Research and Training Institute, led by P. R. Mishra, traced the silt to the bare, overgrazed Shivalik hills above the village of **Sukhomajri** in Haryana.

From 1976, small earthen dams were built in the hill gullies to store runoff. The villagers agreed to stop open grazing on the hills in return for a share of the stored water to irrigate their fields. Each family received an equal share of water, including landless families, who could sell or trade their

share. A water users' society, later the Hill Resource Management Society, managed the dams, collected fees and sold the *bhabbar* grass that grew back on the protected slopes to paper mills.

Within a few years, crop yields and milk production rose, the hills recovered their grass and tree cover through natural regeneration, and silt flow into Sukhna Lake fell sharply. Sukhomajri became the model for participatory watershed development across India. Later disputes over control of the forest land and over sharing benefits weakened the institution, a reminder that restoration also needs lasting institutions.

Chambal ravines

Along the Chambal and its tributaries in Madhya Pradesh, Rajasthan and Uttar Pradesh lies one of the largest ravine (*beehad*) landscapes in India: a maze of deep gullies cut into soft alluvium by runoff. Gully erosion eats into farmland every year. Ravine land has been "reclaimed" by levelling it for agriculture, but levelling is costly and erosion often resumes.

An ecological approach treats ravines as an ecosystem to be stabilised rather than erased. Measures include peripheral bunds along the ravine edge to stop runoff entering the gullies, small check dams and gully plugs, and planting native grasses, shrubs and trees (such as *Acacia*, *Prosopis cineraria*, *Dendrocalamus strictus* bamboo and khus grass) on slopes. Stabilised ravines provide fodder and fuelwood and are themselves habitat for wolves, hyenas, chinkara and birds. They also border the **National Chambal Sanctuary** (1979), a riverine protected area shared by three states that holds the world's largest wild population of the gharial and also shelters Gangetic dolphins and the Indian skimmer. Restoration of the ravines and of the river are therefore linked: less soil erosion means a healthier river.

Mangrove restoration

India is one of the few countries where mangrove cover has broadly stabilised or increased over recent decades, largely because of restoration:

- **Pichavaram, Tamil Nadu:** In the 1990s, the M. S. Swaminathan Research Foundation worked with local fishing communities on a **fishbone canal** technique: digging a network of channels on degraded, hypersaline mudflats so that tidal water could flush them, after which mangrove seedlings planted along the canals survived. Mangrove cover in Pichavaram increased substantially, and villages behind the restored mangroves suffered less damage in the 2004 tsunami.
- **Gujarat and the Sundarbans:** Large-scale planting on new mudflats in the Gulf of Kachchh and the Sundarbans, often by women's self-help groups, has expanded mangrove cover and provided jobs.
- **MISHTI (2023)** extends this effort across the coastline.

Successful mangrove restoration depends on restoring the right tidal flow and planting the right species on the right part of the shore; many failed projects simply planted *Rhizophora* seedlings on mudflats that were too low or too high for them.

2.24.7 Challenges in restoration

- **Planting is not restoring:** Many schemes measure success by the number of saplings planted rather than the number that survive or the ecosystem that develops. Survival rates are often low.

- **Wrong ecosystems, wrong species:** Monoculture plantations, exotic species and trees planted on natural grasslands or wetlands can cause new damage.
- **Unresolved causes:** If grazing, mining, pollution or encroachment continue, restoration fails.
- **Rights and benefits:** Restoration that ignores the needs and rights of local people rarely lasts; where communities gain water, fodder or income, as at Sukhomajri, they protect the land themselves.
- **Climate change:** Reference conditions of the past may not suit a warmer future, so restoration increasingly aims for resilient, functioning ecosystems rather than exact copies of the past.

Key Points

- Ecological restoration assists the recovery of degraded ecosystems towards a native reference state.
- Remove the causes of degradation first; then choose passive or active restoration according to the damage.
- Restoration follows steps: assess, choose a reference, set goals, remove causes, intervene, monitor and adapt.
- India has pledged to restore 26 million hectares of degraded land by 2030 during the UN Decade on Ecosystem Restoration.
- Delhi's biodiversity parks, Sukhomajri, Chambal ravine stabilisation and mangrove restoration at Pichavaram show what restoration can achieve.

Chapter 2 Review

Summary

- An ecosystem is a community of organisms interacting with one another and with their non-living environment, linked by a one-way flow of energy and the cycling of matter. The term was introduced by A. G. Tansley in 1935.
- Ecosystems are open, dynamic and self-regulating systems of every size, from a puddle to the biosphere; ecotones are transition zones between them.
- The abiotic components of an ecosystem are climatic, edaphic, topographic, inorganic and organic factors. The biotic components are producers, consumers, and detritivores and decomposers that return nutrients to the soil and water.
- Liebig's law of the minimum, Blackman's law of limiting factors and Shelford's law of tolerance explain how the scarcest or most extreme factor controls where and how well organisms live.
- Energy flow obeys the laws of thermodynamics. Producers fix only about 1% of incoming sunlight, and on average only about 10% of the energy at one trophic level passes to the next (Lindeman's ten per cent law).
- The single-channel energy flow model shows a linear flow; the Y-shaped model divides it into grazing and detritus channels. In forests and mangroves, most energy flows through the detritus channel.
- Food chains may be grazing, detritus or parasitic; interlinked food chains form food webs, which make communities more stable. The loss of India's vultures shows how changes in a food web can affect human health.
- Persistent, fat-soluble pollutants such as DDT and methylmercury biomagnify, reaching their highest concentrations in top predators, as seen in the decline of fish-eating birds and in Minamata disease.
- Ecological succession is the orderly change of communities over time, from pioneer species to a climax community. Primary succession starts on bare surfaces; secondary succession starts where soil remains. Hydrarch and xerarch seres both tend towards a mesic climax.
- In biogeochemical cycles, elements move between living and non-living pools. Carbon and nitrogen have gaseous cycles, phosphorus a sedimentary cycle, and sulfur both; water cycles through evaporation, transpiration, precipitation, infiltration and runoff.
- Human activities have disturbed these cycles: atmospheric CO_2 has risen from about 280 ppm to about 419 ppm (2023), fertiliser nitrogen and phosphorus cause eutrophication, and coal burning releases sulfur dioxide.
- Gross primary productivity minus plant respiration gives net primary productivity. Coral reefs, wetlands, estuaries and tropical rainforests are the most productive ecosystems per unit area; deserts, tundra and the open ocean are the least.
- Ecological pyramids of number and biomass may be upright or inverted, but the pyramid of energy is always upright.
- Ecosystem homeostasis depends on negative feedback. Positive feedback amplifies change, and stress beyond the homeostatic plateau can push an ecosystem past a tipping point into a new state, as with eutrophic lakes and bleached coral reefs.

- Climate, mainly temperature and rainfall, decides which biome develops on land; salinity, light, oxygen and water movement classify aquatic ecosystems into freshwater (lentic and lotic), marine and transitional types.
- Tundra is the cold, treeless biome of high latitudes and altitudes, with permafrost, a short growing season, low productivity and short food chains. India's alpine tundra lies above the treeline in Ladakh, Spiti, north Sikkim, Uttarakhand and Kashmir and is threatened by warming, overgrazing, medicinal-plant extraction and tourism.
- Forests are layered ecosystems dominated by trees. Champion and Seth (1968) classified Indian forests into 16 type groups, of which tropical dry and moist deciduous forests are the most extensive. Forest and tree cover was 25.17 per cent of India's area in 2023 (ISFR 2023).
- Grasslands (Banni, Deccan savannas, terai, sholas, alpine meadows) are ancient ecosystems shaped by fire, grazing and seasonal drought, but are often misclassified as "wastelands" and lost to conversion, tree planting and invasive species such as *Prosopis juliflora*.
- Deserts receive less than about 250 mm of rain; the Thar is a hot desert and Ladakh a cold desert. Their organisms escape, avoid or tolerate drought, and both face desertification, groundwater loss and large energy projects.
- Lakes show horizontal zones (littoral, limnetic, profundal, benthic) and, in deep lakes, summer thermal stratification into epilimnion, thermocline and hypolimnion. Cultural eutrophication from sewage and fertilisers is the most widespread problem of Indian lakes such as Dal and the urban lakes of Bengaluru.
- Rivers are open, one-way, oxygen-rich systems whose food base and communities change from headwaters to mouth, as described by the river continuum concept. Dams, pollution, sand mining and over-abstraction threaten the Ganga, Brahmaputra, Godavari and other rivers.
- Oceans are zoned by distance from shore and by depth; the sunlit neritic zone, upwelling areas and coral reefs are the most productive. India's reefs in the Gulf of Mannar, Gulf of Kachchh, Lakshadweep and the Andamans are threatened by warming-induced bleaching, overfishing and pollution.
- Estuaries, mangroves and lagoons are highly productive transitional ecosystems that serve as fish nurseries and storm buffers. The Sundarbans and the restored Chilika lagoon are key Indian examples.
- Ecosystem services are the benefits people obtain from ecosystems: provisioning, regulating, cultural and supporting (Millennium Ecosystem Assessment, 2005). Valuation methods, TEEB and India's Net Present Value payments for forest diversion try to make these benefits visible in decisions.
- Conservation strategies at the ecosystem level include the ecosystem approach, connected protected-area networks and OECMs, wetland policies (Wetlands Rules 2017, NPCA, Amrit Dharohar), the Green India Mission, MISHTI, the CRZ Notification and community-based management.
- Ecological restoration assists the recovery of degraded ecosystems. It removes the causes of damage first, then uses passive or active methods, guided by a reference ecosystem and monitoring.
- India has pledged to restore 26 million hectares of degraded land by 2030 during the UN Decade on Ecosystem Restoration (2021–2030). Delhi's Yamuna and Aravalli Biodiversity

Parks, Sukhomajri, Chambal ravine stabilisation and mangrove restoration show what restoration can achieve.

Key Terms

Ecosystem: A community of living organisms together with its physical environment, functioning as a unit through energy flow and nutrient cycling.

Ecotone: A transition zone between two adjacent ecosystems, such as a mangrove belt or the reed fringe of a lake.

Abiotic component: Any non-living physical or chemical factor of an ecosystem, such as light, temperature, water or soil minerals.

Producer (autotroph): An organism that makes its own food from inorganic substances, using light (photosynthesis) or chemical energy (chemosynthesis).

Consumer (heterotroph): An organism that obtains energy by eating other organisms.

Detritivore: An animal, such as an earthworm or termite, that feeds on dead organic matter and breaks it into smaller pieces.

Decomposer: A microorganism, mainly a bacterium or fungus, that breaks down dead organic matter into simple inorganic substances.

Limiting factor: The environmental factor in shortest supply, or furthest from the optimum, that restricts the growth or distribution of an organism.

Law of tolerance: Shelford's principle that each organism can survive only between a minimum and a maximum level of each environmental factor.

Trophic level: The position of an organism in the sequence of energy transfer, such as producer, herbivore or carnivore.

Ten per cent law: The generalisation that, on average, only about 10% of the energy at one trophic level is transferred to the next.

Food chain: A linear sequence of organisms through which energy and nutrients pass as one eats another.

Food web: A network of interconnected food chains in a community.

Biomagnification: The increase in concentration of a persistent substance at each successive trophic level of a food chain.

Ecological succession: The orderly, directional change in the species composition of a community over time.

Pioneer species: The first organisms to colonise a bare or disturbed area.

Climax community: The relatively stable, self-maintaining community at the end of succession.

Hydrarch succession: Succession that begins in water, such as a pond, and ends in a land community.

Xerarch succession: Succession that begins in a dry habitat, such as bare rock.

Biogeochemical cycle: The movement of a chemical element between living organisms and the non-living environment.

Nitrogen fixation: The conversion of atmospheric nitrogen gas into ammonia or nitrate by bacteria, lightning or industry.

Nitrification: The conversion of ammonium to nitrite and then nitrate by bacteria such as *Nitrosomonas* and *Nitrobacter*.

Denitrification: The conversion of nitrate back into nitrogen gas by bacteria in oxygen-poor soils and sediments.

Eutrophication: The excessive growth of algae and plants in a water body caused by added nutrients, often leading to oxygen depletion.

Gross primary productivity (GPP): The total rate at which producers fix energy or carbon in photosynthesis.

Net primary productivity (NPP): Gross primary productivity minus the respiration of the producers themselves.

Ecological pyramid: A diagram showing the number, biomass or energy at successive trophic levels.

Homeostasis: The ability of an ecosystem to maintain a relatively stable state through self-regulating feedback.

Resilience: The ability of an ecosystem to recover its structure and function after a disturbance.

Tipping point: A threshold beyond which an ecosystem shifts suddenly to a different, self-reinforcing state.

Biome: A large terrestrial ecosystem type defined by its climate and dominant vegetation, such as tundra, forest, grassland or desert.

Treeline: The altitude or latitude beyond which trees cannot grow because of cold and wind.

Permafrost: Ground that remains frozen for at least two consecutive years, found beneath Arctic tundra and parts of high mountains.

Stratification (forest): The vertical layering of a forest into emergent, canopy, understorey, shrub, herb and floor layers.

Epiphyte: A plant that grows on another plant for support without taking nutrients from it, such as many orchids.

Forest cover: All land of more than one hectare with a tree canopy density above 10 per cent, irrespective of ownership or legal status (FSI definition).

Savanna: A grassland with scattered trees, maintained by seasonal drought, fire and grazing.

Shola: A patch of stunted montane evergreen forest in the sheltered valleys of the high Western Ghats, set within rolling grassland.

Open natural ecosystem: A naturally treeless or sparsely wooded ecosystem, such as a grassland, savanna, scrubland or rocky outcrop.

Desertification: Land degradation in arid, semi-arid and dry sub-humid areas caused by climatic variations and human activities.

Ephemeral plant: A plant that completes its life cycle in the short period after rain and survives drought as seed.

Lentic ecosystem: A standing-water ecosystem such as a pond, lake or reservoir.

Lotic ecosystem: A flowing-water ecosystem such as a spring, stream or river.

Littoral zone: The shallow margin of a lake where light reaches the bottom and rooted plants grow.

Profundal zone: The deep, dark water of a lake below the compensation level, where photosynthesis is not possible.

Compensation level: The depth at which photosynthesis just balances respiration.

Thermocline: The layer in a stratified water body in which temperature drops sharply with depth, separating the epilimnion from the hypolimnion.

River continuum concept: The idea that physical conditions, food sources and communities change in a predictable way from the headwaters of a river to its mouth.

Environmental flow: The quantity, timing and quality of water flow required to sustain a river ecosystem and the people who depend on it.

Upwelling: The rising of cold, nutrient-rich deep water to the ocean surface, which raises productivity.

Neritic zone: The relatively shallow marine zone above the continental shelf.

Coral bleaching: The whitening of corals when heat or other stress causes them to expel their symbiotic algae (zooxanthellae).

Estuary: A partly enclosed coastal water body where fresh river water mixes with seawater.

Pneumatophore: An upward-growing breathing root of mangroves that takes in air from above the waterlogged mud.

Vivipary: Germination of a seed while it is still attached to the parent plant, as in many mangroves.

Ecosystem services: The benefits people obtain from ecosystems, grouped as provisioning, regulating, cultural and supporting services.

Total Economic Value: The sum of use values (direct, indirect, option) and non-use values (existence, bequest) of an ecosystem.

OECM (Other effective area-based conservation measure): An area outside the formal protected-area system that achieves effective long-term conservation of biodiversity.

Ecological restoration: The process of assisting the recovery of an ecosystem that has been degraded, damaged or destroyed.

Multiple-Choice Questions

1. Who introduced the term "ecosystem" in 1935?

(a) Ernst Haeckel
(b) Arthur G. Tansley
(c) Raymond Lindeman
(d) Charles Elton

2. Which of the following is a chemoautotroph?

(a) *Nitrosomonas*
(b) *Rhizobium*
(c) Earthworm
(d) Mushroom

3. Earthworms and termites that feed on dead leaves are best described as:

(a) Producers
(b) Primary consumers
(c) Detritivores
(d) Parasites

4. Shelford's law of tolerance states that:

(a) Growth is limited by the nutrient present in the largest amount

(b) Energy decreases by 90% at each trophic level

(c) Communities change in an orderly sequence over time

(d) Organisms survive only between a minimum and a maximum level of each factor

5. Energy flow in an ecosystem is unidirectional mainly because of:

(a) The first law of thermodynamics

(b) The law of the minimum

(c) The second law of thermodynamics

(d) The law of tolerance

6. If producers in an ecosystem fix 20,000 kJ of energy, about how much energy would be available to secondary carnivores according to the ten per cent law?

(a) 2,000 kJ

(b) 200 kJ

(c) 20 kJ

(d) 2 kJ

7. In the Y-shaped energy flow model, the two arms represent:

(a) Grazing and detritus channels

(b) Producers and consumers

(c) Land and water ecosystems

(d) Respiration and photosynthesis

8. In a mangrove forest such as the Sundarbans, most plant production first enters:

(a) The grazing food chain through deer

(b) The parasitic food chain

(c) The detritus food chain through fallen leaves

(d) The atmosphere directly as CO_2

9. Which property makes a pesticide likely to biomagnify?

(a) It is water-soluble and breaks down quickly

(b) It is persistent and fat-soluble

(c) It is a gas at room temperature

(d) It is used only in small amounts

10. Minamata disease in Japan was caused by:

(a) DDT in rice

(b) Cadmium in drinking water

(c) Methylmercury in fish and shellfish

(d) Arsenic in groundwater

11. Which is the correct sequence of hydrarch succession?

(a) Phytoplankton → submerged plants → floating plants → reed swamp → sedge meadow → woodland → forest

(b) Lichens → mosses → herbs → shrubs → forest

(c) Submerged plants → phytoplankton → reed swamp → floating plants → forest

(d) Floating plants → phytoplankton → sedge meadow → reed swamp → forest

12. Regrowth of forest on a jhum field left fallow in Meghalaya is an example of:

(a) Primary succession

(b) Secondary succession

(c) Xerarch succession

(d) Heterotrophic succession

13. Which biogeochemical cycle has no significant gaseous phase?

(a) Carbon cycle

(b) Nitrogen cycle

(c) Sulfur cycle

(d) Phosphorus cycle

14. The conversion of nitrate into nitrogen gas by bacteria such as *Pseudomonas* is called:

(a) Nitrogen fixation

(b) Ammonification

(c) Nitrification

(d) Denitrification

15. Why is all urea sold in India coated with neem oil?

(a) To make it dissolve faster in water

(b) To slow the conversion of urea to nitrate and reduce nitrogen losses

(c) To add phosphorus to the soil

(d) To kill nitrogen-fixing bacteria

16. Net primary productivity is equal to:

(a) Gross primary productivity plus plant respiration

(b) Secondary productivity minus consumer respiration

(c) Standing crop divided by time

(d) Gross primary productivity minus plant respiration

17. Which ecosystem has the lowest mean net primary productivity per square metre?

(a) Estuary

(b) Tropical seasonal forest

(c) Desert scrub

(d) Savanna

18. Which ecological pyramid can never be inverted?

(a) Pyramid of number

(b) Pyramid of biomass

(c) Pyramid of energy

(d) Pyramid of number in a parasitic food chain

19. Algal blooms in a lake cause oxygen depletion, fish deaths and the release of more nutrients, which cause further blooms. This is an example of:

(a) Negative feedback
(b) Positive feedback
(c) Resistance
(d) The ten per cent law

20. A grassland that burns easily but regrows from its roots within weeks shows:

(a) Low resistance and high resilience
(b) High resistance and low resilience
(c) High resistance and high resilience
(d) Neither resistance nor resilience

21. Which feature is characteristic of Arctic tundra but generally absent from the alpine tundra of the Himalaya?

(a) Low-growing cushion plants
(b) Continuous permafrost and 24-hour summer daylight
(c) Strong winds
(d) A short growing season

22. The Cold Desert Biosphere Reserve of India is located in

(a) Ladakh
(b) Sikkim
(c) Lahaul–Spiti, Himachal Pradesh
(d) Kachchh, Gujarat

23. According to the India State of Forest Report 2023, India's total forest and tree cover is about what percentage of its geographical area?

(a) 21.76%
(b) 25.17%
(c) 33.33%
(d) 3.41%

24. By area, which forest type group is the most extensive in India?

(a) Tropical wet evergreen
(b) Montane wet temperate
(c) Tropical dry deciduous
(d) Littoral and swamp

25. Why can grasses survive heavy grazing and fire better than most shrubs?

(a) Their growing tissue (meristem) lies at the base of the plant
(b) They have large, broad leaves
(c) They contain toxic chemicals
(d) They flower only once in their lifetime

26. The spread of *Prosopis juliflora* is the major ecological problem of which grassland?

(a) Eravikulam shola grassland
(b) Banni grassland
(c) Kaziranga floodplain grassland
(d) Bugyals of Uttarakhand

27. Which adaptation helps desert plants such as cacti and many euphorbias reduce water loss by keeping stomata closed during the hot day?

(a) C_4 photosynthesis
(b) Vivipary
(c) CAM photosynthesis
(d) Aerenchyma

28. In a deep lake during summer, the layer that may run out of oxygen is the

(a) Epilimnion
(b) Thermocline
(c) Littoral zone
(d) Hypolimnion

29. Arrange the steps of cultural eutrophication in the correct order: (1) algal bloom, (2) nutrient input from sewage, (3) oxygen depletion and fish death, (4) decomposition of dead algae.

(a) 2, 1, 4, 3
(b) 1, 2, 3, 4
(c) 2, 4, 1, 3
(d) 4, 2, 1, 3

30. Keibul Lamjao, the world's only floating national park, protects which species?

(a) Barasingha
(b) Sangai (brow-antlered deer)
(c) Hangul
(d) Nilgiri tahr

31. According to the river continuum concept, the headwater stream of a forested catchment is mainly

(a) Autotrophic, with P/R greater than 1 and grazers dominant
(b) Heterotrophic, with P/R less than 1 and shredders dominant
(c) Autotrophic, with filter-feeding collectors dominant
(d) Heterotrophic, with no invertebrates

32. The Ganges river dolphin finds its prey in muddy water mainly by

(a) Keen eyesight
(b) Smell
(c) Echolocation
(d) Electric fields

33. Which marine zone lies above the continental shelf and supports most of the world's fisheries?

(a) Abyssopelagic zone
(b) Hadal zone
(c) Bathypelagic zone
(d) Neritic zone

34. Pneumatophores in mangroves such as *Avicennia* help the plant to

(a) Excrete salt
(b) Disperse seeds
(c) Obtain oxygen from the air in waterlogged mud
(d) Store water

35. Chilika lagoon was removed from the Montreux Record in 2002 mainly because

(a) A new mouth to the sea restored salinity and fisheries
(b) It was declared a national park
(c) All fishing in the lagoon was banned
(d) A dam was built on the Mahanadi

36. Under the Millennium Ecosystem Assessment, the treatment of Kolkata's sewage by the East Kolkata Wetlands is an example of which ecosystem service?

(a) Provisioning
(b) Regulating
(c) Cultural
(d) Supporting

37. Soil formation and nutrient cycling are classified as

(a) Supporting services
(b) Provisioning services
(c) Cultural services
(d) Option values

38. The MISHTI scheme launched in 2023 aims to

(a) Restore Ramsar sites through community participation
(b) Create urban forests in Indian cities
(c) Restore and plant mangroves along India's coast
(d) Protect grasslands from solar parks

39. Fencing out livestock from a degraded hill and allowing grass and trees to regrow from existing roots and seeds is an example of

(a) Reclamation
(b) Remediation
(c) Active restoration
(d) Passive restoration

40. At the UNCCD COP14 in New Delhi (2019), India raised its target for restoring degraded land by 2030 to

(a) 13 million hectares
(b) 21 million hectares
(c) 26 million hectares
(d) 350 million hectares

Short-Answer Questions

1. Define an ecosystem. Why is an ecosystem described as an open system?
2. Distinguish between detritivores and decomposers, giving two examples of each.
3. State Liebig's law of the minimum and Shelford's law of tolerance. Give one Indian example of a limiting factor.
4. Explain the ten per cent law with a numerical example. Why are food chains usually limited to four or five trophic levels?
5. Draw and explain the Y-shaped model of energy flow. How does it differ from the single-channel model?
6. Differentiate between grazing and detritus food chains with one Indian example of each.
7. What is biomagnification? Explain why fish-eating birds were affected by DDT more than fish.
8. Distinguish between primary and secondary succession with one Indian example of each.
9. Describe the role of bacteria in the nitrogen cycle.
10. Why is phosphorus often the limiting nutrient in freshwater lakes? How does its excess lead to eutrophication?
11. Distinguish between gross primary productivity, net primary productivity and secondary productivity.
12. Explain, with examples, why the pyramid of biomass in the English Channel is inverted while its pyramid of energy is upright.
13. Distinguish between Arctic and alpine tundra. Name two places in India where alpine tundra occurs and two animals adapted to it.
14. Describe the vertical stratification of a tropical evergreen forest and explain why it supports high species diversity.
15. What is the difference between "forest cover" and "tree cover" as defined by the Forest Survey of India? Why does an increase in forest cover not always mean recovery of natural forests?
16. Explain three adaptations that allow grasses to survive grazing, fire and drought.
17. Why are grasslands in India often called "wastelands"? What are the consequences of this classification?
18. Compare the Thar and Ladakh deserts in terms of climate, vegetation and wildlife.
19. With a labelled sketch, describe the littoral, limnetic, profundal and benthic zones of a lake.
20. What is thermal stratification? Explain why a lake's deep water may lose its oxygen in summer and how turnover restores it.
21. Explain the river continuum concept with reference to the Ganga.
22. List four adaptations of mangroves to life in the intertidal zone.

23. Give two examples each of provisioning, regulating and cultural services provided by Indian wetlands.
24. Distinguish between passive and active restoration with one Indian example of each.

Long-Answer Questions

1. Describe the structure of an ecosystem in terms of its abiotic and biotic components. Illustrate your answer with the example of a village pond or a tropical deciduous forest in India.
2. Explain how the laws of thermodynamics govern energy flow in ecosystems. Using the single-channel and Y-shaped models, discuss how energy flow differs between a forest and an open-sea ecosystem.
3. Describe the stages of hydrarch succession. Using examples such as Loktak Lake, urban lakes and jhum fallows in North-East India, discuss how human activities speed up, slow down or deflect succession.
4. Describe the carbon and nitrogen cycles. How have fossil fuel use, deforestation and fertiliser use in India altered these cycles, and what measures has India taken in response?
5. What are ecological pyramids? Describe the pyramids of number, biomass and energy with examples of upright and inverted forms, and explain the limitations of ecological pyramids.
6. Explain ecosystem homeostasis with reference to negative and positive feedback, resistance, resilience and tipping points. Discuss the coral bleaching of Lakshadweep in 2023–24 as an example of an ecosystem under stress.
7. Describe the features, biotic structure and adaptations of forest ecosystems. Using the Champion and Seth classification, explain how forest types change with rainfall and altitude in India, and discuss the major threats to Indian forests.
8. "India's grasslands and deserts are its most neglected ecosystems." Discuss this statement with reference to the Banni grassland, the shola grasslands and the Thar Desert, covering their importance, threats and possible conservation measures.
9. Explain the process of cultural eutrophication. Using the examples of Loktak Lake and Chilika lagoon, discuss how human interventions can degrade or restore aquatic ecosystems.
10. Discuss the ecological importance of estuaries and mangroves. Evaluate the threats to the Sundarbans and suggest a strategy for its long-term conservation.
11. What are ecosystem services? Classify them according to the Millennium Ecosystem Assessment with Indian examples, and explain how economic valuation (including TEEB and India's Net Present Value payments) can help or hinder conservation.
12. Outline the principles and steps of ecological restoration. Critically examine any two Indian restoration examples (such as Delhi's biodiversity parks, Sukhomajri or mangrove restoration) and discuss what made them succeed or limited their success.

Further Reading

- Odum, E. P. and Barrett, G. W. (2005). *Fundamentals of Ecology* (5th ed.). Thomson Brooks/Cole.
- Singh, J. S., Singh, S. P. and Gupta, S. R. (2017). *Ecology, Environmental Science and Conservation*. S. Chand.
- Raven, P. H. and others (2015). *Environment*. Wiley.

- Carson, R. (1962). *Silent Spring*. Houghton Mifflin.
- Forest Survey of India (2024). *India State of Forest Report 2023*. Ministry of Environment, Forest and Climate Change, Government of India. https://fsi.nic.in
- Abrol, Y. P. and others (eds.) (2017). *The Indian Nitrogen Assessment: Sources of Reactive Nitrogen, Environmental and Climate Effects, Management Options, and Policies*. Elsevier.
- Friedlingstein, P. and others (2025). Global Carbon Budget 2024. *Earth System Science Data*, 17, 965–1039. https://globalcarbonbudget.org
- Champion, H. G. and Seth, S. K. (1968). *A Revised Survey of the Forest Types of India*. Manager of Publications, Government of India, Delhi.
- Millennium Ecosystem Assessment (2005). *Ecosystems and Human Well-being: Synthesis*. Island Press, Washington, DC.
- Space Applications Centre, ISRO (2021). *Desertification and Land Degradation Atlas of India*. Ahmedabad.
- Gann, G. D. and others (2019). *International Principles and Standards for the Practice of Ecological Restoration* (2nd ed.). Society for Ecological Restoration, Washington, DC.
- TEEB (2010). *The Economics of Ecosystems and Biodiversity: Mainstreaming the Economics of Nature*. UNEP, Geneva. https://teebweb.org

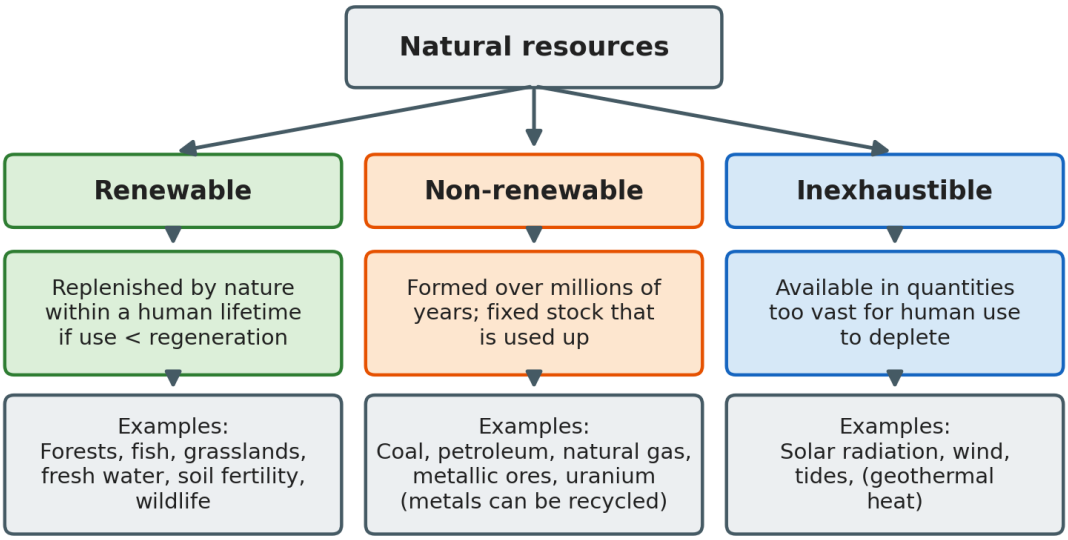
Answers to Multiple-Choice Questions

1. (b)	2. (a)	3. (c)	4. (d)	5. (c)	6. (c)	7. (a)	8. (c)	9. (b)	10. (c)
11. (a)	12. (b)	13. (d)	14. (d)	15. (b)	16. (d)	17. (c)	18. (c)	19. (b)	20. (a)
21. (b)	22. (c)	23. (b)	24. (c)	25. (a)	26. (b)	27. (c)	28. (d)	29. (a)	30. (b)
31. (b)	32. (c)	33. (d)	34. (c)	35. (a)	36. (b)	37. (a)	38. (c)	39. (d)	40. (c)

Chapter 3: Natural Resources

A **natural resource** is any material, energy source or service found in nature that people use to meet their needs: the soil in which crops grow, the water drawn from rivers and aquifers, the coal burnt in power stations, the forests that supply timber and medicines, and the sunlight that drives solar panels. Nothing becomes a resource by itself. A substance turns into a resource when people have the knowledge, technology and economic reason to use it. Bauxite was just reddish rock until the process for making aluminium from it was invented in the 1880s. Uranium was a laboratory curiosity until nuclear reactors were built. What counts as a resource therefore changes with technology, prices and culture.

Resources are commonly grouped by whether, and how fast, they are replaced by nature. **Renewable resources** are replenished by natural processes within a human lifetime, provided they are not used faster than they regenerate. Forests, fish, grasslands, fresh water and soil fertility fall in this group. **Non-renewable resources** form over millions of years, so for practical purposes their stock is fixed. Once used, they are gone. Coal, petroleum, natural gas and metallic ores are examples, although metals (unlike fuels) can be recycled. **Inexhaustible** (or perpetual) resources, such as solar radiation, wind and tides, are available in quantities so vast that human use cannot deplete them. The figure below summarises this classification. The line between the groups is not fixed. Groundwater is renewable in principle, but when an aquifer is pumped far faster than rain can recharge it, as in Punjab and Rajasthan, it behaves like a non-renewable stock. Fertile topsoil takes centuries to form, so soil lost to erosion is not replaced on any timescale that matters to a farmer.



Over-use can turn a renewable resource (e.g., groundwater in Punjab) into a depleting one.

Figure 3.1: *Classification of natural resources by their renewability, with examples. Renewable resources can be exhausted if used faster than they regenerate. Source: Author's diagram.*

Resources can also be classified by origin (**biotic** resources come from living things, such as forests and fisheries; **abiotic** resources come from non-living matter, such as minerals, land and water), by stage of development (actual or proven resources versus potential resources not yet tapped), and by ownership (individual, community, national and international resources such as the high seas).

India's resource situation is shaped by a simple imbalance. The country has about 2.4% of the world's land area and about 4% of its fresh water, but nearly 18% of its people (roughly 1.4 billion, or 140 crore). India is rich in coal, iron ore, bauxite and limestone, has some of the most productive alluvial plains on Earth, and holds a large share of the world's plant and animal diversity. At the same time, it imports most of its crude oil, much of its land is degraded, groundwater is over-drawn in many districts, and hundreds of millions of rural people depend directly on forests, pastures and soils for their living. How these resources are used, and who benefits, is therefore both an environmental and a social question. This chapter examines land resources (minerals, soils, crops, forests and the degradation of land), water resources, energy resources, and several Indian case studies that show how resource use generates conflict and how communities have responded.

Learning Objectives

- Classify natural resources as renewable, non-renewable or inexhaustible, and explain why the distinction depends on the rate of use.
- Describe India's land resources, including minerals, soils, agricultural crops, forest products, medicinal plants and forest-based livelihoods, and the policies that govern them.
- Analyse land-use change, land degradation, soil erosion, desertification and deforestation in India, using official data from ISRO and the Forest Survey of India.
- Evaluate the impacts of mining and dam building on the environment, forests, biodiversity and tribal communities, with Indian examples.
- Explain the natural and man-made sources and uses of water, the over-exploitation of surface and groundwater, floods and droughts, and inter-state and international water conflicts.
- Compare renewable and non-renewable energy sources, the energy content of fuels, and the role of alternative energy and agro-residues in meeting India's growing energy needs.
- Assess Indian case studies such as the National Solar Mission, the Cauvery water dispute, the Sardar Sarovar dam, the Chipko and Appiko movements and Tarun Bharat Sangh.

3.1 Mineral Resources

Minerals underpin almost every part of modern life. A single smartphone contains more than 30 elements, a kilometre of railway track needs about 120 tonnes of steel, and every tonne of cement needs well over a tonne of limestone. India is a major mining nation: it is the world's second-largest producer of coal and of aluminium, and the fourth-largest producer of iron ore (Ministry of Mines). Mining also brings serious environmental and social costs, which are examined later in this chapter. This section explains what minerals are, how they form, where India's deposits lie and how they are extracted and managed.

3.1.1 Formation and types of minerals

A **mineral** is a naturally occurring, inorganic solid with a definite chemical composition and an ordered internal (crystalline) structure. Quartz (SiO_2), haematite (Fe_2O_3) and calcite ($CaCO_3$) are examples. A **rock** is an aggregate of one or more minerals. An **ore** is a rock that contains a valuable mineral in high enough concentration to be mined at a profit. Whether a deposit is an ore therefore depends on its grade, depth and location as well as on market prices and technology.

Geologists distinguish a **mineral resource**, which is any concentration of a mineral that may be extracted now or in the future, from a **mineral reserve**, which is the part of a resource that has been measured in detail and can be mined economically with current technology. Reserves grow when prices rise or new techniques are developed, and shrink as mines are worked out.

Ore deposits form only where geological processes concentrate an element far above its average abundance in the Earth's crust. The main processes are:

- **Magmatic (igneous) processes**: as molten rock (magma) cools, heavy minerals such as chromite and magnetite crystallise and settle in layers. The chromite of the Sukinda valley in Odisha formed this way.
- **Hydrothermal processes**: hot, mineral-rich water circulating through cracks in rock deposits metals such as copper, lead, zinc and gold in veins. The lead-zinc ores of Zawar (Rajasthan) and the gold of the Kolar and Hutti fields (Karnataka) are hydrothermal.
- **Sedimentary processes**: minerals precipitate from water or accumulate as sediments. Coal forms from buried plant matter compressed over millions of years; most Indian coal formed in the Gondwana basins about 250–300 million years ago. Limestone, gypsum and rock salt are also sedimentary.
- **Residual weathering**: intense tropical weathering removes soluble elements and leaves insoluble ones behind. Bauxite (the ore of aluminium) and laterite on the plateaus of Odisha, Jharkhand and the Western Ghats formed this way.
- **Metamorphic processes**: heat and pressure transform existing rocks and minerals, producing deposits such as graphite, marble and some mica.
- **Placer deposits**: running water or waves sort heavy, resistant mineral grains and concentrate them. The beach sands of Kerala, Tamil Nadu and Odisha contain ilmenite, rutile and monazite (a source of thorium).

Minerals are commonly grouped by use, as shown in the table below. India records production of about 95 minerals, made up of 4 fuel minerals, 10 metallic minerals, 23 non-metallic minerals, 3 atomic minerals and 55 minor minerals. In law, the Mines and Minerals (Development and Regulation) Act, 1957 (MMDR Act) divides minerals into **major minerals**, regulated by the Union government, and **minor minerals** such as sand, gravel, building stone and ordinary clay, which are regulated by the states.

Table 3.1: Main groups of minerals with Indian examples

Group	Examples	Main uses	Leading Indian sources
Ferrous metallic	Iron ore, manganese, chromite	Steel and alloys	Odisha, Chhattisgarh, Karnataka, Jharkhand
Non-ferrous metallic	Bauxite, copper, lead, zinc, gold	Aluminium, wiring, batteries, jewellery	Odisha (bauxite), Rajasthan (zinc, copper), Karnataka (gold)
Non-metallic	Limestone, dolomite, mica, gypsum, rock phosphate	Cement, fertilisers, electronics	Rajasthan, Madhya Pradesh, Andhra Pradesh
Fuel minerals	Coal, lignite, petroleum, natural gas	Electricity, transport, industry	Jharkhand, Odisha, Chhattisgarh, Tamil Nadu (lignite)

Group	Examples	Main uses	Leading Indian sources
Atomic minerals	Uranium, thorium (monazite)	Nuclear power	Jharkhand (Jaduguda), Kerala and Odisha beach sands

3.1.2 India's mineral belts and key minerals

India's minerals are unevenly distributed. Most metallic minerals and coal lie in the ancient rocks of the Peninsular Plateau, while the alluvial plains of the north have almost none. Petroleum and natural gas occur in sedimentary basins such as Mumbai High, Assam and the Krishna–Godavari basin. Geographers usually recognise the following mineral belts:

- **North-eastern Peninsular belt (Chota Nagpur–Odisha plateau)**: the richest belt, covering Jharkhand, Odisha, Chhattisgarh and West Bengal. It has most of India's coal (Jharia, Bokaro, Raniganj, Talcher, Korba), high-grade iron ore (Noamundi, Kiriburu, Barbil), manganese, bauxite, chromite, mica and copper (Singhbhum).
- **Central belt**: Chhattisgarh and Madhya Pradesh, with coal (Singrauli), iron ore (Bailadila in Chhattisgarh), bauxite (Amarkantak), manganese (Balaghat), limestone and diamonds (Panna).
- **South-western belt**: Karnataka and Goa, with iron ore (Bellary–Hospet, Goa), manganese and gold (Kolar, Hutti).
- **North-western belt**: Rajasthan and Gujarat, with lead-zinc, copper (Khetri), marble, gypsum, rock phosphate, building stone and, in Gujarat, petroleum and natural gas.
- **Southern belt**: Tamil Nadu and Kerala, with lignite (Neyveli), bauxite and beach-sand minerals.

A few figures show the scale. India produced about 1,048 million tonnes of coal in 2024–25, the first year it crossed one billion tonnes (Ministry of Coal, provisional). Its geological coal resource was about 401 billion tonnes in April 2025, roughly 70% of it in Odisha, Jharkhand and Chhattisgarh (Geological Survey of India). Iron ore production reached a record 289 million tonnes and bauxite about 25 million tonnes in 2024–25 (Ministry of Mines, provisional). India is short of some minerals, however. It imports most of its copper concentrate, almost all its lithium, cobalt and nickel, and much of its rock phosphate and potash. Because these **critical minerals** are needed for batteries, electric vehicles, solar panels and electronics, the government launched the **National Critical Mineral Mission** in 2025 to expand exploration, overseas acquisition and recycling.

> **Coal and energy**
>
> Coal, petroleum and natural gas are minerals in the legal and geological sense, but their role as energy sources, their energy content and their environmental effects are discussed under energy resources later in this chapter.

3.1.3 Extraction and processing of minerals

The method used to extract a mineral depends on the depth, shape and grade of the deposit.

- **Surface (opencast) mining** removes the soil and rock lying above a deposit (the **overburden**) to expose the ore. It is cheaper and safer for workers than underground mining and is used for most Indian coal (over 90% of Coal India's output comes from opencast mines), iron ore,

bauxite and limestone. Its environmental footprint is large: whole hillsides are removed, forests cleared and huge waste dumps created.

- **Strip mining** is a form of surface mining for flat, shallow seams, worked in long strips.
- **Underground (shaft) mining** reaches deep deposits through vertical shafts and horizontal tunnels. It disturbs less land but is costlier and more dangerous because of roof collapse, flooding, toxic gases and dust. Some coal mines in Jharkhand and West Bengal and the Kolar gold fields (closed in 2001) are examples.
- **Placer mining and dredging** separates heavy mineral grains from sand and gravel using water, as in the beach-sand mining of Kerala and Tamil Nadu.
- **Solution mining** dissolves soluble minerals underground and pumps the solution to the surface, as for salt and potash.
- **Quarrying** extracts building stone, marble, sand and gravel (minor minerals). Because it is carried out at thousands of small sites, often illegally, it is hard to regulate. Illegal river-sand mining is a serious problem in many states.

After extraction, ore is processed in stages. It is first crushed and ground (**comminution**), then **concentrated** by separating the valuable mineral from waste rock (called **gangue**) using flotation, magnetic separation or washing. The concentrate is then **smelted** (heated with a reducing agent such as coke) or **leached** with chemicals to produce metal, which is finally **refined**. Each stage uses large amounts of energy and water and produces waste: fine-grained **tailings** are stored behind dams, and smelters release sulfur dioxide and particulate matter. Producing one tonne of aluminium from bauxite, for example, generates about two to three tonnes of a caustic residue called **red mud**.

3.1.4 Sustainable mining and District Mineral Foundations

Minerals are non-renewable, so true sustainability in mining means extracting them efficiently, limiting damage, restoring land after mining, sharing benefits fairly with affected people, and reducing demand through recycling and substitution. The main tools in India are:

- **Environmental clearance**: under the Environmental Impact Assessment (EIA) Notification, 2006, new or expanding mines must assess impacts, hold a public hearing and obtain clearance from the Union Ministry of Environment, Forest and Climate Change (MoEFCC) or a state authority.
- **Forest clearance**: mining on forest land requires approval under the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 (the Forest (Conservation) Act, renamed by the 2023 amendment), with payment for **compensatory afforestation** into the CAMPA fund.
- **Mine closure plans**: the Mineral Conservation and Development Rules, 2017 require every lease to have a progressive and final mine closure plan, backed by a financial guarantee, for backfilling pits, stabilising dumps and replanting.
- **Star rating**: the Indian Bureau of Mines awards one to five stars to mines based on their environmental and social performance.
- **Auctions and transparency**: since the 2015 amendment of the MMDR Act, mineral concessions are granted by public auction rather than discretion. This followed scandals over illegal mining in Bellary (Karnataka), Goa and Odisha.

- **Recycling and substitution**: steel, aluminium and copper can be recycled many times with large energy savings. The Steel Scrap Recycling Policy, 2019 and the Battery Waste Management Rules, 2022 aim to increase recovery of metals.

The 2015 amendment also created **District Mineral Foundations (DMFs)**. A DMF is a non-profit trust set up in every mining-affected district to work for the interest and benefit of people and areas affected by mining. Holders of leases granted after January 2015 pay 10% of their royalty to the DMF; holders of older leases pay 30%. The money is spent under the **Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)** on drinking water, health, education, sanitation, livelihoods and environmental protection in affected villages. By November 2024 DMFs had collected over ₹1 lakh crore (Ministry of Mines). Audits and studies have, however, found that much of the money was spent on roads, buildings and urban projects rather than on the most affected tribal villages, and that local gram sabhas often had little say in decisions.

> **Think and Do: Mining in your phone**
>
> List the main materials in a mobile phone (glass, aluminium casing, copper wiring, lithium battery, rare-earth magnets, gold contacts). For each, find one country or Indian state where it is mined. Discuss how recycling e-waste could reduce the need for new mining.

3.2 Soil

Soil is the thin, loose layer of weathered rock, organic matter, water, air and living organisms that covers much of the land surface. It is where the lithosphere, atmosphere, hydrosphere and biosphere meet. Soil anchors plants and supplies them with water and nutrients, stores more carbon than the atmosphere and all vegetation combined, filters water as it moves into aquifers, and is home to a large share of all species on Earth. Roughly 95% of human food is grown directly or indirectly in soil (FAO). Because a few centimetres of topsoil can take several hundred to a thousand years to form, soil is effectively a non-renewable resource on human timescales.

A healthy loam soil, by volume, is about 45% mineral particles, 5% organic matter, 25% water and 25% air. Its mineral fraction is described by **soil texture**, the relative proportion of **sand** (particles 0.05–2 mm), **silt** (0.002–0.05 mm) and **clay** (smaller than 0.002 mm). Sandy soils drain quickly and hold few nutrients; clay soils hold water and nutrients but can become waterlogged and hard; **loam**, a balanced mixture, suits most crops. The dark, decomposed organic matter in soil, called **humus**, binds particles into crumbs, holds water and nutrients, and feeds soil life.

3.2.1 Soil formation

Soil formation, or **pedogenesis**, begins with the **weathering** of rock. Physical weathering breaks rock into fragments through temperature changes, frost and the wedging action of roots. Chemical weathering alters minerals through reactions with water, oxygen and acids (for example, feldspar turns into clay). Biological weathering is carried out by lichens, roots, microbes and burrowing animals. Organisms then add organic matter, which is decomposed into humus, and water moving down through the soil carries dissolved and fine material from upper to lower layers.

The soil scientist Hans Jenny (1941) identified five **soil-forming factors**:

1. **Parent material**: the rock or sediment from which soil forms. Basalt of the Deccan Traps weathers into clay-rich black soil; granite and gneiss give sandy red soils.
2. **Climate**: rainfall and temperature control the rate of weathering and leaching. Heavy rainfall and heat in the Western Ghats leach out silica and leave iron- and aluminium-rich laterite.
3. **Topography (relief)**: soils are thin on steep slopes, where erosion removes material, and deep in valleys and plains, where it accumulates.
4. **Organisms**: plants, earthworms, termites, fungi and bacteria mix soil, add organic matter and release nutrients. Grassland soils are often richer in humus than forest soils.
5. **Time**: older soils show more clearly developed layers. The alluvium of the Ganga plain ranges from old, well-developed *bhangar* to young, annually renewed *khadar*.

3.2.2 Soil profile

As soil develops, it separates into distinct horizontal layers called **horizons**. A vertical section through all the horizons, from the surface down to bedrock, is the **soil profile** (see the figure below).

- **O horizon**: the surface layer of fresh and partly decomposed leaves and other organic litter. It is thick in forests and may be absent in cultivated fields.
- **A horizon (topsoil)**: dark, crumbly mineral soil mixed with humus. It is the zone of most root growth and biological activity, and the layer lost first to erosion.
- **E horizon (eluviation zone)**: a pale layer from which clay, iron and other materials have been washed down (leached). It is common in high-rainfall forest soils.
- **B horizon (subsoil)**: the zone of accumulation (**illuviation**), where clay, iron and aluminium oxides and sometimes calcium carbonate washed from above are deposited. It is usually denser and contains less organic matter.
- **C horizon**: partly weathered parent material, with rock fragments.
- **R horizon**: unweathered bedrock.

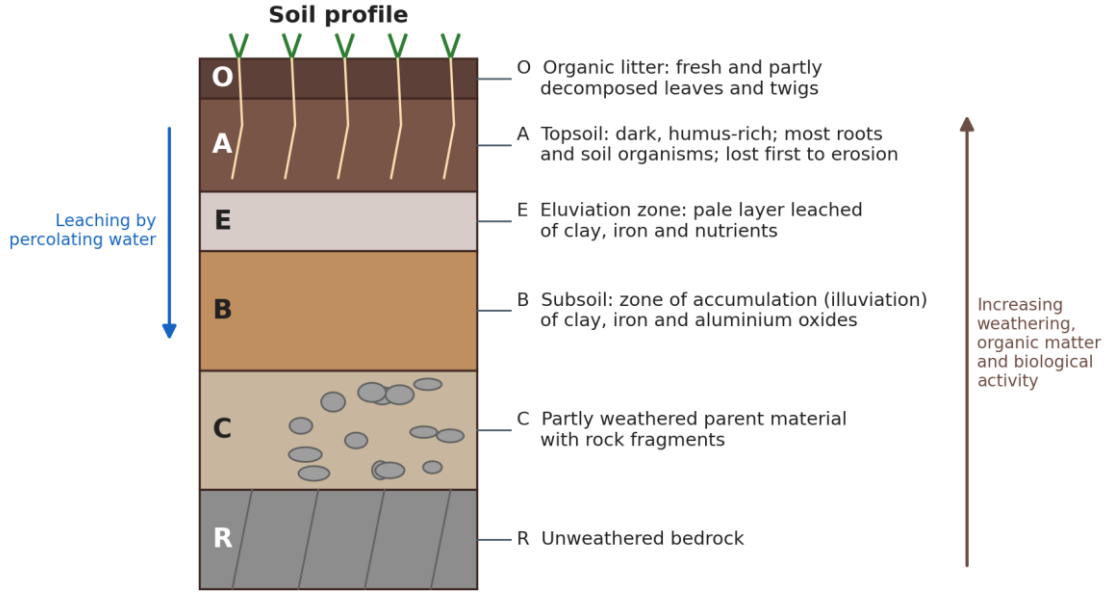


Figure 3.2: *A generalised soil profile showing the main horizons and the processes that operate in each. Not all horizons occur in every soil. Source: Author's diagram, after standard soil-science texts.*

3.2.3 Major soil types of India

India's varied geology, climate and relief produce many soils. The Indian Council of Agricultural Research (ICAR) recognises eight major groups, summarised in the table below. Alluvial soils are the most extensive, covering roughly two-fifths of the country's area.

Table 3.2: Major soil groups of India

Soil group	Where found	Key properties	Typical crops
Alluvial	Indo-Gangetic–Brahmaputra plains, river deltas, coastal plains	Deposited by rivers; fertile; rich in potash, poor in nitrogen and phosphorus	Rice, wheat, sugarcane, pulses, jute
Black (regur)	Deccan Plateau: Maharashtra, Gujarat, Madhya Pradesh, northern Karnataka, Telangana	Formed from basalt; clay-rich; swells when wet and cracks when dry; holds moisture	Cotton, soybean, sorghum, pulses
Red and yellow	Eastern and southern Peninsula: Tamil Nadu, Karnataka, Odisha, Chhattisgarh, Jharkhand	From granite and gneiss; red from iron oxide; porous; low in nitrogen and humus	Millets, groundnut, pulses, rice with irrigation
Laterite	Western Ghats, Kerala, Odisha hills, Meghalaya	Heavily leached in hot, wet climate; rich in iron and aluminium; acidic, infertile	Cashew, tea, coffee, rubber
Arid (desert)	Western Rajasthan, Kutch	Sandy, saline, low in organic matter; high calcium carbonate	Bajra, guar, pulses where irrigated
Saline and alkaline	Parts of Punjab, Haryana, Uttar Pradesh, Gujarat coast, deltas	Excess salts or sodium, often from poor drainage under canal irrigation	Salt-tolerant crops after reclamation
Peaty and marshy	Kerala (Kuttanad), Sundarbans, coastal Odisha	Waterlogged, high in organic matter, acidic	Rice
Forest and mountain	Himalaya, Western and Eastern Ghats	Thin on slopes; humus-rich under forest; acidic at higher altitude	Tea, fruit orchards, spices

3.2.4 Soil health and the Soil Health Card

Soil health is the continued capacity of a soil to function as a living ecosystem that sustains plants, animals and humans. A healthy soil has good structure, adequate organic matter, balanced nutrients, a suitable pH and an active community of organisms. **Soil fertility** is narrower: it refers to the soil's ability to supply nutrients to crops.

Indian soils face several health problems. Many have very low **soil organic carbon** (often below 0.5% in the plains, compared with 1–2% in good agricultural soils) because crop residues are burnt or removed and little manure is returned. Deficiencies of nitrogen, phosphorus, sulfur, zinc and boron are widespread. Decades of unbalanced fertiliser use, with too much nitrogen (urea) relative to phosphorus and potassium, have reduced the yield response to fertiliser. Other problems include salinity and waterlogging in canal-irrigated areas, acidity in high-rainfall regions, compaction by heavy machinery, and contamination by heavy metals near industries and mines.

The **Soil Health Card (SHC) scheme**, launched on 19 February 2015 at Suratgarh (Rajasthan), aims to give every farm holding a card showing the status of its soil and recommending crop-wise doses of fertilisers and soil amendments. Soil samples are tested for twelve parameters: the macronutrients nitrogen (N), phosphorus (P) and potassium (K); the secondary nutrient sulfur (S); the micronutrients zinc, iron, copper, manganese and boron; and pH, electrical conductivity (a measure of salinity) and organic carbon. By July 2025, over 25 crore cards had been distributed and more than 8,000 soil-testing laboratories set up (Ministry of Agriculture and Farmers Welfare). The scheme is now implemented as a component of the Rashtriya Krishi Vikas Yojana.

Evaluations show that farmers who follow card recommendations often cut urea use and raise yields, but many farmers find the cards hard to interpret, receive them late, or do not change their practices. Other measures that support soil health include neem-coated urea (made universal in 2015 to reduce nitrogen losses), the PM-PRANAM programme (2023), which rewards states for reducing chemical fertiliser use, promotion of compost, manures and biofertilisers, and **conservation agriculture**, which minimises tillage, keeps the soil covered and rotates crops.

Key Points

- Soil forms very slowly through weathering and the action of organisms, controlled by parent material, climate, relief, organisms and time.
- A soil profile has O, A, E, B, C and R horizons; topsoil (A horizon) is the most fertile and the most easily lost.
- Alluvial, black, red and yellow, and laterite soils are India's most important groups.
- Low organic carbon, nutrient imbalance, salinity and erosion are the main threats to Indian soil health; the Soil Health Card scheme provides farm-level testing and fertiliser advice.

3.3 Agricultural Crops

Agriculture is the largest use of land in India: about 141 million hectares, or 46% of the reporting area, is sown at least once a year (Land Use Statistics 2022–23). Because many fields carry two or three crops a year, the gross cropped area is about 220 million hectares, a **cropping intensity** of about 156%. Agriculture and allied activities employ about 46% of India's workforce (Periodic Labour Force Survey 2023–24) while contributing under a fifth of gross value added. Crops are therefore both a land resource and the main way most Indians earn a living from land.

3.3.1 Cropping seasons and major crops

India's crops follow the monsoon. **Kharif** crops are sown with the onset of the south-west monsoon (June–July) and harvested in September–October: rice, maize, sorghum (jowar), pearl millet (bajra), cotton, soybean, groundnut, pigeon pea (arhar) and jute. **Rabi** crops are sown in October–December after the monsoon and harvested in March–April, using residual soil moisture, winter rain from western disturbances and irrigation: wheat, barley, chickpea (gram), mustard and peas. **Zaid** crops are short summer crops grown between the rabi and kharif seasons, mainly with irrigation: watermelon, cucumber, vegetables and fodder.

India is the world's largest producer of milk, pulses and millets, and the second-largest producer of rice, wheat, sugarcane, fruits and vegetables. It is also the world's largest exporter of rice. Total

foodgrain production reached a record of about 358 million tonnes in 2024–25 (Ministry of Agriculture and Farmers Welfare, final estimates), roughly seven times the 51 million tonnes of 1950–51. Yet about half of the net sown area is still **rainfed**, depending only on rainfall, and the average farm holding is just over one hectare, so most farmers are vulnerable to droughts, floods and price swings.

3.3.2 The Green Revolution: gains and costs

In the mid-1960s, after two successive droughts and dependence on food imports from the USA, India adopted a strategy now called the **Green Revolution**. It combined **high-yielding varieties (HYVs)** of wheat (semi-dwarf varieties bred by Norman Borlaug in Mexico and adapted in India under M. S. Swaminathan) and rice (such as IR8 from the International Rice Research Institute) with assured irrigation, chemical fertilisers, pesticides, farm credit, a **minimum support price (MSP)** and government procurement through the Food Corporation of India (set up in 1965). The package was first concentrated in Punjab, Haryana and western Uttar Pradesh, which had canal and tubewell irrigation.

The gains were large. Wheat production rose from about 12 million tonnes in 1964–65 to over 26 million tonnes by 1971–72, India stopped depending on food aid, and famine-scale food shortages ended. Higher yields also meant that far less forest had to be cleared for cropland than would otherwise have been needed.

The costs became clear over the following decades, most sharply in Punjab:

- **Groundwater depletion**: the rice–wheat rotation needs large amounts of water, and free electricity for farm pumps encouraged over-pumping. Punjab withdraws about 164% of its annually recharged groundwater, and 117 of its 153 assessment blocks were classed as over-exploited in 2023 (Central Ground Water Board). Water tables in central Punjab are falling by roughly half a metre to a metre a year. The Punjab Preservation of Subsoil Water Act, 2009 delays paddy transplanting to mid-June to reduce evaporation losses.
- **Fertiliser overuse and imbalance**: Punjab applies over 220 kg of nutrients per hectare, among the highest rates in India, with nitrogen far in excess of the recommended N:P:K ratio of 4:2:1. Surplus nitrate leaches into groundwater, and soils have lost organic carbon and micronutrients such as zinc.
- **Pesticides and health**: heavy pesticide use has caused pest resistance, killed beneficial insects and contaminated water and food. Studies in the Malwa region of Punjab have reported pesticide residues in water and blood samples and above-average cancer rates, although the causes are still debated.
- **Stubble burning**: combine harvesters leave tall paddy stubble, and with only two to three weeks between the rice harvest and wheat sowing, many farmers burn it. Punjab produces about 18–20 million tonnes of paddy straw each year. Farm fires fell from 36,663 in 2023 to 10,909 in 2024 (Punjab Remote Sensing Centre), helped by subsidised machines such as the Happy Seeder and by the use of straw in biomass power plants and as pellets co-fired in coal power stations, but smoke from burning still adds to the winter smog of Delhi and the north-western plains (see Chapter 4).
- **Loss of crop diversity**: pulses, oilseeds, millets and thousands of local rice and wheat varieties were displaced by a few HYVs, making the system more vulnerable to pests and climate shocks.

- **Inequality**: the benefits went mainly to irrigated regions and to larger farmers, widening gaps with rainfed eastern and central India.

Case Study: Punjab: from granary to groundwater crisis

Punjab has about 1.5% of India's area but has long supplied a large share of the central pool of wheat and rice. Rice was not a traditional crop of semi-arid Punjab; its area grew from under 0.3 million hectares in 1960–61 to over 3 million hectares today, encouraged by assured procurement at MSP and free farm power. Producing one kilogram of rice in Punjab can consume several thousand litres of water, much of it pumped from aquifers that are not being recharged. As water tables fall, farmers must deepen tubewells and install costlier submersible pumps, raising costs and debt. Punjab's responses include delaying paddy transplanting, promoting direct seeding of rice (which saves water and labour), paying farmers to shift from paddy to maize, cotton or pulses, and encouraging in-field management of stubble. Progress has been slow because no other crop yet offers the same assured price and market as rice and wheat. The Punjab case shows that a strategy which solves one resource problem (food shortage) can create another (water and soil depletion) when environmental limits are ignored.

3.3.3 Towards sustainable cropping: diversification, millets and natural farming

Crop diversification means shifting from a single dominant crop or rotation to a wider mix of cereals, pulses, oilseeds, fruits, vegetables and fodder, matched to local soils and water. It reduces water and fertiliser demand, breaks pest cycles, improves nutrition and spreads income risk. Haryana's Mera Pani Meri Virasat scheme (2020) and Punjab's crop-diversification incentives pay farmers who replace paddy with less thirsty crops.

Millets, such as sorghum (jowar), pearl millet (bajra), finger millet (ragi) and several small millets (kodo, kutki, foxtail, barnyard), are hardy cereals that grow on poor soils with little water and are rich in fibre, iron, calcium and protein. Their area fell sharply after the Green Revolution because policy favoured rice and wheat. At India's proposal, the United Nations declared 2023 the **International Year of Millets**. India has rebranded millets as **Shree Anna** and several states have added them to the public distribution system and school meals; Odisha's Millet Mission (2017) is a well-known state programme.

Natural farming avoids synthetic fertilisers and pesticides and relies on on-farm inputs: preparations of cow dung and urine (such as *jeevamrit*), mulching, mixed cropping and minimal tillage. It was popularised in India by Subhash Palekar and is practised on a large scale through Andhra Pradesh Community-managed Natural Farming (started 2016). In November 2024 the Union Cabinet approved a **National Mission on Natural Farming**, with an outlay of ₹2,481 crore, aiming to reach 1 crore farmers and 7.5 lakh hectares. **Organic farming**, which may use certified organic inputs, is supported through the Paramparagat Krishi Vikas Yojana (2015); Sikkim was declared India's first fully organic state in 2016. Scientists still debate how far natural and organic methods can maintain yields at national scale, so most experts recommend **integrated nutrient management**, which combines organic manures with soil-test-based fertiliser use, along with water-saving micro-irrigation (drip and sprinkler) under the "Per Drop More Crop" component of the Pradhan Mantri Krishi Sinchayee Yojana.

3.4 Natural Forest Products

Forests cover about a fifth of India's land and supply a wide range of goods. These are usually divided into **timber**, the wood of tree trunks used for construction, furniture and industry, and **non-timber forest products (NTFPs)**, which include all other biological products taken from forests: leaves, fruits, seeds, flowers, gums, resins, fibres, honey, medicinal plants, bamboo and cane. In Indian law and administration NTFPs are usually called **minor forest produce (MFP)**. The word "minor" is misleading. For tens of millions of forest dwellers, MFP provides food, medicine, fuel, building material and cash income, and its value to them far exceeds that of timber.

3.4.1 Timber and fuelwood

Under colonial rule, and for some decades after Independence, India's forests were managed mainly to produce timber and revenue: teak, sal, deodar and sissoo for railway sleepers, ships, buildings and furniture, and softwoods for pulp. Large areas of mixed natural forest were clear-felled and replaced by plantations of a single species. The **National Forest Policy, 1988** changed this emphasis. It made environmental stability, conservation and the needs of local people the main goals, with revenue secondary. In 1996, in the *T. N. Godavarman Thirumulpad* case, the Supreme Court halted felling in forests without approved working plans and widened the legal meaning of "forest" to include any area that matches the dictionary meaning of the word, whatever its ownership.

As a result, timber production from government forests has fallen sharply. Most of India's domestic timber now comes from **trees outside forests**: trees on farms, along roads and canals, and in village woodlots, such as poplar, eucalyptus, *Melia* and rubberwood. India also imports large quantities of logs and sawn timber. Wood is still a major household fuel: millions of rural households cook with fuelwood collected from forests and commons, often by women who walk long distances. The spread of LPG connections under the Pradhan Mantri Ujjwala Yojana (2016) has reduced, but not ended, this dependence.

3.4.2 Non-timber forest products

India has hundreds of commercially traded NTFPs. Some of the most important are:

- **Tendu (kendu) leaves** from *Diospyros melanoxylon*, used to roll *bidis*. They are collected in April–May mainly in Madhya Pradesh, Chhattisgarh, Odisha, Jharkhand and Maharashtra, and provide seasonal cash income to several million collectors. In most of these states the trade is nationalised: the state buys leaves at a fixed rate per standard bag and auctions them to traders.
- **Mahua** (*Madhuca longifolia*), whose sweet, fleshy flowers are eaten, dried, and distilled into a traditional liquor, and whose seeds yield an edible and industrial oil. Mahua is central to the food, culture and economy of Gond, Santhal, Bhil and other Adivasi communities in central India.
- **Bamboo**, often called "poor man's timber" or "green gold". India has the second-largest bamboo resource in the world after China, with a bamboo-bearing area of about 1,54,670 km^2 (ISFR 2023). Bamboo is used for housing, baskets, mats, furniture, paper pulp, incense sticks (*agarbatti*) and edible shoots. The **Indian Forest (Amendment) Act, 2017** removed bamboo grown outside forests from the legal definition of "tree", so farmers can now grow, cut and transport it without felling permits. The Forest Rights Act, 2006 classifies bamboo as minor

forest produce owned by forest-dwelling communities, and the restructured National Bamboo Mission (2018) supports planting and processing.

- **Lac**, a resin secreted by the tiny lac insect (*Kerria lacca*) that feeds on host trees such as palas, kusum and ber. It is used in varnishes, bangles, sealing wax, food coatings and cosmetics. India is the world's largest producer of lac, and Jharkhand is the leading state; the ICAR–National Institute of Secondary Agriculture at Ranchi conducts research on lac.
- **Sal seeds and leaves** (*Shorea robusta*): the seeds yield a fat used in chocolate and soap; the leaves are stitched into plates and bowls.
- **Gums and resins**: karaya gum from *Sterculia urens* (used in food and pharmaceuticals), gum from *Anogeissus* (dhawra), and resin tapped from chir pine (*Pinus roxburghii*) in Uttarakhand and Himachal Pradesh for turpentine and rosin.
- **Fruits, seeds and other products**: amla, harra and bahera (the three fruits of the Ayurvedic preparation *triphala*), tamarind, chironji, wild honey, tasar silk cocoons reared on arjun and asan trees, and many medicinal plants (discussed in the next section).

When harvested carefully, NTFP collection can continue for generations without destroying the forest, which is why it is often described as a form of **conservation through use**. When demand rises or rights are insecure, however, collection can become destructive: trees are lopped or girdled, fruits are picked before seeds mature, fires are set to stimulate tendu leaf flush, and roots and bark are stripped, killing the plant.

3.4.3 Minimum Support Price for minor forest produce and Van Dhan

MFP gatherers, many of them Adivasi women, have historically received very low prices from traders and moneylenders, who capture most of the final value. Two national schemes address this:

- **Minimum Support Price for MFP (MSP for MFP)**: started in 2013–14 by the Ministry of Tribal Affairs, it fixes minimum prices for selected MFP items. When market prices fall below the MSP, state agencies buy the produce through the Tribal Cooperative Marketing Development Federation of India (**TRIFED**) network. The list was expanded in 2020 to 87 items, including mahua flowers and seeds, tamarind, sal seeds, lac, honey, chironji, karanj seeds and many medicinal plants.
- **Pradhan Mantri Van Dhan Yojana**: launched on 14 April 2018 at Bijapur (Chhattisgarh), it organises gatherers into **Van Dhan Vikas Kendras**, typically clusters of 15 self-help groups of about 20 members each, which receive training, equipment and working capital to clean, grade, process, package and market forest produce. Value addition, such as turning mahua into candy or tamarind into paste, lets more of the final price reach collectors. Several thousand Kendras have been sanctioned, though studies report that a sizeable share are not yet fully functional.

According to the Ministry of Tribal Affairs, many of India's 10.4 crore Scheduled Tribe people derive 20–40% of their annual income from MFP. These schemes, now implemented together under the Pradhan Mantri Janjatiya Vikas Mission, have helped, but procurement remains small relative to the size of the MFP economy, and benefits depend heavily on how actively each state implements them. Chhattisgarh, for example, procured the bulk of MFP bought under MSP during the 2020 lockdown, when private trade collapsed.

3.5 Medicinal Plants

Medicinal plants are plants, or parts of plants, used to prevent or treat illness. They have been used in India for at least three thousand years. India has one of the richest medicinal floras in the world: about 7,000 or more plant species are used in codified and folk systems of medicine, many of them found only in India. The World Health Organization estimates that a large majority of people in many developing countries use traditional medicine for some of their primary health care, and in 2022 WHO set up its **Global Centre for Traditional Medicine** at Jamnagar, Gujarat.

3.5.1 Traditional systems of medicine

India officially recognises several traditional systems, grouped under the acronym **AYUSH** and overseen by the Ministry of AYUSH (a separate ministry since 2014):

- **Ayurveda**, the oldest, based on the classical texts of Charaka and Sushruta, uses over 1,000 plant species.
- **Yoga and Naturopathy**, which emphasise lifestyle, diet and physical practice.
- **Unani**, of Greek–Arab origin, brought to India in medieval times.
- **Siddha**, practised mainly in Tamil Nadu.
- **Sowa-Rigpa**, the Himalayan (Tibetan) system practised in Ladakh, Sikkim and Himachal Pradesh.
- **Homoeopathy**, of German origin.

Alongside these codified systems, countless **folk healers** (vaidyas, bone-setters, birth attendants) in tribal and rural communities use local plants based on knowledge passed down orally. Modern medicine also depends on plants. Reserpine, an early drug for high blood pressure, was isolated from Indian sarpagandha (*Rauvolfia serpentina*); the anticancer drug paclitaxel (taxol) comes from yew trees, including the Himalayan yew (*Taxus wallichiana*); and vinblastine and vincristine, used against leukaemia, come from the Madagascar periwinkle (*Catharanthus roseus*, *sadabahar*), now widely grown in India.

3.5.2 Trade and overharvesting

Demand for herbal medicines, cosmetics and supplements has grown rapidly in India and abroad. A national study for the National Medicinal Plants Board (Goraya and Ved, 2017) estimated that about 1,178 plant species are traded in India, of which 242 are in high demand (over 100 tonnes a year), and that the herbal industry and households used about 5 lakh tonnes of raw herbal drugs in 2014–15. Most of the high-demand species are still collected from the wild, mainly by poor rural and tribal collectors, rather than cultivated.

Wild collection becomes **overharvesting** when plants are removed faster than they can regenerate. The risk is greatest where:

- the roots, rhizomes, bulbs or bark are used, so that collection kills the plant (for example, kutki, jatamansi and the Himalayan yew);
- the species grows slowly or only in a small area, as many Himalayan alpine herbs do;
- prices are high and rights are insecure, so collectors race to harvest before others do.

Habitat loss, grazing, forest fires, invasive species and climate change add to the pressure. Several Indian medicinal plants are now listed as threatened on the IUCN Red List, and some are in the appendices of CITES, the convention regulating international trade in endangered species (see Chapter 5). The table below gives examples.

Table 3.3: Selected Indian medicinal plants, their uses and conservation concerns

Plant (scientific name)	Part used	Main use	Conservation concern
Sarpagandha (*Rauvolfia serpentina*)	Root	Hypertension, insomnia (source of reserpine)	Heavily over-collected; CITES Appendix II
Jatamansi (*Nardostachys jatamansi*)	Rhizome	Nervous disorders, perfumery	Critically Endangered (IUCN); alpine Himalaya
Kuth (*Saussurea costus*)	Root	Asthma, skin disease	Critically Endangered (IUCN); CITES Appendix I
Guggul (*Commiphora wightii*)	Gum resin	Arthritis, cholesterol	Critically Endangered (IUCN); bark is cut deeply to tap resin
Himalayan yew (*Taxus wallichiana*)	Bark, leaves	Source of taxol (anticancer)	Endangered (IUCN); bark stripping kills trees
Ashwagandha (*Withania somnifera*)	Root	Tonic, stress relief	Mostly cultivated (e.g., Neemuch–Mandsaur, Madhya Pradesh)
Neem (*Azadirachta indica*)	Leaves, seeds, bark	Antiseptic, natural pesticide	Common; subject of a famous patent dispute

3.5.3 Conservation of medicinal plants

The **National Medicinal Plants Board (NMPB)**, set up in November 2000 and now under the Ministry of AYUSH, coordinates policy on conservation, cultivation and trade. It funds in-situ conservation, nurseries, cultivation by farmers, and the development of quality planting material, and runs schemes to link growers with buyers. Key approaches include:

- **In-situ conservation** in forests, including **Medicinal Plant Conservation Areas (MPCAs)**, a network of over a hundred sites, first set up in the southern states in the 1990s with the Foundation for Revitalisation of Local Health Traditions (FRLHT), Bengaluru, where harvesting is controlled.
- **Ex-situ conservation** in herbal gardens, botanical gardens and gene banks.
- **Cultivation** of high-demand species so that industry does not depend on wild stocks. Ashwagandha, isabgol, senna, aloe and safed musli are already mainly cultivated.
- **Legal protection**: some plants, such as kuth and the pitcher plant, are protected under the Wildlife (Protection) Act, 1972 (as amended in 2022), and the export of many wild-collected medicinal plants is restricted.
- **Protection of traditional knowledge**: the **Biological Diversity Act, 2002** (amended in 2023) requires users of biological resources and associated knowledge for commercial purposes to share benefits with local communities, through the National Biodiversity Authority and local Biodiversity Management Committees, which maintain People's Biodiversity Registers. The

Traditional Knowledge Digital Library (TKDL), created by CSIR and the AYUSH department in 2001, documents traditional formulations in a form that patent examiners can search. It followed India's successful challenges to a US patent on the wound-healing use of turmeric (revoked in 1997) and a European patent on a neem-based fungicide (revoked in 2000).

Who gains from medicinal plants?

A collector in the Himalaya may sell a kilogram of dried kutki rhizome for a small fraction of what the same quantity fetches in a city market or as a packaged product. Strengthening collectors' cooperatives, setting sustainable harvest quotas, and paying fair prices for cultivated material can improve both incomes and conservation.

3.6 Forest-Based Industries and Livelihoods

Forests support two linked economies. The first is a formal economy of **forest-based industries**, factories and enterprises that process wood, bamboo and other forest products. The second, much larger in the number of people involved, is the informal **subsistence economy** of forest-dependent households, who collect fuelwood, fodder, food, medicine and MFP for their own use and for sale.

3.6.1 Forest-based industries

The main forest-based industries in India are:

- **Pulp and paper**: Indian paper mills use three main raw materials: wood, agro-residues such as sugarcane bagasse and wheat straw, and recovered (waste) paper, which now provides the largest share of fibre. Most pulpwood (eucalyptus, subabul, casuarina, poplar) now comes from farm and agroforestry plantations, often under contracts between mills and farmers, rather than from natural forests. In earlier decades, paper mills were supplied with bamboo from government forests at very low prices, which led to over-cutting and to conflicts with local communities.
- **Plywood, veneer and wood panels**: Yamunanagar in Haryana is the country's largest plywood cluster, using poplar and eucalyptus grown by farmers in Haryana, Punjab and western Uttar Pradesh. Medium-density fibreboard and particleboard plants use plantation wood and wood waste.
- **Sawmills and furniture**: from large factories to small carpentry workshops. After 1996, the Supreme Court restricted new sawmills near forests to curb illegal felling.
- **Bamboo and cane industries**: furniture, handicrafts, mats, baskets, incense sticks and bamboo flooring, especially in the north-eastern states, Odisha and Karnataka. Higher import duty on bamboo sticks (2019) has encouraged domestic agarbatti stick production.
- **Other industries**: katha (from khair, *Acacia catechu*), tasar silk (Jharkhand is the leading producer), lac processing, bidi rolling, herbal and cosmetic products, natural rubber, sandalwood oil (Karnataka, Kerala) and nature-based tourism.

These industries create jobs in rural areas and add value to forest produce, but they can also drive overexploitation if raw material is taken from natural forests faster than it regrows, as happened with bamboo and sal in the past.

3.6.2 Forest-dependent livelihoods

A World Bank study (2006) estimated that about **275 million** rural people in India depend on forests for at least part of their livelihood. The Forest Survey of India estimates that about 1.7 lakh villages lie in or near forests. For these households, forests provide:

- **fuelwood** for cooking and heating;
- **fodder** and grazing for cattle, goats and sheep;
- **food**: tubers, mushrooms, fruits, leafy greens and honey, which are especially important in the lean months before the harvest;
- **MFP** for sale, which provides cash income;
- **small timber and bamboo** for houses, tools and fencing;
- **medicinal plants** for health care;
- **wage employment** in forestry operations and plantations.

Forest dependence is highest among Adivasi (Scheduled Tribe) communities and among the poorest, landless households. Women do most of the collection work and are most affected when forests are degraded or access is denied. Forest-based livelihoods are often insecure: incomes are seasonal, collectors receive low prices, and until recently most forest dwellers had no legal rights to the land and forests they used.

3.6.3 Community management: Joint Forest Management and Community Forest Resource rights

Two policies have tried to make forest-dependent people partners in managing forests rather than illegal users.

Joint Forest Management (JFM) grew out of an experiment at Arabari in West Bengal in the early 1970s, where forest officials gave villagers a share of produce in return for protecting degraded sal forest, which then regenerated well. Following the National Forest Policy, 1988, a Union government circular of June 1990 asked states to set up JFM committees. Village committees protect and regenerate a patch of state forest and in return receive usufruct rights to grass, fuelwood and NTFPs and a share of final timber revenue. By the mid-2000s, about 85,000 JFM committees were managing roughly a quarter of India's forest area (World Bank, 2006). JFM helped regenerate many degraded forests, but critics point out that committees were often created and controlled by the forest department, could be dissolved at will, and gave communities no legal rights.

The **Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006**, usually called the **Forest Rights Act (FRA)**, went further. It recognises three kinds of rights: individual rights to land cultivated before December 2005; community rights, including ownership of minor forest produce, grazing and fishing; and **Community Forest Resource (CFR)** rights, which give the gram sabha the right and duty to protect, regenerate, conserve and manage its customary forest. By May 2025, about 25 lakh titles had been distributed under the Act, of which about 1.2 lakh were community titles (Ministry of Tribal Affairs). Mendha Lekha in Gadchiroli district, Maharashtra, became one of the first villages to receive CFR rights in 2009 and later earned substantial income by harvesting and selling bamboo itself. The FRA also requires the consent of gram sabhas before forest land is diverted for projects, a provision that proved decisive in the Niyamgiri case discussed later in this chapter. The Act is covered in more detail in Chapter 5.

3.7 Land Cover and Land-Use Change

Land is finite. India's geographical area of about 328.7 million hectares must provide food, water, housing, industry, infrastructure, forests and wildlife habitat for a growing population. How land is divided among these uses, and how that division changes, shapes almost every other environmental issue in this book.

3.7.1 Land cover and land use

Land cover is the physical and biological material on the Earth's surface: forest, grassland, cropland, bare rock, sand, water bodies, snow and built-up surfaces. It is what a satellite sees. **Land use** is the purpose for which people use land: agriculture, grazing, forestry, settlement, industry, mining, transport or conservation. The two are related but not the same. A forest (land cover) may be used for timber production, a tiger reserve or a sacred grove (land uses). A grass-covered area may be a pasture, a golf course or an airport.

Land-use and land-cover change (LULCC) refers to any change in either. It can be a **conversion**, in which one cover type replaces another (forest cleared for farmland, farmland built over by a town), or a **modification**, in which the cover type stays the same but its condition changes (a natural forest thinned by logging, a rainfed field converted to irrigated double-cropping).

India's land records classify the **reporting area** (the area for which land-use records exist, about 306.7 million hectares) into nine categories, shown in the figure below. Net sown area (46%) and forests (23%) dominate. About 5% is barren and unculturable, and around 12% is fallow or culturable waste, land that could be cultivated but currently is not.

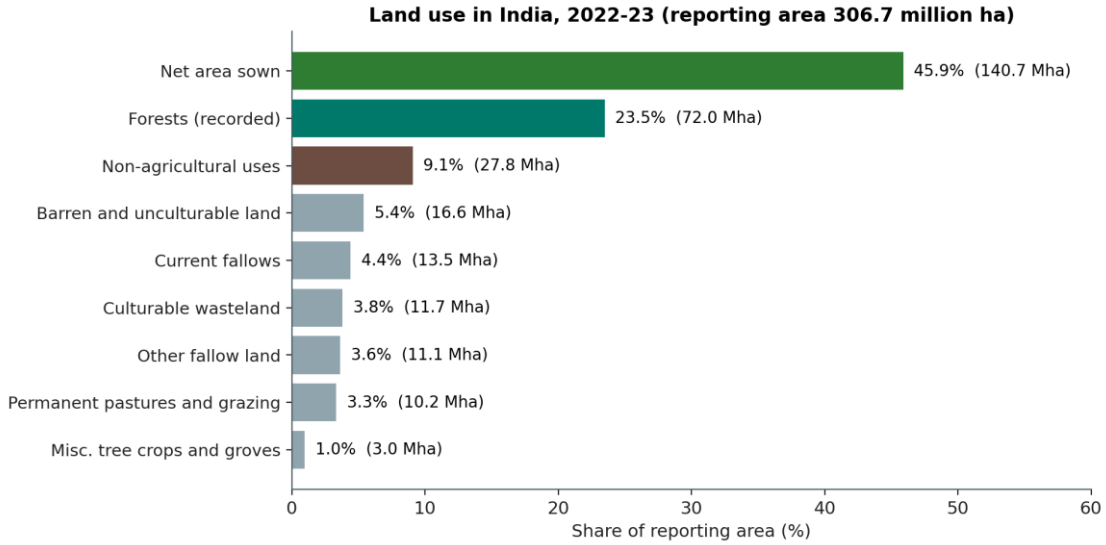


Figure 3.3: *Land-use pattern of India, 2022–23, as a percentage of the reporting area of 306.7 million hectares (nine-fold classification). Source: Land Use Statistics at a Glance 2022–23, Directorate of Economics and Statistics, Ministry of Agriculture and Farmers Welfare.*

Two kinds of "forest"

The 23% "forests" in land records is **recorded forest area**: land legally designated as forest, whether or not it has trees. The 21.76% **forest cover** reported by the Forest Survey of India (ISFR 2023) is land actually covered by tree canopy, measured from satellite images, whatever its legal status. Some recorded forest has no trees, and some tree-covered land, such as orchards and plantations on private land, is not recorded forest. The two figures answer different questions and should not be confused.

3.7.2 Drivers of land-use change in India

Since Independence, India's net sown area has stayed roughly constant at about 140 million hectares, because almost all cultivable land was already under the plough by the 1970s. The big changes have come from intensification and from the growth of non-agricultural uses. Land under non-agricultural uses (settlements, roads, industry, canals) has roughly tripled, from about 9 million hectares in 1950–51 to about 28 million hectares in 2022–23. The main drivers are:

- **Population growth and urbanisation**: cities expand onto surrounding farmland, wetlands and commons. The growth of Gurugram, Noida and Greater Noida has converted large areas of farmland around Delhi, and Bengaluru has lost many of its lakes and tank beds to construction.
- **Infrastructure**: highways, railways, airports, industrial corridors, power plants and transmission lines.
- **Agricultural expansion and intensification**: the spread of irrigation and double cropping, conversion of grasslands (for example, in the Banni of Kutch and the Deccan plateau) and wetlands to farmland, and shifting cultivation (*jhum*) with shortened fallow cycles in the north-east.
- **Mining and dams**, which convert forest and farmland directly and attract roads and settlements (discussed later in this chapter).
- **Plantations**: rubber in Kerala and Tripura, oil palm in the north-east and Andhra Pradesh, tea and coffee in the hills.
- **Policies and markets**: crop prices, subsidies, land laws and tenure security influence what landowners do with their land.
- **Natural factors**: floods, river course changes, landslides, cyclones and droughts also alter land cover, often in combination with human pressure.

3.7.3 Monitoring land with satellites: ISRO and NRSC

India has one of the most advanced land-monitoring systems among developing countries, based on its own Indian Remote Sensing (IRS) satellites, first launched in 1988. Key products include:

- **Land use/land cover maps** prepared by the **National Remote Sensing Centre (NRSC)**, Hyderabad. National maps at 1:250,000 scale are produced every year from AWiFS sensor data, and detailed maps at 1:50,000 scale, using a 54-class system, have been prepared for 2005–06, 2011–12 and 2015–16 from LISS-III data. They are freely viewable on ISRO's **Bhuvan** geoportal.
- **Wastelands Atlas of India** (Department of Land Resources with NRSC). The 2019 edition found 55.76 million hectares of wasteland (16.96% of the geographical area) in 2015–16,

slightly less than the 56.60 million hectares (17.21%) of 2008–09, as about 1.45 million hectares were brought back into productive use.

- **Desertification and Land Degradation Atlas of India** (Space Applications Centre, ISRO, Ahmedabad), discussed in the next section.
- **India State of Forest Report** (Forest Survey of India, Dehradun), published every two years since 1987 using IRS satellite data.

> **Think and Do: Watch your neighbourhood change**
>
> Open Google Earth Pro and use the historical imagery slider to compare your college campus or home area in about 2005 and today. Identify which land covers (fields, trees, water bodies, open ground) have been converted to buildings or roads. Estimate the area that changed and suggest one reason for each change.

3.8 Land Degradation

Land degradation is a reduction or loss of the biological or economic productivity of land, caused mainly by human activities and sometimes worsened by climate. Degraded land produces less food, fodder and wood, stores less water and carbon, and supports fewer species. It includes erosion of soil by water and wind, loss of vegetation, salinisation and waterlogging, loss of soil fertility and organic matter, compaction, and pollution. When land degradation occurs in drylands, it is called **desertification**.

3.8.1 Extent of land degradation in India

The **Desertification and Land Degradation Atlas of India** (Space Applications Centre, ISRO, 2021) mapped the whole country from satellite data. It found that about **97.85 million hectares, or 29.7% of India's geographical area, was undergoing land degradation in 2018–19**, up from 94.53 million hectares in 2003–05 and 96.40 million hectares in 2011–13. In other words, nearly one-third of India's land is degraded, and the area is still growing. The most important processes, as shown in the figure below, are water erosion (11.0% of India's area), loss of vegetation cover (9.2%) and wind erosion (5.5%), followed by salinisation, waterlogging, mass movement, frost shattering and man-made degradation such as mining and industrial dumps. In Jharkhand, Rajasthan, Gujarat, Goa and Delhi, more than half of the land area has been found to be degraded.

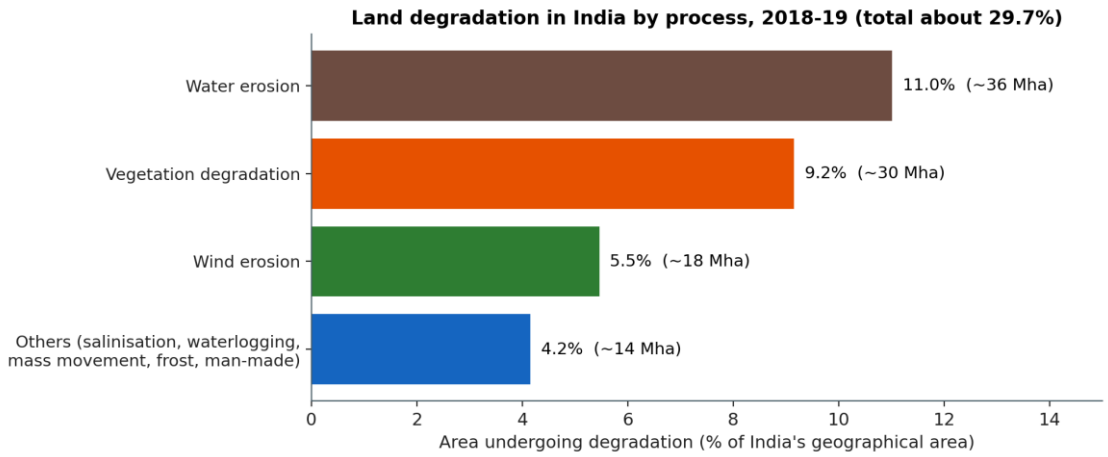


Figure 3.4: *Main processes of land degradation in India, 2018–19, as a percentage of the country's total geographical area. Total degraded land was 97.85 million hectares (29.7%). Source: Desertification and Land Degradation Atlas of India, Space Applications Centre, ISRO (2021).*

3.8.2 Causes of land degradation

The causes act at two levels. **Direct causes** are the physical processes and land-use practices on the ground:

- **Deforestation and loss of vegetation**, which expose soil to rain and wind.
- **Overgrazing** of pastures and forests by large herds of cattle, goats and sheep, which removes grass cover and compacts soil.
- **Unsustainable farming**: cultivation of steep slopes without terraces, ploughing up and down slopes, leaving fields bare between crops, burning crop residues, monocropping and excessive or unbalanced use of fertilisers.
- **Faulty irrigation**: over-irrigation and poor drainage in canal command areas raise the water table, causing **waterlogging**, and as water evaporates it leaves salts at the surface (**salinisation**). Parts of the Indira Gandhi Canal area in Rajasthan, the Sharda Sahayak command in Uttar Pradesh and the Tungabhadra command in Karnataka are affected.
- **Mining, quarrying and industry**, which strip land and leave dumps, pits and contaminated soil.
- **Shifting cultivation with short fallows**, forest fires and invasive species such as *Lantana camara* and *Prosopis juliflora*.

Underlying causes explain why these practices occur: rapid population growth and poverty, which force people to over-use land; insecure land tenure, which discourages long-term investment in soil; subsidies that favour water- and fertiliser-intensive crops; weak management of common lands; and climate change, which brings more intense rainfall and longer droughts.

3.8.3 Consequences

Land degradation reduces crop yields and farm incomes, increases dependence on fertilisers and irrigation, and pushes farmers onto more marginal land. Sediment eroded from catchments silts up reservoirs, canals and rivers, reducing their storage capacity and worsening floods. Loss of vegetation reduces groundwater recharge and increases runoff. Degraded land stores less carbon and supports less biodiversity. The poorest households, which depend most on common lands, forests and rainfed

farming, suffer first, and degradation can drive distress migration. A study by The Energy and Resources Institute (TERI) for the MoEFCC estimated that land degradation and land-use change cost India about 2.5% of its GDP in 2014–15.

3.8.4 Land Degradation Neutrality

Land Degradation Neutrality (LDN) is a state in which the amount and quality of land resources stays stable or increases. It is Target 15.3 of the Sustainable Development Goals: by 2030, combat desertification, restore degraded land and soil, and strive to achieve a land-degradation-neutral world. LDN does not mean no land will be degraded anywhere; it means that new degradation is avoided where possible, reduced where it occurs, and balanced by restoring an equal or greater area elsewhere.

At the fourteenth Conference of the Parties to the UN Convention to Combat Desertification (UNCCD COP14), hosted by India in New Delhi in September 2019, India raised its target to **restore 26 million hectares of degraded land by 2030**. Programmes that contribute to this include the watershed development component of the Pradhan Mantri Krishi Sinchayee Yojana, soil and water conservation works under the Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS), the Green India Mission, the Nagar Van Yojana for urban forests, compensatory afforestation through CAMPA, and the **Aravalli Green Wall Project** (2023), which aims to restore a 5-km-wide green belt around the Aravalli hills across Gujarat, Rajasthan, Haryana and Delhi.

3.9 Soil Erosion and Desertification

3.9.1 Soil erosion: agents and types

Soil erosion is the detachment and removal of soil particles from one place and their deposition elsewhere by water, wind, ice or gravity. **Geological (natural) erosion** occurs slowly under natural vegetation and is roughly balanced by soil formation. **Accelerated erosion** occurs when human activities remove the protective cover of vegetation and litter, so that soil is lost many times faster than it forms. A classic national estimate (Narayana and Ram Babu, 1983) found that India loses about **5.3 billion tonnes of soil a year**, an average of about 16 tonnes per hectare; roughly 29% of this is carried to the sea and about 10% is deposited in reservoirs.

Water erosion is the most widespread type in India and passes through several stages:

- **Splash erosion**: raindrops striking bare soil detach particles and splash them around.
- **Sheet erosion**: a thin, uniform layer of topsoil is washed off by surface runoff. It is barely visible but removes the most fertile soil.
- **Rill erosion**: runoff concentrates into many small channels a few centimetres deep, which can be smoothed by ploughing.
- **Gully erosion**: rills deepen into large channels that cannot be crossed by farm machinery. Extensive gullying produces **ravines**, as in the Chambal valley of Madhya Pradesh, Rajasthan and Uttar Pradesh, the Yamuna and Mahi ravines, and the badlands of the Shivalik foothills.
- **Stream-bank and coastal erosion**: rivers and waves cut into their banks and shores, as on the Brahmaputra at Majuli island in Assam.

- **Landslides and mass movement**: on steep Himalayan and Western Ghat slopes, especially where roads are cut and forests cleared.

Wind erosion dominates in arid and semi-arid areas such as western Rajasthan and Kutch, where dry, loose, sandy soil has little vegetation. Wind moves particles by **suspension** (fine dust carried high into the air, forming dust storms), **saltation** (sand grains bouncing along the surface) and **surface creep** (larger grains rolled along). Wind erosion forms shifting sand dunes that bury fields, roads and villages.

The rate of erosion depends on rainfall intensity, how easily the soil is detached (its **erodibility**), slope length and steepness, plant cover and conservation practices. These factors are combined in the widely used **Universal Soil Loss Equation**, $A = R \times K \times LS \times C \times P$, where A is annual soil loss, R is rainfall erosivity, K is soil erodibility, LS is the slope factor, C is the cover and management factor, and P is the support-practice factor. Farmers and planners can do little about R, K and slope, but they can greatly reduce C and P through better management.

3.9.2 Desertification

The UNCCD defines **desertification** as "land degradation in arid, semi-arid and dry sub-humid areas resulting from various factors, including climatic variations and human activities". It is not the natural advance of existing deserts such as the Thar, but the gradual loss of productivity of dryland soils and vegetation until they take on desert-like conditions. Drylands make up about 69% of India's area (MoEFCC). The ISRO atlas found that about **83.69 million hectares were undergoing desertification in 2018–19**, most of it in Rajasthan, Maharashtra, Gujarat, Karnataka, Ladakh, Jharkhand, Odisha, Madhya Pradesh and Telangana.

Desertification results from the interaction of human pressure and a harsh climate. Overgrazing, over-cultivation of marginal land, cutting of shrubs and trees for fuel, faulty irrigation and over-extraction of groundwater weaken the land; recurrent droughts and rising temperatures then push it past the point of recovery. Its effects include falling yields, shortage of fodder and fuel, dust storms, drying wells, poverty and migration.

The **United Nations Convention to Combat Desertification (UNCCD)** was adopted in Paris in June 1994 and came into force in 1996. It is the only legally binding international agreement linking environment and development to sustainable land management. India signed it in 1994, ratified it in 1996, prepared a National Action Programme to Combat Desertification in 2001, and hosted COP14 in 2019. The MoEFCC is the nodal ministry, and a Centre of Excellence on sustainable land management has been set up at the Indian Council of Forestry Research and Education (ICFRE), Dehradun. 17 June is observed worldwide as Desertification and Drought Day.

3.9.3 Control measures

Soil erosion and desertification are controlled by keeping the soil covered, slowing the flow of water and wind, and allowing water to soak in. Measures fall into three groups, illustrated in the figure below.

- **Agronomic (crop-management) measures**: contour ploughing and sowing across the slope; strip cropping, in which bands of erosion-resistant crops alternate with row crops; cover crops and mixed cropping to keep the ground covered; mulching with crop residues; conservation tillage; and crop rotation with legumes to build organic matter.

- **Mechanical (engineering) measures**: contour bunds and graded bunds on gentle slopes; bench terraces on steep slopes, as in the hills of Uttarakhand and the north-east; contour trenches to catch runoff; gully plugs and check dams of stone, brushwood or masonry to stop gullies growing; and farm ponds to store runoff.
- **Biological and vegetative measures**: afforestation of degraded slopes and catchments; grass strips and vetiver hedges along contours; agroforestry; regulated or rotational grazing; and, in deserts, **shelterbelts** (rows of trees planted at right angles to the prevailing wind) and stabilisation of sand dunes with grasses and shrubs. The ICAR–Central Arid Zone Research Institute (CAZRI) at Jodhpur developed dune-stabilisation and shelterbelt techniques widely used in western Rajasthan, and the ICAR–Indian Institute of Soil and Water Conservation at Dehradun leads research on water erosion.

These measures work best when combined on a **watershed** basis, treating an entire catchment from ridge to valley, with the participation of the people who live in it.

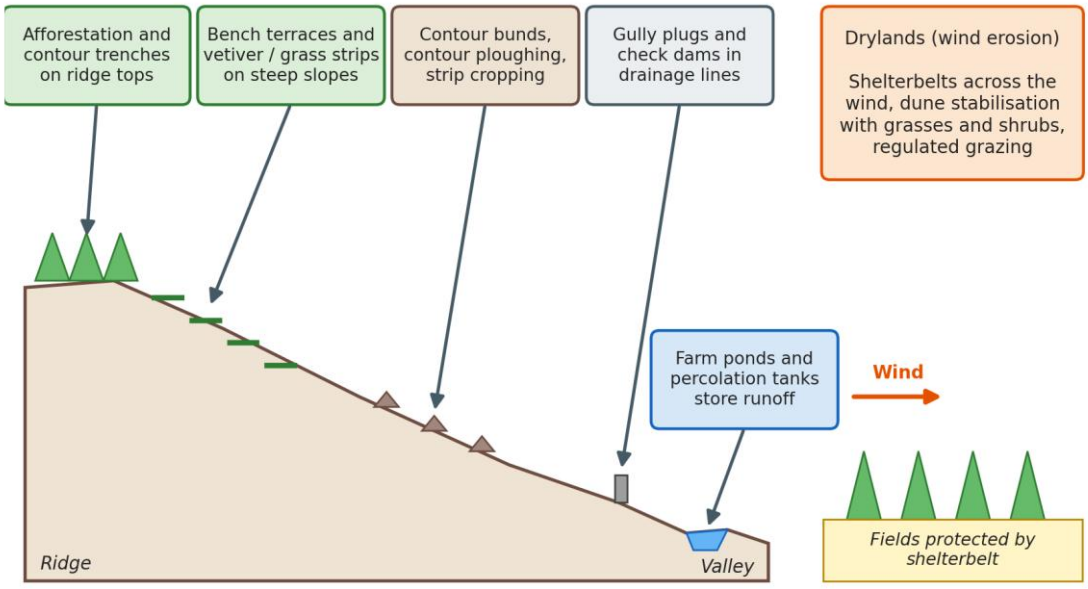


Figure 3.5: *Soil and water conservation measures arranged along a slope, from ridge to valley, and against wind in drylands. Source: Author's diagram.*

Case Study: Sukhomajri and watershed management

In the 1970s, Sukhna Lake in Chandigarh was silting up rapidly because of erosion in the bare, overgrazed Shivalik hills above it. Scientists from the Central Soil and Water Conservation Research and Training Institute worked with the nearby village of Sukhomajri (Haryana) to build small earthen dams in the gullies to catch runoff. The stored water allowed villagers to irrigate their fields and grow more crops and fodder. In return, the village agreed to stop free grazing on the hill slopes and to cut grass instead. A Hill Resource Management Society was formed to share the water fairly, including with landless families, who could sell their share. Within a decade, vegetation on the slopes recovered, sediment flow into Sukhna Lake fell sharply, crop yields and milk production rose, and household incomes increased. Sukhomajri became a model for participatory watershed development across India, along with later examples such as Ralegan Siddhi and Hiware Bazar in Maharashtra. It shows that erosion control succeeds when the people who bear its costs also share its benefits.

3.10 Causes of Deforestation

Deforestation is the permanent clearing of forest and the conversion of the land to another use, such as farming, mining, settlements or reservoirs. It differs from **forest degradation**, in which trees remain but the forest's density, structure, species composition or productivity declines, for example through repeated lopping, overgrazing, fire or selective logging. In India, degradation of existing forests is at least as serious a problem as outright clearance. Deforestation and degradation reduce biodiversity, increase soil erosion and floods, reduce groundwater recharge, release stored carbon, and harm the livelihoods of forest-dependent people.

3.10.1 Forest cover in India

The Forest Survey of India (FSI) measures forest cover every two years from satellite images. It classifies as **forest cover** all land of more than one hectare with tree canopy density above 10%, irrespective of ownership or legal status, and divides it into very dense (canopy density 70% or more), moderately dense (40–70%) and open forest (10–40%). Patches of trees smaller than one hectare outside recorded forests are counted as **tree cover**. The latest figures are given in the table below.

Table 3.4: Forest and tree cover of India (ISFR 2023)

Category	Area (km^2)	Share of geographical area
Forest cover	7,15,343	21.76%
Tree cover	1,12,014	3.41%
Total forest and tree cover	8,27,357	25.17%
Mangrove cover (part of forest cover)	4,992	0.15%

Forest and tree cover together crossed 25% of India's area for the first time in 2023, and increased by about 1,446 km^2 compared with 2021, mostly through tree cover outside forests. Total carbon stock in forests was estimated at about 7,285 million tonnes. These national totals hide important losses. Madhya Pradesh, Karnataka, Ladakh and Nagaland recorded the largest decreases in forest

and tree cover between 2021 and 2023, and forest cover in several north-eastern states has declined over the past decade. Critics also point out that FSI's definition counts plantations, orchards, tea gardens and even urban parks as forest cover, so a gain in plantations can hide the loss of old natural forest. The National Forest Policy, 1988 set a goal of bringing one-third of the country under forest or tree cover, and India's updated Nationally Determined Contribution (2022) commits to creating an additional carbon sink of 2.5–3 billion tonnes of CO_2 equivalent through additional forest and tree cover by 2030.

3.10.2 Direct causes of deforestation

Direct (proximate) causes are the activities on the ground that remove or degrade forest. In India the most important are:

- **Agricultural expansion and encroachment**: from the 1950s to the 1980s, millions of hectares of forest were cleared for farming and resettlement schemes, for example in the *terai* of Uttar Pradesh and Uttarakhand and in central India. Encroachment continues in many places today.
- **Shifting cultivation (jhum)**: in the north-eastern hill states, forest patches are cut and burnt, cultivated for a year or two and then left fallow to regrow. With a fallow cycle of 15–20 years, this is sustainable, but rising population has shortened cycles to as little as 3–5 years in places, so forests do not recover.
- **Diversion of forest land for development projects**: between 2014–15 and 2023–24, about 1.73 lakh hectares of forest land (1,734 km^2, larger than the area of Delhi) were approved for non-forest use under the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980 (MoEFCC, Lok Sabha reply, 2024). As the figure below shows, mining, hydropower and irrigation projects, and roads accounted for the largest shares. Major recent examples include coal mining in central India, the Great Nicobar Island development project, which was granted clearance in 2022 to divert about 130 km^2 of tropical rainforest, and the felling of trees in Mumbai's Aarey forest for a metro car shed.
- **Commercial plantations**: replacement of natural forest by monoculture plantations of teak, eucalyptus, rubber or oil palm, which support far less biodiversity.
- **Overgrazing, fuelwood and fodder collection**: these usually degrade forests rather than clear them, preventing natural regeneration and gradually turning dense forest into open scrub.
- **Forest fires**: most forest fires in India are set by people, to clear undergrowth, encourage fresh grass or tendu leaves, or by carelessness. Dry deciduous forests of central India and the chir pine forests of Uttarakhand are most affected.
- **Illegal logging and smuggling** of valuable timber such as teak, sandalwood and red sanders (in the Seshachalam hills of Andhra Pradesh).
- **Invasive species and pests**: *Lantana camara* now infests a large share of India's forests and prevents native seedlings from establishing.

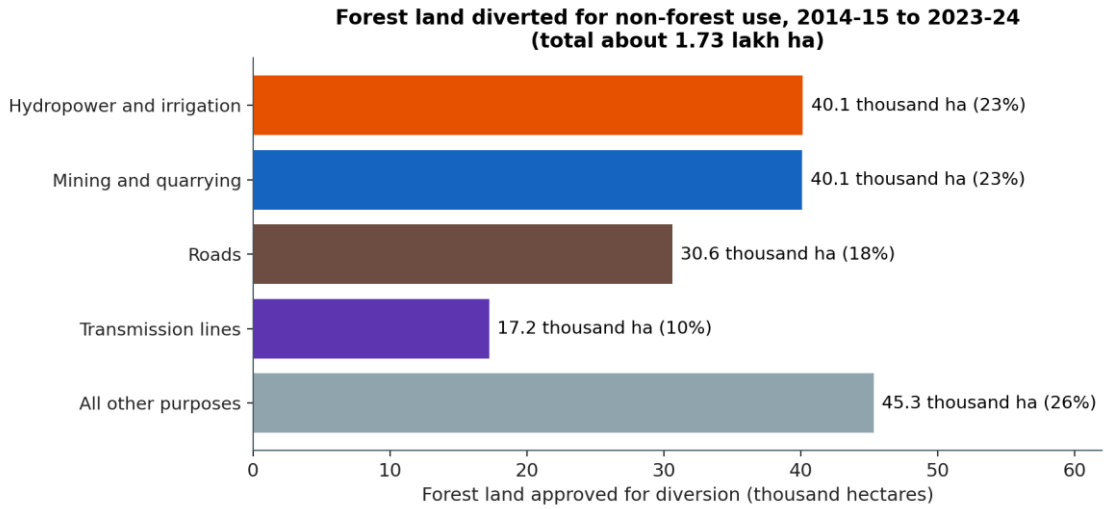


Figure 3.6: *Forest land approved for diversion to non-forest uses in India, 2014–15 to 2023–24 (total about 1.73 lakh hectares), by purpose. Source: MoEFCC reply in the Lok Sabha, 2024.*

3.10.3 Underlying causes of deforestation

Underlying causes are the social, economic and political forces that drive the direct causes. They are less visible but more important for designing solutions. The figure below shows how the two levels of causes combine to produce deforestation and its consequences.

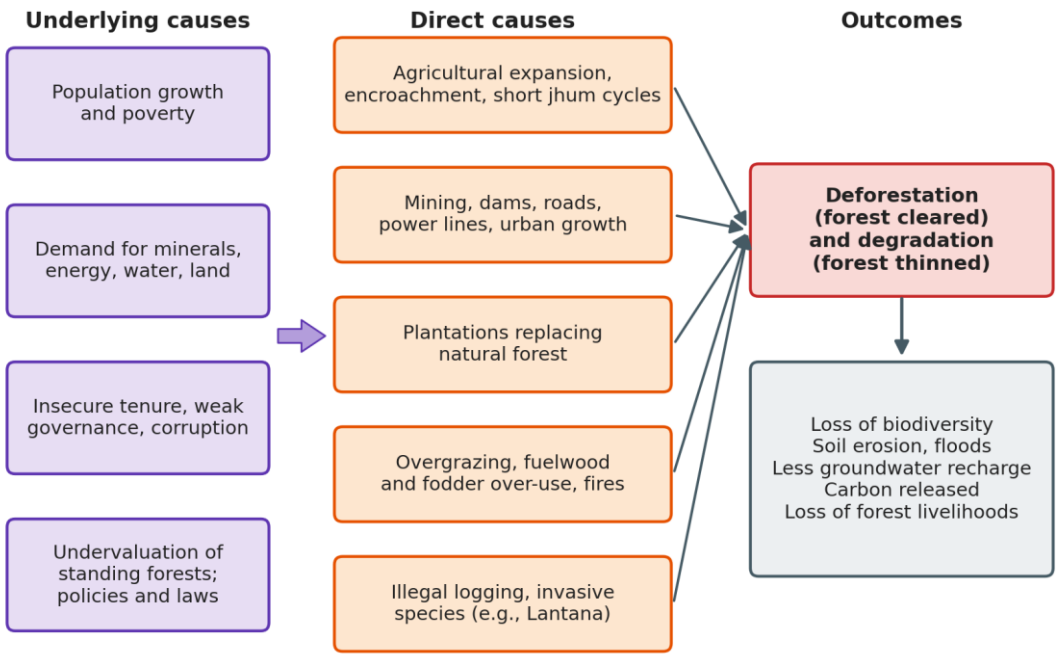


Figure 3.7: *How underlying causes drive the direct causes of deforestation and forest degradation, and their main consequences. Source: Author's diagram, after Geist and Lambin (2002).*

- **Population growth and poverty**: more people need more land, food, fuel and fodder, and poor households have few alternatives to using nearby forests.
- **Economic growth and demand for resources**: rising demand for coal, iron ore, bauxite, electricity, water, roads and real estate puts pressure on forest land, much of which lies over mineral deposits and on river headwaters.

- **Historical policies**: colonial forest laws (especially the Indian Forest Act, 1927) took forests away from community control and managed them for timber and revenue, weakening local institutions that had conserved them. Post-Independence policies continued to supply industries with cheap forest raw material.
- **Insecure tenure and weak governance**: when forest dwellers have no secure rights, they have little incentive to protect forests, while outsiders can exploit them. Corruption and weak enforcement allow illegal felling and encroachment.
- **Undervaluation of forests**: markets pay for timber and minerals but not for the water regulation, soil protection, carbon storage and biodiversity that standing forests provide. Development decisions therefore tend to favour clearance.
- **Changes in law**: the Forest (Conservation) Amendment Act, 2023 limited the Act's coverage to notified forests and land recorded as forest in government records, and exempted some categories, such as strategic linear projects within 100 km of international borders, from prior approval. Critics feared this would leave unrecorded "deemed forests" unprotected. In 2024 the Supreme Court directed that the broad dictionary meaning of forest from the *Godavarman* case must continue to apply while it hears challenges to the amendment.

Compensatory afforestation, required whenever forest land is diverted and funded through the Compensatory Afforestation Fund Act, 2016, is meant to offset losses. In practice, plantations of a few species on scattered patches of land rarely replace the biodiversity and ecosystem services of the natural forest that was lost, and suitable land for afforestation is scarce.

> **Key Points**
>
> - India's forest and tree cover is 25.17% of its area (ISFR 2023), but national gains in tree cover can mask losses of natural forest in particular states.
> - Direct causes include agricultural expansion, shortened jhum cycles, diversion of forest land for mining, hydropower, irrigation and roads, plantations, overgrazing, fires and illegal logging.
> - Underlying causes include population pressure, poverty, demand for minerals and energy, insecure tenure, weak governance and the undervaluation of standing forests.

3.11 Impacts of Mining and Dam Building

Mining and large dams are central to India's development: mines supply coal for most of its electricity and ores for steel, aluminium and cement, and dams store monsoon water for irrigation, drinking water, flood control and hydropower. Both, however, are concentrated in forested, hilly and often tribal regions, and both change land and rivers permanently. This section examines their impacts on the environment, forests, biodiversity and tribal communities.

3.11.1 Impacts of mining

On the environment

- **Land**: opencast mining removes vegetation, topsoil and overburden and leaves deep pits and huge waste dumps. Unless restored, mined land remains barren for decades. Underground mining causes **subsidence** (sinking of the ground surface) that cracks houses, roads and fields.

- **Water**: mines lower local water tables when they pump out groundwater, and dry up springs and streams. Exposure of sulfide minerals to air and water produces sulfuric acid, a process called **acid mine drainage**, which dissolves heavy metals and kills aquatic life. Unscientific "rat-hole" coal mining in the Jaintia Hills of Meghalaya acidified rivers such as the Kopili and Lukha, and the National Green Tribunal banned it in 2014. At Sukinda in Odisha, chromite mining has contaminated water with toxic hexavalent chromium. Silt from waste dumps chokes streams and fields.
- **Air**: blasting, crushing, loading and transport generate dust; coal fires and smelters emit sulfur dioxide, carbon monoxide and particulate matter. Mine workers and nearby residents suffer from respiratory diseases. Silicosis, an incurable lung disease caused by inhaling silica dust, is widespread among stone-quarry and sandstone-mine workers in Rajasthan, which adopted a Pneumoconiosis Policy in 2019 to compensate victims.
- **Noise and vibration** from blasting and heavy machinery disturb people and wildlife.

On forests and biodiversity

Mining and quarrying accounted for about 40,000 hectares of the forest land diverted in India between 2014–15 and 2023–24, roughly as much as hydropower and irrigation projects combined. Its impact goes well beyond the mine itself. Roads, railways, conveyor belts, townships and power lines fragment surrounding forests, while in-migration of workers increases hunting, fuelwood collection and encroachment. India's coal and iron ore belts in Jharkhand, Odisha and Chhattisgarh overlap with some of its best remaining sal forests and with the habitats and migration corridors of Asian elephants and tigers. Elephant herds displaced from mined areas increasingly raid crops and enter villages, intensifying human–elephant conflict in these states. In 2002 the Supreme Court ordered the closure (completed in 2005) of the Kudremukh iron ore mine in Karnataka, because it lay within a national park in the Western Ghats and was silting the Bhadra river.

On tribal communities

Most of India's coal, iron ore, bauxite and other major minerals lie under forested hills inhabited by Adivasi communities, many of them in the **Fifth Schedule areas** that the Constitution protects. Mining affects these communities through:

- **Displacement and loss of land**: families lose their fields, forests and homes, often with inadequate compensation. Those without formal land titles, common among Adivasis who cultivate forest land, have often received nothing.
- **Loss of livelihoods**: forests that supplied MFP, fuelwood, fodder and food are cleared, and water sources dry up or are polluted. Few local people get stable jobs in highly mechanised mines.
- **Social and cultural disruption**: sacred hills, groves and burial grounds are destroyed, communities are scattered, and the influx of outsiders brings alcoholism, exploitation and conflict.
- **Health**: dust, polluted water and occupational diseases harm health, while health care is often poor.

Several laws aim to protect tribal interests. The **Panchayats (Extension to Scheduled Areas) Act, 1996 (PESA)** requires gram sabhas in Scheduled Areas to be consulted before land acquisition and gives them powers over minor minerals and MFP. In the *Samatha* judgment (1997), the Supreme Court held that tribal land in Scheduled Areas cannot be leased to private companies for mining. The

Forest Rights Act, 2006 requires the consent of gram sabhas before forest land is diverted. The **Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013** sets standards for compensation and rehabilitation. District Mineral Foundations are meant to return part of mining revenue to affected areas. Implementation of all these remains uneven, and consent processes have frequently been bypassed or manipulated.

Case Study: Niyamgiri: a gram sabha says no

The Niyamgiri hills in Rayagada and Kalahandi districts of Odisha are covered with dense forest, feed several streams, and are home to the Dongria Kondh, a particularly vulnerable tribal group who revere the hill's summit as the seat of their deity, Niyam Raja. The hills also hold large bauxite deposits. In the 2000s Vedanta, a multinational company, built an alumina refinery at Lanjigarh at the foot of the hills and planned to supply it with bauxite mined from the Niyamgiri plateau through a joint venture with the state-owned Odisha Mining Corporation. The Dongria Kondh and neighbouring communities opposed the mine. In 2010 a committee chaired by N. C. Saxena found serious violations of the Forest Rights Act, and the MoEFCC refused final forest clearance. In April 2013, the Supreme Court ruled that the gram sabhas of the affected villages must decide whether mining would affect their religious, cultural and community rights under the Forest Rights Act. All twelve gram sabhas that were consulted rejected the project, and in January 2014 the ministry formally rejected forest clearance for the mine. Niyamgiri was the first time India's highest court asked local gram sabhas to decide the fate of a major mining project, and it remains a landmark for the rights of forest-dwelling communities.

Case Study: Jharia: a century of coal fires

The Jharia coalfield near Dhanbad, Jharkhand, holds India's main reserves of prime coking coal, used in steelmaking. Fires were first recorded there in 1916. Unscientific mining by private owners before nationalisation in the early 1970s left abandoned workings in which coal came into contact with air and caught fire through spontaneous combustion. Underground fires have since burnt millions of tonnes of coal, while the ground above has subsided, swallowing houses and roads, and toxic gases and smoke have harmed residents' health. Opencast mining by Bharat Coking Coal Limited later exposed and spread many fires. A Jharia Master Plan approved in 2009 aimed to control the fires and resettle over a lakh families living in unstable areas, but progress was slow, and many families resisted moving to resettlement sites far from their livelihoods. In June 2025 the Union Cabinet approved a revised Jharia Master Plan of about ₹5,940 crore, adding livelihood grants and credit support for resettled families. According to the Ministry of Coal, the number of fire sites had been reduced from 77 to 27, and the surface area affected from about 17 km^2 to under 2 km^2. Jharia shows how damage from careless mining can persist for generations.

Case Study: Goa: iron ore boom and bust

Iron ore has been mined in Goa since the 1940s, mostly by opencast methods in the forested foothills of the Western Ghats, and almost all of it exported. Booming demand from China in the 2000s led to a rapid rise in output. In 2012 the Justice M. B. Shah Commission, appointed by the Union government, reported widespread illegal mining: mining outside lease boundaries and without valid clearances, and over-extraction, with huge losses to the public exchequer. Mining in Goa was suspended in late 2012. In 2014 the Supreme Court allowed mining to resume only with an annual cap on output, and in February 2018, ruling on a petition by the Goa Foundation, it quashed the renewal of all 88 mining leases, halting mining again. The environmental record of the boom years was severe: forest loss in the Western Ghats, waste dumps on hillsides, silt washed into rivers such as the Mandovi and Zuari, buried paddy fields and khazan lands, falling water tables, and dust along ore-transport routes. The closure also caused hardship for thousands of truck owners, barge operators and workers who depended on mining. Since 2022–23 the state has begun auctioning mining blocks afresh under the reformed MMDR Act. The Goa case shows the need for regulated extraction limits, honest enforcement and planning for life after mining.

Case Study: Aravalli hills: mining the oldest mountains

The Aravallis, among the oldest mountain systems on Earth, run about 700 km from Gujarat through Rajasthan and Haryana to Delhi. They act as a barrier against the spread of the Thar Desert eastward, recharge groundwater in a water-scarce region, moderate dust storms, and provide habitat for leopards, hyenas and many birds. For decades the hills have been quarried for stone, marble, granite and silica, much of it illegally, to feed construction in the National Capital Region and Rajasthan. The Supreme Court banned mining in the Aravallis of Haryana in 2002 and, in 2009, in the Faridabad, Gurugram and Nuh (Mewat) districts. A report by the Court's Central Empowered Committee led the Court to observe in 2018 that 31 of 128 sampled hills in Rajasthan had vanished. Because there was no uniform legal definition of the range, mining continued in many areas. In November 2025 the Supreme Court accepted a definition based on hills rising 100 metres or more above the surrounding land, together with the land enclosed by them, and in December 2025 the MoEFCC banned new mining leases across the Aravalli landscape until a sustainable mining management plan is prepared. Environmental groups protested that the height-based definition could leave large parts of the range unprotected, and the definition was taken up for fresh review. The Aravalli Green Wall Project aims to restore degraded land around the hills.

Case Study: Hasdeo Arand: coal versus forest

Hasdeo Arand in Chhattisgarh's Korba, Surguja and Surajpur districts is one of the largest unbroken stretches of dense forest in central India, about 1,700 km^2 in extent. It is home to Gond and other Adivasi communities, forms the catchment of the Hasdeo river and the Hasdeo Bango reservoir that irrigates farmland downstream, and is part of an elephant corridor. It also lies over a coalfield with about two dozen coal blocks. In 2010 the MoEFCC classified the area as a "no-go" zone for mining, but in 2012 forest clearance was granted for the Parsa East Kete Basan block, allotted to a Rajasthan state power company, overruling its own Forest Advisory Committee. A study by ICFRE and the Wildlife Institute of India, released in 2021, warned that mining most of the coalfield would cause irreversible damage and intensify human–elephant conflict, and recommended that mining be restricted to a few blocks already opened. Villagers have alleged that gram sabha consent for new blocks was obtained improperly, and have protested for over a decade. In July 2022 the Chhattisgarh Legislative Assembly unanimously passed a resolution urging the Union government to cancel all coal block allocations in Hasdeo. Tree felling for mine expansion has nevertheless continued amid protests. Hasdeo illustrates the conflict between India's coal-based power needs and the protection of forests, wildlife and tribal rights.

3.11.2 Impacts of dam building

India has about 6,000 large dams, the third-largest number in the world after China and the USA (Central Water Commission, National Register of Large Dams, 2023). Dams such as Bhakra Nangal, Hirakud and Nagarjuna Sagar were described by Jawaharlal Nehru as the "temples of modern India", and they made the Green Revolution and much of India's hydropower possible. They also have major costs.

On the environment and rivers

- **Submergence**: reservoirs drown forests, farmland, villages and river valleys.
- **Altered river flows**: downstream flows are reduced and regulated, so seasonal floods that renewed floodplains and recharged groundwater no longer occur, and stretches of river below diversion dams may run almost dry.
- **Sediment trapping**: dams trap silt that would otherwise reach the coast. Reservoirs lose storage capacity as they fill with sediment, while deltas such as those of the Krishna, Godavari and Mahanadi are eroding because they receive less sediment.
- **Water quality**: still reservoir water stratifies, may become low in oxygen at depth, and accumulates nutrients; decomposing submerged vegetation releases methane, a greenhouse gas.
- **Command-area problems**: poorly drained canal irrigation causes waterlogging and salinisation, as discussed under land degradation.
- **Geological hazards**: large reservoirs can trigger earthquakes (reservoir-induced seismicity); the 1967 Koyna earthquake in Maharashtra is widely linked to the Koyna reservoir. Dams and tunnels in the young, seismic and landslide-prone Himalaya face flash floods: the February 2021 Chamoli disaster in Uttarakhand destroyed the Rishiganga project and damaged the Tapovan–

Vishnugad project, killing about 200 people, and an October 2023 glacial lake outburst flood in Sikkim destroyed the Teesta III dam at Chungthang.

On forests and biodiversity

Reservoirs submerge some of the richest riverine forests, which are often the most fertile parts of a landscape. The proposed Silent Valley hydroelectric project in Kerala was abandoned in the early 1980s after a public campaign, and the area became a national park in 1984 (see Chapter 6). The Ken–Betwa river-link project, whose foundation stone was laid in December 2024, is expected to submerge about 9,000 hectares, including a significant part of the core of Panna Tiger Reserve. Dams also block the movement of migratory fish. After the Farakka Barrage was built across the Ganga in 1975, the hilsa fishery upstream of it collapsed, and dams and barrages have split the population of the endangered Ganges river dolphin into isolated groups. Mahseer, the large game fish of Himalayan and peninsular rivers, has declined for similar reasons.

On tribal and other local communities

Large dams have displaced millions of people in India. The World Commission on Dams (2000) estimated that 16 to 38 million people had been displaced by large dams in India, and the sociologist Walter Fernandes estimated that about 40% of all people displaced by development projects were Adivasis, although they make up under 9% of the population. Many were never properly resettled. Thousands of families displaced by the Hirakud dam in Odisha in the 1950s received little or no compensation, and some people have been displaced twice. Displacement breaks up communities, separates them from forests and rivers on which their livelihoods and culture depend, and often pushes them into urban poverty. Downstream fishing communities and those who depended on floodplain farming lose their livelihoods without being counted as project-affected. The Sardar Sarovar dam on the Narmada, the most debated example in India, is examined as a full case study later in this chapter, and the resistance to it by the Narmada Bachao Andolan is discussed in Chapter 7.

3.11.3 Reducing the impacts

Mining and dams will remain part of India's development, so the key questions are where, how and for whose benefit they are built. Approaches that reduce harm include:

- **Rigorous and honest assessment**: genuine Environmental Impact Assessments, cumulative impact studies of whole river basins and mining regions, and meaningful public hearings.
- **"No-go" areas**: keeping mining and dams out of protected areas, critical wildlife corridors, dense natural forests and sacred landscapes.
- **Free, prior and informed consent** of affected gram sabhas, as required by PESA and the FRA, and fair rehabilitation that restores livelihoods, not just cash compensation.
- **Better practices**: progressive mine reclamation, dust and water management, safe tailings dams, environmental flows below dams (such as the minimum flows notified for the upper Ganga in 2018), fish passes, and safety regulation under the Dam Safety Act, 2021.
- **Alternatives**: recycling of metals, more efficient use of materials and water, decentralised water harvesting (such as the work of Tarun Bharat Sangh, discussed later in this chapter), and renewable energy sources that reduce the demand for coal.

3.12 Water Resources: Natural and Man-made Sources

Water is the resource on which every other resource depends. Crops, forests, livestock, industry, power stations and cities all need a reliable supply of fresh water. Yet fresh water is scarce on a planet that looks blue from space. About 97.5% of all water on Earth is salty sea water. Of the remaining 2.5% that is fresh, roughly two-thirds is locked in glaciers and ice caps and almost one-third lies underground. Less than 1% of the world's fresh water is found in rivers, lakes, wetlands, soil moisture and the atmosphere, where people can reach it easily.

India's position is tighter still. The country has about 2.4% of the world's land area and nearly 18% of its people, but only about 4% of its renewable fresh water. This water is also very unevenly spread in space and time, as the rest of this section shows. How India stores, shares and protects its water is therefore one of its central environmental and economic questions. The movement of water through evaporation, condensation, precipitation, infiltration and runoff, the **hydrological cycle**, is described in Chapter 2. This section looks at water as a resource: where it comes from, how much there is, and how people have built systems to capture it.

3.12.1 India's water budget

A **water budget** is an account of the water entering, stored in and leaving a region. For India, the inflow is precipitation (rain and snow) plus a small amount of river water from neighbouring countries. Out of an estimated 4,000 billion cubic metres (BCM) of annual precipitation, a large part evaporates back to the atmosphere or is used by natural vegetation. The rest flows in rivers or soaks into the ground.

The Central Water Commission (CWC) reassessed the country's water availability in 2019 using satellite data and hydrological models for the period 1985–2015. It put the **average annual water resource** of India's river basins at 1,999 BCM. Not all of this can be used. Steep terrain, floods that rush to the sea in a few monsoon weeks, and the lack of storage sites mean that only part of it is **utilisable water**, the amount that can practically be stored or extracted. The CWC's conventional estimate of utilisable water is about 1,137 BCM: 690 BCM of surface water and 447 BCM of replenishable groundwater.

Table 3.5: India's water budget at a glance (approximate annual values)

Component	Volume (BCM per year)	Source
Precipitation (rain and snow)	about 4,000	National Commission for Integrated Water Resources Development
Average annual water resource of river basins	1,999	CWC reassessment, 2019
Utilisable surface water	690	CWC
Replenishable groundwater (utilisable)	447	CWC
Total utilisable water	about 1,137	CWC

Per-capita availability

Because the total resource is roughly fixed while the population keeps growing, the water available per person keeps falling. **Per-capita water availability** is the average annual water resource divided by the population. Hydrologists use the **Falkenmark indicator** to judge stress: below 1,700 m^3 per person per year a country is **water-stressed**, and below 1,000 m^3 it faces **water scarcity**.

India crossed into the water-stressed category around the start of this century. According to the Ministry of Jal Shakti and the CWC, per-capita availability fell from 5,177 m^3 in 1951 to 1,816 m^3 in 2001, 1,545 m^3 in 2011 and 1,486 m^3 in 2021, and is projected to fall to 1,367 m^3 by 2031 (see the figure below). The national average also hides large differences. The Brahmaputra and Barak basins of the North-East have many times the national average per person, whereas basins such as the Sabarmati, the Cauvery and the rivers of Kutch and Saurashtra are already below the scarcity line.

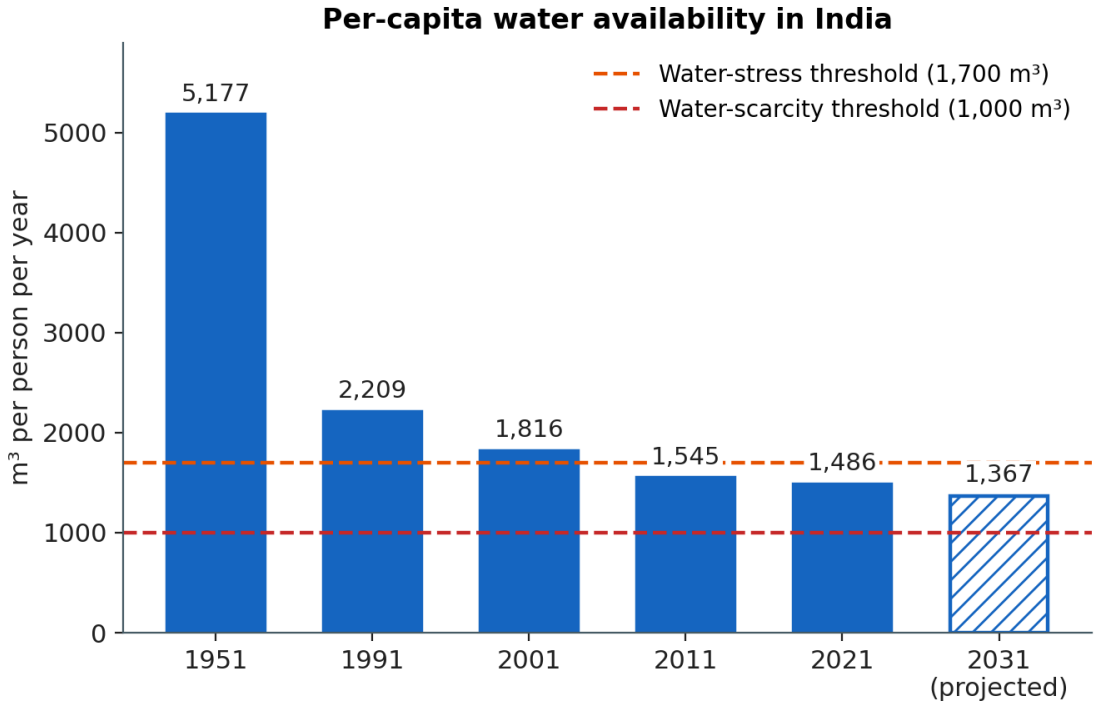


Figure 3.8: *Average annual per-capita water availability in India, 1951–2031 (2031 projected). Dashed lines mark the water-stress (1,700 m^3) and water-scarcity (1,000 m^3) thresholds. Source: Ministry of Jal Shakti / Central Water Commission.*

Uneven in time and space

Most of India's rain falls in about four months. The south-west monsoon (June to September) brings roughly three-quarters of the annual rainfall, often in a few dozen heavy-rain days. Rivers therefore carry floods for part of the year and shrink sharply in the dry season. Spatially, annual rainfall ranges from over 11,000 mm at Mawsynram in Meghalaya to below 150 mm in western Rajasthan and Ladakh. These patterns explain why India needs storage, and why floods and droughts can occur in the same year in different parts of the country.

3.12.2 Natural sources of water

Natural sources are those formed and replenished by the hydrological cycle without human construction. They include surface water, groundwater, glaciers and snow, and springs.

Surface water: rivers, lakes and wetlands

Surface water is water found above ground in rivers, streams, lakes, ponds and wetlands. India has about 20 major river basins. Two broad types matter for water supply:

- **Himalayan rivers** such as the Indus, Ganga and Brahmaputra are **perennial**: they flow throughout the year because snow and glacier melt add water in the dry summer months. They carry heavy loads of sediment and build the vast alluvial plains of northern India.
- **Peninsular rivers** such as the Godavari, Krishna, Cauvery, Mahanadi and Narmada are **rain-fed**. Their flow depends almost entirely on the monsoon, so lean-season flows are low and highly variable from year to year. This is one reason why disputes over peninsular rivers such as the Cauvery are so bitter.

Natural lakes (for example Wular in Jammu and Kashmir, Loktak in Manipur and Chilika, a coastal lagoon in Odisha) and wetlands store water, recharge groundwater, absorb floods and support fisheries. Their ecology is covered in Chapter 2.

Groundwater and aquifers

Groundwater is water that fills the pores and cracks in soil and rock below the land surface. Rainwater and river water seep downward (**infiltration** and **percolation**) until they reach the **zone of saturation**, where all spaces are filled with water. The top of this zone is the **water table**. A rock or sediment layer that stores and transmits enough water to supply wells is called an **aquifer**.

- An **unconfined aquifer** has the water table as its upper surface. It is recharged directly by rain from above, and its level rises and falls with the seasons and with pumping. Most open wells and shallow tube wells tap unconfined aquifers.
- A **confined aquifer** is sandwiched between layers of clay or rock that let little water through (**aquitards** or **aquicludes**). It is recharged only where the permeable layer is exposed, often far away, and the water may be under pressure. If the pressure is enough to push water above the ground when a well is drilled, the well is an **artesian well**.

India's aquifers fall into two very different groups. The **alluvial aquifers** of the Indus–Ganga–Brahmaputra plains, Gujarat and coastal deltas are thick layers of sand and gravel that hold large volumes of water. The **hard-rock aquifers** of the peninsula, formed in granite, gneiss and Deccan basalt, cover roughly two-thirds of the country's area. They store water only in weathered zones and fractures, so wells there yield little and dry up quickly when over-pumped.

Groundwater is India's single most important source of water. It supplies roughly 60% of irrigation, about 85% of rural drinking water and a large share of urban supply. India is the world's largest user of groundwater, extracting more each year than China and the United States combined.

Glaciers and snow

Glaciers are slowly moving masses of ice formed from compacted snow. The Geological Survey of India has catalogued more than 9,500 glaciers in the Indian Himalaya, and the wider Hindu Kush–Himalaya region is often called the **Third Pole** because it holds more ice than any region outside the polar caps. Snow and glacier melt sustain the dry-season flow of the Indus, Ganga and Brahmaputra. The share of meltwater is highest in the Indus basin and lower in the monsoon-fed Brahmaputra. Most Himalayan glaciers are retreating as temperatures rise (see Chapter 5). In the short term, faster melting can increase flows and the risk of **glacial lake outburst floods (GLOFs)**. In the long term, shrinking glaciers threaten dry-season water for hundreds of millions of people.

Springs

A **spring** is a place where groundwater emerges naturally at the surface. In the Himalaya and the hills of the North-East and Western Ghats, springs are the main source of drinking water for millions of people. A 2018 NITI Aayog report on spring-shed management estimated that India has about five million springs, nearly three million of them in the Indian Himalayan Region, and warned that close to half of the region's perennial springs are drying up or turning seasonal because of deforestation, road building and changing rainfall. Spring-shed programmes in Sikkim (the *Dhara Vikas* initiative) and Meghalaya revive springs by recharging their catchments with trenches and ponds.

3.12.3 Man-made sources of water

Man-made sources are structures and systems that people build to capture, store, move or create usable water. They range from giant reservoirs to village tanks, rooftop harvesting systems and desalination plants.

Dams, reservoirs and canals

A **dam** is a barrier across a river that stores water in a lake called a **reservoir**. Multipurpose dams provide irrigation, drinking water, hydroelectric power and flood control. A **barrage** has gates along its whole length and raises the water level for diversion into canals rather than storing large volumes. India has over 6,000 large dams listed in the CWC's National Register of Large Dams, the third-highest number in the world after China and the United States. Examples include Bhakra (Sutlej), Hirakud (Mahanadi), Nagarjuna Sagar (Krishna), Tehri (Bhagirathi) and Sardar Sarovar (Narmada). Reservoirs smooth out the monsoon by holding flood water for use in the dry season.

Canals carry water from rivers or reservoirs to farms and towns. The Indira Gandhi Canal carries water from the Harike barrage in Punjab deep into the Thar desert of Rajasthan, and the Narmada main canal carries water across Gujarat to the Rajasthan border. **Inter-basin water transfer** moves water from a "surplus" basin to a "deficit" one. India's National Perspective Plan proposes 30 river links; the first to begin construction is the Ken–Betwa link in Bundelkhand, whose foundation stone was laid in December 2024. Large dams and links bring real benefits but also submerge forests and villages, trap sediment, alter river ecology and displace people. Their environmental and social impacts are discussed in the first part of this chapter and in the Sardar Sarovar case study later in this part.

Tanks, stepwells and traditional systems

Long before modern dams, Indian communities developed ingenious ways of capturing rain where it falls. These **traditional water-harvesting systems** are adapted to local rainfall, soil and slope, and many are being revived today.

Table 3.6: Some traditional water-harvesting systems of India

System	Region	How it works
Eri (tank)	Tamil Nadu	Earthen embankment stores runoff for irrigation; tanks are linked in chains
Johad	Rajasthan, Haryana	Crescent-shaped earthen check dam that holds runoff and recharges wells

System	Region	How it works
Kund / kundi	Thar desert, Rajasthan	Covered underground pit with a saucer-shaped catchment for drinking water
Baoli / vav (stepwell)	Gujarat, Rajasthan, Delhi	Steps lead down to groundwater; stores and protects water
Kuhl	Himachal Pradesh	Gravity channels divert glacier-fed streams to fields
Zabo	Nagaland	Forest catchment, ponds and cattle yards combined in one farming system
Bamboo drip irrigation	Meghalaya	Bamboo pipes carry spring water downhill to betel and pepper plants

Tanks are the most widespread of these systems in peninsular India, where hard-rock aquifers hold little water. Tamil Nadu alone has tens of thousands of irrigation tanks, many built under the Pallava, Chola and Vijayanagara kingdoms. Tanks irrigate fields, recharge wells, support fish and birds, and reduce floods. Many have been encroached, silted up or cut off from their catchments by roads and housing.

Stepwells (*baoli* in Hindi, *vav* in Gujarati) are deep wells with flights of steps leading down to the water. Rani ki Vav at Patan in Gujarat, an eleventh-century stepwell, is a UNESCO World Heritage Site, and Chand Baori at Abhaneri in Rajasthan has about 3,500 steps. Delhi's Agrasen ki Baoli shows that the city once relied on such structures. Because stepwells reach down to the water table, they became useless wherever groundwater levels fell far below their base.

Rainwater harvesting

Rainwater harvesting (RWH) is the collection and storage of rain where it falls, either for direct use or to recharge groundwater. There are two main forms:

- **Rooftop rainwater harvesting** collects rain from roofs through gutters and pipes. A **first-flush device** discards the first, dirty runoff; a filter of gravel, sand and charcoal removes debris; and the water is either stored in a tank for use or led into a **recharge pit**, trench or abandoned well to refill the aquifer (see the figure below).
- **Surface-runoff harvesting** captures runoff from open land and roads in ponds, check dams, percolation tanks and contour trenches.

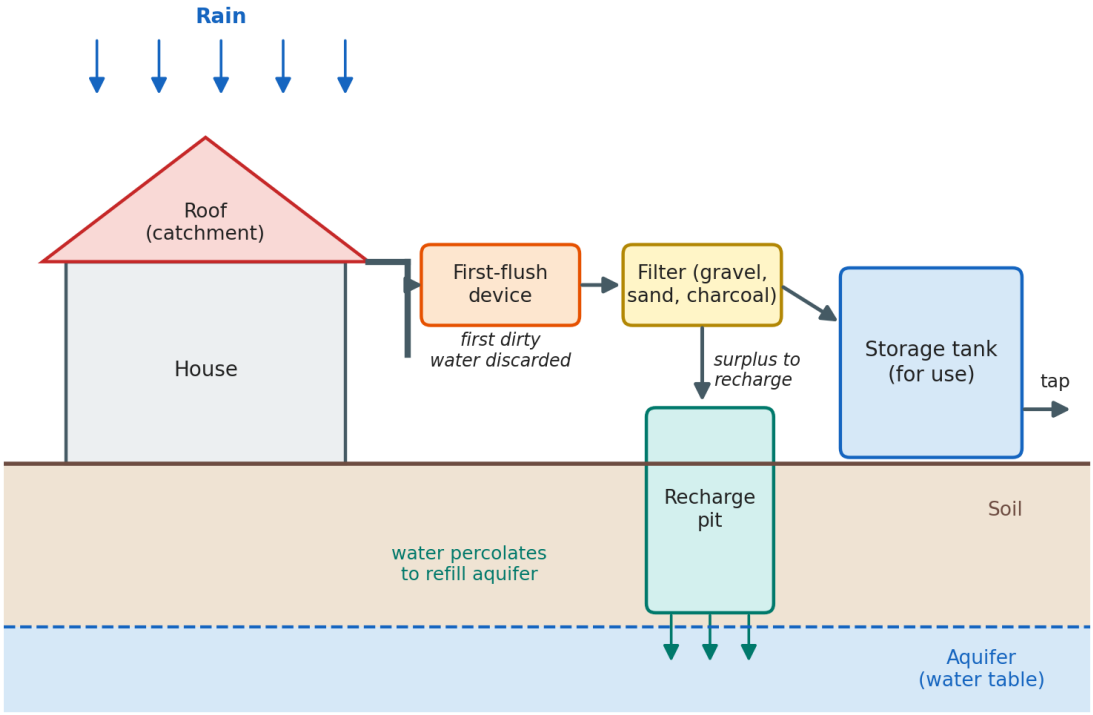


Figure 3.9: *Rooftop rainwater harvesting: rain from the roof passes a first-flush device and filter, then goes either to a storage tank for use or to a recharge pit that refills the aquifer. Source: Original diagram.*

The potential is large. The volume that can be harvested equals roof area × annual rainfall × a **runoff coefficient** (about 0.8 for hard roofs, to allow for losses). A 100 m^2 roof in Delhi, which receives about 750 mm of rain a year, can capture about 100 × 0.75 × 0.8 = 60 m^3, that is 60,000 litres, every year.

Tamil Nadu made rooftop harvesting compulsory for all buildings in 2003, and groundwater levels in Chennai rose in the following years. Delhi's building bye-laws require RWH on plots of 100 m^2 or more, and many states and cities have similar rules, although enforcement is often weak. The Central Ground Water Authority also requires industries and large projects that draw groundwater to recharge it.

> **Think and Do: Harvest the rain on your campus**
>
> Measure (or estimate from Google Earth) the roof area of one building on your campus. Find your city's average annual rainfall from the India Meteorological Department website. Calculate the volume of water the roof could harvest each year using a runoff coefficient of 0.8. Compare this with the building's daily water use (ask the estate office or estimate 45 litres per person per day for a college). How many days of use could the harvested water cover?

Desalination, wastewater reuse and other sources

Desalination removes salt from sea water or brackish groundwater, usually by **reverse osmosis**, in which water is pushed through a membrane that holds back salt. Chennai, which has no perennial river, runs large seawater desalination plants at Minjur and Nemmeli and is building more. Desalination is drought-proof but needs a great deal of energy and produces concentrated brine that can harm coastal ecosystems if discharged carelessly.

Treated wastewater is a growing source. Cities such as Bengaluru pump treated sewage into dry lakes and tanks in neighbouring districts (the Koramangala–Challaghatta Valley project in Kolar is an example), and thermal power stations are required to use treated sewage from sewage treatment plants within 50 km where available. **Managed aquifer recharge**, the deliberate refilling of aquifers through check dams, recharge wells and spreading basins, turns surplus monsoon water into stored groundwater.

Key Points

- India has about 18% of the world's people but only about 4% of its fresh water; per-capita availability fell from 5,177 m^3 (1951) to 1,486 m^3 (2021).
- The CWC (2019) estimates India's average annual water resource at 1,999 BCM, of which about 1,137 BCM is utilisable.
- Natural sources are surface water, groundwater, glaciers and snow, and springs; groundwater is India's most important source.
- Man-made sources include dams, reservoirs, canals, tanks, stepwells, rainwater harvesting, desalination and treated wastewater.

3.13 Uses of Water

People use water in many ways, but a few sectors account for almost all of it. To understand water demand, it helps to separate two ideas. **Water withdrawal** is the total water taken from a river, lake or aquifer. **Water consumption** is the part of that withdrawal that does not return to a usable source because it evaporates, is transpired by crops, or is built into a product. A thermal power station may withdraw a lot of water for cooling but return most of it (warmer) to the river; an irrigated paddy field consumes most of what it receives.

3.13.1 Agriculture

Agriculture is by far the largest user of water in India. According to the FAO's AQUASTAT database, about 90% of India's total water withdrawal (around 690 of 760 BCM in 2010) went to agriculture, mostly for irrigation. Other estimates by Indian agencies put the share between about 80% and 90%, depending on what is counted (see the figure below). Groundwater has become the backbone of irrigation: close to nine-tenths of all groundwater extracted in India is used for irrigation.

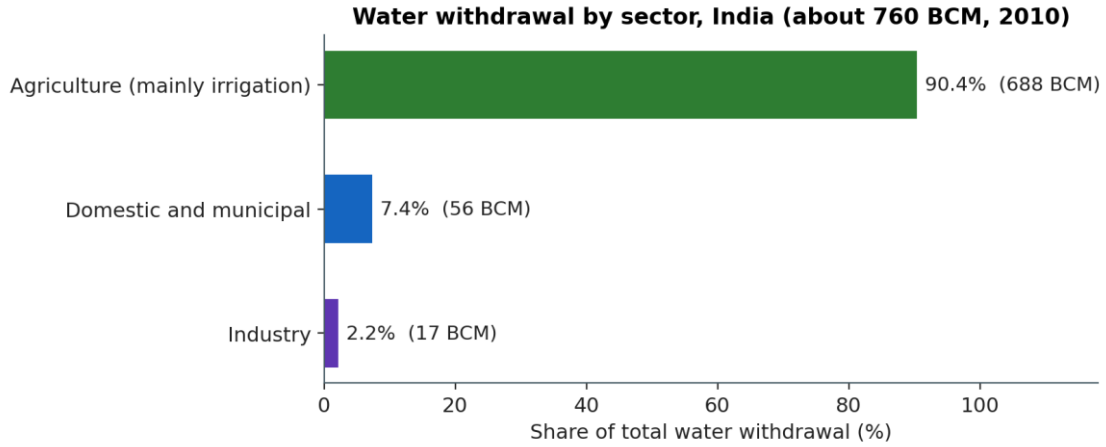


Figure 3.10: *Estimated share of total water withdrawal by sector in India (about 760 BCM, 2010). Source: FAO AQUASTAT.*

The way water is applied matters as much as how much is used. In **flood** or **furrow irrigation**, water is let into the whole field, and only 30–50% of it actually reaches crop roots; the rest evaporates or seeps away. **Sprinkler irrigation** sprays water like rain, with efficiencies of about 70–80%. **Drip irrigation** delivers water drop by drop through pipes and emitters to each plant's root zone and can reach 80–90% efficiency. The Pradhan Mantri Krishi Sinchayee Yojana (2015) promotes drip and sprinkler systems under its "Per Drop More Crop" component. Water-hungry crops grown in dry regions, such as paddy in Punjab and Haryana and sugarcane in Maharashtra, are the main reason irrigation demand is so high.

3.13.2 Domestic use

Domestic water covers drinking, cooking, bathing, washing, sanitation and gardening. Indian planning norms aim for at least 55 litres per capita per day (lpcd) in rural areas under the Jal Jeevan Mission, and 135 lpcd in cities with piped sewerage (Central Public Health and Environmental Engineering Organisation norms). Domestic use is a small share of the total (under a tenth), but it has the highest priority because it is essential for life and health. The National Water Policy (2012) places drinking water and sanitation first among competing uses. Losses are high: many city systems lose 30–50% of treated water through leaking pipes and illegal connections, which water managers call **non-revenue water**.

3.13.3 Industry and energy

Industries use water for processing, washing, cooling, steam raising and as a raw material. Textiles, paper and pulp, chemicals, fertilisers, steel, refineries and beverages are major users. **Thermal power stations** are the largest industrial users because they need water to cool the steam that drives their turbines. To reduce this demand, the Ministry of Environment, Forest and Climate Change in 2015 capped water use at 3.5 m^3 per megawatt-hour for existing coal plants and 3 m^3 per MWh for new ones. Hydroelectric power uses water without consuming much of it, but reservoirs lose water by evaporation and change the timing of river flows.

Water and energy are closely linked: pumping groundwater and moving water through canals and city pipes uses large amounts of electricity, while power generation needs water. This **water–energy nexus** means that saving one often saves the other.

3.13.4 Environmental flows

Rivers also need water for themselves. **Environmental flow (e-flow)** is the quantity, timing and quality of water that must remain in a river to sustain its ecosystem, fish, sediment transport and the people who depend on it downstream. In 2018 the National Mission for Clean Ganga notified minimum e-flows for the upper Ganga and its tributaries, the first such legal requirement in India. Most Indian rivers still have no enforced e-flow.

3.13.5 Virtual water and water footprint

When a product is made, the water used in its production is "embedded" in it even though the product itself may contain little water. This hidden water is called **virtual water**. The **water footprint** of a person, product or country is the total volume of fresh water used to produce the goods and services consumed.

Global average estimates from the Water Footprint Network give a sense of scale: about 2,500 litres of water per kilogram of rice, about 1,800 litres per kilogram of wheat, and roughly 2,700 litres for a cotton T-shirt. Figures for Punjab rice, grown in the hot, dry pre-monsoon months with heavy pumping, are considerably higher. India is the world's largest rice exporter, so it exports large volumes of virtual water, much of it pumped from depleting aquifers in Punjab and Haryana. Thinking in terms of virtual water helps explain why choices about crops and trade are also choices about water.

> **Withdrawal is not the same as use-up**
>
> A coal power station near a river may withdraw a huge volume of water each day, but most of it returns to the river after cooling. A field of paddy may withdraw less, yet consume nearly all of it through evaporation and transpiration. When comparing sectors, ask whether a figure refers to withdrawal or consumption.

3.14 Over-exploitation of Surface and Groundwater

A water source is **over-exploited** when water is taken from it faster than nature replenishes it, or when so much is taken that the source can no longer support its ecosystems and downstream users. India faces both forms: aquifers are being mined in the north-west and the hard-rock peninsula, and many rivers and lakes are being drained, diverted and polluted.

3.14.1 Groundwater over-exploitation

How groundwater status is measured

The Central Ground Water Board (CGWB) and state agencies jointly assess India's **dynamic groundwater resources**, the part of groundwater that is renewed each year by rain and other recharge. The key measure is the **stage of groundwater extraction**:

$$\text{Stage of extraction (\%)} = \text{annual groundwater extraction} \div \text{annual extractable groundwater resource} \times 100$$

Each assessment unit (a block, taluka or mandal) is then classified as **safe** (below 70%), **semi-critical** (70–90%), **critical** (90–100%) or **over-exploited** (above 100%, meaning more water is pumped out each year than is recharged). Units where groundwater is too salty to use are classed as **saline**.

The national picture

The Dynamic Ground Water Resources Assessment 2025 estimated India's total annual groundwater recharge at about 448.5 BCM and the annual extractable resource at about 407.8 BCM. Total annual extraction was about 247.2 BCM, giving a national stage of extraction of 60.63%. The national average looks comfortable, but it hides severe local stress. Of 6,762 assessment units, 730 (10.8%) were over-exploited, 201 (3.0%) critical and 758 (11.2%) semi-critical, while 4,946 (73.1%) were safe. In other words, about one unit in four is under stress.

Over-exploitation is concentrated in the north-west (Punjab, Haryana, Rajasthan and Delhi), in western Uttar Pradesh, in Gujarat, and in the hard-rock regions of Tamil Nadu, Karnataka and Telangana. In 2025 Punjab had the highest stage of extraction of any state at about 156%, followed by Rajasthan (about 147%) and Haryana (about 137%) (see the figure below). Delhi was at about 101% in the 2024 assessment. A stage above 100% means that the aquifer is being "mined": water stored over centuries is being used up.

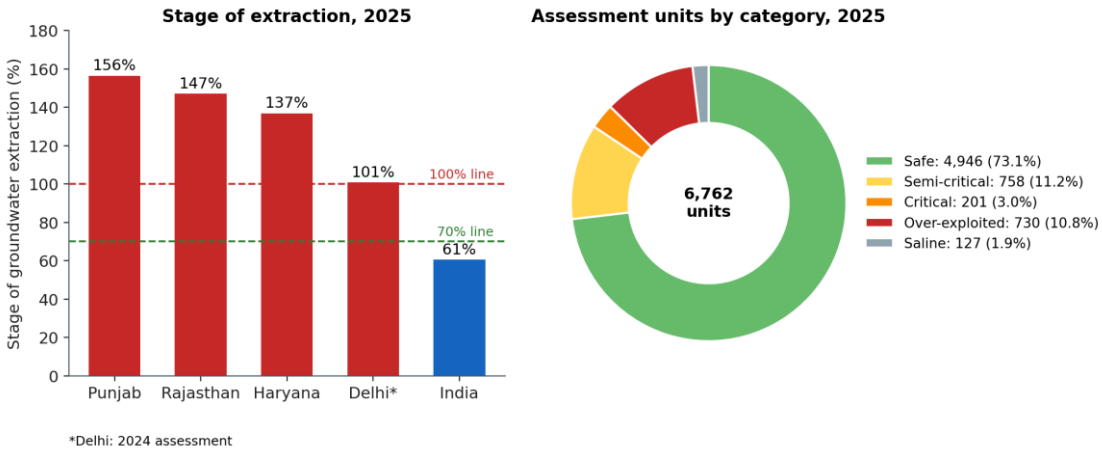


Figure 3.11: *Groundwater status in India. Left: stage of groundwater extraction in the most stressed states compared with the national average (2025; Delhi 2024). Right: classification of 6,762 assessment units (2025). Source: CGWB, Dynamic Ground Water Resources of India 2024 and 2025.*

It is encouraging that the national stage of extraction has fallen slightly since 2017 (from about 63% to about 61%), and the number of over-exploited units has declined, helped by recharge works and good monsoons. But in the worst-hit districts, water tables continue to fall year after year.

Causes of groundwater depletion

- **Cheap pumping technology.** The spread of tube wells and submersible electric pumps since the 1970s let farmers draw water from depths that open wells could never reach.
- **Free or flat-rate electricity for farms.** In many states farmers pay nothing or a flat fee for power, so pumping more costs nothing extra. Pumping is limited only by the hours of supply.
- **Crop choice and procurement policies.** Assured government procurement at minimum support prices makes paddy and wheat the safest crops in Punjab and Haryana, even though paddy needs far more water than the region's rainfall provides.

- **Weak legal control.** Under the Indian Easements Act 1882, groundwater is treated as attached to the land, so a landowner may pump as much as they like. Several states have groundwater laws, but enforcement is limited.
- **Urbanisation.** Cities pave over recharge areas, fill lakes and wetlands, and rely on private borewells when piped supply falls short.

Case Study: Punjab: the water cost of the Green Revolution

Punjab covers about 1.5% of India's area but supplies a large share of the rice and wheat procured for the national food stock. Paddy was not a traditional summer crop in much of the state. With canals, tube wells, free electricity and assured procurement, the rice–wheat rotation spread across the state from the 1970s, and groundwater use soared. Today Punjab extracts far more groundwater each year than is recharged, and most of its blocks are classed as over-exploited.

To reduce pumping in the hottest months, the state passed the Punjab Preservation of Subsoil Water Act 2009, which forbids transplanting paddy before a notified date in June (Haryana passed a similar law). Delaying transplanting until closer to the monsoon cut evaporation losses and saved water. It had an unintended side effect: paddy is now harvested later, leaving farmers only a short window before sowing wheat, which encouraged the burning of paddy stubble (discussed later in this chapter). Haryana's *Mera Pani Meri Virasat* scheme (2020) pays farmers per acre to switch from paddy to less thirsty crops such as maize, pulses and cotton. Breaking the rice–wheat cycle without hurting farm incomes remains the central challenge.

Consequences of over-exploitation

- **Falling water tables.** Wells must be deepened again and again. Small farmers who cannot afford deeper borewells lose access first, so depletion deepens inequality.
- **Rising energy use.** Lifting water from greater depths needs bigger pumps and more electricity.
- **Poorer water quality.** Pumping from deeper layers can draw up naturally occurring contaminants. Arsenic affects parts of the Ganga–Brahmaputra plains in West Bengal, Bihar, Uttar Pradesh and Assam; fluoride affects parts of Rajasthan, Telangana, Andhra Pradesh and Gujarat; and uranium has been reported in parts of Punjab. Nitrate from fertilisers and sewage is the most widespread pollutant. The health effects are discussed in Chapter 4.
- **Seawater intrusion.** In coastal aquifers, over-pumping lets sea water move inland. This has made wells saline along the Saurashtra coast of Gujarat and in the Minjur area north of Chennai.
- **Land subsidence.** When water is removed from clay layers, they compact and the ground sinks. Satellite studies have recorded subsidence of several centimetres a year in parts of the Delhi National Capital Region linked to heavy pumping.
- **Drying of rivers, springs and wetlands.** Groundwater feeds many rivers in the dry season as **base flow**. When the water table falls below the river bed, the flow reverses and the river loses water to the ground.

3.14.2 Over-exploitation of surface water

Rivers and lakes are over-exploited when so much water is diverted that little or none remains downstream. During the dry season, barrages on the upper Yamuna divert almost the entire flow into irrigation and drinking-water canals. The river that enters Delhi is a trickle, and below Wazirabad it

carries mostly sewage, treated and untreated. In some dry years the Cauvery and Krishna carry very little water to their deltas, which allows sea water to push inland.

Surface water is also lost by degradation. Pollution from sewage, industrial effluents and farm runoff makes water unfit for use, effectively reducing supply (see Chapter 4). Urban lakes and wetlands are encroached upon or filled for housing and roads. Bengaluru's chain of tanks and Chennai's marshes and lakes once stored monsoon rain and recharged aquifers; many have shrunk or disappeared. Illegal **sand mining** from river beds lowers river channels, dries nearby wells and destroys habitat. Siltation of reservoirs, caused by soil erosion in catchments, steadily reduces their storage capacity.

3.14.3 Urban water crises

Fast-growing Indian cities depend on distant reservoirs, private borewells and tanker supply. When the monsoon fails, these systems can collapse quickly.

Case Study: Chennai's "Day Zero", 2019

Chennai depends on monsoon-fed reservoirs, groundwater and desalination. After a very weak north-east monsoon in late 2018, the four main reservoirs that supply the city (Puzhal, Chembarambakkam, Cholavaram and Poondi) were virtually dry by June 2019. Piped supply was cut sharply, hotels and offices restricted water use, and residents queued for tankers. The media called it the city's "Day Zero", the day the taps run dry. From July to October 2019, a train of 50 wagons, each carrying about 50,000 litres, brought water daily from Jolarpettai, over 200 km away, until the monsoon refilled the reservoirs.

The irony is that Chennai had suffered devastating floods only four years earlier, in 2015. Both disasters had the same roots: lakes, tanks and wetlands that once stored and slowly released rainwater had been built over, and the city's water bodies were poorly maintained. Chennai has since expanded desalination, revived several lakes and tightened enforcement of rooftop rainwater harvesting.

Case Study: Bengaluru's water crisis, 2024

Bengaluru receives most of its piped water from the Cauvery, pumped about 100 km and lifted roughly 300 metres uphill, and draws much of the rest from borewells. After a weak monsoon in 2023, groundwater levels dropped steeply in early 2024. Thousands of borewells dried up, tanker prices soared, and in March 2024 the water board banned the use of drinking water for washing cars, gardening and construction. The city once had hundreds of interconnected tanks that stored rain and recharged aquifers; many had been lost to urban growth. The crisis led to wider use of treated wastewater, faster work on new Cauvery supply phases, and renewed attention to lake revival and rainwater harvesting.

3.14.4 Responses: managing demand and restoring supply

India's water policies increasingly combine supply-side works with demand management and community participation.

- **Atal Bhujal Yojana (Atal Jal).** Launched in December 2019 and implemented from April 2020, this ₹6,000 crore programme (half financed by a World Bank loan) promotes community-led groundwater management in about 8,200 water-stressed gram panchayats in seven states: Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan and Uttar Pradesh. Villages measure their groundwater, prepare **water budgets** and **water security plans**, and earn incentive funds for reducing demand through micro-irrigation, crop diversification and recharge.
- **Jal Jeevan Mission (JJM).** Launched in August 2019 to provide a functional household tap connection to every rural home, JJM raised tap-water coverage from about 17% of rural households in 2019 to over 15.8 crore households (about 81%) by early 2026. In 2026 the Union Cabinet extended the mission to December 2028, with a stronger focus on source sustainability, water-quality testing, and operation and maintenance by gram panchayats. Without recharge, piped supply can simply shift the burden onto the same over-exploited aquifers.
- **Jal Shakti Abhiyan.** Launched in 2019 as a time-bound campaign in water-stressed districts, it has continued every year since 2021 as *Jal Shakti Abhiyan: Catch the Rain*, with the slogan "catch the rain, where it falls, when it falls". It mobilises districts to build and renovate water-harvesting structures, recharge borewells, map water bodies and plant trees. *Mission Amrit Sarovar* (2022) aimed to build or revive 75 ponds in every district.
- **Regulation and knowledge.** The CGWB's National Aquifer Mapping programme maps aquifers in detail, and the Central Ground Water Authority's 2020 guidelines require permission and abstraction charges for industrial and bulk users. Gujarat's *Jyotigram Yojana* (2003) separated farm power feeders from village feeders, giving pumps reliable but rationed power.
- **Saving water on farms.** Micro-irrigation, crop diversification away from paddy and sugarcane in dry regions, laser land levelling and direct seeding of rice all reduce demand. Care is needed with solar pumps: because sunlight costs nothing once a pump is installed, they can encourage over-pumping unless farmers can sell surplus power to the grid instead.

Key Points

- India is the world's largest groundwater user; the CGWB's 2025 assessment put the national stage of extraction at 60.63%, but about 11% of assessment units are over-exploited.
- Punjab (about 156%), Rajasthan (about 147%) and Haryana (about 137%) extract far more groundwater than is recharged.
- Free power, cheap pumps, paddy procurement, weak laws and urbanisation drive depletion; consequences include falling water tables, contamination, seawater intrusion and subsidence.
- Chennai (2019) and Bengaluru (2024) show how cities that destroy their lakes become vulnerable to both floods and water famine.
- Atal Bhujal Yojana, Jal Jeevan Mission and Jal Shakti Abhiyan combine community action, supply and recharge.

3.15 Floods and Droughts

Floods and droughts are the two extremes of the water cycle. India, with its concentrated monsoon, steep Himalayan rivers and vast plains, experiences both, often in the same year: Assam may be under water while Marathwada waits for rain. Both are natural hazards, but human actions decide how often they turn into disasters and how severe the damage is.

3.15.1 Floods

A **flood** is the overflow of water onto land that is normally dry. India's National Flood Commission (Rashtriya Barh Ayog, 1980) estimated that about 40 million hectares, roughly one-eighth of the country, is flood-prone. The Ganga and Brahmaputra basins account for most of this area.

Types of floods

- **Riverine floods** occur when prolonged heavy rain over a basin raises a river above its banks, as in Assam and Bihar almost every year.
- **Flash floods** rise within minutes or hours after very intense rain, typically in hills. A **cloudburst** (about 100 mm of rain or more in an hour over a small area) can trigger them.
- **Urban floods** happen when rain falls faster than paved cities can drain it, as in Mumbai in July 2005 (about 944 mm in 24 hours at Santacruz), Chennai in 2015 and Delhi in July 2023, when the Yamuna reached its highest recorded level.
- **Coastal floods** are caused by storm surges from cyclones, which push sea water inland in Odisha, Andhra Pradesh, West Bengal and Gujarat.
- **Glacial lake outburst floods (GLOFs)** occur when a lake dammed by ice or loose glacial debris bursts. In October 2023 the South Lhonak lake in Sikkim burst, and the flood destroyed the Teesta-III hydropower dam at Chungthang.

Causes

Natural causes include intense or prolonged monsoon rain, cyclones, snowmelt, the flat gradient of plains rivers, and the huge sediment loads of Himalayan rivers that raise their beds and make them shift course. Human actions make floods worse:

- **Deforestation and land degradation** in catchments increase runoff and soil erosion, which silts up rivers and reservoirs.
- **Encroachment on floodplains, wetlands and drains** removes the space rivers need and places people directly in harm's way.
- **Urbanisation** replaces soil with concrete, so rain runs off instead of soaking in.
- **Embankments** protect land locally but trap sediment within the channel, raising the river bed. When they breach, as the Kosi did at Kusaha in 2008, the flood is sudden and catastrophic.
- **Poor reservoir operation**, such as late or sudden releases from full dams, can add to downstream floods.
- **Climate change** increases the frequency of extreme rainfall because warmer air holds more moisture (see Chapter 5).

Indian examples

> **Case Study: Uttarakhand, June 2013**
>
> In mid-June 2013 the monsoon arrived early and collided with a western disturbance, bringing several days of extremely heavy rain to the upper Ganga valleys while snow was still melting. Above Kedarnath, the Chorabari glacial lake breached, sending a torrent of water and boulders through the temple town. The Mandakini, Alaknanda and Bhagirathi tore away roads, bridges, hotels and villages built close to the rivers. About 6,000 people, many of them pilgrims, died or went missing. Investigations, including an expert committee set up on the Supreme Court's orders, linked the scale of damage to unregulated construction on river banks, road building on fragile slopes and the effect of hydropower projects on river beds.

Kerala, August 2018. Kerala received exceptionally heavy rain in the first three weeks of August 2018, more than double the normal for that period. With reservoirs already full, dam authorities opened the gates of dozens of dams, including Idukki. About 480 people died, and over a million people moved to relief camps. Studies by the CWC and others concluded that the extreme rain was the main cause, but the event raised questions about reservoir operation, construction on paddy wetlands and quarrying in the Western Ghats.

Assam. The Brahmaputra and its tributaries flood Assam almost every year. About two-fifths of the state's area is flood-prone. Floods bring fertile silt, but they also destroy crops and homes, drown animals in Kaziranga National Park and cause severe bank erosion. Majuli, the world's largest river island, has lost a large part of its area to erosion over the last century.

3.15.2 Droughts

A **drought** is a prolonged shortage of water compared with normal conditions. Unlike floods, droughts develop slowly and are often called a "creeping disaster". The National Disaster Management Authority estimates that about two-thirds of India's cultivated area is vulnerable to drought, especially in the rain-shadow regions of the Deccan and in arid western India.

Types of drought

- **Meteorological drought**: rainfall is significantly below normal. The India Meteorological Department (IMD) treats a seasonal deficiency of more than 10% for the country, with over 20% of the area affected, as an all-India drought year, as in 2002, 2009 and 2015.
- **Agricultural drought**: soil moisture is too low for crops during the growing season, so crops wither even if total rain is not very low.
- **Hydrological drought**: rivers, reservoirs and groundwater fall well below normal. It lags behind meteorological drought.
- **Socio-economic drought**: water shortage starts affecting food, fodder, drinking water, jobs and incomes.

Causes

Droughts in India are linked mainly to failure or delay of the monsoon, long breaks within the monsoon season, and climate patterns such as **El Niño**, the warming of the eastern Pacific that often weakens the Indian monsoon. Human actions deepen droughts: over-pumping of groundwater,

cultivation of water-intensive crops in dry areas, loss of tanks and village ponds, and deforestation that reduces infiltration.

Indian examples

Marathwada (Maharashtra). This rain-shadow region suffered back-to-back droughts in 2014 and 2015. By early 2016 most of its dams were nearly empty. Latur received drinking water by train (the "Jaldoot") from Miraj, more than 300 km away, and authorities restricted gatherings near water tanks. The large area under sugarcane, an extremely thirsty crop, in a drought-prone region drew sharp criticism.

Bundelkhand (Uttar Pradesh and Madhya Pradesh). This rocky, semi-arid plateau faced repeated droughts in the 2000s and again in 2014–16. Crop failures led to farmer distress, livestock abandonment and large-scale migration. The region's old chandela tanks had fallen into disrepair. Revival of tanks, farm ponds and the Ken–Betwa river link are among the responses, the last being controversial because it will submerge part of Panna Tiger Reserve.

3.15.3 Managing floods and droughts

Disaster management in India is guided by the Disaster Management Act 2005 and the National Disaster Management Authority (NDMA). The emphasis has shifted from relief after the event to preparedness and risk reduction.

Table 3.7: Measures to manage floods and droughts

Approach	Floods	Droughts
Structural	Dams and reservoirs, embankments, channel improvement, drainage	Reservoirs, canals, check dams, tanks, farm ponds
Land and water management	Catchment afforestation, floodplain zoning, wetland and lake protection, sponge cities	Watershed development, rainwater harvesting, groundwater recharge
Early warning	CWC flood forecasts, IMD rainfall and nowcasts, GLOF monitoring	IMD monsoon forecasts, drought indices from rainfall, vegetation and soil moisture
Social protection	Evacuation plans, shelters, insurance	Contingency crop plans, MGNREGA works, crop insurance, fodder banks

Early warning systems save the most lives. The CWC issues flood forecasts at several hundred river stations across the country, and IMD provides district-level rainfall warnings and impact-based forecasts. Satellites from ISRO map flood extent in near real time. For drought, the Ministry of Agriculture's Manual for Drought Management (2016, updated 2020) uses a set of indicators, including rainfall deficit, satellite vegetation indices, soil moisture, reservoir levels and crop sowing, to declare drought objectively.

Non-structural measures are cheaper and often more sustainable than concrete. **Floodplain zoning** restricts building in areas likely to flood; a model bill was circulated to states as early as 1975, but few have adopted it. Restoring wetlands and lakes gives rivers and rain room to spread. For droughts, **watershed development**, which treats an entire catchment from ridge to valley with trenches, check dams and vegetation, has transformed villages such as Ralegan Siddhi and Hiware Bazar in

Maharashtra. Contingency crop plans prepared by ICAR's Central Research Institute for Dryland Agriculture tell farmers which crops to sow if the monsoon is delayed.

3.16 International and Inter-state Conflicts over Water

Rivers ignore political boundaries. When a river flows through several countries or states, the **upstream** user can affect the quantity, timing and quality of water reaching **downstream** users. About 310 river basins in the world are shared by two or more countries. Outright wars over water are rare, but water often adds to existing tensions and can become a powerful political symbol.

International law offers several competing principles. The **Harmon doctrine** of absolute territorial sovereignty (an upstream state may do what it likes) has been widely rejected. Modern law, set out in the Helsinki Rules (1966) and the UN Watercourses Convention (1997), rests on **equitable and reasonable utilisation** and the **obligation not to cause significant harm** to other riparian states, together with prior notification and data sharing. India is not a party to the 1997 Convention but has signed bilateral treaties with its neighbours.

3.16.1 India's international water relations

Indus Waters Treaty (India and Pakistan)

When the subcontinent was partitioned in 1947, the boundary cut across the Indus irrigation system, leaving headworks in India and canals in Pakistan. After nine years of negotiation brokered by the World Bank, India and Pakistan signed the **Indus Waters Treaty** in Karachi in September 1960.

- The three **eastern rivers** (Ravi, Beas and Sutlej) were allocated to India for unrestricted use.
- The three **western rivers** (Indus, Jhelum and Chenab), which carry about four-fifths of the system's water, were allocated to Pakistan. India may use them for domestic needs, limited irrigation and **run-of-river** hydropower that does not store large volumes or alter flows beyond set limits.
- A **Permanent Indus Commission** with one commissioner from each country meets regularly and exchanges data. Disputes go in stages: questions to the Commission, technical "differences" to a **Neutral Expert**, and legal "disputes" to a **Court of Arbitration**.

The treaty survived the wars of 1965, 1971 and 1999 and was long cited as a model of cooperation. Tensions grew over Indian hydropower projects on the western rivers, such as Baglihar, Kishenganga and Ratle, which Pakistan said violated design limits. In 2023 and 2024 India issued notices seeking to modify the treaty. On 23 April 2025, a day after the terrorist attack at Pahalgam, India announced that it would hold the treaty **"in abeyance"** until Pakistan "credibly and irrevocably abjures" support for cross-border terrorism. India stopped sharing hydrological data and attending Commission meetings. Pakistan called the move unlawful. A Court of Arbitration, whose proceedings India boycotts and whose legitimacy it rejects, ruled in 2025 and again in August 2026 that the treaty could not be unilaterally suspended; India rejected those rulings. As of late 2026 the treaty remains in abeyance from India's side. The episode shows how a long-lasting water agreement can be drawn into wider security conflicts.

Ganga and the Farakka Barrage (India and Bangladesh)

India built the **Farakka Barrage** on the Ganga in West Bengal, commissioned in 1975, to divert water into the Bhagirathi–Hooghly and flush silt from Kolkata port. Bangladesh, downstream, complained

that its share in the dry season fell sharply, harming farming, fisheries and the Sundarbans, and causing seawater to move inland. After short-term agreements in the 1970s and 1980s, the two countries signed the **Ganga Water Treaty** in December 1996. It shares the flow at Farakka between 1 January and 31 May in ten-day periods according to a formula, with a guaranteed minimum for each country in the leanest periods. The treaty runs for 30 years and expires in December 2026, so talks on its renewal are under way. Both sides want the formula updated to reflect current flows and climate change.

Teesta (India and Bangladesh)

The Teesta rises in Sikkim, flows through north Bengal and enters Bangladesh. An interim agreement on sharing its dry-season flow was ready in 2011, but it was not signed because the West Bengal government objected that the state's farmers needed the water. The issue remains unresolved and is a sore point in India–Bangladesh relations. India and Bangladesh share 54 rivers in all, managed through the Joint Rivers Commission.

Brahmaputra and China

The Brahmaputra rises in Tibet as the Yarlung Tsangpo, enters India in Arunachal Pradesh as the Siang, and flows through Assam into Bangladesh. China is the upstream country, and there is no water-sharing treaty. India and China have only memorandums of understanding under which China supplies flood-season hydrological data, and an Expert Level Mechanism for dialogue. China has built several hydropower dams on the main stem. In July 2025 it began construction of a very large hydropower project of about 60,000 MW in Medog county near the Great Bend, close to the Indian border. India worries about sudden releases or reduced lean-season flow, sediment trapping, and risks in an earthquake-prone zone. China says the project will not harm downstream countries. India has proposed its own large storage project on the Siang, partly to buffer against upstream releases, which in turn faces local opposition from the Adi community over displacement.

Other neighbours and global examples

India and Nepal share the Kosi, Gandak and Mahakali rivers under treaties of 1954, 1959 and 1996; disagreements over the costs and benefits of large projects such as Pancheshwar have slowed cooperation. Two global examples show similar tensions. On the **Nile**, Egypt, which depends on the river for almost all its water, has strongly opposed Ethiopia's Grand Ethiopian Renaissance Dam on the Blue Nile, completed in 2025, while Ethiopia insists on its right to develop. On the **Mekong**, a series of Chinese dams upstream and Laotian dams downstream have changed flows, sediment and fisheries in Cambodia and Vietnam; the Mekong River Commission (1995) includes the four lower countries but not China.

3.16.2 Inter-state water disputes in India

Seventeen of India's 18 major rivers flow through more than one state, so inter-state disputes are common. The Constitution divides responsibility:

- **Entry 17 of the State List** makes water supply, irrigation, canals, drainage, embankments, storage and water power state subjects.
- **Entry 56 of the Union List** lets Parliament regulate and develop inter-state rivers where it declares this to be in the public interest.

- **Article 262** allows Parliament to provide for adjudication of inter-state water disputes and to bar courts from hearing them.

Under Article 262, Parliament passed the **Inter-State River Water Disputes Act 1956**. When negotiations fail and a state complains, the Union government constitutes a **water disputes tribunal**, whose award, once published, has the force of a Supreme Court decree. A 2002 amendment set time limits (a tribunal within a year of a request, an award within three years, extendable by two). In practice tribunals have taken decades, and states have often rejected awards and returned to court; the Supreme Court hears appeals against awards through special leave petitions, as it did on the Cauvery. An amendment bill to create a single permanent tribunal with benches and a Disputes Resolution Committee was passed by the Lok Sabha in 2019 but has not become law. The **River Boards Act 1956** allowed river boards for joint management, but no board was ever set up under it.

Table 3.8: Major inter-state water disputes tribunals

River	States	Tribunal set up	Status
Krishna	Maharashtra, Karnataka, Andhra Pradesh, Telangana	1969; second tribunal 2004	First award 1976; second tribunal's award (2010) under challenge; AP–Telangana reference 2023
Godavari	Maharashtra, Andhra Pradesh, Karnataka, Madhya Pradesh, Odisha	1969	Award 1980
Narmada	Gujarat, Madhya Pradesh, Maharashtra, Rajasthan	1969	Award 1979
Ravi–Beas	Punjab, Haryana, Rajasthan	1986	Pending
Cauvery	Karnataka, Tamil Nadu, Kerala, Puducherry	1990	Award 2007; modified by Supreme Court 2018
Mahadayi	Goa, Karnataka, Maharashtra	2010	Award 2018; under challenge
Mahanadi	Odisha, Chhattisgarh	2018	Pending

Krishna. The first Krishna tribunal (1976) divided the river among Maharashtra, Karnataka and undivided Andhra Pradesh. A second tribunal reallocated the water in 2010, but its award was challenged and has not been implemented. After Telangana was created in 2014, sharing between Telangana and Andhra Pradesh became a new dispute, referred to the tribunal in 2023. Karnataka's plans to raise the Almatti dam are also contested.

Mahadayi (Mandovi). This small west-flowing river rises in Karnataka and flows into Goa. Karnataka wants to divert some water through the Kalasa and Banduri canals to the drought-prone Malaprabha basin to supply drinking water to Hubballi–Dharwad. Goa fears harm to its drinking water, estuary and wildlife sanctuaries. The tribunal's 2018 award allocated about 13.4 thousand million cubic feet (TMC) to Karnataka, 24 TMC to Goa and 1.3 TMC to Maharashtra; the states have appealed.

Ravi–Beas and the SYL canal. When Haryana was carved out of Punjab in 1966, the share of Ravi–Beas water due to Haryana had to be carried by a new **Sutlej–Yamuna Link (SYL) canal**. Haryana built its portion, but construction in Punjab stopped in the 1980s amid militancy and political opposition. In 2004 Punjab passed a law terminating its water agreements; the Supreme Court

declared it unconstitutional in 2016 and has repeatedly asked the two states to settle the matter. The canal remains incomplete, and Punjab argues that, with its own groundwater crisis, it has no surplus water to share.

The Cauvery dispute, the most prominent of all, is examined as a case study later in this chapter.

Key Points

- About one-eighth of India is flood-prone and about two-thirds of its cultivated area is drought-prone; human actions such as floodplain encroachment, deforestation and over-pumping turn hazards into disasters.
- Early warning, floodplain zoning, wetland protection and watershed development are as important as dams and embankments.
- The Indus Waters Treaty (1960) has been held in abeyance by India since April 2025; the Ganga Water Treaty (1996) expires in December 2026.
- China's upstream dams on the Yarlung Tsangpo are a growing concern for India.
- Inter-state disputes are adjudicated by tribunals under the Inter-State River Water Disputes Act 1956, but awards often take decades and are challenged.

3.17 Energy Resources: Renewable and Non-renewable

Energy is the capacity to do work. Every economic activity, from cooking a meal to running a steel plant, depends on it, and the way a country obtains its energy shapes its air quality, water use, land use, foreign-exchange bill and greenhouse-gas emissions. An **energy resource** is any natural source from which usable energy can be obtained, such as coal, sunlight or flowing water.

Energy units in brief

Energy is measured in **joules (J)**; 1 megajoule (MJ) = 1 million J. Electricity is usually measured in **kilowatt-hours (kWh)**: 1 kWh = 3.6 MJ, the energy used by a 1,000-watt heater in one hour. Heat values of fuels in India are often given in **kilocalories (kcal)**: 1 kcal = 4.187 kJ, so 1 MJ/kg ≈ 239 kcal/kg. **Power** is the rate of energy use, measured in watts (W). The **installed capacity** of a power station (in megawatts, MW, or gigawatts, GW) is its maximum power output; the energy it actually generates over a year (in kWh) depends on how many hours it runs.

3.17.1 Classification of energy resources

Energy resources are classified in several overlapping ways.

- **Non-renewable resources** exist in fixed amounts that formed over geological time and cannot be replaced within a human lifetime. They include the **fossil fuels** (coal, petroleum and natural gas), formed from buried plants and micro-organisms over millions of years, and **nuclear fuels** such as uranium and thorium.

- **Renewable resources** are replenished naturally at a rate equal to or faster than human use. They include solar, wind, hydro, biomass, geothermal and ocean energy (tidal, wave and thermal).
- **Conventional sources** (coal, oil, gas, large hydro and fuelwood) have been in widespread use for a long time; **non-conventional** or **alternate sources** (solar, wind, small hydro, modern biomass, geothermal, tidal, hydrogen) are newer.
- **Commercial energy** is bought and sold (coal, petroleum products, electricity). **Non-commercial energy** (fuelwood, crop residues, cow dung) is gathered freely and still meets much of the cooking energy need of poor rural households.
- **Primary energy** is energy in its natural form (coal, crude oil, sunlight). **Secondary energy** is a converted form, such as electricity or petrol, that carries energy to the user.

Nuclear power is **non-fossil** and almost carbon-free during operation, but it is not renewable because uranium reserves are finite. Large hydropower is renewable but has major environmental and social costs, which is why Indian policy long treated only small hydro (up to 25 MW) as "renewable" in official statistics.

Table 3.9: Renewable and non-renewable energy resources compared

Feature	Non-renewable (fossil and nuclear)	Renewable
Supply	Finite; will be depleted	Replenished continuously
Availability	Stored; can be used on demand	Solar and wind vary by hour and season; storage needed
Emissions	Fossil fuels release CO_2, SO_2, NO_x and particulates	Very low during operation
Land and water	Mining, ash ponds, cooling water	Large land area for solar and wind; dams submerge land
Cost trend	Fuel prices volatile; import-dependent	Capital-intensive but costs have fallen steeply
Indian examples	Coal in Jharkhand and Odisha; oil from Mumbai High; Kudankulam nuclear plant	Bhadla solar park, Muppandal wind farms, Bhakra hydro

3.17.2 India's energy mix

India is the world's third-largest energy consumer, after China and the United States, but its consumption per person is less than half the world average. Coal supplies more than half of India's primary energy, oil roughly a quarter to a third, and natural gas about 6%, with the remainder from biomass, hydro, nuclear and new renewables. India produced over one billion tonnes of coal for the first time in 2024–25 and has large coal resources in Jharkhand, Odisha, Chhattisgarh, West Bengal, Madhya Pradesh and Telangana. It is far poorer in oil and gas: in 2024–25 it imported close to 90% of its crude oil and about half of its natural gas, which makes the economy vulnerable to price shocks.

Electricity is where the energy transition is most visible. India's installed generating capacity reached about 549 GW on 30 June 2026 (see the table and figure below). Coal remained the largest single source, but solar had grown to almost 30% of capacity, and non-fossil sources together made up about 54%. India first crossed the 50% non-fossil mark in June 2025, five years ahead of the target in its updated Nationally Determined Contribution.

Table 3.10: India's installed electricity generation capacity by source (30 June 2026)

Source	Capacity (GW)	Share of total
Coal	224.2	40.8%
Lignite, gas and diesel	27.3	5.0%
Nuclear	8.8	1.6%
Large hydro (incl. pumped storage)	52.1	9.5%
Solar	162.2	29.5%
Wind	57.4	10.5%
Bio-power and small hydro	16.9	3.1%
Total	548.9	100%

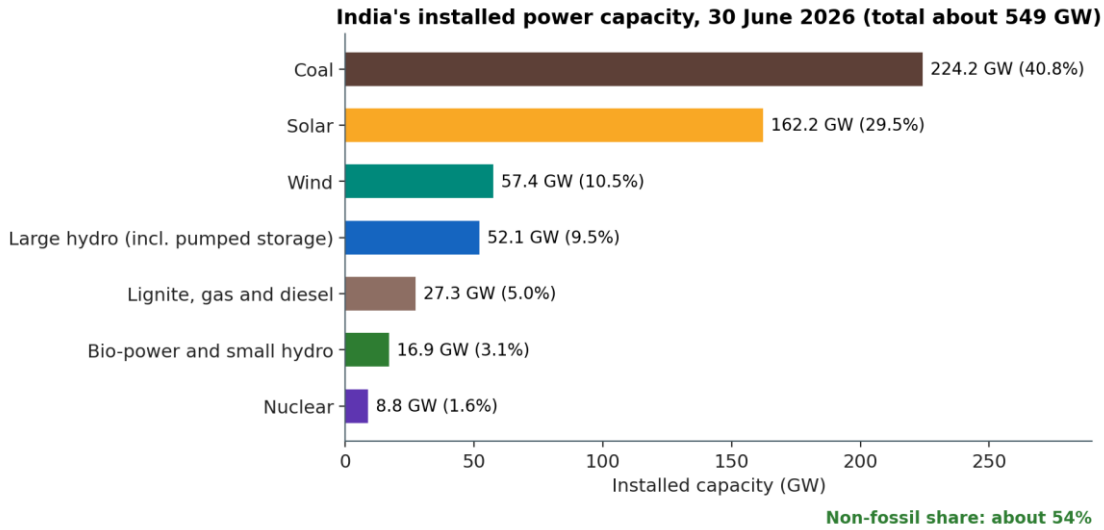


Figure 3.12: *India's installed electricity generation capacity by source, 30 June 2026 (total about 549 GW). Source: Central Electricity Authority; MNRE.*

Capacity is not the same as generation. A solar plant produces power only in daylight, so over a year it delivers about a fifth of what it would at full output all the time; this ratio is the **capacity utilisation factor (CUF)**. A coal plant may run at 60–70%. As a result, coal still generated roughly three-quarters of India's electricity in 2025–26 even though it was well under half of the capacity. Closing this gap requires not only more solar and wind but also **energy storage**, stronger transmission and flexible operation of existing plants.

3.18 Alternate Energy Sources and Growing Energy Needs

3.18.1 India's growing energy needs

India's demand for energy is rising faster than that of any other large economy. The drivers include a population of over 1.4 billion (140 crore), economic growth, urbanisation, industrialisation, near-universal household electrification (achieved by the Saubhagya scheme in 2019), and rising ownership of fans, air conditioners, refrigerators and vehicles. Hotter summers drive cooling demand. India's peak electricity demand rose from 250 GW in May 2024 to a record of about 260 GW on 19

May 2026, and the CEA projects a peak of about 459 GW by 2035–36. Even so, average electricity consumption per person, about 1,400 kWh a year in 2023–24 (CEA), is roughly a third of the world average.

Meeting this demand with coal alone would worsen air pollution (see Chapter 4) and climate change (see Chapter 5). India's commitments therefore aim to add most new capacity from non-fossil sources:

- At the Glasgow climate conference (COP26, 2021), India announced a target of **500 GW of non-fossil electricity capacity by 2030** and **net-zero emissions by 2070**.
- India's updated **Nationally Determined Contribution** (2022) commits to reducing the emissions intensity of GDP by 45% from 2005 levels by 2030 and to obtaining about 50% of installed electricity capacity from non-fossil sources by 2030 (reached in 2025).

3.18.2 Alternate energy sources

Alternate energy sources are sources that can replace fossil fuels with lower environmental impact. India's main options are outlined below.

Solar energy

India receives abundant sunshine, about 300 clear days a year in most regions. **Solar photovoltaic (PV)** cells, made mostly of silicon, convert sunlight directly into electricity. **Solar thermal** systems use sunlight to heat water or air, or concentrate it with mirrors to raise steam. The National Institute of Solar Energy estimated India's solar potential at about 748 GW, assuming only 3% of wasteland is covered with panels; rooftops, canal tops, reservoirs and farms add more. Installed solar capacity grew from about 10 MW in 2010 to about 162 GW by June 2026, the fastest growth of any source (see the National Solar Mission case study). Floating solar plants on reservoirs, such as NTPC's 100 MW plant at Ramagundam in Telangana and the Omkareshwar project in Madhya Pradesh, save land and reduce evaporation.

Wind energy

Wind turbines convert the kinetic energy of moving air into electricity. The National Institute of Wind Energy estimates India's onshore wind potential at about 1,164 GW at 150 metres above ground. Installed capacity was about 57 GW in June 2026, concentrated in Gujarat, Tamil Nadu, Karnataka, Maharashtra and Rajasthan; the Muppandal wind farms near Kanyakumari are among the oldest large clusters. Wind often blows strongly during the monsoon and at night, when solar output is low, so the two complement each other in **hybrid** projects. India has identified offshore wind zones off Gujarat and Tamil Nadu.

Hydropower

Hydropower converts the energy of falling water into electricity. Large hydro (above 25 MW) stood at about 52 GW in 2026, including pumped storage, with major stations at Bhakra, Tehri, Sardar Sarovar and Srisailam. **Small hydro** (up to 25 MW), often run-of-river plants in the Himalaya and Western Ghats, adds about 5 GW. Hydropower can be switched on and off quickly to balance the grid. **Pumped storage** plants pump water uphill to an upper reservoir when power is cheap (for example, at midday when solar is abundant) and release it through turbines when demand peaks, acting as giant batteries. Large dams, however, flood forests and villages and alter rivers, and Himalayan

projects face risks from earthquakes, landslides and floods, as the Chamoli disaster of February 2021 showed.

Biomass energy

Biomass is organic matter from plants and animals: wood, crop residues, cattle dung, bagasse (sugarcane waste), and municipal and food waste. It can be burnt for heat and power, converted to **biogas** by bacteria, or turned into liquid **biofuels** such as ethanol and biodiesel. Bagasse-based **cogeneration** in sugar mills, which produces both steam and electricity, provides most of India's bio-power capacity of about 11–12 GW. India's ethanol blending programme reached its target of 20% ethanol in petrol (E20) in 2025, using sugarcane molasses, cane juice and surplus grain. Biomass is renewable only if it is regrown, and burning it in open fires or traditional stoves causes heavy air pollution. Agro-residues are discussed in detail later in this chapter.

Geothermal energy

Geothermal energy is heat from the Earth's interior, tapped through hot springs or deep wells. The Geological Survey of India has identified over 380 hot springs, at sites such as Puga and Chumathang in Ladakh, Manikaran in Himachal Pradesh, and Tattapani in Chhattisgarh. India's potential is estimated at about 10 GW. There is no commercial geothermal power plant yet, but MNRE issued India's first **National Policy on Geothermal Energy** in September 2025, and ONGC is drilling exploratory wells at Puga for a demonstration plant.

Ocean energy

Tidal energy uses the rise and fall of tides to drive turbines. The Gulf of Khambhat and Gulf of Kutch in Gujarat, where the tidal range exceeds 5 metres, and the Sundarbans in West Bengal are the most promising sites, but proposed projects have not been built because of high costs and ecological concerns about mangroves and mudflats. **Wave energy** and **ocean thermal energy conversion (OTEC)**, which uses the temperature difference between warm surface water and cold deep water, have been tested only in pilot plants, such as the National Institute of Ocean Technology's low-temperature desalination plants in Lakshadweep.

Green hydrogen

Hydrogen releases only water when burnt or used in a fuel cell. **Green hydrogen** is made by splitting water with electricity from renewable sources (**electrolysis**); **grey hydrogen**, made from natural gas, releases CO_2. The **National Green Hydrogen Mission** (January 2023, outlay ₹19,744 crore) aims for 5 million tonnes of green hydrogen production a year by 2030 to replace grey hydrogen in refineries and fertiliser plants, decarbonise steel making, and produce green ammonia. Hydrogen is a way of storing and carrying renewable energy rather than a primary source, and it is still costly to make and transport.

Nuclear energy

Nuclear power generates heat by splitting uranium atoms (**fission**). India had about 8.8 GW of nuclear capacity in 2026 at stations such as Tarapur, Rawatbhata, Kudankulam and Kakrapar. India's three-stage programme aims eventually to use its large reserves of thorium. The government has set a goal of 100 GW of nuclear capacity by 2047. Nuclear power is non-fossil and reliable but expensive and slow to build, and it raises concerns about radioactive waste and accident risk (see Chapter 4).

3.18.3 Integrating alternate energy: storage and efficiency

Solar and wind are **variable**: output changes with the time of day, cloud and wind. A grid with high shares of them needs **battery energy storage systems**, pumped storage, demand that can shift in time (for example, charging electric vehicles at midday), and a strong transmission network, such as India's Green Energy Corridors, to move power from sunny and windy states to cities.

The cheapest unit of energy is the one not used. **Energy efficiency** programmes have made a real difference. The UJALA scheme (2015) distributed over 36 crore LED bulbs, cutting household lighting demand. The Bureau of Energy Efficiency's star labels guide buyers to efficient appliances, and the Perform, Achieve and Trade scheme sets energy-saving targets for heavy industries. The Energy Conservation (Amendment) Act 2022 created the legal basis for a domestic **carbon credit trading scheme**.

Think and Do: Graph your state's energy story

Using the CEA's annual "General Review" or the Ministry of Statistics' "Energy Statistics India" reports, collect your state's (or Delhi's) peak electricity demand and total electricity consumption for the last ten years. Plot both on a line graph. In which years did demand grow fastest? Then find your state's installed capacity of solar, wind and coal in the latest year and draw a pie chart. Discuss what has driven the changes.

3.19 Energy Contents of Coal, Petroleum, Natural Gas and Biogas

The usefulness of a fuel depends largely on how much energy it releases when burnt. The **calorific value (CV)** of a fuel is the amount of heat released by the complete combustion of a unit mass (or, for gases, a unit volume) of the fuel. It is measured in MJ/kg or kcal/kg for solids and liquids, and in MJ/m^3 or kcal/m^3 for gases. In the laboratory, calorific value is measured with a **bomb calorimeter**, in which a weighed sample is burnt in pure oxygen inside a sealed steel vessel surrounded by water, and the rise in water temperature is recorded.

Burning fuels that contain hydrogen produces water vapour. The **gross calorific value (GCV)**, or higher heating value, includes the heat recovered when this vapour condenses. The **net calorific value (NCV)**, or lower heating value, excludes it, because in most boilers and engines the vapour escapes up the chimney. NCV is therefore lower than GCV, by about 5% for coal and about 10% for natural gas. Indian coal is graded on GCV.

3.19.1 Coal

Coal is a black or brown sedimentary rock formed from plants buried in swamps millions of years ago and transformed by heat and pressure (**coalification**). As coalification proceeds, moisture and volatile matter fall and carbon content rises, so energy content increases. This sequence is called **rank**.

Table 3.11: Ranks of coal

Rank	Carbon content (approx.)	Typical GCV (MJ/kg)	Indian occurrence
Peat	below 60%	10–15	Not mined commercially
Lignite (brown coal)	60–70%	10–17	Neyveli (Tamil Nadu), Rajasthan, Gujarat
Sub-bituminous	70–76%	17–24	Many Gondwana coals
Bituminous	76–86%	24–33	Jharia (coking coal), Raniganj
Anthracite	above 86%	30–35	Very small deposits in Jammu and Kashmir

About 99% of India's coal is **Gondwana coal**, around 250–300 million years old, found in the Damodar, Mahanadi, Godavari and Son valleys. The small **Tertiary coal** deposits of the North-East are younger and richer in sulfur. Indian Gondwana coal is low in sulfur, which is an advantage, but it typically contains 30–45% **ash**, the incombustible mineral matter left after burning. High ash lowers calorific value, wears out boilers, and produces vast amounts of **fly ash**, more than 300 million tonnes a year, which must be used in cement and bricks or stored in ash ponds that can pollute water.

Since 2012 non-coking coal in India has been graded by GCV into 17 grades, from **G1** (above 7,000 kcal/kg) to **G17** (2,200–2,500 kcal/kg), each band 300 kcal/kg wide. Much of the coal burnt in power stations falls in the middle grades (roughly G10–G14, about 3,100–4,600 kcal/kg or 13–19 MJ/kg), well below imported bituminous coal at about 5,500–6,500 kcal/kg. **Coking coal**, needed to make coke for steel blast furnaces, is graded separately by ash content; India has limited reserves of good coking coal, mainly at Jharia, and imports much of what it needs.

3.19.2 Petroleum and its products

Petroleum (crude oil) is a liquid mixture of hydrocarbons formed from marine micro-organisms buried in sediments. India's main oil fields are Mumbai High offshore, and onshore fields in Assam (Digboi, the country's oldest refinery, dates from 1901), Gujarat and Rajasthan (Barmer). In a refinery, crude oil is separated by **fractional distillation** into products with different boiling ranges: LPG, petrol, naphtha, kerosene and aviation fuel, diesel, fuel oil and bitumen.

Liquid fuels have high energy content per kilogram (about 42–46 MJ/kg), are easy to store and transport, and burn readily in engines. That is why they dominate transport. Diesel contains more energy per litre than petrol because it is denser. **Liquefied petroleum gas (LPG)**, a mixture of propane and butane stored as a liquid under moderate pressure, is the main clean cooking fuel in India. The Pradhan Mantri Ujjwala Yojana (2016) has provided over 10 crore LPG connections to poor households, reducing indoor smoke from wood and dung fires.

3.19.3 Natural gas

Natural gas is mostly **methane** (CH_4, usually 80–95%), with smaller amounts of ethane, propane, butane, CO_2 and nitrogen. It is found with or near oil, as in the Krishna–Godavari basin and Mumbai High, and also in coal seams (**coal-bed methane**). It has the highest calorific value per kilogram of the fossil fuels (about 50–55 MJ/kg, or about 37–40 MJ per cubic metre, roughly 9,000–9,500 kcal/m^3) because methane has the highest ratio of hydrogen to carbon. It therefore releases the least CO_2 per

unit of energy, about 40% less than coal, and almost no ash or SO_2. Leaks of unburnt methane, a strong greenhouse gas, reduce this advantage.

- **Compressed natural gas (CNG)** is natural gas compressed to about 200–250 times atmospheric pressure in cylinders. Delhi's shift of buses, autos and taxis to CNG after a 1998 Supreme Court order is a well-known clean-air measure (see Chapter 7).
- **Liquefied natural gas (LNG)** is natural gas cooled to about −162 °C, when it shrinks to about 1/600 of its volume, allowing it to be shipped across oceans. India imports LNG at terminals such as Dahej and Kochi.
- **Piped natural gas (PNG)** supplies homes and industries through city gas distribution networks. India aims to raise the share of gas in its primary energy to 15% by 2030.

3.19.4 Biogas

Biogas is a mixture of gases produced when bacteria break down organic matter such as cattle dung, crop residues, kitchen waste and sewage in the absence of oxygen (**anaerobic digestion**). The process occurs in four stages: **hydrolysis** (large molecules broken into sugars, amino acids and fatty acids), **acidogenesis** (conversion to volatile fatty acids), **acetogenesis** (conversion to acetic acid, hydrogen and CO_2) and **methanogenesis**, in which methane-producing archaea form methane:

$$CH_3COOH \rightarrow CH_4 + CO_2$$

Table 3.12: Typical composition of biogas

Component	Share by volume
Methane (CH_4)	50–70%
Carbon dioxide (CO_2)	30–45%
Hydrogen sulfide (H_2S)	traces (up to about 1%)
Nitrogen, hydrogen, water vapour	small amounts

Because about a third of biogas is non-combustible CO_2, its calorific value is lower than that of natural gas, about 20–25 MJ/m^3 (about 5,000–6,000 kcal/m^3). One cubic metre of biogas can replace roughly 0.45 kg of LPG for cooking. When biogas is **upgraded** by removing CO_2, H_2S and moisture, it becomes **compressed biogas (CBG)** or bio-CNG, with at least 90% methane under the Indian standard (IS 16087), and can replace CNG in vehicles.

A household biogas plant has a **digester** in which a slurry of dung and water ferments for about 30–50 days, a **gas holder**, an inlet and an outlet for the spent slurry. India's two common designs are the **floating-drum** (KVIC) type, in which a steel drum rises as gas collects, and the cheaper **fixed-dome** (Deenbandhu) type, built of brick and cement, in which gas collects under the dome and pushes slurry into a displacement chamber (see the figure below). The spent slurry, called **digestate**, is an excellent organic manure. Biogas thus provides clean cooking fuel, reduces the burden of collecting firewood, improves sanitation and recycles nutrients. The government's GOBARdhan initiative supports village and community biogas plants.

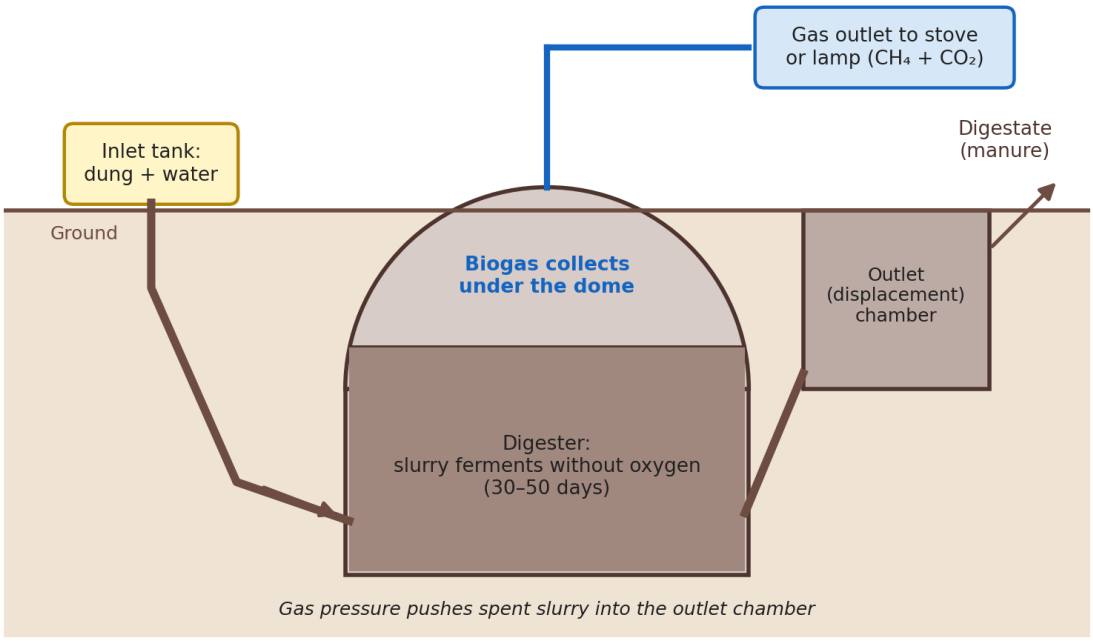


Figure 3.13: *A fixed-dome (Deenbandhu-type) biogas plant. Dung slurry enters the digester, bacteria produce biogas that collects under the dome, and the gas pressure pushes spent slurry into the outlet chamber. Source: Original diagram.*

3.19.5 Comparing energy contents

The table and figure below compare typical calorific values. Values vary with moisture, ash and composition, so they should be read as approximate.

Table 3.13: Approximate gross calorific values of common fuels

Fuel	MJ/kg	kcal/kg	Remarks
Cow-dung cake (dry)	8–10	1,900–2,400	Smoky; better used for biogas
Agro-residues (straw, husk)	13–16	3,100–3,800	Bulky; densified into briquettes and pellets
Fuelwood (air-dry)	15–18	3,600–4,300	Main rural cooking fuel
Lignite (Indian)	10–14	2,400–3,300	High moisture
Indian power-grade coal	13–19	3,100–4,600	High ash (30–45%)
Bituminous coal (high grade)	24–30	5,700–7,200	Imported thermal and coking coal
Crude oil	42–45	10,000–10,700	Refined into fuels
Diesel	45–46	10,700–11,000	About 38 MJ per litre
Petrol	46–47	11,000–11,200	About 34 MJ per litre
LPG	49–50	11,700–11,900	Cooking fuel
Natural gas / CNG	50–55	12,000–13,100	About 37–40 MJ/m^3
Biogas (60% methane)	about 20	about 4,800	About 20–25 MJ/m^3
Hydrogen	about 142	about 34,000	Highest per kg but very bulky as a gas

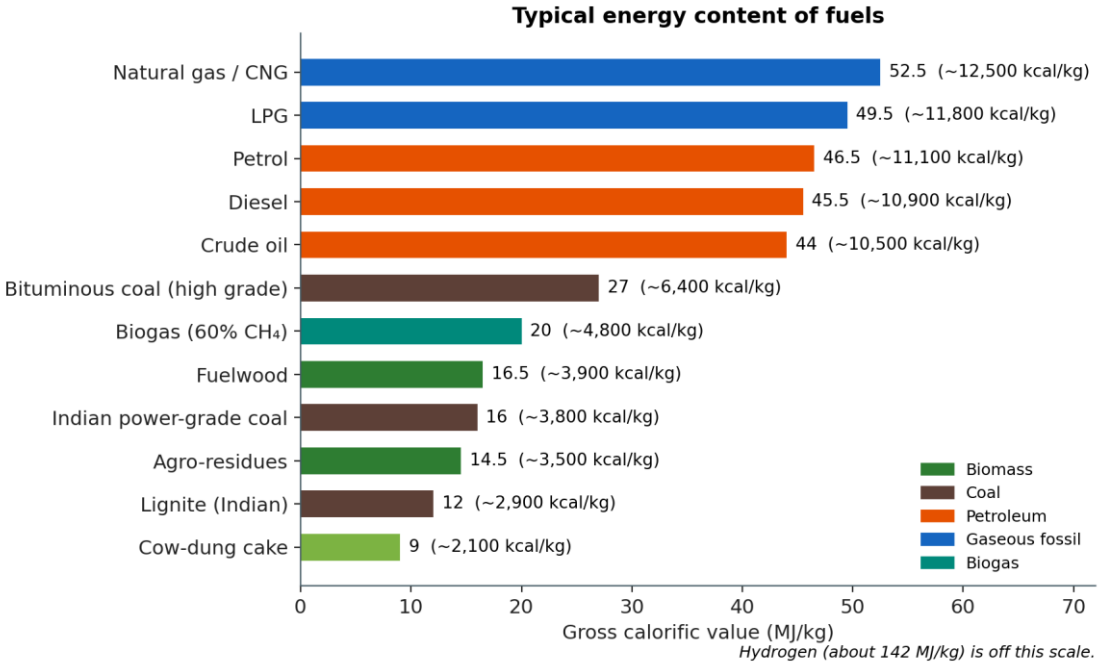


Figure 3.14: *Approximate gross calorific values of common fuels (MJ per kg; bars show typical values, with ranges in the table). Source: compiled from Bureau of Energy Efficiency, Ministry of Coal and IPCC data.*

Three lessons follow. First, fuels richer in hydrogen (natural gas, LPG, petrol) release more energy per kilogram and less CO_2 per unit of energy than carbon-rich coal; IPCC default factors are about 95 kg CO_2 per GJ for bituminous coal against about 56 kg CO_2 per GJ for natural gas. Second, India's coal yields less energy and much more ash per tonne than the coal of most other producers. Third, traditional biomass fuels such as dung cakes and wood have low calorific values and burn inefficiently in open stoves, which is why converting them into biogas, pellets or electricity gives much more useful energy and far less smoke.

3.20 Agro-residues as Biomass Energy

Agro-residues (crop residues) are the parts of crops left after the main product is harvested or processed. **Field residues** remain in the field: straw and stubble of paddy and wheat, stalks of cotton, maize, mustard and pigeon pea, and sugarcane tops and leaves. **Process residues** arise during processing: rice husk at rice mills, bagasse at sugar mills, groundnut shells, coconut shells and coir pith, and cotton gin waste.

As a major agricultural country, India produces very large quantities of residue. Estimates cited by MNRE put total biomass availability at about 750 million tonnes a year, of which roughly 230–250 million tonnes of agricultural residue is **surplus**, that is, left after uses as fodder, bedding, roofing, fuel and manure. This surplus could support about 28 GW of power generation. Rice straw, wheat straw and sugarcane residues make up the largest shares. Not all residue should be removed: leaving part of it on the soil protects against erosion and maintains soil organic carbon.

3.20.1 Stubble burning: a wasted resource

In Punjab, Haryana and western Uttar Pradesh, much paddy straw has traditionally been burnt in the fields in October and November. Combine harvesters leave tall stubble, the gap between the paddy harvest and wheat sowing is only two to three weeks (shortened further by the delayed transplanting

rules described earlier in this chapter), and removing straw by hand is costly. Non-basmati paddy straw is also high in silica and unpalatable to cattle.

Stubble burning releases particulate matter, carbon monoxide, black carbon and other pollutants. On some days in late October and early November it contributes a large share of the PM2.5 in Delhi's air, adding to the region's winter smog (see Chapter 4). Burning also kills soil organisms and wastes nutrients and organic matter.

The responses combine **in-situ management** (keeping straw in the field) and **ex-situ management** (removing it for use):

- The Crop Residue Management scheme (since 2018) subsidises machines such as the Happy Seeder and Super Seeder, which sow wheat directly through the straw, and balers that collect straw into bales.
- The **Pusa decomposer**, a microbial spray developed by the Indian Agricultural Research Institute, speeds up the decomposition of straw in the field.
- The Commission for Air Quality Management in the National Capital Region and Adjoining Areas monitors fires and directs states and industries to use paddy straw.

Satellite counts of farm fires in Punjab fell sharply after 2021, although some of the decline may reflect changes in burning times. Turning residue into energy gives farmers an income from straw instead of a reason to burn it.

3.20.2 Converting agro-residues to energy

Agro-residues can be converted to useful energy by thermochemical or biochemical routes (see the figure below).

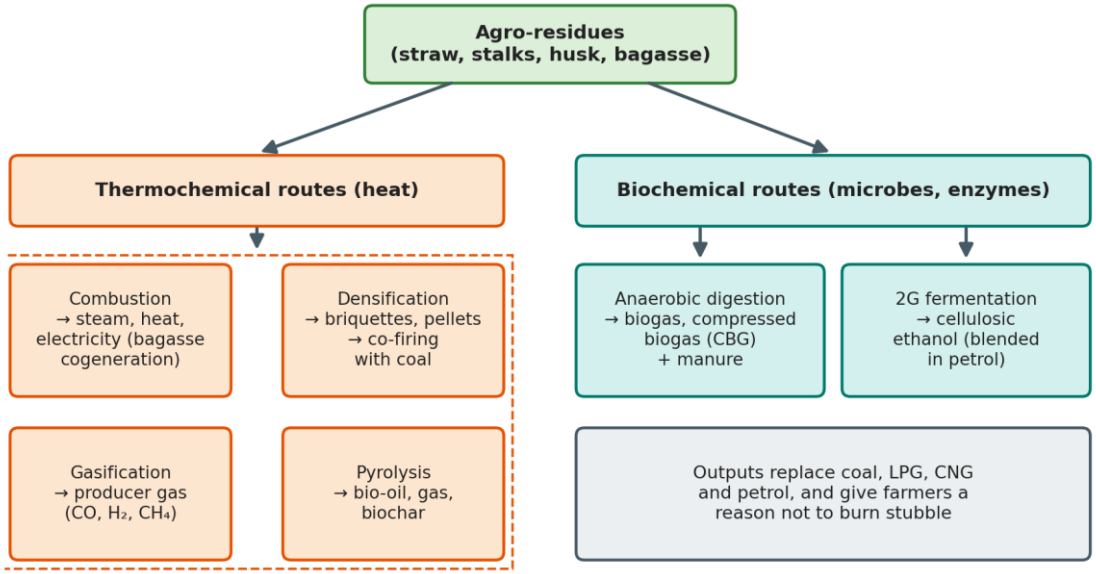


Figure 3.15: *Pathways for converting agro-residues into energy. Source: Original diagram.*

- **Direct combustion** in boilers produces steam for industry or electricity. Sugar mills burn bagasse in cogeneration plants, supplying their own needs and exporting surplus power. Dedicated biomass power plants burn rice husk, straw and cotton stalks.

- **Densification** compresses loose, bulky residues into **briquettes** and **pellets**, which are easier to transport and store and burn more evenly. **Torrefaction**, mild roasting at 200–300 °C, makes pellets more coal-like.
- **Co-firing** burns pellets with coal in existing power stations. The Ministry of Power's biomass policy (2021) and the **SAMARTH** mission (National Mission on Use of Biomass in Thermal Power Plants) required coal plants to co-fire 5% biomass pellets. From 2025–26, a revised policy requires 5% co-firing in coal plants generally and 7% in plants in the National Capital Region. Co-firing creates a large, steady market for straw pellets close to where stubble is burnt.
- **Gasification** heats biomass with limited air to produce **producer gas** (mainly CO, H_2 and CH_4), which can run engines for village-scale power or replace fossil fuel in industrial furnaces.
- **Anaerobic digestion** converts residues, often mixed with dung or press mud, into biogas and then **compressed biogas (CBG)**. The **SATAT** initiative (Sustainable Alternative Towards Affordable Transportation, launched in October 2018) invited entrepreneurs to set up 5,000 CBG plants, with oil companies guaranteeing to buy the gas. Progress has been slower than planned; over 130 plants had been commissioned by the end of 2025. One of the largest, at Lehragaga in Sangrur (Punjab), uses paddy straw. A **CBG blending obligation** requires city gas companies to blend CBG into CNG and piped gas, starting at 1% in 2025–26 and rising to 5% by 2028–29.
- **Second-generation (2G) ethanol.** First-generation ethanol is made from sugar and grain, which compete with food. **2G ethanol** is made from the cellulose in straw, stalks and other inedible biomass, which is broken down into sugars by enzymes and then fermented. Indian Oil's 2G ethanol plant at Panipat (Haryana), dedicated in August 2022, is designed to process about two lakh tonnes of rice straw a year. The PM JI-VAN Yojana (2019) provides financial support for such plants.
- **Pyrolysis** heats biomass without oxygen to produce bio-oil, gas and **biochar**, a charcoal that can be added to soil to store carbon and improve fertility.

3.20.3 Benefits and limitations

Using agro-residues for energy reduces air pollution from open burning, replaces imported fossil fuels, gives farmers extra income, creates rural jobs in collection and processing, and is close to carbon-neutral when crops are regrown. The challenges are practical: residues are scattered, bulky and seasonal, so collection, baling, transport and storage are expensive; quality varies; and removing too much residue depletes soil organic matter. Success depends on reliable supply chains, assured prices for farmers and steady demand from power plants, CBG plants and industries.

Key Points

- Calorific value is the heat released by complete combustion of a unit mass or volume of fuel; gases rich in hydrogen have the highest values, and high-ash Indian coal has relatively low values.
- Natural gas is mostly methane; biogas is 50–70% methane and 30–45% CO_2, with a calorific value of about 20–25 MJ/m^3.
- India produces hundreds of millions of tonnes of surplus crop residue; burning paddy straw pollutes the air over north India.
- Residues can be used for direct combustion, pellets and co-firing (5–7% mandate), gasification, compressed biogas (SATAT) and 2G ethanol.

3.21 Case Studies: Natural Resources in India

The case studies in this section show how the ideas of this chapter play out on the ground: who gains and who loses when rivers are dammed, forests mined, water shared or new energy systems built, and how communities have responded.

3.21.1 Contemporary Indian issues: mining, dams, forests and energy

India must find coal, minerals, water and land for a growing economy while protecting the forests, rivers and people that the same resources come from. Many of today's conflicts arise where these demands overlap. The first part of this chapter describes the general impacts of mining and dams; the examples below show how they appear in current debates.

- **Coal versus forests.** Much of India's coal lies under dense forests in central and eastern India, home to Adivasi communities. In Hasdeo Arand in Chhattisgarh, one of the largest continuous forests of central India and an elephant corridor, the expansion of coal mines has led to tree felling and a long protest by gram sabhas, who say their consent under the Forest Rights Act and PESA was not properly obtained. The Chhattisgarh assembly passed a resolution in 2022 asking that no new blocks be allotted there. The case highlights the tension between energy security and forest rights.
- **Himalayan hydropower and disaster risk.** In February 2021, a rock-and-ice avalanche in Chamoli, Uttarakhand, sent a flood down the Rishiganga and Dhauliganga that destroyed one hydropower project and badly damaged another at Tapovan, killing about 200 people, most of them workers. In January 2023, cracks spread through homes in Joshimath, a town built on old landslide debris, raising questions about tunnels, roads and unplanned construction. The destruction of the Teesta-III dam by a glacial lake outburst flood in 2023 added to concerns about building large dams in a fragile, warming mountain range.
- **Dams and river links versus wildlife.** The Ken–Betwa river link will submerge part of the core of Panna Tiger Reserve in Madhya Pradesh. The proposed Etalin hydropower project in Arunachal Pradesh would require felling a very large number of trees in one of India's richest biodiversity areas.
- **Forest law and large projects.** The Forest (Conservation) Amendment Act 2023 (Van (Sanrakshan Evam Samvardhan) Adhiniyam) exempts some categories of land and projects

from forest clearance, which critics say weakens protection for unrecorded forests (see Chapter 5). The Great Nicobar development project, which includes a port, airport and township, requires the diversion of about 130 km^2 of tropical rain forest on an island inhabited by the Shompen and Nicobarese peoples.

- **Clean energy also has footprints.** Solar parks and wind farms need large areas, and land classified as "wasteland" is often grassland, pasture or village common land (such as the *orans*, or sacred groves, of western Rajasthan). Overhead transmission lines kill the critically endangered Great Indian Bustard by collision. In 2021 the Supreme Court ordered that lines in its habitat be placed underground; in 2024 it modified the order, recognising both the bird's protection and the need for renewable energy, and set up an expert committee to find a balance.
- **A just transition.** Districts in Jharkhand, Chhattisgarh and Odisha depend on coal mining for jobs and public revenue. As the country moves towards cleaner energy, planning is needed so that workers and communities are not left behind. District Mineral Foundations, created in 2015, collect a share of mining royalties for the welfare of mining-affected areas, but their funds are often spent on works unrelated to the most affected people.

The common lesson is that resource decisions involve trade-offs among development, ecology and the rights of local communities. Good decisions need sound environmental impact assessment, honest public consultation, respect for forest rights, and attention to cumulative effects rather than one project at a time.

3.21.2 National Solar Mission

The **Jawaharlal Nehru National Solar Mission (JNNSM)** was launched on 11 January 2010 as one of the eight missions under India's National Action Plan on Climate Change (see Chapter 5). At the time, India had only about 10 MW of grid-connected solar power, and solar electricity cost around ₹17–18 per kWh, several times the cost of coal power.

Targets

The mission's original goal was 20 GW (20,000 MW) of grid-connected solar power by 2022, together with 2 GW of off-grid applications, 20 million m^2 of solar thermal collectors and 20 million solar lights. It was to proceed in three phases: Phase I (2010–13), Phase II (2013–17) and Phase III (2017–22). Early projects were supported by bundling expensive solar power with cheaper coal power and by viability gap funding.

In June 2015 the government raised the target five-fold to **100 GW by 2022**: 60 GW from utility-scale ground-mounted plants and 40 GW from rooftop systems. The same year, India and France launched the **International Solar Alliance** at the Paris climate conference, headquartered in Gurugram, to promote solar energy among sunny countries.

Achievements

India did not reach 100 GW by 2022; installed solar capacity was about 62 GW in March 2022, with rooftop solar lagging far behind its target. But growth then accelerated. India crossed 100 GW of solar capacity in January 2025 and reached about 162 GW by June 2026, making solar the second-largest source of capacity after coal (see the installed-capacity figure earlier in this chapter). India is now among the top three countries in solar capacity.

Several instruments made this possible:

- **Reverse auctions**, in which developers compete to offer the lowest tariff, drove solar tariffs down to below ₹2.50 per kWh, cheaper than power from new coal plants.
- **Solar parks.** Under a scheme launched in 2014, states and central agencies develop large parks with land, roads, water and transmission ready for developers. Major examples are Bhadla in Rajasthan (about 2,245 MW), Pavagada in Karnataka (about 2,050 MW), where farmers lease their land to the park and earn annual rent, and Rewa in Madhya Pradesh (750 MW), which supplies part of the Delhi Metro's power. The Khavda renewable energy park in Kutch, Gujarat, is planned at about 30 GW of solar and wind.
- **PM-KUSUM** (2019) helps farmers to install solar pumps and small solar plants. Component A supports small grid-connected solar plants on barren or fallow land; Component B supports standalone off-grid solar pumps, over ten lakh of which had been installed by early 2026; and Component C solarises existing grid-connected pumps so farmers can sell surplus power. By March 2026 the scheme had added about 13 GW against a target of 34.8 GW.
- **PM Surya Ghar: Muft Bijli Yojana** (2024), with an outlay of ₹75,021 crore, gives subsidies for rooftop solar to one crore households, which can then receive up to 300 units of free electricity a month. By mid-2026 more than 50 lakh households had benefited, and rooftop capacity had reached about 15 GW.
- **Domestic manufacturing** is supported by a production-linked incentive scheme and a list of approved domestic module makers, to reduce dependence on imported panels, especially from China.

Challenges

- **Land and local conflicts.** Large parks need about 1.5–2 ha per MW. Land that looks barren may be grazing commons or wildlife habitat, as the Great Indian Bustard case shows.
- **Grid integration.** Solar output peaks at noon and drops to zero at sunset, just as evening demand rises. Storage and flexible generation are needed, and in some states solar power has been curtailed because the grid could not absorb it.
- **Finances of distribution companies.** Many state electricity distribution companies are in debt and delay payments to generators, which raises risks for investors.
- **Water and waste.** Panels in dusty deserts need regular cleaning, often with scarce water. Old panels will become a large waste stream; solar panels are covered by the E-Waste (Management) Rules 2022.
- **Rooftop and imports.** Rooftop solar grew slowly until 2024, and India still imports most solar cells and wafers.

The National Solar Mission shows how a clear long-term target, supportive policy and falling technology costs can transform an energy system within fifteen years. Its next test is to integrate several hundred gigawatts of variable power reliably and fairly.

3.21.3 Cauvery river water conflict

The **Cauvery** (Kaveri) rises at Talakaveri in the Brahmagiri hills of Kodagu district, Karnataka, and flows about 800 km south-east through Karnataka and Tamil Nadu to the Bay of Bengal, forming a wide, fertile delta in Thanjavur and neighbouring districts. Its basin of about 81,000 km^2 lies mostly in Tamil Nadu (about 54%) and Karnataka (about 42%), with small parts in Kerala and Puducherry (Karaikal). The river has no glaciers; it depends on the south-west monsoon over its upper catchment

in Karnataka and Kerala and on the north-east monsoon over Tamil Nadu. When the monsoon fails, there is simply not enough water for everyone, and the dispute flares.

History

- **1892 and 1924 agreements.** The dispute began between the Madras Presidency (British India, downstream) and the princely state of Mysore (upstream). An 1892 agreement required Mysore to obtain Madras's consent for new irrigation works on the river. After arbitration over Mysore's plan for the Krishnaraja Sagar dam, the two signed the 1924 agreement, which allowed both the Krishnaraja Sagar dam in Mysore and the Mettur dam in Madras (completed in 1934), and fixed limits on irrigated area. Some of its clauses were to be reviewed after 50 years.
- **After 1974.** When the review fell due, Karnataka argued that the colonial agreements were unfair because Mysore had been the weaker party; Tamil Nadu insisted on protecting the irrigation its delta farmers had depended on for centuries. Decades of negotiations failed.
- **The tribunal.** On the Supreme Court's direction, the Union government set up the **Cauvery Water Disputes Tribunal** in June 1990 under the Inter-State River Water Disputes Act. Its 1991 interim order, requiring Karnataka to release 205 TMC a year to Tamil Nadu, triggered violence against Tamil residents in Karnataka. The tribunal's final award came in February 2007.
- **Supreme Court verdict, 2018.** Both states appealed. On 16 February 2018 the Supreme Court upheld the award with modifications. It declared that the river is a national asset, that no state can claim exclusive ownership, and that drinking water has the highest priority. It increased Karnataka's share by 14.75 TMC, partly for Bengaluru's drinking water, and allowed Tamil Nadu to compensate by drawing on groundwater in the basin. The verdict is to apply for 15 years.

Table 3.14: Cauvery water allocations (thousand million cubic feet, TMC, in a normal year)

Party	Tribunal award 2007	Supreme Court 2018
Tamil Nadu	419	404.25
Karnataka	270	284.75
Kerala	30	30
Puducherry	7	7
Environmental protection and inevitable escape to sea	14	14
Total	740	740

One TMC is about 28.3 million cubic metres. Under the 2018 verdict, Karnataka must ensure that 177.25 TMC reaches the inter-state border at Biligundlu in a normal year, in monthly instalments.

The Cauvery Water Management Authority

As directed by the Supreme Court, the Union government notified the Cauvery Water Management Scheme in June 2018, creating the **Cauvery Water Management Authority (CWMA)** and the **Cauvery Water Regulation Committee (CWRC)**. The CWMA supervises reservoir operations and ensures monthly releases according to the award; the CWRC monitors daily flows and rainfall. In "distress years" with poor rainfall, the award provides only that shares be reduced proportionately,

and deciding what is fair remains contentious. In 2016 and again in 2023, orders to release water to Tamil Nadu during a deficit year led to protests and bandhs in Karnataka.

Ongoing issues and lessons

New disputes keep arising. Karnataka's proposed **Mekedatu** reservoir near the Tamil Nadu border, meant mainly for Bengaluru's drinking water and power, is opposed by Tamil Nadu, which fears that upstream storage will reduce its share. Both states grow thirsty crops: paddy in the delta and in Mandya, and sugarcane in Karnataka. Deforestation in the catchment, sand mining and pollution from towns further reduce usable water.

The Cauvery dispute teaches that legal allocation alone cannot create water. Lasting solutions require basin-wide data shared openly, an agreed formula for distress years, efficient irrigation and crop diversification, catchment protection, and treating the river as a shared ecosystem rather than a pipe to be divided.

3.21.4 Sardar Sarovar dam

The **Narmada** rises at Amarkantak in Madhya Pradesh and flows about 1,300 km west through Madhya Pradesh, along the Maharashtra border and through Gujarat into the Gulf of Khambhat. The Narmada Valley Development plan envisaged 30 large, 135 medium and about 3,000 small dams in the basin. **Sardar Sarovar**, the largest and last dam on the main river, stands near Kevadia in Gujarat's Narmada district.

The project

Jawaharlal Nehru laid the foundation stone of an earlier, smaller design in 1961. Disagreement among the riparian states led to the **Narmada Water Disputes Tribunal** (1969), whose 1979 award divided the river's usable flow among Madhya Pradesh, Gujarat, Rajasthan and Maharashtra, fixed the dam's full reservoir level at 138.68 metres, and set rules for resettlement. Construction began in 1987, with a World Bank loan.

- **Dam.** A concrete gravity dam about 1.2 km long and 163 m high from its deepest foundation, among the largest in the world by volume of concrete.
- **Reservoir.** Gross storage of about 9.5 billion m^3, extending about 214 km upstream.
- **Power.** 1,450 MW (a 1,200 MW riverbed power house and a 250 MW canal-head power house), shared Madhya Pradesh 57%, Maharashtra 27% and Gujarat 16%.
- **Canal.** The Narmada main canal runs about 458 km to the Rajasthan border and is one of the largest lined irrigation canals in the world, feeding a planned network of tens of thousands of kilometres of branch and distribution canals.

Benefits claimed

The project was promoted as the lifeline of Gujarat. Official estimates promised irrigation for about 17.9 lakh hectares in Gujarat and 2.5 lakh hectares in Rajasthan, mostly in the drought-prone regions of Saurashtra, Kutch, north Gujarat, Barmer and Jalore; drinking water for over 9,000 villages and more than 130 towns and cities in Gujarat, as well as parts of Rajasthan; hydroelectric power for three states; and flood protection for the lower valley around Bharuch. Narmada water now supplies many towns in Saurashtra and Kutch through pipelines, and the area around the dam has become a tourist destination with the Statue of Unity.

Submergence and displacement

At full reservoir level the dam submerges about 37,690 hectares in 245 villages: 193 in Madhya Pradesh, 33 in Maharashtra and 19 in Gujarat. The submerged area includes farmland, about 13,000–14,000 hectares of forest, and riverbed land. Official figures count about 47,000 project-affected families (around 37,750 in Madhya Pradesh, 4,350 in Maharashtra and 4,750 in Gujarat). Activist estimates, which include those not officially recognised, put the number of affected people at well over two lakh. Many thousands more lost land to the canal network, the dam colony and later tourism projects without being counted as "oustees".

A majority of the displaced in Gujarat and Maharashtra were **Adivasis** (Bhil, Bhilala, Tadvi and Vasava communities) living in hill villages who depended on the forest and river for food, fuel, fodder and fish, often without formal land titles. In Madhya Pradesh's fertile Nimar plains, many large villages of settled farmers were submerged.

The tribunal's resettlement rules were progressive for their time: land for land (at least two hectares of irrigable land per family), resettlement at least six months before submergence, and resettlement as village communities. Implementation fell short. Resettlement sites often had poor or waterlogged land, inadequate drinking water and grazing land, and communities were scattered. In 2017, when the dam's gates were closed and the reservoir filled to full level, thousands of families in Madhya Pradesh were still in their villages, contesting their entitlements.

Environmental impacts

- **Forest and biodiversity loss** in the submerged valley, and fragmentation of habitats in the Satpura and Vindhya ranges. Compensatory afforestation was often carried out far away and with poor survival.
- **Downstream changes.** Reduced freshwater flow below the dam has allowed sea water to advance tens of kilometres up the estuary near Bharuch, increasing salinity in groundwater and harming the estuarine fishery, especially of hilsa.
- **Sedimentation and changed river ecology**, as the reservoir traps silt and turns a flowing river into a lake.
- **Risks in the command area**, such as waterlogging and salinity where canals irrigate poorly drained land.
- **Delays in the canal network.** For many years after the dam was raised, only part of the distribution canals had been built, so the promised irrigation lagged far behind the dam's construction, while costs rose many-fold from about ₹6,400 crore at 1986–87 prices.

Controversy and completion

The **Narmada Bachao Andolan (NBA)**, led by Medha Patkar and supported by Baba Amte, mobilised the affected people from the mid-1980s, questioning both the resettlement record and the project's benefits. An independent review commissioned by the World Bank (the Morse review, 1992) found serious failures in resettlement and environmental planning, and in 1993 India gave up the remaining World Bank loan and continued with its own funds. In 2000 the Supreme Court allowed construction to proceed in stages, with resettlement to keep pace with each increase in height. The dam was raised in steps from 2000 to 2017; its gates were closed in 2017, and the Prime Minister dedicated it to the nation on 17 September 2017. The reservoir reached full level for the first time in 2019. The NBA as

a social movement, and the wider questions of resettlement and rehabilitation, are discussed in Chapter 7.

Sardar Sarovar remains the most debated dam in India. Supporters point to water now reaching drought-prone Saurashtra and Kutch; critics point to the human and ecological costs, delayed benefits, and unfulfilled resettlement promises. The case shaped India's later policies on resettlement and environmental clearance and influenced the global debate on large dams, including the World Commission on Dams (2000).

3.21.5 Chipko and Appiko movements

The **Chipko movement** (from the Hindi word for "to cling") began in 1973 in Chamoli district of present-day Uttarakhand, when villagers organised by Chandi Prasad Bhatt's Dasholi Gram Swarajya Sangh prevented the felling of ash trees allotted to a sports-goods company at Mandal. In March 1974, women of Reni village led by Gaura Devi stopped contractors from felling their forest by hugging the trees. The movement, later championed by Sunderlal Bahuguna, linked commercial logging to floods and landslides in the Himalaya and to the loss of local livelihoods, and in 1980 led to a 15-year ban on green felling in the Uttar Pradesh Himalaya. Its history, its leadership by women and its legacy are covered in Chapter 7.

The **Appiko movement** (from the Kannada word for "to hug") began in September 1983 in the Kalase forest of Uttara Kannada district, Karnataka, inspired by Chipko. Villagers led by Pandurang Hegde hugged trees to stop commercial felling for timber and plywood. Its message of "save, grow and use rationally" combined protest with tree planting and fuel-efficient stoves, and it helped to halt clear-felling in the Western Ghats forests of the district. It too is discussed further in Chapter 7.

3.21.6 Tarun Bharat Sangh

Tarun Bharat Sangh (TBS) is a voluntary organisation founded in 1975 at the University of Rajasthan in Jaipur. In 1985, Rajendra Singh, a young member of TBS, went to Gopalpura village in the Thanagazi area of Alwar district, intending to run health and education work. The villagers told him that their most urgent need was water. Wells had dried, forests in the Aravalli hills had been cut, and young men had migrated to cities.

Reviving johads

Guided by village elders, TBS and the villagers began to desilt and rebuild **johads**, the traditional crescent-shaped earthen check dams that collect monsoon runoff on slopes. A johad holds water for some months, but more importantly it lets the water seep slowly into the ground, recharging wells downstream. Villagers contributed labour and part of the cost, which created a sense of ownership. As wells filled again, farmers could grow a second crop, cattle had fodder, and families that had migrated began to return. The approach spread from village to village; TBS reports helping communities build thousands of water-harvesting structures in over a thousand villages of Rajasthan and neighbouring states.

The Arvari river and its parliament

The best-known result is the revival of the **Arvari river** in Alwar district. The Arvari had stopped flowing outside the monsoon. From the late 1980s, villages in its catchment built hundreds of johads and small dams and protected their forests. By the mid-1990s the river was flowing throughout the year again. When the state government granted a private contractor a licence to fish in the revived

river, the villagers protested that the river belonged to the community that had restored it, and the licence was withdrawn.

In 1999 the villages of the basin formed the **Arvari Sansad** (Arvari River Parliament), with representatives from about 70 villages. It frames rules for the whole catchment: for example, discouraging very water-intensive crops such as sugarcane, prohibiting pumping directly from the river, banning the sale of water and land to outside industries, and protecting forests and grazing land. Other rivers in the region, such as the Ruparel, Sarsa, Bhagani and Jahajwali, were also revived through similar efforts.

Wider impact

TBS also campaigned against mining in and around Sariska Tiger Reserve; in the early 1990s the Supreme Court ordered the closure of mines in the area. Rajendra Singh, often called the "Waterman of India", received the Ramon Magsaysay Award in 2001 and the Stockholm Water Prize in 2015.

The Tarun Bharat Sangh story shows that decentralised, low-cost water harvesting based on traditional knowledge and community institutions can restore groundwater and rivers in semi-arid regions. It also has limits: many small structures can reduce flows to downstream areas, recharge can be wasted if pumping is not controlled, and the model depends on strong local cooperation. These are the same lessons that today's national programmes, such as Atal Bhujal Yojana and Jal Shakti Abhiyan, try to apply at scale.

Key Points

- Contemporary conflicts over coal, Himalayan hydropower, forest law and even clean energy show the trade-offs among development, ecology and local rights.
- The National Solar Mission (2010) raised its target from 20 GW to 100 GW; India crossed 100 GW of solar in January 2025 and about 162 GW by mid-2026, supported by solar parks, PM-KUSUM and PM Surya Ghar.
- The Cauvery dispute, rooted in the 1892 and 1924 agreements, was settled by the 2007 tribunal award as modified by the Supreme Court in 2018 and is administered by the CWMA.
- Sardar Sarovar promised irrigation, drinking water and power to Gujarat and Rajasthan but submerged 245 villages, displaced tens of thousands of mostly Adivasi families and altered the Narmada estuary.
- Chipko (1973) and Appiko (1983) used tree-hugging to defend forests; Tarun Bharat Sangh revived the Arvari river through johads and a community river parliament.

Chapter 3 Review

Summary

- Natural resources are classified as renewable, non-renewable or inexhaustible; a renewable resource such as groundwater behaves like a non-renewable one when it is used faster than it is replenished.
- India has about 2.4% of the world's land and 4% of its fresh water but nearly 18% of its people, so pressure on land resources is intense.
- Minerals form through magmatic, hydrothermal, sedimentary, residual, metamorphic and placer processes; most of India's coal and metallic ores lie in the Chota Nagpur–Odisha plateau and other parts of the Peninsula.
- Mining is regulated through environmental and forest clearances, mine closure plans and auctions; District Mineral Foundations receive a share of royalty for the welfare of mining-affected people, but their funds are often poorly targeted.
- Soil forms very slowly from weathered rock under the influence of parent material, climate, relief, organisms and time, and develops a profile of O, A, E, B, C and R horizons.
- Alluvial, black, red and yellow, and laterite soils are India's main soil groups; low organic carbon and unbalanced fertiliser use threaten soil health, and the Soil Health Card scheme provides farm-level testing.
- The Green Revolution made India self-sufficient in food but caused groundwater depletion, fertiliser and pesticide overuse, stubble burning and loss of crop diversity, most acutely in Punjab; crop diversification, millets and natural farming are responses.
- Non-timber forest products such as tendu, mahua, bamboo and lac support millions of forest dwellers; MSP for minor forest produce and the Van Dhan Yojana aim to raise collectors' incomes.
- India has about 7,000 or more medicinal plant species; overharvesting of roots and bark threatens species such as jatamansi, kuth and guggul, and the National Medicinal Plants Board, the Biological Diversity Act and TKDL support conservation and benefit sharing.
- About 275 million rural Indians depend partly on forests; Joint Forest Management and Community Forest Resource rights under the Forest Rights Act give communities a role in managing forests.
- Land cover is what covers the land; land use is how people use it. ISRO and NRSC monitor land change by satellite.
- About 97.85 million hectares (29.7%) of India's land was undergoing degradation in 2018–19, mainly through water erosion, vegetation loss and wind erosion; India aims to restore 26 million hectares of degraded land by 2030.
- Soil erosion by water (sheet, rill, gully) and wind can be controlled by agronomic, mechanical and biological measures applied on a watershed basis; desertification is land degradation in drylands, addressed internationally by the UNCCD.
- India's forest and tree cover is 25.17% of its area (ISFR 2023), but about 1.73 lakh hectares of forest land were diverted for projects between 2014–15 and 2023–24; deforestation has both direct and underlying causes.

- Mining and dams damage land, water, forests and biodiversity and have displaced large numbers of Adivasis; Niyamgiri, Jharia, Goa, the Aravallis and Hasdeo Arand illustrate these impacts and the importance of gram sabha consent.
- India has about 18% of the world's population but only about 4% of its fresh water. The Central Water Commission (2019) puts India's average annual water resource at 1,999 BCM, of which about 1,137 BCM is utilisable.
- Per-capita water availability fell from 5,177 m^3 in 1951 to 1,486 m^3 in 2021, below the water-stress threshold of 1,700 m^3.
- Natural water sources include rivers, lakes, wetlands, groundwater in aquifers, glaciers and springs. Man-made sources include dams, reservoirs, canals, tanks, stepwells, rainwater harvesting, desalination and treated wastewater.
- Agriculture accounts for roughly 80–90% of India's water withdrawal. Efficient irrigation and crop choice are the biggest levers for saving water, and virtual water links crop exports to aquifer depletion.
- India is the world's largest user of groundwater. In 2025 the national stage of extraction was 60.63%, but about 11% of assessment units were over-exploited, and Punjab, Rajasthan and Haryana extracted far more than was recharged.
- Free farm power, cheap pumps, paddy procurement, weak laws and urbanisation drive depletion. Its effects include falling water tables, contamination, seawater intrusion and land subsidence.
- Chennai (2019) and Bengaluru (2024) show how urban water crises follow the loss of lakes and wetlands. Atal Bhujal Yojana, Jal Jeevan Mission and Jal Shakti Abhiyan are major national responses.
- Floods and droughts are natural hazards made worse by deforestation, floodplain encroachment, urbanisation, over-pumping and climate change. Early warning, floodplain zoning and watershed development reduce the risk.
- India's international water relations include the Indus Waters Treaty (held in abeyance by India since April 2025), the Ganga Water Treaty (1996, due to expire in 2026), the unresolved Teesta issue and concern over Chinese dams on the Brahmaputra.
- Inter-state water disputes are adjudicated by tribunals under the Inter-State River Water Disputes Act 1956. These tribunals are often slow, as on the Krishna, Mahadayi and Ravi–Beas (SYL) disputes.
- Energy resources are non-renewable (coal, oil, gas, nuclear) or renewable (solar, wind, hydro, biomass, geothermal, ocean). India's installed power capacity was about 549 GW in June 2026, with over half from non-fossil sources, but coal still generated most electricity.
- India targets 500 GW of non-fossil capacity by 2030 and net-zero emissions by 2070, using solar, wind, hydro, biomass, storage, green hydrogen, nuclear power and energy efficiency.
- Calorific values range from about 9 MJ/kg for dung cake and about 16 MJ/kg for Indian power-grade coal to about 52 MJ/kg for natural gas. Biogas (50–70% methane) yields about 20–25 MJ/m^3.
- Surplus crop residues could generate large amounts of energy through co-firing, pellets, compressed biogas (SATAT) and 2G ethanol instead of being burnt in fields.

- Case studies of the National Solar Mission, the Cauvery dispute, Sardar Sarovar, Chipko, Appiko and Tarun Bharat Sangh show the trade-offs and community responses that shape India's use of natural resources.

Key Terms

Natural resource: any material, energy source or service from nature that people use to meet their needs.

Renewable resource: a resource replenished by natural processes within a human lifetime, provided it is not used faster than it regenerates.

Non-renewable resource: a resource formed over geological time whose stock is effectively fixed, such as coal or metallic ores.

Mineral: a naturally occurring inorganic solid with a definite chemical composition and ordered crystalline structure.

Ore: a rock containing a valuable mineral in high enough concentration to be mined at a profit.

Overburden: soil and rock lying above a mineral deposit that must be removed in surface mining.

Tailings: finely ground waste left after the valuable mineral has been separated from ore.

District Mineral Foundation: a trust in each mining-affected district, funded by a share of mining royalty, that works for the benefit of people and areas affected by mining.

Pedogenesis: the process of soil formation from weathered rock and organic matter.

Soil profile: a vertical section through the soil showing its horizons from the surface to bedrock.

Humus: dark, stable organic matter in soil formed by the decomposition of plant and animal remains.

Soil Health Card: a card given to farmers that reports the nutrient status of their soil and recommends fertiliser doses.

Green Revolution: the rapid rise in Indian food production from the mid-1960s based on high-yielding seeds, irrigation, fertilisers and price support.

Cropping intensity: the ratio of gross cropped area to net sown area, expressed as a percentage.

Natural farming: chemical-free farming that relies on on-farm inputs, mulching and mixed cropping.

Minor forest produce (MFP): non-timber products of plant origin collected from forests, such as tendu leaves, mahua, bamboo, lac and honey.

Van Dhan Vikas Kendra: a cluster of tribal self-help groups that processes and markets forest produce under the Pradhan Mantri Van Dhan Yojana.

Joint Forest Management: an arrangement in which village committees protect state forests in return for a share of forest produce.

Community Forest Resource rights: rights under the Forest Rights Act, 2006 that allow a gram sabha to protect, regenerate and manage its customary forest.

Land cover: the physical and biological material covering the land surface, such as forest, crops, water or buildings.

Land use: the purpose for which people use land, such as agriculture, settlement or conservation.

Land degradation: a reduction or loss of the biological or economic productivity of land.

Land Degradation Neutrality: a state in which the amount and quality of land resources remains stable or increases (SDG Target 15.3).

Soil erosion: the detachment and removal of soil by water, wind, ice or gravity.

Gully erosion: erosion in which runoff cuts deep channels that cannot be removed by ploughing.

Desertification: land degradation in arid, semi-arid and dry sub-humid areas caused by climatic variations and human activities.

Shelterbelt: rows of trees or shrubs planted across the prevailing wind to reduce wind speed and erosion.

Deforestation: permanent clearing of forest and conversion of the land to a non-forest use.

Forest cover: land of more than one hectare with tree canopy density above 10%, as mapped by the Forest Survey of India.

Acid mine drainage: acidic, metal-rich water formed when sulfide minerals exposed by mining react with air and water.

Aquifer: A layer of rock or sediment that stores and transmits enough groundwater to supply wells.

Artesian well: A well tapping a confined aquifer in which water pressure pushes water up without pumping.

Water table: The upper surface of the zone in which all pores in soil or rock are filled with water.

Per-capita water availability: The average annual water resource of a region divided by its population.

Water stress: A condition in which annual water availability falls below 1,700 m^3 per person.

Rainwater harvesting: Collecting and storing rain where it falls, for direct use or to recharge groundwater.

Johad: A traditional crescent-shaped earthen check dam in Rajasthan that stores runoff and recharges groundwater.

Stepwell (baoli): A traditional well with steps leading down to the water, built to store and give access to groundwater.

Virtual water: The water used to produce a good or service, which is hidden in the product and in trade.

Water footprint: The total volume of fresh water used to produce the goods and services consumed by a person, product or country.

Environmental flow: The quantity, timing and quality of water that must remain in a river to sustain its ecosystem and dependent users.

Stage of groundwater extraction: Annual groundwater extraction expressed as a percentage of the annual extractable groundwater resource.

Over-exploited unit: An assessment unit where annual groundwater extraction exceeds annual extractable recharge (stage above 100%).

Seawater intrusion: The movement of salt water into coastal freshwater aquifers caused by over-pumping.

Meteorological drought: A significant shortfall of rainfall compared with the long-term normal.

Glacial lake outburst flood (GLOF): A sudden flood caused by the failure of a natural dam holding back a glacial lake.

Floodplain zoning: Regulating land use on floodplains according to flood risk to reduce damage.

Riparian state: A state or country through which a river flows or along whose banks it lies.

Equitable and reasonable utilisation: The principle of international water law that shared rivers should be used fairly by all riparian states.

Water disputes tribunal: A quasi-judicial body set up under the Inter-State River Water Disputes Act 1956 to adjudicate disputes between states.

Renewable energy: Energy from sources that are naturally replenished as fast as they are used, such as sunlight, wind and flowing water.

Capacity utilisation factor: The ratio of the actual energy produced by a plant over a period to the energy it would produce at full capacity throughout.

Calorific value: The amount of heat released by complete combustion of a unit mass or volume of a fuel.

Gross calorific value: The calorific value that includes the heat recovered from condensing water vapour formed in combustion.

Anaerobic digestion: The breakdown of organic matter by micro-organisms in the absence of oxygen, producing biogas.

Biogas: A fuel gas of about 50–70% methane and 30–45% carbon dioxide, produced by anaerobic digestion.

Compressed biogas (CBG): Biogas purified to at least 90% methane and compressed for use like CNG.

Co-firing: Burning biomass pellets together with coal in a thermal power station.

Second-generation (2G) ethanol: Ethanol made from cellulose in inedible plant material such as straw, rather than from sugar or grain.

Green hydrogen: Hydrogen produced by electrolysis of water using electricity from renewable sources.

Multiple-Choice Questions

1. Which of the following is an inexhaustible resource?

(a) Groundwater
(b) Coal
(c) Solar radiation
(d) Topsoil

2. Bauxite deposits on the plateaus of Odisha and Jharkhand were formed mainly by which process?

(a) Residual weathering under hot, wet conditions
(b) Crystallisation from cooling magma
(c) Sorting of heavy grains by waves
(d) Evaporation of sea water

3. Under the MMDR Act as amended in 2015, holders of mining leases granted after January 2015 pay what share of their royalty to the District Mineral Foundation?

(a) 2%
(b) 10%
(c) 30%
(d) 50%

4. In a soil profile, the horizon where clay and iron oxides washed down from above accumulate is the

(a) O horizon
(b) A horizon
(c) E horizon
(d) B horizon

5. Black (regur) soil, ideal for cotton, develops mainly from which parent rock?

(a) Granite
(b) Basalt of the Deccan Traps
(c) River alluvium
(d) Limestone

6. Which of the following is NOT one of the parameters tested under the Soil Health Card scheme?

(a) Organic carbon
(b) Electrical conductivity
(c) Soil temperature
(d) Zinc

7. Punjab's stage of groundwater extraction (annual extraction as a percentage of annual extractable recharge) is about

(a) 60%
(b) 100%
(c) 125%
(d) 164%

8. Tendu leaves, an important minor forest produce of central India, are mainly used for

(a) rolling bidis
(b) making paper pulp
(c) producing lac
(d) extracting edible oil

9. The Indian Forest (Amendment) Act, 2017 is significant because it

(a) banned all bamboo harvesting
(b) removed bamboo grown outside forests from the definition of "tree"
(c) made bamboo a Schedule I species
(d) transferred bamboo forests to the Ministry of AYUSH

10. Which medicinal plant is the source of the anticancer drug taxol (paclitaxel) and is threatened by bark stripping in the Himalaya?

(a) Ashwagandha
(b) Neem
(c) Himalayan yew
(d) Sarpagandha

11. Community Forest Resource rights in India are granted under which law?

(a) Indian Forest Act, 1927
(b) Wildlife (Protection) Act, 1972
(c) Forest Rights Act, 2006
(d) Biological Diversity Act, 2002

12. According to the Desertification and Land Degradation Atlas of India (ISRO, 2021), the leading process of land degradation in India is

(a) water erosion
(b) wind erosion
(c) salinisation
(d) waterlogging

13. At UNCCD COP14 in New Delhi (2019), India committed to restore how much degraded land by 2030?

(a) 5 million hectares
(b) 13 million hectares
(c) 26 million hectares
(d) 50 million hectares

14. A farmer builds a series of level steps on a steep hillside in Uttarakhand to reduce runoff. This measure is called

(a) strip cropping
(b) bench terracing
(c) shelterbelt planting
(d) mulching

15. According to ISFR 2023, India's total forest and tree cover is about what share of its geographical area?

(a) 21.76%
(b) 25.17%
(c) 29.7%
(d) 33.3%

16. In the Niyamgiri case (2013), the Supreme Court directed that the decision on bauxite mining be taken by

(a) the Odisha High Court
(b) the Ministry of Mines
(c) the gram sabhas of affected villages
(d) the National Green Tribunal

17. Acid mine drainage is formed when

(a) limestone dissolves in rainwater

(b) sulfide minerals exposed by mining react with air and water

(c) reservoirs trap sediment

(d) coal dust settles on crops

18. Which of these is an underlying, rather than a direct, cause of deforestation?

(a) Felling trees for a highway

(b) Clearing forest for a coal mine

(c) Forest fires set to clear undergrowth

(d) Insecure land tenure of forest dwellers

19. According to the Central Water Commission's 2019 reassessment, India's average annual water resource is about:

(a) 690 BCM

(b) 1,137 BCM

(c) 1,999 BCM

(d) 4,000 BCM

20. A country is described as water-stressed when its annual per-capita water availability falls below:

(a) 500 m^3

(b) 1,000 m^3

(c) 1,700 m^3

(d) 2,500 m^3

21. A well tapping a confined aquifer in which water rises above the ground without pumping is called:

(a) an artesian well

(b) a stepwell

(c) a recharge well

(d) a dug well

22. Which sector withdraws the largest share of water in India?

(a) Industry

(b) Domestic supply

(c) Thermal power

(d) Agriculture

23. An assessment unit is classified as "over-exploited" when its stage of groundwater extraction is:

(a) above 100%

(b) between 90% and 100%

(c) between 70% and 90%

(d) below 70%

24. In the CGWB's 2025 groundwater assessment, which state had the highest stage of groundwater extraction?

(a) Rajasthan
(b) Punjab
(c) Haryana
(d) Tamil Nadu

25. Atal Bhujal Yojana is mainly concerned with:

(a) building large dams in the Himalaya
(b) providing tap connections to urban homes
(c) community-led management of groundwater in water-stressed gram panchayats
(d) cleaning the Ganga

26. The 2009 Punjab law that delays paddy transplanting until June saves groundwater but has also been linked to:

(a) arsenic contamination
(b) the burning of paddy stubble
(c) urban flooding in Chennai
(d) seawater intrusion

27. Under the Indus Waters Treaty, which rivers were allocated to India for unrestricted use?

(a) Indus, Jhelum and Chenab
(b) Ganga, Yamuna and Sutlej
(c) Ravi, Beas and Sutlej
(d) Chenab, Ravi and Beas

28. In its 2018 Cauvery judgment, the Supreme Court fixed Karnataka's annual share at:

(a) 270 TMC
(b) 284.75 TMC
(c) 404.25 TMC
(d) 419 TMC

29. Which constitutional provision allows Parliament to provide for the adjudication of inter-state river water disputes?

(a) Article 21
(b) Article 48A
(c) Article 262
(d) Article 356

30. Which of the following is a non-fossil but non-renewable source of electricity?

(a) Nuclear power
(b) Wind power
(c) Small hydropower
(d) Bagasse cogeneration

31. Which fuel has the highest gross calorific value per kilogram among these?

(a) Indian power-grade coal
(b) Fuelwood
(c) Biogas
(d) Natural gas

32. The main combustible component of biogas is:

(a) carbon monoxide
(b) methane
(c) hydrogen sulfide
(d) propane

33. Indian Gondwana coal is generally characterised by:

(a) high ash and low sulfur
(b) low ash and high sulfur
(c) high carbon content like anthracite
(d) very low moisture and no ash

34. The SATAT initiative promotes:

(a) rooftop solar for households
(b) compressed biogas plants
(c) offshore wind farms
(d) nuclear power plants

35. The original 2010 grid-connected target of the National Solar Mission was raised in 2015 to:

(a) 20 GW
(b) 40 GW
(c) 100 GW
(d) 500 GW

36. The Arvari River Parliament (Arvari Sansad) was formed by villages associated with:

(a) Tarun Bharat Sangh in Alwar, Rajasthan
(b) the Appiko movement in Uttara Kannada
(c) the Narmada Bachao Andolan
(d) the Chipko movement in Chamoli

37. At full reservoir level, the Sardar Sarovar dam submerges villages in which states?

(a) Gujarat and Rajasthan only
(b) Madhya Pradesh, Maharashtra and Gujarat
(c) Madhya Pradesh and Rajasthan
(d) Maharashtra and Karnataka

Short-Answer Questions

1. Distinguish between renewable, non-renewable and inexhaustible resources with two Indian examples of each. Why can a renewable resource become effectively non-renewable?
2. Explain any three processes by which mineral deposits form, giving an Indian example of each.

3. What are District Mineral Foundations? How are they funded, and what criticisms have been made of their functioning?
4. Draw a labelled diagram of a soil profile and describe the characteristics of each horizon.
5. Describe the distribution, properties and major crops of alluvial and black soils of India.
6. What is the Soil Health Card scheme? List the parameters tested and state two limitations of the scheme.
7. Explain three environmental costs of the Green Revolution in Punjab.
8. What is meant by minor forest produce? Explain how MSP for MFP and the Van Dhan Yojana aim to benefit forest-dwelling communities.
9. Why are medicinal plants whose roots or bark are used more vulnerable to overharvesting? Give two Indian examples.
10. Differentiate between land cover and land use, and between deforestation and forest degradation.
11. Describe the stages of water erosion from splash erosion to gully formation.
12. What is Land Degradation Neutrality? State India's target and two programmes that contribute to it.
13. Distinguish between an unconfined and a confined aquifer, and explain why hard-rock aquifers of peninsular India are easily over-exploited.
14. Calculate the volume of rainwater that can be harvested each year from a 200 m^2 roof in a city receiving 800 mm of rain, using a runoff coefficient of 0.8. Show your working.
15. Describe any four traditional water-harvesting systems of India and the regions where they are found.
16. Explain the concepts of virtual water and water footprint with an Indian example.
17. How is the stage of groundwater extraction calculated, and how are assessment units categorised?
18. List the causes and consequences of groundwater depletion in Punjab.
19. Differentiate between meteorological, agricultural and hydrological drought.
20. What are the main provisions of the Ganga Water Treaty of 1996, and why is its renewal significant?
21. Explain the difference between gross and net calorific value. Why does Indian coal have a relatively low calorific value?
22. Describe the composition of biogas and the working of a fixed-dome biogas plant.
23. Why do farmers in Punjab and Haryana burn paddy stubble? Suggest three ways of turning this residue into energy.
24. What were the main achievements of the National Solar Mission and the main challenges it still faces?

Long-Answer Questions

1. Using data from the Desertification and Land Degradation Atlas of India, discuss the extent, causes and consequences of land degradation and desertification in India. Suggest an integrated strategy, including agronomic, mechanical and biological measures, to combat them.

2. "The Green Revolution solved one resource problem but created others." Critically examine this statement with reference to Punjab, and evaluate crop diversification, millets and natural farming as alternatives.
3. Discuss the role of non-timber forest products, medicinal plants and forest-based industries in the livelihoods of India's forest-dependent people. How have Joint Forest Management and the Forest Rights Act changed the relationship between communities and forests?
4. Analyse the direct and underlying causes of deforestation in India. Why might national forest-cover figures from ISFR 2023 not fully reflect the loss of natural forests?
5. Evaluate the impacts of mining on the environment, forests, biodiversity and tribal communities in India, with reference to any two of the following: Niyamgiri, Jharia, Goa, the Aravalli hills and Hasdeo Arand. What legal safeguards exist, and how effective have they been?
6. Large dams have been called the "temples of modern India". Assess their environmental and social costs, with Indian examples, and suggest ways to reduce these costs.
7. "India is not short of water; it is short of water management." Discuss this statement with reference to India's water budget, the uses of water and the over-exploitation of groundwater, using examples from Punjab, Chennai and Bengaluru.
8. Explain the natural and human causes of floods and droughts in India. Using the Uttarakhand (2013), Kerala (2018) and Marathwada droughts as examples, evaluate structural and non-structural approaches to managing these hazards.
9. Critically examine India's international and inter-state water conflicts. Discuss the Indus Waters Treaty and its suspension, the Cauvery dispute and the working of the Inter-State River Water Disputes Act 1956, and suggest reforms.
10. Compare renewable and non-renewable energy resources. Using India's installed capacity and generation mix, discuss how the country can meet its growing energy needs while achieving its 500 GW non-fossil target.
11. Analyse the Sardar Sarovar project as a case study of the costs and benefits of large dams, covering its claimed benefits, submergence, displacement and environmental impacts.
12. "Agro-residues are a resource, not a waste." Discuss with reference to stubble burning, calorific values, conversion technologies and government programmes such as SAMARTH, SATAT and 2G ethanol.

Further Reading

- Gadgil, M. and Guha, R. (1993). *This Fissured Land: An Ecological History of India*. Oxford University Press.
- McCully, P. (1996). *Silenced Rivers: The Ecology and Politics of Large Dams*. Zed Books.
- Singh, J. S., Singh, S. P. and Gupta, S. R. (2017). *Ecology, Environmental Science and Conservation*. S. Chand.
- Raven, P. H., Hassenzahl, D. M., Hager, M. C., Gift, N. Y. and Berg, L. R. (2015). *Environment* (9th ed.). Wiley.
- Forest Survey of India (2024). *India State of Forest Report 2023*. Ministry of Environment, Forest and Climate Change. https://fsi.nic.in
- Space Applications Centre, ISRO (2021). *Desertification and Land Degradation Atlas of India*. https://www.sac.gov.in

- Goraya, G. S. and Ved, D. K. (2017). *Medicinal Plants in India: An Assessment of their Demand and Supply*. National Medicinal Plants Board and ICFRE.
- Ministry of Mines (2024–25). *Annual Report*. Government of India. https://mines.gov.in
- Central Ground Water Board (2025). *Dynamic Ground Water Resources of India, 2025*. Ministry of Jal Shakti, Government of India.
- Central Water Commission (2019). *Reassessment of Water Availability in India Using Space Inputs*. Ministry of Jal Shakti, Government of India.
- Central Electricity Authority. *Installed Capacity Reports* (monthly) and *General Review*. Ministry of Power. https://cea.nic.in
- Ministry of New and Renewable Energy. *Physical Progress* and *Annual Reports*. https://mnre.gov.in
- Agarwal, A. and Narain, S. (eds.) (1997). *Dying Wisdom: Rise, Fall and Potential of India's Traditional Water Harvesting Systems*. Centre for Science and Environment.

Answers to Multiple-Choice Questions

1. (c)	2. (a)	3. (b)	4. (d)	5. (b)	6. (c)	7. (d)	8. (a)	9. (b)	10. (c)
11. (c)	12. (a)	13. (c)	14. (b)	15. (b)	16. (c)	17. (b)	18. (d)	19. (c)	20. (c)
21. (a)	22. (d)	23. (a)	24. (b)	25. (c)	26. (b)	27. (c)	28. (b)	29. (c)	30. (a)
31. (d)	32. (b)	33. (a)	34. (b)	35. (c)	36. (a)	37. (b)			

Chapter 4: Environmental Pollution and Control

Pollution is an undesirable change in the physical, chemical or biological characteristics of air, water or soil that harms, or is likely to harm, human health, other living organisms, property or natural processes. The substance or form of energy that causes the change is a **pollutant**. Pollutants can be material (smoke, sewage, pesticides, plastics), or they can be energy, such as heat, noise and ionising radiation. In India, pollution is now among the largest causes of illness and early death: an Indian Council of Medical Research (ICMR) study published in *The Lancet Planetary Health* (2021) attributed about 1.67 million deaths in 2019, or 17.8% of all deaths in the country, to air pollution alone. This chapter explains where pollutants come from, how they behave in air, water and soil, how they affect people and ecosystems, how they are measured against standards, and how they can be controlled.

Learning Objectives

- Define pollution and pollutants, and classify pollutants by source, persistence and form.
- Explain the causes, effects and control of air, water, soil, thermal and noise pollution, with Indian examples.
- Distinguish primary from secondary air pollutants and describe how photochemical smog and secondary particles form.
- Interpret India's air and water quality standards (NAAQS, AQI, CPCB water-use classes, IS 10500) and compare them with WHO guidelines.
- Describe the types, units, sources and health effects of ionising radiation, and evaluate nuclear hazards using global and Indian cases.
- Explain the management of urban, industrial, hazardous and electronic waste, including segregation and disposal.
- Compare legal, biological and physico-chemical methods of pollution control and assess their role in sustainability.
- Analyse Indian case studies such as the Ganga Action Plan, Delhi's air pollution, the Plastic Waste Management Rules and the Bhopal gas tragedy.

4.1 Pollution and Pollutants

Every environment receives some waste naturally. Volcanoes release sulfur dioxide, dust storms lift soil into the air, and dead leaves decay in streams. Natural systems can absorb, dilute and break down a certain amount of such material; this capacity is called **assimilative capacity**. Pollution occurs when the input of a substance exceeds this capacity, so that it builds up to harmful levels. The idea explains why the same sewage discharge may be harmless in a large, fast river during the monsoon and severely polluting in a small, slow stream in May.

A pollutant usually moves through three stages before it causes harm. It is released at a **source**, it is carried and changed in the environment (**transport and transformation**), and it finally reaches a **receptor**, which may be a person, a crop, a fish, a building or an ecosystem. Pollution control can act at any of these stages, but stopping the pollutant at its source is almost always cheaper and more effective than treating its effects. The figure below shows this source–pathway–receptor chain.

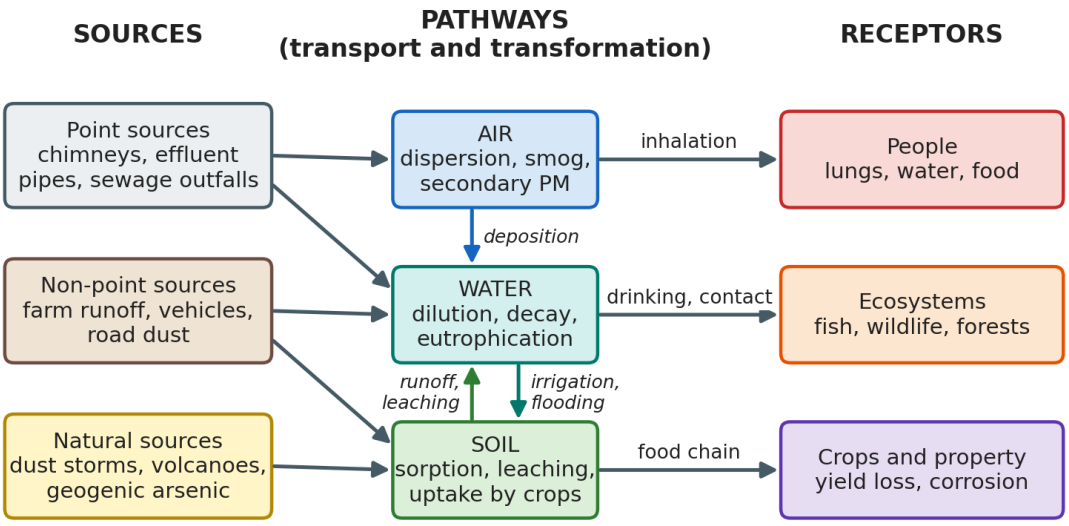


Figure 4.1: *The source–pathway–receptor model of pollution. Pollutants released from point and non-point sources are carried through air, water and soil, may be transformed on the way, and finally reach human and ecological receptors. Source: Author.*

4.1.1 Classification of pollutants

Pollutants can be grouped in several ways. Each classification answers a different practical question, such as where to place a control device or how long a pollutant will last.

By source: point and non-point

A **point source** discharges pollutants from a single, identifiable location, such as a factory chimney, the outfall pipe of a sewage treatment plant or the drain of a tannery. Point sources are relatively easy to monitor and regulate because the discharge can be measured and a permit can be attached to it. A **non-point source** releases pollutants over a wide area with no single outlet. Fertiliser and pesticide runoff from farmland, dust from unpaved roads, and the exhaust of millions of vehicles are non-point sources. They are much harder to control because responsibility is spread across many people. In India, the share of river pollution coming from non-point sources such as agricultural runoff and open defecation remains poorly measured, which is one reason river clean-up programmes that focus only on sewage pipes have had limited success.

By persistence: biodegradable and non-biodegradable

Biodegradable pollutants are broken down by bacteria, fungi and other decomposers into simple, harmless substances such as carbon dioxide, water and mineral salts. Sewage, kitchen waste, paper and animal dung are examples. They become pollutants only when their quantity exceeds the decomposing capacity of the environment; for instance, untreated sewage in a river uses up dissolved oxygen as it decays. **Non-biodegradable pollutants** are not broken down, or are broken down extremely slowly, by natural processes. Heavy metals (lead, mercury, cadmium, arsenic), most plastics, many synthetic pesticides such as DDT, and radioactive waste belong to this group. They persist for years to centuries, and some of them accumulate in living tissue.

The build-up of a persistent pollutant inside an organism faster than it can be excreted is called **bioaccumulation**. When such a pollutant becomes more concentrated at each higher level of a food

chain, the process is called **biomagnification**. Biomagnification explains why fish-eating birds and humans who eat contaminated fish carry far higher concentrations of mercury or DDT than the water itself (see Chapter 2 for food chains).

Other useful classifications

- **By origin**: **natural** pollutants (volcanic gases, pollen, wind-blown dust, geogenic arsenic in groundwater) and **anthropogenic** pollutants (vehicle exhaust, industrial effluents, pesticides).
- **By the medium affected**: air, water, soil, and the energy forms of noise, heat and radiation. These form the main sections of this chapter.
- **By formation in air**: **primary** pollutants emitted directly and **secondary** pollutants formed by reactions in the atmosphere (explained later in this chapter).
- **By form**: gases (SO_2, CO), particles (dust, soot), liquids (oil), and energy (heat, sound, radiation).
- **By existence in nature**: **quantitative pollutants**, which occur naturally but become harmful in excess (CO_2, nitrate, fluoride), and **qualitative pollutants**, which are entirely human-made (DDT, plastics, chlorofluorocarbons).

Table 4.1: Classification of pollutants with examples

Basis	Category	Indian example
Source	Point	Effluent pipe of a Kanpur tannery
Source	Non-point	Fertiliser runoff from Punjab paddy fields
Persistence	Biodegradable	Domestic sewage entering the Yamuna
Persistence	Non-biodegradable	Lead and cadmium from battery recycling
Origin	Natural	Geogenic arsenic in Bengal groundwater
Formation in air	Secondary	Ozone over Delhi on summer afternoons

> **Pollution is about dose, not just presence**
>
> Fluoride in small amounts (about 0.5–1.0 mg/L in drinking water) helps prevent tooth decay, but above 1.5 mg/L it causes dental and skeletal fluorosis. Many substances are harmful only above a threshold. Standards, explained later in this chapter, are attempts to define that threshold for each pollutant. For some pollutants, such as fine particles and ionising radiation, no completely safe threshold has been found.

4.2 Air Pollution

Air pollution is the presence in the atmosphere of one or more substances (gases, particles or biological materials) in concentrations and for durations that harm human health, living organisms or property, or that interfere with the comfortable enjoyment of life.

Air pollution is the most serious pollution problem in India. According to the World Health Organization (WHO), 99% of the world's population breathes air that exceeds WHO guideline levels, and ambient and household air pollution together cause about 6.7 million premature deaths a year

worldwide (WHO, 2022). India carries a large share of this burden: the *State of Global Air 2024* report estimated about 2.1 million deaths in India in 2021 linked to air pollution. The Air Quality Life Index (AQLI 2025) found that every Indian lives in an area where annual fine-particle levels exceed the WHO guideline, and that meeting the guideline would add several years to average life expectancy in the Indo-Gangetic Plain.

4.2.1 Sources of air pollution

Sources of air pollution are either natural or anthropogenic.

Natural sources include volcanic eruptions (sulfur dioxide, ash), forest fires (smoke, carbon monoxide), dust storms (coarse particles; the pre-monsoon dust storms that blow from the Thar Desert across north India are an example), sea spray (salt particles), pollen and fungal spores, lightning (nitrogen oxides) and the decay of organic matter in wetlands (methane). Vegetation also releases volatile organic compounds such as isoprene, which can take part in ozone formation.

Anthropogenic sources are those created by human activity. In India they fall into the following broad groups:

- **Power generation**: coal-based thermal power plants generate about three-quarters of India's electricity and are the largest source of sulfur dioxide, as well as major sources of nitrogen oxides, fly ash and mercury. Large plant clusters occur at Singrauli (Madhya Pradesh–Uttar Pradesh), Korba (Chhattisgarh) and Talcher (Odisha).
- **Industry**: steel, cement, fertilisers, refineries and chemicals, and small units such as foundries, metal-recycling workshops and brick kilns. India has more than one lakh brick kilns, most of them in the Indo-Gangetic Plain.
- **Transport**: petrol and diesel vehicles emit nitrogen oxides, carbon monoxide, hydrocarbons and fine particles at street level, where people breathe. India has more than 30 crore registered motor vehicles.
- **Households**: burning of wood, dung cakes, crop residue, kerosene and coal for cooking and heating. This is the largest source of exposure for rural women and children (discussed under indoor air pollution below).
- **Agriculture**: open burning of paddy straw in Punjab and Haryana in October–November, and ammonia from fertilisers and livestock, which helps form secondary particles.
- **Dust**: re-suspended road dust and construction and demolition activity, which are major contributors to coarse particles in cities.
- **Waste burning**: open burning of municipal solid waste and fires at landfills such as Ghazipur and Bhalswa in Delhi.

The GBD MAPS study by the Health Effects Institute and IIT Bombay (2018) found that household burning of solid fuels was the largest single contributor to deaths from outdoor fine particles nationally, followed by coal burning and agricultural burning. Delhi's sources are examined as a case study later in this chapter.

4.2.2 Major air pollutants

Six pollutants dominate air-quality concerns worldwide and in India: particulate matter, sulfur dioxide, nitrogen oxides, carbon monoxide, ground-level ozone and lead. Ammonia, benzene and polycyclic aromatic hydrocarbons are also monitored in India.

Particulate matter (PM) is a mixture of solid particles and liquid droplets suspended in air. Particles are classified by their aerodynamic diameter. **PM10** (particles 10 micrometres (μm) or smaller) can enter the nose, throat and upper lungs. **PM2.5** (2.5 μm or smaller, about one-thirtieth the width of a human hair) can penetrate deep into the air sacs of the lungs, and the smallest of them pass into the bloodstream. PM includes soot (black carbon), dust, sulfates, nitrates, ammonium salts, organic compounds and metals. PM2.5 is the pollutant most strongly linked to deaths from heart disease, stroke, lung cancer and chronic lung disease.

Sulfur dioxide (SO_2) is a colourless gas with a sharp smell, formed when sulfur-containing fuels such as coal and heavy fuel oil are burned and when sulfide ores are smelted. Indian coal is low in sulfur, but the huge quantities burned make power plants the dominant SO_2 source. SO_2 irritates the airways and is a precursor of sulfate particles and acid rain.

Nitrogen oxides (NO_x), mainly nitric oxide (NO) and nitrogen dioxide (NO_2), form when nitrogen and oxygen in air combine at the high temperatures inside engines and furnaces. Vehicles and power plants are the main sources. NO_2 is a reddish-brown, toxic gas that inflames the airways. NO_x is a precursor of ozone, nitrate particles and acid rain.

Carbon monoxide (CO) is a colourless, odourless gas produced by the incomplete combustion of carbon-containing fuels in vehicles, chulhas, charcoal braziers and generators. It binds to haemoglobin about 200 times more strongly than oxygen, forming carboxyhaemoglobin and reducing the blood's ability to carry oxygen. Low doses cause headache and dizziness; high doses in closed rooms, such as a coal brazier left burning in a sealed room on a winter night, can be fatal.

Ground-level ozone (O_3) is not emitted directly. It forms when NO_x and volatile organic compounds react in sunlight, and it is therefore a secondary pollutant (explained in the next section). Ozone is a strong oxidant that damages lung tissue and reduces crop yields. It should not be confused with the protective ozone layer in the stratosphere (see Chapter 5).

Lead (Pb) was once emitted mainly from leaded petrol. India completed the phase-out of leaded petrol in 2000, and airborne lead in cities fell sharply. Exposure now comes mainly from informal lead-acid battery recycling, smelters, some paints and adulterated spices, and contaminated soil and dust. Lead damages the developing brain, and no safe level of exposure is known for children.

Volatile organic compounds (VOCs) are carbon-based chemicals that evaporate easily, such as benzene, toluene and formaldehyde. Sources include petrol evaporation at fuel stations, solvents, paints and vehicle exhaust. Benzene causes leukaemia. **Ammonia (NH_3)**, from fertilisers and animal waste, reacts with acidic gases to form ammonium sulfate and ammonium nitrate particles.

Table 4.2: Major air pollutants, their main sources and effects

Pollutant	Main sources in India	Main health effects	Main environmental effects
PM2.5 and PM10	Biomass and coal burning, vehicles, dust, industry	Heart and lung disease, lung cancer, low birth weight	Haze, poor visibility, soot on glaciers
SO_2	Coal power plants, smelters, refineries	Airway irritation, asthma attacks	Acid rain, sulfate particles
NO_x	Vehicles, power plants, industry	Airway inflammation, reduced lung function	Ozone, nitrate particles, acid rain

Pollutant	Main sources in India	Main health effects	Main environmental effects
CO	Vehicles, chulhas, braziers	Less oxygen to tissues; fatal at high doses	Helps form ozone
O_3 (ground level)	Formed from NO_x and VOCs in sunlight	Chest pain, coughing, asthma	Crop and forest damage
Lead	Battery recycling, smelters, old paint	Brain damage in children, anaemia	Soil contamination

4.2.3 Effects of air pollution on human health

Air pollution enters the body mainly through the lungs, but its effects reach almost every organ. The WHO lists ischaemic heart disease, stroke, chronic obstructive pulmonary disease (COPD), lung cancer and lower respiratory infections as the main causes of death linked to air pollution.

- **Respiratory system**: short-term exposure to high levels of PM, SO_2, NO_2 and ozone triggers coughing, wheezing, asthma attacks and hospital admissions. Long-term exposure slows lung growth in children and causes COPD and lung cancer. The International Agency for Research on Cancer (IARC) classified outdoor air pollution and particulate matter as Group 1 carcinogens (known to cause cancer in humans) in 2013.
- **Cardiovascular system**: fine particles cause inflammation and stiffening of blood vessels and increase the risk of heart attacks, irregular heartbeat and stroke. Globally, more air-pollution deaths are due to heart disease and stroke than to lung disease.
- **Pregnancy and child development**: exposure during pregnancy is linked to low birth weight and preterm birth.

Some groups are more vulnerable than others. **Children** breathe more air per kilogram of body weight than adults, and their lungs are still developing. **Elderly people** and those with heart or lung disease are more easily tipped into serious illness. **Outdoor workers**, such as traffic police, construction workers, street vendors and rickshaw pullers, and **poor households** living near busy roads, industrial areas or landfills receive higher doses with fewer means of protection. The ICMR-led study cited at the start of this chapter estimated that lost output due to air pollution cost India about 1.4% of its GDP in 2019.

4.2.4 Effects of air pollution on the environment

- **Vegetation and crops**: ozone enters leaves through stomata and damages cells, reducing photosynthesis; studies estimate that ozone reduces wheat and rice yields in India by several per cent each year. SO_2 causes chlorosis (yellowing) and necrosis (death) of leaf tissue, and dust deposits block sunlight.
- **Acid rain**: SO_2 and NO_x are converted in the atmosphere into sulfuric and nitric acids, which fall as acid rain and acidify soils and lakes (see Chapter 5).
- **Buildings and monuments**: acidic gases react with marble and limestone. Concern over yellowing and corrosion of the Taj Mahal by emissions from the Mathura refinery and the foundries of Agra led to the Supreme Court's 1996 judgment in *M.C. Mehta v. Union of India* (the

Taj Trapezium case), which ordered polluting units in the area to switch to natural gas or relocate.

- **Visibility and climate**: particles scatter and absorb sunlight, producing haze and smog. Winter smog in the Indo-Gangetic Plain disrupts rail, road and air traffic. Black carbon deposited on Himalayan snow and ice darkens the surface and speeds melting. Several air pollutants, including black carbon, methane and ozone, also warm the climate, so many air-quality measures bring climate benefits (see Chapter 5).

4.2.5 Indoor air pollution

Indoor (household) air pollution is pollution of the air inside homes, schools and workplaces. In India, its main cause is the burning of solid fuels such as firewood, crop residue, dung cakes and coal in traditional **chulhas** (simple clay or brick stoves) that have no chimney. Incomplete combustion in these stoves releases large amounts of PM2.5, carbon monoxide, benzene, formaldehyde and polycyclic aromatic hydrocarbons into the kitchen. During cooking, PM2.5 levels in such kitchens commonly reach several hundred to a few thousand $\mu g/m^3$, many times the national 24-hour standard of 60 $\mu g/m^3$.

The WHO estimates that household air pollution from cooking with polluting fuels caused about 3.2 million deaths worldwide in 2020, mainly from pneumonia in young children, COPD, heart disease, stroke and lung cancer. Women, who do most of the cooking, and infants kept close to the stove receive the highest exposure. Collecting fuelwood also costs women and girls many hours each week. Smoke from chulhas escapes outdoors too, adding substantially to outdoor PM2.5 in rural India.

Other indoor pollutants include tobacco smoke, mosquito coils, incense, formaldehyde from plywood furniture, moulds, and **radon**, a natural radioactive gas that seeps from soil and rocks into poorly ventilated buildings and is the second most important cause of lung cancer after smoking.

Solutions include switching to clean fuels (liquefied petroleum gas (LPG), piped natural gas, biogas, electricity), using improved cookstoves with chimneys or forced draught, cooking in well-ventilated spaces, and keeping homes smoke-free.

Case Study: Pradhan Mantri Ujjwala Yojana: clean cooking fuel for poor households

The Pradhan Mantri Ujjwala Yojana (PMUY) was launched by the Ministry of Petroleum and Natural Gas on 1 May 2016 at Ballia, Uttar Pradesh, to provide free LPG connections to women from poor households. Its aim was to replace chulhas burning wood and dung with a cleaner fuel, and so reduce household air pollution, drudgery and the health burden on women and children. The first target of 5 crore connections was raised to 8 crore and met in 2019. Ujjwala 2.0 (2021) added more connections, including for migrant families who could not produce proof of address. By 2025 the scheme had issued about 10.3 crore connections.

The scheme sharply increased LPG access. The National Family Health Survey (NFHS-5, 2019–21) found that about 59% of Indian households used clean fuel for cooking, up from about 44% in NFHS-4 (2015–16). However, owning a connection is not the same as using it every day. Many poor households refill their cylinders only a few times a year because of the cost, and continue to use free firewood and dung alongside LPG, a pattern called **fuel stacking**. To encourage regular use, the government gives PMUY beneficiaries a targeted subsidy on refills. The Ujjwala experience shows that clean-air policies must address affordability and habits, not just supply.

4.2.6 Control of air pollution

Air pollution can be controlled at three levels: by preventing pollutants from forming (**source control**), by capturing them before they leave a chimney or exhaust pipe (**end-of-pipe control**), and by reducing people's exposure. The legal and policy framework for these measures, including the role of the Central Pollution Control Board (CPCB), the State Pollution Control Boards (SPCBs) and the National Clean Air Programme, is discussed later in this chapter.

Control at source

- **Cleaner fuels**: switching from coal and diesel to natural gas, LPG or electricity; using low-sulfur fuels (BS-VI petrol and diesel contain no more than 10 ppm sulfur); and washing coal to reduce its ash content. Delhi's conversion of buses, autorickshaws and taxis to compressed natural gas (CNG) after a Supreme Court order in 1998 is a well-known example (see Chapter 7).
- **Cleaner processes and equipment**: efficient combustion, low-NO_x burners in boilers, zig-zag firing in brick kilns (which burns fuel more completely and reduces soot), and replacement of old equipment.
- **Banning or limiting polluting activities**: bans on open burning of waste and crop residue, restrictions on old diesel vehicles, and rules on firecrackers.
- **Alternatives to crop-residue burning**: machines such as the Happy Seeder that sow wheat directly into paddy stubble, microbial decomposers sprayed on stubble, and use of straw as fuel in biomass power plants or as feedstock for compressed biogas.

End-of-pipe control of particles

Industries and power plants remove particles from their waste gases (called **flue gases**) before releasing them through the stack. MoEFCC emission norms for coal-based thermal power plants (December 2015) limit particulate emissions to between 30 and 100 mg/Nm^3, depending on the age of the plant. The main devices are described below.

- **Cyclone separators** make the dusty gas spin rapidly inside a cone-shaped chamber. Centrifugal force throws particles outward against the wall, where they slide down into a collecting hopper, while clean gas leaves through a central outlet at the top. Cyclones are cheap and robust and remove coarse particles (above about 10 μm) efficiently, but they capture fine particles poorly.
- **Electrostatic precipitators (ESPs)** pass flue gas between **discharge electrodes**, which carry a high negative voltage (typically tens of thousands of volts), and earthed **collecting plates**. The strong electric field ionises the gas, the particles pick up a negative charge, and they are attracted to the collecting plates, where they stick. The plates are struck periodically by hammers (**rapping**) and the dust layer falls into hoppers. ESPs remove more than 99% of particles by mass, including fine particles, and can treat very large gas volumes. Almost all coal-based power plants in India use ESPs to capture fly ash. The figure below shows how an ESP works.
- **Fabric filters (bag filters or baghouses)** pass dusty gas through hundreds of long bags made of woven or felted fabric. Particles are trapped on the fabric as the gas passes through, much as in a vacuum-cleaner bag. The dust cake is removed by shaking or by pulses of compressed air. Bag filters capture more than 99% of particles, including very fine ones, and are common in cement, steel and pharmaceutical plants. Very hot or wet gases damage the fabric.
- **Wet scrubbers** spray water or another liquid into the gas stream. Particles are captured by the droplets and removed as a slurry. Venturi scrubbers, which accelerate the gas through a narrow throat where water is injected, can capture fine particles. Scrubbers also remove soluble gases, but they produce wastewater that must itself be treated.

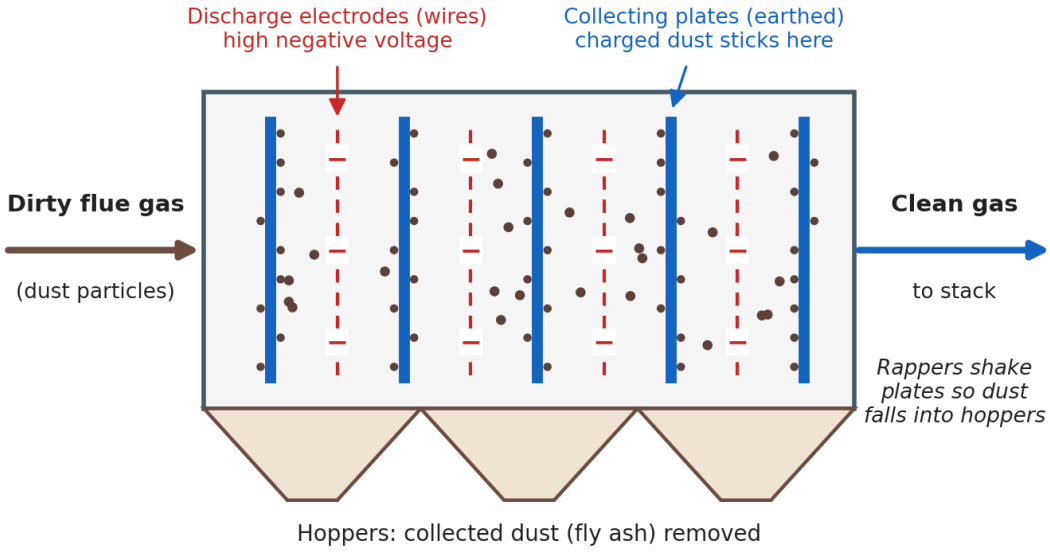


Figure 4.2: *Working of an electrostatic precipitator. Dust particles in the flue gas are given a negative charge by the high-voltage discharge electrodes and collect on the earthed plates; rapping drops the collected dust into hoppers. Source: Author.*

End-of-pipe control of gases

Flue gas desulfurisation (FGD) removes SO_2 by scrubbing the flue gas with a slurry of limestone (calcium carbonate) or lime. The SO_2 reacts to form calcium sulfite, which is oxidised to calcium

sulfate dihydrate (gypsum). The gypsum can be sold to the cement and plasterboard industries. Wet limestone FGD can remove more than 90% of SO_2.

$$CaCO_3 + SO_2 \rightarrow CaSO_3 + CO_2$$

$$2CaSO_3 + O_2 + 4H_2O \rightarrow 2CaSO_4 \cdot 2H_2O$$

The 2015 norms required all coal power plants to meet SO_2 limits, which in practice meant installing FGD units. After several deadline extensions, a July 2025 MoEFCC notification limited the FGD requirement mainly to plants within 10 km of the National Capital Region or of cities with more than one million people (deadline December 2027), left plants near critically polluted areas and non-attainment cities to case-by-case decisions, and exempted the remaining plants, the large majority, from the SO_2 standard. The decision is debated, because SO_2 is an important precursor of secondary fine particles over north India.

- **Selective catalytic reduction (SCR)** reduces NO_x by injecting ammonia or urea into the flue gas over a catalyst, converting NO_x into harmless nitrogen and water.
- **Adsorption** on activated carbon and **incineration** (thermal or catalytic oxidation) remove VOCs and odours.

Control of vehicle emissions

A **catalytic converter** is a device in a vehicle's exhaust system that uses a catalyst to convert harmful gases into less harmful ones. The **three-way catalytic converter** used in petrol vehicles contains a ceramic honeycomb coated with platinum, palladium and rhodium. It does three jobs at once: it oxidises carbon monoxide to CO_2, oxidises unburnt hydrocarbons to CO_2 and water, and reduces NO_x to nitrogen.

$$2CO + O_2 \rightarrow 2CO_2$$

$$2NO + 2CO \rightarrow N_2 + 2CO_2$$

Diesel vehicles use a **diesel oxidation catalyst**, a **diesel particulate filter (DPF)** that traps soot, and selective catalytic reduction using a urea solution (diesel exhaust fluid) to cut NO_x. Lead and sulfur poison these catalysts, which is why clean fuels and clean engines must be introduced together.

Bharat Stage (BS) emission standards are India's vehicle emission norms, modelled on the European (Euro) standards and notified under the Central Motor Vehicles Rules. BS-I came into force nationally in 2000. India moved directly from BS-IV to **BS-VI** across the country on 1 April 2020, skipping BS-V. BS-VI cut the permitted sulfur content of fuel from 50 ppm to 10 ppm and, for diesel cars, reduced the NO_x limit by about 68% and the particulate limit by over 80% compared with BS-IV. BS-VI Stage 2 (2023) added real-driving emission tests and on-board diagnostics.

Other measures include Pollution Under Control (PUC) certificates, the phase-out of old vehicles under the Vehicle Scrappage Policy (2021), promotion of electric vehicles through the FAME and PM E-DRIVE schemes, better public transport (metro rail, electric buses), and safe infrastructure for walking and cycling.

Reducing exposure

Tall stacks lower ground-level concentrations near a source but only spread pollutants over a wider area. **Green belts** of trees around industries trap some dust, and land-use planning keeps polluting

industries away from homes and schools. The Air Quality Index (explained later in this chapter) warns people to limit outdoor activity on bad days.

Key Points

- PM2.5 is the deadliest air pollutant; in India it comes from biomass and coal burning, vehicles, industry, dust and crop-residue burning.
- Air pollution causes heart disease, stroke, lung disease and cancer, and harms crops, monuments and the climate.
- Household chulhas are the largest source of exposure for poor rural families; Ujjwala expanded LPG access, but regular use depends on affordability.
- Cyclones, bag filters, ESPs and scrubbers remove particles; FGD and SCR remove SO_2 and NO_x; catalytic converters and BS-VI norms clean vehicle exhaust.

4.3 Primary and Secondary Air Pollutants

Air pollutants are classified into two groups according to how they enter the atmosphere. The distinction matters for control: a primary pollutant can be cut by controlling its source, whereas a secondary pollutant can be reduced only by controlling the **precursors** (the primary pollutants from which it forms) and often depends on weather and sunlight.

A **primary air pollutant** is emitted directly into the air from a source and is harmful in the form in which it is released. Examples are SO_2, NO, CO, black carbon (soot), dust, fly ash, lead, ammonia and most VOCs.

A **secondary air pollutant** is not emitted directly but forms in the atmosphere through chemical reactions among primary pollutants, or between primary pollutants and natural components of air such as oxygen and water vapour, often driven by sunlight. Examples are ground-level ozone, peroxyacetyl nitrate (PAN), sulfuric acid and nitric acid droplets, sulfate and nitrate particles, and secondary organic aerosols. Some pollutants, notably NO_2 and PM2.5, are both: some NO_2 is emitted directly, but most forms from the oxidation of NO, and PM2.5 contains both primary soot and dust and secondary sulfates, nitrates and organic particles.

Table 4.3: Primary and secondary air pollutants compared

Feature	Primary pollutants	Secondary pollutants
Origin	Emitted directly from a source	Formed in the atmosphere by reactions
Examples	SO_2, NO, CO, soot, dust, NH_3, VOCs	O_3, PAN, H_2SO_4, HNO_3, sulfate and nitrate particles
Where highest	Close to the source	Often downwind of the source
Depends on sunlight	No	Often yes (photochemical)
Main control approach	Control the source directly	Control the precursor emissions

4.3.1 Photochemical smog

The word **smog** (smoke + fog) was first used for the thick, sooty, sulfurous fogs of industrial cities such as London, where the "Great Smog" of December 1952 killed about 4,000 people in a few days and possibly many more over the following months. This **classical** or **London-type smog** forms in cold, damp, still weather from coal smoke (soot and SO_2) and is chemically **reducing**. A different kind of smog was recognised in Los Angeles in the 1940s. **Photochemical smog** is a brownish, oxidising haze formed when NO_x and VOCs from vehicle exhaust and industry react in strong sunlight. Its main products are ozone, PAN, aldehydes and fine particles. It is typical of warm, sunny cities with heavy traffic, and it peaks in the early afternoon.

Photochemical smog forms in three steps, shown in the figure below.

1. **Photolysis of NO_2**: sunlight with wavelengths shorter than about 420 nm splits NO_2 into nitric oxide and a free oxygen atom (written as hν for a photon of light).
2. **Ozone formation**: the oxygen atom combines with an oxygen molecule to form ozone. (M is any third molecule, such as N_2, that carries away excess energy.)
3. **Ozone removal by NO**: in clean air, ozone reacts with NO and is converted back into NO_2. These three reactions form a cycle that keeps ozone at low levels.

$$NO_2 + \text{sunlight } (h\nu) \rightarrow NO + O$$

$$O + O_2 + M \rightarrow O_3 + M$$

$$O_3 + NO \rightarrow NO_2 + O_2$$

VOCs upset this balance. Hydroxyl radicals (OH), produced naturally in sunlit air, attack VOCs and convert them into **peroxy radicals** (RO_2, where R stands for an organic group). Peroxy radicals convert NO into NO_2 *without using up ozone*. Because NO is no longer available to destroy ozone, ozone accumulates through the day.

$$VOC + OH \rightarrow RO_2 + \text{other products}$$

$$RO_2 + NO \rightarrow RO + NO_2$$

Some organic radicals also combine with NO_2 to form **peroxyacetyl nitrate (PAN)**, $CH_3C(O)OONO_2$, which irritates the eyes and damages plants. In polluted air, ozone therefore builds up on sunny afternoons downwind of traffic, then falls at night when there is no sunlight and fresh NO from vehicles removes it. This is why ozone is often highest in the suburbs and outskirts of a city rather than at its busiest junctions. Ozone exceedances are common in Indian cities from March to June, when sunlight is strongest, even though the winter months have worse particulate pollution.

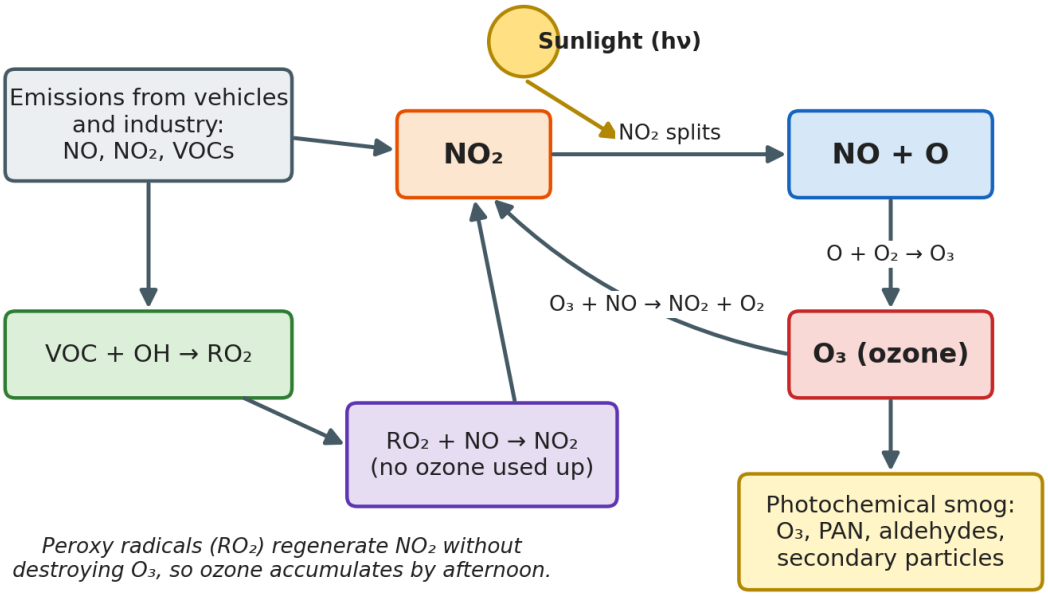


Figure 4.3: *Formation of photochemical smog. Sunlight splits NO_2 into NO and an oxygen atom, which forms ozone; VOC-derived peroxy radicals convert NO back to NO_2 without consuming ozone, so ozone and PAN build up. Source: Author.*

Factors that favour photochemical smog include strong sunlight, high temperature, heavy traffic, and a **temperature inversion**. Normally air temperature falls with height, so warm polluted air near the ground rises and disperses. In an inversion, a layer of warmer air lies above cooler air near the ground and acts like a lid, trapping pollutants. Winter inversions, low wind speeds and a shallow mixing layer are major reasons for the severe pollution episodes over Delhi and the Indo-Gangetic Plain in November–January (see the Delhi case study later in this chapter).

4.3.2 Secondary particulate matter

A large share of PM2.5 is secondary. Gases emitted from chimneys, vehicles and farms are oxidised in the air and converted into particles, a process called **gas-to-particle conversion**. The three main types of **secondary particles** are:

- **Sulfate particles**: SO_2 is oxidised (by OH radicals in air, or by hydrogen peroxide and ozone in cloud droplets) to sulfuric acid, which then reacts with ammonia to form ammonium sulfate.

$$2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$$

$$H_2SO_4 + 2NH_3 \rightarrow (NH_4)_2SO_4$$

- **Nitrate particles**: NO_2 reacts with OH radicals to form nitric acid, which combines with ammonia to form ammonium nitrate. Ammonium nitrate forms most readily in cool, humid conditions, which is one reason winter PM2.5 in north India is so high.

$$NO_2 + OH \rightarrow HNO_3$$

$$HNO_3 + NH_3 \rightleftharpoons NH_4NO_3$$

- **Secondary organic aerosol (SOA)**: VOCs from vehicles, solvents, biomass burning and vegetation are oxidised to less volatile compounds that condense into particles.

Source-apportionment studies in Delhi and other north Indian cities have found that secondary inorganic particles (ammonium sulfate and ammonium nitrate) often make up a quarter or more of winter PM2.5. Ammonia from fertiliser and livestock, SO_2 from power plants and NO_x from vehicles all contribute, even though they may be emitted hundreds of kilometres apart. The practical lesson is that secondary particles cannot be controlled city by city: they require cuts in SO_2, NO_x, ammonia and VOCs across the whole airshed, which in north India extends across several states.

Think and Do: Watch ozone and PM2.5 through the day

Open the CPCB "Sameer" app or the CPCB real-time air-quality website and select a monitoring station near your college. Note the hourly concentrations of PM2.5, NO_2 and ozone over one sunny day. At what time does each pollutant peak? Explain why ozone peaks in the afternoon while NO_2 often peaks during morning and evening traffic, using the reactions given in this section.

4.4 Water Pollution

Water pollution is any change in the physical, chemical or biological quality of water that makes it harmful to living organisms or unfit for drinking, bathing, agriculture, industry or other uses.

India has about 18% of the world's population but only about 4% of its freshwater, so pollution of the water it has is especially costly (see Chapter 3 for water resources). The CPCB's 2025 assessment identified 296 polluted river stretches on 271 rivers, down from 351 stretches in 2018, with Maharashtra having the largest number. Groundwater, which supplies most of India's drinking water and irrigation, is contaminated in many districts by nitrate, fluoride, arsenic and other substances.

4.4.1 Sources of water pollution

Water pollution comes from point and non-point sources, and from both human activity and natural (geogenic) processes.

- **Domestic sewage**: wastewater from toilets, kitchens and bathrooms carries organic matter, pathogens, nutrients (nitrogen and phosphorus from excreta and detergents) and, increasingly, medicines and personal-care products. It is the largest source of river pollution in India. CPCB's national inventory (2021) estimated that urban India generates about 72,368 million litres per day (MLD) of sewage, against an installed treatment capacity of about 31,841 MLD, and that actual treatment was lower still. The rest flows untreated into drains, rivers, lakes and the sea.
- **Industrial effluents**: tanneries, textile dyeing and printing units, distilleries, sugar mills, paper mills, pharmaceutical units, electroplating units and chemical industries discharge organic matter, acids and alkalis, dyes, heavy metals (chromium, cadmium, mercury, lead), oils and toxic organic chemicals. Tannery clusters at Kanpur–Unnao on the Ganga and Vellore–Ranipet on the Palar are long-standing examples of chromium pollution.
- **Agricultural runoff**: rain and irrigation water wash fertilisers (nitrate, phosphate), pesticides, eroded soil and animal waste from fields into streams and down into groundwater. This is a non-point source and is very difficult to regulate.
- **Solid waste and leachate**: garbage dumped on riverbanks, and **leachate** (the dark, polluted liquid that seeps out of landfills), contaminate surface and groundwater.

- **Religious and cultural practices**: immersion of idols painted with lead- and chromium-based paints, disposal of flowers and other offerings, and cremation of bodies along rivers add organic load and toxic substances.
- **Oil spills**: accidents such as the 2010 collision of two ships off Mumbai and the 2017 Ennore oil spill near Chennai polluted coastal waters and mangroves.
- **Thermal discharges**: heated cooling water from power plants (discussed under thermal pollution).
- **Geogenic sources**: natural release of arsenic, fluoride, iron and uranium from rocks into groundwater, often worsened by heavy pumping.

4.4.2 Water quality parameters

Water quality is measured by a set of physical, chemical and biological **parameters**. The most important ones, used in the standards discussed later in this chapter, are described below.

- **Dissolved oxygen (DO)** is the amount of oxygen gas dissolved in water, in mg/L. Fish and other aquatic animals need DO to breathe. Clean water at 20 °C holds about 9 mg/L when saturated; most fish are stressed below about 4–5 mg/L, and water with less than 2 mg/L is called **hypoxic**. DO falls when organic waste decomposes and when water is warmed.
- **Biochemical oxygen demand (BOD)** is the amount of oxygen consumed by micro-organisms while decomposing the biodegradable organic matter in a water sample, usually measured over 5 days at 20 °C (Indian standards often use 3 days at 27 °C). BOD is the most widely used measure of organic pollution: the higher the BOD, the more polluted the water. Clean river water has a BOD below 3 mg/L, whereas raw domestic sewage typically has a BOD of 150–300 mg/L.
- **Chemical oxygen demand (COD)** is the amount of oxygen needed to oxidise all the organic matter in a sample chemically, using a strong oxidant (potassium dichromate) in acid. COD measures both biodegradable and non-biodegradable organic matter and takes only about two hours. COD is always higher than BOD; a high COD:BOD ratio suggests industrial waste that bacteria cannot easily break down.
- **pH** measures acidity or alkalinity on a scale of 0 to 14. Most aquatic life and drinking water require a pH between about 6.5 and 8.5. Acidic mine drainage and some industrial effluents lower pH; alkaline effluents raise it.
- **Total dissolved solids (TDS)** is the total amount of dissolved salts and minerals, in mg/L. High TDS makes water taste salty or bitter, causes scaling in pipes, and makes water unsuitable for irrigation. TDS in groundwater is high in parts of Rajasthan, Haryana and Gujarat.
- **Turbidity** is the cloudiness of water caused by suspended particles, measured in nephelometric turbidity units (NTU). Turbid water shields pathogens from disinfection.
- **Coliform bacteria** are bacteria that live in the intestines of warm-blooded animals. Their presence, especially of **faecal coliforms** and *Escherichia coli*, indicates contamination by faeces and the likely presence of pathogens. They are counted as the **most probable number (MPN)** per 100 mL. Safe drinking water must contain no detectable *E. coli* in 100 mL.
- **Nitrate (NO_3^-)** comes mainly from fertilisers, sewage and animal waste. Above 45 mg/L (as NO_3) in drinking water, it can cause **methaemoglobinaemia** (**blue baby syndrome**) in infants, in which nitrite formed from nitrate prevents haemoglobin from carrying oxygen.

- **Fluoride** and **arsenic** are mostly geogenic and are discussed below.
- **Heavy metals** such as lead, mercury, cadmium and chromium are toxic at very low concentrations (often in micrograms per litre).

4.4.3 Fluoride and arsenic: India's geogenic groundwater problem

Fluoride dissolves into groundwater from fluoride-bearing minerals such as fluorite and apatite in granite and other rocks. Drinking water with more than 1.5 mg/L of fluoride over many years causes **dental fluorosis** (mottled, brown-stained teeth) and **skeletal fluorosis** (stiff, painful and deformed bones and joints). Fluoride contamination occurs in parts of at least 20 states, with severe areas in Rajasthan, Telangana (Nalgonda district became known for crippling skeletal fluorosis), Andhra Pradesh, Gujarat, Karnataka, Madhya Pradesh and Punjab. The CGWB's *Annual Ground Water Quality Report 2024* found fluoride above the permissible limit in about 9% of samples tested.

Arsenic occurs naturally in the sediments of the Ganga–Brahmaputra–Meghna delta and other young alluvial plains. When groundwater is pumped heavily from these aquifers, arsenic is released from iron oxide coatings on sediment grains. Long-term drinking of water with arsenic above 0.01 mg/L causes **arsenicosis**: skin lesions (dark spots and hardened palms and soles), damage to the liver, lungs and blood vessels, and cancers of the skin, lungs and bladder. Arsenic in groundwater was first reported in West Bengal in 1983. Affected areas now include West Bengal, Bihar, Uttar Pradesh, Jharkhand, Assam and Punjab. The CGWB 2024 report found arsenic above the limit in about 3.5% of samples, and nitrate above the limit in about 20% of samples, mainly from agricultural and sewage sources.

The Jal Jeevan Mission (2019), which aims to give every rural household a tap connection, gives priority to fluoride- and arsenic-affected habitations, supplying treated surface water through piped schemes and community purification plants.

4.4.4 Organic pollution and the oxygen sag curve

When sewage enters a river, bacteria begin decomposing its organic matter and consume dissolved oxygen faster than the river can replace it from the air (a process called **reaeration**). DO therefore falls downstream of the outfall, reaches a minimum (the **critical point**), and then recovers as the organic load is used up and reaeration catches up. Plotted against distance downstream, DO forms a characteristic **oxygen sag curve**, shown in the figure below.

The river can be divided into zones along the curve. Just below the outfall lies a **zone of degradation**, followed by a **zone of active decomposition**, where DO may fall to near zero, fish disappear, and pollution-tolerant organisms such as sludge worms (*Tubifex*) and red bloodworm (*Chironomus*) larvae dominate. Downstream, in the **zone of recovery**, DO rises and sensitive species return, until a **clean-water zone** is reached. If sewage enters at many points along the river, as along the Yamuna in Delhi, recovery never occurs, and the river stays in a near-anoxic state for long stretches.

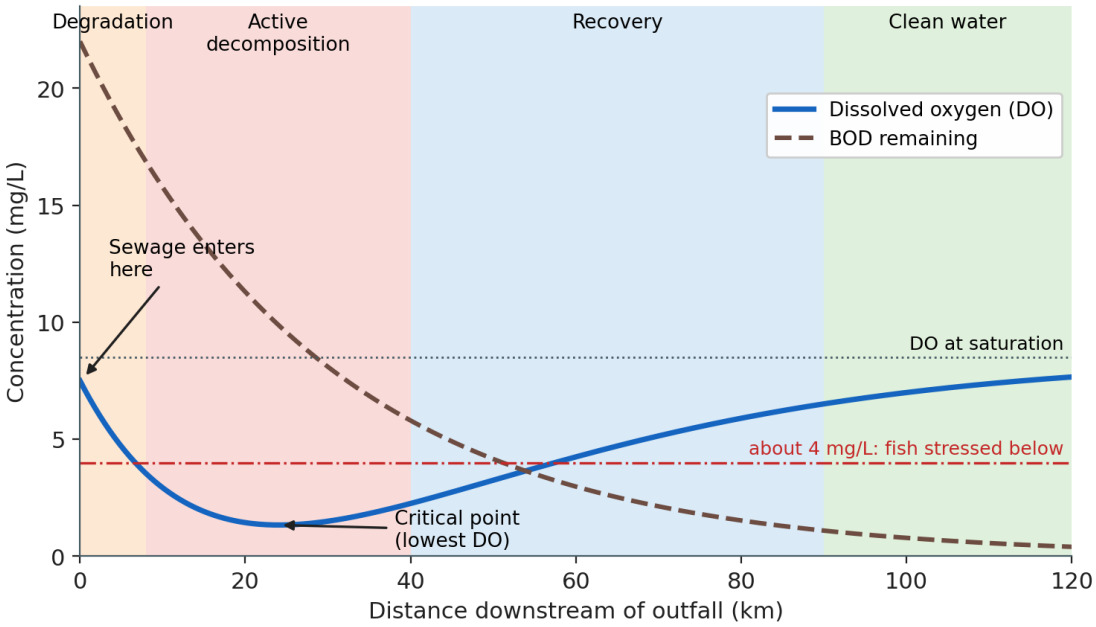


Figure 4.4: *The dissolved oxygen sag curve downstream of a sewage outfall. DO falls as bacteria decompose organic matter, reaches a critical minimum, and recovers as reaeration exceeds oxygen consumption, while BOD declines steadily. Source: Author, based on the Streeter–Phelps model.*

4.4.5 Eutrophication

Eutrophication is the enrichment of a water body with plant nutrients, mainly nitrogen and phosphorus, leading to excessive growth of algae and aquatic plants and a decline in water quality. It happens naturally over centuries as lakes age, but human inputs of sewage, detergents and fertiliser runoff speed it up to a few years; this is called **cultural eutrophication**. It proceeds in stages:

1. Nutrients from sewage and farmland enter a lake or slow-moving river.
2. Algae, including toxic cyanobacteria (blue-green algae), multiply rapidly, forming **algal blooms** that turn the water green and cut off light to submerged plants. Floating weeds such as water hyacinth (*Eichhornia crassipes*) cover the surface.
3. When the algae die, bacteria decompose them, consuming large amounts of dissolved oxygen.
4. Oxygen in deeper water drops to very low levels (hypoxia), fish die in large numbers, and the water smells of hydrogen sulfide.

Indian examples include Bengaluru's Bellandur Lake, where untreated sewage rich in detergents produced toxic foam and even caught fire several times between 2015 and 2018; the foam on the Yamuna at Kalindi Kunj in Delhi, caused by surfactants and phosphates in untreated sewage; and nutrient loading in Hussain Sagar (Hyderabad), Dal Lake (Srinagar) and Loktak Lake (Manipur).

4.4.6 Effects of water pollution

- **Human health**: water-borne diseases such as diarrhoea, cholera, typhoid, hepatitis A and E remain major causes of illness in India, especially among children under five. Chemical contamination causes fluorosis, arsenicosis, blue baby syndrome, and heavy-metal poisoning; the classic global example is **Minamata disease** in Japan (1950s), a neurological disorder caused by eating fish contaminated with methylmercury released by a chemical factory.

- **Aquatic ecosystems**: loss of DO kills fish and invertebrates; toxic substances bioaccumulate and biomagnify; sediment smothers spawning grounds; and species that need clean water, such as the Gangetic river dolphin (*Platanista gangetica*) and the mahseer fish, decline.
- **Agriculture**: irrigation with polluted or saline water contaminates soils and crops with metals and pathogens. Vegetables grown with untreated wastewater on the Yamuna floodplain have been found to contain heavy metals.

4.4.7 Control of water pollution

Water pollution is best controlled by reducing what enters the water in the first place, and by treating the wastewater that must be discharged.

- **Sewage collection and treatment**: building sewer networks and sewage treatment plants (STPs), and ending open defecation. The Swachh Bharat Mission and the Namami Gange programme have invested heavily in toilets, sewers and STPs.
- **Industrial effluent control**: every polluting industry must obtain consent from its SPCB and treat its effluent in an **effluent treatment plant (ETP)** to meet discharge standards (see standards later in this chapter). Clusters of small units share a **common effluent treatment plant (CETP)**. Highly polluting industries in the Ganga basin must aim for **zero liquid discharge (ZLD)**, recycling all their wastewater.
- **Agricultural measures**: balanced and efficient use of fertilisers (soil-health cards, neem-coated urea), integrated pest management, organic and natural farming, and buffer strips of vegetation along streams to trap runoff.
- **Protecting water bodies**: banning dumping of waste and idol immersion in natural water bodies, restoring wetlands that filter nutrients, and removing water hyacinth.

Sewage treatment: an overview

A conventional STP treats sewage in stages (methods are described in more detail later in this chapter):

1. **Preliminary treatment**: screens remove rags, plastics and large objects; grit chambers allow sand and gravel to settle.
2. **Primary treatment**: sewage is held in large sedimentation tanks where about 50–70% of suspended solids and 25–40% of BOD settle out as **primary sludge**.
3. **Secondary (biological) treatment**: micro-organisms decompose the dissolved and fine organic matter. In the **activated sludge process**, air is bubbled through the sewage in aeration tanks to keep a large population of aerobic bacteria active; the bacterial flocs are then settled out. Other methods include trickling filters, sequencing batch reactors (SBR), up-flow anaerobic sludge blanket (UASB) reactors and waste stabilisation ponds. Secondary treatment removes 85–95% of BOD.
4. **Tertiary (advanced) treatment**: filtration, removal of nitrogen and phosphorus, and disinfection with chlorine, ozone or ultraviolet light, so that the water can be reused for gardening, industry or groundwater recharge.

The sludge produced is digested anaerobically to produce biogas and a stabilised residue that can be used as manure if free of heavy metals.

Key Points

- Untreated sewage is the largest source of river pollution in India; industrial effluents, agricultural runoff and geogenic contaminants are also major sources.
- DO, BOD, COD, pH, TDS, coliforms and nitrate are the key parameters; high BOD and coliforms signal sewage pollution.
- Fluoride and arsenic in groundwater cause fluorosis and arsenicosis across large parts of India.
- Nutrient loading causes eutrophication, algal blooms and fish kills; sewage treatment and nutrient control are the main remedies.

4.5 Soil Pollution

Soil pollution is the presence in soil of toxic chemicals or other harmful materials at concentrations high enough to harm soil organisms, crops, animals or human health, or to reduce the soil's fertility. Soil is the base of food production and filters the water that recharges aquifers, so its pollution spreads quickly to food and water. Unlike air and water, soil does not move or dilute pollutants rapidly; persistent pollutants such as heavy metals can remain for centuries. The FAO's *Global Assessment of Soil Pollution* (2021) described soil pollution as a largely hidden but growing threat. Soil degradation through erosion and desertification is covered in Chapter 3; this section deals with contamination.

4.5.1 Sources of soil pollution

- **Agrochemicals**: overuse and misuse of chemical fertilisers and pesticides. India's fertiliser use is heavily skewed towards nitrogen (urea), far from the recommended N:P:K ratio of about 4:2:1, especially in Punjab and Haryana. Excess nitrogen acidifies soil and leaches to groundwater as nitrate. Some phosphate fertilisers contain cadmium and uranium as impurities. Persistent organochlorine pesticides such as DDT, lindane and endosulfan, many of which are now banned, remain in soils for decades.
- **Industrial waste**: dumping of solid wastes and sludge, and irrigation with industrial effluents, add heavy metals and toxic organic chemicals to soil. Chromium-bearing waste from tanneries at Rania and Khan Chandpur near Kanpur has contaminated soil and groundwater with hexavalent chromium, a carcinogen. Mining areas such as the Sukinda valley chromite mines in Odisha suffer similar contamination.
- **Heavy metals**: lead, cadmium, mercury, arsenic, chromium and nickel come from smelting, mining, battery recycling, electroplating, irrigation with sewage, sewage sludge and atmospheric fallout. Metals are not broken down; they bind to soil particles and are slowly taken up by plants.
- **Electronic waste (e-waste)**: discarded computers, phones and appliances contain lead, mercury, cadmium, brominated flame retardants and other toxic substances. India generated about 17.5 lakh tonnes of e-waste in 2023–24 (MoEFCC). Much of it is dismantled in informal clusters such as Seelampur and Mustafabad in Delhi and Moradabad in Uttar Pradesh, where

open burning of cables and acid baths used to recover gold and copper release metals and dioxins into soil and nearby water. E-waste management is discussed later in this chapter.

- **Plastics**: plastic bags, packaging and agricultural plastic mulch fragment into **microplastics** (particles smaller than 5 mm) that alter soil structure and water movement and may carry toxic chemicals. Sewage sludge and wastewater irrigation also add microplastics.
- **Municipal solid waste and landfills**: unscientific dumpsites such as Delhi's Ghazipur, Bhalswa and Okhla landfills, and Mumbai's Deonar, leak leachate into surrounding soil and groundwater.
- **Other sources**: oil spills and leaking fuel tanks, irrigation with saline water (secondary salinisation), and radioactive fallout from nuclear accidents and tests.

Case Study: Endosulfan in Kasaragod, Kerala

From the late 1970s until 2001, the Plantation Corporation of Kerala sprayed the organochlorine insecticide endosulfan from helicopters over its cashew plantations in Kasaragod district, two to three times a year. The spray fell on villages, streams and wells as well as on the cashew trees. Over the years, residents reported unusually high rates of birth defects, mental and physical disabilities, cancers and reproductive disorders, especially in children. Studies by the National Institute of Occupational Health (2002) found higher levels of endosulfan in residents' blood and a higher prevalence of several disorders in sprayed villages than in comparison villages, although scientific debate continued about how much of the harm was caused by endosulfan alone.

Sustained protests by residents and activists led the Kerala government to stop aerial spraying in 2001 and to ban endosulfan in the state. In 2011, endosulfan was listed for global elimination under the **Stockholm Convention on Persistent Organic Pollutants**, and the Supreme Court of India banned its production and use across the country. In 2017, the Court directed the Kerala government to pay ₹5 lakh compensation to each identified victim. The case shows how a persistent pesticide can contaminate soil and water, how poor people near plantations bear the cost, and how public action can force regulation.

4.5.2 Effects of soil pollution

- **On soil fertility and organisms**: toxic substances kill or inhibit earthworms, soil bacteria, fungi and other organisms that decompose organic matter, cycle nutrients and maintain soil structure. Excess fertiliser changes soil pH and depletes organic carbon.
- **On crops and food**: plants absorb heavy metals and pesticides, which then enter the food chain. Cadmium and lead have been found in vegetables grown with untreated wastewater around many Indian cities. Crop yields may decline on contaminated or salinised land.
- **On water**: pollutants leach into groundwater (nitrate, pesticides, chromium) or are washed into rivers with eroded soil.
- **On human health**: people are exposed by eating contaminated food, drinking contaminated groundwater, and by direct contact with or inhalation of contaminated dust. Children playing outdoors are especially exposed. Effects range from lead poisoning and cadmium kidney damage to cancers. In Punjab's Malwa region, high cancer rates have been widely linked in public debate to heavy pesticide use and contaminated groundwater, giving the overnight Bathinda–Bikaner train the nickname "cancer train", though the causes are still being studied.

- **On wildlife**: biomagnification of pesticides such as DDT caused thinning of eggshells in birds of prey; the near-collapse of India's vulture populations was caused by a different chemical, the veterinary drug diclofenac in cattle carcasses (see Chapter 6).

4.5.3 Control and remediation of soil pollution

Prevention is the most effective approach:

- Balanced and need-based use of fertilisers guided by **Soil Health Cards** (2015), use of organic manure, compost, biofertilisers and neem-coated urea, and promotion of natural and organic farming (for example, Sikkim became a fully organic state in 2016).
- **Integrated pest management (IPM)**, which combines biological control, resistant varieties, crop rotation and need-based pesticide use, and bans on persistent and hazardous pesticides.
- Proper collection, segregation and scientific disposal of municipal, industrial, hazardous and electronic waste (see later in this chapter), and treatment of effluents before they are used for irrigation.
- Reducing single-use plastics and recovering agricultural plastic.

Remediation means cleaning up soil that is already contaminated. The main approaches are:

- **Physical methods**: excavation and removal of contaminated soil to a secure landfill, soil washing, and capping (covering) contaminated sites.
- **Chemical methods**: **immobilisation**, in which lime, phosphate or biochar is added to bind metals so that plants cannot take them up, and chemical oxidation of organic pollutants.
- **Biological methods**: **bioremediation**, which uses micro-organisms to break down organic pollutants such as oil and pesticides, and **phytoremediation**, which uses plants to extract, stabilise or degrade pollutants. Indian mustard (*Brassica juncea*) accumulates lead and cadmium, and vetiver grass (*Chrysopogon zizanioides*) stabilises contaminated soil.

Biological and physico-chemical methods of pollution control are discussed in more detail later in this chapter.

4.6 Thermal Pollution

Thermal pollution is the degradation of water quality by any process that changes its natural temperature, usually by the discharge of heated water into rivers, lakes or the sea. It differs from other water pollution in that no foreign substance need be added; the pollutant is heat itself.

4.6.1 Sources of thermal pollution

- **Thermal and nuclear power plants**: these are the largest source. A coal-based or nuclear plant converts only about a third of the heat from its fuel into electricity; most of the rest must be removed by cooling water that condenses the steam leaving the turbines. In **once-through cooling**, water is drawn from a river, lake or the sea, passed through the condenser, and returned several degrees warmer (typically 5–10 °C). Indian coastal plants that use seawater for once-through cooling include the Tarapur (Maharashtra), Madras (Kalpakkam, Tamil Nadu) and Kudankulam (Tamil Nadu) nuclear power stations and several coal-based plants in Gujarat and Tamil Nadu.

- **Industries**: steel plants, oil refineries, paper mills, textile units and chemical plants that use water for cooling.
- **Urban runoff**: rainwater flowing over hot roads, roofs and pavements in summer warms streams.
- **Removal of shade trees**: deforestation along stream banks exposes water to direct sunlight.
- **Cold-water pollution**: release of very cold water from the bottom of deep reservoirs lowers downstream river temperatures, which also harms fish adapted to warmer water.

4.6.2 Effects of thermal pollution

- **Lower dissolved oxygen**: warm water holds less oxygen. Saturated DO falls from about 14.6 mg/L at 0 °C to about 9.1 mg/L at 20 °C and about 7.6 mg/L at 30 °C. At the same time, warmer temperatures raise the metabolic rate and oxygen demand of fish and of decomposing bacteria (roughly doubling with every 10 °C rise). The combination of less oxygen supply and greater demand can suffocate aquatic animals.
- **Thermal shock**: sudden temperature changes when a plant starts up or shuts down can kill fish that cannot adjust quickly.
- **Disrupted life cycles**: many fish and invertebrates use temperature as a cue for spawning, migration and hatching. Warm water can cause early hatching, failed reproduction and loss of cold-water species.
- **Changes in species composition**: heat-tolerant species, including nuisance algae and cyanobacteria, replace sensitive ones; warm water favours algal blooms and eutrophication.
- **Greater toxicity and disease**: higher temperature increases the toxicity of many pollutants (for example, a larger share of ammonia is present in its toxic un-ionised form at higher temperature and pH) and makes fish more vulnerable to parasites and disease.
- **Coral bleaching**: warm discharges in tropical coastal waters can bleach corals, adding to the stress caused by ocean warming.

Fishing communities near the Tata Mundra coal power plant in Kutch, Gujarat, complained to the World Bank Group's Compliance Advisor Ombudsman in 2011 that warm-water discharge from the plant's seawater cooling outfall had damaged fish catches; the Ombudsman's 2013 audit found shortcomings in the assessment and monitoring of these impacts.

4.6.3 Control of thermal pollution

- **Cooling towers**: tall structures in which warm water is sprayed or trickled down while air flows upward, cooling the water mainly by evaporation. The large hyperboloid towers seen at many Indian power plants are **natural-draught** cooling towers; **mechanical-draught** towers use fans. The cooled water is recirculated, so only a small amount of make-up water is needed.
- **Cooling ponds and spray ponds**: warm water is held in a large shallow pond, or sprayed into the air above it, and loses heat by evaporation, convection and radiation before reuse or release.
- **Artificial lakes and canals**: long discharge channels allow water to cool before it reaches the main water body.
- **Cogeneration and waste-heat recovery**: using waste heat for industrial processes, space heating, or desalination reduces the heat that must be dumped.

- **Improved efficiency**: supercritical and ultra-supercritical coal plants convert more of their fuel's energy into electricity and therefore reject less heat per unit of power.

In India, the Environment (Protection) Rules require that condenser cooling water discharged by thermal power plants using once-through cooling should not be more than 5 °C warmer than the intake water. MoEFCC norms of December 2015 required existing plants with once-through cooling to install cooling towers, and limited specific water consumption to 3.5 m^3 per megawatt-hour for existing plants and 3 m^3/MWh for plants installed from 2017. Seawater-cooled coastal plants are exempt from cooling-tower requirements but must meet discharge-temperature conditions.

4.7 Noise Pollution

Noise is unwanted or excessive sound. **Noise pollution** is noise at levels or durations that harm health, disturb sleep, communication or work, or reduce the quality of life. Noise was brought under the Air (Prevention and Control of Pollution) Act, 1981 through an amendment in 1987, and is regulated specifically by the Noise Pollution (Regulation and Control) Rules, 2000, framed under the Environment (Protection) Act, 1986.

4.7.1 Measuring noise: the decibel scale

Sound is a pressure wave travelling through air. The human ear responds to an enormous range of sound pressures, so loudness is measured on a logarithmic scale in **decibels (dB)**. On this scale, 0 dB is the faintest sound a healthy young person can hear. An increase of 10 dB means a tenfold increase in sound intensity and is heard as roughly a doubling of loudness; an increase of 3 dB doubles the sound energy. Thus 90 dB carries ten times the energy of 80 dB and a hundred times that of 70 dB. Because the ear is less sensitive to very low and very high pitches, noise meters apply a weighting that mimics human hearing, giving readings in **dB(A)**. Since noise levels fluctuate, standards use the **equivalent continuous sound level (Leq)**, the steady sound level that would carry the same total energy as the fluctuating noise over the measuring period.

The figure below shows typical sound levels from common sources and compares them with the limits under Indian rules.

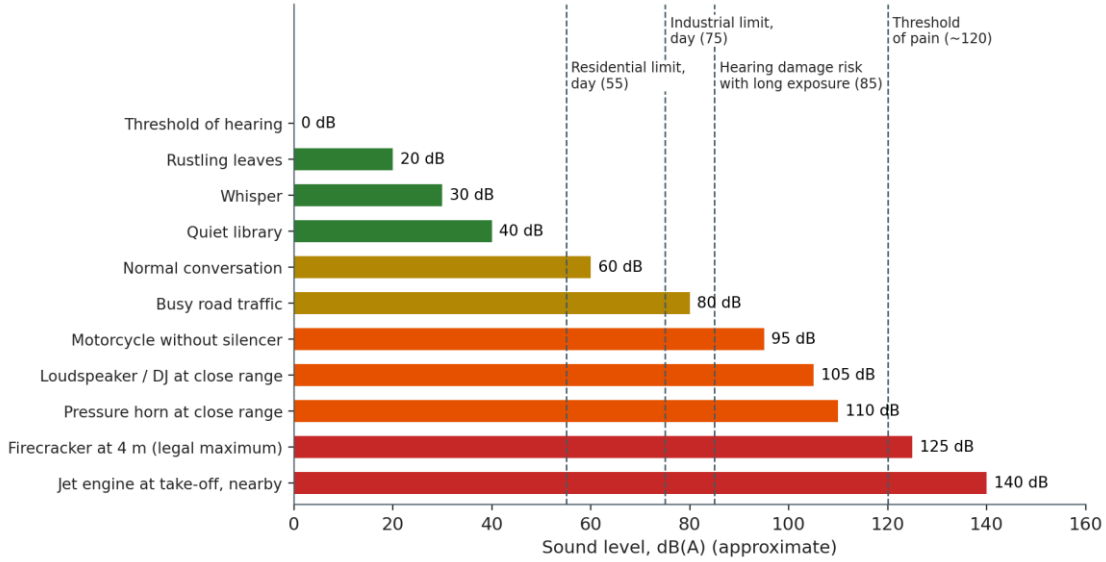


Figure 4.5: *Typical sound levels in decibels from everyday sources, compared with the Indian daytime ambient noise limits for silence and residential zones and with the threshold of pain. Values are approximate. Source: Author, compiled from CPCB and WHO.*

4.7.2 Sources of noise pollution

- **Road traffic**: the dominant source in Indian cities, worsened by constant horn use, poorly maintained vehicles, pressure horns and modified silencers on motorcycles. CPCB's National Ambient Noise Monitoring Network has repeatedly recorded levels above the limits at most monitoring locations in major cities, including in silence zones near hospitals.
- **Aircraft and railways**: residents near airports and railway lines are exposed to high, intermittent noise.
- **Industry and construction**: generators, compressors, presses, looms, pile drivers, stone crushers and diesel generator sets.
- **Social events**: loudspeakers, DJ systems, processions and festivals.
- **Firecrackers**: some crackers produce well over 125 dB at close range. The Environment (Protection) Rules limit cracker noise to 125 dB(AI) or 145 dB(C) peak at 4 m distance.

4.7.3 Effects of noise pollution

- **Hearing damage**: prolonged exposure to sound above about 85 dB(A) damages the hair cells of the inner ear and causes permanent **noise-induced hearing loss**; very loud sudden sounds (such as firecrackers close by) can rupture the eardrum. **Tinnitus** (ringing in the ears) often accompanies hearing loss. The WHO estimates that over 1 billion young people worldwide are at risk of hearing loss from unsafe listening, including through earphones.
- **Sleep disturbance**: night-time noise causes difficulty falling asleep and repeated awakenings, leading to fatigue and reduced performance.
- **Cardiovascular and stress effects**: chronic noise raises stress hormones, blood pressure and the risk of heart disease. The WHO considers environmental noise one of the top environmental health risks in Europe after air pollution.
- **Mental and social effects**: annoyance, irritability, anxiety, difficulty in concentrating, and interference with conversation.

- **Wildlife**: noise masks animal calls, disturbs breeding and feeding of birds, drives animals away from roads and mines, and harms marine mammals through ship noise and sonar.

4.7.4 Noise Pollution (Regulation and Control) Rules, 2000

The Noise Rules classify areas into four zones and set ambient noise limits for each, shown in the table below. **Daytime** is from 6 a.m. to 10 p.m. and **night-time** from 10 p.m. to 6 a.m. A **silence zone** is an area of at least 100 metres around hospitals, educational institutions and courts, or any other area declared so by the authority.

Table 4.4: Ambient air quality standards in respect of noise, Noise Pollution (Regulation and Control) Rules, 2000

Area code	Category of area or zone	Day time limit, dB(A) Leq	Night time limit, dB(A) Leq
A	Industrial area	75	70
B	Commercial area	65	55
C	Residential area	55	45
D	Silence zone	50	40

Other provisions of the Rules include:

- Loudspeakers and public-address systems may not be used without permission, and not at all between 10 p.m. and 6 a.m. except in closed premises such as auditoriums and banquet halls. A state government may relax this to midnight for cultural or religious festivals on up to 15 days a year.
- Horns, sound-emitting construction equipment and the bursting of crackers are not allowed in silence zones or residential areas at night.
- Noise at the boundary of a public place where a loudspeaker is used must not exceed the ambient limit by more than 10 dB(A), or 75 dB(A), whichever is lower.

In *In Re: Noise Pollution* (2005), the Supreme Court held that freedom from noise is part of the right to life under Article 21 of the Constitution, and upheld the night-time ban on loudspeakers and the bursting of firecrackers between 10 p.m. and 6 a.m. For workplaces, the Factories Act framework limits workers' exposure to 90 dB(A) over an eight-hour shift.

4.7.5 Control of noise pollution

Noise can be reduced at the source, along its path, or at the receiver.

- **At the source**: quieter machines, regular vehicle maintenance, silencers and mufflers, enclosures around generators (CPCB norms require acoustic enclosures for diesel generator sets), restrictions on pressure horns, "No Honking" campaigns, and a shift to electric vehicles, which are much quieter at low speeds.
- **Along the path**: noise barriers along highways, flyovers and metro lines; thick green belts of trees and shrubs, which reduce noise modestly and screen the source; and greater distance between sources and homes through land-use planning (for example, keeping airports, highways and industries away from residential areas, schools and hospitals).

- **At the receiver**: sound insulation of buildings, double-glazed windows, and ear plugs or ear muffs for workers in noisy industries.
- **Legal and social measures**: enforcement of the Noise Rules, awareness campaigns, and voluntary restraint during festivals, such as the shift to "green crackers" with lower noise and emissions developed by CSIR–NEERI.

Think and Do: Map the noise around your college

Download a free sound-level meter app on a smartphone. Record the noise level for five minutes each at the college gate, a classroom, the library and the nearest main road, in the morning and in the evening. Note the highest and average readings, and compare them with the limits in the table above. Is your college in a silence zone? Suggest three practical measures to reduce noise on campus.

4.8 Air and Water Quality Standards

An **environmental quality standard** is a legally notified limit on the concentration of a pollutant in the environment (an **ambient standard**) or in a discharge (an **emission** or **effluent standard**). Standards turn scientific evidence about harm into enforceable numbers. They tell regulators when air or water is unacceptably polluted, set targets for clean-up programmes, and give industries a clear limit to meet. In India, ambient air and water standards are notified by the Central Government under the Environment (Protection) Act, 1986 on the advice of the CPCB; drinking-water standards are set by the Bureau of Indian Standards (BIS). International guidelines, chiefly those of the WHO, are not binding but are used as a health benchmark.

4.8.1 National Ambient Air Quality Standards (NAAQS), 2009

India's current **National Ambient Air Quality Standards**, notified on 18 November 2009, set limits for 12 pollutants. Unlike earlier standards, which allowed dirtier air in industrial areas, the 2009 standards apply the same limits to industrial, residential, rural and other areas, with stricter limits for SO_2 and NO_2 in **ecologically sensitive areas** notified by the central government. Annual limits are averages of measurements taken over a year; 24-hour and 8-hour limits must be met 98% of the time.

Table 4.5: National Ambient Air Quality Standards, 2009 (industrial, residential, rural and other areas)

Pollutant	Annual average	24-hour average	Shorter period
Sulfur dioxide (SO_2), µg/m³	50 (ES: 20)	80	—
Nitrogen dioxide (NO_2), µg/m³	40 (ES: 30)	80	—
PM10, µg/m³	60	100	—
PM2.5, µg/m³	40	60	—
Ozone (O_3), µg/m³	—	—	100 (8 h); 180 (1 h)
Lead (Pb), µg/m³	0.5	1.0	—

Pollutant	Annual average	24-hour average	Shorter period
Carbon monoxide (CO), mg/m³	—	—	2 (8 h); 4 (1 h)
Ammonia (NH_3), µg/m³	100	400	—
Benzene (C_6H_6), µg/m³	5	—	—
Benzo(a)pyrene (particulate), ng/m³	1	—	—
Arsenic (As), ng/m³	6	—	—
Nickel (Ni), ng/m³	20	—	—

ES: stricter limit for ecologically sensitive areas notified by the central government. Source: CPCB notification, 2009.

4.8.2 National Air Quality Index (AQI)

Concentrations of many pollutants are difficult for the public to interpret, so in April 2015 India launched the **National Air Quality Index**. The AQI converts the measured concentrations of up to eight pollutants (PM10, PM2.5, NO_2, SO_2, CO, O_3, NH_3 and Pb) into sub-indices on a common scale of 0 to 500. The highest sub-index becomes the overall AQI, and the pollutant responsible is reported as the **prominent pollutant**. At least three pollutants, one of which must be PM10 or PM2.5, are needed to calculate the index. The AQI is divided into six categories, each with a colour and a health message, shown in the table and figure below. CPCB publishes the AQI for cities daily in its national AQI bulletin, and real-time values through the "Sameer" app.

Table 4.6: Categories of India's National Air Quality Index with PM breakpoints (24-hour average, µg/m³)

AQI category (range)	Colour	PM2.5	PM10	Likely health impact
Good (0–50)	Dark green	0–30	0–50	Minimal impact
Satisfactory (51–100)	Light green	31–60	51–100	Minor breathing discomfort to sensitive people
Moderate (101–200)	Yellow	61–90	101–250	Discomfort to people with lung or heart disease, children and older adults
Poor (201–300)	Orange	91–120	251–350	Breathing discomfort to most people on prolonged exposure
Very Poor (301–400)	Red	121–250	351–430	Respiratory illness on prolonged exposure
Severe (401–500)	Maroon	above 250	above 430	Affects healthy people; serious impact on those with existing disease

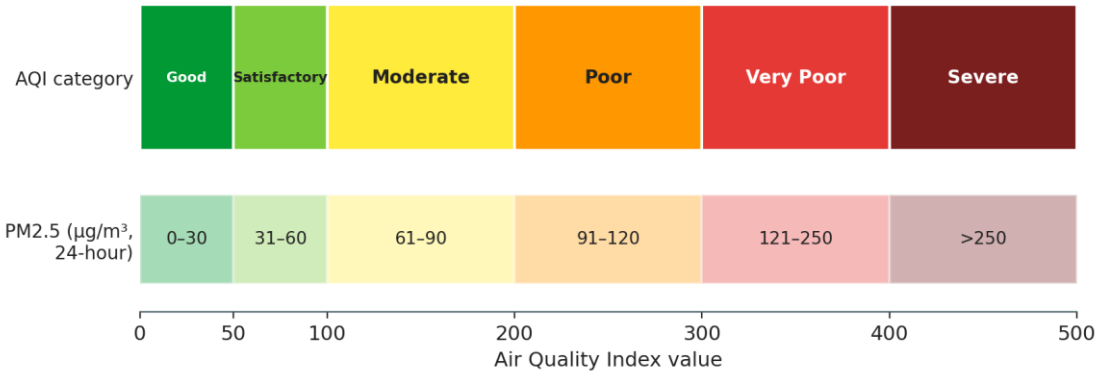


Figure 4.6: *India's National Air Quality Index categories with their colour bands and the corresponding 24-hour PM2.5 breakpoints. Source: Author, based on CPCB (2014) National Air Quality Index.*

The AQI is used to trigger emergency measures. In Delhi-NCR, the **Graded Response Action Plan (GRAP)** imposes progressively stricter restrictions, such as bans on construction or on older diesel vehicles, as the AQI moves from Poor to Severe (see the Delhi case study later in this chapter).

4.8.3 WHO Air Quality Guidelines (2021) compared with NAAQS

In September 2021 the WHO released updated **Global Air Quality Guidelines**, the first revision since 2005. Based on evidence that harm occurs even at low concentrations, it halved the annual PM2.5 guideline from 10 to 5 µg/m^3 and tightened most other values. The guidelines are not legal standards; they indicate levels below which health risks are low. For countries far above them, the WHO also sets **interim targets**; India's annual PM2.5 standard of 40 µg/m^3 is close to the WHO's first interim target (35 µg/m^3). The comparison below shows how far India's standards are from the health-based guidelines.

Table 4.7: India's NAAQS (2009) compared with WHO Air Quality Guidelines (2021)

Pollutant and averaging time	India NAAQS	WHO AQG 2021	NAAQS as multiple of WHO
PM2.5 annual (µg/m^3)	40	5	8
PM2.5 24-hour (µg/m^3)	60	15	4
PM10 annual (µg/m^3)	60	15	4
PM10 24-hour (µg/m^3)	100	45	2.2
NO_2 annual (µg/m^3)	40	10	4
NO_2 24-hour (µg/m^3)	80	25	3.2
SO_2 24-hour (µg/m^3)	80	40	2
O_3 8-hour (µg/m^3)	100	100	1

Many experts argue that India's NAAQS, unchanged since 2009, should be revised towards the WHO interim targets. Even the existing national standards are far from being met in many Indian cities, especially in the Indo-Gangetic Plain, where annual PM2.5 levels are often two to three times the standard of 40 µg/m^3.

4.8.4 Water quality standards

Designated best use classification of surface water

The CPCB classifies inland surface waters according to their **designated best use**, the most demanding use to which a water body or stretch is to be put. Each use has water quality criteria. A river stretch that fails to meet the criteria for its designated use is identified as polluted and targeted for action. Bathing water (Class B) is the target class for the Ganga under the Namami Gange programme.

Table 4.8: CPCB water quality criteria for designated best uses

Class	Designated best use	Key criteria
A	Drinking water source without conventional treatment but after disinfection	Total coliforms ≤ 50 MPN/100 mL; pH 6.5–8.5; DO ≥ 6 mg/L; BOD ≤ 2 mg/L
B	Outdoor bathing (organised)	Total coliforms ≤ 500 MPN/100 mL; pH 6.5–8.5; DO ≥ 5 mg/L; BOD ≤ 3 mg/L
C	Drinking water source after conventional treatment and disinfection	Total coliforms ≤ 5,000 MPN/100 mL; pH 6–9; DO ≥ 4 mg/L; BOD ≤ 3 mg/L
D	Propagation of wildlife and fisheries	pH 6.5–8.5; DO ≥ 4 mg/L; free ammonia (as N) ≤ 1.2 mg/L
E	Irrigation, industrial cooling, controlled waste disposal	pH 6.0–8.5; electrical conductivity ≤ 2,250 µS/cm; sodium absorption ratio ≤ 26; boron ≤ 2 mg/L

Several heavily polluted urban stretches, such as the Yamuna in Delhi downstream of Wazirabad, have near-zero DO and very high coliform counts for much of the year and fail even the criteria for bathing and for wildlife and fisheries.

Drinking water: BIS IS 10500:2012

India's standard for drinking water is **IS 10500:2012** of the Bureau of Indian Standards. For each parameter it gives an **acceptable limit**, which water should meet, and for some parameters a higher **permissible limit** that may be tolerated only when no alternative source is available. Parameters with "no relaxation" must always meet the acceptable limit. Key limits are shown below.

Table 4.9: Selected drinking-water limits under IS 10500:2012 (with amendments)

Parameter	Acceptable limit	Permissible limit (no alternative source)	Concern
pH	6.5–8.5	No relaxation	Corrosion, taste
Turbidity (NTU)	1	5	Shields pathogens
Total dissolved solids (mg/L)	500	2,000	Taste, scaling
Total hardness as $CaCO_3$ (mg/L)	200	600	Scaling, soap use
Chloride (mg/L)	250	1,000	Taste, corrosion
Nitrate as NO_3 (mg/L)	45	No relaxation	Blue baby syndrome
Fluoride (mg/L)	1.0	1.5	Fluorosis

Parameter	Acceptable limit	Permissible limit (no alternative source)	Concern
Arsenic (mg/L)	0.01	No relaxation (since 2015 amendment)	Arsenicosis, cancer
Lead (mg/L)	0.01	No relaxation	Nerve and brain damage
Total coliforms and *E. coli*	Not detectable in 100 mL	No relaxation	Water-borne disease

Effluent discharge standards

Ambient standards describe the quality of the receiving environment; **effluent** (or **discharge**) **standards** limit what a source may release. The Environment (Protection) Rules, 1986 contain **general standards** for the discharge of effluents into inland surface water, public sewers, land for irrigation and marine coastal areas. For discharge into inland surface water, for example, they include pH 5.5–9.0, total suspended solids up to 100 mg/L, BOD (3 days at 27 °C) up to 30 mg/L, COD up to 250 mg/L and oil and grease up to 10 mg/L. In addition, **industry-specific standards** apply to more than 100 industrial categories, such as tanneries, distilleries, thermal power plants and cement plants, and separate standards apply to sewage treatment plants, which have been made progressively stricter. SPCBs enforce these standards through the consent system and monitoring, including **online continuous emission and effluent monitoring systems** that 17 categories of highly polluting industries must install and connect to CPCB servers. The consent system and enforcement are explained under legal methods of control later in this chapter.

4.9 Nuclear Hazards and Human Health Risks

A **nuclear hazard** is the risk of harm to people and the environment from **ionising radiation** released by radioactive materials. **Radioactivity** is the spontaneous breakdown (decay) of unstable atomic nuclei, such as those of uranium, thorium, radium and caesium-137, during which they emit energy as radiation. The radiation is called ionising because it carries enough energy to knock electrons out of atoms, creating charged ions and breaking chemical bonds, including those in DNA. Ionising radiation cannot be seen, heard or smelt, which makes it particularly feared. It is also extremely useful, in medicine, industry, research and electricity generation, so the aim of radiation protection is to obtain the benefits while keeping the risks low.

4.9.1 Types of ionising radiation

- **Alpha (α) particles** consist of two protons and two neutrons (a helium nucleus). They are heavy and lose their energy quickly, so they travel only a few centimetres in air and are stopped by a sheet of paper or the dead outer layer of skin. Alpha emitters such as radon, polonium-210 and plutonium-239 are very dangerous if inhaled or swallowed, because then all their energy is deposited in a small volume of living tissue.
- **Beta (β) particles** are high-speed electrons. They travel up to a few metres in air and penetrate a few millimetres of skin, and are stopped by a few millimetres of aluminium or plastic. Strontium-90 and iodine-131 are beta emitters.

- **Gamma (γ) rays and X-rays** are high-energy electromagnetic waves with no mass or charge. They are very penetrating and can pass through the body; they are reduced, not completely stopped, by thick layers of lead, concrete or water. Cobalt-60 and caesium-137 emit gamma rays.
- **Neutrons**, released mainly inside nuclear reactors and explosions, are also highly penetrating and are best shielded by water, concrete or materials rich in hydrogen.

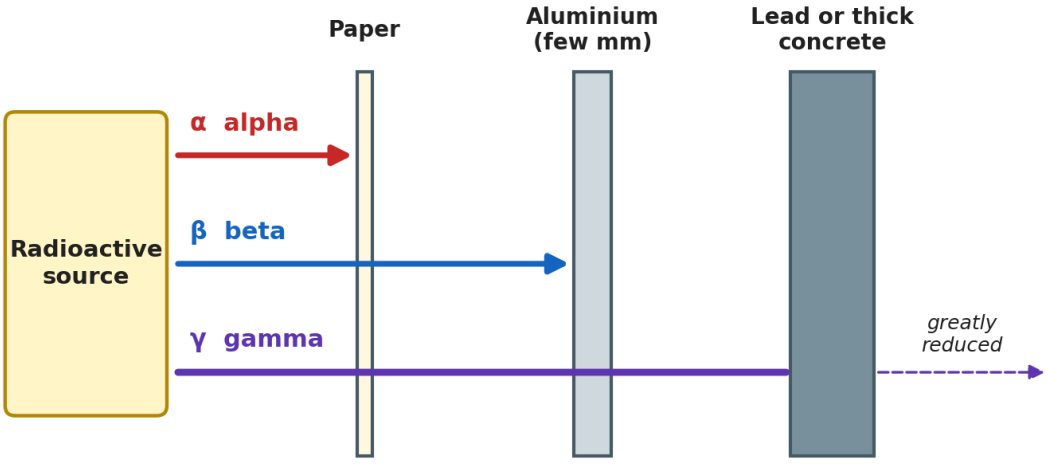


Figure 4.7: *Penetrating power of different types of ionising radiation. Alpha particles are stopped by paper, beta particles by a few millimetres of aluminium, while gamma rays need thick lead or concrete to be greatly reduced. Source: Author.*

4.9.2 Units of radiation

- The **becquerel (Bq)** measures **activity**, the amount of radioactive material: one becquerel is one nuclear decay per second.
- The **gray (Gy)** measures **absorbed dose**, the energy deposited in a material: one gray is one joule of energy absorbed per kilogram.
- The **sievert (Sv)** measures **equivalent** and **effective dose**, which take into account how damaging each type of radiation is and how sensitive each organ is. For gamma rays and beta particles, 1 Gy gives about 1 Sv; alpha particles are about 20 times more damaging per gray. Doses to people are usually expressed in millisieverts (mSv; one-thousandth of a sievert).

The **half-life** of a radioactive substance is the time taken for half of its atoms to decay. It ranges from 8 days for iodine-131, about 30 years for caesium-137 and strontium-90, to 24,100 years for plutonium-239. Substances with long half-lives remain hazardous for very long periods.

4.9.3 Sources of radiation

Natural sources give most people most of their dose. The world average natural background dose is about 2.4 mSv a year (UNSCEAR). It comes from **radon** gas seeping from soil and building materials (about half of the total), **cosmic rays** from space (higher at high altitude and during air travel), **terrestrial radiation** from uranium, thorium and potassium-40 in rocks and soil, and radioactive substances inside the body, such as potassium-40 in food. Some coastal areas of Kerala and Tamil Nadu, such as Karunagappally in Kollam district, have unusually high natural background radiation

from thorium-rich **monazite** sands; long-term studies of residents there have not found a clear increase in cancer rates.

Anthropogenic sources include:

- **Medical uses**: X-rays, CT scans, nuclear medicine and radiotherapy. This is the largest human-made source; a CT scan of the abdomen gives about 8–10 mSv, while a chest X-ray gives less than 0.1 mSv.
- **The nuclear fuel cycle**: uranium mining and milling, fuel fabrication, reactor operation, reprocessing of spent fuel, and waste storage. India has nearly 9,000 MW of installed nuclear capacity (2025) and plans major expansion.
- **Nuclear weapons tests**: atmospheric tests between 1945 and 1980 spread radioactive fallout worldwide.
- **Accidents and lost sources**: reactor accidents, and industrial or medical radioactive sources that are lost, stolen or sold as scrap.
- **Industrial and research uses**: industrial radiography of welds, food irradiation, gauges, and research reactors.

4.9.4 Effects of radiation on human health

Radiation damages cells either directly, by breaking DNA strands, or indirectly, by splitting water molecules into reactive **free radicals** that then attack DNA and other molecules. Cells that divide rapidly, such as those in bone marrow, the lining of the gut, the skin, the reproductive organs and the developing foetus, are the most sensitive. Health effects are of two kinds.

Deterministic (tissue) effects occur only above a **threshold dose**, usually a high dose received over a short time, and their severity increases with dose. Examples are radiation burns of the skin, cataracts, temporary or permanent sterility, and **acute radiation syndrome (ARS)**. ARS appears after whole-body doses above about 1 Gy and causes nausea, vomiting, hair loss, bleeding and infection as bone marrow and gut lining fail; without treatment, about half of those receiving 4–5 Gy die within weeks.

Stochastic (chance) effects have no known threshold; the *probability* of the effect, not its severity, increases with dose. The main stochastic effects are **cancers**, especially leukaemia and thyroid, breast and lung cancers, which may appear years to decades after exposure, and **heritable effects** (genetic damage passed to offspring), which have been seen in animals but not clearly demonstrated in humans. For protection purposes, regulators assume a **linear no-threshold** relationship: any dose, however small, carries some risk, roughly in proportion to the dose. On this basis, the International Commission on Radiological Protection (ICRP) estimates the lifetime risk of fatal cancer at about 5% per sievert.

Table 4.10: Deterministic and stochastic effects of radiation

Feature	Deterministic effects	Stochastic effects
Threshold dose	Yes	No (assumed)
Dose increases	Severity of effect	Probability of effect
Typical dose	High, short-term	Any dose, including low
Time to appear	Hours to weeks (cataract: years)	Years to decades

Feature	Deterministic effects	Stochastic effects
Examples	ARS, skin burns, cataract, sterility	Cancer, heritable genetic effects

4.9.5 Major nuclear accidents

The International Nuclear and Radiological Event Scale (INES) rates nuclear and radiological events from level 1 (anomaly) to level 7 (major accident). Three accidents illustrate the range of consequences.

- **Kyshtym, USSR (1957)**: at the Mayak plutonium production complex in the southern Urals, the cooling system of a tank of high-level liquid waste failed, and the tank exploded. The release contaminated a strip of land about 300 km long (the East Urals Radioactive Trace), more than 10,000 people were evacuated, and the accident was kept secret for decades. It is rated INES level 6.
- **Chernobyl, Ukraine, USSR (26 April 1986)**: during a poorly planned safety test, a power surge destroyed reactor 4, and the graphite fire that followed released radioactive iodine, caesium and other materials over much of Europe. Of 134 plant workers and firefighters who developed acute radiation syndrome, 28 died within four months. About 3.5 lakh people were eventually relocated, and a 30 km exclusion zone remains. The clearest long-term health effect has been thousands of cases of thyroid cancer among people who were children at the time and drank milk contaminated with iodine-131. Rated INES 7.
- **Fukushima Daiichi, Japan (11 March 2011)**: an earthquake and tsunami knocked out power to the cooling systems, causing meltdowns in three reactors and hydrogen explosions. More than 1.5 lakh people were evacuated. No deaths from acute radiation occurred, but the stress and disruption of evacuation contributed to many deaths among elderly and ill people. Rated INES 7. Since 2023, Japan has been releasing treated, diluted water from the site into the Pacific, a step reviewed by the International Atomic Energy Agency (IAEA) but opposed by some fishing communities and neighbouring countries.

These accidents show the need for safe design, protection against natural hazards, a culture of safety and transparency, emergency planning and an independent regulator.

4.9.6 Nuclear hazards in India

India's nuclear programme, under the Department of Atomic Energy (DAE), has had no accident with significant off-site release, but it has faced concerns over uranium mining, plant siting and the security of radioactive sources.

Case Study: Mayapuri, Delhi (2010): a radioactive source in a scrap market

In February 2010, the University of Delhi auctioned an old **Gammacell 220** irradiator, which its chemistry department had bought in 1968 and had not used since 1985, to a scrap dealer in Mayapuri, an industrial area in west Delhi. The irradiator contained cobalt-60, a strong gamma emitter. In April 2010, workers at a scrap shop dismantled it and handled the small, shiny pencils containing the source, unaware of the danger. Within days, several of them developed burns, vomiting and hair loss.

Seven people were hospitalised with high radiation exposure, and one worker died of acute radiation syndrome at AIIMS, New Delhi. The Atomic Energy Regulatory Board (AERB), with scientists of the Bhabha Atomic Research Centre (BARC) and the National Disaster Response Force, traced and recovered the scattered sources using radiation detectors. The IAEA rated the event at level 4 on the INES scale.

The incident revealed serious gaps: the university had not informed the regulator before disposal, AERB did not have a complete inventory of radioactive sources acquired before its creation in 1983, and scrap traders had no means of detecting radioactivity. Since then AERB has created an inventory and an online licensing system (e-LORA) for radiation sources, made disposal through authorised agencies mandatory, and promoted radiation detectors at major scrap and steel-recycling yards and ports.

Uranium mining at Jaduguda, Jharkhand: the Uranium Corporation of India Limited (UCIL) has mined uranium at Jaduguda in East Singhbhum district since the late 1960s, later opening more mines at Bhatin, Narwapahar, Turamdih and other sites. Radioactive mill wastes (tailings) are stored in large tailings ponds near Adivasi villages. Activist groups such as the Jharkhandi Organisation Against Radiation (JOAR) and some independent health surveys have reported higher rates of birth defects, infertility and cancers in nearby villages, and leaks from tailings pipelines have occurred. UCIL and DAE state that public radiation doses are within regulatory limits and attribute illness to poverty and malnutrition. The dispute shows the need for independent, transparent health monitoring.

Kudankulam, Tamil Nadu: the Kudankulam Nuclear Power Project, built with Russian collaboration, has two 1,000 MW pressurised water reactors in operation (Unit 1 from 2014, Unit 2 from 2017) and four more under construction. After the 2011 Fukushima accident, the People's Movement Against Nuclear Energy, centred on the fishing village of Idinthakarai, led prolonged protests over the plant's safety, evacuation plans, the impact of hot-water discharge on fisheries and the storage of spent fuel. In 2013, the Supreme Court allowed the plant to operate but directed authorities to implement safety upgrades, to plan a deep geological repository for spent fuel, and to monitor the plant's impact.

4.9.7 Nuclear waste

Nuclear waste is classified by its level of radioactivity:

- **Low-level waste**: contaminated clothing, gloves, tools and filters, which are compacted and buried in engineered near-surface trenches.
- **Intermediate-level waste**: resins, chemical sludges and reactor components that need shielding.

- **High-level waste**: spent fuel or the liquid waste left after spent fuel is reprocessed. It is intensely radioactive, generates heat, and remains hazardous for thousands of years.

India follows a **closed fuel cycle**, in which spent fuel is reprocessed at Trombay, Tarapur and Kalpakkam to recover plutonium for the next stage of its three-stage nuclear programme (see Chapter 3). High-level liquid waste is **vitrified**, that is, mixed with glass-forming materials and melted into stable glass blocks sealed in steel canisters, which are stored under surveillance at Tarapur and Kalpakkam. The long-term plan, as elsewhere, is disposal in a **deep geological repository** in stable rock. Finland, whose Onkalo repository is the first in the world to approach operation, is furthest along.

4.9.8 Regulation and radiation safety in India

The **Atomic Energy Regulatory Board (AERB)**, set up in 1983, regulates nuclear power plants, fuel-cycle facilities and every use of radiation sources in India, including X-ray machines in hospitals, radiotherapy units and industrial radiography cameras. It issues licences, inspects facilities, sets safety codes and investigates incidents. The AERB long functioned under the DAE, which also operates the plants it regulates, raising concerns about its independence; the SHANTI Act, 2025, which replaced the Atomic Energy Act, 1962 and opened the nuclear power sector to private participation, gave AERB statutory status.

AERB dose limits, based on ICRP recommendations, exclude natural background and medical exposure:

- **Radiation workers**: 20 mSv per year averaged over five years, and not more than 30 mSv in any single year.
- **Members of the public**: 1 mSv per year from all regulated sources.

4.9.9 Principles of radiation safety

The ICRP system, followed by AERB, rests on three principles:

1. **Justification**: no practice involving radiation should be adopted unless it produces more benefit than harm.
2. **Optimisation (ALARA)**: doses should be kept **as low as reasonably achievable**, taking economic and social factors into account.
3. **Dose limitation**: individual doses must not exceed the prescribed limits.

In practice, individuals protect themselves by three simple means: **time** (reduce the time spent near a source), **distance** (dose falls sharply with distance; doubling the distance from a point source cuts the dose rate to a quarter) and **shielding** (lead aprons in X-ray rooms, concrete walls around reactors and radiotherapy units). Radiation workers wear personal dosimeters, radioactive areas are marked with the trefoil warning symbol, and nuclear plants maintain off-site emergency plans for a zone of about 16 km around them, including the stockpiling of potassium iodide tablets that protect the thyroid from radioactive iodine.

Key Points

- Alpha, beta and gamma radiation differ in penetrating power; alpha emitters are most dangerous inside the body.
- Activity is measured in becquerels, absorbed dose in grays, and health-relevant dose in sieverts.
- Deterministic effects occur above a threshold; stochastic effects such as cancer are assumed to have no threshold.
- Chernobyl, Fukushima and Kyshtym show the scale of reactor and waste accidents; Mayapuri shows the danger of poorly tracked radioactive sources.
- AERB regulates radiation safety in India using the principles of justification, optimisation (ALARA) and dose limitation.

4.10 Solid Waste: Types, Sources and Generation in India

Solid waste is any unwanted or discarded material that is not a liquid or a gas and that its owner no longer needs. It includes food scraps and packaging from homes, ash from power stations, slag from steel plants, used syringes from hospitals and old mobile phones. The term is used broadly: semi-solid sludge from treatment plants is also handled as solid waste. **Solid waste management (SWM)** is the planned control of waste from the point where it is generated, through storage, collection, transport and processing, to its final disposal, in ways that protect health and the environment and recover as much value as possible.

Waste is a pollution problem because nothing truly disappears. Food waste dumped on open land rots without air and gives off **methane** (CH_4), a greenhouse gas about 80 times more powerful than CO_2 over 20 years. Rainwater trickling through a dump dissolves salts, organic acids and heavy metals to form a dark, foul liquid called **leachate**, which seeps into groundwater. Uncollected plastic chokes drains and causes urban flooding, and dumps burn for weeks, releasing smoke laden with fine particles and dioxins. Poorly managed waste therefore pollutes air, water and soil at the same time.

4.10.1 Classification of solid waste

Solid waste is usually classified by its **source** and by the **hazard** it poses. The two schemes overlap: a household produces mostly harmless kitchen waste but also small amounts of hazardous items such as batteries and medicines.

Table 4.11: Classification of solid waste by source

Category	Main sources	Typical components	Main concerns
Municipal solid waste (MSW)	Homes, shops, offices, markets, hotels, street sweeping	Food and garden waste, paper, plastic, glass, metal, textiles, inert dust	Methane, leachate, disease vectors, open burning
Industrial waste	Factories, power plants, mines	Fly ash, slag, red mud, sludge, process residues	Large volumes, land use, toxic leachate

Category	Main sources	Typical components	Main concerns
Hazardous waste	Chemical, pharmaceutical, metal-plating, refinery and other industries	Spent solvents, acids, heavy-metal sludge, oily waste	Toxicity, fire, explosion, groundwater contamination
Biomedical waste	Hospitals, clinics, laboratories, blood banks	Body parts, soiled dressings, sharps, used culture media	Infection, needle-stick injuries
E-waste	Homes, offices, IT industry	Discarded phones, computers, appliances, batteries	Lead, mercury, cadmium, brominated flame retardants
Construction and demolition (C&D) waste	Building, renovation and demolition	Concrete, bricks, sand, steel, wood	Dust, illegal dumping on floodplains and roadsides
Agricultural waste	Farms, dairies	Crop residue, husk, animal dung	Open burning of residue, water pollution

By **biodegradability**, waste is either **biodegradable** (broken down by microorganisms within weeks or months, such as food, leaves and paper) or **non-biodegradable** (persisting for decades or centuries, such as most plastics, glass and metals). This simple division is the basis of the "wet" and "dry" streams that households are asked to keep apart.

4.10.2 Composition of Indian municipal waste

Indian MSW differs from that of rich countries: it is wetter, denser and richer in organic matter, and it contains less paper and packaging. It also varies between rich and poor neighbourhoods and between cities, so planning starts with a local **waste characterisation** study in which samples are sorted and weighed. National studies, including those reviewed by the Planning Commission's Task Force on Waste to Energy (2014), place the biodegradable (compostable) fraction at roughly 40–60% of urban waste, with recyclables (paper, plastic, metal, glass) at about 15–20% and inerts (street dust, silt, construction debris) making up most of the rest. Moisture content is often 45–55%, and the **calorific value** (heat released on burning) of mixed Indian waste is low. These facts matter: wet, organic waste is ideal for composting and biogas but burns poorly, so technologies imported from Europe often fail in Indian conditions.

4.10.3 How much waste does India generate?

The **Central Pollution Control Board (CPCB)** compiles annual reports from State Pollution Control Boards on the implementation of the Solid Waste Management Rules. According to its *Annual Report on Solid Waste Management 2021–22*, India's urban areas generated about **1,70,339 tonnes per day (TPD)** of solid waste. About 92% of this was collected, 54% was processed or treated, and 24% went to sanitary landfills. The remaining 22%, about 37,000 TPD, was unaccounted for, which in practice means it lay in open dumps, drains, vacant plots and riverbanks. The same report counted 2,452 dumpsites in the country, of which only 366 had been reclaimed.

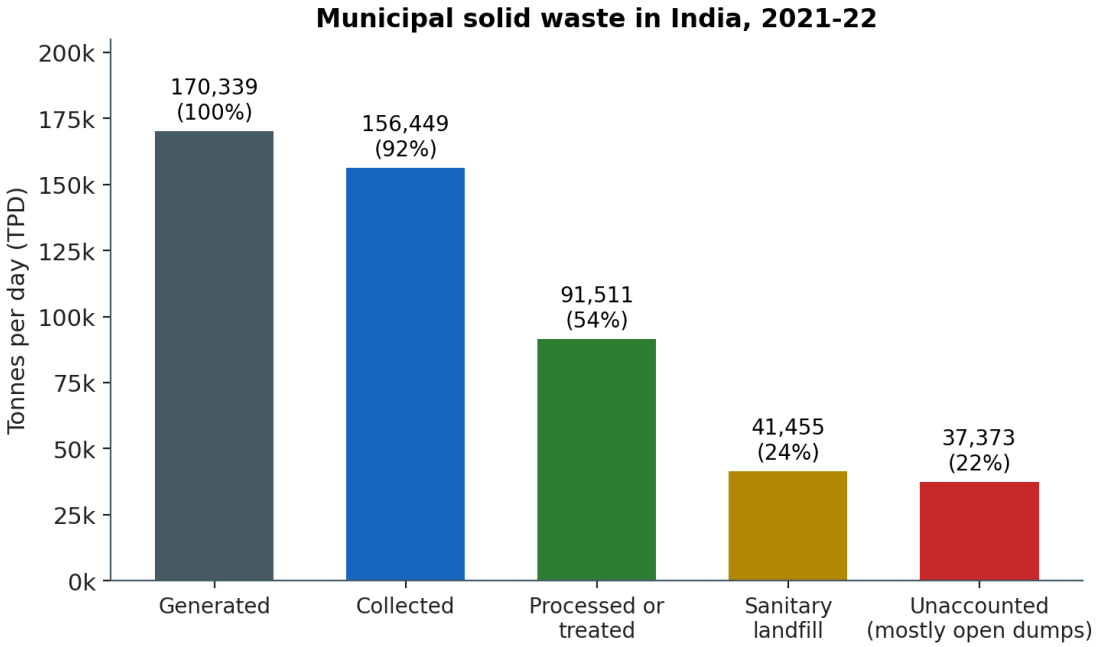


Figure 4.8: *Fate of municipal solid waste generated in India's urban areas in 2021–22. Processing has risen steadily since 2016, but more than one-fifth of waste is still unaccounted for. Source: CPCB, Annual Report on Solid Waste Management 2021–22.*

Per-person generation in India is low by world standards, roughly 0.3–0.5 kg per urban resident per day compared with over 2 kg in the United States, but the total is large and growing with urbanisation and rising incomes. The World Bank's *What a Waste 2.0* (2018) projects that global MSW will rise from about 2 billion tonnes in 2016 to 3.4 billion tonnes by 2050, with the fastest growth in South Asia and Africa.

Mountains of garbage

Delhi's Ghazipur dumpsite, started in 1984, had grown to about 65 metres by 2019, almost the height of the Qutub Minar (73 m). In September 2017 a portion of it collapsed onto a road, sweeping vehicles into a canal and killing two people. Garbage mountains are not just unsightly; they are unstable, they catch fire, and they leak leachate into the Yamuna floodplain.

4.11 The Waste Hierarchy and Integrated Waste Management

The **waste hierarchy** ranks waste-management options from the most to the least desirable for the environment. Its logic is simple: it is better not to create waste at all than to manage it well, and better to keep a material in use than to burn it or bury it. The older slogan of the "3Rs" (Reduce, Reuse, Recycle) has been widened into longer lists such as Refuse, Reduce, Reuse, Repair, Repurpose, Recycle and Recover, but the order of priority is the same.

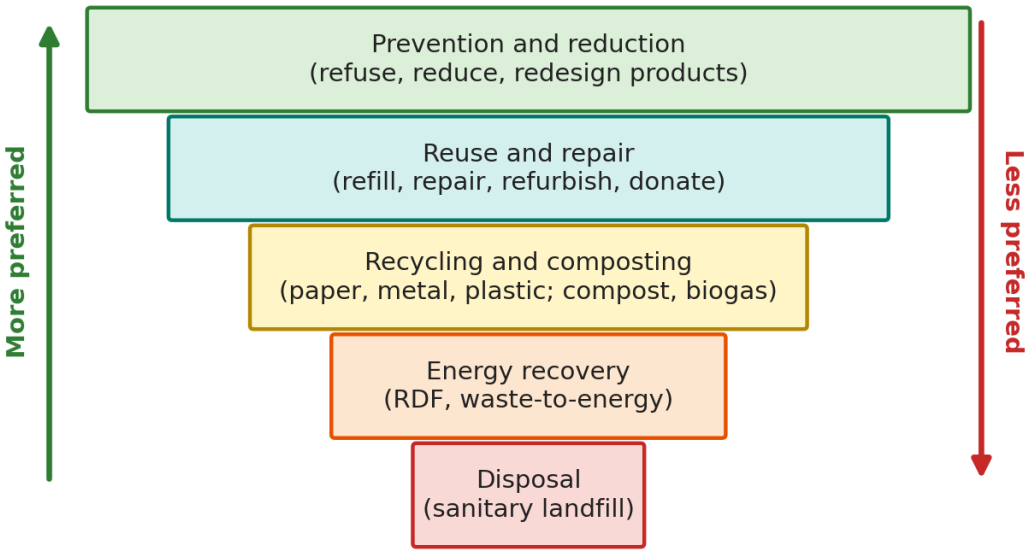


Figure 4.9: *The waste hierarchy. Options near the top save the most resources and cause the least pollution; disposal is the last resort for residues that cannot be dealt with in any other way. Source: Original diagram based on the UNEP and MoEFCC waste hierarchy.*

- **Prevention and reduction**: refusing unnecessary items (a plastic bag for a single packet of biscuits), buying in bulk, designing products that use less material or last longer. In industry this is called **source reduction**.
- **Reuse and repair**: using an item again for the same or a new purpose, such as refilling glass milk bottles, repairing a mixer, donating old clothes or using a textile shop's cloth bag. India's tradition of repair shops, *kabadiwalas* and hand-me-downs is a strong base for this level.
- **Recycling and composting**: reprocessing a discarded material into a new product (paper into cardboard, PET bottles into polyester fibre) or turning organic waste into compost or biogas. Recycling saves raw materials and energy; making aluminium from scrap uses only about 5% of the energy needed to make it from bauxite.
- **Energy recovery**: burning waste that cannot be recycled to produce heat or electricity, or converting it into a fuel. Energy is recovered, but the material is lost and the residues must still be disposed of.
- **Disposal**: placing residual, inert waste in an engineered **sanitary landfill**. Open dumping does not count as disposal at all; it is simply uncontrolled pollution.

Integrated solid waste management (ISWM) combines several options at once, matched to the composition of local waste. A city may home-compost some wet waste, run decentralised biogas plants for markets, send dry waste to a **material recovery facility (MRF)** for sorting, convert non-recyclable plastics into fuel for cement kilns and landfill only the inert residue. ISWM also looks beyond technology to finance, laws, institutions and public participation. Experience across India shows that cities fail not for lack of machines but for lack of segregation, money for operation and maintenance, and accountability.

Think and Do: Audit your own bin

For one week, weigh (or estimate) the waste your household or hostel room produces each day, keeping wet waste, dry waste and other items separate. Calculate grams per person per day and the share of each stream. Which items could you have refused, reused or composted? Compare your figures with the national average discussed above.

4.12 Segregation at Source and Collection

Segregation at source means separating waste into different streams at the point where it is generated, such as a kitchen, office or shop, instead of mixing everything in one bin. It is the single most important step in waste management. Once food scraps are mixed with paper, plastic, batteries and broken glass, the paper becomes soggy and unrecyclable, the compost is contaminated with glass and heavy metals, and the plastic is too dirty to sell. Clean, separated streams have value; mixed waste is a costly nuisance.

4.12.1 Streams and colour-coded bins

The Solid Waste Management Rules, 2016 required every household to keep waste in three streams: biodegradable (wet), non-biodegradable (dry) and domestic hazardous waste. The **Solid Waste Management Rules, 2026**, which replaced the 2016 Rules from 1 April 2026, made **four-stream segregation** mandatory by adding a separate stream for sanitary waste.

Table 4.12: Four-stream segregation of household waste under the SWM Rules, 2026

Stream	Examples	Usual bin colour	Where it should go
Wet (biodegradable) waste	Vegetable and fruit peels, cooked food, tea leaves, eggshells, flowers, garden leaves	Green	Home or community composting, biomethanation plant
Dry waste	Paper, cardboard, plastic, metal cans, glass, rubber, wood, cloth	Blue	Material recovery facility, recyclers, waste pickers
Sanitary waste	Used diapers, sanitary pads, tampons, condoms	Wrapped securely, kept separately	Separate collection; incineration or approved processing
Special care (domestic hazardous) waste	Paint and pesticide cans, CFL and tube lights, mercury thermometers, expired medicines, used batteries	Red or black, or a separate bag	Authorised collection centres and hazardous-waste facilities

Bin colours are not identical in every city, but green for wet and blue for dry is the national convention promoted under the Swachh Bharat Mission. **Bulk waste generators**, defined in the 2026 Rules as entities producing at least 100 kg of waste a day (or meeting thresholds of water use or floor area), such as hotels, hostels, malls and large housing societies, must process their wet waste on-site where possible.

4.12.2 Collection and transport

Collection is the most visible and most expensive part of SWM; it typically uses well over half of a city's waste budget. It also decides whether segregation survives: if a vehicle tips separated bins into one compartment, residents conclude that their effort is pointless. The main systems are:

- **Door-to-door collection**: a vehicle or handcart visits each house daily, often with a jingle to signal its arrival. This replaced the old system of community bins (*dhalaos*), which overflowed and attracted animals.
- **Secondary collection and transfer**: small vehicles bring waste to a **transfer station**, where it is compacted into large trucks for the journey to processing sites. GPS tracking of vehicles is now common.
- **Collection of special streams**: separate rounds or drop-off points for e-waste, household hazardous waste, bulky items and C&D waste.

India's informal sector deserves special mention. An estimated 1.5–4 million **waste pickers**, many of them women, Dalits and migrants, recover recyclables from bins and dumps and sell them through a chain of *kabadiwalas* and traders. They divert large quantities of material from landfills at no cost to the city, yet work without protective gear, health care or social security. The SWM Rules require urban local bodies to recognise and integrate waste pickers. Pune's waste-picker cooperative **SWaCH**, which provides door-to-door collection under an agreement with the municipal corporation, is a well-known model of such integration.

4.13 Processing and Disposal of Municipal Solid Waste

Processing changes the physical, chemical or biological character of waste to recover materials or energy and to reduce what must be disposed of. The choice of technology depends on the waste stream.

4.13.1 Composting

Composting is the controlled aerobic (oxygen-using) decomposition of organic waste by bacteria, fungi and other microorganisms into **compost**, a dark, crumbly, humus-like material that improves soil structure, water retention and fertility. It is the most suitable method for the large wet fraction of Indian waste.

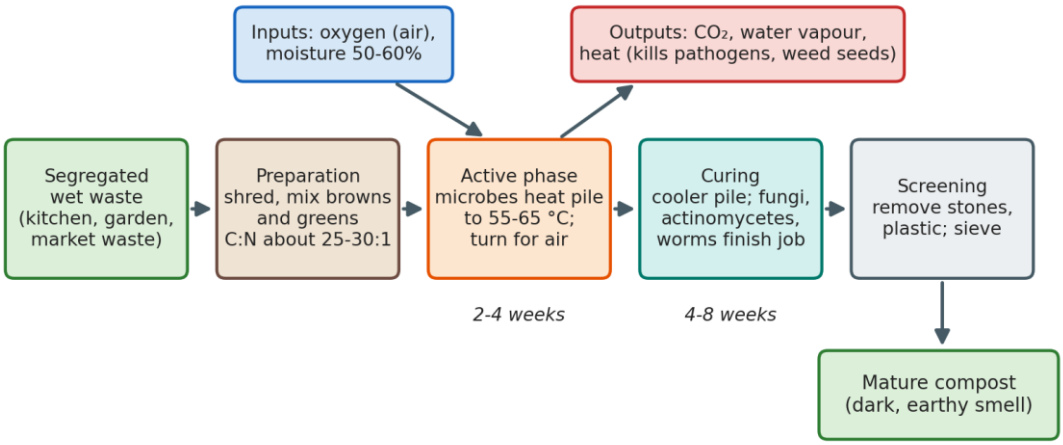


Figure 4.10: *Main stages of aerobic composting. Heat generated by microbes in the active phase kills most pathogens and weed seeds; the pile then cools and matures. Durations are typical for windrow composting. Source: Original diagram.*

Four factors control composting: the **carbon-to-nitrogen (C:N) ratio** (about 25–30:1 is ideal, achieved by mixing carbon-rich "browns" such as dry leaves with nitrogen-rich "greens" such as vegetable peels), moisture (about 50–60%), aeration (turning the pile or blowing air through it) and particle size. During the active, **thermophilic** phase, temperatures reach 55–65 °C, which destroys most pathogens and weed seeds. Compost can be made in household pots and pits (the three-tier terracotta *khamba* is popular in Bengaluru and Delhi), in community **windrows** (long piles turned by machines) or in enclosed in-vessel composters for hotels and campuses.

The government promotes city compost through a market development assistance scheme for fertiliser companies that co-market it with chemical fertilisers. The main obstacle is quality: compost made from mixed waste often contains glass, plastic and heavy metals, and farmers rightly reject it. Good compost needs good segregation.

4.13.2 Vermicomposting

Vermicomposting uses earthworms, usually the red wiggler *Eisenia fetida* or the African nightcrawler *Eudrilus eugeniae*, to break down organic waste. The worms eat the waste and excrete **vermicast**, a fine, nutrient-rich manure containing plant-growth substances and beneficial microbes. The process works at normal temperatures, in shaded beds kept moist, and is ideal for households, colleges, dairies and temple flower waste. Worms cannot tolerate oil, salt, citrus peels in bulk, meat or high heat, so pre-composted or carefully selected waste works best. Many Delhi University colleges run vermicomposting pits for campus leaves and canteen waste, which is one of the practicals in this course.

4.13.3 Biomethanation

Biomethanation (anaerobic digestion) is the breakdown of organic waste by microorganisms in the absence of oxygen, inside a sealed tank called a **digester**. It produces **biogas**, about 55–65% methane and the rest mostly CO_2, which can be used for cooking, electricity or, after cleaning and compression, as **compressed biogas (CBG or bio-CNG)** for vehicles. The leftover **digestate** is a good manure. Biomethanation suits wet, food-rich waste from markets, hotels and hostels, and it is a cleaner use for methane than letting it escape from dumps.

$$\text{Organic matter (e.g., } C_6H_{12}O_6) \rightarrow 3CH_4 + 3CO_2 \text{ (anaerobic microbes)}$$

India's largest example is Indore's bio-CNG plant, commissioned in 2023 under the GOBARdhan scheme, which can treat about 550 tonnes of segregated wet waste per day and supplies bio-CNG to city buses. The **SATAT** (Sustainable Alternative Towards Affordable Transportation) initiative of the petroleum ministry encourages private CBG plants by guaranteeing purchase by oil-marketing companies.

4.13.4 Refuse-derived fuel

Refuse-derived fuel (RDF) is a fuel made from the dry, combustible, non-recyclable fraction of waste, such as low-grade plastic, textiles, multilayered packaging and paper, after removing wet matter, metals and inerts and shredding the rest. RDF is burned as a substitute fuel in cement kilns, whose very high temperatures (about 1,400 °C) and alkaline conditions destroy most pollutants, and in some power plants. Co-processing RDF in cement kilns is one of the few outlets for plastics that cannot be recycled, and the Plastic Waste Management Rules allow it. Cement companies demand RDF of consistent quality, which again depends on segregation.

4.13.5 Waste-to-energy incineration

Incineration is the controlled burning of waste at high temperature (850 °C or more) in a furnace. A **waste-to-energy (WtE)** plant uses the heat to raise steam and generate electricity. Incineration reduces waste volume by about 90% and weight by about 75%, which is attractive for land-scarce cities. Delhi has four WtE plants, at Okhla (Timarpur), Ghazipur, Narela–Bawana and Tehkhand, which together handle a large share of the city's waste.

WtE has had a troubled history in India for several reasons:

- **Low calorific value**: mixed Indian waste is wet and full of inert dust, so it burns poorly. Several early plants, including one at Timarpur in Delhi (1987), shut down soon after opening because the waste would not sustain combustion without added fuel.
- **Air pollution**: burning plastics and mixed waste can release fine particles, acid gases (HCl, SO_2), heavy metals and highly toxic **dioxins and furans** unless the plant has good flue-gas cleaning and keeps furnace temperatures steady. Residents near the Okhla plant have complained for years about smoke and ash, and the matter has reached the National Green Tribunal.
- **Ash**: about a fifth to a quarter of the input remains as bottom ash and fly ash; the fly ash is often hazardous and needs a secure landfill.
- **Cost and conflict with recycling**: power from WtE is expensive, and plants that are paid by the tonne have an incentive to burn recyclables and wet waste that should go elsewhere.

The waste hierarchy therefore places WtE below recycling and composting. It is a sensible option only for segregated, non-recyclable, combustible residues and only with strict emission control.

4.13.6 Sanitary landfills versus open dumps

An **open dump** is land where waste is simply tipped without any engineering. It has no liner, no leachate or gas control and no daily soil cover. Rain percolates through the waste, and the leachate pollutes groundwater; methane builds up and the dump catches fire; flies, rats, stray animals and birds spread disease. Most Indian cities still rely on such dumps.

A **sanitary landfill** is an engineered site for the safe, final disposal of residual and inert waste. Its key features are shown below. Under the SWM Rules, only non-usable, non-recyclable, non-biodegradable, non-combustible and non-reactive inert waste and pre-processing rejects should reach a landfill, not mixed raw waste.

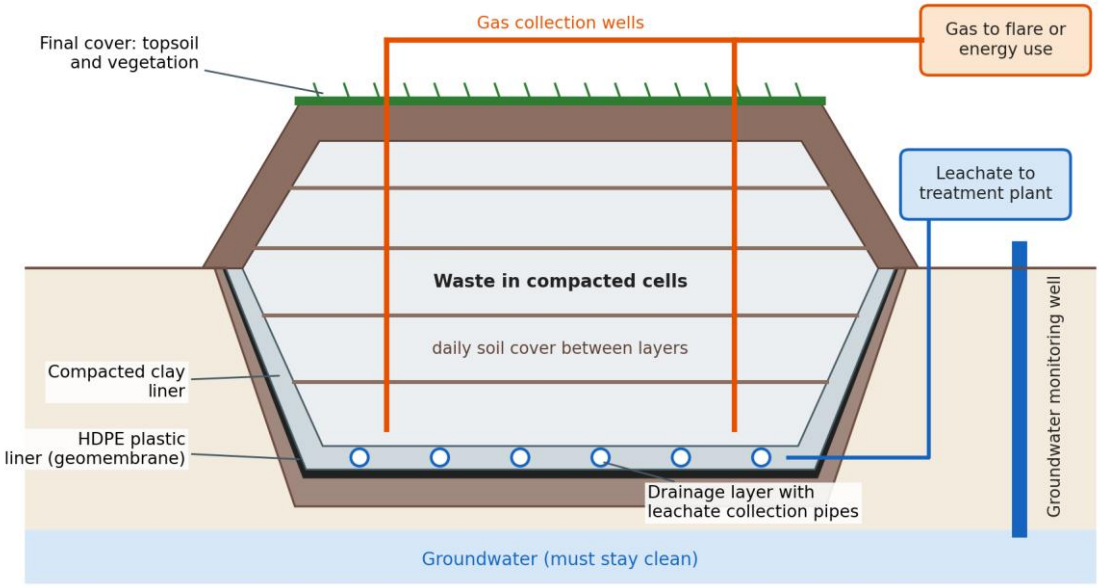


Figure 4.11: *Cross-section of a sanitary landfill. A double liner of compacted clay and HDPE plastic keeps leachate out of groundwater; pipes collect leachate for treatment; wells draw off landfill gas; and a final cover of soil and vegetation seals the site. Source: Original diagram.*

Table 4.13: Open dump compared with a sanitary landfill

Feature	Open dump	Sanitary landfill
Liner	None	Compacted clay plus HDPE geomembrane
Leachate	Seeps into soil and groundwater	Collected through pipes and treated
Landfill gas	Escapes, causes fires and odour	Collected; flared or used for energy
Daily cover	None; waste exposed	Soil or other cover at the end of each day
Waste accepted	Everything, mixed	Only inert residues and processing rejects
Monitoring and closure	None	Groundwater monitoring; capping and post-closure care

Delhi and Mumbai: lessons from the garbage mountains

Delhi's three old dumpsites, **Ghazipur** (opened 1984), **Bhalswa** (1994) and **Okhla** (1996), exceeded their capacity long ago but continued to receive waste. They have repeatedly caught fire; in the spring of 2022 major blazes at Ghazipur (March) and Bhalswa (April) burned for days and covered nearby neighbourhoods in smoke. In Mumbai, the **Deonar** dumpsite, in use since 1927 and one of the oldest and largest in Asia, burned through January–March 2016. The smoke was visible from satellites, schools in nearby areas were closed, and the Bombay High Court restricted new construction in the city until waste management improved. A similar fire at the **Brahmapuram** dump in Kochi in March

2023 smouldered for nearly two weeks. Such fires release fine particles, carbon monoxide and dioxins over dense neighbourhoods, often home to poorer families and waste pickers.

4.13.7 Biomining of legacy waste

Legacy waste is the old, mixed waste that has accumulated in dumpsites over years or decades. **Biomining** (or bioremediation of dumpsites) is the process of excavating this waste, stabilising it, often by spraying microbial cultures and turning it in windrows to dry and decompose it, and then sieving it in **trommels** (rotating drum screens) into fractions. The fine soil-like fraction is used to fill low-lying land, the combustible fraction is sent as RDF to cement plants, and inert material such as stones and debris is used in construction. The cleared land can then be reused.

The Swachh Bharat Mission–Urban 2.0 (2021–26) set the goal of remediating all legacy dumpsites. Indore cleared the legacy waste from its roughly 100-acre Devguradia dump by about 2019–20 and converted part of the land into a city forest. In Delhi, the Municipal Corporation reported in 2026 that over 200 lakh tonnes of legacy waste had been processed at the three sites, with targets of clearing Okhla and Bhalswa by the end of 2026 and Ghazipur by 2027. Biomining makes sense only if fresh waste stops arriving; otherwise the mountain is merely being shuffled.

Key Points

- Segregation at source decides the success of every later step; mixed waste has little value.
- Wet waste suits composting, vermicomposting and biomethanation; dry waste suits recycling; combustible rejects can become RDF.
- Waste-to-energy needs dry, high-calorie, segregated waste and strict emission control; it is not a substitute for segregation.
- Only inert residues should reach an engineered sanitary landfill; open dumps pollute air, water and soil and catch fire.
- Biomining can reclaim old dumpsites, but only if fresh dumping stops.

4.14 Control Measures for Urban Waste

Technology alone cannot solve the urban waste problem. Cities also need clear rules that assign responsibility, money for operation and maintenance, monitoring, and the cooperation of citizens. India's approach has evolved through three generations of rules, backed by national missions.

4.14.1 From the 2000 Rules to the SWM Rules, 2016

India's first specific law on city waste, the **Municipal Solid Wastes (Management and Handling) Rules, 2000**, followed a Supreme Court case (Almitra Patel v. Union of India) on the filthy state of Indian cities. The rules required collection, segregation and scientific landfilling, but compliance was poor. They were replaced by the **Solid Waste Management Rules, 2016**, framed under the Environment (Protection) Act, 1986. Their main features were:

- **Wider coverage**: beyond municipal areas to urban agglomerations, census towns, notified industrial townships, railways, airports, ports, defence establishments, special economic zones, places of pilgrimage and places of religious or historical importance.

- **Duties of waste generators**: every household and establishment had to segregate waste into wet, dry and domestic hazardous streams, hand it over to authorised collectors and pay a **user fee**; littering, burning and burying waste were prohibited, and local bodies could levy **spot fines**.
- **Bulk generators** (more than 100 kg of waste per day) had to process wet waste on-site where possible.
- **Event organisers** of gatherings of more than 100 people had to ensure segregation and handover.
- **Brand owners** of packaged products were asked to support take-back of packaging waste.
- **Processing priorities**: composting and biomethanation for wet waste; recycling for dry waste; RDF to be used by cement plants and other industries located within 100 km of an RDF plant; WtE only for non-recyclable waste with a calorific value of at least 1,500 kcal/kg.
- **Landfill restrictions**: only rejects and inerts to be landfilled; old dumpsites to be capped or biomined.
- **Recognition of waste pickers** and their integration into door-to-door collection.

Separate rules were notified in 2016 for **construction and demolition waste**, requiring large generators to segregate and deliver C&D waste to recycling plants, and for plastic, biomedical, hazardous and e-waste, discussed below.

4.14.2 The Solid Waste Management Rules, 2026

On the basis of a decade's experience, the Ministry of Environment, Forest and Climate Change (MoEFCC) notified the **Solid Waste Management Rules, 2026**, which came into force on 1 April 2026 and replaced the 2016 Rules. The key changes are:

- **Four-stream segregation** at source (wet, dry, sanitary and special-care waste), with clear routes for each stream.
- **Extended bulk waste generator responsibility**: bulk generators (100 kg of waste a day or more, or above thresholds of water use or built-up area) must process wet waste on-site or obtain certificates showing that it has been processed by an authorised agency elsewhere.
- **Environmental compensation** on the **polluter-pays principle** for violations such as operating without registration, false reporting or forged documents; the CPCB frames guidelines and State Pollution Control Boards levy the compensation.
- Greater emphasis on registration of processing facilities, digital reporting and time-bound remediation of legacy dumpsites.

4.14.3 National missions: Swachh Bharat and Swachh Survekshan

The **Swachh Bharat Mission–Urban** (2014) and its second phase, **SBM-U 2.0** (2021–26), with the goal of "Garbage Free Cities", funded door-to-door collection, MRFs, compost and biogas plants and dumpsite remediation. Its role as a national awareness campaign is discussed in Chapter 7.

Swachh Survekshan, conducted annually since 2016 by the Ministry of Housing and Urban Affairs, ranks cities on cleanliness through document checks, field inspections and citizen feedback. It has turned sanitation into a matter of civic pride and healthy competition between cities. Indore topped the survey for eight consecutive years, and in the 2024–25 edition it was placed in a new **Super**

Swachh League along with Surat, Navi Mumbai and Vijayawada, while Ahmedabad, Bhopal and Lucknow led the remaining cities with populations above 10 lakh. Critics note that rankings reward visible cleanliness more than reduction of waste, and that some cities clean up only for the survey period.

Case Study: The Indore model

Until 2015 Indore was an ordinary, littered city with overflowing community bins and a large dump at Devguradia. Within about two years it transformed its system. The Indore Municipal Corporation removed street bins and introduced daily door-to-door collection by vehicles with separate compartments for wet, dry, sanitary and hazardous waste, later extended to six streams including plastics and e-waste. Non-governmental organisations went from house to house to explain segregation, songs played from collection vehicles, and spot fines were imposed on people who littered or mixed waste. Collected waste goes through compactor transfer stations to processing: wet waste to composting units and a 550-tonnes-per-day bio-CNG plant, and dry waste to a mechanised material recovery facility that employs former waste pickers. The old dump was biomined and the land reclaimed. Indore also earns revenue by selling compost, recyclables and bio-CNG. The lessons are that leadership, sustained citizen engagement, a closed chain from segregation to processing and steady funding matter more than any single technology.

4.15 Industrial Waste: Types and Control Measures

Industries generate far larger quantities of solid waste than cities, but much of it comes in a few large streams from a few sectors: thermal power, steel, aluminium, cement, fertilisers, chemicals and mining. Some industrial wastes are bulky but relatively harmless; others are hazardous and are discussed in the next section. The aim is to turn bulk wastes into raw materials for other industries.

4.15.1 Fly ash

Fly ash is the fine, powdery residue left after pulverised coal is burned in thermal power plants. It is captured from flue gases by electrostatic precipitators; the coarser ash that falls to the bottom of the furnace is **bottom ash**. Indian coal has a high ash content (often 30–45%), so Indian power stations produce enormous quantities of ash, of the order of 300 million tonnes a year. Ash stored as slurry in **ash ponds** occupies vast areas, blows about as dust in dry weather and can leach metals such as arsenic, lead and mercury into water. Ash-pond walls have breached, as at the Sasan power plant in Singrauli in April 2020, when a slurry flood killed six people and buried farmland.

Fly ash is also a useful material. It reacts with lime in the presence of water to form cement-like compounds (it is **pozzolanic**), so it is used:

- in **Portland pozzolana cement** and as a partial replacement for cement in concrete;
- to make **fly ash bricks and blocks**, saving fertile topsoil from brick kilns;
- in road and highway embankments and in filling of mine voids and low-lying areas;
- in agriculture as a soil amendment, in limited doses.

MoEFCC's fly ash notifications since 1999, most recently that of 31 December 2021, require coal-based plants to achieve 100% utilisation of current ash and to use up legacy ash over a set period, on

pain of environmental compensation; construction agencies within 300 km of a plant must use ash-based products. Utilisation has risen sharply, from under 10% in the mid-1990s to about 96% in 2024–25 according to the Central Electricity Authority (CEA), led by road construction, cement and mine filling.

4.15.2 Red mud, slag and other bulk wastes

Red mud (bauxite residue) is the alkaline sludge left when alumina is extracted from bauxite by the Bayer process; roughly one to one-and-a-half tonnes is produced for every tonne of alumina. It is strongly alkaline (pH 10–13) and contains iron oxides (which give its red colour), aluminium, titanium and traces of heavy metals. It is stored in large ponds near refineries in Odisha, Jharkhand, Chhattisgarh and Uttar Pradesh, and pond failures can devastate surrounding land, as at Ajka in Hungary in 2010, when a burst reservoir flooded villages with caustic sludge and killed ten people. Research and pilot plants are developing ways to use red mud in cement, bricks and road construction and to recover iron and rare elements from it.

Slag is the glassy by-product of iron and steel making. **Blast-furnace slag**, when quenched with water, forms granulated slag, a valued raw material for slag cement. **Steel slag** is harder to use because it can swell, but processed steel slag is now used as road aggregate; a road built with processed steel slag at Hazira in Surat with the help of the CSIR–Central Road Research Institute is a recent example.

Other bulk industrial wastes include **phosphogypsum** from phosphate fertiliser plants (used in cement and as a soil conditioner), **lime sludge** from paper mills, **mine overburden and tailings**, and **press mud** from sugar mills (used for compost and biogas).

4.15.3 Cleaner production and industrial symbiosis

Cleaner production is the continuous application of a preventive strategy to processes and products so that less raw material, water and energy are used and less waste is generated. It replaces costly **end-of-pipe** treatment (cleaning pollutants after they are produced) with prevention at source. Typical measures are:

- **Good housekeeping**: fixing leaks, preventing spills, proper storage of materials.
- **Input substitution**: water-based paints instead of solvent-based ones; less toxic dyes.
- **Process change**: dry instead of wet processes, counter-current rinsing in electroplating, membrane cell technology in chlor-alkali plants (which ended the use of mercury cells in Indian caustic soda production).
- **On-site recovery and reuse**: chrome recovery from tannery liquor, chemical recovery boilers in paper mills, condensate and water recycling.

Industrial symbiosis treats one industry's waste as another's raw material. Fly ash going to cement plants, granulated slag to slag cement and RDF to cement kilns are Indian examples; the Kalundborg eco-industrial park in Denmark, where a power station, refinery, pharmaceutical plant and others exchange steam, water and by-products, is the classic global case.

4.15.4 Common effluent treatment plants and zero liquid discharge

India has many clusters of small and medium industries, such as tanneries in Kanpur and Vellore, textile dyeing units in Tiruppur and Pali, and chemical units in Vapi and Ankleshwar, that cannot each

afford a proper effluent treatment plant. A **common effluent treatment plant (CETP)** treats the combined wastewater of such a cluster, with costs shared among member units. CETPs have been supported by central schemes since the early 1990s. Their performance has often been poor: member units send more effluent than the plant was designed for, or toxic effluents it cannot treat, and operators lack funds and skilled staff. CPCB inspections have repeatedly found many CETPs not meeting discharge standards.

Zero liquid discharge (ZLD) is a treatment approach in which all wastewater is treated and recycled within the plant, and the dissolved salts are recovered as solids through evaporation and crystallisation, so that no liquid effluent leaves the site. After the Madras High Court in 2011 ordered the closure of polluting dyeing and bleaching units that were poisoning the Noyyal river, Tiruppur's textile cluster adopted ZLD on a large scale. The CPCB has since directed distilleries and textile, tannery and pulp and paper units in the Ganga basin and elsewhere to move towards ZLD. ZLD protects rivers but is energy-intensive and costly, and the solid salts it produces must themselves be managed safely.

4.16 Hazardous Waste and Biomedical Waste

4.16.1 What makes a waste hazardous?

Hazardous waste, as defined in India's Hazardous and Other Wastes Rules, 2016, is any waste that, because of its physical, chemical, biological, reactive, toxic, flammable, explosive or corrosive characteristics, causes or is likely to cause danger to health or the environment, whether alone or in contact with other wastes or substances. Four main characteristics are used worldwide to identify hazardous wastes:

Table 4.14: Characteristics of hazardous waste

Characteristic	Meaning	Examples
Ignitability	Catches fire easily (low flash point) or burns vigorously	Waste solvents, paint thinners, oily sludge
Corrosivity	Strongly acidic or alkaline; eats into metal or skin	Spent pickling acids, battery acid, caustic wash
Reactivity	Unstable; may explode or release toxic gas on contact with water or air	Cyanide and sulfide wastes, peroxides
Toxicity	Harmful even in small amounts; may leach into water	Heavy-metal sludges (chromium, lead, mercury), pesticide residues

Biological hazards (infectious waste) and environmental persistence are further grounds for concern. India's main generators of hazardous waste are chemical, petrochemical and pharmaceutical industries, pesticide manufacture, metal plating, refineries and tanneries; Gujarat and Maharashtra account for a large share. Households, too, produce small amounts in the form of used batteries, CFLs, paints and expired medicines.

4.16.2 Hazardous and Other Wastes Rules, 2016

The **Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016**, framed under the Environment (Protection) Act, 1986, replaced earlier rules of 1989 and 2008. "Other wastes" covers materials such as used tyres, paper waste, metal scrap and used electronic items that are traded for reuse or recycling. The main provisions are:

- **Hierarchy**: prevention, minimisation, reuse, recycling, recovery (including co-processing in cement kilns) and only then safe disposal.
- **Authorisation**: every occupier who generates, stores, treats, recycles or disposes of hazardous waste must obtain authorisation from the State Pollution Control Board and file annual returns.
- **Storage**: generators may store waste for a limited period (normally up to 90 days) in safe conditions.
- **Packaging, labelling and transport** in prescribed containers, with a **manifest system**, a multi-copy tracking document that follows the waste from generator to transporter to final facility, creating a "cradle-to-grave" record.
- **Import and export** only with permission, and a ban on importing certain wastes for disposal; India follows the prior-informed-consent procedure of the Basel Convention.
- **Liability**: the occupier is liable for damage to the environment or third parties caused by improper handling, and must pay for remediation.

4.16.3 Treatment, storage and disposal facilities

A **treatment, storage and disposal facility (TSDF)** is a common facility where hazardous wastes from many industries are received, analysed, treated and disposed of. A typical TSDF has:

- a laboratory to analyse each consignment;
- **stabilisation and solidification** units, where wastes are mixed with cement, lime or fly ash to lock heavy metals into a solid mass;
- a **secured landfill**, with double composite liners, leak detection, leachate collection and a permanent cap, for waste that cannot be destroyed;
- a high-temperature **incinerator** with gas cleaning for organic wastes that should not be landfilled;
- facilities for recovering solvents, oils and metals.

India has around 50 common TSDFs, spread over about 20 states and union territories, with the largest concentrations in industrial states such as Gujarat, Maharashtra and Telangana. The facility at **Pithampur** near Indore incinerated the long-stored toxic waste from the Bhopal disaster site in 2025 (see the Bhopal case study at the end of this chapter). Many cement plants are also authorised to **co-process** hazardous wastes such as spent solvents and paint sludge as alternative fuel. Gaps remain: some states have no TSDF, and illegal dumping of chemical waste on vacant land and into drains still occurs.

4.16.4 Biomedical waste

Biomedical waste is any waste generated during the diagnosis, treatment or immunisation of human beings or animals, or in related research or production of biologicals. According to the World Health Organization, only about 15% of health-care waste is infectious, toxic or radioactive; the rest is

general waste like paper and food. If the two are mixed, everything must be treated as infectious. Improperly managed biomedical waste spreads infections such as hepatitis B and C and HIV through needle-stick injuries and illegal reuse of syringes, and open burning of plastic medical items releases dioxins.

The **Bio-Medical Waste Management Rules, 2016** (amended in 2018 and 2019) replaced the rules of 1998. They require every health-care facility, from a large hospital to a small clinic, veterinary hospital or blood bank, to segregate waste at the point of generation into colour-coded containers, pre-treat laboratory and highly infectious waste on-site, and hand everything over to a **common bio-medical waste treatment facility (CBWTF)**. Bags and containers carry **bar codes** so that waste can be tracked. Chlorinated plastic bags and gloves were to be phased out, incinerator emission standards were tightened (including limits on dioxins and furans), and deep burial is allowed only in remote or rural areas with approval.

Table 4.15: Colour coding of biomedical waste under the Bio-Medical Waste Management Rules, 2016

Category and container	Types of waste	Treatment and disposal
Yellow (non-chlorinated plastic bags)	Human and animal anatomical waste, soiled dressings and swabs, expired or discarded medicines, chemical waste, microbiology and laboratory waste after pre-treatment	Incineration or plasma pyrolysis; deep burial only where permitted
Red (non-chlorinated plastic bags or containers)	Contaminated recyclable items such as tubing, bottles, IV sets, catheters, urine bags, syringes without needles, gloves	Autoclaving or microwaving, then shredding and recycling
White (translucent, puncture-proof, leak-proof container)	Waste sharps: needles, syringes with fixed needles, scalpels, blades	Autoclaving or dry-heat sterilisation, shredding, then encapsulation or recycling
Blue (cardboard box with blue marking)	Broken or discarded glassware, including vials and ampoules; metallic body implants	Disinfection or autoclaving, then recycling

India has roughly 200 CBWTFs, each expected to serve health-care facilities within a radius of about 75 km. The COVID-19 pandemic suddenly multiplied biomedical waste in the form of masks, gloves and PPE kits from hospitals, quarantine centres and homes. The CPCB issued special guidelines in March 2020 and launched a tracking app, and the episode exposed shortages of treatment capacity in smaller towns.

4.16.5 The Basel Convention

The **Basel Convention** on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (adopted 1989, in force 1992) was a global response to scandals in the 1980s in which ships carried toxic waste from industrialised countries to poorer ones in Africa and elsewhere. It requires **prior informed consent**: a shipment of hazardous waste may move only if the importing country agrees in writing, and parties must minimise waste generation and manage waste as close to its source as possible. India became a party in 1992. Amendments have brought plastic waste (from 2021) and all e-waste, hazardous or not (from January 2025), under its control procedures. The

related Rotterdam and Stockholm Conventions on hazardous chemicals and persistent organic pollutants are discussed with other international agreements in Chapter 5.

4.17 E-Waste: Composition, Hazards and Management

E-waste (electronic waste) is discarded electrical and electronic equipment (EEE), whole or in part, including mobile phones, computers, televisions, refrigerators, washing machines, air conditioners, lighting equipment, toys and medical devices and, increasingly, solar panels and lithium-ion batteries. It is the fastest-growing solid waste stream in the world, driven by falling prices, short product lifetimes and the rapid spread of digital devices.

According to the *Global E-waste Monitor 2024* (UN Institute for Training and Research and the International Telecommunication Union), the world generated a record **62 million tonnes** of e-waste in 2022, about 7.8 kg per person, and only **22.3%** was documented as formally collected and recycled. India is the third-largest generator after China and the United States. The CPCB estimated that India generated about **17.5 lakh tonnes (1.75 million tonnes)** of e-waste in 2023–24, up from about 10 lakh tonnes in 2019–20, of which about 7.6 lakh tonnes, roughly 43%, was processed by registered recyclers.

4.17.1 Composition: an "urban mine"

By weight, typical e-waste consists of about half iron and steel, about one-fifth plastics, more than a tenth non-ferrous metals such as copper and aluminium, and smaller amounts of glass, circuit boards and other materials. Circuit boards contain gold, silver, palladium and copper at concentrations far higher than in natural ores; a tonne of discarded mobile phones contains much more gold than a tonne of typical gold ore. Recovering these metals, called **urban mining**, saves energy and reduces the environmental damage of mining.

Table 4.16: Hazardous substances in e-waste and their effects

Substance	Where found	Health and environmental effects
Lead	Solder, old CRT television and monitor glass, batteries	Damages nervous system and kidneys; lowers children's IQ
Mercury	Switches, old flat-screen backlights, CFLs	Damages brain and nervous system; bioaccumulates in fish
Cadmium	Nickel-cadmium batteries, some semiconductors	Damages kidneys and bones; carcinogen
Hexavalent chromium	Corrosion protection of metal parts	Carcinogen; skin and lung damage
Brominated flame retardants	Plastic casings, circuit boards	Hormone disruption; form dioxins and furans when burned
PVC plastic	Cable insulation	Releases dioxins and hydrochloric acid when burned
Lithium-ion cells	Phones, laptops, power banks, electric vehicles	Fire and explosion risk when crushed or punctured

4.17.2 The informal sector: Seelampur and Moradabad

Most of India's e-waste has historically been handled by the **informal sector**: unregistered collectors, dismantlers and recyclers who buy old equipment from *kabadiwalas* and dealers. **Seelampur** in north-east Delhi is one of Asia's largest e-waste dismantling markets, where thousands of workers, including children, break open devices with hammers and bare hands to sort parts and metals. In **Moradabad** in Uttar Pradesh, workers have extracted copper and traces of gold from circuit boards by dipping them in acid baths and burning them, and the waste "black powder" and spent acid have been dumped along the banks of the Ramganga river. Wires are burned in the open to recover copper.

These crude methods recover only the most valuable metals and lose the rest. Workers and nearby residents inhale fumes containing lead, cadmium and dioxins, suffer skin and eye injuries from acids and show higher rates of respiratory illness; soil and water around such clusters are contaminated with heavy metals. At the same time, the informal sector provides livelihoods to lakhs of poor workers and achieves high collection rates. Policy therefore aims to formalise it, by training workers, linking them to authorised recyclers and helping them register, rather than simply shutting it down.

4.17.3 E-Waste (Management) Rules, 2022 and extended producer responsibility

Extended producer responsibility (EPR) is a policy principle under which producers are made responsible, financially and physically, for the safe collection and recycling of their products after consumers discard them. It shifts the cost of end-of-life management from municipalities and taxpayers to producers and gives them a reason to design products that last longer, contain fewer toxic substances and are easier to recycle.

India's first e-waste rules came into force in 2012, and EPR collection targets were added in the E-Waste (Management) Rules, 2016. The **E-Waste (Management) Rules, 2022**, in force since 1 April 2023, created a new system:

- **Wider coverage**: 106 categories of EEE, including solar photovoltaic panels and cells, are covered.
- **Registration**: manufacturers, producers, refurbishers and recyclers must register on a central online **EPR portal** run by the CPCB; trade with unregistered entities is prohibited.
- **EPR certificates**: registered recyclers generate certificates for the quantity they recycle, and producers buy these certificates to meet annual recycling targets. Targets, calculated on the quantity of products a producer has sold in earlier years, rise from 60% for 2023–24 and 2024–25 to 70% for 2025–26 and 2026–27 and 80% from 2027–28 onwards.
- **Environmental compensation** is imposed on producers who fail to meet targets and on anyone who handles e-waste in violation of the rules.
- **Reduction of hazardous substances**: new EEE must not contain lead, mercury, hexavalent chromium, polybrominated biphenyls or polybrominated diphenyl ethers above 0.1% by weight, or cadmium above 0.01%, except in listed exemptions.
- **Bulk consumers** such as government offices, banks and companies must hand their e-waste only to registered recyclers or refurbishers.

A 2024 amendment set a price band for EPR certificates, between 30% and 100% of the environmental compensation for non-compliance, so that cheap or fake certificates do not undercut genuine recycling; some large manufacturers have challenged this in court. Challenges remain: estimates of e-waste generation are uncertain, some recyclers have been found to issue certificates

for e-waste they never processed, and much e-waste still flows to informal dismantlers because they pay consumers more.

> **Think and Do: Where do old phones go?**
>
> Ask ten people how many unused mobile phones, chargers and cables they have at home and what they intend to do with them. Find the nearest authorised e-waste collection point or producer take-back scheme in your city. Prepare a short poster on safe e-waste disposal for your college.

4.18 Pollution Control Measures: Legal Methods

Pollution control measures fall into three broad groups. **Legal (regulatory) methods** set standards and make polluters responsible for meeting them. **Biological methods** use living organisms to remove or break down pollutants. **Physico-chemical methods** use physical forces and chemical reactions to separate, transform or destroy pollutants. In practice all three work together: a law sets a discharge standard, and a treatment plant combining biological and physico-chemical steps helps a factory meet it.

This section introduces India's regulatory architecture for pollution control. The environmental Acts themselves, their history and amendments are covered fully in Chapter 5.

4.18.1 Constitutional and legal foundation

The Constitution, as amended in 1976, directs the State to "protect and improve the environment" (Article 48A) and makes it a fundamental duty of every citizen to protect the natural environment (Article 51A(g)). The Supreme Court has read the right to life under **Article 21** to include the right to a pollution-free environment, which allows citizens to approach the courts through **public interest litigation (PIL)**. On this base Parliament enacted the core pollution laws.

Table 4.17: Main Indian laws for pollution control (details in Chapter 5)

Law	Year	Main purpose for pollution control
Water (Prevention and Control of Pollution) Act	1974	Created the CPCB and State Boards; consent system for discharges into water
Air (Prevention and Control of Pollution) Act	1981	Air pollution control areas; consent and emission standards for industries
Environment (Protection) Act	1986	Umbrella law after Bhopal; Centre can set standards, issue directions and frame rules
Public Liability Insurance Act	1991	Compulsory insurance for hazardous industries to give immediate relief to accident victims
National Green Tribunal Act	2010	Specialised court for environmental disputes
Jan Vishwas (Amendment of Provisions) Act and Water (Amendment) Act	2023, 2024	Replaced imprisonment for many minor violations with monetary penalties imposed by adjudicating officers

The **National Green Tribunal (NGT)**, set up in 2010, hears civil cases involving substantial questions of environmental law and can order compensation and restoration; it has issued many orders on waste dumps, river pollution and air quality, and is discussed in Chapter 7.

4.18.2 Regulatory institutions

- The **Ministry of Environment, Forest and Climate Change (MoEFCC)** frames policy, notifies standards and rules under the Environment (Protection) Act and grants environmental clearance to large projects.
- The **Central Pollution Control Board (CPCB)**, constituted in 1974, advises the Centre, sets and recommends standards, coordinates State Boards, runs national air and water quality monitoring networks, publishes data such as the National Air Quality Index and issues directions under the Environment (Protection) Act.
- **State Pollution Control Boards (SPCBs)**, and **Pollution Control Committees (PCCs)** in union territories, grant consents, inspect industries, collect samples, monitor compliance and take action against violators. The Delhi Pollution Control Committee (DPCC) is an example.
- **Urban local bodies** implement waste rules, and special bodies such as the **Commission for Air Quality Management (CAQM)** in Delhi–NCR coordinate action across states.

4.18.3 The consent system

The heart of everyday pollution control in India is the **consent** mechanism under the Water and Air Acts. An industry must obtain:

1. **Consent to Establish (CTE)** from the SPCB before setting up or expanding a unit; the Board checks the location, process, expected pollution load and proposed control equipment.
2. **Consent to Operate (CTO)** before starting production, after the pollution control equipment has been installed; the CTO carries conditions such as discharge limits, monitoring and reporting, and must be renewed periodically.

Since 2016 industries have been grouped by their pollution potential into **Red** (highly polluting, such as thermal power, cement, distilleries, tanneries and pesticides), **Orange**, **Green** and **White** (practically non-polluting, exempt from consent) categories. Red-category units face the strictest siting rules, the shortest consent validity and the most frequent inspection. Large polluting industries must install **online continuous emission and effluent monitoring systems** that send data directly to the CPCB and SPCBs. If an industry violates standards, the Board can issue directions, including closure or disconnection of power and water, and can levy **environmental compensation**.

4.18.4 Environmental Impact Assessment

Environmental Impact Assessment (EIA) is a process for predicting the environmental and social effects of a proposed project before it is approved, so that harmful projects can be redesigned, relocated or rejected. Under the **EIA Notification, 2006**, listed projects such as mines, thermal power plants, dams, ports, highways, large construction projects and many industries need **prior environmental clearance**. Category A projects are appraised by expert committees of the MoEFCC; Category B projects by State-level authorities (SEIAAs). The process has four stages:

1. **Screening**: deciding whether a project needs a full EIA.
2. **Scoping**: fixing the terms of reference for the study.

3. **Public consultation**: a public hearing near the site and written comments from affected people.
4. **Appraisal**: examination by an expert committee, which recommends clearance with conditions or rejection.

Clearance conditions are then monitored. EIA is a preventive legal tool, but it has been criticised for consultant-driven reports of poor quality, weak public hearings and the practice of granting clearance after a project has already started, which courts have repeatedly criticised.

4.18.5 Guiding legal principles

Indian courts have built pollution law on several principles, most of them drawn from international environmental law:

- **Polluter-pays principle**: the polluter must bear the cost of preventing pollution and of repairing any damage it causes. In the Bichhri case (Indian Council for Enviro-Legal Action v. Union of India, 1996), chemical units that had poisoned groundwater in Rajasthan were ordered to pay for its remediation.
- **Precautionary principle**: where there is a threat of serious or irreversible damage, lack of full scientific certainty is not a reason to postpone measures to prevent it; the burden of proof lies on the developer or industry. It was adopted in the Vellore Citizens' Welfare Forum case (1996) on tannery pollution in Tamil Nadu.
- **Absolute liability**: an enterprise engaged in a hazardous activity is absolutely liable for any harm caused by an accident, with no exceptions, as held in the oleum gas leak case (M.C. Mehta v. Union of India, 1986–87).
- **Public trust doctrine**: the State holds natural resources such as rivers, forests and the sea-shore in trust for the public and cannot hand them over for private use against the public interest.
- **Sustainable development**: development must meet present needs without compromising the ability of future generations to meet theirs.

4.18.6 Comprehensive Environmental Pollution Index

The **Comprehensive Environmental Pollution Index (CEPI)**, developed by the CPCB with IIT Delhi in 2009, scores industrial clusters on a scale of 0–100 on the basis of the pollutants present, the pathways by which people are exposed, and the evidence of damage to health and ecosystems. Clusters scoring 70 or more are labelled **critically polluted**. The first assessment in 2009 covered 88 clusters and identified 43, including Vapi, Ankleshwar, Singrauli, Korba, Ghaziabad and Ludhiana, as critically polluted. The MoEFCC then imposed a temporary moratorium on new projects in these areas until action plans were prepared. Later assessments have continued to use CEPI to target enforcement.

4.18.7 Waste and chemical safety rules

Most detailed obligations are contained in rules framed under the Environment (Protection) Act. Those discussed in this chapter include the Solid Waste Management Rules (2016, replaced in 2026), Plastic Waste Management Rules (2016, with later amendments), Bio-Medical Waste Management Rules (2016), Hazardous and Other Wastes Rules (2016), E-Waste (Management) Rules (2022) and

Construction and Demolition Waste Rules (2016). Others include the **Battery Waste Management Rules, 2022**, which apply EPR to all kinds of batteries; the **Manufacture, Storage and Import of Hazardous Chemical Rules, 1989** and **Chemical Accidents (Emergency Planning, Preparedness and Response) Rules, 1996**, written after Bhopal; the **Noise Pollution (Regulation and Control) Rules, 2000**; and rules notified in 2025 for identifying and remediating **contaminated sites**.

4.18.8 Strengths and weaknesses of the legal approach

India's legal framework is extensive and in places innovative, but implementation is weak. Many SPCBs have large numbers of vacant technical posts and limited laboratories, so inspections are infrequent. Penalties have often been too small to deter large companies, and court cases take years. Boards depend on state governments and can be reluctant to act against influential industries. Data from self-monitoring can be manipulated. Courts and the NGT have filled some gaps, but judicial orders are no substitute for strong, independent and well-funded regulators. Economic instruments, such as environmental compensation, tradable permits (India's first **emissions trading scheme** for particulate matter was started in Surat in 2019) and EPR, are increasingly used to complement command-and-control rules.

4.19 Pollution Control Measures: Biological Methods

Biological methods of pollution control use living organisms, mainly bacteria and fungi but also algae, plants and earthworms, to absorb, break down or immobilise pollutants. Microorganisms have evolved to digest almost every natural organic compound, and many can be adapted to man-made ones. Biological methods are usually cheaper and use less energy and chemicals than physico-chemical ones, and they often convert pollutants completely into harmless products such as CO_2, water and biomass. Their limitations are that they work slowly, need suitable temperature, pH, oxygen and nutrients, and can be poisoned by toxic shocks.

4.19.1 Bioremediation

Bioremediation is the use of microorganisms to remove or detoxify pollutants in soil, water or sediments. It may be done **in situ** (at the contaminated site, for example by adding nutrients to polluted groundwater) or **ex situ** (after excavating soil or pumping water to a treatment area). Two strategies are common:

- **Biostimulation**: adding nutrients (nitrogen, phosphorus), oxygen or moisture to encourage the native microbes that already degrade the pollutant.
- **Bioaugmentation**: adding selected or specially cultured microbes that can degrade a particular pollutant.

A well-known Indian example is **Oilzapper**, a consortium of oil-degrading bacteria developed by The Energy and Resources Institute (TERI) with Indian Oil Corporation. It has been used to clean oily sludge at refineries and soil contaminated by oil spills, converting hydrocarbons into CO_2 and water. Under Namami Gange and city programmes, microbial cultures and enzyme preparations have been applied to open drains to reduce their organic load and odour before they meet rivers, although such in-drain treatment is a stopgap rather than a substitute for sewage treatment. Dumpsite biomining, described earlier, also relies on microbes to stabilise old waste.

4.19.2 Phytoremediation

Phytoremediation is the use of green plants and their associated soil microbes to remove, degrade or contain pollutants. It is slow and limited to the depth of roots, but it is cheap, solar-powered and improves the appearance of degraded land. Its main forms are:

- **Phytoextraction**: plants absorb metals from soil and store them in their shoots, which are then harvested. Indian mustard (*Brassica juncea*) accumulates lead and other metals, and the Chinese brake fern (*Pteris vittata*) is a **hyperaccumulator** of arsenic.
- **Phytostabilisation**: deep-rooted plants hold contaminated soil in place and reduce the spread of pollutants by wind and water. Vetiver grass (*Chrysopogon zizanioides*) is widely used on fly ash dumps, mine spoil and riverbanks.
- **Rhizofiltration**: plant roots in water absorb or precipitate metals and nutrients; water hyacinth (*Eichhornia crassipes*) and duckweed can strip nutrients and some metals from wastewater ponds, though they must be harvested before they decay.
- **Phytodegradation and rhizodegradation**: plants and root-zone microbes break down organic pollutants such as pesticides and petroleum compounds.
- **Phytovolatilisation**: plants take up certain pollutants and release them into the air in a less toxic form; this simply moves the pollutant and is used cautiously.

4.19.3 Constructed wetlands and natural treatment systems

A **constructed wetland** is an engineered shallow basin planted with wetland plants such as reeds (*Phragmites*), cattails (*Typha*) and cannas, through which wastewater flows slowly. Suspended solids settle, plants take up nutrients, and microbes living on roots and gravel break down organic matter and pathogens. Wetlands are well suited to small towns, villages, campuses and as a final "polishing" step after a sewage treatment plant. India's National Environmental Engineering Research Institute (NEERI) has developed a constructed-wetland technology called **Phytorid**, used for treating drains and small sewage flows in several cities. Bengaluru's Jakkur lake receives treated sewage that passes through a constructed wetland before entering the lake.

The **East Kolkata Wetlands**, a Ramsar site of about 12,500 hectares, are the world's best-known natural example. For more than a century, sewage from Kolkata has flowed into shallow fish ponds (*bheris*), where sunlight, algae and bacteria purify it while producing large quantities of fish and vegetables. The system treats a large part of the city's sewage at almost no cost and supports thousands of livelihoods, but it is threatened by urban expansion.

Waste stabilisation ponds (oxidation ponds) are a related low-cost system: sewage is held for several days in a series of shallow ponds where algae produce oxygen and bacteria break down organic matter. They need a lot of land but very little energy and are used in many small Indian towns.

4.19.4 Activated sludge and other engineered biological processes

The **activated sludge process** is the most widely used secondary treatment method for sewage. Wastewater is mixed in an **aeration tank** with a dense population of microbes (the "activated sludge") while air is blown in. The microbes consume dissolved organic matter, reducing the **biochemical oxygen demand (BOD)**, the amount of oxygen microbes need to decompose the organic matter in water, and they clump together into **flocs**. In the next tank, the **secondary clarifier**, the flocs settle; part of the settled sludge is returned to the aeration tank to keep the microbial population

high, and the excess is removed for treatment. Modern variants include the **sequencing batch reactor (SBR)**, **moving bed biofilm reactor (MBBR)** and **membrane bioreactor (MBR)**, all used in new Indian STPs.

Other biological processes include:

- **Trickling filters**: wastewater trickles over a bed of stones or plastic media coated with a microbial film (biofilm) that consumes organic matter.
- **Upflow anaerobic sludge blanket (UASB) reactors**: wastewater flows upward through a blanket of granular anaerobic sludge that converts organic matter into biogas. India's first large UASB plant for sewage was built at Kanpur under the Ganga Action Plan; UASB is energy-saving but needs a further aerobic step to meet discharge standards.
- **Anaerobic digesters**: used to stabilise sewage sludge and high-strength industrial effluents such as distillery spent wash, producing biogas.

4.19.5 Biofilters for air and odour

A **biofilter** passes polluted air through a moist bed of compost, wood chips, peat or soil that supports microbes. The microbes oxidise odorous and volatile compounds such as hydrogen sulfide, ammonia and solvent vapours to harmless products. Biofilters are used at compost plants, sewage pumping stations, food-processing units and rendering plants. A **bioscrubber** works similarly but circulates water containing microbes over the gas stream.

4.20 Pollution Control Measures: Physico-Chemical Methods

Physico-chemical methods remove or transform pollutants through physical processes such as settling, filtration and adsorption, and chemical processes such as coagulation, precipitation, oxidation and disinfection. They act quickly, work for pollutants that microbes cannot digest (metals, salts, many synthetic chemicals) and are less sensitive to toxic shocks, but they often need chemicals and energy and may generate sludge that must itself be managed. Control devices for air pollution, such as cyclones, bag filters and electrostatic precipitators, are discussed with air pollution earlier in this chapter; this section concentrates on water and wastewater, with a brief note on gaseous pollutants.

4.20.1 Sedimentation

Sedimentation is the settling of suspended particles under gravity in a quiet tank. **Grit chambers** remove sand and gravel; **primary clarifiers** in sewage plants remove about half the suspended solids and a third of the BOD; and settling tanks after coagulation remove the flocs formed by chemicals. Sedimentation is the oldest and cheapest treatment step; people have long stored river water in pots to let the silt settle.

4.20.2 Coagulation and flocculation

Very fine particles of clay, colour and microbes carry similar electric charges, repel each other and do not settle. **Coagulation** neutralises these charges by adding a **coagulant**, usually alum (aluminium sulfate), ferric chloride or poly-aluminium chloride, with rapid mixing. **Flocculation** then follows with slow, gentle stirring that brings the destabilised particles together into larger, heavier **flocs** that

settle easily. With alum, the reaction with the natural alkalinity of water forms a sticky precipitate of aluminium hydroxide:

$$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \rightarrow 2Al(OH)_3\downarrow + 3CaSO_4 + 6CO_2$$

The **Nalgonda technique**, developed by NEERI for fluoride-affected villages, uses alum and lime in a similar way to remove excess fluoride from drinking water.

4.20.3 Filtration

Filtration passes water through a porous medium that traps particles. **Rapid sand filters**, used in city water works, remove flocs left after settling and are cleaned by backwashing. **Slow sand filters** work at much lower rates; a biologically active layer called the *schmutzdecke* forms on the sand surface and removes most bacteria as well. In wastewater plants, filters are used as a tertiary step before reuse.

4.20.4 Adsorption

Adsorption is the attachment of dissolved molecules to the surface of a solid. **Activated carbon**, made by heating coconut shells, wood or coal to create an enormous internal surface area (several hundred square metres per gram), adsorbs organic chemicals, pesticides, colour, taste and odour compounds. It is used in household water purifiers, in tertiary treatment and for cleaning industrial gases. **Activated alumina** adsorbs fluoride and arsenic, and low-cost adsorbents such as fly ash, rice husk ash and biochar are being studied for removing dyes and metals.

4.20.5 Ion exchange

Ion exchange swaps unwanted ions in water for harmless ones held on a resin. In water softening, a resin loaded with sodium ions removes calcium and magnesium (which cause hardness):

$$2R\text{–}Na + Ca^{2+} \rightarrow R_2\text{–}Ca + 2Na^{+}$$

When the resin is exhausted it is regenerated with strong brine. Ion exchange is also used to demineralise water for boilers and to recover metals such as chromium, nickel and copper from plating rinse waters.

4.20.6 Membrane processes

Membrane processes push water through thin synthetic membranes that block particles or dissolved substances according to their size. In order of decreasing pore size, they are **microfiltration** (removes suspended solids and bacteria), **ultrafiltration** (also removes viruses and large molecules), **nanofiltration** (removes hardness and some salts) and **reverse osmosis (RO)**, in which high pressure forces water through a dense membrane against its natural osmotic flow, leaving almost all dissolved salts behind. RO is used in household purifiers, in desalination plants such as those near Chennai at Minjur and Nemmeli, and in ZLD systems. Its drawbacks are high energy use, membrane fouling and a concentrated **reject** stream (often 50–80% of the input water in small domestic units) that must be disposed of carefully.

4.20.7 Chemical precipitation, neutralisation and oxidation

- **Neutralisation** adjusts the pH of acidic or alkaline effluents with lime, caustic soda or acid before further treatment.

- **Chemical precipitation** converts dissolved metals into insoluble compounds that settle. In tannery and electroplating effluents, toxic hexavalent chromium is first reduced to trivalent chromium and then precipitated as chromium hydroxide by adding lime.
- **Oxidation** destroys cyanide, phenols and colour using chlorine, hydrogen peroxide or ozone. **Advanced oxidation processes** combine ozone, hydrogen peroxide or ultraviolet light to generate highly reactive hydroxyl radicals that break down stubborn chemicals such as pharmaceuticals and dyes.

4.20.8 Disinfection

Disinfection kills or inactivates disease-causing microorganisms, the last step before water is supplied for drinking or treated wastewater is reused. **Chlorination** is cheapest and leaves a protective residual in pipes; chlorine reacts with water to form hypochlorous acid, the active disinfectant:

$$Cl_2 + H_2O \rightarrow HOCl + HCl$$

Chlorine can, however, react with organic matter to form by-products such as trihalomethanes. **Ultraviolet (UV) radiation** damages the DNA of microbes without adding chemicals, and **ozone** is a powerful disinfectant that also removes colour and odour but leaves no residual.

4.20.9 Scrubbing of gaseous pollutants

A **scrubber** removes gases or particles from an exhaust stream by bringing it into contact with a liquid. In **wet scrubbers** and packed towers, the choice of liquid matches the pollutant, which is a physico-chemical principle: water absorbs soluble gases, alkaline solutions neutralise acid gases such as SO_2 and HCl, and acid solutions capture ammonia. The best-known large-scale example, **flue gas desulfurisation (FGD)** in coal power plants, captures SO_2 with limestone slurry as gypsum, as described with air pollution control earlier in this chapter:

$$CaCO_3 + SO_2 + ½O_2 \rightarrow CaSO_4 + CO_2$$

Scrubbing transfers the pollutant from air to a liquid or sludge, so the scrubber effluent must itself be treated, an example of why prevention at source is preferred.

4.20.10 Sewage and effluent treatment plants

A **sewage treatment plant (STP)** treats domestic wastewater; an **effluent treatment plant (ETP)** treats industrial wastewater. Both combine physical, biological and chemical steps in stages, as shown below.

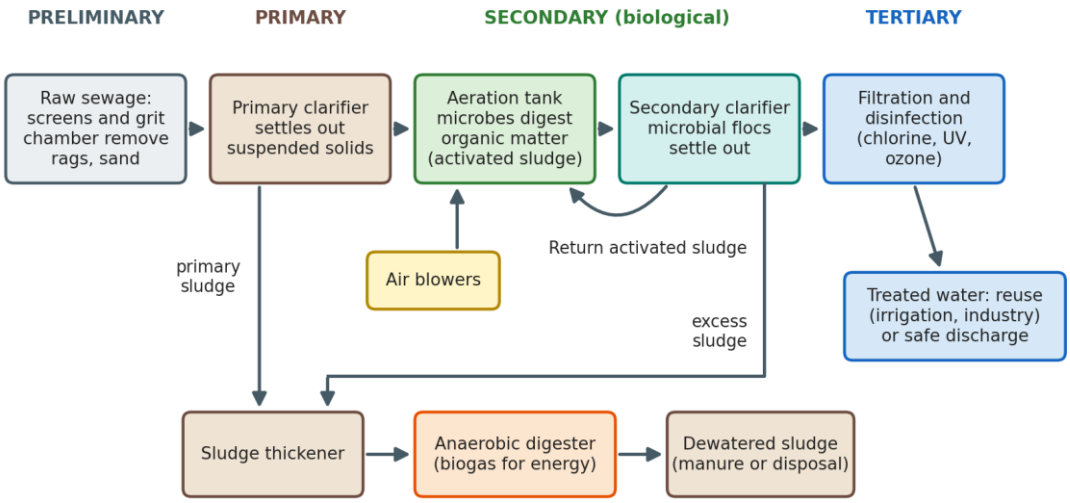


Figure 4.12: *Stages of a conventional sewage treatment plant using the activated sludge process. Preliminary and primary treatment are physical; secondary treatment is biological; tertiary treatment uses filtration and disinfection. The sludge line produces biogas and manure. Source: Original diagram.*

Table 4.18: Stages of wastewater treatment

Stage	Main processes	What is removed
Preliminary	Screens, grit chambers, oil and grease traps	Rags, plastics, sand, floating oil
Primary	Sedimentation in primary clarifiers	Settleable solids, part of BOD
Secondary	Activated sludge, trickling filters, UASB, ponds	Dissolved and colloidal organic matter (most of BOD)
Tertiary (advanced)	Filtration, nutrient removal, adsorption, membranes, disinfection	Remaining solids, nitrogen and phosphorus, pathogens, trace chemicals
Sludge treatment	Thickening, anaerobic digestion, dewatering, drying	Stabilised sludge for manure; biogas

India's sewage treatment capacity lags far behind its needs. The CPCB's national inventory of STPs (2021) estimated that urban India generated about 72,000 million litres per day (MLD) of sewage, while installed treatment capacity was about 32,000 MLD, and only about 20,000 MLD was actually being treated. The gap, together with poorly maintained plants, is the main reason why most urban rivers in India are polluted.

Visit an STP

The syllabus recommends field visits to sewage treatment plants. Delhi Jal Board operates many STPs, including a very large plant at Okhla commissioned in recent years. On a visit, notice the smell change between inlet and outlet, ask the operator the inlet and outlet BOD values, and find out where the treated water and sludge go.

4.21 Role of Pollution Control in Sustainability

Sustainability means meeting human needs today without degrading the natural systems on which future generations depend (Chapter 1). Pollution control is central to it: polluted air, water and soil damage health, reduce farm and fish production, destroy ecosystem services and impose costs on the economy. The *Lancet* Commission on Pollution and Health has estimated that pollution of all kinds causes about nine million premature deaths worldwide each year, and India bears a large share of this burden. Pollution control is therefore not a luxury to be postponed until a country is rich; it is a condition for durable development.

4.21.1 From end-of-pipe to prevention

Pollution control has moved through stages. Early efforts relied on **dilution** (taller chimneys, discharge into large rivers). Then came **end-of-pipe treatment**: scrubbers, filters and treatment plants that capture pollutants after they are formed. The modern approach emphasises **prevention**: cleaner production, cleaner fuels, efficient processes and products designed to be reused and recycled. Prevention is usually cheaper over time, saves raw materials and energy, and does not merely shift pollution from one medium to another, as when a scrubber turns an air problem into a sludge problem.

4.21.2 Circular economy

A **circular economy** is an economic system that keeps products and materials in use for as long as possible through durable design, maintenance, reuse, remanufacturing and recycling, and regenerates natural systems, in contrast to the **linear economy** of "take, make, dispose". Waste is treated as a design flaw and as a resource. India's policies increasingly reflect this idea: NITI Aayog and line ministries have prepared circular economy action plans for priority streams such as municipal waste, plastics, e-waste, lithium-ion batteries, tyres, end-of-life vehicles (under the 2021 vehicle scrappage policy), metal scrap, used oil and gypsum. The government's **Mission LiFE** (Lifestyle for Environment, launched in 2021–22) promotes mindful consumption and reuse at the level of individuals and communities.

4.21.3 Extended producer responsibility

EPR, applied in India to plastic packaging, e-waste, batteries, waste tyres and used oil, puts the circular economy into law. By making producers pay for collection and recycling, EPR creates markets for recyclers, encourages lighter and recyclable designs and moves the cost of waste from taxpayers to those who profit from the products. Its success depends on credible tracking, so that recycling certificates represent real material.

4.21.4 Links with the Sustainable Development Goals

Pollution control contributes directly to several targets of the Sustainable Development Goals (introduced in Chapter 1):

- **SDG 3.9**: substantially reduce deaths and illnesses from hazardous chemicals and air, water and soil pollution.
- **SDG 6.3**: improve water quality by reducing pollution and halving the proportion of untreated wastewater.

- **SDG 11.6**: reduce the adverse environmental impact of cities, with special attention to air quality and municipal waste management.
- **SDG 12.4 and 12.5**: achieve sound management of chemicals and wastes and substantially reduce waste generation through prevention, reduction, recycling and reuse.
- **SDG 14.1**: reduce marine pollution, including plastic debris.

There are also strong co-benefits with **SDG 13 (climate action)**: capturing methane from landfills, composting instead of dumping, and replacing coal with clean energy cut both local pollution and greenhouse gases.

4.21.5 Green jobs and social equity

Pollution control and waste management create **green jobs**: in collection and recycling, compost and biogas plants, e-waste recovery, renewable energy, environmental monitoring and consulting. In India, the most numerous "green workers" are the informal waste pickers and recyclers whose labour keeps millions of tonnes of material in circulation. Making their work safer and better paid, through identity cards, cooperatives, protective equipment and inclusion in EPR systems, links environmental goals with social justice. Pollution also falls hardest on the poor, who live near dumps, drains, highways and factories and cannot buy air purifiers or bottled water. Fair pollution control is therefore part of **environmental justice**.

Key Points

- Legal methods set standards and assign responsibility; in India they rest on the Water, Air and Environment (Protection) Acts, implemented through the CPCB, SPCBs, the consent system and EIA.
- Polluter pays, precaution and absolute liability are core legal principles developed by Indian courts.
- Biological methods (bioremediation, phytoremediation, wetlands, activated sludge, biofilters) are low-energy but slower and sensitive to conditions.
- Physico-chemical methods (sedimentation, coagulation, filtration, adsorption, ion exchange, membranes, disinfection, scrubbing) are fast and versatile but use chemicals and energy and produce sludge.
- Prevention, circular economy and EPR link pollution control with sustainability, the SDGs and green jobs.

4.22 Case Study: The Ganga Action Plan and Namami Gange

The Ganga flows about 2,525 km from Gangotri glacier in Uttarakhand to the Bay of Bengal. Its basin covers about a quarter of India's land area, spans 11 states and is home to over 40% of the country's people. The river supplies drinking water, irrigation, fisheries and navigation, supports endangered species such as the Ganges river dolphin (*Platanista gangetica*), India's National Aquatic Animal, and is sacred to hundreds of millions of people. It is also one of the most heavily used and polluted rivers in the world, and India's efforts to clean it are the country's longest-running experiment in pollution control.

4.22.1 Sources of pollution

- **Municipal sewage** is the largest source, contributing roughly three-quarters of the organic pollution load. Most of the 97 towns on the main stem discharged untreated sewage through open drains, such as the Sisamau *nala* in Kanpur and the Assi and Varuna in Varanasi.
- **Industrial effluents** are smaller in volume but often far more toxic. The tannery cluster at Jajmau in Kanpur discharged chromium-laden effluent, and sugar mills, distilleries, paper mills, textile units and chemical plants added organic load and chemicals.
- **Non-point sources**: fertiliser and pesticide runoff from farms, solid waste dumped on banks, open defecation, cattle wallowing.
- **Religious and cultural practices**: mass bathing during festivals, immersion of idols painted with toxic colours, floral offerings, and the disposal of partly burnt bodies and animal carcasses.
- **Reduced flow**: dams and barrages, such as those at Haridwar and Narora, divert much of the dry-season flow into canals, so the river has less water to dilute and assimilate wastes.

4.22.2 Ganga Action Plan, Phase I and Phase II

In 1985 the Government of India constituted the **Central Ganga Authority**, chaired by the Prime Minister, and launched the **Ganga Action Plan (GAP)**, India's first major river cleaning programme; the plan was formally inaugurated by Prime Minister Rajiv Gandhi at Varanasi in 1986. Its main aim was to improve water quality to bathing standard by **intercepting** sewage flowing in drains, **diverting** it away from the river and **treating** it in sewage treatment plants. It also funded electric crematoria, low-cost toilets, riverfront (ghat) improvements and research, including the release of captive-bred turtles to scavenge on carcasses and the building of India's first large UASB sewage plant at Kanpur.

GAP Phase I covered 25 Class I towns (population above one lakh) in Uttar Pradesh, Bihar and West Bengal. Planned for five years, it was declared complete only in March 2000, after spending roughly ₹450 crore and creating about 865 MLD of treatment capacity, close to its target of about 880 MLD. **GAP Phase II**, approved in stages from 1993, extended the approach to tributaries such as the Yamuna, Gomti, Damodar and Mahananda and was merged in 1995–96 into a new **National River Conservation Plan (NRCP)**, which applied the same model to other rivers across India.

4.22.3 Why the Ganga Action Plan fell short

Despite some improvement in dissolved oxygen and BOD at certain stretches, the river remained heavily polluted, and faecal coliform counts at cities such as Kanpur, Allahabad (Prayagraj), Varanasi and Patna stayed far above bathing standards. Evaluations, including reports of the Comptroller and Auditor General, identified several reasons:

- **Under-estimated sewage**: capacity was planned for the sewage of the 1980s, but urban populations and sewage flows grew much faster; a large share of sewage was never intercepted.
- **Plants that did not run**: many STPs and pumping stations stood idle or worked below capacity because of frequent power cuts, poor maintenance and the lack of money and staff in municipal bodies, which had been handed assets they could not afford to operate.
- **Incomplete sewer networks**: plants were built without the house connections and sewers needed to bring sewage to them.

- **Weak control of industry**: tanneries and other industries continued to bypass treatment, and the common effluent plant at Jajmau could not handle their chromium load.
- **Neglect of river flow and the basin**: GAP focused on pollution in the main stem and ignored the need for adequate flow, the role of tributaries and non-point pollution.
- **Top-down, engineering-centred approach**: there was little public participation and little attention to cultural practices or to the people who live along the river.

4.22.4 From river basin authority to Namami Gange

Learning from these failures, the government declared the Ganga a **National River** in 2008 and, in 2009, set up the **National Ganga River Basin Authority (NGRBA)**, chaired by the Prime Minister, to plan for the whole basin rather than individual towns. The **National Mission for Clean Ganga (NMCG)** was registered in 2011 as its implementing agency, with World Bank support.

In 2014 the government announced the **Namami Gange** programme and, in 2015, approved ₹20,000 crore for it; the programme was later extended, with a second phase running to 2026. It is implemented by NMCG under what is now the Ministry of Jal Shakti. Its main components are:

- **Sewerage infrastructure**: new and rehabilitated STPs, sewers and interception of drains in towns along the Ganga and its tributaries, increasingly built under a **hybrid annuity model**, in which a private operator builds the plant and is paid over 15 years only if the plant performs, to solve the maintenance problem that crippled GAP.
- **Industrial pollution control**: inspection of grossly polluting industries, online effluent monitoring, and directions to distilleries, sugar, pulp and paper and textile units to cut water use and move towards zero liquid discharge.
- **Riverfront development and river surface cleaning**: ghats, crematoria and trash skimmers.
- **Ecological flow**: in 2018 the government notified minimum environmental flows that dams and barrages on the upper Ganga must release in each season.
- **Biodiversity conservation and afforestation**: protection of dolphins, turtles and gharials, and planting along the banks.
- **Public participation**: volunteer "Ganga Praharis", awareness campaigns and the "Arth Ganga" concept of linking river health with local livelihoods such as natural farming and eco-tourism.

In 2016 NMCG was given legal powers as an **authority**, allowing it to issue directions to polluters, and the programme now includes a **Ganga Knowledge Centre** and a ranking of cities on river-related work.

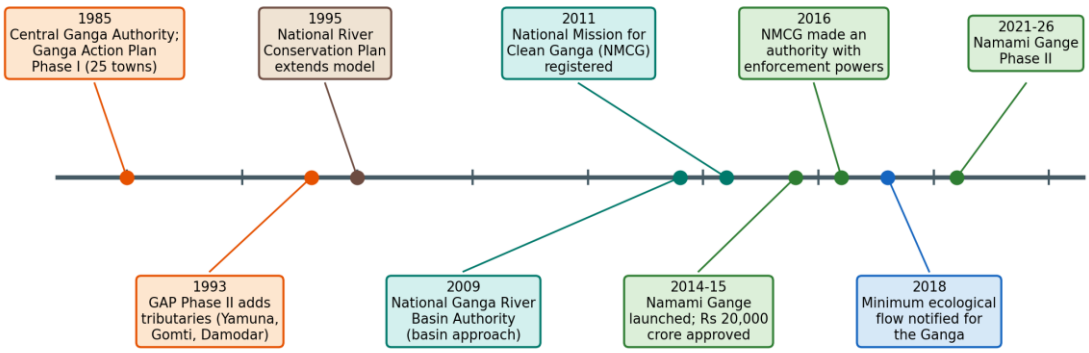


Figure 4.13: *Timeline of India's programmes to clean the Ganga, from the Ganga Action Plan (1985) to Namami Gange Phase II (2021–26). Source: Compiled from NMCG and MoEFCC records.*

4.22.5 Current status

By early 2026, more than 500 projects worth about ₹43,000 crore had been sanctioned under Namami Gange, and about two-thirds of them had been completed; sewerage projects sanctioned for about 6,600 MLD of treatment capacity had created and commissioned roughly 4,000 MLD. Government monitoring reports that dissolved oxygen and BOD now meet bathing criteria along most of the main stem, and surveys of the Ganges river dolphin, including the first national count of river dolphins reported in 2025 (about 6,300 animals, nearly all of them Ganges dolphins), suggest that the river can still support a healthy ecosystem. Yet faecal coliform levels continue to exceed bathing standards at many locations, especially downstream of large cities; during the 2025 Maha Kumbh at Prayagraj, CPCB monitoring reported faecal coliform above permissible limits at several points. Many STPs have been delayed, some new plants do not meet standards, and drains in smaller towns remain untapped.

The Ganga story shows that river cleaning is a long-term task that needs adequate flow, a whole-basin view, plants that are actually operated and maintained, strict control of industry and the involvement of the people who live along the river. It also shows that there is no "one-time" cleaning: every growing city must keep expanding and running its sewage system.

> **Think and Do: River–society–economy nexus**
>
> Using news reports and NMCG or CPCB data, prepare a poster on one stretch of the Ganga or Yamuna near a city you know. Show who depends on the river (farmers, fishers, boatmen, pilgrims, industries), the main sources of pollution, and one practical step that could improve it.

4.23 Case Study: Delhi Air Pollution and Public Health

Delhi, India's capital region with over 3 crore (30 million) people in its urban agglomeration, regularly ranks among the most polluted major cities in the world. Its annual average PM2.5 concentration was about 100 µg/m^3 in 2025 according to the Delhi Pollution Control Committee, the lowest level in a decade apart from the 2020 lockdown year, but still two-and-a-half times India's national standard of 40 µg/m^3 and twenty times the World Health Organization's 2021 guideline of 5 µg/m^3. Each winter,

daily PM2.5 frequently exceeds 300 µg/m^3, the Air Quality Index enters the "severe" category, schools close and flights are delayed. The basics of air pollutants, the AQI and standards are explained earlier in this chapter; this case study examines why Delhi's air is so bad, what it does to health, and how governments have responded.

4.23.1 Sources of pollution

Delhi's air pollution comes from many sources, both inside the city and far beyond its borders. **Source apportionment** studies identify these sources either by building emission inventories or by chemically "fingerprinting" particles collected on filters (receptor modelling). The most widely cited study, by IIT Kanpur for the Delhi government (2016), found that in winter the main contributors to PM2.5 were secondary particles, biomass burning and vehicles.

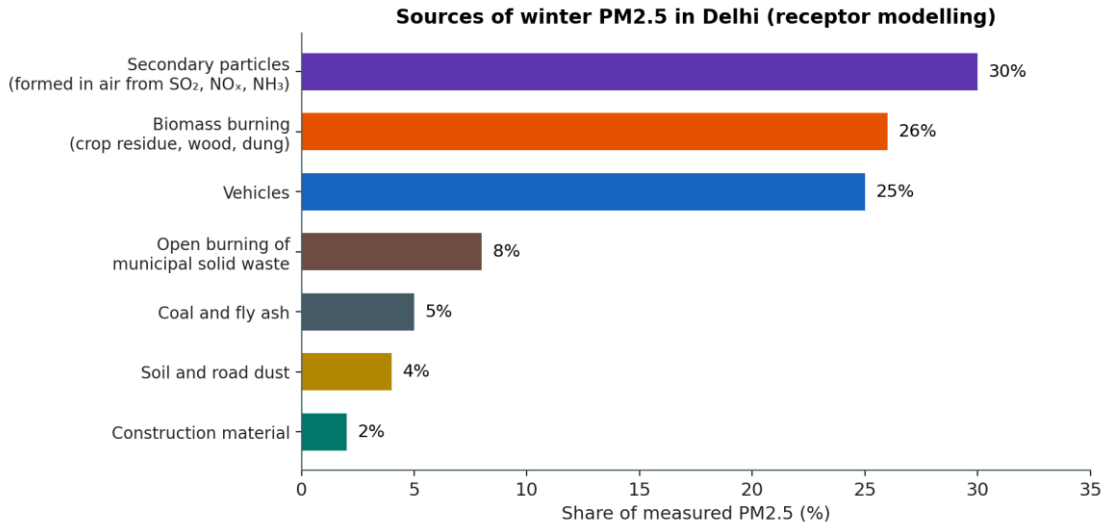


Figure 4.14: *Contribution of different sources to winter PM2.5 in Delhi estimated by receptor modelling. Secondary particles form in the air from gases such as SO_2 and NO_x emitted by vehicles, power plants and industry, and ammonia from agriculture. Source: IIT Kanpur (2016), Comprehensive Study on Air Pollution and Green House Gases in Delhi.*

- **Vehicles**: Delhi has one of India's largest vehicle fleets. Diesel trucks and older vehicles emit fine particles and NO_x; two-wheelers and cars add NO_x, carbon monoxide and hydrocarbons; congestion increases emissions per kilometre. Vehicle exhaust is also a major source of secondary nitrate particles.
- **Dust**: road dust, construction and demolition and dust blown from dry land dominate PM10, especially in summer. Emission inventories have attributed over half of Delhi's PM10 emissions to road dust.
- **Industry and power**: industrial units in and around Delhi, brick kilns and coal-based power plants in the region emit particles and SO_2; the city's last coal power station, at Badarpur, closed in 2018.
- **Biomass and waste burning**: burning of wood, dung and coal for cooking and heating in poorer households and by night guards, and open burning of garbage and dumpsite fires.
- **Crop residue burning**: in October and November, farmers in Punjab and Haryana burn the stubble left after mechanical harvesting of paddy to clear fields quickly for wheat. North-westerly winds carry the smoke to Delhi; on peak days it can contribute a third or more of the city's PM2.5, although its contribution averaged over the whole winter is smaller. Fire counts in

Punjab fell sharply between 2021 and 2024, helped by subsidised machines that manage residue in the field, and by enforcement.

- **Festivals**: firecrackers during Diwali produce sharp, short-lived spikes in PM2.5 and toxic metals, despite court-ordered restrictions on sales and use.

4.23.2 Why winter is worst: meteorology

Emissions in Delhi do not change dramatically between seasons, yet winter air is far worse. The reason is **meteorology**. On cold, clear winter nights the ground cools rapidly, creating a **temperature inversion**, a layer of warm air above cool air near the ground that acts like a lid. The **mixing height**, the depth of air into which pollutants can disperse, falls to a few hundred metres, and winds are light. Delhi also lies in the land-locked Indo-Gangetic Plain, bounded by the Himalaya to the north, where pollution from across the plain accumulates. High humidity and fog favour the formation of secondary particles. In summer, by contrast, strong heating and winds lift the mixing height to several kilometres and dilute pollution, although dust storms raise PM10.

4.23.3 Health evidence

Fine particles (PM2.5) penetrate deep into the lungs, and the smallest enter the bloodstream. They cause inflammation and oxidative stress that damage the lungs, heart, blood vessels and brain. The evidence from India is now substantial:

- The *India State-Level Disease Burden Initiative* (*Lancet Planetary Health*, 2020) estimated that air pollution caused about 1.67 million deaths in India in 2019, nearly 18% of all deaths, from heart disease, stroke, chronic obstructive pulmonary disease (COPD), lung cancer, lower respiratory infections, diabetes and neonatal conditions.
- A 2024 study in *Lancet Planetary Health* covering ten Indian cities estimated that short-term exposure to PM2.5 above the WHO guideline accounted for about 11.5% of daily deaths in Delhi, roughly 12,000 deaths a year, the highest share among the cities studied.
- The **Air Quality Life Index** (University of Chicago, 2025) estimates that if Delhi's PM2.5 met the WHO guideline, an average resident would live about 8 years longer.
- A CPCB-sponsored study of Delhi schoolchildren by the Chittaranjan National Cancer Institute found reduced lung function in a much larger share of children than in cleaner areas. Children breathe more air relative to their body weight, and exposure during growth can permanently reduce lung capacity.
- Air pollution is also linked to low birth weight, premature birth, asthma, cognitive impairment and dementia. The International Agency for Research on Cancer classifies outdoor air pollution and particulate matter as **Group 1 carcinogens**.

The burden is unequal. Street vendors, traffic police, construction workers, rickshaw pullers and people living near highways, industrial areas and dumpsites are most exposed, and poorer households cannot afford air purifiers or medical care.

4.23.4 Responses

Delhi's experience shows a pattern of strong, often court-driven action that is followed by renewed growth in emissions.

- **CNG mandate (1998–2003)**: in the M.C. Mehta case, the Supreme Court ordered in 1998 that Delhi's buses, autorickshaws and taxis switch to **compressed natural gas (CNG)**. Despite resistance, the conversion was largely completed by 2002–03, and Delhi then had one of the world's largest CNG public transport fleets. Air quality improved for a few years before growth in private vehicles and other sources overtook the gains. The CNG transition as a lesson in public communication is discussed in Chapter 7.
- **Cleaner vehicles and fuels**: India leapfrogged from Bharat Stage IV to **BS-VI** emission norms, equivalent to Euro 6, in April 2020 (Delhi received BS-VI fuel in 2018); old diesel vehicles (over 10 years) and petrol vehicles (over 15 years) are barred from Delhi's roads; the Delhi Metro, electric buses and an electric vehicle policy aim to reduce dependence on private vehicles. Peripheral expressways completed in 2018 divert through-trucks around the city.
- **Industry and power**: closure of the Badarpur coal plant, a ban on pet coke and furnace oil in the NCR, a shift of industries to piped natural gas and of brick kilns to cleaner zig-zag technology.
- **Graded Response Action Plan (GRAP)**: first notified in 2017 and revised by CAQM from 2022, GRAP is a set of emergency measures that are triggered automatically as the AQI worsens, in four stages: Stage I "Poor" (AQI 201–300), Stage II "Very Poor" (301–400), Stage III "Severe" (401–450) and Stage IV "Severe+" (above 450). Measures escalate from dust control and stopping open burning to banning diesel generators, halting construction, restricting older vehicles and trucks, moving schools online and allowing work from home.
- **Commission for Air Quality Management (CAQM)**: created by ordinance in 2020 and by the **CAQM Act, 2021**, it is a statutory body for the NCR and adjoining states with powers to coordinate action, issue directions and penalise violators, replacing the court-appointed Environment Pollution (Prevention and Control) Authority (EPCA, 1998). Having a single body for the whole airshed recognises that Delhi cannot clean its air alone.
- **Odd-even scheme**: in January and April 2016 and November 2019, private cars with odd- and even-numbered plates were allowed on alternate days. Studies found small and uncertain effects on PM2.5, partly because of exemptions and because cars are only one source, although the scheme raised public awareness.
- **Smog towers**: two large air-filtering towers were installed in 2021 at Connaught Place and Anand Vihar. Monitoring showed that their effect was limited to a few hundred metres, and experts have argued that the money is better spent on cutting emissions at source.
- **National Clean Air Programme (NCAP)**: launched in January 2019, NCAP covers about 130 cities that do not meet national standards. Its original target was a 20–30% cut in particulate concentrations by 2024 compared with 2017; this was revised to up to 40% by 2025–26, with central funds linked to performance. Independent analyses of official data show Delhi's PM10 only about 15–17% below the 2017–18 baseline in 2024–25 and 2025–26, short of the target. Critics note that NCAP is not legally binding, focuses on PM10 (much of it dust) rather than the more harmful PM2.5, and spends most of its funds on dust control.
- **Stubble management**: subsidised machines (such as the Happy Seeder and super seeder) that sow wheat without burning, a microbial "bio-decomposer" spray developed by the Indian Agricultural Research Institute, and a requirement that coal plants in the region co-fire crop residue pellets.

The Delhi case shows that air pollution is a regional problem requiring action across sectors and states, that emergency measures relieve the worst episodes but do not replace sustained cuts in emissions, and that clear health communication can build public pressure for change.

4.24 Case Study: Plastic Waste Management Rules

Plastics are cheap, light, strong and versatile, which is why their use has grown so rapidly. But most plastics are not biodegradable; they break into smaller fragments called **microplastics** (less than 5 mm), which have been found in rivers, soil, sea salt, fish and human blood. Single-use plastics such as carry bags, sachets, cutlery and wrappers are used for minutes and persist for centuries. In Indian cities, plastic bags choke drains and worsen urban flooding (as in the Mumbai floods of 2005), are eaten by cows on the streets, and are burned in the open, releasing toxic fumes. According to CPCB's annual report, India generated about 4.1 million tonnes of plastic waste in 2020–21, and the true figure is likely higher because many cities do not report. A 2024 study published in *Nature* estimated that India was the world's largest emitter of unmanaged plastic pollution, because of uncollected waste and open burning, although such estimates carry large uncertainties.

India's response has been a series of rules framed under the Environment (Protection) Act, tightening step by step.

4.24.1 Evolution of the rules

Table 4.19: Main milestones in India's plastic waste regulation

Year	Rule or amendment	Key provisions
1999	Recycled Plastics Manufacture and Usage Rules	Carry bags at least 20 microns thick; recycled plastic not to be used for food
2011	Plastic Waste (Management and Handling) Rules	Minimum thickness raised to 40 microns; carry bags not to be given free
2016	Plastic Waste Management Rules	Minimum thickness 50 microns; rules extended to rural areas; EPR introduced for producers, importers and brand owners; phase-out of non-recyclable multilayered plastic
2018	Amendment	Phase-out limited to multilayered plastic that is non-recyclable, non-energy-recoverable or with no alternate use; central registration of large producers and brand owners
2021	Amendment	Carry bags at least 75 microns from 30 September 2021 and 120 microns from 31 December 2022; ban on identified single-use plastic items from 1 July 2022
2022	Amendment with EPR Guidelines for plastic packaging	Packaging categories, collection and recycling targets, mandatory recycled content, EPR certificates and a central online EPR portal
2024	Amendment	Clear definitions and labelling for compostable and biodegradable plastics; reporting duties for their manufacturers; stronger responsibilities of producers

Further amendments in 2025–26 refined the recycled-content obligations, for example allowing food-contact packaging to carry forward shortfalls where food-safety rules restrict the use of recycled plastic.

4.24.2 The single-use plastic ban of July 2022

From **1 July 2022**, India banned the manufacture, import, stocking, distribution, sale and use of identified single-use plastic items that have **low utility and high littering potential**:

- earbuds with plastic sticks, plastic sticks for balloons, plastic flags, candy sticks and ice-cream sticks;
- polystyrene (thermocol) for decoration;
- plates, cups, glasses and cutlery such as forks, spoons, knives, straws, trays and stirrers;
- wrapping or packing films around sweet boxes, invitation cards and cigarette packets;
- plastic or PVC banners less than 100 microns thick.

Together with the 120-micron minimum for carry bags, which makes them thicker, more durable and worth reusing and recycling, this was the most visible step against plastic litter. The CPCB set up a national control room and a mobile app for citizens to report violations, and states formed special task forces. Some states had already gone further: Sikkim and Himachal Pradesh were early pioneers in banning plastic bags, and Maharashtra imposed a wide ban in 2018.

4.24.3 Extended producer responsibility for plastic packaging

Bans cover only a few items; most plastic waste is **packaging**, which cannot simply be banned. The **EPR Guidelines of 2022** made **producers, importers and brand owners (PIBOs)** responsible for collecting and recycling the plastic packaging they put on the market. Packaging is divided into four categories: **Category I** rigid packaging (bottles, containers); **Category II** flexible single-layer or multilayer packaging made of different plastics (carry bags, pouches); **Category III** multilayered packaging with at least one non-plastic layer (chips and biscuit packets, sachets); and **Category IV** compostable plastic packaging. PIBOs register on the CPCB's **centralised EPR portal**, report the quantity of packaging they introduce, and meet annual targets by buying **EPR certificates** from registered plastic waste processors. Rigid packaging also carries targets for reuse.

The guidelines also require brand owners to use **recycled plastic** in their packaging, rising year by year:

Table 4.20: Minimum recycled plastic content required in packaging (EPR Guidelines, 2022)

Packaging category	2025–26	2026–27	2027–28	2028–29 onwards
Category I (rigid)	30%	40%	50%	60%
Category II (flexible)	10%	10%	20%	20%
Category III (multilayered)	5%	5%	10%	10%

Producers that fail to meet their obligations pay **environmental compensation**, and the money is to be used for plastic waste collection and management.

4.24.4 Enforcement gaps

The rules are among the most comprehensive in the developing world, but implementation is uneven.

- **Weak local enforcement**: urban local bodies and SPCBs lack staff for regular inspection; banned items, especially thin carry bags and disposable cutlery, are widely available in markets. A survey by the Delhi-based organisation Toxics Link a year after the ban found banned items on sale in most of the markets it surveyed.
- **Informal manufacturing**: many small units making banned items are unregistered and simply shift location.
- **Lack of cheap alternatives**: street vendors and small shops find paper, cloth or leaf-based alternatives costlier and less convenient.
- **Multilayered packaging**: chips, biscuit, shampoo and gutka sachets are not banned and are hard to recycle; most end up in dumps, burned, or as RDF.
- **EPR integrity**: there have been reports of fake or inflated EPR certificates from processors that did not actually recycle the material, prompting the CPCB to cancel registrations and tighten audits.
- **Role of waste pickers**: informal collectors, who gather most recyclable plastics, are largely left out of formal EPR payments.

International negotiations on a global plastics treaty, begun in 2022 under the UN Environment Assembly, had not concluded by 2025, with countries divided on whether to limit plastic production. For India, the lesson of the Plastic Waste Management Rules is that bans must be backed by enforcement and affordable alternatives, that EPR needs transparent tracking, and that citizens' habits of refusing and reusing remain the first line of defence.

> **Think and Do: Plastic diary**
>
> For three days, list every single-use plastic item you receive or throw away, and classify each as banned, recyclable (rigid), flexible or multilayered. Identify three items you could refuse or replace with reusable alternatives. Share your findings with your class.

4.25 Case Study: The Bhopal Gas Tragedy

On the night of **2–3 December 1984**, a cloud of highly toxic **methyl isocyanate (MIC)** gas escaped from a pesticide factory run by **Union Carbide India Limited (UCIL)**, a subsidiary of the US-based **Union Carbide Corporation (UCC)**, in Bhopal, Madhya Pradesh. The gas drifted over densely populated neighbourhoods, many of them slums built next to the factory, while people slept. It remains the world's worst industrial disaster, and it transformed India's environmental and industrial safety law.

4.25.1 What happened: the chemistry

The Bhopal plant manufactured the insecticide carbaryl (sold as Sevin), using MIC as an intermediate. MIC (CH_3NCO) is a volatile, highly reactive and extremely toxic liquid that boils at about 39 °C. It must be kept cold, dry and in small quantities. Tank **E610** held about 42 tonnes of MIC, more than the recommended limit.

Late on 2 December, a large quantity of water entered tank E610. Whether it came through poorly isolated pipes during washing, as most independent investigators concluded, or through deliberate sabotage, as UCC claimed, is still debated. Water reacts with MIC in an **exothermic** (heat-releasing) reaction:

$$CH_3NCO + H_2O \rightarrow CH_3NH_2 + CO_2 + \text{heat}$$

The heat speeded up further reactions, including polymerisation of MIC, which were also catalysed by iron from corroded steel. Temperature and pressure in the tank rose rapidly until the safety valve burst, and over about two hours roughly 40 tonnes of MIC and its reaction products escaped through the vent stack. Because MIC vapour is heavier than air, the cloud stayed close to the ground and flowed into homes.

4.25.2 Why the safety systems failed

The disaster resulted from a chain of failures, not a single error:

- The **refrigeration system** meant to keep MIC near 0 °C had been switched off months earlier to save money, and its coolant drained.
- The **vent gas scrubber**, which should have neutralised escaping MIC with caustic soda, was not working effectively and was in any case too small for a release of this size.
- The **flare tower**, which could have burned off the gas, was out of service for repairs.
- The **water curtain** meant to knock down escaping gas could not reach the height of the release.
- Staff and training had been cut, instruments were unreliable, maintenance was deferred, and the plant, which was losing money, was being run down for sale.
- Earlier leaks at the plant between 1981 and 1984, including one that killed a worker, had not led to corrective action, and warnings by journalists were ignored.
- The public siren was switched off soon after it sounded, and there was no community emergency plan; doctors did not know what antidote to use, because the company did not share full information on the gas.
- Hazardous industry had been allowed close to dense settlements, a failure of land-use planning.

4.25.3 The human toll

Exposed people suffered burning eyes, choking, vomiting and fluid in the lungs (pulmonary oedema); many died in their sleep or while fleeing. The Madhya Pradesh government initially confirmed about 3,800 deaths. In its 2010 curative petition to the Supreme Court, the Union government placed the number of deaths at **5,295** and injury cases at over **5.27 lakh**. Survivors' organisations and some researchers estimate many more deaths over the following years, of the order of 15,000–25,000. Over half a million people were exposed.

Survivors have suffered chronic lung disease, eye damage, neurological and psychological disorders, weakened immunity, menstrual and reproductive problems and higher rates of some cancers. Studies have reported growth and health problems among children born to exposed parents. The Bhopal Memorial Hospital and Research Centre was set up to treat survivors, but they have complained for decades of inadequate care.

4.25.4 The legal saga

- In 1985 Parliament passed the **Bhopal Gas Leak Disaster (Processing of Claims) Act**, making the Government of India the sole representative of all victims in litigation. India first sued UCC in a US court, which in 1986 sent the case back to India.
- In **February 1989**, during hearings in the Supreme Court, a **settlement** was reached under which UCC paid **US$470 million** (about ₹715 crore at the time) in "full and final" settlement of all claims, and criminal proceedings were quashed. The settlement was based on an assumption of about 3,000 deaths and far fewer injuries than were later recognised, and victims had not been consulted. After petitions by survivors, the Supreme Court in 1991 upheld the settlement but restored the criminal cases. Individual compensation, paid through claims tribunals, was small, typically a few lakh rupees for a death and much less for injuries.
- In **June 2010**, a Bhopal court convicted seven former senior officials of UCIL of causing death by negligence and sentenced them to two years in prison, the maximum for that offence; they were released on bail. UCC's chairman at the time, Warren Anderson, never appeared before an Indian court and died in the United States in 2014. Dow Chemical, which acquired UCC in 2001, maintains that it has no liability for Bhopal.
- In 2010 the Union government filed a **curative petition** seeking an additional ₹7,844 crore from UCC's successors. On **14 March 2023** a five-judge bench of the Supreme Court dismissed it, holding that a settlement could be reopened only on grounds of fraud, which had not been alleged, and that the government itself was responsible for making good any shortfall in compensation.

4.25.5 Legacy contamination and the 2025 waste disposal

Long before the disaster, the factory had dumped chemical wastes on its premises and into **solar evaporation ponds** nearby. After the plant closed, these wastes were never properly cleaned up. Studies by government laboratories and independent groups have found pesticides, chlorinated benzenes, mercury and other toxic chemicals in the soil and in groundwater in surrounding residential colonies; the Supreme Court in 2004 ordered the state to supply safe piped water to these areas.

About **337 tonnes** of packed toxic waste lay in a warehouse at the factory for four decades. After repeated court orders, and a December 2024 deadline set by the Madhya Pradesh High Court, the waste was moved in sealed containers in **January 2025** to the hazardous waste TSDF at **Pithampur** in Dhar district, about 250 km away. Residents of Pithampur protested, fearing pollution of their own air and water. After trial runs monitored by the CPCB, the waste was fully **incinerated by the end of June 2025**. The process produced roughly 900 tonnes of ash and residue, whose safe disposal in a secured landfill was still pending at the end of 2025. Removing the stored waste is only a partial step: the contaminated soil and groundwater around the site have not yet been remediated.

4.25.6 Lessons and legal reforms

Bhopal exposed the absence of a comprehensive environmental law in India and led directly to major reforms:

- The **Environment (Protection) Act, 1986**, an umbrella law giving the Central Government wide powers to set standards, regulate hazardous substances and close polluting industries.

- The **Factories (Amendment) Act, 1987**, which added special provisions for hazardous processes, including site appraisal committees, disclosure of information to workers and the public, and emergency plans.
- The **Manufacture, Storage and Import of Hazardous Chemical Rules, 1989** and the **Chemical Accidents (Emergency Planning, Preparedness and Response) Rules, 1996**, requiring safety reports, on-site and off-site emergency plans, and crisis groups at central, state, district and local levels.
- The **Public Liability Insurance Act, 1991**, which requires every owner of a hazardous installation to take insurance so that victims of an accident receive immediate, no-fault relief.
- The doctrine of **absolute liability**, laid down by the Supreme Court in the oleum gas leak case of 1986 in the shadow of Bhopal, under which a hazardous enterprise is liable for all harm caused by an accident, without any exceptions.

The wider lessons are as relevant today as in 1984: store the minimum of hazardous intermediates (**inherent safety**); keep hazardous plants away from homes and enforce zoning; maintain safety systems and train workers; tell communities what is stored nearby and what to do in an emergency (**right to know**); hold parent companies responsible for their subsidiaries; and ensure that compensation and clean-up are fair and swift. The Visakhapatnam styrene gas leak at LG Polymers in May 2020, which killed at least 12 people, showed that these lessons are not yet fully learned.

Key Points

- The Ganga Action Plan built treatment capacity but fell short because of under-estimated sewage, idle plants, weak industrial control and neglect of river flow; Namami Gange has expanded the effort with a basin approach and performance-linked contracts.
- Delhi's air pollution comes from vehicles, dust, industry, biomass and waste burning and regional crop-residue fires, worsened by winter meteorology; responses range from the CNG mandate and BS-VI norms to GRAP, CAQM and NCAP.
- The Plastic Waste Management Rules combine thickness limits, a ban on identified single-use items and EPR with recycled-content targets, but enforcement remains weak.
- Bhopal (1984) was caused by multiple safety failures; it led to the Environment (Protection) Act, the Public Liability Insurance Act and chemical-accident rules, but justice and clean-up remain incomplete.

Chapter 4 Review

Summary

- Pollution is an undesirable change in air, water or soil that harms living things or property; it occurs when inputs exceed the environment's assimilative capacity.
- Pollutants can be classified as point or non-point, biodegradable or non-biodegradable, natural or anthropogenic, and primary or secondary; persistent pollutants can bioaccumulate and biomagnify.
- Air pollution, led by PM2.5, is India's largest environmental health risk; its main sources are household solid fuels, coal combustion, vehicles, industry, dust and crop-residue burning.
- Household chulhas expose women and children to very high smoke levels; PM Ujjwala Yojana has issued about 10.3 crore LPG connections, but regular use depends on affordability.
- Particles are controlled by cyclones, bag filters, electrostatic precipitators and scrubbers; SO_2 by flue gas desulfurisation; NO_x by selective catalytic reduction; and vehicle exhaust by catalytic converters and BS-VI norms.
- Primary pollutants are emitted directly; secondary pollutants such as ozone, PAN and sulfate and nitrate particles form in the atmosphere, so they are controlled by cutting precursor emissions.
- Photochemical smog forms when sunlight splits NO_2 and VOC-derived radicals allow ozone to accumulate; temperature inversions trap pollutants near the ground.
- Untreated sewage is the main cause of river pollution in India; BOD, COD, DO, pH, TDS, coliforms and nitrate are key water quality parameters.
- Geogenic fluoride and arsenic in groundwater cause fluorosis and arsenicosis in many states; nutrient enrichment causes eutrophication and fish kills.
- Soil is polluted by agrochemicals, industrial waste, heavy metals, e-waste and plastics; prevention is cheaper than remediation by physical, chemical or biological methods.
- Thermal pollution from power-plant cooling water lowers dissolved oxygen and stresses aquatic life; cooling towers and ponds reduce it.
- Noise is measured on the logarithmic decibel scale; the Noise Rules, 2000 set day and night limits for industrial, commercial, residential and silence zones.
- India's NAAQS (2009) cover 12 pollutants; the National AQI reports air quality in six colour-coded categories; India's standards are several times weaker than the WHO 2021 guidelines.
- Surface waters are classified by designated best use (Classes A–E), and drinking water must meet IS 10500:2012.
- Ionising radiation (alpha, beta, gamma) causes deterministic effects above a threshold and stochastic effects such as cancer; Chernobyl, Fukushima, Kyshtym and Mayapuri show the consequences of failures, and AERB regulates radiation safety in India.
- Solid waste includes municipal, industrial, hazardous, biomedical, electronic, construction and agricultural waste; poorly managed waste pollutes air (methane, smoke, dioxins), water (leachate) and soil at the same time.

- India's urban areas generated about 1.7 lakh tonnes of solid waste per day in 2021–22 (CPCB); about 54% was processed, 24% landfilled and 22% remained unaccounted for, mostly in open dumps.
- The waste hierarchy ranks prevention and reduction first, followed by reuse, recycling and composting, energy recovery, and disposal in sanitary landfills as the last resort.
- Segregation at source is the foundation of waste management; the Solid Waste Management Rules, 2026 require four streams (wet, dry, sanitary and special-care waste) and place extra duties on bulk waste generators.
- Wet waste suits composting, vermicomposting and biomethanation; dry waste suits recycling; combustible rejects can become refuse-derived fuel; waste-to-energy incineration struggles with India's wet, low-calorie mixed waste and needs strict emission control.
- Open dumps such as Ghazipur, Bhalswa, Okhla and Deonar catch fire and contaminate groundwater; engineered sanitary landfills and biomining of legacy waste are the remedies.
- Bulk industrial wastes such as fly ash, slag and red mud can become raw materials; cleaner production, industrial symbiosis, CETPs and zero liquid discharge reduce industrial pollution.
- Hazardous waste is ignitable, corrosive, reactive or toxic; it is regulated by the Hazardous and Other Wastes Rules, 2016 and treated at TSDFs, while biomedical waste is segregated into yellow, red, white and blue categories under the 2016 Rules.
- E-waste contains both valuable metals and toxic substances; informal recycling in places like Seelampur and Moradabad harms health, and the E-Waste (Management) Rules, 2022 use EPR certificates and rising recycling targets.
- Legal methods of pollution control in India rest on the Water, Air and Environment (Protection) Acts, implemented by the CPCB and SPCBs through consents, EIA, standards and principles such as polluter pays and precaution.
- Biological methods (bioremediation, phytoremediation, constructed wetlands, activated sludge, biofilters) and physico-chemical methods (sedimentation, coagulation, filtration, adsorption, ion exchange, membranes, disinfection, scrubbing) are combined in treatment plants.
- Pollution control supports sustainability through the circular economy, extended producer responsibility, the SDGs and green jobs.
- The Ganga Action Plan fell short because of under-estimated sewage, idle plants and neglect of river flow; Namami Gange has taken a basin-wide, performance-linked approach, but faecal contamination persists.
- Delhi's air pollution arises from many local and regional sources and is worsened by winter inversions; it causes thousands of premature deaths each year, and responses include CNG, BS-VI, GRAP, CAQM and NCAP.
- The Plastic Waste Management Rules and the 2022 single-use plastic ban, together with EPR and recycled-content targets, mark major progress, while the Bhopal gas tragedy of 1984 remains a lesson in industrial safety, corporate accountability and environmental justice.

Key Terms

Pollutant: a substance or form of energy that causes an undesirable change in the environment.

Assimilative capacity: the ability of an environment to absorb, dilute or break down a pollutant without harm.

Point source: a single, identifiable outlet of pollution, such as a chimney or effluent pipe.

Non-point source: pollution released over a wide area with no single outlet, such as farm runoff.

Biodegradable pollutant: a pollutant that decomposers break down into harmless substances.

Biomagnification: the increase in concentration of a persistent pollutant at each higher level of a food chain.

Particulate matter (PM2.5): airborne particles 2.5 micrometres or smaller that penetrate deep into the lungs.

Primary air pollutant: a pollutant emitted directly into the air from a source.

Secondary air pollutant: a pollutant formed in the atmosphere by reactions among primary pollutants and natural air components.

Photochemical smog: an oxidising haze of ozone, PAN and particles formed when NO_x and VOCs react in sunlight.

Temperature inversion: a layer of warm air above cooler surface air that traps pollutants near the ground.

Electrostatic precipitator: a device that charges dust particles electrically and collects them on earthed plates.

Flue gas desulfurisation: removal of SO_2 from flue gas, usually by reaction with limestone to form gypsum.

Catalytic converter: an exhaust device that uses a catalyst to convert CO, hydrocarbons and NO_x into less harmful gases.

Dissolved oxygen (DO): the amount of oxygen dissolved in water, needed by aquatic animals.

Biochemical oxygen demand (BOD): the oxygen consumed by micro-organisms decomposing organic matter in a water sample.

Chemical oxygen demand (COD): the oxygen needed to chemically oxidise all organic matter in a water sample.

Coliform bacteria: intestinal bacteria whose presence in water indicates faecal contamination.

Eutrophication: nutrient enrichment of a water body causing excessive algal growth and oxygen depletion.

Oxygen sag curve: the fall and recovery of dissolved oxygen in a river downstream of an organic waste discharge.

Phytoremediation: the use of plants to remove, stabilise or degrade pollutants in soil or water.

Thermal pollution: degradation of water quality by a change in its natural temperature, usually by heated discharges.

Decibel (dB): a logarithmic unit of sound level in which a 10 dB rise is a tenfold increase in intensity.

Silence zone: an area of at least 100 m around hospitals, educational institutions and courts with the strictest noise limits.

Air Quality Index (AQI): a single number and colour category that summarises air pollution levels for the public.

Designated best use: the most demanding use assigned to a water body, which determines its water quality criteria.

Sievert (Sv): the unit of radiation dose that accounts for the biological harm caused by different radiation types.

Half-life: the time taken for half of the atoms of a radioactive substance to decay.

Stochastic effect: a radiation effect, such as cancer, whose probability rather than severity increases with dose.

ALARA: the principle that radiation doses should be kept as low as reasonably achievable.

Solid waste management: The planned control of waste generation, storage, collection, transport, processing and disposal to protect health and the environment.

Leachate: The contaminated liquid formed when water percolates through waste, carrying dissolved and suspended pollutants.

Waste hierarchy: A ranking of waste-management options from prevention (most preferred) to disposal (least preferred).

Segregation at source: Separation of waste into different streams at the point where it is generated.

Bulk waste generator: An entity generating large amounts of waste (100 kg or more a day under the SWM Rules) with special duties to process its own waste.

Composting: Controlled aerobic decomposition of organic waste by microorganisms into humus-like compost.

Vermicomposting: Conversion of organic waste into nutrient-rich manure (vermicast) using earthworms.

Biomethanation: Anaerobic digestion of organic waste in a sealed digester to produce biogas and digestate.

Refuse-derived fuel (RDF): Fuel made from the dry, combustible, non-recyclable fraction of waste, often burned in cement kilns.

Sanitary landfill: An engineered disposal site with liners, leachate and gas collection, daily cover and monitoring.

Biomining: Excavation, stabilisation and sieving of legacy waste to recover soil, fuel and inert fractions and reclaim land.

Fly ash: Fine ash produced by burning pulverised coal in thermal power plants, usable in cement, bricks and roads.

Red mud: Highly alkaline residue left after extracting alumina from bauxite.

Cleaner production: A preventive strategy that reduces raw material, energy and water use and waste generation at source.

Common effluent treatment plant (CETP): A shared plant that treats the combined wastewater of a cluster of small industries.

Zero liquid discharge (ZLD): A system in which all wastewater is treated and recycled and dissolved salts are recovered as solids, so no effluent is released.

Hazardous waste: Waste that is ignitable, corrosive, reactive, toxic or otherwise likely to harm health or the environment.

Treatment, storage and disposal facility (TSDF): A common facility for the safe treatment and secure disposal of hazardous waste.

Extended producer responsibility (EPR): A policy making producers responsible for the collection and recycling of their products after use.

E-waste: Discarded electrical and electronic equipment, whole or in parts.

Consent to Operate: Permission from a State Pollution Control Board, with conditions, required before an industry starts production.

Environmental Impact Assessment (EIA): A process for predicting and evaluating the environmental effects of a proposed project before it is approved.

Polluter-pays principle: The principle that the polluter must bear the cost of preventing and remedying the pollution it causes.

Precautionary principle: The principle that lack of full scientific certainty must not delay measures to prevent serious or irreversible environmental harm.

Bioremediation: The use of microorganisms to remove or detoxify pollutants in soil or water.

Activated sludge process: Biological wastewater treatment in which aerated microbial flocs consume organic matter and are then settled and partly recycled.

Coagulation: Addition of chemicals such as alum to neutralise charges on fine particles so that they clump into settleable flocs.

Circular economy: An economic system that keeps materials in use through durable design, reuse, repair and recycling instead of take-make-dispose.

Multiple-Choice Questions

1. Which of the following is a non-point source of water pollution?

(a) Effluent pipe of a tannery
(b) Outfall of a sewage treatment plant
(c) Fertiliser runoff from farmland
(d) Discharge channel of a thermal power plant

2. Which pollutant is most strongly linked to deaths from heart disease, stroke and lung disease?

(a) PM2.5
(b) Carbon dioxide
(c) Nitrous oxide
(d) Ammonia

3. Carbon monoxide is dangerous mainly because it

(a) dissolves in rainwater to form acid
(b) binds to haemoglobin and reduces oxygen transport
(c) destroys stratospheric ozone
(d) causes eutrophication

4. Which of the following is a secondary air pollutant?

(a) Sulfur dioxide
(b) Carbon monoxide
(c) Black carbon
(d) Peroxyacetyl nitrate (PAN)

5. In photochemical smog, ozone accumulates mainly because peroxy radicals formed from VOCs

(a) convert NO to NO_2 without consuming ozone

(b) absorb ultraviolet light

(c) react with ozone to form oxygen

(d) remove ammonia from the air

6. Which device removes particles by giving them an electric charge and collecting them on earthed plates?

(a) Cyclone separator

(b) Fabric filter

(c) Electrostatic precipitator

(d) Venturi scrubber

7. Flue gas desulfurisation using limestone produces which useful by-product?

(a) Fly ash

(b) Gypsum

(c) Ammonium nitrate

(d) Activated carbon

8. India implemented BS-VI vehicle emission norms nationwide from

(a) 1 April 2017

(b) 1 April 2018

(c) 1 April 2020

(d) 1 April 2023

9. A river sample has a BOD of 2 mg/L and a DO of 7 mg/L. The sample is most likely

(a) heavily polluted with raw sewage

(b) relatively clean

(c) anoxic

(d) highly eutrophic

10. Blue baby syndrome in infants is caused by excess of which substance in drinking water?

(a) Fluoride

(b) Arsenic

(c) Nitrate

(d) Chloride

11. The acceptable limit of arsenic in drinking water under IS 10500:2012 is

(a) 0.01 mg/L

(b) 0.05 mg/L

(c) 0.1 mg/L

(d) 1.0 mg/L

12. The pesticide whose aerial spraying on cashew plantations caused a health crisis in Kasaragod, Kerala, is

(a) Endosulfan
(b) Malathion
(c) DDT
(d) Lindane

13. Thermal pollution harms fish mainly because warmer water

(a) holds more dissolved oxygen
(b) holds less dissolved oxygen while raising oxygen demand
(c) increases water hardness
(d) lowers BOD

14. Under the Noise Pollution (Regulation and Control) Rules, 2000, the daytime limit for a residential area is

(a) 45 dB(A)
(b) 50 dB(A)
(c) 65 dB(A)
(d) 55 dB(A)

15. An increase in sound level from 70 dB to 90 dB means the sound intensity has increased

(a) 2 times
(b) 20 times
(c) 100 times
(d) 1,000 times

16. The annual PM2.5 limit under India's NAAQS (2009) and the WHO 2021 guideline are, respectively,

(a) 40 and 5 $\mu g/m^3$
(b) 60 and 15 $\mu g/m^3$
(c) 40 and 10 $\mu g/m^3$
(d) 25 and 5 $\mu g/m^3$

17. A surface water body designated for organised outdoor bathing belongs to CPCB class

(a) A
(b) B
(c) C
(d) E

18. Which type of radiation is stopped by a sheet of paper but is most harmful when inhaled or swallowed?

(a) Gamma rays
(b) Beta particles
(c) Neutrons
(d) Alpha particles

19. The 2010 Mayapuri radiation incident in Delhi involved

(a) a leak from a nuclear reactor
(b) a cobalt-60 source from a discarded irradiator sold as scrap
(c) uranium mill tailings
(d) a medical X-ray machine

20. Which statement about stochastic effects of radiation is correct?

(a) They occur only above a threshold dose.
(b) Their probability increases with dose, and cancer is an example.
(c) Their severity increases with dose.
(d) They appear within hours of exposure.

21. According to the CPCB annual report for 2021–22, roughly what share of urban solid waste generated in India was processed or treated?

(a) 24%
(b) 38%
(c) 54%
(d) 92%

22. Which option sits at the top (most preferred level) of the waste hierarchy?

(a) Recycling
(b) Sanitary landfilling
(c) Energy recovery
(d) Prevention and reduction

23. The Solid Waste Management Rules, 2026 require household waste to be segregated into how many streams?

(a) Two
(b) Three
(c) Four
(d) Six

24. Mixed municipal waste in India is often unsuitable for incineration mainly because it

(a) has high moisture and low calorific value
(b) contains too much metal
(c) is mostly glass
(d) has too little organic matter

25. The dark liquid that forms when rainwater percolates through a dumpsite is called

(a) leachate
(b) digestate
(c) slurry
(d) vermicast

26. Which earthworm species is commonly used in vermicomposting?

(a) Pheretima posthuma
(b) Lumbricus terrestris only
(c) Eisenia fetida
(d) Nereis virens

27. Under the Bio-Medical Waste Management Rules, 2016, used needles and scalpel blades should be placed in

(a) a yellow bag
(b) a red bag
(c) a white translucent puncture-proof container
(d) a blue cardboard box

28. Which of the following is NOT one of the four standard characteristics of hazardous waste?

(a) Ignitability
(b) Biodegradability
(c) Corrosivity
(d) Reactivity

29. Seelampur in Delhi and Moradabad in Uttar Pradesh are known for

(a) informal e-waste dismantling and recycling
(b) fly ash brick manufacturing
(c) constructed wetlands
(d) waste-to-energy plants

30. Under the E-Waste (Management) Rules, 2022, producers meet their recycling obligations mainly by

(a) paying a fixed tax to municipalities
(b) purchasing EPR certificates generated by registered recyclers
(c) exporting e-waste under the Basel Convention
(d) burying e-waste in secured landfills

31. Fly ash is useful in making cement and bricks because it is

(a) radioactive
(b) pozzolanic
(c) highly acidic
(d) biodegradable

32. The principle that "lack of full scientific certainty shall not be a reason for postponing measures to prevent environmental degradation" is the

(a) polluter-pays principle
(b) public trust doctrine
(c) precautionary principle
(d) absolute liability principle

33. In the activated sludge process, part of the sludge settled in the secondary clarifier is returned to the aeration tank in order to

(a) add chlorine for disinfection
(b) produce biogas
(c) remove grit and sand
(d) keep a high population of microbes for treatment

34. Water softening by a resin that swaps calcium ions for sodium ions is an example of

(a) adsorption
(b) coagulation
(c) ion exchange
(d) reverse osmosis

35. The East Kolkata Wetlands are a well-known example of

(a) natural treatment of city sewage through fish ponds
(b) a hazardous waste TSDF
(c) a waste-to-energy plant
(d) a desalination plant

36. Which statement about the Ganga Action Plan is correct?

(a) It was launched in 2014 as Namami Gange.
(b) Its first phase focused on intercepting and treating sewage in 25 towns.
(c) It fully achieved bathing-quality water along the river by 2000.
(d) It was designed mainly to control crop residue burning.

37. Winter air pollution in Delhi is worse than summer pollution mainly because

(a) vehicles are banned in summer
(b) industries close in summer
(c) temperature inversions and low mixing height trap pollutants near the ground
(d) monsoon winds blow from the Himalaya in winter

38. Under the Graded Response Action Plan as revised by CAQM, Stage IV ("Severe+") is triggered when the AQI is

(a) above 200
(b) between 301 and 400
(c) between 401 and 450
(d) above 450

39. Since 31 December 2022, plastic carry bags in India must be at least

(a) 50 microns thick
(b) 75 microns thick
(c) 100 microns thick
(d) 120 microns thick

40. The gas that leaked from the Union Carbide plant in Bhopal in December 1984 was

(a) methyl isocyanate
(b) phosgene
(c) oleum
(d) hydrogen sulfide

Short-Answer Questions

1. Distinguish between point and non-point sources of pollution, giving two Indian examples of each.
2. What are primary and secondary air pollutants? Give three examples of each.
3. Explain, with reactions, how a three-way catalytic converter reduces vehicle emissions.
4. Describe the working of an electrostatic precipitator with a labelled sketch.
5. Why are women and young children in rural India most exposed to indoor air pollution? What has the Ujjwala Yojana achieved?
6. Define BOD and COD. Why is COD always higher than BOD for the same sample?
7. Explain the stages of eutrophication with an Indian example.
8. What causes fluorosis and arsenicosis? Name two states affected by each.
9. Write a note on the effects of thermal pollution on aquatic life and how cooling towers reduce it.
10. Tabulate the ambient noise limits for the four zones under the Noise Pollution (Regulation and Control) Rules, 2000.
11. Explain the six categories of India's National Air Quality Index and the PM2.5 range for each.
12. Distinguish between deterministic and stochastic effects of ionising radiation, with examples.
13. Distinguish between an open dump and a sanitary landfill, listing at least four engineering features of a sanitary landfill.
14. Why is segregation at source considered the foundation of solid waste management? Describe the four streams required by the Solid Waste Management Rules, 2026.
15. Explain the main stages of composting and the factors that control it.
16. What is biomethanation? Give one Indian example of its use for urban waste.
17. Give three reasons why waste-to-energy incineration has had limited success in India.
18. What is fly ash? How has India tried to increase its utilisation?
19. Tabulate the colour-coding categories of biomedical waste under the Bio-Medical Waste Management Rules, 2016, with one example of each.
20. Explain extended producer responsibility with reference to the E-Waste (Management) Rules, 2022.
21. Differentiate between Consent to Establish and Consent to Operate. Which bodies grant them?
22. Describe two forms of phytoremediation with an example plant for each.
23. What are coagulation and flocculation? Write the reaction of alum with water containing alkalinity.
24. List the main components of the Namami Gange programme.

Long-Answer Questions

1. Discuss the major sources, health effects and environmental effects of air pollution in India. Describe source-control and end-of-pipe measures for power plants, industries and vehicles.
2. Explain the formation of photochemical smog and secondary particulate matter with chemical reactions. Why must secondary pollutants be controlled at the level of an entire airshed, such as the Indo-Gangetic Plain?
3. Describe the sources of water pollution in India and the key parameters used to assess water quality. Explain the oxygen sag curve and outline how sewage is treated before discharge.
4. Compare India's National Ambient Air Quality Standards with the WHO Air Quality Guidelines of 2021. Should India revise its standards? Present arguments on both sides.
5. Discuss the sources and effects of soil pollution in India, with reference to agrochemicals, heavy metals and e-waste, and describe the main methods of prevention and remediation.
6. What are nuclear hazards? Describe the types and units of radiation, its health effects and the lessons of the Chernobyl, Fukushima and Mayapuri incidents for radiation safety in India.
7. Describe the generation, composition and fate of municipal solid waste in India. Explain how the waste hierarchy and integrated solid waste management can guide a city like Delhi, with reference to the Indore model and the problem of legacy dumpsites.
8. Discuss the control measures for industrial, hazardous and electronic waste in India, covering the relevant rules, treatment facilities and the role of the informal sector.
9. Compare legal, biological and physico-chemical methods of pollution control, using Indian examples, and explain how they work together in a sewage or effluent treatment plant.
10. Critically evaluate the Ganga Action Plan and Namami Gange. Why did the earlier programme fall short, and what lessons have been applied?
11. "Delhi's air pollution is a regional and seasonal problem that cannot be solved by the city alone." Discuss this statement with reference to sources, health effects and policy responses.
12. Analyse the Bhopal gas tragedy as a case of industrial pollution and environmental injustice. What legal and institutional reforms followed, and what issues remain unresolved?

Further Reading

- Central Pollution Control Board (2009). *National Ambient Air Quality Standards*. CPCB, New Delhi. https://cpcb.nic.in
- Central Pollution Control Board (2014). *National Air Quality Index*. CPCB, New Delhi.
- Bureau of Indian Standards (2012). *IS 10500:2012 Drinking Water — Specification* (with amendments). BIS, New Delhi.
- World Health Organization (2021). *WHO Global Air Quality Guidelines: Particulate Matter (PM2.5 and PM10), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide*. WHO, Geneva.
- India State-Level Disease Burden Initiative Air Pollution Collaborators (2021). Health and economic impact of air pollution in the states of India: the Global Burden of Disease Study 2019. *The Lancet Planetary Health*, 5(1), e25–e38.
- Brusseau, M. L., Pepper, I. L. & Gerba, C. P. (2019). *Environmental and Pollution Science* (3rd ed.). Academic Press.

- Central Ground Water Board (2024). *Annual Ground Water Quality Report 2024*. Ministry of Jal Shakti, Government of India.
- Atomic Energy Regulatory Board. *Radiation Protection and Safety* (official website). https://www.aerb.gov.in
- Central Pollution Control Board (2023). *Annual Report on Solid Waste Management 2021–22*. CPCB, Delhi. https://cpcb.nic.in
- Ministry of Environment, Forest and Climate Change (2026). *Solid Waste Management Rules, 2026* and related waste-management rules. Government of India. https://moef.gov.in
- Kaza, S., Yao, L., Bhada-Tata, P. & Van Woerden, F. (2018). *What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050*. World Bank.
- Baldé, C.P. et al. (2024). *Global E-waste Monitor 2024*. ITU and UNITAR.
- National Mission for Clean Ganga. *Namami Gange* programme website. https://nmcg.nic.in
- Divan, S. & Rosencranz, A. (2002). *Environmental Law and Policy in India: Cases, Materials and Statutes*, 2nd edition. Oxford University Press.
- Lapierre, D. & Moro, J. (2002). *Five Past Midnight in Bhopal*. Warner Books.

Answers to Multiple-Choice Questions

1. (c)	2. (a)	3. (b)	4. (d)	5. (a)	6. (c)	7. (b)	8. (c)	9. (b)	10. (c)
11. (a)	12. (a)	13. (b)	14. (d)	15. (c)	16. (a)	17. (b)	18. (d)	19. (b)	20. (b)
21. (c)	22. (d)	23. (c)	24. (a)	25. (a)	26. (c)	27. (c)	28. (b)	29. (a)	30. (b)
31. (b)	32. (c)	33. (d)	34. (c)	35. (a)	36. (b)	37. (c)	38. (d)	39. (d)	40. (a)

Part II: Year 2

Units 5–7 · Semester I/II

Chapter 5: Global Environmental Issues and Policies

Some environmental problems do not stop at a city limit or a national border. Carbon dioxide released from a power plant in Singrauli mixes into the same atmosphere as carbon dioxide from a car in Chicago, and the chlorine from a discarded refrigerator anywhere can thin the ozone layer over Antarctica. Climate change, ozone depletion and acid rain are global or regional in scale, and only international cooperation backed by national law can control them. This chapter explains the science behind these problems and their impacts on people, biodiversity, the economy and agriculture, with a focus on India. It then describes the main international agreements and programmes, India's climate policy and the Sustainable Development Goals, and the six Acts that form the backbone of India's environmental law.

Learning Objectives

- Explain the natural and human causes of climate change, the greenhouse effect, radiative forcing and climate feedbacks.
- Describe the evidence for global warming, with Indian data on temperature and the monsoon.
- Explain the chemistry of ozone depletion and acid rain, and the measures used to control them.
- Analyse the impacts of climate change on human communities, biodiversity, the global economy and agriculture in India.
- Compare the aims, mechanisms and India's role in major agreements: the Earth Summit, UNFCCC, Kyoto Protocol, Paris Agreement, Montreal Protocol, CBD, Ramsar, CITES, the Chemical Weapons Convention and UNEP.
- Evaluate India's National Action Plan on Climate Change, its missions, and India's progress on the Sustainable Development Goals.
- Summarise the objectives, provisions, institutions, amendments and landmark cases of India's main environmental Acts from 1972 to 2006.

5.1 Earth's Climate System and the Causes of Climate Change

Weather is the state of the atmosphere at a place over hours or days. **Climate** is the average weather, including its variability and extremes, over a long period, conventionally 30 years. "Delhi has hot summers and a short monsoon" describes climate; "it rained in Delhi yesterday" describes weather. **Climate change** is a lasting change in the statistics of climate, such as average temperature or the frequency of heavy rain, that persists for decades or longer.

5.1.1 The climate system

The **climate system** is made up of five interacting parts. The **atmosphere** carries heat and moisture around the planet. The **hydrosphere**, especially the ocean, stores enormous amounts of heat and carbon and releases them slowly. The **cryosphere** (glaciers, ice sheets, sea ice, snow and permafrost) reflects sunlight and stores fresh water. The **lithosphere** shapes climate through mountains,

volcanoes and the slow drift of continents. The **biosphere** exchanges carbon dioxide and water vapour with the air through photosynthesis, respiration and decay.

These parts are linked through the water cycle, the carbon cycle and the flow of energy. The Indian summer monsoon is a good example. It is driven by the difference in heating between the Indian landmass and the Indian Ocean, steered by the Himalaya and the Tibetan Plateau, and influenced by ocean patterns in the Pacific and Indian Oceans.

Climate also varies naturally from year to year without any change in its long-term state. This **natural variability** includes the **El Niño–Southern Oscillation (ENSO)**, a periodic warming (El Niño) or cooling (La Niña) of the eastern tropical Pacific. El Niño years such as 2009, 2015 and 2023 often bring a weaker Indian monsoon, and they also raise global average temperature for a year or so. Climate change must be detected against this background of natural ups and downs.

5.1.2 Natural causes of climate change

Earth's climate has changed many times in the past, long before humans existed. The main natural drivers are listed below.

- **Orbital (Milankovitch) cycles.** Slow changes in the shape of Earth's orbit (about 100,000 years), the tilt of its axis (about 41,000 years) and the wobble of the axis (about 23,000 years) change how sunlight is distributed between seasons and latitudes. These cycles paced the ice ages of the past million years. They act far too slowly to explain warming over decades.
- **Solar variability.** The Sun's output varies by about 0.1% over the 11-year sunspot cycle. A period of low solar activity, the Maunder Minimum (about 1645–1715), coincided with part of the cool "Little Ice Age" in Europe. Satellite measurements since 1978 show no upward trend in solar output, so the Sun cannot explain recent warming.
- **Volcanic eruptions.** Large eruptions inject sulfur dioxide into the stratosphere, where it forms tiny sulfate droplets that reflect sunlight. The 1991 eruption of Mount Pinatubo in the Philippines cooled the globe by about 0.5 °C for one to two years. The effect is temporary.
- **Plate tectonics.** Over millions of years, moving continents change ocean currents and wind patterns. The rise of the Himalaya and Tibetan Plateau, for example, strengthened the Asian monsoon.
- **Internal variability.** Exchanges of heat between ocean and atmosphere, such as ENSO and the Indian Ocean Dipole, shift climate for months to years.

5.1.3 Anthropogenic causes: the enhanced greenhouse effect

Anthropogenic means caused by humans. The main human cause of present-day climate change is the strengthening of the natural greenhouse effect by gases released from burning fossil fuels, clearing forests, farming and industry.

The Sun emits mostly **shortwave radiation** (visible light and some ultraviolet). About 30% of the incoming sunlight is reflected back to space by clouds, ice and bright surfaces; this reflectivity is called **albedo**. The rest is absorbed and warms the land and ocean. The warm surface then emits **longwave (infrared) radiation**. Certain gases in the atmosphere, called **greenhouse gases (GHGs)**, let sunlight through but absorb much of the outgoing infrared radiation and re-emit it in all directions, including back towards the surface. This is the **greenhouse effect**. Without it, Earth's average surface temperature would be about −18 °C instead of about +15 °C, and the planet would be frozen.

The main natural greenhouse gases are water vapour, carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O) and ozone (O_3). Since the Industrial Revolution (taken as about 1750), human activities have raised the concentrations of CO_2, CH_4 and N_2O and have added entirely synthetic gases such as chlorofluorocarbons (CFCs), hydrofluorocarbons (HFCs) and sulfur hexafluoride (SF_6). The extra gases trap more heat. This is the **enhanced greenhouse effect**, and it is the main cause of the warming observed since the mid-20th century.

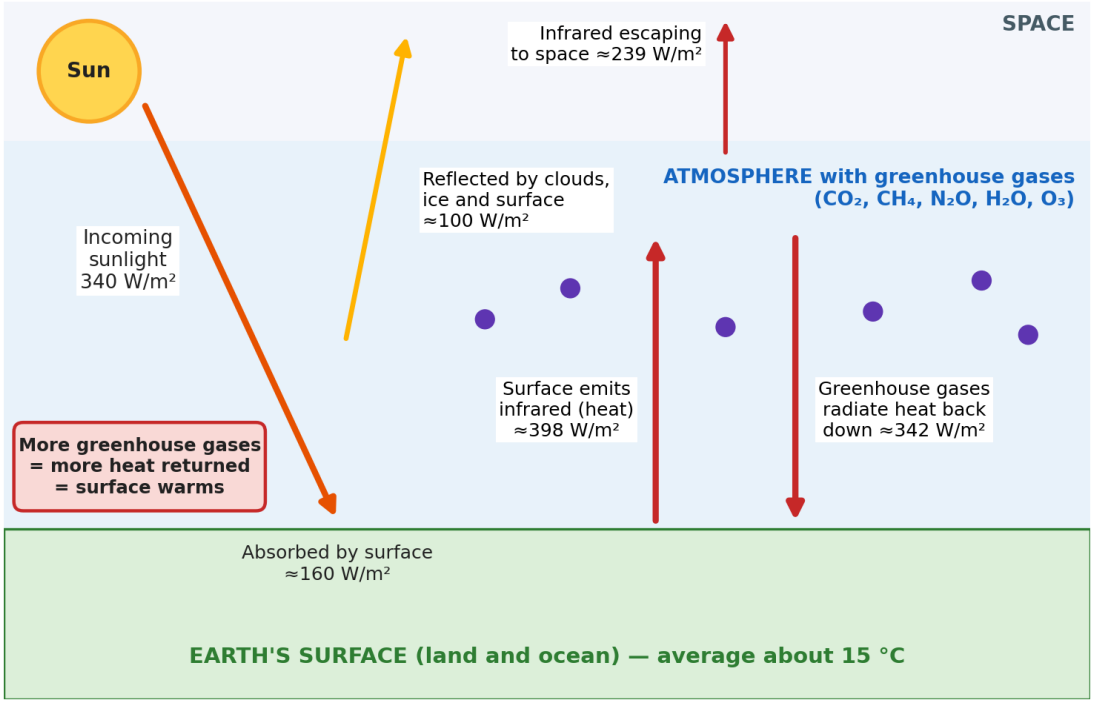


Figure 5.1: *The natural and enhanced greenhouse effect. Incoming sunlight warms the surface; greenhouse gases absorb part of the outgoing infrared radiation and send some of it back down. Adding more greenhouse gases strengthens this return flow and warms the surface. Values are approximate global averages. Source: Prepared by the author, based on IPCC AR6 WG I (2021).*

5.1.4 Greenhouse gases and their sources

Each greenhouse gas differs in how strongly it absorbs heat and how long it stays in the air. Scientists compare them using the **global warming potential (GWP)**, the warming caused by releasing 1 kg of a gas over a chosen period (usually 100 years) relative to 1 kg of CO_2. Emissions of different gases can then be added together as **carbon dioxide equivalent (CO_2e)**. The table below uses values from the IPCC Sixth Assessment Report (AR6) and concentrations from the World Meteorological Organization (WMO).

Table 5.1: Major greenhouse gases: sources, lifetimes and 100-year global warming potentials (IPCC AR6, 2021; concentrations from WMO Greenhouse Gas Bulletin, 2025)

Gas	Main human sources	Lifetime in atmosphere	GWP (100 years)	Concentration (pre-industrial → 2024)
Carbon dioxide (CO_2)	Burning coal, oil and gas; cement; deforestation	Centuries to millennia (no single value)	1	278 → 423.9 ppm

Gas	Main human sources	Lifetime in atmosphere	GWP (100 years)	Concentration (pre-industrial → 2024)
Methane (CH_4)	Livestock, paddy fields, landfills, coal mines, gas leaks	About 12 years	About 27–30	722 → 1942 ppb
Nitrous oxide (N_2O)	Nitrogen fertilisers, manure, industry	About 109 years	273	270 → 338 ppb
HFC-134a	Refrigerators, car air-conditioners	About 14 years	About 1,530	Synthetic, rising
CFC-12	Old refrigerants and aerosol sprays (now banned)	About 100 years	About 12,500	Synthetic, slowly falling
Sulfur hexafluoride (SF_6)	Electrical switchgear	About 3,200 years	About 25,200	Synthetic, rising

Note: ppm = parts per million; ppb = parts per billion. The methane GWP is about 30 for fossil methane and about 27 for methane from farms and wetlands.

CO_2 is by far the most important human-made greenhouse gas because it is released in huge quantities and a large part of each emission stays in the air for centuries. Methane is the second most important. It is short-lived but powerful, so cutting methane slows warming quickly. India's methane comes mainly from enteric fermentation in its large cattle and buffalo population, from flooded rice paddies and from waste. This is why India has not joined the voluntary Global Methane Pledge launched at COP26 (2021), arguing that the pledge targets the livelihoods of small farmers.

Two other human influences matter. **Aerosols** are tiny particles of sulfate, nitrate, dust and soot. Sulfate aerosols reflect sunlight and help form brighter clouds, so they cool the planet and hide part of the greenhouse warming. **Black carbon** (soot) from diesel engines, brick kilns and biomass burning does the opposite: it absorbs sunlight and darkens snow and ice when it settles on Himalayan glaciers. **Land-use change** such as deforestation releases stored carbon and changes the albedo of the surface.

5.1.5 Radiative forcing

Radiative forcing is the change in Earth's energy balance, measured in watts per square metre (W/m^2), caused by a change in a driver such as a greenhouse gas or aerosol. A positive forcing means the Earth system gains energy and warms; a negative forcing means it loses energy and cools. IPCC AR6 estimates that the total human-caused effective radiative forcing in 2019, relative to 1750, was about +2.72 W/m^2. Well-mixed greenhouse gases and ozone together contributed about +3.84 W/m^2, while aerosols offset about −1.1 W/m^2.

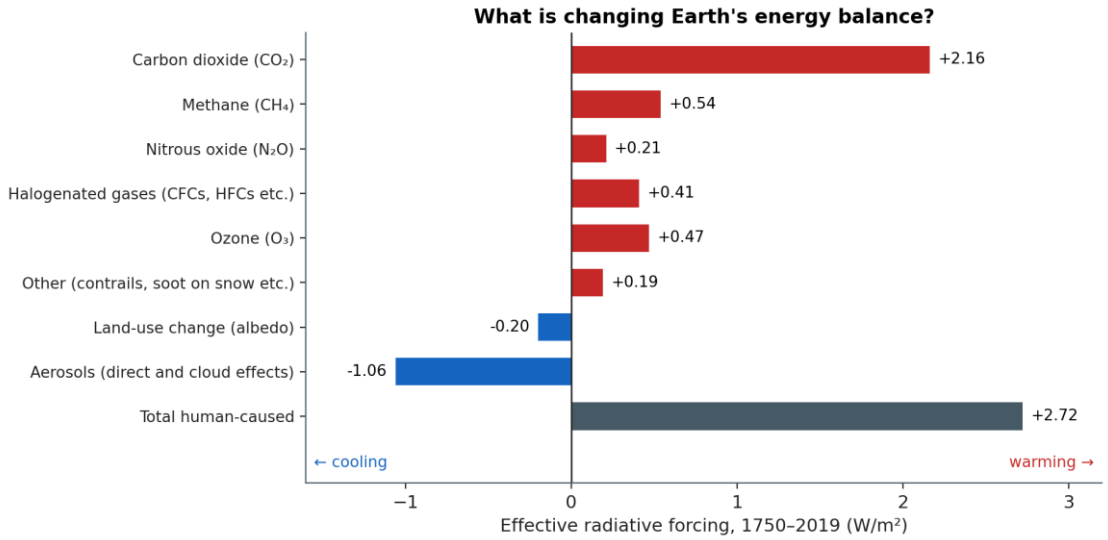


Figure 5.2: *Effective radiative forcing in 2019 relative to 1750 for the main climate drivers. Bars to the right warm the climate; bars to the left cool it. Source: IPCC AR6 WG I (2021), Chapter 7.*

Two lessons follow from the chart. First, CO_2 alone accounts for most of the human-caused warming. Second, air pollution by aerosols has been masking part of the warming. As countries, including India, clean their air by cutting sulfur dioxide, some of this hidden warming will be revealed. This is not a reason to tolerate pollution, which kills over a million Indians every year (see Chapter 4); it is a reason to cut greenhouse gases at the same time.

The IPCC's AR6 concluded in 2021 that "it is unequivocal that human influence has warmed the atmosphere, ocean and land." The evidence for human causation includes several independent fingerprints:

- The extra CO_2 in the air carries the chemical (isotopic) signature of fossil carbon, which is ancient plant material poor in carbon-13 and lacking carbon-14.
- The lower atmosphere (troposphere) is warming while the upper atmosphere (stratosphere) is cooling. Extra greenhouse gases produce this pattern; a brighter Sun would warm both layers.
- Nights are warming faster than days and winters faster than summers, as expected from heat trapping rather than more sunlight.
- Climate models can reproduce the observed warming only when human emissions are included.

5.1.6 Climate feedbacks

A **feedback** is a process that changes in response to warming and in turn amplifies (positive feedback) or dampens (negative feedback) the original change. Feedbacks determine how much the climate finally warms. The **equilibrium climate sensitivity**, the long-term warming from doubling CO_2, is very likely between 2 °C and 5 °C, with a best estimate of 3 °C (IPCC AR6).

- **Water vapour feedback (positive).** Warmer air holds more water vapour, which traps more heat. This roughly doubles the warming from CO_2 alone.
- **Ice–albedo feedback (positive).** Melting snow and ice expose darker land or sea, which absorbs more sunlight and causes more melting. This is one reason the Arctic is warming about four times faster than the global average.

- **Cloud feedback (probably positive).** Changes in cloud amount and height can either warm or cool. AR6 judges the net effect likely to be warming.
- **Carbon-cycle feedbacks (positive).** Warming thaws permafrost, releasing CO_2 and methane, and can weaken the ability of oceans and forests to absorb CO_2.
- **Planck response (negative).** A warmer Earth radiates more energy to space, which limits warming. This is the basic stabilising response of the climate system.

Some parts of the climate system may cross **tipping points**, thresholds beyond which change becomes self-sustaining and hard to reverse, such as the loss of the Greenland and West Antarctic ice sheets or dieback of the Amazon rainforest.

> **Global warming versus climate change**
>
> The two terms are related but not the same. **Global warming** is the rise in Earth's average surface temperature. **Climate change** is the wider set of changes that warming causes: shifting rainfall, stronger extremes, melting ice, rising seas and acidifying oceans. Warming is one symptom; climate change is the whole condition.

5.2 Global Warming

Global warming is the long-term rise in the average temperature of Earth's surface caused mainly by the enhanced greenhouse effect. It is measured by combining readings from thousands of weather stations, ships and buoys into a global average, and comparing it with the 1850–1900 average, which is taken as the "pre-industrial" baseline.

5.2.1 Observed evidence of a warming world

Many independent lines of evidence show that Earth is warming.

- **Surface temperature.** According to IPCC AR6, global surface temperature in 2011–2020 was 1.09 °C higher than in 1850–1900, and about 1.07 °C of this was caused by humans. Warming has continued since. The WMO reported that 2024 was the warmest year on record, about 1.55 °C above the pre-industrial level, and the first calendar year above 1.5 °C. It confirmed in January 2026 that 2025 was the second or third warmest year (about 1.44 °C above pre-industrial), and that the eleven years 2015–2025 were the eleven warmest on record. A single year above 1.5 °C does not mean the Paris Agreement limit has been breached, because that limit refers to a 20-year average.
- **Ocean heat.** The ocean has absorbed about 90% of the extra heat trapped since 1970. Argo floats, a global network of robotic instruments, show ocean heat content rising year after year.
- **Sea level.** Global mean sea level rose by about 20 cm between 1901 and 2018, and the rate has nearly tripled to about 3.7 mm per year in 2006–2018 (IPCC AR6). Two processes cause this: water expands as it warms (**thermal expansion**), and melting glaciers and ice sheets add water to the ocean.
- **Ice.** Arctic sea ice in late summer has shrunk by roughly 40% since satellite records began in 1979. Greenland and Antarctica are losing ice, and almost all mountain glaciers, including those of the Himalaya, are retreating.

- **Ocean acidification.** The ocean absorbs about a quarter of the CO_2 emitted each year. The dissolved CO_2 forms carbonic acid, and surface ocean pH has fallen from about 8.2 to about 8.1 since the Industrial Revolution, a 30% increase in acidity. This makes it harder for corals, shellfish and some plankton to build shells and skeletons of calcium carbonate.
- **Extremes.** Hot extremes have become more frequent and intense, and heavy rainfall events have become more frequent over most land areas. Scientists can now estimate how much climate change increased the odds of a particular heatwave or flood; this is called **extreme event attribution**.

5.2.2 Rising carbon dioxide: the Keeling Curve

The most famous record in climate science began in 1958, when Charles David Keeling started measuring CO_2 at the Mauna Loa Observatory in Hawaii, far from local pollution. The record, now called the **Keeling Curve**, shows two features: a yearly saw-tooth caused by Northern Hemisphere plants absorbing CO_2 in summer and releasing it in winter, and a steady, accelerating rise caused by fossil fuel burning.

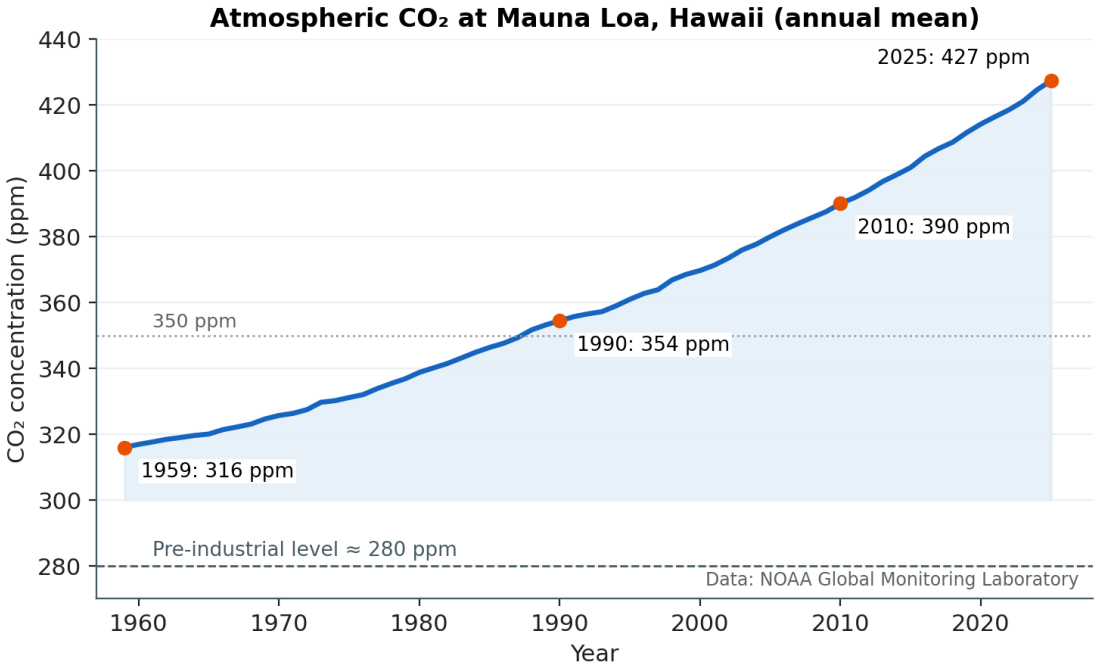


Figure 5.3: *Annual mean atmospheric CO_2 concentration at Mauna Loa Observatory, 1959–2025. The concentration rose from about 316 ppm in 1959 to 427 ppm in 2025, and the rate of increase has itself grown. Source: NOAA Global Monitoring Laboratory (2026).*

Air bubbles trapped in Antarctic ice cores show that CO_2 stayed between about 180 and 300 ppm for the past 800,000 years. The global average reached 423.9 ppm in 2024, the highest in at least 2 million years. The rise from 2023 to 2024, about 3.5 ppm, was the largest annual increase since modern measurements began (WMO Greenhouse Gas Bulletin, 2025). Because a large fraction of CO_2 remains in the air for centuries, warming depends on **cumulative emissions**, the total CO_2 emitted over time. This is why historical responsibility matters in climate negotiations: the United States and Europe together account for a far larger share of cumulative CO_2 since 1850 than India, even though India is now the third-largest annual emitter after China and the United States.

The **carbon budget** is the total amount of CO_2 that can still be emitted while keeping warming below a given limit. Updated estimates by IPCC authors (Indicators of Global Climate Change, 2025) suggest that only about 130 billion tonnes of CO_2 remained from the start of 2025 for a 50% chance of limiting warming to 1.5 °C. At current global emission rates of about 40 billion tonnes a year, this budget would be used up within a few years.

5.2.3 Warming in India and changes in the monsoon

India's first national climate assessment, *Assessment of Climate Change over the Indian Region* (Ministry of Earth Sciences, 2020), summarised the observed changes:

- India's average temperature rose by about 0.7 °C between 1901 and 2018, with stronger warming in recent decades.
- The India Meteorological Department (IMD) reported that 2024 was India's warmest year since national records began in 1901, about 0.65 °C above the 1991–2020 average.
- The frequency of heatwaves in pre-monsoon months has increased, especially over north-west, central and south-east India, and is projected to be three to four times higher by 2100 under high emissions.
- Summer monsoon rainfall (June–September) declined by about 6% between 1951 and 2015, with notable drops over the Indo-Gangetic plains and the Western Ghats.
- At the same time, rainfall has become more concentrated in short, intense bursts. Very heavy rainfall events (more than 150 mm in a day) rose by about 75% over central India between 1950 and 2015, while the number of moderate rain days fell.
- Droughts have become more frequent and widespread since 1951.
- Sea level in the north Indian Ocean rose at about 1.06–1.75 mm per year during 1874–2004, accelerating to about 3.3 mm per year during 1993–2017.
- The frequency of very severe cyclones in the Arabian Sea has increased in the post-monsoon season since 1980.

This pattern of "longer dry spells broken by extreme downpours" matters for a country where about half the net sown area is rainfed. Flash floods such as those in Kerala (2018), Uttarakhand (2013 and 2021) and Himachal Pradesh (2023) show what intense rain does in steep or densely built terrain. The warming of the Indian Ocean, one of the fastest-warming oceans, also feeds more moisture into monsoon storms and cyclones.

Think and Do: Temperature and rainfall trend of your district

Download monthly temperature and rainfall data for your district or nearest city for at least 30 years from the IMD data portal or the NASA POWER website. Plot the annual mean temperature and the annual (or June–September) rainfall against year in a spreadsheet and add a linear trend line. Is your area warming? Has the number of very wet days changed? Compare your result with the national trends described above.

Key Points

- The greenhouse effect is natural; the enhanced greenhouse effect caused by human emissions is the main driver of recent warming.
- CO_2 accounts for most human-caused forcing; methane is short-lived but potent; aerosols mask part of the warming.
- Earth has warmed by about 1.1–1.3 °C since 1850–1900; 2024 was the warmest year on record.
- India has warmed by about 0.7 °C since 1901, with more heatwaves, a weaker but more erratic monsoon and rising seas.

5.3 Ozone Layer Depletion

Ozone (O_3) is a molecule of three oxygen atoms. Near the ground it is a harmful pollutant formed in smog (see Chapter 4). High in the **stratosphere**, between about 15 and 35 km above the surface, it forms the **ozone layer**, which absorbs most of the Sun's harmful ultraviolet (UV) radiation. Even here ozone is scarce: if all the ozone in a column of air were brought down to sea-level pressure, it would form a layer only about 3 mm thick. This amount is expressed as 300 **Dobson units (DU)**.

5.3.1 The ozone layer and the Chapman cycle

Ultraviolet radiation is divided by wavelength. **UV-C** (100–280 nm) is the most energetic and is completely absorbed by oxygen and ozone. **UV-B** (280–315 nm) is mostly absorbed by ozone; the small part that reaches the surface causes sunburn and skin cancer. **UV-A** (315–400 nm) is least harmful and largely passes through.

In 1930 the British physicist Sydney Chapman explained how ozone is continuously made and destroyed by sunlight. The **Chapman cycle** has four steps.

1. High-energy UV (wavelength below about 240 nm) splits an oxygen molecule into two oxygen atoms.
2. Each oxygen atom joins an oxygen molecule to form ozone. A third molecule (M, usually N_2 or O_2) carries away the extra energy.
3. Ozone absorbs UV-B and UV-C and splits. This is the step that shields the surface, and it converts UV energy into heat that warms the stratosphere.
4. Ozone is naturally removed when it meets an oxygen atom.

The four reactions, in order, are:

$$O_2 + \text{UV} \rightarrow O + O$$

$$O + O_2 + M \rightarrow O_3 + M$$

$$O_3 + \text{UV} \rightarrow O_2 + O$$

$$O + O_3 \rightarrow 2O_2$$

In an undisturbed atmosphere, production and destruction balance and the amount of ozone stays roughly steady. Natural catalysts, such as nitrogen oxides and hydroxyl radicals, also destroy some ozone.

5.3.2 Ozone-depleting substances and their chemistry

Ozone-depleting substances (ODS) are human-made chemicals that release chlorine or bromine in the stratosphere. The most important are:

- **Chlorofluorocarbons (CFCs)** such as CFC-11 and CFC-12, used from the 1930s as refrigerants, aerosol propellants, foam-blowing agents and cleaning solvents. They are non-toxic and non-flammable, which made them popular.
- **Halons**, used in fire extinguishers. They contain bromine, which is about 60 times more destructive to ozone than chlorine, atom for atom.
- **Carbon tetrachloride (CTC)** and **methyl chloroform**, used as solvents and in chemical manufacture.
- **Methyl bromide**, a soil and grain fumigant.
- **Hydrochlorofluorocarbons (HCFCs)**, introduced as transitional substitutes for CFCs. They are less damaging but still deplete ozone.

The very stability that made CFCs useful is the problem. They do not dissolve in rain or react in the lower atmosphere, so over several years they drift up into the stratosphere. There, intense UV light breaks them apart and frees chlorine atoms. In 1974 Mario Molina and F. Sherwood Rowland showed that each chlorine atom acts as a **catalyst**, a substance that speeds up a reaction without being used up. With Paul Crutzen, who had shown the role of nitrogen oxides, they shared the 1995 Nobel Prize in Chemistry. The chlorine cycle runs as follows:

$$CFCl_3 + UV \rightarrow CFCl_2 + Cl$$

$$Cl + O_3 \rightarrow ClO + O_2$$

$$ClO + O \rightarrow Cl + O_2$$

$$\text{Net: } O_3 + O \rightarrow 2O_2$$

The chlorine atom comes out of the cycle unchanged and can repeat it. A single chlorine atom can destroy tens of thousands of ozone molecules before it is locked into a less reactive "reservoir" compound such as hydrogen chloride (HCl) or chlorine nitrate ($ClONO_2$).

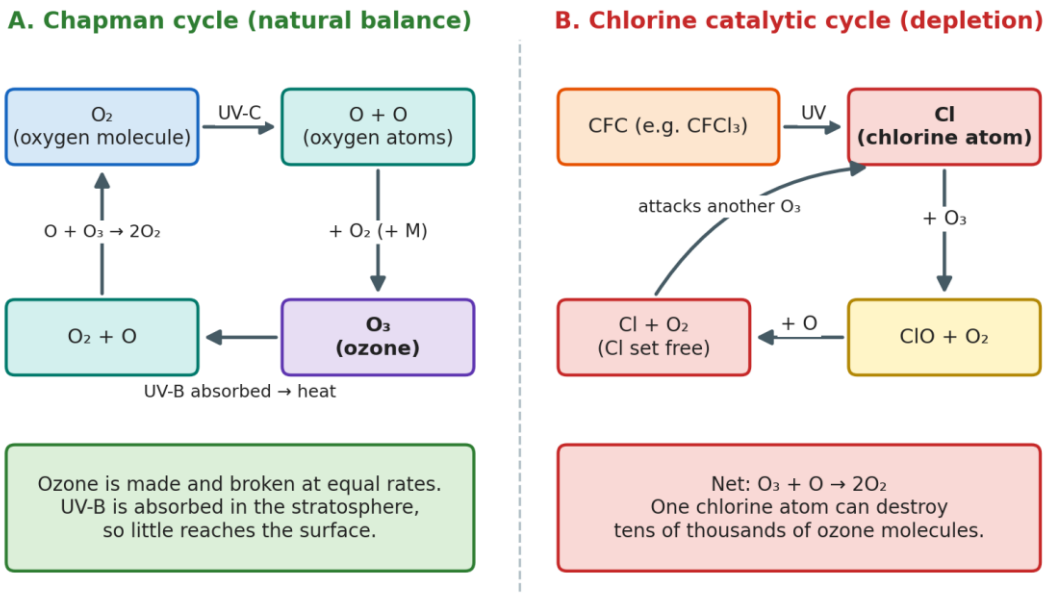


Figure 5.4: *Ozone chemistry in the stratosphere. Left: the natural Chapman cycle, in which UV light makes and breaks ozone in balance. Right: the chlorine catalytic cycle, in which a chlorine atom released from a CFC destroys ozone and is regenerated to attack again. Source: Prepared by the author, based on WMO/UNEP Scientific Assessment of Ozone Depletion (2022).*

5.3.3 The Antarctic ozone hole

In May 1985, Joseph Farman, Brian Gardiner and Jonathan Shanklin of the British Antarctic Survey reported that springtime ozone over Halley Bay in Antarctica had fallen by about a third since the 1960s. NASA satellite data, re-examined, confirmed a vast region of depleted ozone. The **ozone hole** is defined as the area where total ozone falls below 220 DU. In its worst years, around 2000 and 2006, it covered about 28–29 million km^2, more than eight times the area of India.

The hole forms over Antarctica every September–October because of a special combination of conditions.

1. **Polar vortex.** In the dark Antarctic winter, a ring of strong winds isolates the air over the continent, so it becomes extremely cold.
2. **Polar stratospheric clouds (PSCs).** Below about −78 °C, thin clouds of ice and nitric acid form in the stratosphere.
3. **Chlorine activation.** On the surfaces of PSC particles, the stable reservoir compounds react and release chlorine gas, which accumulates in the dark.
4. **Spring sunlight.** When the Sun returns in September, light splits chlorine gas into chlorine atoms, which destroy ozone rapidly through catalytic cycles. Within weeks, up to 60% of the total ozone column, and nearly all the ozone at some altitudes, disappears.

The key activation reaction on cloud surfaces is:

$$ClONO_2 + HCl \rightarrow Cl_2 + HNO_3$$

The Arctic warms more in winter and has a weaker, less stable vortex, so Arctic ozone loss is smaller and more variable; a large Arctic loss occurred in spring 2020. Outside the poles, ozone over middle latitudes declined by a few per cent between 1980 and the late 1990s.

5.3.4 Effects of ozone depletion

More UV-B at the surface harms living things.

- **Human health.** UV-B damages DNA in skin cells and is the main cause of non-melanoma skin cancers and a contributor to melanoma. It also causes **cataracts** (clouding of the eye lens) and suppresses the immune system. The US Environmental Protection Agency estimates that, for Americans born between 1890 and 2100, the Montreal Protocol will prevent about 443 million cases of skin cancer and 63 million cases of cataract.
- **Ecosystems.** UV-B reduces the growth of **phytoplankton**, the microscopic algae at the base of marine food webs, and harms fish larvae and amphibian eggs.
- **Agriculture.** Many crops, including rice, wheat and soybean, show reduced growth and yield under higher UV-B.

Because India lies in the tropics, where the natural ozone layer is thinner and the Sun is high overhead, surface UV levels are already high. Polar-type ozone holes do not form over India, but any global loss of ozone would raise UV exposure for a very large population.

5.3.5 The global response and recovery

The response to ozone depletion is widely regarded as the most successful international environmental action to date. The **Vienna Convention for the Protection of the Ozone Layer** (1985) set up a framework for cooperation, and the **Montreal Protocol on Substances that Deplete the Ozone Layer** (1987) set binding phase-out schedules. The Protocol's design is described later in this chapter, among the international agreements. Its results are clear:

- About 99% of the production and consumption of controlled ODS has been phased out worldwide.
- The amount of ozone-depleting chlorine and bromine in the stratosphere has been falling since the late 1990s.
- The 2022 WMO/UNEP Scientific Assessment of Ozone Depletion projects that ozone will return to 1980 values by about 2040 over most of the world, about 2045 over the Arctic and about 2066 over Antarctica.
- Because most ODS are also powerful greenhouse gases, the Protocol has avoided more warming than the Kyoto Protocol was designed to.

5.3.6 HFCs and the Kigali Amendment

As CFCs and HCFCs were phased out, industry turned to **hydrofluorocarbons (HFCs)**. HFCs contain no chlorine and do not harm ozone, but many are powerful greenhouse gases; HFC-134a, for example, has a GWP of about 1,530. With air-conditioner sales rising rapidly in hot, growing economies such as India, HFC emissions were set to soar.

In October 2016, at Kigali in Rwanda, the Parties agreed the **Kigali Amendment**, which adds HFCs to the Montreal Protocol and requires their production and consumption to be cut by 80–85% by the late 2040s. It entered into force on 1 January 2019. Full implementation is expected to avoid up to about 0.4 °C of global warming by 2100. Countries are divided into groups with different schedules. India, in the group of developing countries with later deadlines, will freeze HFC use in 2028 (against

a 2024–2026 baseline) and then reduce it by 10% in 2032, 20% in 2037, 30% in 2042 and 85% in 2047. India ratified the amendment in 2021.

5.3.7 India's ODS phase-out and the India Cooling Action Plan

India joined the Vienna Convention and the Montreal Protocol in 1992. An **Ozone Cell** in the Ministry of Environment, Forest and Climate Change (MoEFCC) coordinates implementation, and the **Ozone Depleting Substances (Regulation and Control) Rules, 2000**, framed under the Environment (Protection) Act 1986, control the production, trade and use of ODS through licensing. With help from the Protocol's Multilateral Fund, India:

- ended production of CFCs in August 2008 and completed the phase-out of CFCs, halons and carbon tetrachloride by 1 January 2010, the deadline for developing countries;
- converted refrigerator, foam and aerosol factories, many of them small and medium enterprises, to ozone-safe technologies;
- phased out HCFC-141b, a foam-blowing agent, on 1 January 2020 under its HCFC Phase-out Management Plan, and is phasing out the remaining HCFCs by 2030.

In March 2019 India became one of the first countries to release a national **India Cooling Action Plan (ICAP)**. Recognising that demand for fans, air-conditioners, refrigerated storage and vehicle cooling will grow several-fold, it set targets for 2037–38 relative to a 2017–18 baseline:

- reduce cooling demand across sectors by 20–25%;
- reduce refrigerant demand by 25–30%;
- reduce cooling energy requirements by 25–40%;
- train and certify 100,000 refrigeration and air-conditioning service technicians;
- promote research on low-GWP refrigerants and **passive cooling** through building design, such as shading, ventilation and reflective roofs.

ICAP links the ozone and climate agendas with public health, since access to cooling is essential in a warming country.

Table 5.2: Ozone depletion and global warming compared

Feature	Ozone depletion	Global warming
Main location	Stratosphere (15–35 km)	Troposphere and surface
Main cause	Chlorine and bromine from CFCs, halons and other ODS	CO_2, CH_4, N_2O and other greenhouse gases
Mechanism	Catalytic destruction of ozone	Absorption of outgoing infrared radiation
Main harm	More UV-B at the surface	Higher temperatures, sea-level rise, extremes
Key treaty	Montreal Protocol (1987)	UNFCCC (1992) and Paris Agreement (2015)
Status	Recovering; full recovery expected 2040–2066	Still worsening

5.4 Acid Rain

Acid rain is precipitation that is more acidic than normal because it contains sulfuric and nitric acids formed from air pollutants. Scientists prefer the broader term **acid deposition**, which includes **wet deposition** (acidic rain, snow, fog and hail) and **dry deposition** (acidic gases and particles that settle on surfaces). The term "acid rain" was first used by the Scottish chemist Robert Angus Smith in 1872 to describe rain around industrial Manchester.

5.4.1 The pH scale and natural acidity

Acidity is measured on the **pH scale**, which runs from 0 (very acidic) through 7 (neutral) to 14 (very alkaline). The scale is logarithmic: rain of pH 4.6 is ten times more acidic than rain of pH 5.6. Clean rain is naturally slightly acidic, with a pH of about 5.6, because carbon dioxide dissolves in water droplets to form weak carbonic acid:

$$CO_2 + H_2O \rightleftharpoons H_2CO_3$$

Rain with a pH below 5.6 is therefore usually called acid rain. Polluted regions have recorded rain with a pH between 4 and 4.5, and acid fogs below pH 3.

5.4.2 Formation of acid rain

The main precursors are **sulfur dioxide (SO_2)** and **nitrogen oxides (NO_x,** a mixture of NO and NO_2). SO_2 comes mainly from burning sulfur-containing coal and oil in power plants, refineries and industries, and from smelting metal ores. NO_x forms when nitrogen and oxygen in air combine at the high temperatures in vehicle engines, power plants and furnaces:

$$N_2 + O_2 \rightarrow 2NO$$

These primary pollutants are converted in the atmosphere into secondary pollutants, the strong acids, over hours to days, while winds carry them hundreds of kilometres. Sulfur dioxide is oxidised to sulfur trioxide, mainly with the help of hydroxyl radicals (OH) or, inside cloud droplets, hydrogen peroxide and ozone. Sulfur trioxide then dissolves in water. The overall steps are:

$$2SO_2 + O_2 \rightarrow 2SO_3$$

$$SO_3 + H_2O \rightarrow H_2SO_4$$

Nitric oxide is oxidised to nitrogen dioxide, which reacts with hydroxyl radicals or water to form nitric acid:

$$2NO + O_2 \rightarrow 2NO_2$$

$$NO_2 + OH \rightarrow HNO_3$$

$$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$$

Sulfuric and nitric acids are strong acids that release hydrogen ions (H^+) into rainwater and lower its pH. In much of Asia, sulfuric acid contributes most of the acidity, while in Europe and North America the share of nitric acid has grown as SO_2 has been controlled.

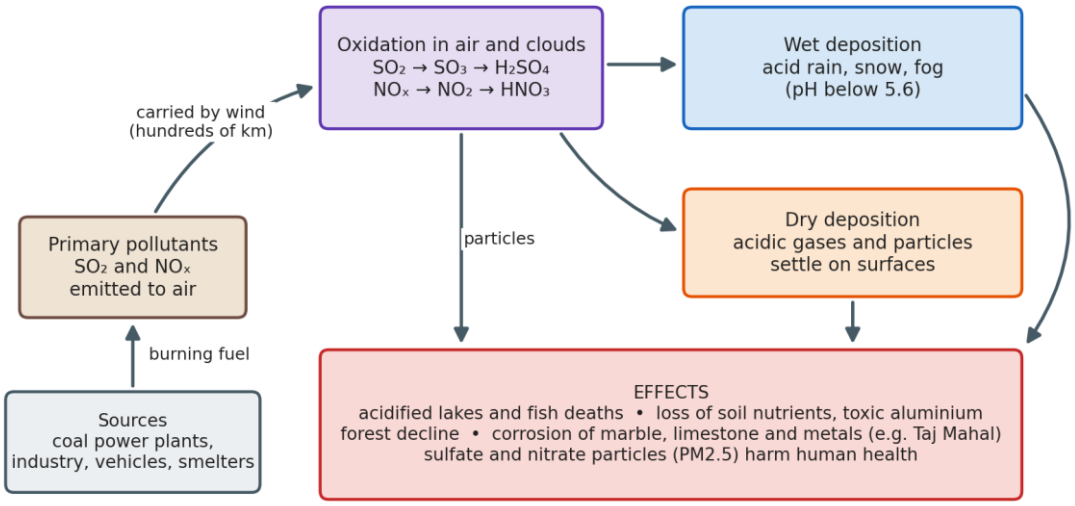


Figure 5.5: *Formation and effects of acid rain. Sulfur dioxide and nitrogen oxides from power plants, industry and vehicles are carried by winds, oxidised and dissolved in cloud water to form sulfuric and nitric acids, which return to the surface as wet or dry deposition. Source: Prepared by the author.*

5.4.3 Effects of acid rain

- **Lakes and streams.** Lakes on hard rocks such as granite, with little natural buffering capacity, become acidic. Fish eggs and young fish die below about pH 5, and acid water releases toxic aluminium from soils, which damages fish gills. Thousands of lakes in Scandinavia, Canada and the north-eastern United States lost their fish in the 1970s and 1980s.
- **Soils and forests.** Acid deposition leaches nutrients such as calcium and magnesium from soil and mobilises aluminium, which damages fine roots. Weakened trees become vulnerable to drought, frost and pests. Forest dieback in central Europe, called *Waldsterben* in German, was linked to acid deposition.
- **Buildings and monuments.** Acids dissolve limestone and marble (calcium carbonate), turning the surface into gypsum that crumbles or flakes away, a process sometimes called "stone cancer" (reaction below).
- **Human health.** Acid rain itself is not harmful to touch, but the sulfate and nitrate particles that form from SO_2 and NO_x are a large part of fine particulate matter (PM2.5), which causes heart and lung disease.

$$CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$$

Acid rain is a classic **transboundary** problem, because pollution from one country falls on another. Emissions from Britain and Germany acidified Scandinavian lakes, and those from the United States affected Canada. These disputes produced the 1979 **Convention on Long-range Transboundary Air Pollution**, the first regional air pollution treaty.

5.4.4 Acid rain in India

In most of India, rain is not strongly acidic because wind-blown dust from soils rich in calcium carbonate neutralises the acids. Rainwater pH at most monitoring stations lies near or above 5.6. There are still reasons for concern:

- India is one of the world's largest emitters of anthropogenic SO_2, mainly from coal-fired power plants.
- Lower rainwater pH has been recorded near industrial clusters and power plants, for example around Chembur in Mumbai, and in parts of the north-east, where less dust is available to neutralise rain.
- Soils in the north-east, the Western Ghats and parts of Odisha and Jharkhand are already acidic and poorly buffered, so they are more sensitive to acid deposition.

Case Study: The Taj Mahal, the Mathura refinery and the Taj Trapezium

The Taj Mahal in Agra, built of white Makrana marble in the 17th century, is India's best-known monument. When the Indian Oil Corporation planned a refinery at Mathura, about 40 km upwind, conservationists feared that its SO_2 emissions would damage the marble. The government set up an expert committee under Dr S. Varadarajan in the late 1970s, and the refinery, commissioned in 1982, was required to use pollution controls and to limit its sulfur emissions. The bigger problem turned out to be the many foundries, glass factories, brick kilns, diesel generators and vehicles around Agra, which released SO_2 and soot.

In 1984 the lawyer M.C. Mehta filed a public interest petition in the Supreme Court. In *M.C. Mehta v. Union of India* (the Taj Trapezium case, judgment of 30 December 1996), the Court applied the precautionary and polluter-pays principles. It ordered 292 coal- and coke-using industries in the **Taj Trapezium Zone (TTZ)**, an area of about 10,400 km^2 around the Taj, to switch to natural gas or relocate. The government supplied piped gas, banned new polluting industries and restricted traffic near the monument.

Later research added a twist. A 2015 study by Indian and American scientists found that the yellowing and browning of the marble is caused mainly by deposits of black carbon, brown carbon from burning biomass and refuse, and dust, rather than by acid alone. Protecting the Taj therefore requires control of all forms of air pollution in the region, not only SO_2. The Supreme Court has continued to monitor the TTZ, and in 2018 criticised the authorities over the monument's discolouration.

5.4.5 Control of acid rain

Acid rain can be controlled because its sources are known and limited.

- **Cleaner fuels.** Switching from coal to natural gas, and using low-sulfur coal and fuels, cuts SO_2 at source. India moved to Bharat Stage VI (BS-VI) vehicle fuel with 10 ppm sulfur in April 2020.
- **Flue gas desulfurisation (FGD).** "Scrubbers" spray limestone slurry into power plant exhaust and remove over 90% of SO_2, producing gypsum that can be used in cement and wallboard.
- **NO_x controls.** Low-NO_x burners, selective catalytic reduction (which converts NO_x to nitrogen and water using ammonia or urea) and catalytic converters in vehicles.
- **Energy efficiency and renewable energy**, which reduce the amount of fuel burnt.
- **Market-based tools.** The US Acid Rain Program (1990) capped SO_2 emissions from power plants and allowed trading of emission allowances. Emissions fell faster and more cheaply than expected. In Europe, SO_2 emissions have fallen by more than 80% since 1980, and many lakes are slowly recovering.

The main FGD reaction is:

$$CaCO_3 + SO_2 \rightarrow CaSO_3 + CO_2$$

India notified SO_2 emission limits for coal-fired thermal power plants in December 2015, which in practice required FGD units. Deadlines were extended several times, and by 2025 only a minority of plants had installed FGD. In July 2025 the MoEFCC revised the rules. Plants within 10 km of Delhi-NCR or cities with over a million people (Category A) must comply by the end of 2027; plants near critically polluted areas or non-attainment cities (Category B) will be decided case by case; and the remaining plants (Category C, roughly three-quarters of the total) were exempted from SO_2 limits, subject to stack-height norms. The government cited the low sulfur content of Indian coal and the cost and extra CO_2 emissions of running FGD. Critics argued that SO_2 forms sulfate particles that harm health far from the plant.

5.5 Impacts of Climate Change

Climate change acts as a **threat multiplier**: it worsens existing problems such as poverty, water scarcity, disease and habitat loss. Its impacts depend on three factors: the **hazard** (such as a heatwave or flood), **exposure** (people and assets in harm's way) and **vulnerability** (their sensitivity and lack of capacity to cope). India, with a large and mostly low-income population, heavy dependence on monsoon-fed farming, a long and densely populated coastline and the Himalayan glaciers, is among the most exposed and vulnerable countries.

5.5.1 Impacts on human communities

Heatwaves and health. The IMD declares a **heatwave** at a plains station when the maximum temperature is at least 40 °C and at least 4.5 °C above normal, or when it reaches 45 °C or more. Heatwaves are becoming longer, more frequent and more humid. March 2022 was India's warmest March since 1901, and during the long heatwave of May–June 2024 Phalodi in Rajasthan reached 50 °C. Such events cause heatstroke deaths, power cuts and crop losses. The combination of heat and humidity, measured as **wet-bulb temperature**, is especially dangerous because sweat can no longer cool the body. Outdoor workers, the elderly, infants and people living under tin roofs in slums suffer most. The **Ahmedabad Heat Action Plan** (2013), the first in South Asia, introduced early warnings, cool shelters, drinking-water points and hospital preparedness; the National Disaster Management Authority has since promoted heat action plans in heat-prone states and cities.

Disease. Warmer and wetter conditions extend the season and range of mosquitoes that carry malaria and dengue, including into hill districts that were once too cool for them. Floods spread water-borne diseases such as cholera and leptospirosis, and heat worsens air pollution episodes. Climate change also harms mental health through the loss of homes, crops and livelihoods. India's **National Programme on Climate Change and Human Health** (2019) prepares the health system for these risks.

Water security and the Himalaya. The glaciers of the Hindu Kush Himalaya feed the Indus, Ganga and Brahmaputra, on which hundreds of millions of people depend for water, food and electricity. The International Centre for Integrated Mountain Development (ICIMOD) reported in 2023 that these glaciers lost mass about 65% faster in 2011–2020 than in the previous decade, and could lose 30–50% of their volume by 2100 even if warming is limited to 1.5–2 °C, and much more under higher warming. Glacier-fed rivers may first carry more water and then less later in the century. Melting also

creates and swells **glacial lakes**, whose sudden failure causes a **glacial lake outburst flood (GLOF)**. On the night of 3–4 October 2023, South Lhonak lake in north Sikkim burst; the flood down the Teesta destroyed the Chungthang dam of the Teesta III hydropower project and killed dozens of people. In February 2021, a rock and ice avalanche in Chamoli, Uttarakhand, swept away the Rishiganga and Tapovan hydro projects, killing about 200 people.

Sea-level rise and coasts. India has a coastline of over 7,500 km, and many of its large cities, including Mumbai, Kolkata and Chennai, are on the coast. Rising seas increase flooding during storms and high tides, push salt water into rivers and aquifers, and erode beaches.

- In the **Sundarbans** delta of West Bengal, relative sea level is rising faster than the global average because the delta is also sinking as river sediment supply declines. The island of Lohachara has been submerged, and Ghoramara has lost about half its area since the 1960s. Cyclones Aila (2009), Amphan (2020) and Yaas (2021) breached embankments and flooded farmland with salt water, forcing many families to migrate to Kolkata and other cities for work.
- **Mumbai** is built largely on land reclaimed between seven former islands. On 26 July 2005, 944 mm of rain fell in 24 hours at Santacruz, and the flood killed hundreds of people. IPCC AR6 identifies Mumbai among the world's coastal megacities most exposed to sea-level rise and extreme rainfall. The city prepared a Mumbai Climate Action Plan in 2022.

Migration and inequality. The World Bank's *Groundswell* report (2021) estimated that, without strong action, climate change could force up to about 40 million people in South Asia to move within their own countries by 2050. The poorest people, who have contributed least to emissions, are hit hardest. This unequal burden is the basis of **climate justice**, the idea that responsibility for causing climate change and the capacity to respond should determine who acts and who pays.

In *M.K. Ranjitsinh v. Union of India* (2024), a case about protecting the Great Indian Bustard from power lines, the Supreme Court recognised a right against the adverse effects of climate change as part of the rights to life (Article 21) and equality (Article 14).

5.5.2 Impacts on biodiversity

Climate change is one of the five main direct drivers of biodiversity loss, alongside land- and sea-use change, direct exploitation, pollution and invasive species (IPBES Global Assessment, 2019). The same assessment estimated that about one million species face extinction, many within decades.

- **Range shifts.** Species move towards the poles or uphill to stay within their preferred temperature range. Treelines and many plant and butterfly species in the Himalaya are moving to higher altitudes. Species already living on mountaintops, such as those of the shola–grassland "sky islands" of the Western Ghats (for example the Nilgiri pipit and many endemic frogs), have nowhere higher to go.
- **Phenology.** The timing of flowering, fruiting, breeding and migration is changing. Rhododendrons in the Himalaya and some fruit trees now flower earlier; when plants and their pollinators shift at different rates, the relationship can break down.
- **Coral bleaching.** Corals live in partnership with algae (zooxanthellae) in their tissues that supply food and colour. When sea temperature stays about 1 °C above the normal summer maximum for weeks, corals expel the algae and turn white; prolonged bleaching kills them. India's reefs in the Gulf of Mannar, Gulf of Kachchh, Lakshadweep and the Andaman and Nicobar Islands bleached severely in 1998, 2010, 2016 and 2024. The 2023–2025 marine

heatwave was declared the fourth global bleaching event, affecting more than 80% of the world's reef area.

- **Coastal habitats.** Mangroves and salt marshes can be squeezed between rising seas and embankments. In the Sundarbans, home to the world's only mangrove tiger population, rising salinity is killing the freshwater-loving sundari tree (*Heritiera fomes*) after which the forest is named.
- **Fire, drought and invasive species.** Hotter, drier conditions increase forest fires, as in Uttarakhand in 2016 and 2024, and can favour invasive species such as *Lantana camara*.
- **Ocean acidification.** Lower pH threatens shell-building organisms and, through them, fisheries.

Biodiversity and climate are linked in both directions. Healthy forests, mangroves, wetlands and grasslands store large amounts of carbon and protect people from floods and storms. Conserving them is therefore both a climate solution and an adaptation strategy (see Chapter 6).

5.5.3 Impacts on the global economy

Climate change damages economies through several channels: destruction of property and infrastructure by extreme events, lost labour productivity in heat, lower crop and fishery yields, disrupted supply chains, higher health costs and losses to tourism. The *Stern Review* (2006) for the British government called climate change "the greatest market failure the world has ever seen", because polluters do not pay for the damage their emissions cause. It estimated that unchecked climate change could cost 5–20% of global output every year, while strong action would cost about 1% of GDP.

More recent estimates point in the same direction. A 2024 study in *Nature* (Kotz and colleagues) estimated that warming already locked in by past emissions will reduce average global income by about 19% by 2049, relative to a world without climate change, with the largest losses in tropical, low-income countries.

For India, heat is the largest economic threat. The International Labour Organization (2019) projected that by 2030 India would lose about 5.8% of its working hours to heat stress, equivalent to about 34 million full-time jobs, mostly in agriculture and construction. The Reserve Bank of India's *Report on Currency and Finance 2022–23* warned that up to 4.5% of India's GDP could be at risk by 2030 from lost labour hours due to extreme heat and humidity, and that climate shocks push up food inflation. Cyclone Amphan (2020) alone caused losses estimated at about US$ 13 billion in India and Bangladesh.

The shift to a low-carbon economy also creates **transition risks**: coal mines, power plants and oil reserves may become **stranded assets**. India's coal-dependent districts in Jharkhand, Chhattisgarh and Odisha will need a **just transition** that protects workers and state revenues.

5.5.4 Impacts on agriculture and food security

Agriculture is the sector most directly exposed to climate, and it employs about 45% of India's workforce.

- **Heat stress.** Wheat is very sensitive to heat while its grains fill in March. The heatwave of March 2022 shrivelled grains across Punjab, Haryana and Uttar Pradesh. The government cut its estimate of the wheat harvest from about 111 million tonnes to about 106 million tonnes, and banned wheat exports in May 2022.

- **Erratic monsoon.** Delayed onset, long dry spells and intense rain at harvest damage rice, pulses, cotton and oilseeds, especially on rainfed land.
- **Projected yields.** Simulations under the Indian Council of Agricultural Research's National Innovations in Climate Resilient Agriculture (NICRA) project that, without adaptation, rainfed rice yields could fall by about 20% by 2050 and 47% by 2080, irrigated rice by 3.5% and 5%, wheat by about 19.3% and 40%, and kharif maize by 18% and 23% respectively.
- **Water.** Higher evaporation, falling groundwater and changing river flows reduce water for irrigation.
- **Pests and diseases.** Warmer winters help pests survive. The fall armyworm, an invasive pest of maize, spread across India from 2018, and the desert locust outbreak of 2019–2020 was fuelled by unusual cyclone rains over the Arabian Peninsula.
- **Nutrition.** Crops grown under high CO_2 contain less protein, zinc and iron, which could worsen "hidden hunger".
- **Livestock and fisheries.** Heat reduces milk yield in cattle and buffalo, and warming seas are shifting fish such as the Indian oil sardine along the west coast.

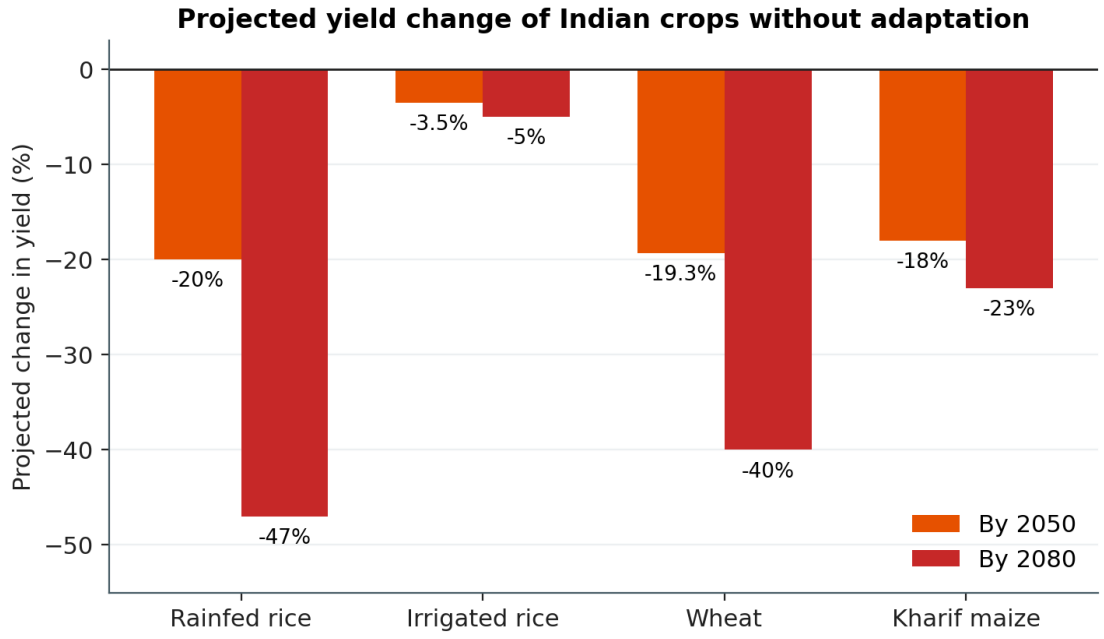


Figure 5.6: *Projected change in yields of major Indian crops without adaptation, for 2050 and 2080, relative to current yields. Source: ICAR National Innovations in Climate Resilient Agriculture (NICRA), as reported by the Government of India (2023).*

Adaptation can reduce these losses. India's responses include heat- and drought-tolerant crop varieties developed by ICAR, climate-resilient villages under NICRA, micro-irrigation under the Pradhan Mantri Krishi Sinchayee Yojana, crop insurance under the Pradhan Mantri Fasal Bima Yojana, weather-based agro-advisories, and changes in sowing dates (for example, earlier sowing of wheat to escape March heat).

Think and Do: Survey on climate impacts in your community

Design a short questionnaire (10–15 questions) on how people in your locality have experienced climate change: hotter summers, water shortages, flooding, changes in crop timing, health problems or higher electricity bills. Survey at least 20 people of different ages and occupations, including outdoor workers. Summarise the results in a chart and compare them with the scientific trends described in this chapter.

5.6 International Agreements and Programmes

Global problems need global rules. A **multilateral environmental agreement (MEA)** is a legally binding treaty among several countries on an environmental issue. Most MEAs follow a common pattern. A **convention** (or framework convention) sets out general objectives, principles and institutions. A **protocol** adds specific, often binding, obligations. An **amendment** changes a treaty's obligations. Countries that have ratified a treaty are its **Parties**, and they meet regularly at a **Conference of the Parties (COP)** to review progress and take decisions. A **secretariat** handles day-to-day work. Treaties have no world police to enforce them; they rely on national laws, reporting, peer pressure, financial help and, in a few cases, trade measures.

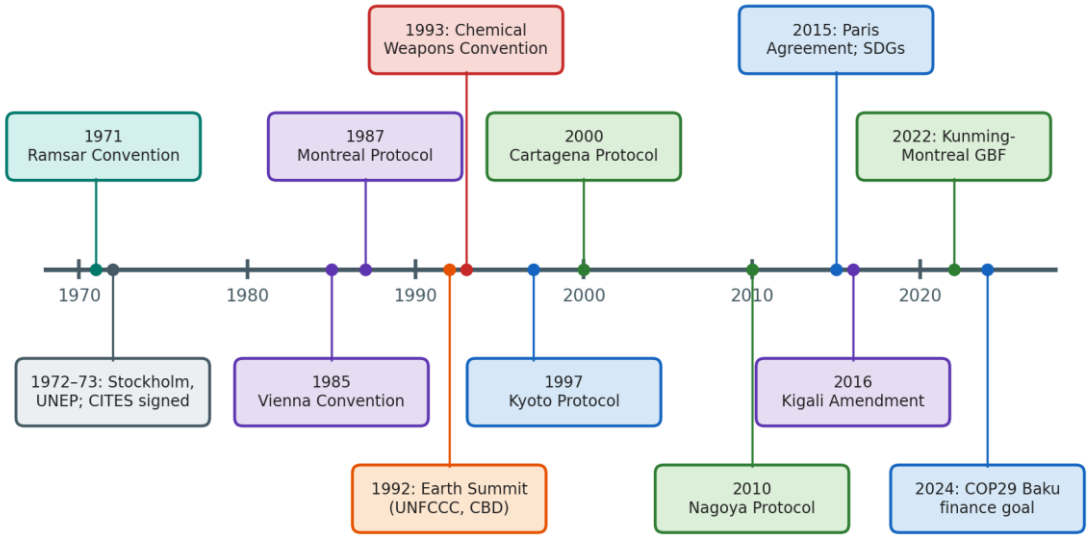


Figure 5.7: *Timeline of major international environmental conferences and agreements, 1971–2024. Source: Prepared by the author from treaty secretariat records.*

5.6.1 Stockholm 1972: the starting point

The **United Nations Conference on the Human Environment**, held in Stockholm, Sweden, from 5 to 16 June 1972, was the first world conference to make the environment a major international issue. Delegates from 113 countries adopted the **Stockholm Declaration** of 26 principles and an Action Plan, and recommended the creation of the **United Nations Environment Programme (UNEP)**. The opening day, 5 June, is celebrated every year as **World Environment Day**.

India's Prime Minister Indira Gandhi was one of the very few heads of government to attend. In her speech she asked, "Are not poverty and need the greatest polluters?", arguing that environmental protection must go together with development for the world's poor. This tension between environment and development, and between rich and poor countries, has shaped every negotiation since. Stockholm also led directly to Indian law: the Water Act 1974, the Air Act 1981 and the Environment (Protection) Act 1986 were all enacted partly to implement its decisions, and the 42nd Constitutional Amendment (1976) added environmental protection to the Directive Principles and Fundamental Duties.

5.7 The Earth Summit, 1992

The **United Nations Conference on Environment and Development (UNCED)**, popularly called the **Earth Summit**, was held in Rio de Janeiro, Brazil, from 3 to 14 June 1992, twenty years after Stockholm. It brought together 172 governments, 108 of them represented by heads of state or government, along with thousands of non-governmental organisations at a parallel Global Forum. It built on the 1987 Brundtland Report, *Our Common Future*, which had defined **sustainable development** as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (see Chapter 1).

The Earth Summit produced five major outcomes.

1. **The Rio Declaration on Environment and Development**, a set of 27 non-binding principles. Among the most influential are:
   - Principle 7, **common but differentiated responsibilities**: all states share responsibility for the global environment, but developed countries bear a greater responsibility because of the pressure their societies place on it and the resources they command;
   - Principle 10, public participation and access to information and justice;
   - Principle 15, the **precautionary approach**: where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures;
   - Principle 16, the **polluter pays principle**: the polluter should bear the cost of pollution;
   - Principle 17, environmental impact assessment of proposed activities.
2. **Agenda 21**, a non-binding action plan of 40 chapters covering poverty, consumption, health, human settlements, forests, oceans, waste, the role of women, youth, Indigenous peoples and local governments. It inspired "Local Agenda 21" plans in thousands of towns.
3. **The Forest Principles**, a non-binding statement on the management of all types of forests.
4. Two legally binding conventions opened for signature: the **United Nations Framework Convention on Climate Change (UNFCCC)** and the **Convention on Biological Diversity (CBD)**.
5. Agreement to negotiate a third convention, the **United Nations Convention to Combat Desertification (UNCCD)**, adopted in Paris in 1994.

The UNFCCC, CBD and UNCCD are together called the **Rio Conventions**. India is a Party to all three, and hosted UNCCD COP14 at Greater Noida in 2019, where it pledged to restore 26 million hectares of degraded land by 2030.

5.8 The United Nations Framework Convention on Climate Change (UNFCCC)

The UNFCCC was adopted in New York on 9 May 1992, opened for signature at the Earth Summit in June 1992 and entered into force on 21 March 1994. With 198 Parties, it has near-universal membership. India ratified it in November 1993. Its secretariat is in Bonn, Germany.

5.8.1 Objective and principles

Article 2 states the **ultimate objective**: to stabilise greenhouse gas concentrations "at a level that would prevent dangerous anthropogenic interference with the climate system", within a time-frame that allows ecosystems to adapt naturally, ensures food production is not threatened and allows economic development to proceed sustainably.

Article 3 sets out the principles that guide the Parties:

- **Equity and common but differentiated responsibilities and respective capabilities (CBDR-RC).** All countries must act, but developed countries "should take the lead", because they caused most past emissions and have greater financial and technological capacity.
- **Special needs** of developing countries, especially those most vulnerable to climate change, such as small island states and least developed countries.
- **Precaution**: lack of full scientific certainty should not be used to postpone action.
- The **right to sustainable development** and the need for economic growth in developing countries.
- A **supportive and open international economic system**, so that climate measures do not become disguised trade barriers.

5.8.2 Groups of countries

The Convention divides countries into groups with different obligations.

- **Annex I Parties**: industrialised countries that were members of the OECD in 1992, plus "economies in transition" such as Russia and the countries of eastern Europe. They were expected to return their emissions to 1990 levels.
- **Annex II Parties**: the OECD members of Annex I. They must provide finance and technology to developing countries.
- **Non-Annex I Parties**: mostly developing countries, including India and China. They report on their emissions and actions but had no binding emission limits under the Convention.

This "firewall" between developed and developing countries was central to the Kyoto Protocol but was replaced by a more flexible approach in the Paris Agreement.

5.8.3 Institutions

The **Conference of the Parties (COP)** is the supreme decision-making body and meets every year, usually in November or December, rotating among the five UN regions. Two subsidiary bodies, the Subsidiary Body for Scientific and Technological Advice (SBSTA) and the Subsidiary Body for Implementation (SBI), meet twice a year. The **Intergovernmental Panel on Climate Change (IPCC)**, set up by WMO and UNEP in 1988, is not part of the UNFCCC but supplies the scientific assessments

on which negotiations rely; it shared the 2007 Nobel Peace Prize. The **Green Climate Fund** (2010) and the **Global Environment Facility** channel finance to developing countries.

India hosted COP8 in New Delhi in 2002, whose **Delhi Declaration** stressed that adaptation and sustainable development deserve as much attention as cutting emissions. India has offered to host COP33 in 2028.

Table 5.3: Milestone Conferences of the Parties to the UNFCCC

COP	Year and place	Main outcome
COP1	1995, Berlin	Berlin Mandate to negotiate binding targets for developed countries
COP3	1997, Kyoto	Kyoto Protocol adopted
COP8	2002, New Delhi	Delhi Declaration on climate change and sustainable development
COP13	2007, Bali	Bali Action Plan for long-term cooperative action
COP15	2009, Copenhagen	Copenhagen Accord; pledge of US$ 100 billion a year by 2020
COP16	2010, Cancún	Green Climate Fund established; 2 °C goal recognised
COP19	2013, Warsaw	Warsaw International Mechanism for Loss and Damage
COP21	2015, Paris	Paris Agreement adopted
COP26	2021, Glasgow	Glasgow Climate Pact; coal "phase-down"; India's Panchamrit
COP27	2022, Sharm el-Sheikh	Decision to set up a loss and damage fund
COP28	2023, Dubai	First global stocktake; "transitioning away from fossil fuels"
COP29	2024, Baku	New climate finance goal of US$ 300 billion a year by 2035
COP30	2025, Belém	"Global Mutirão" package; tripling of adaptation finance by 2035

5.9 The Kyoto Protocol

The **Kyoto Protocol** was adopted at COP3 in Kyoto, Japan, on 11 December 1997 and entered into force on 16 February 2005, after Russia's ratification pushed it past the required threshold. It was the first treaty to set **legally binding emission targets**. India ratified it in August 2002.

5.9.1 Targets and Annex B

The Protocol covered a "basket" of six greenhouse gases: CO_2, CH_4, N_2O, HFCs, PFCs and SF_6 (nitrogen trifluoride was added later). Developed countries and economies in transition, listed in its **Annex B**, agreed to reduce their combined emissions by at least 5% below 1990 levels during the **first commitment period, 2008–2012**. Individual targets differed: the European Union accepted −8%, the United States −7% and Japan −6%, while Russia was allowed 0% and Australia +8%. Developing countries, including India and China, had no binding targets, in line with CBDR-RC.

The **Doha Amendment** (2012) set a **second commitment period, 2013–2020**, in which participating developed countries aimed to cut emissions by at least 18% below 1990 levels. It came into force only on 31 December 2020, the last day of the period. India ratified it in 2017.

5.9.2 Flexibility mechanisms

To lower the cost of meeting targets, the Protocol created three market-based **flexibility mechanisms**.

- **International Emissions Trading (ET).** Annex B countries could buy and sell parts of their allowed emissions ("assigned amount units") among themselves. The European Union Emissions Trading System, launched in 2005, grew out of this idea.
- **Joint Implementation (JI).** An Annex B country could earn credits by funding an emission-reduction project in another Annex B country, often in eastern Europe.
- **Clean Development Mechanism (CDM).** An Annex B country or company could fund an emission-reducing project in a developing country and receive **certified emission reductions (CERs)**, each equal to one tonne of CO_2e, which it could count towards its own target. The CDM was meant to help developing countries achieve sustainable development while lowering the cost of compliance for developed countries.

5.9.3 India and the CDM

India was among the most active CDM host countries. The MoEFCC set up a **National CDM Authority** in 2003 to approve projects and certify that they contributed to sustainable development. Indian projects included wind farms, small hydropower, biomass power from bagasse and rice husk, waste-heat recovery in cement and steel plants, and energy-efficient lighting. By the time the CDM wound down, India had registered roughly one-fifth of all CDM projects in the world, second only to China. The CDM brought investment and experience in carbon accounting to Indian industry, but it was also criticised: some projects would have happened anyway (they lacked **additionality**), and CER prices collapsed after 2012 when demand from Europe fell.

5.9.4 Assessment

Kyoto had mixed results. Countries that stayed in the Protocol as a group met their first-period target, partly because of the economic collapse of eastern Europe after 1990. But the United States, then the largest emitter, signed and never ratified it, Canada withdrew in 2012, and the Protocol did not limit the rapidly growing emissions of large developing economies. Global emissions rose by about 50% between 1990 and 2012. These weaknesses led to the different design of the Paris Agreement. The CDM itself is being replaced by a new mechanism under Article 6.4 of the Paris Agreement, and eligible CDM projects can transfer to it. India's own **Carbon Credit Trading Scheme** (2023), set up under the Energy Conservation (Amendment) Act 2022, builds on this experience.

5.10 The Paris Agreement and Recent Climate Conferences

The **Paris Agreement** was adopted at COP21 on 12 December 2015 and entered into force on 4 November 2016. India ratified it on 2 October 2016, Gandhi Jayanti. Almost every country is a Party. The United States joined, withdrew (effective 2020), rejoined in 2021 and withdrew again with effect from January 2026.

5.10.1 Main features

- **Temperature goal (Article 2).** Hold the increase in global average temperature to "well below 2 °C" above pre-industrial levels and pursue efforts to limit it to 1.5 °C. Parties aim for global

emissions to peak as soon as possible and to reach a balance between emissions and removals (net zero) in the second half of the century.

- **Nationally Determined Contributions (NDCs) (Article 4).** Every Party, developed or developing, must prepare and communicate its own climate action plan, an NDC, every five years. Each new NDC should be more ambitious than the last; this is the **ratchet mechanism**. The targets themselves are not legally binding, but the obligations to prepare, report and update them are.
- **Differentiation.** Developed countries should continue to take the lead with economy-wide absolute targets, while developing countries are encouraged to move towards them over time. CBDR-RC is retained, "in the light of different national circumstances".
- **Carbon markets (Article 6).** Countries may cooperate by transferring emission reductions (Article 6.2) or through a UN-supervised crediting mechanism (Article 6.4).
- **Adaptation (Article 7)** with a **Global Goal on Adaptation**, and recognition of **loss and damage (Article 8)**.
- **Finance (Article 9).** Developed countries shall provide financial resources to developing countries.
- **Enhanced Transparency Framework (Article 13).** All Parties report their emissions and progress using common rules; the first Biennial Transparency Reports were due at the end of 2024.
- **Global stocktake (Article 14).** Every five years, starting in 2023, Parties assess collective progress towards the goals.

At COP21, India and France also launched the **International Solar Alliance**, now headquartered in Gurugram, to promote solar energy in sunny countries.

5.10.2 From Glasgow to Belém

- **COP26, Glasgow (2021).** The Glasgow Climate Pact called for the "phase-down" of unabated coal power, wording India and China pressed for in place of "phase-out". It finalised the rulebook for Article 6 trading. India announced its **Panchamrit** ("five nectar elements") goals and a target of **net-zero emissions by 2070**.
- **COP27, Sharm el-Sheikh (2022).** After decades of demands from vulnerable countries, Parties agreed to create a fund for **loss and damage**, meaning the unavoidable harm from climate change that adaptation cannot prevent.
- **COP28, Dubai (2023).** The Fund for responding to Loss and Damage was made operational on the first day. The **first global stocktake** found the world off track and called on Parties to triple global renewable energy capacity and double the rate of energy-efficiency improvement by 2030, and to contribute to "transitioning away from fossil fuels in energy systems, in a just, orderly and equitable manner". This was the first time a COP decision named fossil fuels.
- **COP29, Baku (2024).** Called the "finance COP", it agreed a **New Collective Quantified Goal (NCQG)** on climate finance: developed countries are to take the lead in mobilising at least **US$ 300 billion a year by 2035** for developing countries, from a wide range of public and private sources, and all actors are called on to scale up finance to US$ 1.3 trillion a year by 2035. This replaced the earlier US$ 100 billion a year goal, which the OECD reported was first met in 2022. Rules for the Article 6.4 carbon market were also completed. India strongly objected to the

finance decision, calling the amount "abysmally small" and the adoption "stage-managed", a view shared by many developing countries.

- **COP30, Belém (2025).** Held at the edge of the Amazon, ten years after Paris. The final "Global Mutirão" decision (from an Indigenous Tupi-Guarani word for collective effort) called for **adaptation finance to be tripled by 2035**, adopted a set of indicators to track the Global Goal on Adaptation and launched a mechanism for a just transition. It did not include a roadmap away from fossil fuels; the Brazilian presidency instead announced voluntary roadmaps on fossil fuels and deforestation outside the formal text. Brazil also launched the **Tropical Forest Forever Facility** to pay countries for keeping forests standing.

The next conference, COP31, is scheduled to be held in Antalya, Türkiye, in November 2026. In July 2025, the International Court of Justice gave an advisory opinion that states have legal obligations under international law to protect the climate system from greenhouse gas emissions, which may strengthen climate litigation worldwide.

Despite these steps, UNEP's annual *Emissions Gap Report* shows that current NDCs put the world on course for roughly 2.5–3 °C of warming by 2100, well above the Paris goals. Closing this **ambition gap** and the finance gap remains the central challenge.

Think and Do: Comparing Nationally Determined Contributions

Download the latest NDCs of India and of one developed country (for example the European Union, the United Kingdom or Japan) from the UNFCCC NDC Registry. Compare them in a table: type of target (absolute or intensity), base year, target year, sectors covered, adaptation content, and whether any part is conditional on international finance. Calculate per-capita emissions for both using World Bank or Our World in Data figures. Discuss whether each NDC is fair and ambitious given the principle of CBDR-RC.

Think and Do: Model UNFCCC negotiation

Divide the class into country groups: India, China, the United States, the European Union, a small island state (such as Tuvalu), a least developed country (such as Bangladesh) and an oil exporter (such as Saudi Arabia). Each group prepares a two-minute opening statement on a negotiating text about climate finance and fossil fuel transition. Negotiate for one class period and try to agree a one-page decision by consensus. Afterwards, reflect on why consensus is so difficult.

5.11 The Montreal Protocol

The science and results of ozone protection are described earlier in this chapter; this section explains how the treaty works and why it succeeded.

The **Vienna Convention for the Protection of the Ozone Layer** was adopted on 22 March 1985 as a framework treaty, before the ozone hole was even announced. The **Montreal Protocol on Substances that Deplete the Ozone Layer** was adopted on 16 September 1987, now marked as the International Day for the Preservation of the Ozone Layer, and entered into force on 1 January 1989.

In 2009 the Vienna Convention and the Montreal Protocol became the first treaties in UN history to be ratified by every UN member state; they now have 198 Parties. India acceded in 1992.

Key features of the Protocol are listed below.

- **Phase-out schedules** for about 100 chemicals, with deadlines for developed countries and a grace period of about ten years for developing countries, called **Article 5 countries** after the article that defines them.
- **The Multilateral Fund** (set up by the London Amendment, 1990, operating since 1991), financed by developed countries, pays the agreed extra costs of developing countries in switching to alternatives. It has provided several billion US dollars, including substantial support to India's conversion projects.
- **Adjustments and amendments.** The Protocol was strengthened repeatedly as science advanced: London (1990), Copenhagen (1992), Montreal (1997), Beijing (1999) and Kigali (2016). "Adjustments" to phase-out dates for already controlled substances apply automatically to all Parties.
- **Scientific, environmental and technology assessment panels** review the evidence every four years.
- **Trade measures.** Parties may not trade controlled substances with non-Parties, which gave every country an incentive to join.
- **A non-compliance procedure** that helps rather than punishes countries that fall behind.

The Protocol succeeded because the science was clear and visible (the ozone hole), a few companies made the chemicals, substitutes became available, developing countries received finance and time, and the treaty could be tightened quickly. Climate change is harder because CO_2 comes from almost every activity in every economy.

5.12 The Convention on Biological Diversity

The **Convention on Biological Diversity (CBD)** was opened for signature at the Earth Summit in June 1992 and entered into force on 29 December 1993. It has 196 Parties; the United States has signed but not ratified it. Its secretariat is in Montreal, Canada. India ratified the CBD in 1994 and hosted COP11 in Hyderabad in 2012.

5.12.1 Objectives and principles

Article 1 sets three objectives:

1. the **conservation** of biological diversity;
2. the **sustainable use** of its components;
3. the **fair and equitable sharing of benefits** arising from the use of genetic resources, known as **access and benefit-sharing (ABS)**.

The CBD made several important shifts. It recognised that states have **sovereign rights** over their biological resources, ending the older idea that wild genetic resources were a free "common heritage of mankind". It treated conservation of biodiversity as "a common concern of humankind". It covered all levels of biodiversity (genes, species and ecosystems) and both **in-situ** conservation (in natural habitats, such as protected areas) and **ex-situ** conservation (outside natural habitats, such as seed banks and zoos). Article 8(j) asks Parties to respect and protect the knowledge, innovations and

practices of Indigenous and local communities. Each Party must prepare a **National Biodiversity Strategy and Action Plan (NBSAP)**.

The ABS objective was a response to **biopiracy**, the use of biological resources or traditional knowledge from developing countries for commercial products without permission or payment. India successfully challenged a US patent on turmeric for wound healing (revoked in 1997) and a European patent on a neem-based fungicide (revoked in 2000). To prevent such cases, the Council of Scientific and Industrial Research built the **Traditional Knowledge Digital Library (TKDL)**, which documents traditional medicinal knowledge in a form patent examiners can search.

5.12.2 The Cartagena Protocol on Biosafety

The **Cartagena Protocol on Biosafety** was adopted on 29 January 2000 (negotiations begun in Cartagena, Colombia, were concluded in Montreal) and entered into force on 11 September 2003. It regulates the movement across borders of **living modified organisms (LMOs)**, organisms with a new combination of genetic material obtained through modern biotechnology, such as genetically modified seeds.

- An **Advance Informed Agreement (AIA)** procedure requires the exporter to notify the importing country and obtain its consent before the first shipment of an LMO intended for release into the environment, such as seeds for planting. The importer must decide within 270 days, based on a risk assessment.
- LMOs intended for direct use as **food, feed or processing**, such as GM soybean grain, follow a simpler procedure.
- The Protocol explicitly allows the **precautionary approach**: lack of scientific certainty does not prevent an importing country from refusing an LMO.
- Shipments must be labelled and documented, and countries share decisions through the **Biosafety Clearing-House**.

India ratified the Cartagena Protocol in 2003. In India, the **Genetic Engineering Appraisal Committee (GEAC)** under the MoEFCC, functioning under the Rules of 1989 framed under the Environment (Protection) Act, approves the release of GM organisms. Bt cotton, approved in 2002, remains the only GM crop approved for commercial cultivation in India; the environmental release of GM mustard (DMH-11), approved by the GEAC in 2022, has been the subject of litigation in the Supreme Court.

5.12.3 The Nagoya Protocol on Access and Benefit-Sharing

The **Nagoya Protocol** was adopted in Nagoya, Japan, on 29 October 2010 and entered into force on 12 October 2014. It gives legal shape to the CBD's third objective.

- **Prior informed consent (PIC).** A person or company wishing to use genetic resources or associated traditional knowledge must obtain permission from the provider country and, where relevant, from Indigenous and local communities.
- **Mutually agreed terms (MAT).** The user and provider agree how benefits will be shared. Benefits can be monetary (fees, royalties) or non-monetary (sharing research results, training, technology transfer).
- **User compliance.** Countries where resources are used must check that they were obtained legally, for example through "checkpoints" at patent offices or research-funding agencies.

- An **ABS Clearing-House** publishes national rules and internationally recognised certificates of compliance.

India ratified the Nagoya Protocol in 2012, and its Biological Diversity Act already contained ABS rules. A frequently cited Indian example of benefit-sharing came before both, in the 1990s: scientists of the Tropical Botanic Garden and Research Institute, Thiruvananthapuram, developed a health tonic, Jeevani, from the plant arogyapacha (*Trichopus zeylanicus*) using knowledge shared by the Kani tribe of Kerala, and half of the licence fee and royalties went to a trust fund for the Kani community.

A new challenge is **digital sequence information (DSI)**, genetic data stored in online databases that can be used without touching a physical sample. In 2024, CBD COP16 in Cali, Colombia, set up the **Cali Fund**, to which large companies that profit from DSI are expected to contribute.

5.12.4 The Kunming-Montreal Global Biodiversity Framework

The CBD's earlier targets were not met: none of the 20 **Aichi Biodiversity Targets** for 2011–2020 was fully achieved globally. In December 2022, COP15 (held in Montreal under China's presidency) adopted the **Kunming-Montreal Global Biodiversity Framework (GBF)**, with 4 long-term goals for 2050 and 23 targets for 2030. Important targets include:

- **30×30**: effectively conserve at least 30% of the world's land, inland waters, coasts and oceans by 2030, through protected areas and "other effective area-based conservation measures", while respecting the rights of Indigenous peoples and local communities (Target 3);
- restore at least 30% of degraded ecosystems (Target 2);
- halve the rate of introduction of invasive alien species (Target 6);
- halve excess nutrients lost to the environment and the overall risk from pesticides and highly hazardous chemicals (Target 7);
- reduce subsidies harmful to biodiversity by at least US$ 500 billion a year (Target 18);
- mobilise at least US$ 200 billion a year for biodiversity from all sources (Target 19).

India submitted an updated NBSAP with 23 national targets aligned with the GBF at COP16 in 2024.

5.12.5 India's Biological Diversity Act 2002 and its 2023 amendment

India implements the CBD through the **Biological Diversity Act, 2002**, and the Biological Diversity Rules. The Act sets up a three-tier structure:

- the **National Biodiversity Authority (NBA)**, Chennai, which regulates access to biological resources by foreigners and foreign companies and approves applications for patents based on Indian biological resources;
- **State Biodiversity Boards (SBBs)**, which regulate commercial use by Indian entities;
- **Biodiversity Management Committees (BMCs)** in local bodies, which prepare **People's Biodiversity Registers (PBRs)** recording local species and traditional knowledge, and may levy collection fees.

Benefits from commercial use, including a share of sales, are paid into National, State or Local Biodiversity Funds for conservation and for the communities concerned. The Act also allows states to notify **Biodiversity Heritage Sites**, such as Nallur Tamarind Grove in Karnataka, the first, notified in 2007.

The **Biological Diversity (Amendment) Act, 2023** made major changes. It replaced imprisonment with monetary penalties for most offences, exempted registered practitioners of Indian systems of medicine (AYUSH) and cultivated medicinal plants from certain approval requirements, eased procedures for Indian companies, including some with foreign investment, and strengthened the role of the NBA. The government argued that this would promote research, investment and the AYUSH sector. Critics warned that the exemptions could allow companies to escape benefit-sharing, that it is hard to tell cultivated from wild-collected plants, and that the amendment shifts the law's focus from conservation to commerce while weakening the voice of local communities.

5.13 The Ramsar Convention on Wetlands

The **Convention on Wetlands of International Importance especially as Waterfowl Habitat** was signed at Ramsar, a town on the Caspian Sea coast of Iran, on 2 February 1971, and entered into force in 1975. It is the oldest of the modern global environmental treaties and the only one devoted to a single type of ecosystem. 2 February is celebrated as **World Wetlands Day**. The Convention has 172 Parties, and its secretariat shares offices with IUCN in Gland, Switzerland.

The Convention defines **wetlands** very broadly as "areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary, with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide does not exceed six metres". This includes lakes, rivers, swamps, mangroves, coral reefs, rice paddies and reservoirs.

5.13.1 The three pillars

Parties commit to three "pillars":

1. **Wise use** of all wetlands in their territory, defined as the maintenance of their ecological character within the context of sustainable development. The obligation is in the 1971 text; guidance on applying it was adopted from 1987 onwards.
2. Designating suitable wetlands for the **List of Wetlands of International Importance** (the Ramsar List) and ensuring their effective management.
3. **International cooperation** on transboundary wetlands, shared wetland systems and shared species such as migratory waterbirds.

5.13.2 Criteria for Ramsar sites

A wetland qualifies as a Ramsar site if it meets at least one of nine criteria in two groups.

- **Group A, representative, rare or unique wetland types**: Criterion 1, a wetland that contains a representative, rare or unique example of a natural or near-natural wetland type in its region.
- **Group B, importance for conserving biodiversity**:
  - Criterion 2: supports vulnerable, endangered or critically endangered species or threatened ecological communities;
  - Criterion 3: supports populations important for maintaining the biodiversity of a region;
  - Criterion 4: supports species at a critical stage of their life cycle (such as breeding or migration stopovers) or provides refuge in adverse conditions;
  - Criterion 5: regularly supports 20,000 or more waterbirds;

- Criterion 6: regularly supports 1% of the individuals in a population of one species or subspecies of waterbird;
- Criterion 7: supports a significant proportion of indigenous fish species or families;
- Criterion 8: is an important source of food, spawning ground, nursery or migration path for fish;
- Criterion 9: regularly supports 1% of the individuals in a population of a non-avian, wetland-dependent animal species.

Worldwide there are over 2,500 Ramsar sites covering more than 2.5 million km^2. Designation does not itself create a protected area, but the country commits to maintaining the site's ecological character and to reporting changes.

5.13.3 The Montreux Record

The **Montreux Record**, created in 1990, is a register of Ramsar sites where changes in ecological character have occurred, are occurring or are likely to occur because of human interference, and which need priority conservation attention. It is maintained with the consent of the country concerned and is meant to attract help rather than to shame. India currently has two sites on the Record: **Keoladeo National Park** in Rajasthan (listed in 1990, because of water shortage and the spread of invasive grass) and **Loktak Lake** in Manipur (listed in 1993, because of the Ithai barrage, siltation, pollution and spread of floating vegetation called *phumdis*). **Chilika Lake** in Odisha was placed on the Record in 1993 because its mouth to the sea had silted up, reducing salinity and fish catch. After a new mouth was opened in 2000 and fisheries and birds recovered, Chilika was removed in 2002, becoming the first Asian site taken off the Record; it later won the Ramsar Wetland Conservation Award.

5.13.4 India's Ramsar sites

India acceded to the Convention on 1 February 1982. Its first two Ramsar sites, Chilika Lake and Keoladeo National Park, were designated in 1981. The number of sites remained small for decades but has grown rapidly since 2019. As of August 2026 India had **101 Ramsar sites**; the 100th was the Jai Prakash Narayan Bird Sanctuary (Surha Tal) in Uttar Pradesh (June 2026) and the 101st was Glaw Lake in Arunachal Pradesh (August 2026). Tamil Nadu has the most sites (20), followed by Uttar Pradesh. The largest Indian Ramsar site is the Sundarban Wetland in West Bengal; others include Wular Lake and Tso Moriri in Jammu and Kashmir and Ladakh, Vembanad-Kol in Kerala, Bhitarkanika mangroves in Odisha, Sultanpur National Park in Haryana and Point Calimere in Tamil Nadu. Because the count changes every few months, check the Ramsar Sites Information Service or the MoEFCC website for the latest figure.

Within India, wetlands are governed by the **Wetlands (Conservation and Management) Rules, 2017** under the Environment (Protection) Act, which set up State Wetlands Authorities and prohibit activities such as reclamation, encroachment and dumping of waste in notified wetlands. The **National Plan for Conservation of Aquatic Ecosystems** funds restoration, and the **Amrit Dharohar** scheme (2023) promotes eco-tourism and community stewardship at Ramsar sites. Yet Indian wetlands remain under heavy pressure from urban encroachment, sewage, invasive weeds and changes in water flow: Bengaluru has lost most of its historical lakes, and East Kolkata Wetlands, a Ramsar site that treats the city's sewage naturally, faces constant encroachment.

5.14 The Chemical Weapons Convention

The **Convention on the Prohibition of the Development, Production, Stockpiling and Use of Chemical Weapons and on their Destruction**, known as the **Chemical Weapons Convention (CWC)**, is a disarmament treaty with direct environmental consequences. It was opened for signature in Paris in January 1993 and entered into force on 29 April 1997. With 193 States Parties, it covers about 98% of the world's population.

5.14.1 Background and obligations

Chemical weapons such as chlorine, phosgene and mustard gas caused about 1.3 million casualties in the First World War. The 1925 **Geneva Protocol** banned their use in war but not their production or stockpiling.

Under the CWC, each State Party undertakes never, under any circumstances, to develop, produce, acquire, stockpile, retain, transfer or use chemical weapons, and to destroy any chemical weapons and production facilities it owns. Toxic chemicals are controlled by a **general purpose criterion**: any toxic chemical is banned for weapons purposes, while peaceful uses in industry, agriculture, medicine and research are allowed. Chemicals are listed in three **schedules** by risk: Schedule 1 includes nerve agents such as sarin and VX and blister agents such as sulfur mustard, which have almost no peaceful uses; Schedules 2 and 3 include precursors and dual-use chemicals such as phosgene, which also has industrial uses.

5.14.2 The OPCW

The **Organisation for the Prohibition of Chemical Weapons (OPCW)**, headquartered in The Hague, Netherlands, implements the Convention. It receives declarations from States Parties, inspects military and industrial sites, verifies the destruction of stockpiles and can carry out challenge inspections at short notice. It received the **Nobel Peace Prize in 2013**. In July 2023, the OPCW verified that all chemical weapons stockpiles declared by States Parties, about 72,000 tonnes of agent, had been destroyed, after the United States completed destruction of its last declared munitions. Challenges remain: chemical weapons were used in Syria, and the nerve agent Novichok was used in attacks on individuals in the United Kingdom (2018) and Russia (2020).

5.14.3 India and the CWC

India signed the CWC on 14 January 1993, during the opening signing ceremony in Paris, and ratified it in 1996. In 1997 it declared a stockpile of about 1,044 tonnes of sulfur mustard, revealing for the first time that it possessed chemical weapons. India passed the **Chemical Weapons Convention Act, 2000**, and set up the **National Authority, Chemical Weapons Convention** under the Cabinet Secretariat to manage declarations and inspections of Indian chemical industries. India completed the destruction of its entire stockpile under OPCW verification in March 2009, becoming the third State Party (after Albania and South Korea) to do so. As a major chemical producer, India receives regular OPCW inspections of its industries.

5.14.4 Why the CWC is an environmental treaty

The CWC is included in environmental studies for several reasons.

- **Safe destruction.** Destroying stockpiles by incineration or chemical neutralisation must not release toxic gases or contaminate soil and water. The Convention requires States Parties to give the highest priority to the safety of people and the protection of the environment during destruction.
- **Old and abandoned weapons.** Tens of thousands of tonnes of chemical munitions dumped at sea after the World Wars, for example in the Baltic Sea, are corroding and leaking toxic compounds into marine sediments and fisheries.
- **Industrial safety.** Monitoring of dual-use chemicals strengthens the safe management of hazardous chemicals, a lesson India learned painfully from the Bhopal gas tragedy of 1984, which involved methyl isocyanate made from phosgene.

5.15 CITES: Regulating Trade in Endangered Species

The **Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)** was signed in Washington, DC, on 3 March 1973 (now **World Wildlife Day**) and entered into force on 1 July 1975. It has about 185 Parties, and its secretariat, administered by UNEP, is in Geneva. CITES does not deal with habitat protection or domestic trade; its aim is to ensure that **international trade** in specimens of wild animals and plants does not threaten their survival. It covers live animals and plants and products made from them, such as ivory, skins, timber, medicines and pet reptiles.

5.15.1 The appendices

Over 40,000 species are listed in three appendices, according to the degree of protection they need.

Table 5.4: The three CITES appendices

Appendix	Which species	Trade rules	Examples relevant to India
I	Threatened with extinction	Commercial trade banned; non-commercial trade needs export and import permits	Tiger, Asian elephant, snow leopard, one-horned rhinoceros, Indian pangolin, Indian star tortoise
II	Not yet threatened, but could become so without trade control; also look-alike species	Trade allowed with an export permit if it is legal and "non-detrimental" to the species' survival	Red sanders, many corals, seahorses, several sharks, many orchids
III	Protected in at least one country that asks others for help	Export permit from the listing country; certificate of origin from others	Species listed by India, such as the Bengal fox, golden jackal and several mongooses

Species are added, removed or moved between Appendices I and II by a two-thirds majority at the **Conference of the Parties (CoP)**, held about every three years. CoP20 was held in Samarkand, Uzbekistan, in late 2025. Each Party must designate a **Management Authority** to issue permits and a **Scientific Authority** to advise whether trade is non-detrimental.

5.15.2 India and CITES

India became a Party in 1976 and hosted CoP3 in New Delhi in 1981. Domestically, CITES is implemented through the **Wildlife (Protection) Act 1972**, which since the 2022 amendment

contains a dedicated chapter and Schedule IV listing CITES species, and through the Foreign Trade Policy and Customs Act. The Director of Wildlife Preservation is the Management Authority, and the Zoological Survey of India, Botanical Survey of India and other institutes serve as Scientific Authorities. The **Wildlife Crime Control Bureau (WCCB)** works with customs to stop smuggling.

India has generally supported strict protection. It successfully proposed moving the Indian star tortoise and the smooth-coated and small-clawed otters to Appendix I at CoP18 (2019), both because of heavy illegal trade for pets. India is also a source and transit country for illegal trade in tiger parts, pangolin scales, red sanders timber, star tortoises, and birds such as parakeets. Ivory trade has been banned in India since 1991, and India has supported keeping the African elephant in Appendix I.

5.16 The United Nations Environment Programme (UNEP)

The **United Nations Environment Programme (UNEP)**, also known as UN Environment, was established by UN General Assembly Resolution 2997 in December 1972, following the Stockholm Conference. Its headquarters are in **Nairobi, Kenya**, making it the first UN agency headquartered in the Global South. Its mission is "to provide leadership and encourage partnership in caring for the environment by inspiring, informing, and enabling nations and peoples to improve their quality of life without compromising that of future generations."

5.16.1 Governance and mandate

Since 2014, UNEP's governing body has been the **United Nations Environment Assembly (UNEA)**, in which all 193 UN member states are represented. It meets every two years in Nairobi and is often called the "world's parliament on the environment". UNEP is a programme rather than a specialised agency; it is funded mostly by voluntary contributions and works mainly by assessing, advising and convening rather than by running large projects. It frames its work around the **triple planetary crisis** of climate change, biodiversity and nature loss, and pollution and waste.

5.16.2 Major functions and programmes

- **Assessment and early warning.** The *Global Environment Outlook* (GEO) reports assess the state of the planet; the annual *Emissions Gap Report* compares countries' pledges with what is needed to meet the Paris goals; the *Adaptation Gap Report* does the same for adaptation; the *Frontiers* reports flag emerging issues.
- **Treaty-making.** UNEP has incubated many MEAs and hosts several of their secretariats, including the Vienna Convention and Montreal Protocol (Ozone Secretariat), the CBD, CITES, the Convention on Migratory Species, and the Basel (hazardous waste), Rotterdam (prior informed consent for hazardous chemicals) and Stockholm (persistent organic pollutants) Conventions. It co-founded the IPCC with WMO in 1988. It also supported the **Minamata Convention on Mercury** (2013) and, since UNEA-5 in 2022, negotiations towards a global treaty on plastic pollution, which had not reached agreement by the end of 2025.
- **Awareness.** World Environment Day (5 June) is led by UNEP. India hosted the global celebrations in 2011 (theme: forests) and 2018 (theme: "Beat Plastic Pollution").

5.16.3 Champions of the Earth

UNEP's **Champions of the Earth** award, created in 2005, is the UN's highest environmental honour. Indian recipients include:

- **Afroz Shah** (2016), a Mumbai lawyer who led a volunteer clean-up that removed thousands of tonnes of plastic and waste from Versova beach;
- **Prime Minister Narendra Modi** (2018, Policy Leadership, jointly with French President Emmanuel Macron), for the International Solar Alliance and the pledge to eliminate single-use plastic;
- **Cochin International Airport** (2018, Entrepreneurial Vision), the world's first airport fully powered by solar energy;
- **Purnima Devi Barman** (2022, Entrepreneurial Vision), an Assamese wildlife biologist who leads the all-women "Hargila Army" protecting the endangered greater adjutant stork;
- **Madhav Gadgil** (2024, Lifetime Achievement), ecologist and author of the 2011 Western Ghats Ecology Expert Panel report.

In addition, Vidyut Mohan, co-founder of Takachar, which turns crop residue into fuel and fertiliser, was named a **Young Champion of the Earth** in 2020.

Critics argue that UNEP is underfunded and lacks authority over the many separate treaty bodies; some have proposed upgrading it to a **World Environment Organization**.

5.17 India and the International Agreements: An Overview

The table below summarises the agreements described in this chapter and India's status in each.

Table 5.5: Major international environmental agreements and India's status

Agreement	Year adopted	Main aim	India's status
Ramsar Convention	1971	Conservation and wise use of wetlands	Party since 1982; 101 sites (August 2026)
CITES	1973	Control international trade in wild species	Party since 1976; hosted CoP3 (1981)
Vienna Convention and Montreal Protocol	1985 and 1987	Protect the ozone layer by phasing out ODS	Party since 1992; ratified Kigali Amendment 2021
UNFCCC	1992	Prevent dangerous climate change	Party since 1993; hosted COP8 (2002)
CBD	1992	Conserve biodiversity, sustainable use, benefit-sharing	Party since 1994; hosted COP11 (2012)
Chemical Weapons Convention	1993	Ban and destroy chemical weapons	Party since 1996; stockpile destroyed 2009
UNCCD	1994	Combat desertification and land degradation	Party since 1996; hosted COP14 (2019)
Kyoto Protocol	1997	Binding emission cuts for developed countries	Ratified 2002; major CDM host

Agreement	Year adopted	Main aim	India's status
Cartagena Protocol	2000	Safe transboundary movement of LMOs	Party since 2003
Nagoya Protocol	2010	Access and benefit-sharing for genetic resources	Party since 2012
Paris Agreement	2015	Limit warming to well below 2 °C, pursuing 1.5 °C	Ratified 2016; NDCs submitted 2016, 2022 and 2026

Key Points

- Framework conventions set principles; protocols and amendments add binding obligations; COPs keep the process moving.
- The Earth Summit (1992) produced the Rio Declaration, Agenda 21 and the three Rio Conventions; its principles of CBDR, precaution and polluter pays are now part of Indian law.
- Kyoto set binding targets only for developed countries and created the CDM, in which India was a leading host; Paris requires NDCs from all countries.
- The Montreal Protocol is the most successful environmental treaty; its Kigali Amendment now tackles HFCs.
- The CBD, Ramsar Convention and CITES protect biodiversity through national plans, site designation and trade control; the CWC and UNEP show the wider reach of international environmental governance.

5.18 Sustainable Development Goals and India

The 17 **Sustainable Development Goals (SDGs)**, adopted by all UN member states in 2015 as part of the 2030 Agenda, are introduced in Chapter 1. Several goals deal directly with the environment: clean water and sanitation (SDG 6), affordable and clean energy (SDG 7), sustainable cities (SDG 11), responsible consumption and production (SDG 12), climate action (SDG 13), life below water (SDG 14) and life on land (SDG 15). Others, such as zero hunger (SDG 2) and good health (SDG 3), depend on a healthy environment.

5.18.1 The SDG India Index

NITI Aayog coordinates SDG implementation in India, and the Ministry of Statistics and Programme Implementation maintains a National Indicator Framework to measure progress. Since 2018, NITI Aayog has published the **SDG India Index**, which scores every state and union territory from 0 to 100 on each goal and on a composite score, and classifies them as Aspirant (0–49), Performer (50–64), Front Runner (65–99) or Achiever (100). Ranking states against each other is an example of **competitive federalism**.

The fourth edition, *SDG India Index 2023–24*, released in July 2024, used 113 indicators. Its main findings were:

- India's composite score rose to 71, from 66 in 2020–21 and 57 in 2018.

- Kerala and Uttarakhand jointly topped the states with a score of 79; Bihar (57) and Jharkhand had the lowest scores.
- State scores ranged from 57 to 79, compared with 42 to 69 in 2018, and all states were now Performers or Front Runners.
- Large gains were made in reducing poverty, providing decent work and climate action, while gender equality (SDG 5) remained the weakest goal.

Internationally, the UN Sustainable Development Solutions Network's *Sustainable Development Report 2025* ranked India 99th of 167 countries, the first time it entered the top 100. Environmental goals, especially SDG 13 (climate action), SDG 14 and SDG 15, remain among India's weaker areas, which is why climate policy is closely linked to India's SDG strategy.

> **Think and Do: Status of three SDGs in your neighbourhood**
>
> Choose three SDGs, for example clean water (SDG 6), sustainable cities (SDG 11) and climate action (SDG 13). Collect evidence on their status in your neighbourhood through observation, photographs, short interviews and ward-level data. Then write a short project proposal to improve one of them, with the problem statement, objectives, methodology, a simple budget and expected outcomes, and present it to your class.

5.19 India's National Action Plan on Climate Change

India's climate policy rests on two ideas. First, India's historical responsibility for climate change is small: its per-capita emissions are less than half the world average, and its share of cumulative global CO_2 emissions since 1850 is only about 3–4%. Second, India is highly vulnerable to climate impacts and has a large energy deficit, so it needs both development and climate action. The **National Action Plan on Climate Change (NAPCC)**, released by the Prime Minister's Council on Climate Change on 30 June 2008, expresses this approach. It states that India's per-capita emissions will at no point exceed those of developed countries, and it seeks **co-benefits**, actions that advance development while also addressing climate change.

5.19.1 The eight national missions

The NAPCC set up eight **national missions**, each led by a ministry.

Table 5.6: The eight national missions under the NAPCC

Mission	Lead ministry	Main aims
National Solar Mission (2010)	New and Renewable Energy	Expand solar power; original target 20 GW by 2022, raised to 100 GW in 2015
National Mission for Enhanced Energy Efficiency	Power (Bureau of Energy Efficiency)	Cut energy use in industry and appliances through market-based schemes
National Mission on Sustainable Habitat	Housing and Urban Affairs	Energy-efficient buildings, urban transport, waste management, resilient cities
National Water Mission	Jal Shakti	Conserve water, cut waste and raise water-use efficiency by 20%

Mission	Lead ministry	Main aims
National Mission for Sustaining the Himalayan Ecosystem	Science and Technology	Monitor glaciers and protect Himalayan ecosystems and communities
National Mission for a Green India	Environment, Forest and Climate Change	Increase forest and tree cover and improve forest quality on 10 million hectares
National Mission for Sustainable Agriculture	Agriculture and Farmers Welfare	Make farming climate-resilient through soil health, water use and dryland practices
National Mission on Strategic Knowledge for Climate Change	Science and Technology	Build research capacity, climate models and centres of excellence

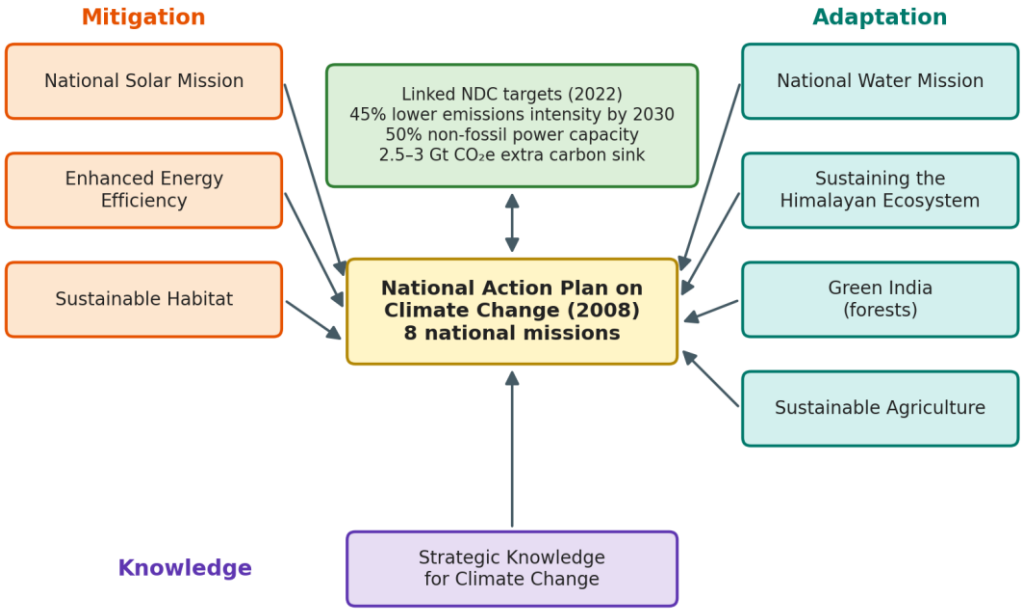


Figure 5.8: *The eight national missions of the National Action Plan on Climate Change (2008), grouped into mitigation, adaptation and knowledge missions, and their links with India's NDC targets. Source: Prepared by the author from the NAPCC (Government of India, 2008).*

The progress of the main missions is summarised below; the others are described in the table above.

- **National Solar Mission.** Launched in January 2010 as the Jawaharlal Nehru National Solar Mission, it used auctions, solar parks, rooftop subsidies and the PM-KUSUM scheme for solar pumps. Solar tariffs fell from over ₹10 per kWh in 2010 to below ₹3 per kWh by 2020, and India's installed solar capacity crossed 100 GW in early 2025. The PM Surya Ghar Muft Bijli Yojana (2024) aims to put rooftop solar on 1 crore (10 million) homes.
- **National Mission for Enhanced Energy Efficiency.** It has four parts: **Perform, Achieve and Trade (PAT)**, which sets energy-saving targets for energy-intensive industries and lets those that exceed them sell certificates; **Market Transformation for Energy Efficiency**, which includes the UJALA scheme that distributed over 36 crore LED bulbs; the **Energy Efficiency Financing Platform**; and the **Framework for Energy Efficient Economic Development**. Star labelling of appliances and the Energy Conservation Building Code support it.
- **National Mission on Sustainable Habitat.** It promotes energy-efficient buildings (the Energy Conservation Building Code for commercial buildings and Eco-Niwas Samhita for homes),

public transport, and better solid waste and wastewater management, through programmes such as AMRUT and the Smart Cities Mission.

- **National Water Mission.** Its goal is to raise water-use efficiency by 20%, with focus on over-exploited groundwater areas. The "Catch the Rain" campaign and the Jal Shakti Abhiyan promote rainwater harvesting.
- **Green India Mission.** Approved in 2014, it aims to increase forest and tree cover on 5 million hectares and improve the quality of forest on another 5 million hectares, increase carbon sequestration, and improve livelihoods of forest-dependent households. It has been held back by limited funding, and its plan has since been revised to give more weight to restoring degraded landscapes such as the Aravalli hills, the Western Ghats and mangroves. It supports India's NDC target of an additional carbon sink of 2.5–3 billion tonnes of CO_2e.

5.19.2 Newer missions and programmes

In 2015 the government announced that new missions would be added on **wind energy**, **human health**, **coastal resources** and **waste to energy**, and that the water and agriculture missions would be redesigned. The health component was launched in 2019 as the **National Programme on Climate Change and Human Health**, run by the National Centre for Disease Control, which prepares state health action plans for heat, vector-borne diseases and air pollution. Other major initiatives outside the original eight include the **National Green Hydrogen Mission** (2023), which aims to produce 5 million tonnes of green hydrogen a year by 2030; the **Carbon Credit Trading Scheme** (2023); and the **Green Credit Programme** (2023), which rewards voluntary environmental actions such as tree planting and water conservation. A revised NAPCC has been under preparation to align the missions with the NDC and net-zero goal.

5.19.3 State Action Plans on Climate Change

Because many climate-related subjects, including agriculture, water, health and land, are under state governments, the Centre asked every state and union territory in 2009 to prepare a **State Action Plan on Climate Change (SAPCC)**. Almost all states and union territories have prepared one, identifying their main vulnerabilities and priority actions. For example, Odisha's plan focuses on coastal and cyclone risks, Himachal Pradesh's on glaciers, horticulture and hydropower, and Delhi's on heat, air pollution and urban flooding. Since 2019, states have been revising their plans (often called SAPCC 2.0) to align them with the NDC and to strengthen vulnerability assessment and financing. Implementation has been uneven, because many plans lack dedicated budgets and the institutional capacity to integrate climate into departmental planning.

5.19.4 India's NDC, Panchamrit, net zero and LiFE

India submitted its first NDC in October 2015, before the Paris conference. At COP26 in Glasgow (November 2021), Prime Minister Narendra Modi announced the **Panchamrit**, five commitments for India:

1. reach 500 GW of non-fossil electricity capacity by 2030;
2. meet 50% of its energy requirements from renewable energy by 2030;
3. reduce total projected carbon emissions by one billion tonnes by 2030;
4. reduce the emissions intensity of its economy by 45% by 2030, compared with 2005;

5. achieve **net-zero emissions by 2070**.

The **updated NDC**, approved by the Union Cabinet in August 2022, turned these into formal targets:

- reduce the **emissions intensity of GDP** (emissions per unit of GDP) by 45% from the 2005 level by 2030;
- achieve about 50% of cumulative installed electric power capacity from non-fossil sources by 2030, with the help of technology transfer and low-cost international finance;
- create an additional carbon sink of 2.5–3 billion tonnes of CO_2e through additional forest and tree cover by 2030;
- spread a healthy and sustainable way of living, including through a mass movement for **LiFE – Lifestyle for Environment**;
- adopt a climate-friendly, cleaner development path and strengthen adaptation in vulnerable sectors.

India's long-term low-emission development strategy, submitted at COP27 in 2022, sets out pathways to net zero by 2070.

India has made fast progress on some of these targets. Emissions intensity of GDP fell by about 36% between 2005 and 2020, and non-fossil sources (renewables, large hydro and nuclear) reached 50% of installed electricity capacity in June 2025, five years ahead of the 2030 target. Coal, however, still produces about three-quarters of India's electricity, because solar and wind plants generate power for fewer hours than coal plants, and electricity demand is growing rapidly. In 2026 India communicated its third NDC for 2031–2035, with targets including a 47% reduction in emissions intensity from 2005 levels and 60% non-fossil installed capacity by 2035, and an additional carbon sink of 3.5–4 billion tonnes of CO_2e.

Mission LiFE was launched by the Prime Minister with the UN Secretary-General at Ekta Nagar (Kevadia), Gujarat, in October 2022. It argues that the climate crisis is driven by wasteful consumption, and asks individuals and communities to adopt "mindful and deliberate utilisation" instead of "mindless and destructive consumption". It lists simple actions such as saving energy and water, reducing single-use plastic, adopting sustainable food systems, reducing waste and e-waste, and choosing healthy lifestyles. LiFE was included in the G20 New Delhi Leaders' Declaration (2023) and in a UNEA resolution on sustainable lifestyles in 2024.

Key Points

- The NAPCC (2008) set up eight national missions covering solar energy, energy efficiency, habitat, water, the Himalaya, forests, agriculture and knowledge.
- Almost all states and union territories have State Action Plans on Climate Change.
- India's updated NDC (2022) targets a 45% cut in emissions intensity and 50% non-fossil power capacity by 2030; India reached the capacity target in 2025.
- India aims for net zero by 2070 and promotes sustainable lifestyles through Mission LiFE.

5.20 Environment Legislation in India

India has one of the most extensive bodies of environmental law in the developing world. Its foundations lie in the Constitution. The **42nd Amendment (1976)** added **Article 48A**, a Directive Principle requiring the State to "endeavour to protect and improve the environment and to safeguard the forests and wild life of the country", and **Article 51A(g)**, a Fundamental Duty of every citizen "to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures". The same amendment moved "forests" and "protection of wild animals and birds" from the State List to the **Concurrent List**, so that Parliament could legislate on them. **Article 253** allows Parliament to make laws to implement international agreements, and was used for the Air Act and the Environment (Protection) Act after the Stockholm Conference. The Supreme Court has also read the right to a clean and healthy environment into the **right to life under Article 21**, starting with cases such as *Subhash Kumar v. State of Bihar* (1991).

The courts have developed key principles of environmental law, including **absolute liability** for hazardous industries, the **polluter pays** and **precautionary** principles, the **public trust doctrine** (the State holds natural resources such as rivers and forests in trust for the public) and **sustainable development**. **Public interest litigation (PIL)**, which allows any citizen to approach the High Courts or Supreme Court on behalf of the public, made these cases possible. Since 2010, the **National Green Tribunal** has heard civil cases arising under the main environmental Acts; its role is discussed in Chapter 7. The regulatory machinery of pollution control boards, consents and waste management rules is described in Chapter 4.

Table 5.7: Major environmental Acts of India

Act	Main purpose	Key institutions	Major amendments
Wild Life (Protection) Act, 1972	Protect wild animals and plants; protected areas; control trade	NBWL, Chief Wildlife Wardens, NTCA, WCCB, CZA	1991, 2002, 2006, 2022
Water (Prevention and Control of Pollution) Act, 1974	Prevent and control water pollution	CPCB, SPCBs	1978, 1988, 2024
Forest (Conservation) Act, 1980 (Van (Sanrakshan Evam Samvardhan) Adhiniyam since 2023)	Restrict use of forest land for non-forest purposes	Central Government, Forest Advisory Committee	1988, 2023
Air (Prevention and Control of Pollution) Act, 1981	Prevent and control air pollution	CPCB, SPCBs	1987, 2023 (Jan Vishwas)
Environment (Protection) Act, 1986	Umbrella law to protect and improve the environment	Central Government, authorities under Section 3(3)	1991, 2023 (Jan Vishwas)
Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006	Recognise forest rights of forest-dwelling communities	Gram sabhas, sub-divisional and district committees, Ministry of Tribal Affairs	Rules amended 2012

Think and Do: Moot court in a green tribunal

Form three groups: petitioners (residents near a river), respondents (an industry discharging effluent and the State Pollution Control Board) and a bench of three judges. Using the Water Act and the Environment (Protection) Act described below, the petitioners argue for closure and compensation, and the respondents argue for time to comply. The bench gives a reasoned order that applies the polluter pays and precautionary principles.

5.21 The Wild Life (Protection) Act, 1972

Background. By the early 1970s, hunting, the fur and skin trade and habitat loss had pushed many Indian species towards extinction; the 1972 tiger census counted only about 1,800 tigers. Existing laws, such as the Wild Birds and Animals Protection Act 1912, were weak and varied between states. Parliament passed the **Wild Life (Protection) Act** on 9 September 1972, using Article 252 after several states consented. Project Tiger followed in 1973 (see Chapter 6).

Objectives. To protect wild animals, birds and plants; to set up a network of protected areas; and to regulate trade in wildlife and its products.

Key provisions.

- **Prohibition of hunting** (Section 9) of any wild animal listed in the schedules, except in narrowly defined cases, such as an animal that has become dangerous to human life, under permit from the Chief Wildlife Warden.
- **Protection of specified plants** (Chapter IIIA, added in 1991): picking, uprooting or trading in them is prohibited.
- **Protected areas.** The Act provides for **wildlife sanctuaries** and **national parks**, and, since 2002, **conservation reserves** (government land next to protected areas) and **community reserves** (community or private land, managed with local people). National parks have stricter protection than sanctuaries: no grazing or other rights are allowed inside them. **Tiger reserves** are notified under Chapter IVB (2006). In 2023 India had 1,014 protected areas (106 national parks, 573 wildlife sanctuaries, 115 conservation reserves and 220 community reserves) covering about 5.3% of its area.
- **Trade controls.** Trade in trophies, animal articles and products of scheduled animals is regulated, and trade in many products, including ivory, is banned (Chapter VA, 1986, strengthened in 1991).
- **Penalties.** Offences involving the most protected species carry imprisonment of three to seven years and a fine; repeat offences carry heavier punishment.

Schedules after the 2022 amendment. The **Wild Life (Protection) Amendment Act, 2022**, in force from 1 April 2023, reduced the schedules from six to **four**:

- **Schedule I**: animal species with the highest protection, such as the tiger, Asiatic lion, snow leopard, Great Indian Bustard and blackbuck;
- **Schedule II**: animal species with a lower level of protection;
- **Schedule III**: protected plant species;

- **Schedule IV**: specimens listed in the CITES appendices.

The old Schedule V (vermin) was removed, although the central government can still declare a species as vermin in a particular area for a period (Section 62), as has been done for wild boar and nilgai in some states. The old Schedule VI (specified plants such as the red vanda and pitcher plant) was merged into the new plant schedule.

Institutions.

- **National Board for Wild Life (NBWL)**, chaired by the Prime Minister, advises on wildlife policy; its **Standing Committee**, chaired by the environment minister, decides on projects inside protected areas and eco-sensitive zones.
- **State Boards for Wild Life**, chaired by Chief Ministers, and a **Chief Wildlife Warden** in each state who manages protected areas.
- **Central Zoo Authority** (1991 amendment), which recognises and regulates zoos.
- **National Tiger Conservation Authority (NTCA)** and **Wildlife Crime Control Bureau (WCCB)**, both created by the 2006 amendment.

Other changes in 2022. The amendment added a chapter to implement CITES (with a Management Authority and Scientific Authority), allowed the central government to regulate or prohibit the import, trade and possession of **invasive alien species**, raised fines (for general violations from ₹25,000 to ₹1 lakh, and the minimum fine for offences involving specially protected animals from ₹10,000 to ₹25,000), and provided for voluntary surrender of captive animals.

Significance and criticisms. The Act has helped tiger, lion, one-horned rhinoceros and elephant populations recover, and India's estimated tiger population reached about 3,680 in 2022. Critics point to weak enforcement against organised wildlife crime, rising human–wildlife conflict, the poor treatment of marine species, and the exclusion of forest communities from protected areas, which the Forest Rights Act tried to address. The 2022 amendment was criticised for allowing transfer and transport of captive elephants for "religious or any other purpose", which wildlife groups fear may enable trade in wild-caught elephants, and for centralising powers in the NBWL Standing Committee.

Landmark cases.

- *State of Bihar v. Murad Ali Khan* (1988): the Supreme Court stressed that the Act must be strictly enforced, in a case about the killing of an elephant.
- *Centre for Environmental Law, WWF-India v. Union of India* (2013): the Court ordered that some Asiatic lions be translocated from Gir in Gujarat to Kuno in Madhya Pradesh to create a second population that would protect the species from a single disaster.
- *Salman Khan blackbuck case*: in 2018 a Jodhpur court convicted the film actor of killing two blackbuck in 1998, showing that the law applies to the powerful (appeals have continued).
- *M.K. Ranjitsinh v. Union of India* (2021 and 2024): orders to place overhead power lines underground or fit them with bird diverters in Great Indian Bustard habitat in Rajasthan and Gujarat, later modified after an expert committee review.

5.22 The Water (Prevention and Control of Pollution) Act, 1974

Background. Water was a state subject, so Parliament could legislate only after the legislatures of several states asked it to under **Article 252**. The Act was passed on 23 March 1974 and applies in

states that adopted it; all states have since done so. It was India's first major pollution law, and it created the pollution control boards.

Objectives. To prevent and control water pollution and to maintain or restore the **wholesomeness of water** in streams, wells, sewers and on land, and to set up boards to carry out these functions.

Key provisions and institutions.

- The **Central Pollution Control Board (CPCB)** and **State Pollution Control Boards (SPCBs)** (in union territories, Pollution Control Committees) plan programmes, set effluent and water-quality standards, collect data, inspect industries and advise governments.
- **Consent mechanism.** No one may set up any industry or treatment plant likely to discharge sewage or trade effluent into a stream, well, sewer or on land without the previous **consent** of the SPCB (Sections 25–26). In practice this means a **Consent to Establish** before construction and a **Consent to Operate** before production begins. The CPCB classifies industries as Red, Orange, Green or White by their pollution potential, which decides how strictly they are regulated.
- **Prohibition** on knowingly discharging poisonous, noxious or polluting matter into water bodies beyond prescribed standards (Section 24).
- **Powers of entry, inspection and sampling**, with a legal procedure so that samples can be used as evidence.
- **Directions** (Section 33A, added in 1988): boards can order the closure, prohibition or regulation of any industry, and the stoppage of its electricity or water supply.
- **Citizen suits** (Section 49, as amended in 1988): courts can take up a complaint by any person who has given the board 60 days' notice.
- A companion law, the Water (Prevention and Control of Pollution) Cess Act, 1977, levied a cess on water used by industries to fund the boards; it was repealed in 2017 when the Goods and Services Tax was introduced.

Amendments. The 1988 amendment strengthened the boards' powers. The **Water (Prevention and Control of Pollution) Amendment Act, 2024**, notified in February 2024, initially applies to Himachal Pradesh, Rajasthan and the union territories, and other states can adopt it by resolution. It **decriminalised** many minor violations, replacing imprisonment with penalties from ₹10,000 up to ₹15 lakh decided by an **adjudicating officer**, with appeals to the National Green Tribunal and penalties credited to the Environment Protection Fund. It also allows the central government to exempt certain categories of industry from the consent requirement and to issue guidelines on consents and on the selection of board chairpersons.

Significance and criticisms. The Act created India's pollution control institutions and the consent system that still regulate industry. However, the boards are often understaffed, lack scientific capacity and independence, and rely on self-reported data. Penalties were rarely imposed in practice, and municipal sewage, the largest source of river pollution, has been hard to control. Most stretches of rivers such as the Yamuna in Delhi still fail bathing-water standards (see Chapter 4). Critics of the 2024 amendment fear that removing imprisonment and allowing exemptions from consent will weaken deterrence; supporters argue that faster civil penalties will be more effective than criminal cases that took years.

Landmark cases.

- *M.C. Mehta v. Union of India* (Ganga pollution cases, 1987 and 1988): the Supreme Court ordered tanneries at Kanpur to set up effluent treatment or close, and directed municipalities to treat sewage, holding that water pollution affects the right to life.
- *Vellore Citizens' Welfare Forum v. Union of India* (1996): about tanneries polluting the Palar river in Tamil Nadu; the Court held that the precautionary and polluter pays principles are part of Indian law.
- *A.P. Pollution Control Board v. Prof. M.V. Nayudu* (1999): the Court placed the burden of proving that an activity is environmentally safe on the person proposing it.

5.23 The Forest (Conservation) Act, 1980

Background. Between 1951 and 1980, India diverted more than 4 million hectares of forest land to agriculture, river valley projects, industries and roads, largely through state government orders. The colonial **Indian Forest Act, 1927** regulated forest use but did not prevent states from de-reserving forests. The Forest (Conservation) Ordinance was issued on 25 October 1980 and replaced by the Act in December 1980, which took effect from the date of the ordinance.

Objective. To check deforestation by requiring **prior approval of the central government** for any use of forest land for non-forest purposes.

Key provisions. Section 2 provides that no state government or other authority shall, without prior approval of the central government, order:

- the de-reservation of a reserved forest;
- the use of any forest land for any non-forest purpose (such as mining, dams, roads or industry);
- the lease of forest land to any private person or body not controlled by the government (added in 1988);
- the clearing of naturally grown trees in order to replant (added in 1988).

Institutions and procedure. The central government decides on the advice of a **Forest Advisory Committee** (Section 3) and regional offices. Approval is usually given in two stages: in-principle (Stage I) and final (Stage II) after conditions are met. Conditions include **compensatory afforestation** on equal non-forest land or on twice the area of degraded forest land, and payment of the **net present value** of the forest lost. These funds are managed under the **Compensatory Afforestation Fund Act, 2016** through national and state CAMPA (Compensatory Afforestation Fund Management and Planning Authority) bodies. Violations are punishable with simple imprisonment of up to 15 days, and the head of a government department can be held responsible for its offences.

The 2023 amendment. The **Forest (Conservation) Amendment Act, 2023**, in force from 1 December 2023, renamed the law the **Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980** ("Forest (Conservation and Augmentation) Act"). Its preamble refers to India's net-zero and carbon-sink targets. Main changes include:

- **Scope.** The Act applies to land declared or notified as forest under the Indian Forest Act or other laws, and to land recorded as forest in government records on or after 25 October 1980. Land officially changed to non-forest use before 12 December 1996 is excluded.
- **Exemptions.** Exempted categories include strategic linear projects of national importance within 100 km of international borders, the Line of Control or the Line of Actual Control;

security-related infrastructure up to 10 hectares; defence or public-utility projects up to 5 hectares in areas affected by left-wing extremism; and small strips of up to 0.10 hectare for access to habitations and roadside or railside amenities.

- **Activities allowed without approval.** Zoos, safaris, eco-tourism facilities, silvicultural operations and surveys are added to the list of activities treated as forestry purposes.

Controversies. In *T.N. Godavarman Thirumulpad v. Union of India* (order of 12 December 1996), the Supreme Court had held that "forest" must be understood in its **dictionary meaning**, covering all forests regardless of ownership or official classification, including unrecorded "deemed forests". Critics argued that the 2023 amendment's narrower definition could remove protection from large areas of such unrecorded forests, especially in the north-east and the Aravalli hills, and that the border-area exemptions affect ecologically fragile Himalayan and north-eastern forests. They also said the amendment weakened the Forest Rights Act by not requiring gram sabha consent. The Forest Conservation Rules, 2022, had already allowed Stage I clearance before the verification of forest rights. In an interim order of 19 February 2024 in *Ashok Kumar Sharma v. Union of India*, the Supreme Court directed governments to continue applying the Godavarman definition until states complete their records of forest land, and required its approval before new zoos or safaris on forest land. The challenge to the amendment's validity was still pending in 2026.

Significance and criticisms. The Act sharply slowed the rate of forest diversion: the area diverted fell from roughly 1.4 lakh hectares a year before 1980 to a fraction of that afterwards. However, the clearance process has been criticised both as too slow by industry and as a "rubber stamp" by conservationists, since the vast majority of proposals are eventually approved. Compensatory plantations, often monocultures on poor land, cannot replace the biodiversity of old natural forests.

Landmark cases. Besides *Godavarman* (1996), which led to the creation of the Central Empowered Committee and the CAMPA funds, important cases include *Lafarge Umiam Mining v. Union of India* (2011), which laid down guidelines for forest clearance, and *Orissa Mining Corporation v. Ministry of Environment and Forests* (2013), the Niyamgiri case discussed under the Forest Rights Act.

5.24 The Air (Prevention and Control of Pollution) Act, 1981

Background. The Air Act was passed on 29 March 1981 under **Article 253**, to implement the decisions of the Stockholm Conference on preserving air quality. Because the Water Act had already created pollution control boards, the Air Act gave the same boards responsibility for air pollution.

Objectives. To provide for the prevention, control and abatement of air pollution, and to confer air-pollution powers on the CPCB and SPCBs.

Key provisions.

- An **air pollutant** is any solid, liquid or gaseous substance, **including noise** (added in 1987), present in the atmosphere in a concentration that may be injurious to human beings, other living creatures, plants, property or the environment.
- **Air pollution control areas** (Section 19): state governments can declare any area a control area and prohibit the use of polluting fuels or the burning of materials there. Today the whole of most states is declared a control area.

- **Consent** (Section 21): no one may establish or operate an industrial plant in a control area without consent from the SPCB, which can impose conditions such as stack height and control equipment.
- **Emission standards** (Section 22): industries may not emit pollutants above standards set by the boards. The CPCB set the **National Ambient Air Quality Standards**, revised in 2009 to cover 12 pollutants (see Chapter 4).
- **Directions** (Section 31A, added in 1987): boards can order the closure of polluting units and cut their power and water supply.
- **Citizen suits** (Section 43) after 60 days' notice to the board.

Amendments. The 1987 amendment added noise, powers of closure and citizen suits. The **Jan Vishwas (Amendment of Provisions) Act, 2023**, which amended 42 laws to reduce minor criminal offences, replaced imprisonment for most violations of the Air Act with monetary penalties of ₹10,000 to ₹15 lakh, decided by an adjudicating officer, with effect from 2024.

Significance and criticisms. The Air Act provides the legal basis for regulating industrial emissions and has been used, together with court orders, to shift Delhi's buses and autos to CNG and to close or relocate polluting industries (see Chapters 4 and 7). The **National Clean Air Programme** (2019), which targets a 40% reduction in particulate matter in over 130 non-attainment cities by 2025–26 compared with 2017, works within this framework. However, the Act focuses on industry and is weak in dealing with diffuse sources such as vehicles, construction dust, crop-residue burning and household fuels, and with pollution that crosses state boundaries. To address the Delhi region, Parliament set up the **Commission for Air Quality Management in National Capital Region and Adjoining Areas** under a separate Act in 2021.

Landmark cases. *M.C. Mehta v. Union of India* (Taj Trapezium, 1996, described earlier in this chapter) and the *M.C. Mehta* vehicular pollution case (orders from 1998 onwards that led to CNG in Delhi) both relied on the Air Act and Article 21. In *Subhash Kumar v. State of Bihar* (1991), the Supreme Court recognised that the right to life includes the right to pollution-free water and air.

5.25 The Environment (Protection) Act, 1986

Background. On the night of 2–3 December 1984, methyl isocyanate gas leaked from the Union Carbide pesticide plant in Bhopal, killing thousands of people and injuring hundreds of thousands, India's worst industrial disaster (see Chapter 4). The tragedy exposed gaps in the existing laws, which covered only water and air and lacked strong central powers over hazardous substances. Parliament passed the **Environment (Protection) Act** under Article 253 on 23 May 1986; it came into force on 19 November 1986.

Objectives. To provide for the protection and improvement of the environment, to implement the decisions of the Stockholm Conference, to coordinate the activities of authorities under existing laws, and to regulate hazardous substances. It is an **umbrella** or framework law that fills gaps left by sectoral laws.

Key provisions.

- **Definitions.** "Environment" includes water, air and land and the inter-relationship among them and human beings, other living creatures, plants, micro-organisms and property. An "environmental pollutant" is any substance present in a concentration that may be injurious to

the environment, and a "hazardous substance" is one that may cause harm because of its chemical or physico-chemical properties.

- **General powers** (Section 3). The central government may take all measures it considers necessary, including setting standards for environmental quality and emissions, restricting where industries may operate, laying down procedures for handling hazardous substances, and setting up **authorities** under Section 3(3). Such authorities have included the Central Ground Water Authority (1997) and the Environment Pollution (Prevention and Control) Authority for Delhi-NCR (1998).
- **Directions** (Section 5). The government can direct the closure, prohibition or regulation of any industry, operation or process, and stop its electricity or water supply.
- **Rules** (Section 6). The government can make rules on any environmental matter.
- **Duties** (Sections 7–9). No person may discharge pollutants beyond standards or handle hazardous substances except according to procedure, and accidental releases must be reported.
- **Powers of entry and sampling** (Sections 10–11).
- **Liability.** Officers in charge of a company are liable for its offences, and heads of government departments for departmental offences (Sections 16–17).
- **Citizen suits** (Section 19) after 60 days' notice to the government.

Rules and notifications. Most of India's environmental regulation is made under this Act. Examples include the **Environment (Protection) Rules, 1986** (emission and effluent standards); the **Environmental Impact Assessment Notification**, 1994, replaced in 2006, which requires prior environmental clearance for listed projects after public hearings; **Coastal Regulation Zone** notifications (1991, 2011, 2019); notifications of **eco-sensitive zones** such as the Doon Valley (1989) and around protected areas; the Hazardous and Other Wastes Rules (2016); the Bio-Medical, Plastic and Solid Waste Management Rules (2016) and E-Waste (Management) Rules (2022) (see Chapter 4); the Noise Pollution Rules (2000); the Ozone Depleting Substances Rules (2000); the Wetlands Rules (2017); the GM organism rules (1989); and the Green Credit Rules (2023).

Amendments. The original Section 15 provided imprisonment of up to five years, or a fine of up to ₹1 lakh, or both, for violations, with up to seven years if the violation continued. Few cases ever reached conviction. The **Jan Vishwas Act, 2023** replaced imprisonment for contraventions with **monetary penalties** of ₹10,000 to ₹15 lakh, imposed by an adjudicating officer, with appeals to the National Green Tribunal, and created an **Environment Protection Fund** for penalties collected. These provisions took effect in 2024. In October 2024, the Supreme Court remarked that the Act had been rendered toothless because the rules and officers needed to impose the new penalties had not yet been put in place; the government notified them shortly afterwards.

Significance and criticisms. The EP Act gave the central government wide powers to act quickly and created the framework for environmental clearance, coastal protection and waste regulation. Critics argue that it concentrates power in the executive, that environmental clearances have often been granted with poor-quality impact assessments and token public hearings, and that the government has repeatedly sought to regularise projects started without clearance. The Supreme Court struck down such **ex post facto** clearances in *Vanashakti v. Union of India* (May 2025), recalled that judgment in November 2025 by a 2:1 majority, and has continued to hear the matter in 2026, showing how contested this issue remains.

Landmark cases.

- *M.C. Mehta v. Union of India* (Oleum gas leak, 1987): after a leak of oleum gas from a Shriram factory in Delhi in December 1985, the Supreme Court laid down the rule of **absolute liability**: an enterprise engaged in a hazardous activity is fully liable for any harm caused, with no exceptions.
- *Indian Council for Enviro-Legal Action v. Union of India* (1996): chemical factories at Bichhri village in Rajasthan, which had poisoned wells with H-acid sludge, were ordered to pay for remediation under the polluter pays principle.
- *Vellore Citizens' Welfare Forum v. Union of India* (1996): the Court declared the precautionary principle, polluter pays and sustainable development to be part of the law of the land, and directed the government to set up an authority under Section 3(3).

5.26 The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006

Background. Under colonial forest laws, large areas of forest were declared state property without properly recording the rights of the people who lived in and used them. After independence, this process continued, and millions of forest dwellers were treated as **encroachers** on land their families had used for generations. The problem became acute after a 2002 eviction drive that followed Supreme Court orders. The **Forest Rights Act (FRA)** was passed on 18 December 2006, received assent on 29 December 2006, and came into force with its rules on 31 December 2007. Its preamble acknowledges the "historical injustice" done to forest-dwelling communities and recognises that they are integral to the survival and sustainability of forest ecosystems. The Ministry of Tribal Affairs is the nodal ministry.

Who is eligible.

- **Forest-dwelling Scheduled Tribes (FDST)**: members of Scheduled Tribes who primarily resided in and depended on forest land for bona fide livelihood needs before 13 December 2005.
- **Other Traditional Forest Dwellers (OTFD)**: any member or community that primarily resided in and depended on forests for at least three generations (75 years) before 13 December 2005.

Rights recognised.

- **Individual forest rights (IFR)**: the right to hold and live on forest land under occupation for habitation or self-cultivation, up to a maximum of **4 hectares** per family, for land occupied before 13 December 2005. The rights are heritable but cannot be sold or transferred.
- **Community rights**: ownership of and access to **minor forest produce** (such as tendu leaves, bamboo, honey and medicinal plants), grazing, fishing, use of water bodies, seasonal access for pastoralists and nomads, and protection of traditional knowledge and sacred sites.
- **Community Forest Resource (CFR) rights**: the right to protect, regenerate, conserve and manage any community forest resource that the community has traditionally protected (Section 3(1)(i)).
- **Rights of particularly vulnerable tribal groups** to their habitats, and rights to rehabilitation in case of illegal eviction.
- **Development facilities** (Section 3(2)): diversion of up to one hectare of forest land for public facilities such as schools, dispensaries, drinking water and roads.

Process and the gram sabha. The **gram sabha** (village assembly of all adults) is at the centre of the Act. It initiates the process, elects a **Forest Rights Committee** that receives and verifies claims on the ground, and passes resolutions on claims. These go to a **Sub-Divisional Level Committee** and then to a **District Level Committee**, which takes the final decision. Each committee includes officials of the revenue, forest and tribal departments and elected local representatives. The gram sabha is also empowered (Section 5) to protect wildlife, forests, water sources and biodiversity, and to stop activities that harm them. In **critical wildlife habitats** inside national parks and sanctuaries, rights can be modified or people resettled only with their free, informed consent and fair rehabilitation (Section 4(2)).

Implementation. By 31 December 2025, about 51.9 lakh claims had been filed, of which about 25.4 lakh were approved (about 24.2 lakh individual and 1.2 lakh community claims), about 18.9 lakh rejected and about 7.6 lakh pending (Ministry of Tribal Affairs, reply in Parliament, 2026). Community forest resource rights have been recognised far less than individual rights, although their potential area is much larger. A pioneering example is **Mendha-Lekha** village in Gadchiroli, Maharashtra, which in 2009 became one of the first villages to receive CFR rights, and later earned income by harvesting and selling its own bamboo, previously controlled by the forest department.

Significance. The FRA is unusual in combining **social justice** with **conservation**. It gives legal security to some of India's poorest people, democratises forest governance through gram sabhas, and recognises communities as protectors of forests. Studies of CFR villages in Maharashtra and Odisha show improved forest protection and incomes.

Criticisms and challenges. Implementation has been slow and uneven. High rejection rates often result from lack of awareness, demands for documents that forest dwellers cannot provide, and resistance from forest departments. Some conservationists argued that the Act would encourage forest clearance; in *Wildlife First v. Ministry of Environment and Forests*, the Supreme Court in February 2019 ordered the eviction of people whose claims had been rejected, which could have affected over a million families, but stayed its own order about two weeks later after the central government sought time to review the rejections. Tribal rights groups criticise the Forest Conservation Rules, 2022, and the 2023 amendment for weakening the requirement that forest rights be settled and gram sabha consent obtained before forest land is diverted.

Case Study: Niyamgiri: the gram sabha decides

The Niyamgiri hills in Kalahandi and Rayagada districts of Odisha are the home of the Dongria Kondh, a particularly vulnerable tribal group who worship the hill-top as the seat of their deity Niyam Raja, and depend on its forests and streams. In the mid-2000s, the Odisha Mining Corporation and Vedanta proposed to mine bauxite on the hill-top to supply an alumina refinery at Lanjigarh.

The project received in-principle forest clearance, but a committee appointed by the MoEFCC in 2010 found that forest rights had not been settled and that the mine would violate the FRA. The ministry refused final clearance. In *Orissa Mining Corporation v. Ministry of Environment and Forests* (18 April 2013), the Supreme Court held that the religious, cultural and community rights of the Dongria Kondh were protected under the FRA, and directed that the gram sabhas of the affected villages decide whether the mining would affect those rights.

In July–August 2013, all twelve gram sabhas selected for consultation unanimously rejected the project. In January 2014 the ministry formally rejected forest clearance. Niyamgiri became the first case in India where a forest community's own assembly decided the fate of a major mining project, and it remains a landmark for the principle of **free, prior and informed consent**. It also shows the tension between mineral-led development and the rights of Adivasi communities, which is discussed further in Chapter 7.

Key Points

- The Constitution (Articles 21, 48A, 51A(g) and 253) and court-made principles form the base of India's environmental law.
- The Wild Life Act 1972 protects species and protected areas; since 2022 it has four schedules and implements CITES.
- The Water Act 1974 and Air Act 1981 created the pollution control boards and the consent system; both have been partly decriminalised (2023–2024).
- The Forest (Conservation) Act 1980 requires central approval to divert forest land; its 2023 amendment (Van Adhiniyam) narrowed its scope and remains contested.
- The Environment (Protection) Act 1986 is an umbrella law under which most environmental rules, including EIA and coastal regulation, are made.
- The Forest Rights Act 2006 recognises individual and community rights of forest dwellers and gives gram sabhas a central role, as Niyamgiri showed.

Chapter 5 Review

Summary

- The greenhouse effect keeps Earth habitable; human emissions of CO_2, methane, nitrous oxide and fluorinated gases have strengthened it, producing a total human-caused radiative forcing of about +2.7 W/m^2 (IPCC AR6).
- Natural factors such as orbital cycles, solar variability and volcanoes have changed climate in the past, but they cannot explain the rapid warming since the mid-20th century; feedbacks such as water vapour and ice–albedo amplify human-caused warming.
- Earth has warmed by about 1.1–1.3 °C since 1850–1900; 2024 was the warmest year on record (about 1.55 °C), and CO_2 at Mauna Loa reached 427 ppm in 2025.
- India has warmed by about 0.7 °C since 1901; heatwaves, extreme rainfall, droughts and sea-level rise have increased, while monsoon rainfall has become more erratic.
- Chlorine and bromine from CFCs, halons and other ODS destroy stratospheric ozone catalytically; polar stratospheric clouds and spring sunlight create the Antarctic ozone hole.
- The Montreal Protocol has phased out about 99% of ODS; the ozone layer is expected to recover between about 2040 and 2066, and the Kigali Amendment (2016) is phasing down HFCs. India completed its CFC phase-out in 2010 and launched the India Cooling Action Plan in 2019.
- Acid rain forms when SO_2 and NO_x are oxidised to sulfuric and nitric acids; it damages lakes, soils, forests and monuments. The Taj Trapezium case shows how courts protected the Taj Mahal from air pollution.
- Climate change harms human health, water security, coasts and livelihoods, threatens biodiversity (for example through coral bleaching and range shifts), reduces economic output and lowers crop yields; India is among the most vulnerable countries.
- The Earth Summit (1992) produced the Rio Declaration, Agenda 21 and the Rio Conventions; the UNFCCC rests on the principle of common but differentiated responsibilities and respective capabilities.
- The Kyoto Protocol set binding targets for developed countries and created the CDM, in which India was a leading host; the Paris Agreement requires all countries to submit NDCs and aims to keep warming well below 2 °C.
- Recent COPs created a loss and damage fund (2022–23), called for a transition away from fossil fuels (2023), set a US$ 300 billion finance goal (2024) and agreed to triple adaptation finance by 2035 (2025).
- The CBD and its Cartagena and Nagoya Protocols address biodiversity, biosafety and benefit-sharing; the Kunming-Montreal framework sets the 30×30 target; India implements them through the Biological Diversity Act 2002 (amended 2023).
- The Ramsar Convention protects wetlands (India had 101 Ramsar sites in August 2026), CITES regulates wildlife trade, the CWC bans chemical weapons (India destroyed its stockpile by 2009), and UNEP leads global environmental assessment and treaty-making.
- India's NAPCC (2008) has eight national missions; its updated NDC (2022) targets a 45% cut in emissions intensity and 50% non-fossil power capacity by 2030, with net zero by 2070 and Mission LiFE.

- India's environmental law rests on Articles 21, 48A and 51A(g) and six key Acts (1972–2006), shaped by amendments such as the Wild Life Amendment 2022, Jan Vishwas 2023, the Water Amendment 2024 and the Van Adhiniyam 2023, and by landmark cases such as Godavarman, Vellore and Niyamgiri.

Key Terms

Greenhouse effect: The warming of Earth's surface caused by atmospheric gases that absorb outgoing infrared radiation and re-emit part of it downwards.

Enhanced greenhouse effect: The additional warming caused by human-added greenhouse gases.

Global warming potential (GWP): The warming caused by 1 kg of a gas over a set period, usually 100 years, relative to 1 kg of CO_2.

Radiative forcing: The change in Earth's energy balance, in W/m^2, caused by a change in a climate driver.

Climate feedback: A process that amplifies or dampens an initial climate change, such as the water vapour or ice–albedo feedback.

Tipping point: A threshold beyond which a part of the climate system changes in a self-sustaining and hard-to-reverse way.

Carbon budget: The total amount of CO_2 that can still be emitted while keeping warming below a given limit.

Chapman cycle: The natural set of reactions by which UV light forms and destroys ozone in the stratosphere.

Ozone-depleting substance (ODS): A chemical, such as a CFC or halon, that releases chlorine or bromine that destroys stratospheric ozone.

Ozone hole: The region over Antarctica where total ozone falls below 220 Dobson units each spring.

Acid deposition: The fall of acidic rain, snow, fog, gases and particles formed from SO_2 and NO_x.

Flue gas desulfurisation (FGD): A technology that removes SO_2 from power plant exhaust, usually using limestone.

Common but differentiated responsibilities and respective capabilities (CBDR-RC): The principle that all countries must act on climate change but developed countries should take the lead.

Precautionary principle: The rule that lack of full scientific certainty should not be a reason to postpone measures against serious or irreversible environmental harm.

Polluter pays principle: The principle that whoever causes pollution should bear the cost of preventing and repairing it.

Clean Development Mechanism (CDM): A Kyoto Protocol mechanism that let developed countries earn credits by funding emission-reducing projects in developing countries.

Nationally Determined Contribution (NDC): A country's own climate action plan submitted every five years under the Paris Agreement.

Loss and damage: Harm from climate change that cannot be avoided through mitigation or adaptation.

Access and benefit-sharing (ABS): Rules that require permission for using genetic resources and fair sharing of the resulting benefits with providers.

Living modified organism (LMO): A living organism with a new combination of genetic material obtained through modern biotechnology.

Wise use (Ramsar): Maintaining the ecological character of wetlands within the context of sustainable development.

Montreux Record: A register of Ramsar sites whose ecological character is changing and that need priority attention.

CITES Appendix I: The list of species threatened with extinction, in which commercial international trade is banned.

Emissions intensity: Greenhouse gas emissions per unit of GDP.

Net zero: A state in which human greenhouse gas emissions are balanced by removals from the atmosphere.

Mission LiFE: India's campaign for Lifestyle for Environment, promoting mindful consumption by individuals and communities.

Consent to Operate: Permission from a State Pollution Control Board required before an industry starts production under the Water and Air Acts.

Compensatory afforestation: Tree planting on non-forest or degraded land required when forest land is diverted for other uses.

Community Forest Resource (CFR) rights: Rights of a forest-dwelling community under the Forest Rights Act to protect, regenerate and manage its traditional forest.

Absolute liability: The rule that an enterprise engaged in a hazardous activity is fully liable for any harm it causes, without exceptions.

Multiple-Choice Questions

1. Which greenhouse gas contributes the largest share of human-caused radiative forcing?

(a) Carbon dioxide
(b) Methane
(c) Nitrous oxide
(d) Water vapour

2. According to IPCC AR6, the 100-year global warming potential of nitrous oxide is about:

(a) 1
(b) 28
(c) 273
(d) 25,200

3. Which of the following is a negative (stabilising) climate feedback?

(a) Water vapour feedback
(b) Ice–albedo feedback
(c) Permafrost carbon release
(d) Increased infrared emission from a warmer Earth (Planck response)

4. Warming of the troposphere together with cooling of the stratosphere indicates that recent warming is caused mainly by:

(a) Increased solar output
(b) Volcanic eruptions
(c) Greenhouse gases
(d) Milankovitch cycles

5. In the chlorine catalytic cycle, which species is regenerated and destroys ozone again?

(a) Cl
(b) ClO
(c) HCl
(d) O_2

6. Polar stratospheric clouds help create the Antarctic ozone hole because they:

(a) Block sunlight in spring
(b) Convert reservoir chlorine compounds into reactive chlorine
(c) Produce CFCs
(d) Absorb UV-B radiation

7. The Kigali Amendment to the Montreal Protocol controls:

(a) Chlorofluorocarbons
(b) Methyl bromide
(c) Hydrofluorocarbons
(d) Carbon tetrachloride

8. Rain is generally called acid rain when its pH is below:

(a) 7.0
(b) 6.5
(c) 4.0
(d) 5.6

9. In the Taj Trapezium case (1996), the Supreme Court ordered polluting industries around the Taj Mahal mainly to:

(a) Pay a carbon tax
(b) Switch to natural gas or relocate
(c) Install solar panels
(d) Plant trees around the monument

10. Under the Kyoto Protocol, a mechanism that allowed developed countries to earn credits from projects in developing countries such as India was:

(a) Joint Implementation
(b) International Emissions Trading
(c) Enhanced Transparency Framework
(d) Clean Development Mechanism

11. Which statement about the Paris Agreement is correct?

(a) Only developed countries must submit climate plans.
(b) Every Party must submit an NDC every five years.
(c) It sets binding emission targets for each country in its text.
(d) It replaced the Montreal Protocol.

12. The New Collective Quantified Goal on climate finance agreed at COP29 in Baku (2024) calls for developed countries to lead in mobilising at least:

(a) US$ 100 billion a year by 2025
(b) US$ 300 billion a year by 2035
(c) US$ 1.3 trillion a year by 2030
(d) US$ 500 billion a year by 2050

13. Prior informed consent and mutually agreed terms for the use of genetic resources are central to the:

(a) Nagoya Protocol
(b) Cartagena Protocol
(c) Ramsar Convention
(d) Kyoto Protocol

14. Which two Indian wetlands are currently on the Montreux Record?

(a) Chilika Lake and Loktak Lake
(b) Keoladeo National Park and Loktak Lake
(c) Wular Lake and Sambhar Lake
(d) Sundarban Wetland and Chilika Lake

15. The headquarters of UNEP is located in:

(a) Geneva
(b) The Hague
(c) Bonn
(d) Nairobi

16. Which NAPCC mission is led by the Ministry of Environment, Forest and Climate Change?

(a) National Solar Mission
(b) National Mission for a Green India
(c) National Water Mission
(d) National Mission on Sustainable Habitat

17. After the Wild Life (Protection) Amendment Act, 2022, Schedule IV of the Act lists:

(a) Vermin species
(b) Protected plant species
(c) Specimens listed in the CITES appendices
(d) Animals with the highest protection

18. The Supreme Court held that "forest" must be understood in its dictionary meaning, regardless of ownership, in:

(a) M.C. Mehta v. Union of India
(b) T.N. Godavarman Thirumulpad v. Union of India
(c) Vellore Citizens' Welfare Forum v. Union of India
(d) Orissa Mining Corporation v. MoEF

19. Under the Forest Rights Act 2006, individual forest rights over land under cultivation are limited to a maximum of:

(a) 1 hectare
(b) 2 hectares
(c) 4 hectares
(d) 10 hectares

20. The Environment (Protection) Act, 1986 was enacted mainly in response to:

(a) The Bhopal gas tragedy
(b) The Chipko movement
(c) The Earth Summit
(d) The Narmada dam protests

Short-Answer Questions

1. Distinguish between the natural greenhouse effect and the enhanced greenhouse effect. Why is water vapour considered a feedback rather than a driver of climate change?
2. Explain radiative forcing with the help of IPCC AR6 values for CO_2, methane and aerosols.
3. Describe any two positive and one negative climate feedback.
4. Summarise the main observed changes in India's temperature and monsoon rainfall reported by the Ministry of Earth Sciences (2020).
5. Write the Chapman reactions and explain how chlorine atoms disturb this balance.
6. Why does the ozone hole form over Antarctica and not over India?
7. Write the reactions by which SO_2 and NO_x form acid rain. Name two methods of controlling SO_2 emissions from power plants.
8. Explain the principle of common but differentiated responsibilities and respective capabilities with reference to the Kyoto Protocol and the Paris Agreement.
9. What is the Clean Development Mechanism? Describe India's role in it.
10. List the three objectives of the CBD and explain how the Biological Diversity Act 2002 implements the benefit-sharing objective.
11. What are the criteria for designating a Ramsar site? What is the Montreux Record?
12. Explain the role of the gram sabha under the Forest Rights Act 2006, with reference to the Niyamgiri case.

Long-Answer Questions

1. Discuss the natural and anthropogenic causes of climate change. Using evidence from IPCC AR6 and Indian data, explain why scientists conclude that recent warming is caused by human activities.
2. Analyse the impacts of climate change on human communities, biodiversity, the economy and agriculture in India, with specific examples such as heatwaves, Himalayan glaciers, the Sundarbans, coral reefs and wheat production. Suggest adaptation measures.
3. "The Montreal Protocol is the most successful environmental treaty." Evaluate this statement, describing the science of ozone depletion, the design of the Protocol and India's phase-out of ODS and HFCs. What lessons does it offer for the climate regime?
4. Trace the evolution of the international climate regime from the UNFCCC (1992) through the Kyoto Protocol and the Paris Agreement to COP30 (2025). How has India's position and contribution changed over this period?
5. Critically examine India's National Action Plan on Climate Change and its missions, India's updated NDC and its net-zero target. To what extent do they support the Sustainable Development Goals?
6. Compare the objectives, key provisions and amendments of the Forest (Conservation) Act 1980 and the Forest Rights Act 2006. How do the 2023 amendment (Van Adhiniyam) and landmark cases such as Godavarman and Niyamgiri shape the balance between forest conservation and the rights of forest dwellers?

Further Reading

- IPCC (2021). *Climate Change 2021: The Physical Science Basis. Summary for Policymakers*. Cambridge University Press. https://www.ipcc.ch/report/ar6/wg1/
- Krishnan, R. et al. (eds) (2020). *Assessment of Climate Change over the Indian Region*. Ministry of Earth Sciences, Government of India; Springer.
- WMO/UNEP (2022). *Scientific Assessment of Ozone Depletion: 2022*. World Meteorological Organization.
- Divan, S. and Rosencranz, A. (2002). *Environmental Law and Policy in India*, 2nd edition. Oxford University Press.
- Government of India (2008). *National Action Plan on Climate Change*. Prime Minister's Council on Climate Change. https://moef.gov.in
- NITI Aayog (2024). *SDG India Index 2023–24*. https://www.niti.gov.in
- UNEP (annual). *Emissions Gap Report*. United Nations Environment Programme. https://www.unep.org
- Raven, P.H., Hassenzahl, D.M., Hager, M.C., Gift, N.Y. and Berg, L.R. (2015). *Environment*, 9th edition. Wiley.

Answers to Multiple-Choice Questions

1. (a)	2. (c)	3. (d)	4. (c)	5. (a)	6. (b)	7. (c)	8. (d)	9. (b)	10. (d)
11. (b)	12. (b)	13. (a)	14. (b)	15. (d)	16. (b)	17. (c)	18. (b)	19. (c)	20. (a)

Chapter 6: Biodiversity and Conservation

Every glass of water, plate of rice and dose of medicine depends on living things. The variety of life on Earth, from the bacteria that fix nitrogen in a paddy field to the tigers of the Sundarbans, is called biodiversity. It took more than three billion years to build up and it is now being lost faster than at any time in human history. India holds an unusually large share of the world's biodiversity on a small share of its land, and it also carries 18% of the world's people. How a crowded, fast-growing country protects tigers, frogs, crop varieties and sacred groves while meeting the needs of its people is one of the central environmental questions of this century. This chapter explains what biodiversity is, why it matters, what threatens it and how it can be conserved, and ends with case studies of major Indian conservation projects and people's movements.

Learning Objectives

- Define biodiversity and distinguish genetic, species and ecosystem diversity, including alpha, beta and gamma diversity.
- Explain why India is a mega-biodiversity nation and describe its ten biogeographic zones and four biodiversity hotspots.
- Identify important endemic and endangered species of India and apply the IUCN Red List categories and criteria.
- Evaluate the ecological, economic, social, ethical, aesthetic and informational values of biodiversity, including the role of sacred groves.
- Analyse the major threats to biodiversity: habitat loss and fragmentation, poaching, man–wildlife conflict, biological invasion and the current extinction crisis.
- Compare in-situ and ex-situ conservation, and distinguish national parks, wildlife sanctuaries and biosphere reserves.
- Explain the use of keystone, flagship, umbrella and indicator species and the role of reintroduction and translocation.
- Assess Indian conservation programmes and movements such as Project Tiger, Project Elephant, vulture and Great Indian Bustard recovery, crocodile conservation, Silent Valley and Save Western Ghats.

6.1 What Is Biodiversity?

The word **biodiversity** is a short form of "biological diversity". The American scientist Walter G. Rosen coined it in 1985 while planning a national forum on the subject, and the biologist E. O. Wilson made it popular through the book *BioDiversity* (1988), which came out of that forum. Within a few years the word had entered law and international treaties. It is used today by ecologists, farmers, economists and lawmakers alike, and each group stresses a different side of the same idea: life is varied, and that variety is valuable.

6.1.1 The Convention on Biological Diversity definition

The most widely used definition comes from Article 2 of the **Convention on Biological Diversity (CBD)**, which was opened for signature at the Earth Summit in Rio de Janeiro in 1992 (see Chapter 5). It defines biological diversity as "the variability among living organisms from all sources including, *inter alia*, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."

Three points in this definition deserve attention:

- It speaks of **variability**, not simply a count of organisms. A forest of a million identical eucalyptus trees has many organisms but little diversity.
- It covers **all habitats**: land, sea, rivers, lakes, and also farms and cities.
- It names three levels: diversity **within species** (genetic diversity), **between species** (species diversity) and **of ecosystems** (ecosystem diversity).

India's **Biological Diversity Act, 2002**, passed to put the CBD into effect, uses almost the same wording. The Act set up the National Biodiversity Authority, State Biodiversity Boards and local Biodiversity Management Committees, which maintain People's Biodiversity Registers recording local species and traditional knowledge. It was amended in 2023 to ease research and the use of traditional medicine while keeping benefit-sharing with local communities (the Act is discussed in Chapter 5).

6.1.2 How many species are there?

Nobody knows exactly how many species live on Earth. Scientists have formally described and named roughly 2.1–2.2 million species so far, a total kept by global databases such as the Catalogue of Life. Most of these are insects; vertebrates (fish, amphibians, reptiles, birds and mammals) make up only about 3–4% of the named total.

The number of species still undescribed is far larger. A widely cited study by Camilo Mora and colleagues (2011) used patterns in the classification of known groups to estimate about **8.7 million species of eukaryotes** (organisms with a nucleus in their cells), give or take about 1.3 million. On that estimate, around 86% of land species and 91% of marine species have not yet been described. Bacteria and archaea are harder still to count because they rarely fit the usual definition of a species; some estimates run into the millions or even billions. A useful working figure for students is "about 8–9 million species of plants, animals and fungi, of which fewer than a quarter have names".

New species are still found every year, including in India. In 2024 the Zoological Survey of India (ZSI) recorded 683 additions to the Indian fauna (459 species new to science and 224 first records for India), and the Botanical Survey of India (BSI) added 433 plant taxa. Kerala led both lists. Many of these discoveries are small invertebrates, fungi and amphibians from the Western Ghats and the North-East, which shows how much remains to be documented even in a well-studied country.

6.2 Levels of Biological Diversity

Biodiversity is studied at three connected levels: **genetic diversity**, **species diversity** and **ecosystem diversity**. They are like three lenses of increasing width. Genetic diversity looks inside a species, species diversity looks at the variety of species living together, and ecosystem diversity looks at the variety of habitats and communities across a landscape. A change at one level affects the others:

when an ecosystem is destroyed, its species disappear, and with them the genetic variation they carried.

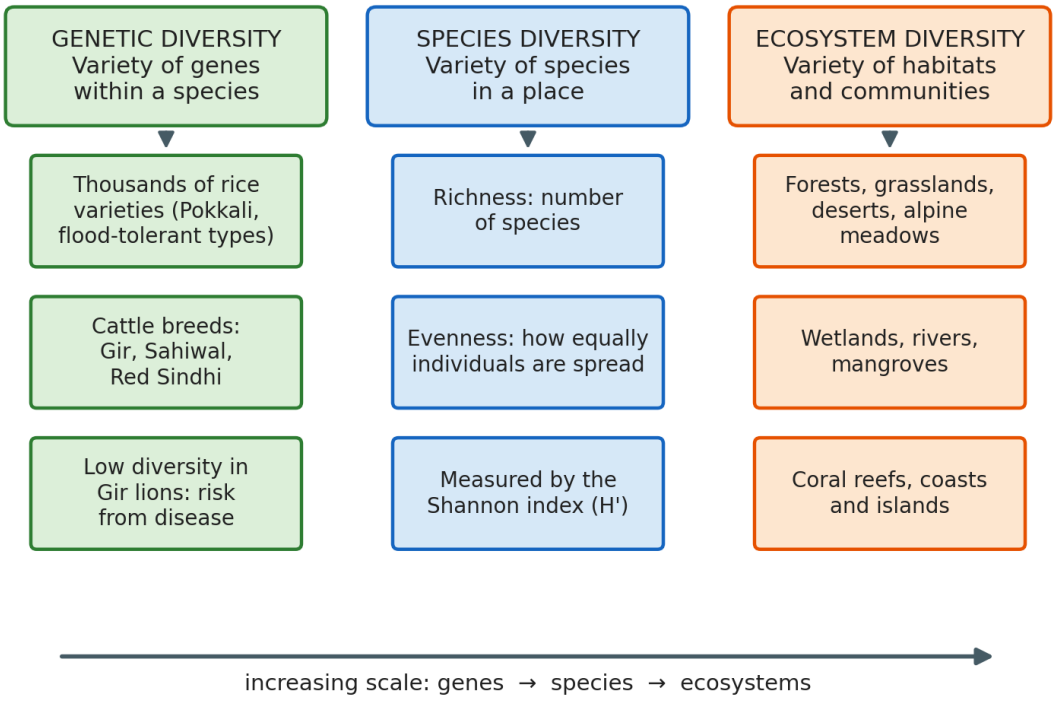


Figure 6.1: *The three levels of biodiversity, from genes within a species to the variety of ecosystems in a landscape, with Indian examples. Source: Author.*

6.2.1 Genetic diversity

Genetic diversity is the variety of genes within a species. Individuals of the same species carry different versions of genes (called **alleles**), and populations living in different places carry different combinations of them. This variation arises through mutation and is reshuffled in every generation by sexual reproduction. It is the raw material on which natural selection acts, so it decides whether a species can adapt to new diseases, pests or a changing climate.

India offers many examples:

- **Rice**: Farmers in India have developed tens of thousands of traditional rice varieties, adapted to floods, drought, salty coastal soils and cold hill slopes. The plant scientist R. H. Richharia alone collected about 19,000 rice samples in the Chhattisgarh region. Salt-tolerant varieties such as Pokkali of Kerala and flood-tolerant varieties of eastern India carry genes that modern breeders now use.
- **Mango**: More than 1,000 named mango varieties are grown in India, from Alphonso and Langra to Dasheri and Banganapalli.
- **Cattle**: Indigenous cattle breeds such as Gir, Sahiwal, Red Sindhi and Tharparkar tolerate heat and resist ticks better than many exotic breeds. More than 200 breeds of livestock and poultry have been registered by the ICAR–National Bureau of Animal Genetic Resources.
- **Wild plants**: Sarpagandha (*Rauvolfia serpentina*) collected from different parts of India differs in the amount of reserpine, the medicinal compound in its roots.

Low genetic diversity makes a species fragile. The **Asiatic lion** (*Panthera leo persica*) of Gir descends from a very small surviving population of the early twentieth century, so the lions are genetically similar to one another. When canine distemper virus spread among lions of Gir in 2018, more than twenty died within a few weeks. The same weakness affects crops: when large areas are sown with one high-yielding variety, a single pest or disease can damage the whole harvest, as happened with the Irish potato famine in the 1840s and the southern corn leaf blight in the United States in 1970.

6.2.2 Species diversity

Species diversity is the variety of species in a place. A **species** is a group of organisms that can interbreed in nature and produce fertile offspring. Species diversity has two components:

- **Species richness** is the number of species present. A hectare of evergreen forest in the Western Ghats may hold more than 100 tree species; a hectare of a eucalyptus plantation may hold one.
- **Species evenness** is how equally individuals are spread among those species. A community where every species is equally common is more even than one where one species dominates.

Two communities can have the same richness but very different evenness. A grassland with ten grass species, each covering a tenth of the ground, is more diverse than one where a single invasive grass covers nine-tenths and nine others share the rest. Ecologists combine richness and evenness in **diversity indices**. The most widely used is the **Shannon index** (also called the Shannon–Wiener index), H′:

$$H' = -\sum p_i \times \ln(p_i)$$

Here p_i is the proportion of all individuals that belong to species i, ln is the natural logarithm, and the sum (Σ) is taken over all species. H′ is zero when only one species is present and rises with both richness and evenness. In most real communities it lies between about 1.5 and 3.5.

Worked example: comparing two ponds

A student counts 100 aquatic insects in each of two ponds on a college campus. Both ponds contain the same four species (A, B, C, D).

Table 6.1: Insect counts and Shannon index calculation for two ponds

Species	Pond 1 count (p_i)	Pond 1: $-p_i \ln p_i$	Pond 2 count (p_i)	Pond 2: $-p_i \ln p_i$
A	25 (0.25)	0.347	85 (0.85)	0.138
B	25 (0.25)	0.347	5 (0.05)	0.150
C	25 (0.25)	0.347	5 (0.05)	0.150
D	25 (0.25)	0.347	5 (0.05)	0.150
Total (H′)	100	1.386	100	0.588

Both ponds have a species richness of 4, but Pond 1 has H′ = 1.39 while Pond 2 has only H′ = 0.59, because one species dominates Pond 2. **Evenness** (E) can be calculated by dividing H′ by its maximum possible value, ln S, where S is the number of species. Here ln 4 = 1.386, so Pond 1 has E = 1.00 (perfectly even) and Pond 2 has E = 0.42. A sudden fall in evenness, such as one pollution-tolerant species taking over, is often an early sign that a habitat is under stress.

Think and Do: Measure diversity on your campus

Mark out four 1 m × 1 m squares (quadrats) on a lawn and four in an unmown patch of ground. In each, count the number of individual plants of each kind (you may label them "species 1", "species 2" if you cannot name them). Calculate species richness and the Shannon index for the lawn and the unmown patch. Which is more diverse, and why? Appendix A gives a full protocol for campus biodiversity sampling.

6.2.3 Ecosystem diversity

Ecosystem diversity is the variety of ecosystems, habitats and ecological processes in a region. An **ecosystem** is a community of organisms together with its non-living surroundings (see Chapter 2). Each ecosystem type holds its own set of species and performs its own services: mangroves protect coasts from cyclones, wetlands store floodwater, grasslands feed livestock and store carbon underground.

India's ecosystem diversity is among the greatest of any country. Within its borders are alpine meadows and glaciers, cold deserts of Ladakh, temperate and subtropical forests of the Himalaya, wet evergreen forests of the Western Ghats and the North-East, dry deciduous forests and thorn scrub of the peninsula, the hot Thar Desert, the salt flats of the Rann of Kachchh, grasslands such as the shola grasslands of the Nilgiris and the terai grasslands of the Himalayan foothills, large river floodplains, freshwater and brackish wetlands, mangroves (about 4,990 km^2, with the Sundarbans the largest block), and coral reefs in the Gulf of Kachchh, the Gulf of Mannar, Lakshadweep and the Andaman and Nicobar Islands. The classic forest classification of Champion and Seth (1968) recognises 16 major forest type groups and about 200 forest types in India.

6.2.4 Alpha, beta and gamma diversity

Diversity also changes with the scale at which it is measured. The ecologist Robert Whittaker proposed three terms that are now standard:

- **Alpha (α) diversity** is the diversity within a single habitat or community, for example the number of bird species in one patch of forest.
- **Beta (β) diversity** is the change in species composition between habitats, for example between a riverbank forest and a hill-top grassland. High beta diversity means the habitats share few species.
- **Gamma (γ) diversity** is the total diversity of a whole region or landscape that contains many habitats.

A simple way to link them is Whittaker's formula, $\beta = \gamma \div \text{mean } \alpha$. Suppose a student records 10 butterfly species in a shola forest and 12 in the neighbouring grassland of the Nilgiris, and 4 species occur in both. The landscape then holds 18 species in all ($\gamma = 18$). The mean alpha diversity is $(10 + 12) \div 2 = 11$, so $\beta = 18 \div 11 = 1.64$. If the two habitats had shared no species, γ would be 22 and β would be 2.0, its maximum value for two habitats; if they had been identical, β would be 1.

Beta diversity explains why conservation cannot focus only on a few species-rich sites. The Western Ghats have high beta diversity because each hill range, valley and rainfall zone holds its own frogs,

fishes and plants. Protecting one large reserve would save many species but would miss others found only a few valleys away.

Key Points

- Biodiversity is the variability of life at the level of genes, species and ecosystems (CBD, 1992).
- Roughly 2.1–2.2 million species are named; around 8–9 million eukaryotic species are estimated to exist.
- Species diversity combines richness and evenness; the Shannon index measures both.
- Alpha diversity is within a habitat, beta diversity is the turnover between habitats and gamma diversity is the total for a region.

6.3 India as a Mega-Biodiversity Nation

A small number of countries, mostly in the tropics, hold most of the world's species. In 1997 the conservation scientist Russell Mittermeier and colleagues identified 17 **megadiverse countries** that together hold about 70% of the world's species. India is one of them, along with Brazil, Indonesia, China, Mexico, Colombia, Australia and others. In 2002 many of these countries formed the **Group of Like-Minded Megadiverse Countries** to press for fair sharing of the benefits from their genetic resources; India hosted the group's ministerial meeting in New Delhi in 2005.

6.3.1 India's share of the world's species

India covers only about **2.4% of the world's land area** but has recorded about **7–8% of the world's known species**. The Zoological Survey of India's national checklist, launched in 2024, listed 1,04,561 animal species and subspecies, and later counts have pushed the total past 1,05,000 (about 6.3% of the world's described animals). The Botanical Survey of India records more than 50,000 species of plants, fungi, lichens, algae and other plant-like organisms; flowering plants alone number about 21,000–22,000 species, about 6–7% of the world total. Endemism is high: about 28% of India's plants are found nowhere else, and more than half of its amphibians are endemic.

Table 6.2: India's biodiversity in numbers (rounded; ZSI and BSI data, 2023–2025)

Group	Species recorded in India	Notes
All animals	about 1,05,000	About 6–6.5% of world fauna
Insects	over 60,000	Largest animal group
Fishes (freshwater and marine)	about 3,500	Many Western Ghats endemics
Amphibians	about 450	Over half endemic
Reptiles	about 750	High endemism in Western Ghats
Birds	about 1,350	About 12% of world's birds
Mammals	about 430	Includes tiger, elephant, rhino, lion
Flowering plants	about 21,000–22,000	About 28% of all plants endemic
All plants and fungi	over 50,000	Includes fungi, algae, lichens, mosses

India also supports some of the largest remaining populations of the world's best-known large animals. It holds about three-quarters of the world's wild tigers (3,682 estimated in 2022), most of the world's Asian elephants (22,446 in a DNA-based estimate released in 2025), more than four-fifths of the greater one-horned rhinoceros, and the only wild population of the Asiatic lion (891 in 2025). The Gangetic river dolphin, India's national aquatic animal, numbered 6,324 in India's first full river survey (2025).

6.3.2 Why is India so rich?

Several factors together explain India's biological wealth:

- **Position at the meeting point of three biogeographic realms.** India lies where the **Indo-Malayan** realm (South and South-East Asia) meets the **Palaearctic** realm (Europe and northern Asia) in the Himalaya, with strong links to the **Afrotropical** realm (Africa) through the dry west. Species from all three have mixed here. The tiger and the hornbills have Malayan affinities, the snow leopard and the ibex are Palaearctic, and the Asiatic lion, the hyena and the caracal reflect African links.
- **Geological history.** India was once part of the southern supercontinent Gondwana. It drifted north as an island for tens of millions of years before colliding with Asia about 50 million years ago. Some ancient Gondwanan lineages survived the journey; the purple frog of the Western Ghats, whose closest relatives live in the Seychelles, is a famous example.
- **Variety of climate and landform.** Altitude in India ranges from sea level to over 8,500 m, and rainfall from under 150 mm a year in western Rajasthan to over 11,000 mm at Mawsynram in Meghalaya. Each combination of temperature and rainfall supports its own community.
- **Isolation of hills and islands.** Isolated hill ranges such as the Western Ghats and the Andaman and Nicobar Islands have allowed species to evolve separately, creating many endemics.
- **Farming and cultural traditions.** Millennia of farming produced many crop varieties, and cultural traditions protected sacred groves and sacred species.

6.3.3 Agrobiodiversity and centres of origin

Agrobiodiversity (agricultural biodiversity) is the variety of crops, livestock, fish, pollinators and soil organisms used in or supporting farming, together with their wild relatives. India is exceptionally rich in it.

In the 1920s the Russian botanist **Nikolai Vavilov** identified eight major **centres of origin** of cultivated plants, regions where crops were first domesticated and where the greatest variety of their forms and wild relatives still exists. One of these, the **Hindustan centre**, covers most of India; India also shares the **Indo-Malayan** (Indo-Burma) centre in the North-East. Crops that originated or diversified greatly in India include rice, pigeon pea (arhar), black gram (urad), green gram (moong), sesame, sugarcane (*Saccharum barberi*), mango, jackfruit, many citrus fruits (the North-East is a centre of citrus diversity), banana, brinjal, cucumber, black pepper, cardamom, turmeric, ginger and jute.

The National Bureau of Plant Genetic Resources (NBPGR) lists more than 300 **crop wild relatives** native to India, such as wild rice (*Oryza nivara* and *O. rufipogon*), wild sugarcane, wild brinjal and wild relatives of pulses and millets. Wild relatives carry genes for resistance to pests, diseases and stress.

A classic example is *Oryza nivara*, a wild rice collected from Gonda in Uttar Pradesh in the 1960s, which supplied resistance to the grassy stunt virus that was devastating rice in Asia; the gene was bred into high-yielding varieties such as IR36 that were then planted over millions of hectares.

Agrobiodiversity is shrinking as farmers switch to a few high-yielding varieties. The Green Revolution greatly raised food production, but it narrowed the genetic base of Indian farming. Conserving traditional varieties in farmers' fields (**on-farm conservation**), in community seed banks and in national gene banks is therefore a priority. The Protection of Plant Varieties and Farmers' Rights Act, 2001 recognises farmers as conservers and breeders of varieties and gives awards to communities that conserve them.

India and the neem and turmeric patents

In 1995 the United States granted a patent on using turmeric to heal wounds. The Council of Scientific and Industrial Research (CSIR) challenged it by producing ancient Sanskrit texts and a 1953 research paper showing the use was already known, and the patent was revoked in 1997. A European patent on the fungicidal use of neem was also revoked in 2000 after a long challenge. These cases led India to build the **Traditional Knowledge Digital Library** (2001), which records traditional medical knowledge so that patent offices can check it.

6.4 Biogeographic Zones of India

Biogeography is the study of how species and ecosystems are distributed across space and why. For conservation planning, a country needs a classification of its natural regions so that every major type of habitat is represented in its protected-area network. India's standard scheme was prepared by W. A. Rodgers and H. S. Panwar of the Wildlife Institute of India (WII) in 1988 and revised by Rodgers, Panwar and V. B. Mathur in 2002. It divides India into **10 biogeographic zones**, which are further split into 27 **biogeographic provinces**. Each zone has a distinct combination of climate, landform, vegetation and animal life.

The zones differ greatly in size. The Deccan Peninsula alone covers about two-fifths of India, while the islands make up less than half a per cent of the land but hold some of its most unusual species.

Table 6.3: The ten biogeographic zones of India (after Rodgers, Panwar and Mathur, 2002)

Zone	Share of India's area	Main areas	Characteristic species
Trans-Himalaya	5.6%	Ladakh, Lahaul–Spiti, north Sikkim	Snow leopard, kiang (Tibetan wild ass), wild yak, black-necked crane
Himalaya	6.4%	Jammu & Kashmir to Arunachal Pradesh	Hangul, musk deer, Himalayan tahr, red panda, Himalayan monal
Desert	6.6%	Thar Desert, Rann of Kachchh	Great Indian bustard, Indian wild ass, chinkara, desert fox
Semi-arid	16.6%	Punjab plains, Gujarat, Rajasthan, Aravallis	Asiatic lion, blackbuck, nilgai, Indian wolf

Zone	Share of India's area	Main areas	Characteristic species
Western Ghats	4.0%	Hill chain from Gujarat to Kanyakumari	Lion-tailed macaque, Nilgiri tahr, Malabar grey hornbill, purple frog
Deccan Peninsula	42.0%	Central Highlands, Chota Nagpur, Eastern Ghats, Deccan plateau	Tiger, gaur, sloth bear, dhole, hard-ground barasingha
Gangetic Plain	10.8%	Plains of the Ganga and its tributaries, terai	One-horned rhino, swamp deer, gharial, Gangetic dolphin
North-East	5.2%	Brahmaputra valley and north-eastern hills	Hoolock gibbon, golden langur, pygmy hog, clouded leopard
Coasts	2.5%	West and east coasts, Sundarbans	Saltwater crocodile, olive ridley turtle, dugong, mangroves
Islands	0.3%	Andaman & Nicobar, Lakshadweep	Nicobar megapode, Narcondam hornbill, coral reefs, robber crab

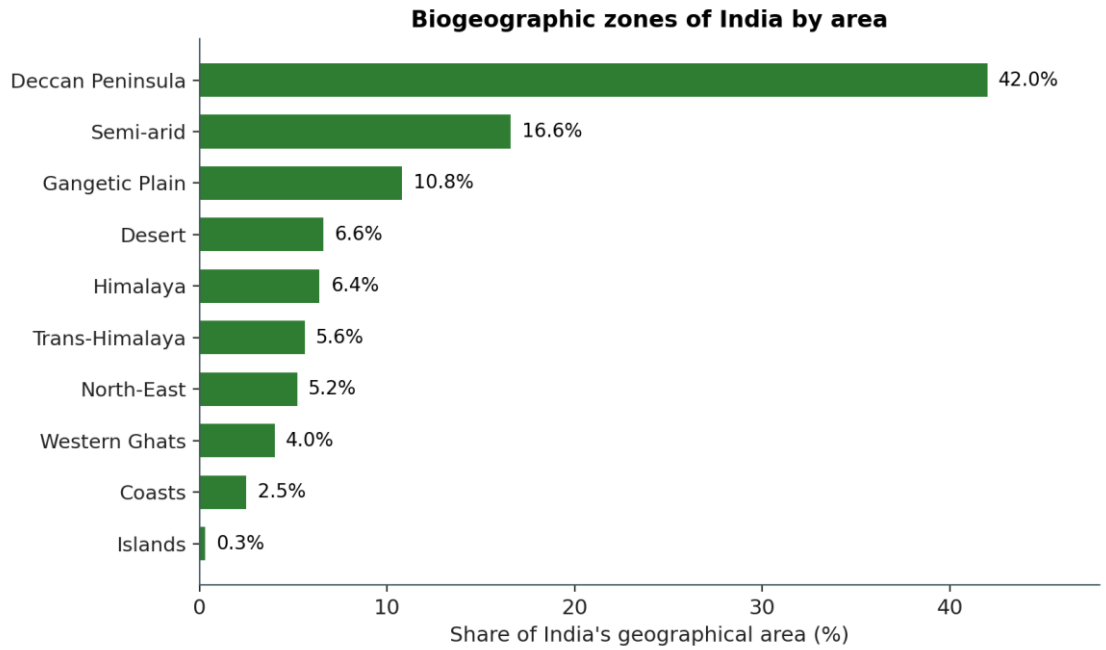


Figure 6.2: *Share of India's land area in each biogeographic zone. Source: Rodgers, Panwar and Mathur (2002), Wildlife Institute of India.*

The zones reflect India's physical geography. The **Trans-Himalaya** is a cold, high desert in the rain shadow of the Great Himalaya, whose wild sheep, goats and snow leopards are related to those of Tibet and Central Asia. The **Himalaya** packs subtropical, temperate and alpine zones into short distances, with the eastern Himalaya far wetter and richer than the western. The **Desert** and **Semi-arid** zones hold grasslands, thorn forest and salt flats; the Little Rann of Kachchh is the last home of the Indian wild ass and Gir is the last home of the Asiatic lion. The **Western Ghats** intercept the south-west monsoon and are rich in endemics. The **Deccan Peninsula**, the largest zone, is mostly deciduous teak and sal forest and holds most of India's tiger reserves. The densely settled **Gangetic Plain** retains little natural habitat outside reserves such as Dudhwa. The **North-East** is the wettest zone and shares many species with South-East Asia. The **Coasts** include mangroves, estuaries, turtle-nesting beaches

and lagoons such as Chilika, and the **Islands** are oceanic, with the Andamans linked biologically to Myanmar and the Nicobars to Sumatra.

Biogeographic zones are used to check whether the protected-area network is representative. WII analyses have repeatedly found that the Semi-arid zone, the Deccan Peninsula grasslands, the Gangetic Plain and the Coasts are under-represented compared with forests in the hills.

6.5 Biodiversity Hotspots

6.5.1 The concept and criteria

Conservation money is limited, so scientists have tried to identify places where it will save the most species. In 1988 the British ecologist **Norman Myers** identified ten tropical forest "hotspots", regions that had very high numbers of endemic plant species and were also losing their habitat rapidly. In 2000 Myers and colleagues, working with **Conservation International**, set two formal criteria. To qualify as a **biodiversity hotspot**, a region must:

1. contain at least **1,500 species of vascular plants as endemics** (more than 0.5% of the world's total), and
2. have lost at least **70% of its original natural vegetation**.

The first criterion measures irreplaceability (species found nowhere else) and the second measures threat. Plants are used because they are relatively well known and because most animals depend on them.

Myers listed 25 hotspots in 2000. Later additions brought the total to 34 (2004), 35 (2011) and **36** (2016, when the North American Coastal Plain was added). Together the 36 hotspots once covered about 15.7% of the Earth's land surface, but their remaining intact habitat covers only about **2.5%**. That small area holds more than half of the world's plant species as endemics and nearly 43% of bird, mammal, reptile and amphibian species as endemics.

The hotspot approach has critics. It is based mainly on plants and vertebrates, it leaves out the sea and species-poor but threatened habitats such as dry grasslands, and its large boundaries include cities and farmland. It is therefore one priority-setting tool among several, alongside Key Biodiversity Areas, Important Bird and Biodiversity Areas, and national systems such as India's biogeographic zones.

6.5.2 India's four hotspots

Parts of four of the 36 global hotspots fall within India.

Table 6.4: India's four biodiversity hotspots (figures for the whole hotspot; source: Conservation International and CEPF profiles)

Hotspot	Indian part	Biodiversity highlights	Key endemic or flagship species
Himalaya	Indian Himalaya from Jammu & Kashmir to Arunachal Pradesh	About 10,000 plant species, about 3,160 endemic; nearly 1,000 bird species	Red panda, Himalayan tahr, Himalayan monal, many rhododendrons

Hotspot	Indian part	Biodiversity highlights	Key endemic or flagship species
Indo-Burma	North-East India south and east of the Brahmaputra, and the Andaman Islands	About 13,500 vascular plants, about half endemic; rich in freshwater turtles	Hoolock gibbon, sangai deer, Andaman wild pig, many orchids
Western Ghats and Sri Lanka	Western Ghats from Gujarat to Kanyakumari	About 5,900 vascular plants, about half endemic; about 180 amphibians, over 70% endemic	Lion-tailed macaque, Nilgiri tahr, purple frog, Malabar civet
Sundaland	Nicobar Islands	About 25,000 vascular plants, about 15,000 endemic	Nicobar megapode, Nicobar long-tailed macaque, Nicobar tree shrew

The **Himalaya** hotspot stretches about 3,000 km from northern Pakistan through India, Nepal and Bhutan to the edges of Myanmar and China. Rapid changes in altitude pack terai grasslands, broadleaved and pine forests, fir and rhododendron forests and alpine meadows into short distances; the wetter eastern Himalaya of Sikkim and Arunachal Pradesh is especially rich in orchids, rhododendrons and birds. Hydropower, roads on fragile slopes, overgrazing, over-collection of medicinal plants and a warming climate that pushes species uphill are the main threats.

Indo-Burma covers more than 2 million km^2 of tropical Asia, from north-eastern India and the Andaman Islands through Myanmar, Thailand, Laos, Cambodia and Vietnam to southern China. It is known for freshwater fishes and turtles, primates and large mammals discovered only recently, such as the saola (described in 1993). In India its forests face shortened shifting-cultivation cycles, hunting, dams and the spread of oil palm, rubber and tea.

The **Western Ghats** run about 1,600 km along India's west coast, from the Tapi river to Kanyakumari, and with the wet zone of Sri Lanka form one hotspot. They cover less than 6% of India's area but hold a large share of its flowering plants, freshwater fishes and amphibians. UNESCO inscribed 39 sites in the Ghats as a World Heritage Site in 2012, noting at least 325 globally threatened species. Dense human settlement, plantations, mining, dams, roads and tourism press on the remaining forest; the debate over how to protect them is the subject of the Save Western Ghats case study later in this chapter.

Sundaland covers the Malay Peninsula, Sumatra, Java, Borneo and Bali. India's share is limited to the **Nicobar Islands**, which are biologically closer to Sumatra than to mainland India. Great Nicobar, a UNESCO biosphere reserve since 2013, has evergreen forest, mangroves and nesting beaches of leatherback turtles. A large port, airport and township project approved for the island in the 2020s would clear a large area of this forest and has been debated intensely for its effects on endemic species and on the Shompen and Nicobarese people.

> **A hotspot is a warning, not an honour**
>
> A hotspot is defined by two numbers: endemic plant species and habitat lost. A region qualifies only because it is both irreplaceable and badly damaged. Being labelled a hotspot is therefore a sign that urgent action is needed.

6.6 Endemic and Endangered Species of India

6.6.1 Endemism

An **endemic species** is one whose natural range is confined to a particular area and that occurs nowhere else in the world. Endemism can be described at any scale: the Nilgiri tahr is endemic to the southern Western Ghats, while the Indian peafowl is endemic to the Indian subcontinent. Endemics are common on islands, isolated mountains and in old, stable habitats, where populations are cut off from relatives elsewhere and evolve separately.

Endemic species with small ranges are especially vulnerable, because a single dam, mine or road can destroy a large part of their total habitat. The Kondana soft-furred rat, for instance, is known only from a small plateau near Pune.

Three regions hold most of India's endemics:

- **Western Ghats**: most of India's endemic amphibians (including the purple frog and many night frogs and bush frogs), endemic freshwater fishes, reptiles such as shieldtail snakes, mammals such as the lion-tailed macaque, Nilgiri tahr, Nilgiri marten and Malabar civet, and well over a thousand endemic flowering plants.
- **North-East and the eastern Himalaya**: endemic orchids, the pitcher plant *Nepenthes khasiana* of Meghalaya, the golden langur, the pygmy hog and the Namdapha flying squirrel.
- **Andaman and Nicobar Islands**: endemic birds such as the Nicobar megapode, the Narcondam hornbill and the Andaman woodpecker, the Andaman wild pig and many reptiles and plants.

6.6.2 Endangered species

A species is commonly called **endangered** when it faces a high risk of extinction. In strict usage, Endangered (EN) is one particular category of the IUCN Red List, explained in the next section, lying between Critically Endangered and Vulnerable. India also has its own legal lists: species in **Schedule I** of the Wild Life (Protection) Act, 1972 (as amended in 2022) receive the highest legal protection, whatever their global Red List category.

Table 6.5: Selected endemic and threatened species of India (IUCN Red List category as last assessed)

Species	Where found	IUCN category	Main threats
Lion-tailed macaque (*Macaca silenus*)	Rainforests of Western Ghats	EN	Forest loss and fragmentation, roads
Nilgiri tahr (*Nilgiritragus hylocrius*)	Montane grasslands, southern Western Ghats	EN	Habitat loss, invasive plants, small isolated herds
Purple frog (*Nasikabatrachus sahyadrensis*)	Western Ghats streams	EN	Forest clearing, collection of tadpoles for food
Malabar large-spotted civet (*Viverra civettina*)	Lowland forests of Western Ghats	CR	Forest conversion; no confirmed record for decades
Pygmy hog (*Porcula salvania*)	Tall grasslands of Assam	EN	Grassland burning and conversion, grazing

Species	Where found	IUCN category	Main threats
Hangul (*Cervus hanglu hanglu*)	Dachigam and nearby forests, Kashmir	CR (as subspecies)	Very small population, grazing, poaching
Namdapha flying squirrel (*Biswamoyopterus biswasi*)	Namdapha, Arunachal Pradesh	CR	Known from one specimen; hunting, habitat loss
Gangetic river dolphin (*Platanista gangetica*)	Ganga and Brahmaputra river systems	EN	Dams and barrages, pollution, fishing nets
Great Indian bustard (*Ardeotis nigriceps*)	Grasslands of Rajasthan and Gujarat	CR	Power lines, loss of grasslands
Gharial (*Gavialis gangeticus*)	Chambal, Girwa and Gandak rivers	CR	Sand mining, dams, fishing nets
Sangai (*Rucervus eldii eldii*)	Keibul Lamjao, Manipur	EN (as Eld's deer)	Single site on a shrinking floating marsh
Red sanders (*Pterocarpus santalinus*)	Southern Eastern Ghats, Andhra Pradesh	EN	Illegal felling for the timber trade

Two of these species show how narrow some ranges are. The **purple frog** spends most of the year underground feeding on termites and surfaces for only a few days in the monsoon to breed; described only in 2003, it belongs to an ancient family whose closest living relatives are frogs of the Seychelles. The **hangul**, the only surviving red deer of the Indian subcontinent, fell from several thousand animals a century ago to fewer than 300 in the Jammu and Kashmir census of 2023. Others are recovering with help: the **pygmy hog**, thought extinct until its rediscovery in Assam in 1971, has been bred in captivity near Guwahati and released into Assam's grasslands, including Manas National Park, since 2008.

6.6.3 Species already lost from India

India has already lost some large animals. The **Asiatic cheetah** was hunted out and declared extinct in India in 1952, the only large mammal known to have vanished from independent India. The **Javan** and **Sumatran rhinoceroses**, which once lived in eastern India and the Sundarbans, disappeared from India in the twentieth century. The **pink-headed duck** of the Gangetic floodplain wetlands has not been reliably seen since 1949, and the **Himalayan quail** not since 1876; both are listed as Critically Endangered (possibly extinct). The return of the cheetah to India through translocation is discussed later in this chapter.

6.7 IUCN Red List Criteria and Categories

6.7.1 What the Red List is

The **International Union for Conservation of Nature (IUCN)**, founded in 1948, is a union of governments and non-governmental organisations. Its **Red List of Threatened Species**, started in 1964, is the world's most widely used system for judging the risk of extinction of species. Assessments are made by expert groups of the IUCN Species Survival Commission and partner organisations, and are reviewed before publication. By 2025 more than 1,70,000 species had been assessed and more

than 48,000 of them (about 28%) were classed as threatened. Around 41% of amphibians, 37% of sharks and rays, 26% of mammals and 12% of birds assessed are threatened.

The Red List is a scientific tool, not a law. It does not by itself protect a species in India; legal protection comes from the Wild Life (Protection) Act and other national laws. But governments, courts, development banks and conservation groups use it to set priorities, and its **Red List Index**, which tracks changes in the extinction risk of whole groups of species over time, is used to measure progress towards global biodiversity targets.

6.7.2 The nine categories

The current system (version 3.1, adopted in 2001) places every species in one of nine categories:

- **Extinct (EX)**: there is no reasonable doubt that the last individual has died.
- **Extinct in the Wild (EW)**: survives only in captivity, in cultivation or as a population well outside its past range.
- **Critically Endangered (CR)**: facing an extremely high risk of extinction in the wild.
- **Endangered (EN)**: facing a very high risk of extinction in the wild.
- **Vulnerable (VU)**: facing a high risk of extinction in the wild.
- **Near Threatened (NT)**: close to qualifying for a threatened category, or likely to qualify soon.
- **Least Concern (LC)**: evaluated and found not to qualify for any of the above; includes widespread and abundant species.
- **Data Deficient (DD)**: not enough information to make an assessment. DD is not a category of threat; many DD species may in fact be threatened.
- **Not Evaluated (NE)**: not yet assessed against the criteria.

CR, EN and VU together are called the **threatened** categories. A species that is Critically Endangered but probably already extinct may be tagged "Possibly Extinct", as with the Malabar civet.

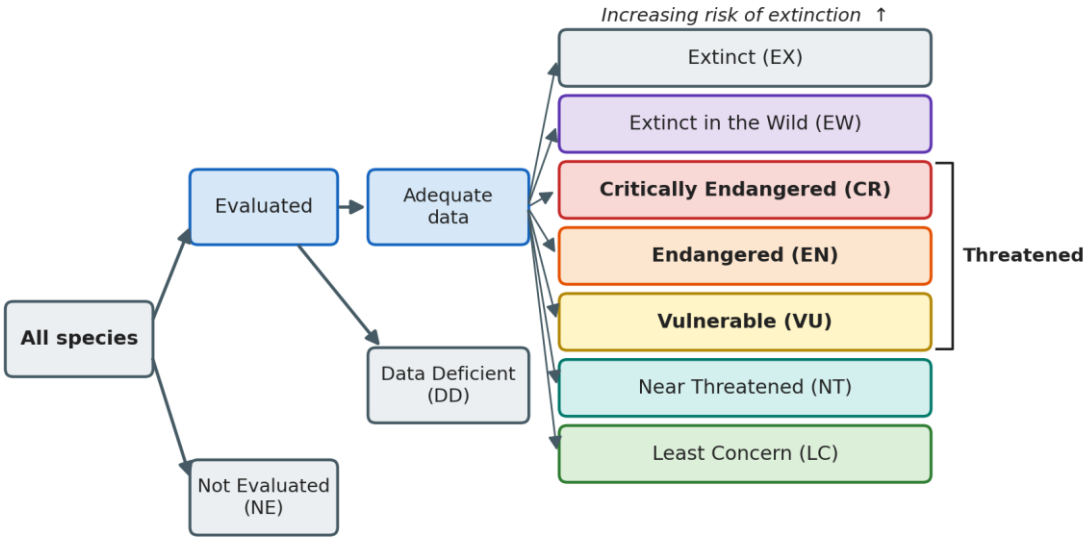


Figure 6.3: *The structure of the IUCN Red List categories, from Not Evaluated to Extinct. Source: Based on IUCN Red List Categories and Criteria, version 3.1.*

6.7.3 The five criteria (A–E)

A species is placed in a threatened category if it meets **any one** of five quantitative criteria, each designed to catch a different kind of risk. The thresholds become stricter from Vulnerable to Critically Endangered.

Table 6.6: Simplified IUCN Red List criteria for the threatened categories

Criterion	What it measures	Critically Endangered	Endangered	Vulnerable
A. Population reduction	Decline over 10 years or 3 generations	≥ 80% (≥ 90% if causes have ceased)	≥ 50% (≥ 70%)	≥ 30% (≥ 50%)
B. Small geographic range	Extent of occurrence, plus fragmentation, decline or fluctuation	< 100 km²	< 5,000 km²	< 20,000 km²
C. Small, declining population	Mature individuals, with continuing decline	< 250	< 2,500	< 10,000
D. Very small population	Mature individuals (or tiny range for VU)	< 50	< 250	< 1,000
E. Quantitative analysis	Probability of extinction in the wild	≥ 50% in 10 years or 3 generations	≥ 20% in 20 years or 5 generations	≥ 10% in 100 years

Some details explain how the criteria work in practice:

- **Criterion A** catches species that may still be numerous but are falling fast. Three Indian *Gyps* vultures were listed as Critically Endangered under criterion A after their numbers fell by more than 95% in about a decade (see the vulture case study later in this chapter).
- **Criterion B** catches species with small ranges. Besides the **extent of occurrence** (the area inside the shortest boundary drawn around all known sites), assessors may use the **area of occupancy** (the area actually occupied: under 10 km² for CR, 500 km² for EN and 2,000 km² for VU). The range test must be combined with at least two of three conditions: severe fragmentation or very few locations, continuing decline, or extreme fluctuations.
- **Criteria C and D** catch species with few adults. Only **mature individuals** (those capable of breeding) are counted, so a herd of 300 hangul that includes calves and yearlings has far fewer mature animals.
- **Criterion E** uses a **population viability analysis**, a computer model that estimates the chance of extinction from data on births, deaths and threats. It is used rarely because good data are scarce.

Think and Do: Classify a species

A frog is known from three streams within a single valley of the Western Ghats, with an extent of occurrence of about 60 km^2. Surveys show that the forest around the streams is being cleared every year for plantations. Using the criteria table, which category would you suggest, and under which criterion? (Hint: check criterion B and its extra conditions.) Then look up a real Western Ghats frog on the IUCN Red List website and compare.

6.8 Value of Biodiversity

Why should a country with pressing needs for food, jobs, roads and electricity spend money and land on frogs, grasslands and tigers? The answer is that biodiversity has many kinds of value, some easy to measure in rupees and some not measurable at all. Environmental economists sort these values into a framework called **total economic value**, while ethicists argue that species also matter for their own sake. The syllabus groups the values into six types: ecological, economic, social, ethical, aesthetic and informational. They overlap, and a single species, such as the neem tree or the honeybee, can have all six.

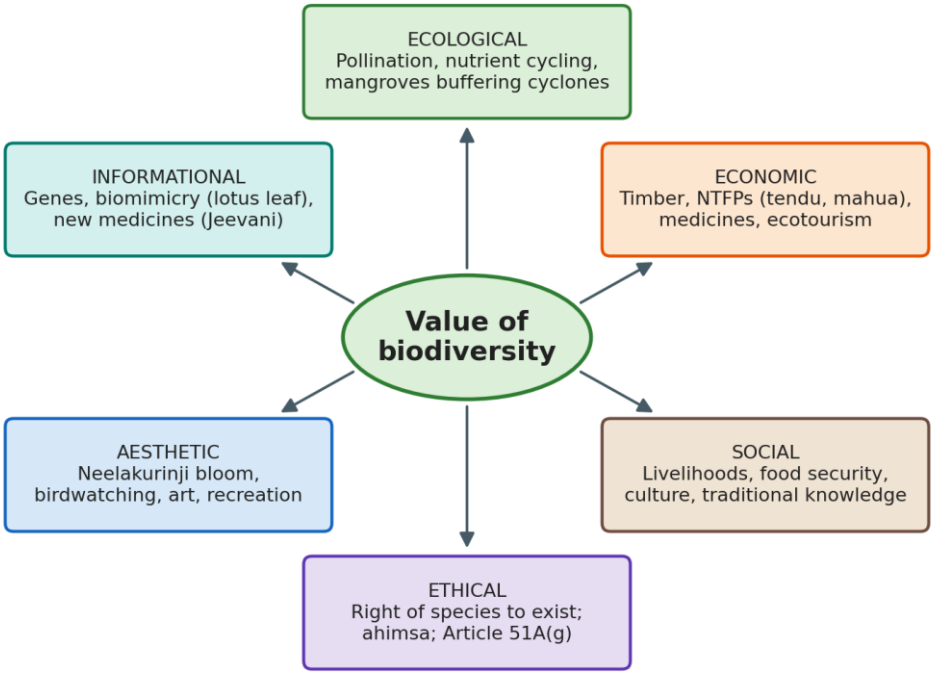


Figure 6.4: *The six kinds of value of biodiversity, with an Indian example of each. Source: Author.*

6.8.1 Ecological value

The **ecological value** of biodiversity lies in the work living things do to keep ecosystems functioning. These benefits are called **ecosystem services**: provisioning, regulating, cultural and supporting services (see Chapter 2). Biodiversity underpins all of them.

- **Pollination**: About three-quarters of the world's leading food crop types benefit at least partly from animal pollination. The Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) estimated in 2016 that US$235–577 billion of the world's annual

crop output depends directly on pollinators. In India, mustard, sunflower, cucurbits, cardamom, litchi and apples all yield more when bees visit their flowers; apple growers in Himachal Pradesh rent honeybee colonies during flowering.

- **Nutrient cycling and soil formation**: Earthworms, termites, fungi and bacteria break down dead matter and release nutrients. Nitrogen-fixing bacteria in the roots of pulses and in paddy fields supply much of the nitrogen in traditional Indian farming.
- **Pest control**: Birds, bats, spiders, frogs and predatory insects eat crop pests.
- **Scavenging and disease control**: Vultures once cleaned up the carcasses of millions of dead cattle in India. When diclofenac poisoning wiped out the vultures in the 1990s, carcasses rotted, feral dog numbers rose and rabies risk increased (see the vulture case study later in this chapter).
- **Protection from disasters**: Mangroves and coastal forests reduce the force of storm surges. A study of the 1999 Odisha super-cyclone found that villages behind wider belts of mangroves near Bhitarkanika suffered fewer deaths than villages without them. Forests on slopes reduce landslides, and wetlands absorb floodwaters.
- **Climate regulation**: Forests, grasslands, peatlands, mangroves and seagrass beds store large amounts of carbon. India's forest and tree cover stores over 7,000 million tonnes of carbon (ISFR 2023).

Biodiversity also gives ecosystems **stability and resilience**. Experiments in grasslands, such as the long-running Cedar Creek plots in the United States, have shown that plots with more plant species produce more biomass and recover better from drought. Diverse communities contain species that do similar jobs, so if one is lost another can take over; ecologists call this the **insurance effect**. The ecologist Paul Ehrlich compared species to the rivets of an aeroplane wing: a plane can lose a few rivets and still fly, but nobody can say which loss will be the one that brings it down.

6.8.2 Economic value

The **economic value** of biodiversity is its contribution to income, production and consumption. It is usually divided into three parts.

Consumptive use value is the value of products consumed directly by local people without passing through a market: fuelwood, fodder, wild fruits, leafy greens, fish, honey, bamboo, thatch and medicinal herbs. In India an estimated 25–30 crore people depend on forests for at least part of their livelihood. Tribal households in central India may earn a large share of their cash and food from **non-timber forest products (NTFPs)** such as tendu leaves (used to wrap bidis), mahua flowers and seeds, sal seeds, chironji, lac, gums and tamarind. The Forest Rights Act, 2006 gives forest dwellers the right to own, collect and sell such minor forest produce.

Productive use value is the value of biological products sold in markets: timber, commercial fisheries, rubber, spices, silk, cotton, and raw materials for medicines and industry. India's pharmaceutical and AYUSH (Ayurveda, Yoga, Unani, Siddha, Sowa-Rigpa and Homoeopathy) industries use several thousand plant species. Some important medicines from plants found in India are:

- **Reserpine**, from the roots of sarpagandha (*Rauvolfia serpentina*), was one of the first modern drugs for high blood pressure.

- **Vincristine and vinblastine**, anti-cancer drugs, come from the Madagascar periwinkle (*Catharanthus roseus*, called sadabahar), widely grown in Indian gardens.
- **Taxol (paclitaxel)**, another anti-cancer drug, was first obtained from yew bark; the Himalayan yew (*Taxus wallichiana*) was heavily overharvested for it in the 1990s and is now listed as Endangered.

Productive value also includes **genetic resources** for crop and livestock breeding, as in the *Oryza nivara* example earlier in this chapter.

Non-consumptive use value comes from using nature without removing anything from it, mainly through **ecotourism**. Wildlife tourism in tiger reserves such as Ranthambore, Corbett, Kanha and Bandhavgarh, birdwatching at Keoladeo and Chilika, and whale-shark and coral-reef tourism support hotels, drivers, guides and local craftspeople. Several states share part of park entry fees with nearby villages through eco-development committees.

Economists also speak of **indirect use value** (the value of ecosystem services such as flood control and pollination), **option value** (the value of keeping a species in case it becomes useful in future), **bequest value** (the value of passing nature on to future generations) and **existence value** (the satisfaction people get simply from knowing that tigers or whales exist, even if they never see one).

Table 6.7: The total economic value of biodiversity, with Indian examples

Type of value	Meaning	Indian example
Consumptive use	Used directly, not sold	Fuelwood, mahua, fish caught for the household
Productive use	Sold in markets	Teak timber, cardamom, medicinal plants, fisheries
Non-consumptive use	Enjoyed without removal	Tiger safaris, birdwatching at Keoladeo
Indirect use	Ecosystem services	Mangroves reducing cyclone damage
Option value	Possible future use	Wild rice genes for future disease resistance
Existence and bequest	Knowing it exists; leaving it for the future	Public support for saving the Great Indian bustard

Large studies have tried to put a price on these services. *The Economics of Ecosystems and Biodiversity* (TEEB, 2010) and the British government's *Dasgupta Review* (2021) concluded that nature is a form of capital that national accounts ignore, so its depletion shows up as growth rather than loss. The World Economic Forum estimated in 2020 that more than half of global GDP depends moderately or highly on nature. Putting a number on nature can help decision-makers, but many values, such as the sacredness of a grove, cannot be priced.

6.8.3 Social value

The **social value** of biodiversity lies in its role in the lives, livelihoods, health, culture and identity of communities. For millions of Indians, especially Adivasi and other forest-dependent communities, pastoralists such as the Raika, Gaddi and Van Gujjar, and fishing communities, biodiversity is the basis of daily survival and of social organisation.

- **Food and nutrition security**: Wild foods such as mushrooms, tubers, leafy greens and fruits fill seasonal food gaps in tribal areas, and millets and pulses grown in diverse traditional farms supply nutrients that a diet based on rice or wheat alone lacks.
- **Health**: India's traditional systems of medicine, and countless household remedies, depend on plants such as tulsi, neem, turmeric, giloy and ashwagandha.
- **Culture and identity**: Many communities treat particular species as totems, name clans after animals and plants, and celebrate festivals tied to forests, harvests and rivers. Sacred groves, discussed below, are one expression of this.
- **Traditional knowledge**: Communities hold detailed knowledge of local plants and animals, built up over generations. People's Biodiversity Registers under the Biological Diversity Act record this knowledge village by village.

Loss of biodiversity hurts the poor most, because they depend most directly on it and have the fewest alternatives.

6.8.4 Ethical value

The **ethical value** of biodiversity rests on the belief that other species have a right to exist and that humans have a duty not to destroy them, whether or not they are useful. This is called **intrinsic value**. Hindu, Jain and Buddhist traditions teach *ahimsa* (non-violence towards living beings); the Bishnois of Rajasthan have protected blackbuck, chinkara and khejri trees for over five centuries (see Chapter 7).

The American forester Aldo Leopold expressed a similar idea as the **land ethic** (1949): a thing is right when it tends to preserve the integrity, stability and beauty of the biotic community. Ethics also includes **intergenerational equity**, the idea that the present generation holds the Earth in trust for future generations, which lies at the heart of sustainable development.

The Indian Constitution reflects these values. Article 48A, added in 1976, directs the State to protect and improve the environment and to safeguard forests and wildlife, and Article 51A(g) makes it a fundamental duty of every citizen "to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures". Indian courts have increasingly adopted an **ecocentric** approach, which places nature, and not only human benefit, at the centre of decisions. In the Asiatic lion case of 2013 the Supreme Court held that the interest of the species, not the pride of one state, must decide where the lions should live.

6.8.5 Aesthetic value

The **aesthetic value** of biodiversity is the beauty, wonder and inspiration that nature provides. A tiger walking out of tall grass, flamingos on a salt pan of Kachchh, the blue haze of Neelakurinji (*Strobilanthes kunthiana*) flowering on the hills of Munnar once every twelve years, the carpet of wildflowers on the Kaas plateau near Satara or in the Valley of Flowers in Uttarakhand, and the synchronised flashing of fireflies in the forests of Maharashtra all draw thousands of visitors.

Nature has shaped Indian art and literature for millennia, from Harappan animal seals to Kalidasa's *Meghaduta*, Mughal miniatures and Warli folk art. Time spent in green, species-rich places also reduces stress and improves mental health, which E. O. Wilson linked to **biophilia**, an inborn human attraction to living things. Aesthetic value is also an economic value, because it drives tourism.

6.8.6 Informational value

The **informational value** of biodiversity is the knowledge stored in living things. Every species carries a unique set of genes refined by millions of years of evolution, and every ecosystem holds lessons about how nature solves problems. Much of this information has not yet been read. When a species becomes extinct, its information is lost for ever, like a book burnt before anyone opened it.

- **Science and medicine**: Organisms are models for research. The enzyme *Taq* polymerase, taken from a bacterium living in hot springs, made the polymerase chain reaction (PCR) possible; PCR was the basis of the RT-PCR tests used to detect COVID-19.
- **Biotechnology**: The insect-killing protein in Bt cotton comes from a gene of the soil bacterium *Bacillus thuringiensis*.
- **Biomimicry**: Engineers copy designs from nature. The lotus leaf, whose surface repels water and dirt, has inspired self-cleaning paints and fabrics; the kingfisher's beak shaped the nose of Japan's quieter bullet trains; and the cooling of termite mounds inspired natural ventilation in buildings.
- **Bioprospecting**: Searching wild species for useful chemicals, which rewards the communities who hold the knowledge only if benefits are shared fairly, as the case below shows.

Case Study: Jeevani and the Kani tribe: sharing the benefits of knowledge

In 1987, scientists of the Tropical Botanic Garden and Research Institute (TBGRI), Thiruvananthapuram, on an expedition in the Agasthyamalai hills of Kerala, noticed that their Kani tribal guides stayed energetic on long treks by eating the fruits of a small plant called *arogyappacha* (*Trichopus zeylanicus*). The Kanis shared their knowledge with the scientists. TBGRI developed an anti-fatigue herbal formulation from the plant's leaves, named it **Jeevani**, and in 1995 licensed its manufacture to an Ayurvedic pharmaceutical company. TBGRI agreed to share half of the licence fee and royalties with the Kanis, and a trust run by the community was set up in 1997 to receive and use the money.

The arrangement was hailed internationally as one of the first working examples of **access and benefit-sharing**, a principle of the Convention on Biological Diversity that was later written into India's Biological Diversity Act, 2002 and the Nagoya Protocol (2010). It also showed the difficulties: the wild plant was overharvested, cultivation by the Kanis was hampered by forest rules, and not all Kani settlements agreed on how the benefits should be shared. The case is still taught as a model and as a lesson in how hard fair benefit-sharing is in practice.

6.9 Sacred Groves

6.9.1 What sacred groves are

Sacred groves are patches of natural vegetation, from a few trees to several hundred hectares of forest, that local communities protect because they are dedicated to a deity, a spirit or ancestors. Felling trees, hunting and often even collecting dead leaves or twigs is forbidden by religious belief and customary rules. They are among the oldest forms of community conservation in the world, and India has more of them than almost any other country.

Estimates of their number vary because no complete survey exists. Studies compiled by the C.P.R. Environmental Education Centre (CPREEC), Chennai, have documented about 13,000–14,000 sacred groves across India, but experts estimate the true number at between 1,00,000 and 1,50,000. Most are small, less than a hectare, but some, especially in Meghalaya and Rajasthan, cover large areas.

Table 6.8: Sacred groves in different parts of India

State or region	Local name	Associated communities or deities
Kerala	Kavu (sarpa kavu)	Serpent deities, Bhagavathi; maintained by families and temples
Karnataka	Devara kadu, Nagabana	Village deities; Kodagu alone has over a thousand
Maharashtra and Goa	Devrai, Devrahati	Mother goddesses; studied by Gadgil and Vartak in the 1970s
Tamil Nadu	Kovil kadu	Village guardian deities such as Ayyanar
Jharkhand, Chhattisgarh, Odisha, West Bengal	Sarna, Jaher than, Jahera, Garamthan	Munda, Oraon, Santal and Ho communities; usually sal groves
Rajasthan	Oran, Dev bani, Rundh	Folk deities; also used as common grazing land
Meghalaya	Law Kyntang, Law Lyngdoh	Khasi and Jaintia clans and village priests
Manipur	Umang Lai	Meitei forest deities
Himachal Pradesh and Uttarakhand	Dev van	Village deities (devtas)

In Kerala, most old households and temples kept a **kavu**, often dedicated to serpent gods; larger ones such as Iringole Kavu near Perumbavoor preserve fragments of lowland evergreen forest that have vanished from the surrounding plains. The **devrais** of Maharashtra, studied by Madhav Gadgil and V. D. Vartak in the 1970s, hold old trees, climbers and medicinal plants no longer found in the degraded land around them; more than 2,000 have been recorded. For the Munda, Oraon, Santal and Ho people of Jharkhand and neighbouring states, the **sarna** is a sal grove where the village deity resides and where the spring festival of Sarhul is celebrated; it is central to Adivasi identity.

The **orans** of western Rajasthan are large community lands dedicated to folk deities, where khejri (*Prosopis cineraria*), rohida (*Tecomella undulata*) and ber are protected and grazing is managed by custom. In the Thar they provide fodder, water from village ponds and shelter for wildlife, including chinkara and the Great Indian bustard. Because many orans are recorded as government "wasteland", some have been allotted to solar parks and power lines. In December 2024 the Supreme Court directed Rajasthan to map its orans and other sacred groves and give them legal protection, including as community reserves.

Case Study: Mawphlang sacred forest, Meghalaya

About 25 km from Shillong, in the East Khasi Hills, lies the Mawphlang sacred forest, one of the best known *law lyngdoh* (sacred groves) of the Khasi people. It covers roughly 75–80 hectares of dense subtropical forest that stands in sharp contrast to the grassy, deforested hills around it.

According to Khasi belief, the forest is the abode of the deity Labasa, who protects the village. Nothing may be taken out of the grove, not even a fallen leaf or twig; anyone who breaks the rule is believed to fall ill or die. Ceremonies and sacrifices are still performed at old stone monoliths inside the forest, and decisions about the grove are taken by the traditional village institution.

Because of this protection, Mawphlang holds tall oak, rhododendron and other broadleaved trees, a thick layer of mosses, ferns and epiphytic orchids, and medicinal plants that have disappeared from the surrounding hills. Its soils hold water and feed springs used by the villages. Meghalaya has many such groves (*law kyntang* and *law lyngdoh*), and they are among the last reservoirs of the original forest of the Khasi and Jaintia Hills. In recent years, Mawphlang and neighbouring villages have also worked with conservation organisations on a community forest-carbon project that pays them for protecting and restoring forests, showing how traditional belief and modern finance can work together.

6.9.2 Ecological and social importance

Sacred groves are important for several reasons:

- **Refuges for biodiversity**: In landscapes converted to farms and plantations, groves are often the only patches of old-growth vegetation left. Surveys in the Western Ghats have found rare and endemic trees, *Myristica* swamp forests, climbers, orchids, fungi and lichens confined to sacred groves.
- **Seed banks and stepping stones**: Groves supply seeds for regeneration and act as stepping stones for birds, bats and insects moving across the landscape.
- **Water security**: Many groves protect springs, streams and ponds. In Kerala, ponds attached to kavus help recharge groundwater; in Meghalaya, groves feed the springs that supply villages.
- **Medicinal plants and genetic resources**: Groves hold wild relatives of crops and plants used in traditional medicine.
- **Cultural continuity**: Festivals, rituals and community decision-making centred on groves keep traditional knowledge alive and strengthen social bonds.
- **Model for community conservation**: Groves show that conservation can work through local institutions and beliefs, without fences or guards.

6.9.3 Threats to sacred groves

Sacred groves are shrinking in number and size. The main pressures are:

- **Weakening beliefs and changing social structure**: As younger generations adopt new beliefs or migrate to cities, the taboos that protected groves lose force.
- **"Templeisation"**: Small shrines are being replaced by large concrete temples, and the surrounding trees cleared for buildings, parking and roads.

- **Encroachment and land conversion**: Rising land prices lead to groves being sold, encroached or converted to plantations, especially where they are private family property or recorded as government wasteland.
- **Overuse**: Grazing, fuelwood cutting and collection of leaf litter degrade groves, especially where customary rules have weakened.
- **Invasive species**: *Lantana*, *Chromolaena* and other weeds invade small, degraded groves.
- **Development projects**: Roads, mines, dams and solar parks have cleared groves, particularly in central India and Rajasthan.

6.9.4 Conservation of sacred groves

Several legal tools can now support sacred groves:

- The **Wild Life (Protection) Act** allows land, including community land, to be declared a **community reserve** or **conservation reserve** with the consent of the community.
- The **Biological Diversity Act, 2002** allows state governments to notify **Biodiversity Heritage Sites**; the Nallur tamarind grove in Karnataka became India's first in 2007, and several sacred groves have since been notified.
- The **Forest Rights Act, 2006** lets gram sabhas claim **community forest resource rights** over forests they have traditionally protected.

Some states, such as Kerala, give grants to owners of sacred groves for their maintenance, and forest departments and NGOs have helped document groves and restore degraded ones with native species. The most important requirement is to support the communities and institutions that have protected these groves for centuries, rather than replace them with government control.

6.10 Threats to Biodiversity

Species have always gone extinct, but today the main causes are human activities. E. O. Wilson summed up the major threats with the acronym **HIPPO**: **H**abitat destruction, **I**nvasive species, **P**ollution, human **P**opulation growth and **O**verharvesting. The IPBES Global Assessment (2019) ranked the direct drivers of biodiversity loss worldwide, in order of impact, as changes in land and sea use, direct exploitation of organisms, climate change, pollution and invasive alien species. Behind these direct drivers lie indirect ones: growing population and consumption, trade, technology, weak governance and economic policies that do not count the value of nature.

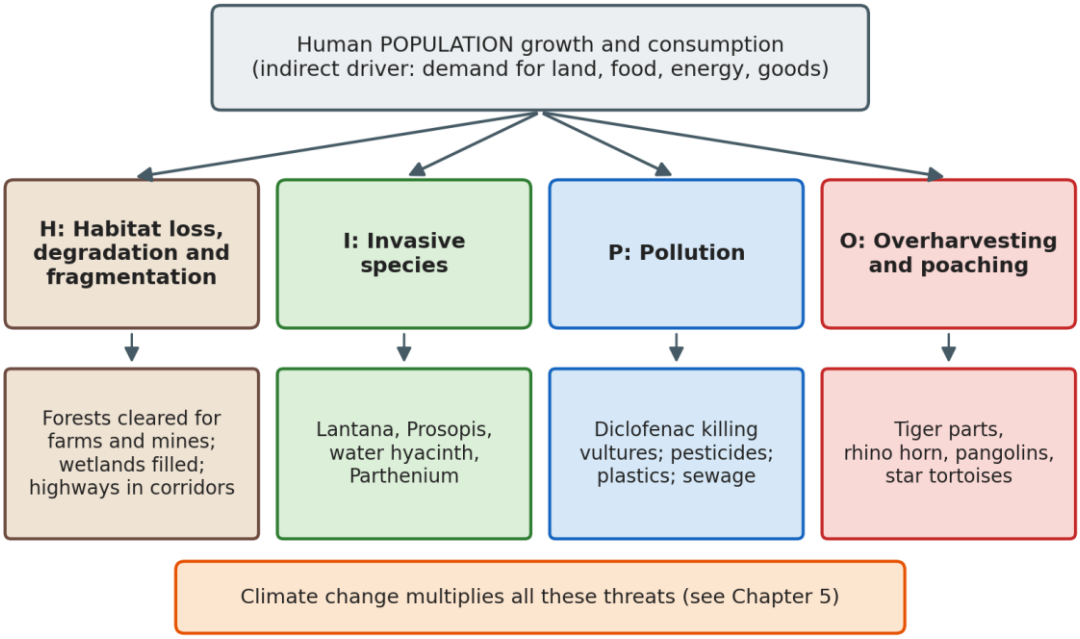


Figure 6.5: *The HIPPO framework of threats to biodiversity, with Indian examples. Human population and consumption drive the other four threats. Source: Author, after E. O. Wilson and IPBES (2019).*

In India these threats act together. A highway through a forest destroys habitat, splits wildlife populations, brings in weeds along its edges, makes poaching easier and kills animals in traffic. The sections below look at the threats named in the syllabus; pollution is covered in Chapter 4 and climate change in Chapter 5.

6.10.1 Habitat loss, degradation and fragmentation

Habitat loss is the complete conversion of a natural habitat to another use, such as a forest cleared for farmland, a wetland filled for housing or a grassland ploughed for crops. It is the single biggest cause of biodiversity loss worldwide and in India. Examples include:

- **Forests**: Forests have been cleared for agriculture, mining, dams, roads and settlements (see Chapter 3). Even where total forest cover is stable, natural forests are often replaced by plantations of teak, eucalyptus or rubber, which support far fewer species.
- **Wetlands**: India is estimated to have lost about a third of its natural wetlands between 1970 and 2014, many of them in and around cities. Chennai, Mumbai, Bengaluru and Kolkata have filled in lakes, marshes and mangroves for housing and roads, worsening urban floods.
- **Grasslands and other "open natural ecosystems"**: Savannas, grasslands, scrub and rocky plateaus cover roughly a tenth of India's land, but many are classified as "wasteland" in revenue records and are diverted to plantations, industries, solar farms and agriculture. The decline of the Great Indian bustard, lesser florican, Indian wolf and blackbuck reflects this loss.

Habitat degradation means that a habitat remains but its quality declines through overgrazing, repeated fires, fuelwood and fodder removal, invasive weeds, pollution or altered water flow. A degraded forest may look green from a satellite yet lack old trees and its original undergrowth.

Habitat fragmentation is the breaking up of a large, continuous habitat into smaller, isolated patches separated by farms, settlements, roads or other barriers. Fragmentation reduces biodiversity in several ways:

- **Smaller area**: Small patches can support only small populations, which are more likely to die out through chance events, disease or inbreeding. Large animals such as tigers and elephants need very large areas.
- **Isolation**: Animals cannot move between patches to find mates, food or new territories, so gene flow stops and populations lose genetic diversity.
- **Edge effects**: The boundary of a habitat patch differs from its interior. Edges receive more sunlight, wind and heat and are drier; they are invaded by weeds such as *Lantana*; and they are more exposed to grazing, fire, hunting and predators such as feral dogs. Species of the deep forest interior retreat from edges. The smaller and more irregular a fragment, the larger the proportion of it that is edge. A square fragment of 1 km^2 has no deep interior left if edge effects reach 500 m inside.

Studies in the rainforest fragments of the **Valparai plateau** in the Anamalai hills of Tamil Nadu, surrounded by tea and coffee estates, have shown that small and disturbed fragments lose rainforest trees with large seeds and their dispersers, such as hornbills and lion-tailed macaques, and that ecological restoration with native trees can help bring them back.

Linear infrastructure

Linear infrastructure means roads, railways, canals, power lines and pipelines. These cut through habitats in long strips. A study by the Centre for Wildlife Studies and colleagues (2020) found linear intrusions in most of India's protected areas. Their effects include:

- **Roadkill and train collisions**: Leopards, deer, reptiles, amphibians and even tigers die on highways, and elephants are killed by trains. MoEFCC data show that between 2009–10 and 2020–21 nearly 190 elephants were killed by trains in India, and more than 700 by electrocution, mostly from illegal electric fences and sagging power lines.
- **Power lines**: Overhead power lines in the Thar Desert are now the main cause of death of the Great Indian bustard (see the case study later in this chapter).
- **Barriers and access**: Busy highways and canals stop animal movement, while new roads open forests to logging, poaching, settlement and weeds.

Mitigation measures include routing projects away from critical habitat, **animal underpasses and overpasses**, speed limits and night-time traffic restrictions, elevated railway tracks or speed restrictions through elephant corridors, and insulating or burying power lines. The Wildlife Institute of India published guidelines on such "eco-friendly measures" in 2016.

Case Study: Underpasses on National Highway 44 at Pench

National Highway 44, India's longest highway, runs between the Pench and Kanha tiger reserves in Madhya Pradesh and Maharashtra, through one of central India's most important tiger corridors. When the road was widened to four lanes in the 2010s, conservationists and courts insisted on wildlife crossings. The final design included several animal underpasses, the longest stretching over a kilometre, built by raising the highway on pillars over forest and stream beds.

Camera traps set up by the Wildlife Institute of India in the first years of operation recorded tigers, leopards, sloth bears, gaur, deer, wild dogs and many other animals using the underpasses thousands of times. The project showed that well-designed crossings can keep a corridor open while allowing development, but also that they work only when the land on either side remains forest, noise and lighting are controlled, and the underpasses are wide and high enough.

6.10.2 Poaching and illegal wildlife trade

Poaching is the illegal killing or capture of wild animals, or the illegal collection of wild plants. It is driven by local demand for meat and traditional uses, and, more dangerously, by organised international trade in high-value parts. The global illegal wildlife trade is worth several billion US dollars a year and is often run by the same networks that smuggle drugs and arms.

Major targets in India include:

- **Tigers and leopards**: for skins, and for bones, claws and other parts used in traditional East Asian medicine and as luxury goods. Tiger parts are usually smuggled through Nepal, Myanmar and the North-East.
- **Rhinoceros**: for horn, which is made of keratin like fingernails but is falsely believed to have medicinal value. Poaching in Assam reached dozens of rhinos a year in the early 2010s; stronger protection, intelligence work and community support cut it to almost nil by the 2020s.
- **Elephants**: males for ivory; poaching has skewed sex ratios in some southern populations.
- **Pangolins**: Pangolins are among the most trafficked mammals in the world, for their scales and meat. TRAFFIC and WWF-India recorded at least 1,203 pangolins seized in illegal trade in India between 2018 and 2022, in 342 seizure incidents across 24 states. Both Indian species, the Indian pangolin (Endangered) and the Chinese pangolin (Critically Endangered), are in Schedule I of the Wild Life (Protection) Act.
- **Indian star tortoise**: Collected in thousands from the dry scrub of southern and western India for the exotic pet trade, especially in South-East Asia. At India's proposal, it was moved to Appendix I of CITES in 2019, banning commercial international trade.
- **Other species**: Parakeets, munias and owls for pets and black magic; sea cucumbers, seahorses and shark fins (in 2020 Lakshadweep notified the world's first conservation reserve for sea cucumbers); red sanders, sandalwood, orchids and medicinal plants; and shahtoosh wool from the Tibetan antelope, which must be killed to obtain it.

Response to wildlife crime

India's main law against poaching is the **Wild Life (Protection) Act, 1972**. Hunting of scheduled animals is banned, and trade in their parts is a serious offence. The **2022 amendment** reorganised

the schedules into four (Schedule I and II for animals with different levels of protection, Schedule III for plants and Schedule IV for species listed in CITES appendices), increased penalties and gave legal effect to India's obligations under **CITES** (the Convention on International Trade in Endangered Species of Wild Fauna and Flora; see Chapter 5).

The **Wildlife Crime Control Bureau (WCCB)**, set up by the central government in 2007 under the Act, collects intelligence on organised wildlife crime, coordinates action by police, forest, customs and border agencies, and trains enforcement staff. **TRAFFIC**, a global wildlife trade monitoring network that works in India with WWF-India, studies trade routes and supports enforcement. Other tools include DNA forensics (the Wildlife Institute of India and the Centre for Cellular and Molecular Biology can identify species and individual tigers from seized parts), sniffer dogs trained to detect wildlife products, and community-based intelligence networks. Low conviction rates in wildlife cases remain a weakness.

6.10.3 Man–wildlife conflict

Man–wildlife conflict (also called **human–wildlife conflict**) is any interaction between people and wild animals that harms people, their crops, livestock or property, or harms the animals in return. India, with its dense human population living alongside large populations of elephants, tigers, leopards, bears, wild pigs, nilgai and monkeys, faces conflict on a very large scale.

Scale of the problem

Data placed before Parliament by MoEFCC show that elephants killed 2,853 people in India in the five years from 2019–20 to 2023–24, an average of nearly 600 people every year, with Odisha, Jharkhand, West Bengal, Assam, Chhattisgarh and Tamil Nadu among the worst-affected states. Tigers killed 378 people in the five calendar years 2020–2024, with a peak of 110 in 2022, many of them around the Tadoba–Andhari landscape of Maharashtra and the Sundarbans. Leopards, sloth bears, wolves and crocodiles kill or injure many more people, and snakebite, though not always counted as conflict, kills tens of thousands of Indians every year.

Animals pay a heavy price too. Elephants die from electrocution by illegal fences, poisoning and train collisions; leopards and tigers are poisoned or beaten to death after attacks on livestock; and nilgai and wild pigs are shot or poisoned by farmers.

Table 6.9: Main forms of man–wildlife conflict in India

Species	Typical conflict	Example areas
Asian elephant	Crop raiding, house damage, human deaths	Odisha, Jharkhand, Chhattisgarh, Assam, Kerala, Karnataka
Tiger	Livestock killing, occasional attacks on people	Tadoba (Maharashtra), Sundarbans, terai of Uttar Pradesh
Leopard	Livestock and dog killing, attacks in villages and city edges	Junnar (Maharashtra), Uttarakhand, Mumbai's Sanjay Gandhi National Park
Nilgai and wild pig	Crop damage over large areas	Bihar, Uttar Pradesh, Madhya Pradesh, Rajasthan, Kerala
Rhesus and bonnet macaques	Crop damage, snatching food, bites	Himachal Pradesh, Uttarakhand, Delhi, temple towns

Species	Typical conflict	Example areas
Sloth bear	Sudden attacks on people collecting forest produce	Central India, Karnataka

Causes

- **Loss and fragmentation of habitat** push animals into farms and villages, and corridors linking forests are blocked by settlements, roads and canals.
- **Changes in land use near forests**: Crops such as sugarcane, paddy and banana are rich food for elephants and wild pigs; sugarcane fields in western Maharashtra give leopards ideal cover close to villages.
- **Declining natural prey and fodder**: Invasive weeds such as *Lantana* and *Senna spectabilis* in Wayanad and Mudumalai replace the grasses eaten by herbivores, pushing them out of forests.
- **Recovering wildlife populations**: Successful protection has increased numbers of tigers, elephants and leopards in some landscapes beyond what the remaining habitat can hold.
- **Human behaviour**: Open defecation in the dark, collecting forest produce alone, leaving garbage that attracts animals and feeding monkeys all increase risks.

Mitigation and compensation

There is no single solution; successful programmes combine several measures:

- **Barriers**: Solar-powered electric fences, elephant-proof trenches, and stone walls; in Karnataka's Kodagu district, the Khadi and Village Industries Commission has tested fences of bee boxes, since elephants avoid bees.
- **Early warning**: In the Valparai plateau, the Nature Conservation Foundation sends SMS alerts and switches on flashing red lights in villages when elephants are nearby, which has helped reduce deaths. Radio-collaring of elephant herds and leopards and thermal cameras along railway tracks serve the same purpose.
- **Corridors**: Securing corridors between forests lets elephants move without passing through villages. A 2023 report by MoEFCC identified 150 elephant corridors in India.
- **Land use and crops**: Growing crops that elephants and wild pigs do not like (such as chillies, citrus and ginger) near forest edges, and better livestock enclosures (predator-proof corrals) at night.
- **Rapid response and awareness**: Trained teams to drive away or capture animals (under the Act, the Chief Wildlife Warden may allow capture or, as a last resort, killing of an animal dangerous to human life), and teaching people how to behave near wildlife.
- **Compensation**: Quick and fair payment for losses reduces anger and retaliatory killing. In December 2023 the central government raised the ex gratia relief for a human death or permanent disability caused by wildlife to ₹10 lakh, with lower amounts for injuries; states set their own rates for crop and livestock losses. Delays and paperwork remain a major complaint, and programmes such as Wild Seve in Karnataka, which helps villagers file claims, have shown how much faster payment helps.

Some states have tried legal changes. Kerala declared man–wildlife conflict a state-specific disaster in 2024, so that disaster-management funds and powers could be used. In several states, wild pigs, nilgai

or rhesus macaques have been declared "vermin" in particular districts for limited periods so that they can be culled, a measure that is controversial because culling rarely solves conflict for long and can be misused.

Coexistence is possible

Not all interactions end in conflict. In Mumbai, about 50 leopards live in and around Sanjay Gandhi National Park, surrounded by a city of over 2 crore people. After a spate of attacks in the early 2000s, the forest department, researchers and residents started the "Mumbaikars for SGNP" programme to teach people how to live safely with leopards. Attacks on people have since fallen sharply, while the leopards continue to feed largely on stray dogs.

6.10.4 Biological invasion

An **alien** (exotic or non-native) **species** is one that has been introduced by humans, deliberately or accidentally, outside its natural range. Most aliens fail to survive, and many, such as most crops, are useful. A small fraction become **invasive alien species**: they establish wild populations, spread rapidly and harm native biodiversity, the economy or human health. This process is called **biological invasion**. Invasive species are a leading cause of extinctions on islands and are among the five main direct drivers of biodiversity loss.

Why invaders succeed

- **Enemy release**: In their new home they are free from the predators, parasites and diseases that kept them in check in their native range.
- **Rapid reproduction and spread**: Many produce huge numbers of seeds or young, grow fast and tolerate a wide range of conditions.
- **Disturbed habitats**: Degraded forests, road edges, overgrazed land and polluted water are easiest to invade.
- **Chemical weapons**: Some plants release chemicals that suppress the growth of others (**allelopathy**), as *Lantana* and *Parthenium* do.

Major invasive species in India

India has recorded well over a thousand alien species, of which a few hundred are considered invasive. A 2022 study estimated that invasive species cost the Indian economy between US$127 billion and US$182 billion from 1960 to 2020, mostly through crop losses, while spending on managing them was tiny by comparison.

Table 6.10: Some major invasive alien species in India

Species	Native region and arrival	Impacts in India
Lantana camara (lantana)	Central and South America; brought as a garden plant in the early 1800s	Dense thickets in forests; replaces fodder grasses; fire hazard; occupies about two-fifths of India's tiger range
Prosopis juliflora (vilayati babool, vilayati kikar)	Central and South America; planted from the late 1800s and widely in the 1960s–70s to green deserts	Has taken over the Banni grasslands of Kachchh and the Delhi Ridge; dries out soil, suppresses native plants

Species	Native region and arrival	Impacts in India
Eichhornia crassipes (water hyacinth)	South America; introduced to Bengal as an ornamental in the 1890s	Chokes lakes, rivers and canals; blocks light and oxygen, harms fish, breeds mosquitoes
Parthenium hysterophorus (congress grass, gajar ghas)	Tropical America; arrived in the 1950s, probably with imported wheat	Invades farms and roadsides; causes skin allergies and asthma in people and harms livestock
Clarias gariepinus (African catfish, African magur)	Africa; brought for fish farming in the 1990s	Voracious predator of native fishes; its culture is banned but continues illegally
Tilapia (*Oreochromis mossambicus* and *O. niloticus*)	Africa; introduced for fish culture from the 1950s	Dominates reservoirs and lakes, out-competes native carps and other fishes

Other important invaders include *Chromolaena odorata* and *Mikania micrantha* in the Western Ghats and the North-East, *Senna spectabilis* in the Nilgiri Biosphere Reserve, and chital deer introduced to the Andaman Islands.

Case Study: Lantana: from garden hedge to India's worst forest weed

Lantana camara, with its colourful clusters of small flowers, was brought from tropical America to the Calcutta Botanic Garden in the early nineteenth century as an ornamental plant and spread across the country as a hedge. Birds eat its berries and spread the seeds widely. Today it forms dense, thorny thickets in dry and moist deciduous forests across India. A 2020 study by scientists of the Wildlife Institute of India estimated that lantana had invaded about two-fifths of India's tiger range and was spreading faster in degraded forests.

Lantana harms forests in several ways. It shades out tree seedlings and fodder grasses, so herbivores such as chital and gaur lose food and carnivores in turn lose prey. Its leaves are toxic to livestock, its dry stems make forest fires hotter, and animals that lose fodder inside forests move into farms, adding to conflict.

Burning or cutting lantana simply makes it re-sprout. The most effective method, developed and tested in Indian forests, is the **cut-rootstock method**, in which each plant is cut just below the point where the stem meets the root, after which it seldom regrows. Cleared areas must then be replanted with native grasses and trees. In places such as the Biligiri Rangaswamy Temple Tiger Reserve in Karnataka, Soliga artisans and local craftspeople make furniture and handicrafts from lantana stems, turning a pest into income while removing it from the forest.

Control of invasive species

Managing invasions is far cheaper at the start than after a species has spread. The main approaches are:

- **Prevention**: Quarantine checks at ports and airports, risk assessment before importing new species, and public awareness so that people do not release aquarium fish or plant invasive ornamentals.

- **Early detection and mechanical control**: Removing small new populations before they spread; uprooting, cutting and dredging established ones.
- **Biological control**: Releasing natural enemies from the invader's home range after careful testing. The Mexican beetle *Zygogramma bicolorata*, released in India from the 1980s, eats *Parthenium*, and *Neochetina* weevils have been used against water hyacinth.
- **Chemical control and utilisation**: Herbicides, used sparingly because they harm native species; and use of invasive biomass for crafts, compost, biogas or charcoal, which helps only if it reduces the invader.
- **Restoration**: Replanting cleared land with native species so that the invader does not simply return.

The Kunming–Montreal Global Biodiversity Framework of 2022 sets a target of reducing the rate of introduction and establishment of invasive alien species by at least half by 2030, and India's updated National Biodiversity Strategy and Action Plan (2024) includes a matching national target.

6.10.5 The current mass-extinction crisis

Past mass extinctions

Extinction is the death of the last individual of a species. Throughout the history of life, species have arisen and gone extinct at a slow **background rate**. A **mass extinction** is a geologically short period in which a large share of all species, conventionally at least three-quarters, disappears. The fossil record shows five such events, called the "Big Five".

Table 6.11: The five past mass extinctions

Event	Approximate time (million years ago)	Estimated species lost	Probable causes
End-Ordovician	445	About 86%	Glaciation and falling sea levels
Late Devonian	372	About 75%	Loss of oxygen in oceans, climate change
End-Permian	252	About 90–96%	Massive volcanic eruptions in Siberia, global warming, ocean acidification
End-Triassic	201	About 80%	Volcanic eruptions as the supercontinent Pangaea split
End-Cretaceous	66	About 76%	Asteroid impact near Mexico, with volcanism (Deccan Traps of India)

The end-Cretaceous event, which wiped out the non-bird dinosaurs, coincided with the outpouring of lava that formed the **Deccan Traps** of western and central India, and scientists still debate how much each cause contributed. After each mass extinction it took life millions of years to recover its former diversity.

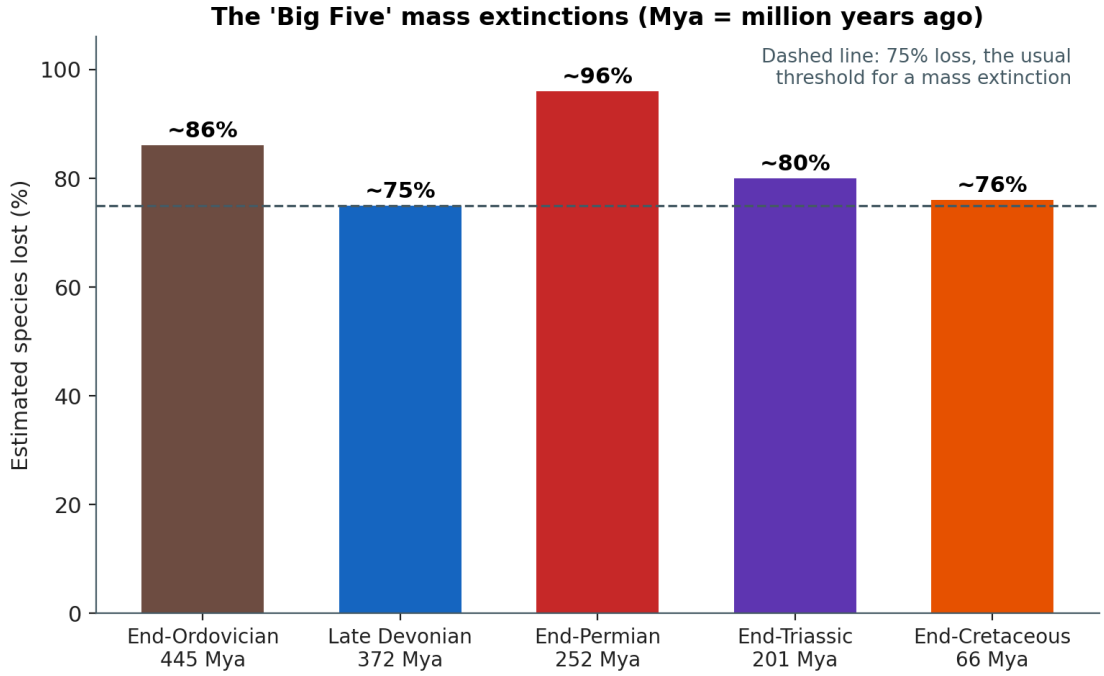


Figure 6.6: *Estimated percentage of species lost in the five past mass extinctions. Source: Compiled from Barnosky et al. (2011, Nature) and other palaeontological estimates.*

Is a sixth mass extinction under way?

Many scientists argue that human activity has started a sixth mass extinction. The evidence includes:

- **Recorded extinctions**: The IUCN lists about 900 species as extinct since 1500, including the dodo, the passenger pigeon and the Tasmanian tiger. The true number is much larger, because most species are never described before they vanish.
- **Rate of extinction**: Current extinction rates are estimated to be tens to hundreds of times higher than the average rate over the past ten million years, and they are accelerating (IPBES, 2019).
- **Species at risk**: The IPBES Global Assessment (2019) estimated that about **1 million** of the world's estimated 8 million species of plants and animals are threatened with extinction, many within decades, unless action is taken. About 28% of species assessed by the IUCN are threatened.
- **Population decline**: The **Living Planet Index**, compiled by the Zoological Society of London and published in WWF's *Living Planet Report*, tracks tens of thousands of populations of vertebrate species. The 2024 report found an average decline of **73%** in monitored wildlife populations between 1970 and 2020, with the steepest fall in freshwater populations (85%), followed by land (69%) and marine (56%) populations; a 2026 update covering 1970–2022 again reported a 73% average decline. Note that this is the average change in population size, not the share of species lost.

Some scientists caution against calling the present crisis a mass extinction. A 2025 study by Wiens and Saban found that only a small fraction of genera (groups of related species) have become extinct in the past 500 years, mostly on islands, far below the losses of the Big Five. Both sides agree, however, that the speed of population decline and habitat loss is alarming, and that if current trends continue for a few more centuries a true mass extinction is likely. The difference from the past is that this crisis has a single cause, human activity, and so can still be slowed or stopped.

Causes of the present crisis

The causes are the HIPPO threats described above: habitat loss, overexploitation, invasive species, pollution and, increasingly, climate change, all driven by human population growth and rising consumption. They act together.

In response, 196 countries adopted the **Kunming–Montreal Global Biodiversity Framework** in December 2022 under the CBD. Its 23 targets for 2030 include protecting at least 30% of the world's land and seas (the "30 by 30" target), restoring 30% of degraded ecosystems, halting human-induced extinctions of known threatened species, and cutting harmful subsidies (see Chapter 5).

Key Points

- HIPPO: habitat destruction, invasive species, pollution, human population growth and overharvesting are the main threats to biodiversity.
- Fragmentation shrinks and isolates populations; edge effects and linear infrastructure degrade what remains.
- Poaching for international trade targets tigers, rhinos, pangolins and star tortoises; the WCCB and the amended Wild Life (Protection) Act are India's main responses.
- Man–wildlife conflict kills about 600 people a year through elephants alone; barriers, early warning, corridors and fair compensation reduce it.
- Invasive species such as lantana, Prosopis, water hyacinth and Parthenium cost India billions of dollars and degrade habitats.
- IPBES (2019) estimates about 1 million species are threatened; monitored vertebrate populations fell 73% on average since 1970.

6.11 Biodiversity Conservation Strategies: In-situ and Ex-situ

Conservation strategies fall into two broad groups. **In-situ conservation** ("on site") protects species in their natural habitats, together with the ecosystems and processes they depend on. **Ex-situ conservation** ("off site") keeps species, or their seeds, cells and genes, outside their natural habitats, in places such as zoos, botanical gardens and gene banks. The CBD treats in-situ conservation as the main approach and ex-situ methods as a support to it.

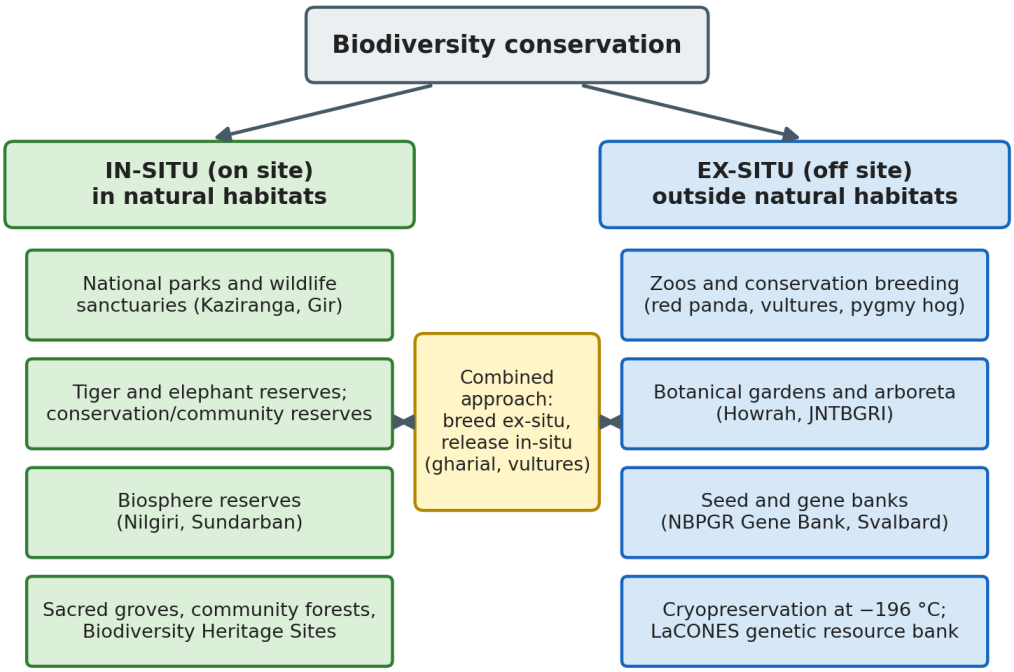


Figure 6.7: *In-situ and ex-situ methods of biodiversity conservation, with Indian examples. Source: Author.*

6.11.1 In-situ conservation

In-situ conservation keeps whole ecosystems working and allows species to go on evolving in response to their environment. It is the only practical way to conserve the vast majority of species, most of which are unknown. Its main forms in India are:

- **Protected areas** under the Wild Life (Protection) Act: national parks, wildlife sanctuaries, conservation reserves and community reserves (described in the next section), with tiger reserves and elephant reserves overlaid on them.
- **Biosphere reserves**, which combine strict protection with sustainable use (also described below).
- **Wetlands of international importance (Ramsar sites)**, of which India had more than 90 by 2025 (see Chapter 5), and other specially managed areas such as marine protected areas.
- **Community conserved areas**: sacred groves, village forests and *van panchayats* in Uttarakhand, community forests protected by villages in Odisha and the North-East, and Biodiversity Heritage Sites under the Biological Diversity Act.
- **Conservation outside protected areas**: corridors, reserved forests, Biodiversity Management Committees, on-farm conservation of crop varieties by farmers, and restoration of degraded land.

In-situ conservation faces its own problems: conflict with local people over land and resources, poaching, invasive species, inadequate funding and staff, and climate change, which may move suitable habitat outside the boundaries of a protected area.

6.11.2 Ex-situ conservation

Ex-situ conservation is essential when a species' numbers in the wild are so low, or its threats so severe, that it may not survive without help. It also stores genetic diversity as insurance, supports

research and education, and produces animals and plants for reintroduction. Its main forms are described below.

Zoos and conservation breeding centres

Modern zoos aim to be centres for conservation and education rather than menageries. In India, zoos are regulated by the **Central Zoo Authority**, set up in 1992 under the Wild Life (Protection) Act. Through its **conservation breeding programme**, selected zoos and centres breed endangered species such as the red panda (Padmaja Naidu Himalayan Zoological Park, Darjeeling), the lion-tailed macaque (Mysuru zoo), the Asiatic lion (Sakkarbaug zoo, Junagadh) and the Manipur brow-antlered deer. Specialised centres breed the pygmy hog (near Guwahati), *Gyps* vultures (Pinjore in Haryana and other centres) and gharials (Kukrail in Lucknow and centres in Madhya Pradesh). Captive breeding needs careful records of ancestry (a **studbook**) so that inbreeding is avoided. Animals bred over many generations in captivity can lose wild behaviour, so programmes aimed at release keep animals in large, natural enclosures with minimal human contact.

Botanical gardens and arboreta

Botanical gardens grow living collections of plants, including rare and threatened species, and are centres of taxonomy and propagation. Important Indian examples include the Acharya Jagadish Chandra Bose Indian Botanic Garden at Howrah (founded in 1787, with its famous giant banyan), the Lloyd Botanic Garden at Darjeeling, the CSIR–National Botanical Research Institute garden at Lucknow and the Jawaharlal Nehru Tropical Botanic Garden and Research Institute near Thiruvananthapuram, which holds large collections of Western Ghats plants and orchids. **Arboreta** specialise in trees, and **field gene banks** maintain living collections of crops that cannot be stored as seed, such as mango, banana, coconut and tea.

Seed banks and gene banks

A **seed bank** stores seeds in cold, dry conditions so that they stay alive for decades. Seeds that tolerate drying and freezing (**orthodox seeds**, such as those of wheat, rice and pulses) are dried to low moisture and stored at about −18 to −20 °C. Seeds that die when dried (**recalcitrant seeds**, such as those of mango, jackfruit, cocoa and many tropical trees) must be kept as living plants, as tissue cultures, or by cryopreservation.

- India's **National Gene Bank** at the ICAR–National Bureau of Plant Genetic Resources (NBPGR), New Delhi, set up in 1996, is one of the largest in the world. It holds about 4.5 lakh accessions (samples) of crops and their wild relatives in long-term storage, along with in vitro and cryopreserved collections.
- The **Svalbard Global Seed Vault**, built in 2008 inside a mountain on a Norwegian Arctic island, stores duplicate "back-up" samples from gene banks around the world, including India's, as insurance against war or disaster. It holds over a million seed samples.

Sister ICAR bureaux conserve the genetic resources of livestock (NBAGR, Karnal) and fish (NBFGR, Lucknow).

Cryopreservation and genetic resource banks

Cryopreservation is storage of living cells, tissues, embryos, sperm, eggs, pollen or shoot tips in liquid nitrogen at −196 °C, where biological activity stops almost completely. Material can be stored

for very long periods and revived later. NBPGR's cryo-gene bank stores recalcitrant seeds, embryos, pollen and buds.

For wild animals, the **Laboratory for the Conservation of Endangered Species (LaCONES)**, part of the CSIR–Centre for Cellular and Molecular Biology (CCMB) in Hyderabad, set up with the Central Zoo Authority, maintains a **genetic resource bank** (sometimes called a "frozen zoo") of sperm, eggs, cells and DNA from endangered Indian animals. It works on assisted reproduction (artificial insemination and in vitro fertilisation) for species such as blackbuck and spotted deer, and on **wildlife forensics**, identifying species from seized skins, meat or ivory to help prosecute poachers.

Comparing the two approaches

Table 6.12: In-situ and ex-situ conservation compared

Feature	In-situ conservation	Ex-situ conservation
Where	In natural habitats	Outside natural habitats
What is conserved	Whole ecosystems, species and processes	Selected species or their genetic material
Evolution	Continues in response to the environment	Largely frozen; captive animals may adapt to captivity
Cost per species	Low, as many species are saved together	High for each species
Main risk	Habitat loss, poaching, conflict, climate change	Inbreeding, disease outbreaks, loss of wild behaviour
Indian examples	Kaziranga National Park, Nilgiri Biosphere Reserve, sacred groves	National Gene Bank, Pinjore vulture centre, LaCONES

The two approaches work best together. The recovery of the gharial, the pygmy hog and the vultures depended on breeding animals ex-situ while making the wild habitat safe for their return.

6.12 National Parks, Wildlife Sanctuaries and Biosphere Reserves

6.12.1 Protected areas under the Wild Life (Protection) Act

A **protected area** is a clearly defined space recognised and managed, through legal or other effective means, to conserve nature over the long term. The IUCN classifies protected areas into categories from strict nature reserves (Category Ia) to areas with sustainable use of resources (Category VI). In India, protected areas are created under the **Wild Life (Protection) Act, 1972**, which provides four types.

- A **wildlife sanctuary** is declared by the state government to protect wildlife and its environment. Some rights of local people, such as grazing or collecting produce, may be allowed if they do not harm wildlife.
- A **national park** is an area of greater ecological importance where all private rights are settled and extinguished; grazing and the removal of forest produce are not allowed. A sanctuary may be upgraded to a national park.

- A **conservation reserve** is government-owned land, often adjoining a protected area or linking two of them, declared after consulting local communities to protect landscapes, seascapes and habitats.
- A **community reserve** is private or community land where the community or individuals volunteer to conserve wildlife and its habitat.

Conservation reserves and community reserves were added by the 2002 amendment to the Act, to bring community land and corridors into the network without taking away people's rights. **Tiger reserves**, created under the National Tiger Conservation Authority (added to the Act in 2006), consist of a core area, which is usually a national park or sanctuary, and a buffer area where people live (see the Project Tiger case study).

Table 6.13: Types of protected areas in India (WII National Wildlife Database, 2025; figures are revised often)

Type	Number	Human rights and use	Example
National parks	107	No private rights; no grazing or extraction	Kaziranga (Assam), Hemis (Ladakh, the largest)
Wildlife sanctuaries	573	Some rights may be allowed	Rann of Kachchh Wildlife Sanctuary (Gujarat, the largest), Bhitarkanika (Odisha)
Conservation reserves	115	Government land; local people consulted	Many in Rajasthan, Karnataka and Uttarakhand
Community reserves	220	Community or private land managed by the community	Keshopur–Miani (Punjab), many in Meghalaya
All protected areas	About 1,015	About 5.3% of India's geographical area	Far below the global "30 by 30" target

India's first national park, Hailey National Park, was set up in 1936 in what is now Uttarakhand; it was renamed **Corbett National Park** in the 1950s and became the first site of Project Tiger in 1973.

6.12.2 Biosphere reserves

A **biosphere reserve** is a large area of land or coast, recognised under UNESCO's **Man and the Biosphere (MAB) Programme** (launched in 1971) or by the national government, that aims to reconcile conservation of biodiversity with its sustainable use. A biosphere reserve is not a separate legal category under the Wild Life (Protection) Act; it is a planning and management framework laid over existing protected areas and the lands around them. Each biosphere reserve has three functions:

1. **Conservation** of landscapes, ecosystems, species and genetic variation.
2. **Development** that is economically and culturally sustainable for local people.
3. **Logistic support** for research, monitoring, education and training.

These functions are carried out in three zones:

- The **core zone** is strictly protected and undisturbed, usually a national park or wildlife sanctuary. Only monitoring and non-destructive research are allowed.
- The **buffer zone** surrounds the core. Activities compatible with conservation, such as research, education, ecotourism and regulated traditional use, are allowed.

- The **transition zone** (or area of cooperation) is the outer zone where people live and farm, and where sustainable resource use, such as organic farming, fisheries and forestry, is promoted in partnership with communities.

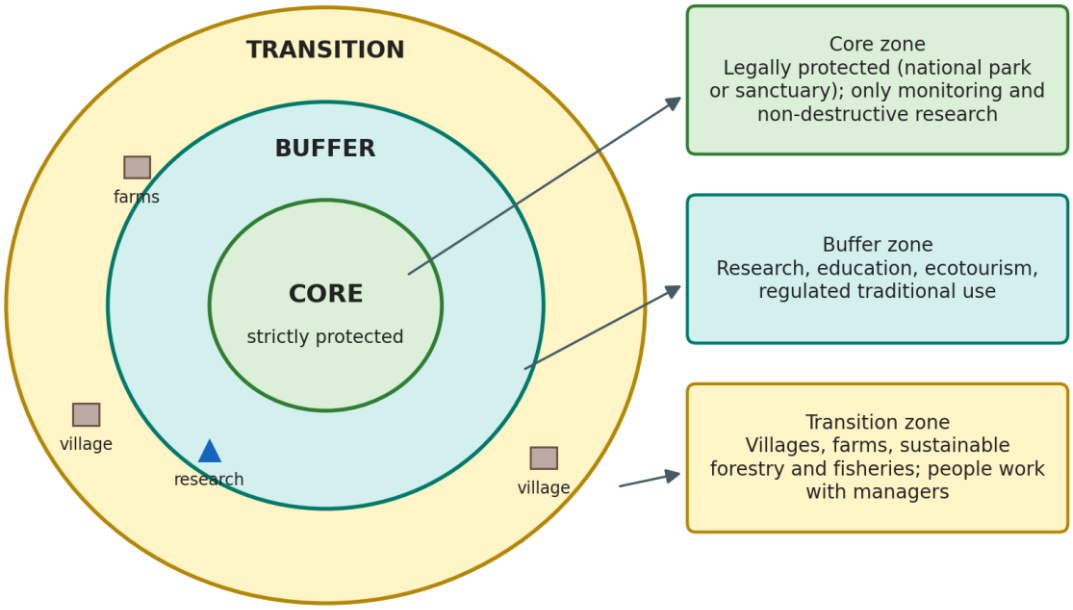


Figure 6.8: *Zonation of a biosphere reserve into core, buffer and transition zones, with the activities permitted in each. Source: Adapted from UNESCO Man and the Biosphere Programme.*

India has designated **18 biosphere reserves**. The first was the **Nilgiri Biosphere Reserve** (1986), spread across Tamil Nadu, Kerala and Karnataka. The largest is the Kachchh Biosphere Reserve in Gujarat. Thirteen Indian biosphere reserves are part of UNESCO's **World Network of Biosphere Reserves**, which links about 780 sites in more than 140 countries.

Table 6.14: Indian biosphere reserves in UNESCO's World Network (year of inclusion)

Biosphere reserve	State or UT	Year	Notable features
Nilgiri	Tamil Nadu, Kerala, Karnataka	2000	Elephants, tigers, Nilgiri tahr, tribal communities
Gulf of Mannar	Tamil Nadu	2001	Coral reefs, seagrass beds, dugong
Sundarban	West Bengal	2001	World's largest mangrove forest, tigers
Nanda Devi	Uttarakhand	2004	High Himalaya, Valley of Flowers
Nokrek	Meghalaya	2009	Wild citrus, red panda
Pachmarhi	Madhya Pradesh	2009	Satpura hills, sal and teak forests
Similipal	Odisha	2009	Tigers, elephants, sal forests
Achanakmar–Amarkantak	Madhya Pradesh, Chhattisgarh	2012	Source of the Narmada and Son rivers
Great Nicobar	Andaman and Nicobar Islands	2013	Tropical rainforest, leatherback turtles
Agasthyamala	Kerala, Tamil Nadu	2016	Western Ghats endemics, Kani tribe

Biosphere reserve	State or UT	Year	Notable features
Khangchendzonga	Sikkim	2018	High-altitude forests, snow leopard, red panda
Panna	Madhya Pradesh	2020	Tiger reintroduction success, Ken river
Cold Desert	Himachal Pradesh	2025	Trans-Himalayan cold desert of Spiti and Lahaul

The other five national biosphere reserves are Manas, Dibru-Saikhowa, Dehang-Dibang, Kachchh and Seshachalam.

Table 6.15: National parks, wildlife sanctuaries and biosphere reserves compared

Feature	National park	Wildlife sanctuary	Biosphere reserve
Legal basis	Wild Life (Protection) Act	Wild Life (Protection) Act	UNESCO MAB / national scheme
Main aim	Strict protection of ecosystems and wildlife	Protection of wildlife, often particular species	Conservation with sustainable development
People's rights	Extinguished	Some may continue	People live in buffer and transition zones
Size	Small to large	Small to large	Usually very large, includes several PAs
Example	Kaziranga	Keoladeo was a sanctuary before becoming a national park	Nilgiri

6.13 Keystone, Flagship, Umbrella and Indicator Species

Conservation agencies cannot study and manage every species. They therefore use a few well-chosen **surrogate species** to guide decisions. The four main kinds differ in what they are chosen for: ecological role, public appeal, area needs or sensitivity.

A **keystone species** has an effect on its community far larger than its numbers suggest; if it is removed, the community changes dramatically. The idea comes from the American ecologist Robert Paine, who in 1969 showed that removing a predatory starfish from rocky shores caused mussels to take over and the number of species to fall. Keystones may be top predators, ecosystem engineers, seed dispersers or providers of key resources.

A **flagship species** is a popular, charismatic species chosen to attract public attention, funds and political support for conservation. It need not be ecologically special.

An **umbrella species** needs such large areas of habitat that protecting enough land for it automatically protects many other species living there.

An **indicator species** is one whose presence, absence, numbers or condition reveals the health of an environment, because it is sensitive to pollution or habitat change.

Table 6.16: Keystone, flagship, umbrella and indicator species with Indian examples

Type	Chosen because	Indian examples	How it helps conservation
Keystone	Large ecological role relative to its numbers	Fig trees (fruit for many birds and mammals in lean seasons), tiger (controls herbivores), Asian elephant (opens forest, disperses seeds), hornbills (disperse seeds of large-fruited trees), vultures (dispose of carcasses)	Protecting it maintains the structure of the whole community
Flagship	Charisma and public appeal	Bengal tiger, Asian elephant, one-horned rhino, snow leopard, Gangetic dolphin, olive ridley turtle	Raises awareness, funds and political will
Umbrella	Very large area needs	Tiger, Asian elephant, snow leopard, Great Indian bustard (grasslands), Asiatic lion	Protecting its range protects many co-occurring species
Indicator	Sensitivity to environmental change	Lichens (air pollution), frogs and stream insects such as mayflies and stoneflies (water quality), Gangetic dolphin (river health), corals (sea warming), house sparrow (urban habitat quality)	Gives early warning of environmental damage

The same species can play several roles. The **tiger** is a flagship that draws public support, an umbrella whose reserves protect thousands of other species, and a top predator that shapes its ecosystem. **Fig trees** are keystones in Indian forests because they fruit at different times of year, feeding hornbills, barbets, bulbuls, monkeys, bats and civets when little other fruit is available. **Lichens** are classic indicators: most species disappear where sulfur dioxide levels are high, and lichen surveys have been used to map air pollution in Indian cities.

Surrogate species have limits. Flagships draw money to a few large mammals while less charismatic species such as frogs, fishes, insects and plants are neglected. Umbrella species do not protect everything: a tiger reserve in a forest does not protect a grassland bird. Indicators must be chosen and calibrated carefully. Surrogates are therefore best used together and alongside habitat-based planning.

6.14 Species Reintroduction and Translocation

6.14.1 Definitions and IUCN guidelines

A **conservation translocation** is the deliberate movement of living organisms from one place to release them in another, for a conservation purpose. The IUCN's *Guidelines for Reintroductions and Other Conservation Translocations* (2013) distinguish:

- **Population restoration**, within the species' natural range:
  - **Reinforcement** (or supplementation): adding individuals to an existing population to increase its size or genetic diversity.

  - **Reintroduction**: releasing a species in part of its natural range from which it has disappeared.
- **Conservation introduction**, outside the natural range:
  - **Assisted colonisation**: moving a species to a new area to save it from extinction, for example because climate change is making its present range unsuitable.
  - **Ecological replacement**: releasing a related species or subspecies to perform the ecological role of one that has become extinct.

The guidelines stress that translocations are risky and costly and should be undertaken only when the original causes of decline have been removed. They require a **feasibility assessment** (habitat, prey, genetics, disease risks, social acceptance), careful choice of source population, **soft release** where appropriate (animals first held in enclosures at the release site to settle), long-term **monitoring**, an exit strategy if things go wrong, and involvement of local people. In India, translocations of scheduled species need the approval of the Chief Wildlife Warden, the National Board for Wildlife and, for tigers, the National Tiger Conservation Authority.

6.14.2 Indian experience

India has carried out many translocations, with mixed results:

- **Tigers in Sariska and Panna**: Tigers were lost to poaching in Sariska (by 2005) and Panna (by 2009) and were reintroduced from other reserves from 2008 and 2009; Panna's population recovered strongly (see the Project Tiger case study).
- **Hard-ground barasingha**: Swamp deer from Kanha, their last stronghold, have been moved to Satpura and other reserves to create second populations.
- **Gharial**: Eggs collected from the wild are hatched and the young reared until they are large enough to escape most predators, then released, a method called **head-starting** (see the crocodile conservation case study).
- **Pygmy hog and vultures**: Captive-bred animals are released into restored habitats in Assam and, for vultures, into "vulture safe zones".

Case Study: Indian Rhino Vision 2020

In 2005 the Assam government, with WWF-India, the International Rhino Foundation and the Bodoland Territorial Council, launched **Indian Rhino Vision 2020 (IRV 2020)**. Most of Assam's greater one-horned rhinos lived in just two areas, Kaziranga National Park and Pobitora Wildlife Sanctuary, so a single disease outbreak, flood or poaching wave could devastate the species. The aim was to raise Assam's rhino population to about 3,000 spread over seven protected areas by 2020.

Manas National Park had lost its entire rhino population to poaching during civil unrest in the 1980s and 1990s. Between 2008 and 2021, IRV 2020 moved 22 wild rhinos from Kaziranga and Pobitora to Manas, while rhinos rescued as orphans and hand-reared at a rehabilitation centre near Kaziranga were also released there. Protection was strengthened with local communities, many former poachers among them. Some translocated rhinos were poached in the early years, but the population took hold and had grown to around 50 by the early 2020s, and Manas was removed from UNESCO's list of World Heritage in Danger in 2011.

The programme fell short of its target of seven populations, but it showed that rhinos could be moved safely and could re-establish themselves where the threats were controlled. Assam's 2022 census counted 2,613 rhinos in Kaziranga alone.

The Asiatic lion relocation debate

All wild Asiatic lions live in one landscape, the Gir forest and surrounding areas of Saurashtra in Gujarat. Their numbers have risen from a few dozen in the early twentieth century to **891 in 2025**, and more than half now live outside the protected area, among villages and farms. Having the whole species in one place exposes it to epidemics, natural disasters and inbreeding.

In the 1990s the Wildlife Institute of India identified **Kuno** in Madhya Pradesh as a second home, and about two dozen villages, mostly of the Sahariya tribe, were relocated from the sanctuary. Gujarat refused to part with its lions. In April 2013 the Supreme Court ordered that some lions be translocated to Kuno within six months, stressing that the survival of the species mattered more than any state's pride. The order was never carried out. The canine distemper outbreak of 2018 at Gir, which killed more than twenty lions, showed that the danger is real. Gujarat has instead promoted the expansion of lions into other parts of Saurashtra, such as Barda Wildlife Sanctuary, as a second home within the state, while Kuno received cheetahs.

Case Study: Project Cheetah at Kuno

On 17 September 2022, eight cheetahs from Namibia were released into enclosures at **Kuno National Park**, Madhya Pradesh, seventy years after the cheetah was declared extinct in India. Twelve more arrived from South Africa in February 2023. It was the world's first intercontinental translocation of a large carnivore. Because the African cheetah is a different subspecies from the extinct Asiatic cheetah, the project is an example of **ecological replacement**. Its aims were to restore a top predator of open grasslands and savannas, to use the cheetah as a flagship for these neglected ecosystems, and to boost local ecotourism.

The project ran into serious problems. Several adults and most cubs of the first Indian-born litter (March 2023) died; among the causes were infections under radio-collars, where the cheetahs' thick African winter coats trapped moisture during the Indian monsoon, fights, and extreme heat. Many animals were therefore kept for long periods in fenced enclosures. Critics pointed to the limited size of Kuno (a national park of about 750 km^2), the presence of leopards, inadequate prey in some areas, the risk of conflict when cheetahs wandered into villages, and the lack of independent scientific review. Supporters noted that early losses are normal in large-carnivore reintroductions.

By 2025 cheetahs had bred repeatedly at Kuno, and a second site, **Gandhi Sagar Wildlife Sanctuary** in Madhya Pradesh, received its first cheetahs in April 2025. A further batch was brought from Botswana in 2026. By the project's fourth anniversary in September 2026, the population was reported at about 50, more than half of them born in India, though many were still in enclosures rather than free-ranging. Whether a self-sustaining wild population can be established remains the key test.

Reintroduction is one of the most visible tools of conservation, but it treats the symptom (a missing species) rather than the cause (loss of habitat and persecution). The lesson from India's experience is the one stressed by the IUCN: translocations succeed when habitat, prey, protection and local support are in place before the animals arrive.

The case studies that follow show these principles at work in India's best-known conservation programmes and people's movements.

6.15 Conservation in Practice: Indian Case Studies

The ideas discussed so far in this chapter — flagship and umbrella species, in-situ and ex-situ conservation, protected areas, reintroduction — are easiest to understand when you see them at work. India offers some of the world's best-known examples. Since the Wildlife (Protection) Act of 1972, the country has run species programmes for its tigers, elephants, crocodiles, vultures and bustards, and it has seen citizens' movements, such as those in Silent Valley and the Western Ghats, change government decisions about forests and dams. Some of these efforts are celebrated successes; others are still a race against extinction. All of them show that conservation is as much about people, money, law and politics as it is about biology.

The case studies in this section follow a common pattern: the background to each crisis, the objectives set, the strategies used, the results achieved (with data), the problems and criticisms, and the lessons

learned. The table below gives a quick comparison. Notice the shift over time — from protecting single charismatic animals inside reserves (1970s) to managing whole landscapes, corridors, veterinary drugs and power lines (2000s onwards).

Table 6.17: Major conservation programmes and movements discussed in this section

Programme or movement	Began	Species or area	Lead agency	Key outcome
Project Tiger	1973	Tiger and its forests	MoEFCC; NTCA since 2006	9 reserves grew to 58; about 3,682 tigers (2022)
Crocodile Conservation Project	1975	Gharial, mugger, saltwater crocodile	MoEFCC with UNDP/FAO; state forest departments	All three species pulled back from the brink
Silent Valley movement	1973–1984	Rainforest in Palakkad, Kerala	KSSP, scientists, writers	Dam shelved (1983); National Park (1984)
Save Western Ghats movement	1987–88	Entire Western Ghats range	Voluntary groups; later WGEEP (Gadgil)	Landscape-level thinking; ESA debate still open
Project Elephant	1992	Asian elephant, habitats, corridors	MoEFCC; state forest departments	33 elephant reserves; 150 corridors mapped
Vulture conservation breeding	2001/2004	Three *Gyps* vultures	BNHS with state forest departments	Diclofenac banned (2006); captive-bred birds released
Great Indian Bustard programme	2013 (Rajasthan); breeding 2018–19	Great Indian Bustard and grasslands	Rajasthan Forest Dept, WII, MoEFCC	First captive-bred chicks; court limits on power lines

The timeline below places these efforts alongside the laws and institutions that made them possible.

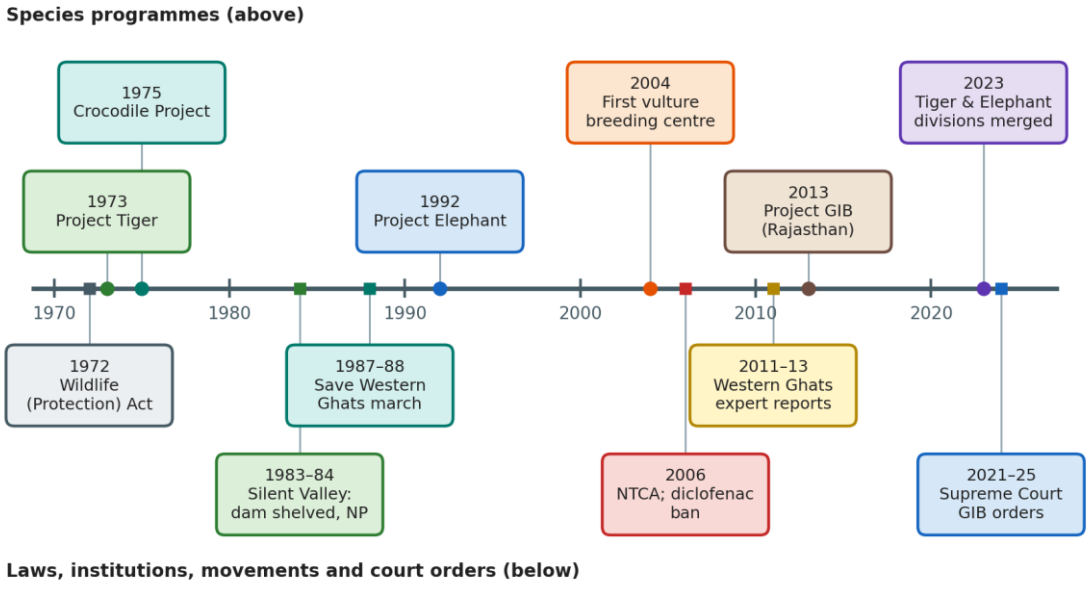


Figure 6.9: *Timeline of major Indian conservation programmes, laws and movements, 1972–2025. Source: compiled from MoEFCC, NTCA and BNHS records.*

6.16 Project Tiger

Project Tiger is India's oldest and best-known species conservation programme. It was launched on 1 April 1973 at Corbett National Park to save the tiger (*Panthera tigris*) from extinction in India. Fifty years later, India holds about three-quarters of the world's wild tigers. The project also shows how a single flagship species can be used to protect entire forests, rivers and the many species living in them.

6.16.1 Background: the collapse of India's tigers

Around 1900, India is often estimated to have had several tens of thousands of tigers (figures of 40,000 or more are commonly quoted, though no count was made). Tigers were treated as vermin and as trophies. Royal and colonial **shikar** (hunting) parties killed thousands; one Maharaja of Surguja is said to have shot more than 1,100 tigers by himself. After Independence, the loss of forests to farms, dams and roads, and a booming international trade in tiger skins, made matters worse.

The first all-India tiger count in 1972, based on pugmarks, gave a figure of only about 1,800. The shock pushed the government into action. Export of tiger skins and hunting had already been restricted, and the **Wildlife (Protection) Act, 1972** gave the legal tools to create protected areas (see Chapter 5 for the Act). A task force of the Indian Board for Wildlife recommended a special programme, and Prime Minister Indira Gandhi launched Project Tiger in 1973 with **Kailash Sankhala** as its first director.

6.16.2 Objectives

The original objectives were:

- To maintain a viable population of tigers in India for scientific, economic, aesthetic, cultural and ecological values.
- To preserve, for all time, areas of biological importance as a national heritage for the benefit, education and enjoyment of the people.

Over time, further aims were added: reducing human–tiger conflict, securing corridors between reserves, supporting forest-dwelling communities, and monitoring tiger numbers scientifically.

6.16.3 Strategy: the tiger reserve

The core idea of Project Tiger is the **tiger reserve**, a protected landscape managed mainly for tigers and their prey. The first nine reserves (1973) were Corbett, Manas, Melghat, Palamau, Ranthambore, Similipal, Sundarbans, Bandipur and Kanha, covering about 9,000 km^2. By March 2025, when Madhav Tiger Reserve in Madhya Pradesh was notified, India had **58 tiger reserves** in 18 states, together covering more than 80,000 km^2, or roughly 2.5% of the country's land area.

Each reserve is planned as two zones:

- **Core or Critical Tiger Habitat (CTH)**: usually a national park or sanctuary, to be kept **inviolate**, that is, free of settlements, grazing, mining and other human use, so that tigers can breed undisturbed.
- **Buffer zone**: the surrounding area where people live and work. Land use is allowed but managed to reduce conflict, and villagers are supported through **eco-development** (alternative fuel, livelihoods, jobs in tourism and protection).

Reserves are linked to one another by **corridors**, strips of forest that let tigers disperse and keep populations genetically healthy. Tiger conservation is therefore now planned at the scale of

landscapes, such as the Central Indian landscape (Kanha–Pench–Satpura–Melghat) or the Terai Arc along the Himalayan foothills.

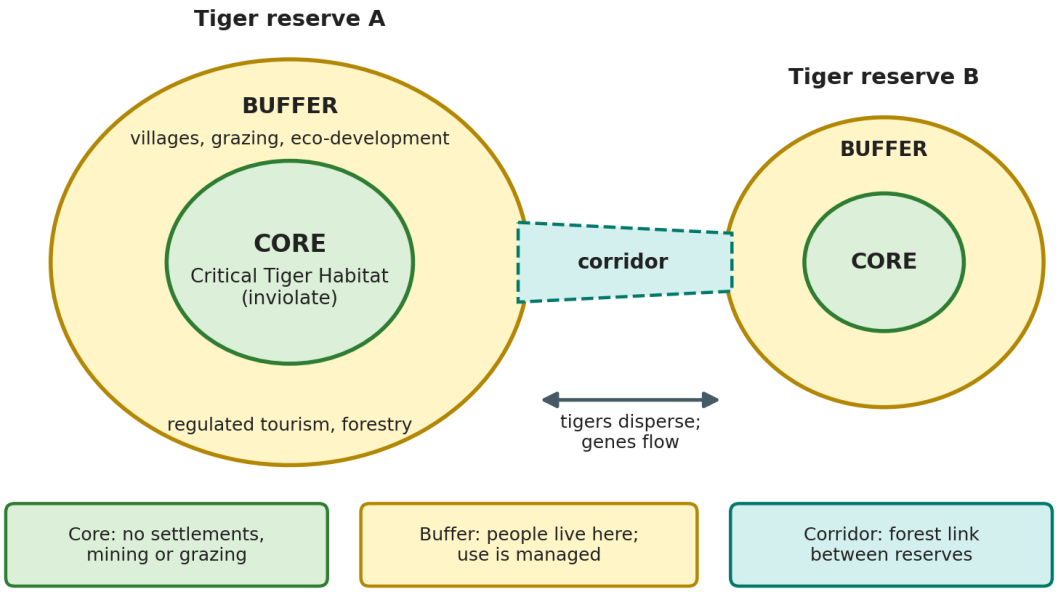


Figure 6.10: *Zoning of a tiger reserve: an inviolate core surrounded by a buffer of managed multiple use, with corridors linking it to other reserves. Source: based on NTCA guidelines.*

Inside reserves, management focuses on:

- **Protection**: anti-poaching camps, patrolling, wireless networks and intelligence-led enforcement, supported by the **Wildlife Crime Control Bureau** (also created in 2006).
- **Habitat and prey**: grassland management, water holes, removal of invasive weeds such as *Lantana camara*, and recovery of prey such as chital, sambar and gaur.
- **Voluntary relocation**: families in villages inside cores are offered land or a cash package (raised to ₹15 lakh per eligible unit in 2021) to move out. Kanha, Sariska, Bhadra and Satpura are examples where relocation has freed large areas for wildlife.

6.16.4 Institutional evolution: the Sariska crisis and the NTCA

For three decades Project Tiger was run by a small directorate in the Environment Ministry, and tiger numbers were reported through pugmark counts that were easy to inflate. In 2004–05 the weakness was exposed. Field surveys found that **Sariska Tiger Reserve** in Rajasthan had **no tigers left**, every one having been poached, even though official records had shown tigers there. Panna Tiger Reserve in Madhya Pradesh lost all its tigers a few years later (2008–09).

The Prime Minister set up a **Tiger Task Force** (chaired by Sunita Narain), whose 2005 report, *Joining the Dots*, called for better science, honest monitoring and a new deal with forest communities. Parliament amended the Wildlife (Protection) Act in 2006 to create the **National Tiger Conservation Authority (NTCA)**, a statutory body chaired by the Union Environment Minister. The NTCA:

- approves the **Tiger Conservation Plan** that each reserve must prepare;
- lays down standards for tourism, relocation and protection;
- evaluates reserves through periodic **Management Effectiveness Evaluation (MEE)**;

- runs the four-yearly **All India Tiger Estimation (AITE)** with the Wildlife Institute of India (WII).

In 2010 the NTCA and WII introduced **M-STrIPES** (Monitoring System for Tigers – Intensive Protection and Ecological Status), software and a mobile app that record patrol routes, wildlife signs, poaching evidence and fires on GPS-enabled phones. It makes patrolling accountable, because managers can see where guards actually walked.

In 2023 the Environment Ministry merged the centrally sponsored schemes of Project Tiger and Project Elephant into a single scheme, run by a new **Project Tiger & Elephant division**, with effect from 1 April 2023. The ministry argued that tigers and elephants share many landscapes; critics worried that the combined budget (about ₹331 crore in 2023–24) would be divided between the two species without transparency.

Case Study: Sariska: losing and regaining tigers

Sariska, in the Aravalli hills of Rajasthan, was one of the early tiger reserves. In early 2005 a survey by the Wildlife Institute of India confirmed that not a single tiger remained. Investigators found that poachers had killed the cats and sold their skins and bones into the illegal trade, while villages, temple traffic, grazing and a busy state highway inside the reserve had weakened protection.

The loss became a national scandal and led directly to the Tiger Task Force and the NTCA. In June 2008 India carried out its first **tiger reintroduction**: a male tiger was flown by Indian Air Force helicopter from Ranthambore to Sariska, followed by females. The relocated tigers were radio-collared and watched closely. The first cubs were born in 2012, and, helped by further translocations and the relocation of villages from the core, Sariska again holds a breeding population of several dozen tigers.

Panna Tiger Reserve followed a similar path. After losing all its tigers by 2009, it received tigresses from Kanha and Bandhavgarh and a male from Pench. With strong local support, Panna's population recovered within a decade. Both cases show that reintroduction works only when the original cause of loss (here, poaching) is fixed first.

6.16.5 Achievements

Tiger numbers

Since 2006, India has estimated tigers scientifically. **Camera traps** photograph tigers, each of which has a unique stripe pattern, and statistical models (capture–recapture and occupancy) extend the results to areas without cameras. The 2018 estimate entered the *Guinness World Records* as the largest camera-trap wildlife survey. The table below gives the results.

Table 6.18: All India Tiger Estimation results (NTCA–WII)

Year	Estimate	Range	Note
2006	1,411	1,165–1,657	First camera-trap based estimate
2010	1,706	1,520–1,909	Sundarbans added
2014	2,226	1,945–2,491	Steady growth

Year	Estimate	Range	Note
2018	2,967	2,603–3,346	Global TX2 goal met four years early
2022	3,682	3,167–3,925	About 75% of the world's wild tigers

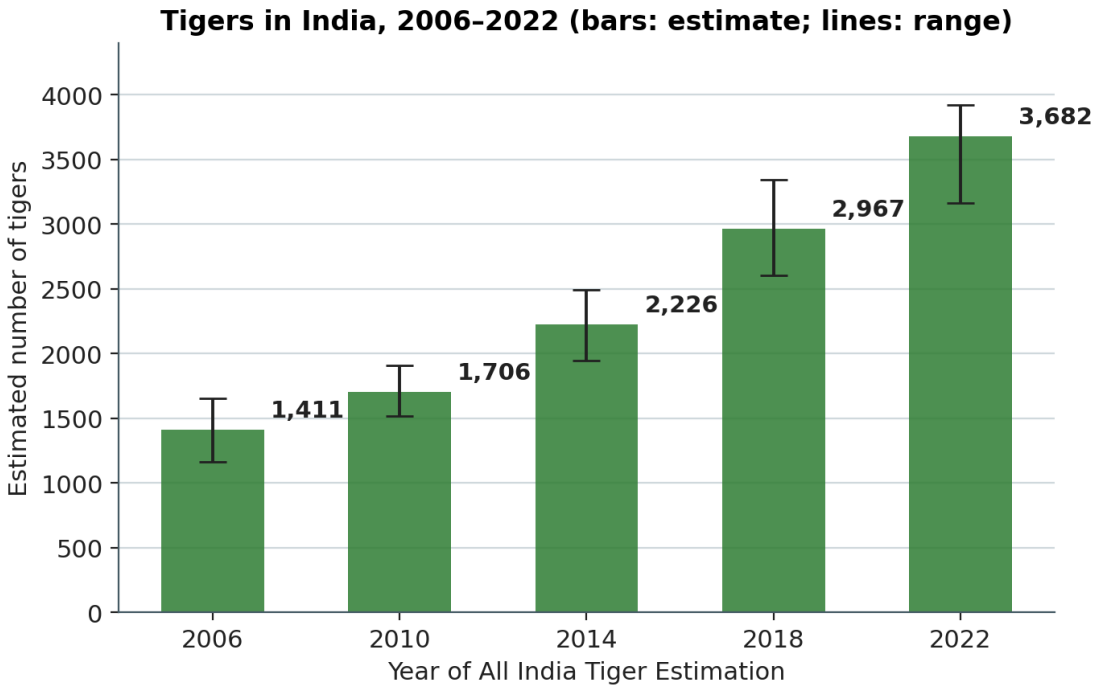


Figure 6.11: *Estimated tiger population of India, 2006–2022, with lower and upper limits. Source: NTCA and WII, Status of Tigers reports (2006–2022).*

The 2022 estimate shows growth of about 6% a year. Madhya Pradesh had the most tigers (785), followed by Karnataka (563), Uttarakhand (560) and Maharashtra (444); Corbett was the reserve with the largest population. The sixth cycle of estimation began in 2025–26, with results expected in 2027.

Doubling the world's tigers

At the 2010 St Petersburg Tiger Summit, the 13 tiger-range countries pledged to double wild tiger numbers by 2022 (the "TX2" goal). India reached this target in 2018, four years early. In 2023, marking 50 years of Project Tiger, India launched the **International Big Cat Alliance** to share conservation experience on seven big cats with other range countries.

Wider benefits

Because tigers need large, prey-rich forests, protecting them protects whole ecosystems; the tiger works as an **umbrella species**. Tiger reserves contain the sources of many rivers and store large amounts of carbon. Studies commissioned by the NTCA have valued the ecosystem services of individual reserves (water supply, carbon storage, pollination, fodder) at thousands of crores of rupees a year. Many species, from the hard-ground barasingha (swamp deer) of Kanha to the wild buffalo of Udanti–Sitanadi, survive largely because they live inside tiger reserves.

6.16.6 Challenges and criticisms

- **Uneven recovery**: Most growth is in central India, the Western Ghats and the Shivalik–Terai. Several reserves in Jharkhand, Odisha, Chhattisgarh and the North-East hold very few tigers; Palamau, one of the original nine, has been almost empty.
- **Tigers outside reserves**: Roughly a third of India's tigers live outside tiger reserves, in other forests, sugarcane fields and tea gardens, where they come into conflict with people. Tigers kill several dozen people a year in India; the Sundarbans and the Tadoba–Brahmapuri region of Maharashtra are hotspots.
- **Habitat fragmentation**: Highways, railways, canals and mines cut through corridors. Some new roads include animal underpasses (for example, NH-44 through the Pench–Kanha corridor), but many corridors remain unprotected.
- **Poaching**: Organised poaching for skins and bones, which feeds demand in East Asia, has fallen from its peak in the 2000s but has not disappeared.
- **People and rights**: The inviolate-core model has been criticised as **fortress conservation**, because it can exclude Adivasi and other forest communities who lived there first. The **Forest Rights Act, 2006** requires their free and informed consent before relocation (see Chapter 5). In 2024 the NTCA asked states to speed up the relocation of about 590 villages from core areas, which forest-rights groups opposed.
- **Tourism**: Heavy tourist traffic in reserves such as Ranthambore and Bandhavgarh disturbs animals, and resort construction on reserve boundaries has reached the courts.

6.16.7 Lessons from Project Tiger

1. A clear legal basis and a dedicated authority (the NTCA) matter as much as money.
2. Honest, scientific monitoring is essential: Sariska showed that inflated counts can hide a local extinction.
3. Protecting a flagship species protects whole ecosystems.
4. Long-term success depends on the goodwill of people living around reserves, which requires fair compensation, secure rights and a share in benefits such as tourism jobs.

6.17 Project Elephant

Project Elephant was launched by the Government of India in February 1992 as a centrally sponsored scheme to protect the Asian elephant (*Elephas maximus*), its habitats and the corridors it uses to move between them. India holds roughly 60% of the world's wild Asian elephants, a species listed as **Endangered** on the IUCN Red List. The elephant is woven into Indian culture, from Ganesha to temple festivals, and in 2010 the Government declared it India's **National Heritage Animal**.

6.17.1 Background: why elephants needed a project

Elephants need very large areas. A herd may range over several hundred square kilometres in a year, moving with the seasons in search of water and fodder. As forests were broken up by farms, dams, canals, mines, roads and railways, herds found their old routes blocked and began entering fields and villages. At the same time, ivory poaching was killing tuskers (male elephants with tusks), especially

in southern India in the 1980s and 1990s; the forest brigand Veerappan alone is believed to have killed hundreds.

Project Tiger had protected some elephant habitat, but elephants range far beyond tiger reserves and cross state boundaries. A separate programme was needed that would treat elephants at the scale of whole landscapes and would deal directly with **human–elephant conflict (HEC)**.

6.17.2 Objectives

- To protect elephants, their habitats and **migration corridors**.
- To address human–elephant conflict.
- To improve the welfare of **captive (domesticated) elephants**.

Under these broad aims, the project funds states for habitat improvement, anti-poaching work, compensation for crop and property loss and loss of life, veterinary care, research, and awareness.

6.17.3 Strategies

Elephant reserves

The main tool is the **elephant reserve (ER)**. Unlike a national park, an ER is not a separate legal category under the Wildlife (Protection) Act; it is an administrative landscape that may include protected areas, reserve forests, plantations and even villages, managed together for elephants. The first, Singhbhum in Jharkhand, was notified in 2001. By 2023 India had **33 elephant reserves covering about 80,777 km² in 14 states** (WII atlas, 2023), the most recent being Terai (Uttar Pradesh) and Agasthiyarmalai (Tamil Nadu), both in 2022. Major elephant landscapes are the North-East (Assam, Meghalaya, Arunachal Pradesh), the East-Central region (Odisha, Jharkhand, Chhattisgarh, West Bengal), the North-West (the Shivalik foothills of Uttarakhand and Uttar Pradesh), and the South (the Western and Eastern Ghats of Karnataka, Kerala and Tamil Nadu).

Securing corridors

An **elephant corridor** is a narrow strip of habitat that links two larger habitats and allows elephants to move between them. If corridors are cut, populations become isolated, inbreeding increases, and herds stray into villages. The Wildlife Trust of India (WTI) and the Ministry documented 88 corridors in 2005 and 101 in the second edition of *Right of Passage: Elephant Corridors of India* (2017). The Ministry's *Elephant Corridors of India 2023* report, compiled by WII, identified **150 corridors in 15 states**: 52 in the East-Central region, 48 in the North-East, 32 in the South and 18 in the North, with West Bengal having the most of any state. Of these, 15 were reported as damaged and 29 showed declining use by elephants.

Securing a corridor often means buying land from willing owners, relocating a few houses, re-routing roads or removing fences. The Thirunelli–Kudrakote corridor in Kerala, for instance, was secured through voluntary relocation of families with WTI support. In the Nilgiris, the Supreme Court in 2020 upheld the Tamil Nadu government's notification of the Sigur plateau corridor and ordered resorts blocking it to be removed.

Case Study: Death on the tracks

Railway lines through forests are among the deadliest threats to elephants. According to the Environment Ministry's reply in Parliament (2024), 81 elephants died in train collisions in the five years up to 2023–24. Most deaths occur in Assam, West Bengal (especially the Alipurduar–Siliguri line through the Dooars), Odisha, Jharkhand and Uttarakhand. In December 2025 a Rajdhani Express ran into a herd in Assam's Hojai district, killing seven elephants.

Responses have included speed limits on forest stretches, joint patrols by railway and forest staff, elevated sections and underpasses on new lines, clearing vegetation along tracks to improve the driver's view, and the "Plan Bee" device that plays the sound of a swarm of bees to drive elephants away from the line. Since 2022 Indian Railways has also tested an intrusion detection system (often called "Gajraj") that uses optical-fibre cables along the track to detect the vibrations of approaching elephants and alert train drivers and station masters.

Reducing human–elephant conflict

Conflict is now the central problem of elephant conservation. According to Ministry data given in Parliament, **2,853 people were killed by elephants between 2019 and 2023**, with 628 deaths in 2023 alone, the highest in five years. Odisha, Jharkhand, West Bengal, Assam, Chhattisgarh and Karnataka are the worst affected. Elephants also destroy crops worth crores of rupees each year. In return, hundreds of elephants die from electrocution (often from illegal fences), poisoning, train hits and retaliatory killing; the Ministry reported 528 elephant deaths from such unnatural causes over five years to 2023–24.

Tools used to reduce conflict include:

- **Early-warning systems**: radio-collaring a member of a problem herd, SMS alerts, community WhatsApp groups and warning lights. Collar-based alert systems in the Mahasamund area of Chhattisgarh and in Odisha have issued thousands of alerts and reduced surprise encounters.
- **Barriers**: solar-powered electric fences, elephant-proof trenches, and **bee-fences**, a method in which beehives strung along a field edge deter elephants. A pilot by the Khadi and Village Industries Commission in Kodagu, Karnataka (2021) reported fewer crop raids.
- **Community response teams**: groups such as Hathi Mitra Dal (elephant-friend teams) and Rapid Response Teams who drive elephants away safely and help families claim compensation.
- **Faster, fairer compensation**, so that farmers do not take revenge on elephants.
- **Land-use planning**: avoiding crops that attract elephants near forest edges, and keeping corridors clear.

Fighting poaching

India takes part in the CITES programme **MIKE (Monitoring the Illegal Killing of Elephants)**, which began in South Asia in 2003. Because only male Asian elephants carry tusks, poaching skews sex ratios: in parts of Periyar, studies in the 1990s found as few as one adult male for every 100 adult females, which reduces breeding. Better protection has reduced ivory poaching since then, though it has not ended.

Captive elephants

India has about 2,500–3,000 captive elephants, owned by forest departments (for patrolling), temples, private owners, circuses and zoos. Their welfare is often poor: chaining for long hours, festival work in heat and noise, and begging on city streets. The **Wildlife (Protection) Amendment Act, 2022** allowed the transfer of captive elephants for "religious or any other purpose", and rules for such transfers were notified in 2024. Animal-welfare groups criticised this, fearing it could legalise trade in wild-caught calves.

6.17.4 Achievements

The project has kept India's elephant range largely intact and built a national system of reserves and corridors. Population figures from synchronised counts are summarised below. Note that the methods changed, so the figures cannot be compared directly.

Table 6.19: Estimates of wild elephants in India

Year	Estimate	Method	Source
2012	29,391–30,711	Direct sightings plus dung counts	Synchronised Elephant Census
2017	29,964	Mainly direct counts, first synchronised all-India estimation	MoEFCC
2021–25	22,446	DNA from dung samples (genetic mark–recapture)	WII, SAIEE report (Oct 2025)

Counting elephants by their DNA

The 2021–25 **Synchronous All India Elephant Estimation (SAIEE)** identified individual elephants from DNA in their dung, much as tigers are identified from stripes. It estimated 22,446 wild elephants, with the Western Ghats holding about 11,934 and Karnataka the largest state population (6,013). Because the method is new, the lower figure is not proof of a crash, but the report did flag sharp losses in central and eastern India, where only about 1,891 elephants remain amid mining and conflict.

Other achievements include the creation of state elephant cells, a veterinary network for captive elephants, the Gaj Yatra awareness campaign (2017) on corridors, and the **Haathi Mere Saathi** campaign launched with WTI at the 2011 meeting of the "Elephant 8" range countries in Delhi.

6.17.5 Challenges and criticisms

- Elephant reserves have **no separate legal protection**, so mining, roads and dams can be approved inside them. Coal and iron-ore mining in Odisha, Jharkhand and Chhattisgarh (for example, in the Hasdeo Arand forests) overlap with key habitat.
- Conflict is rising as elephants move into new areas, such as northern Chhattisgarh and southern Bengal, where people have little experience of living with them.
- Funding has been small compared with the size of the problem, and the 2023 merger with Project Tiger raised fears that elephants would lose out.

- Corridors identified on paper are often not legally notified, and many are being lost to resorts, fences and farms.

6.17.6 Lessons from Project Elephant

1. Wide-ranging species cannot be saved inside small protected areas alone; corridors and the land between reserves matter.
2. Conservation succeeds only when people who bear the cost of living with large animals are protected, warned and compensated.
3. Infrastructure (roads, railways, power lines) must be planned with wildlife in mind from the start, not fixed afterwards.

6.18 Vulture Conservation Breeding Programme

Until the early 1990s, vultures were among the commonest large birds in India. Flocks of hundreds gathered at village carcass dumps, and the White-rumped Vulture was described as perhaps the most abundant large bird of prey in the world. Within about 15 years, three species had fallen by more than 95%. The story of how scientists found the cause, how the government banned a veterinary drug, and how vultures are now being bred in captivity and returned to the wild is one of the most important conservation case studies of our time.

6.18.1 Background: vultures and the services they provide

India has nine species of vulture. The three most affected belong to the genus *Gyps*:

- **White-rumped Vulture** (*Gyps bengalensis*)
- **Indian or Long-billed Vulture** (*Gyps indicus*)
- **Slender-billed Vulture** (*Gyps tenuirostris*)

All three are now **Critically Endangered** on the IUCN Red List, as is the Red-headed Vulture (*Sarcogyps calvus*); the Egyptian Vulture is Endangered.

Vultures are **obligate scavengers**: they feed only on carcasses. In a country with hundreds of millions of cattle, where many people do not eat beef, dead cattle were traditionally left in the open or at carcass dumps, and vultures stripped them within hours. In doing so they removed rotting flesh that would otherwise breed bacteria and feed rats and feral dogs. Vultures also had a cultural role: the Parsi community of Mumbai has long placed its dead in the **Towers of Silence** (dokhmas) for vultures to consume.

6.18.2 The crisis: a mystery and its solution

In the late 1990s, Vibhu Prakash of the **Bombay Natural History Society (BNHS)** noticed that nesting vultures at Keoladeo National Park, Bharatpur, were disappearing, and that birds were found dead or sitting with drooping necks before dying. Surveys along roads across northern India, repeated over the years, showed the scale of the collapse.

Table 6.20: Decline of *Gyps* vultures in India, 1992–2007 (road-transect surveys)

Species	Decline 1992–2007	IUCN status
White-rumped Vulture	99.9%	Critically Endangered
Long-billed and Slender-billed Vultures (counted together)	about 97%	Critically Endangered

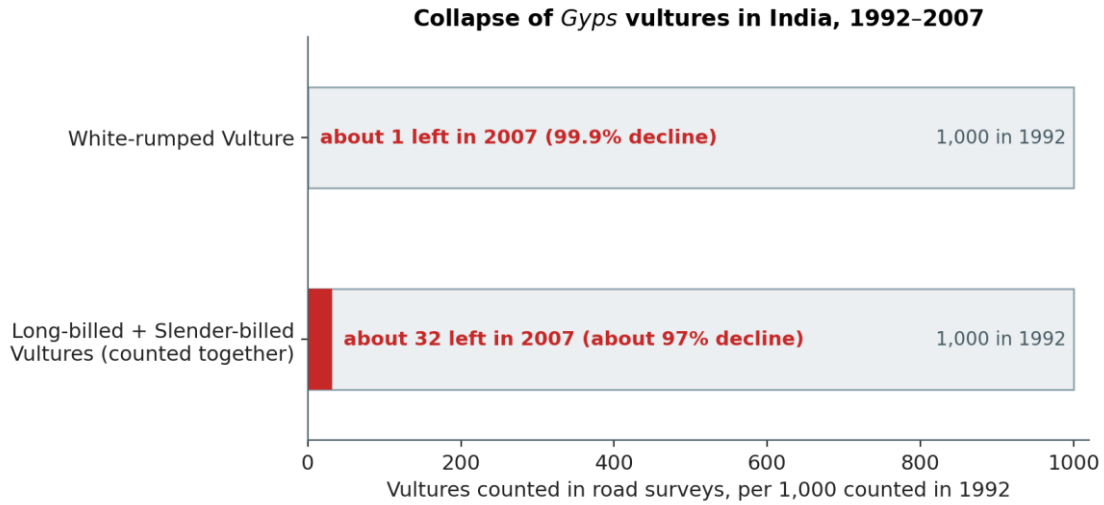


Figure 6.12: *Decline of Gyps vultures in India between 1992 and 2007: of every 1,000 vultures counted in 1992, about one White-rumped and about 32 Long-billed or Slender-billed Vultures remained. Source: Prakash et al. (2007), Biological Conservation.*

At first, scientists suspected a virus or pesticides. The breakthrough came in 2003–04, when Lindsay Oaks and colleagues of The Peregrine Fund, working in Pakistan, showed that dead vultures had **diclofenac** residues and suffered **kidney failure** and **visceral gout** (white crusts of uric acid on the internal organs). Diclofenac is a cheap **non-steroidal anti-inflammatory drug (NSAID)**, a painkiller used in human medicine, that had become popular in the 1990s for treating lame or sick cattle. A vulture feeding on a cow that died within a day or two of treatment swallowed a fatal dose. Modelling showed that contamination of fewer than 1 in 100 carcasses was enough to cause the observed collapse, because a single carcass feeds dozens of birds. Surveys in India found diclofenac in a large share of dead vultures and in many cattle carcasses.

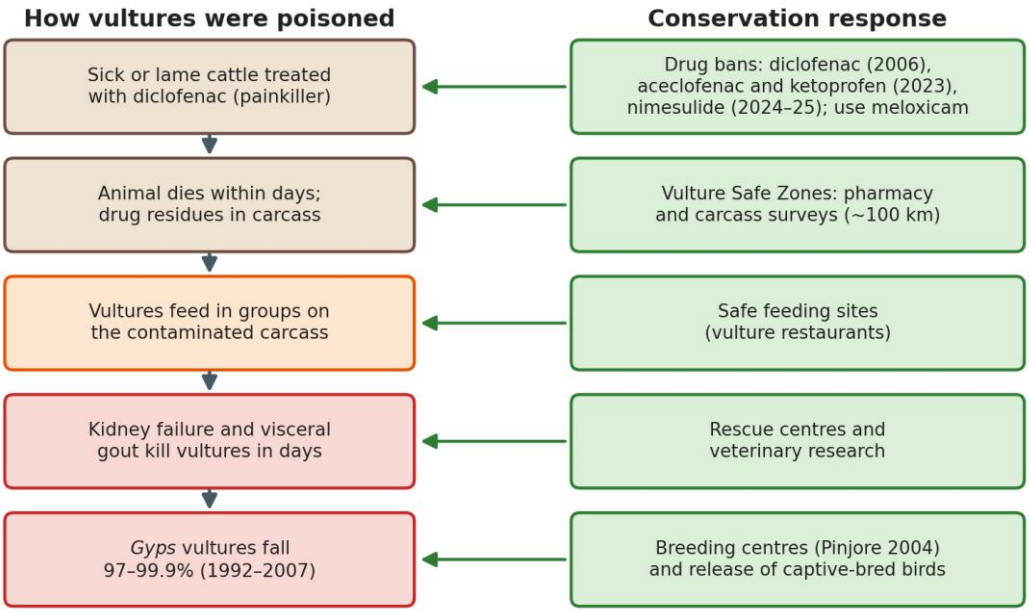


Figure 6.13: *How veterinary diclofenac killed vultures, and the conservation responses at each step. Source: based on Oaks et al. (2004) and the Action Plan for Vulture Conservation in India 2020–25.*

A costly loss of free service

When vultures vanished, uneaten carcasses fed a boom in feral dogs. A 2008 economic study (Markandya and colleagues) estimated that India's feral dog population grew by about 5.5 million between 1992 and 2006, causing tens of thousands of extra deaths from rabies at a cost of about US$34 billion. A 2024 study in the *American Economic Review* went further, estimating that districts that lost their vultures saw human death rates rise, adding up to around half a million extra deaths between 2000 and 2005, partly through polluted water. Vultures had been providing a public-health service free of charge.

6.18.3 Objectives of the conservation response

The **South Asian Vulture Recovery Plan** (2004) and India's first **Vulture Action Plan** (2006) set three goals:

1. Remove the cause: stop vultures from being exposed to diclofenac and other toxic drugs.
2. Prevent extinction: bring wild birds into captivity and breed them, creating a safe **insurance population**.
3. Restore wild populations: release captive-bred vultures into areas made safe from toxic drugs.

6.18.4 Strategies

Banning toxic drugs

After strong advocacy by BNHS and others, the **Drugs Controller General of India** banned the manufacture of diclofenac for veterinary use in **2006**, and the ban was formally notified under the Drugs and Cosmetics Act in 2008. Farmers then began misusing large vials of the human formulation

on cattle, so in **2015** the government restricted human diclofenac to single-dose vials of 3 ml. Vets were encouraged to use **meloxicam**, an NSAID that is safe for vultures.

Other NSAIDs proved toxic too. In **August 2023**, the Ministry of Health banned **aceclofenac** (which turns into diclofenac inside cattle) and **ketoprofen** for animal use. In late 2024 and early 2025, following a recommendation of the Drugs Technical Advisory Board, the government also prohibited veterinary formulations of **nimesulide**. Conservationists want a system in which every new veterinary painkiller is tested for vulture safety before it is sold.

Conservation breeding centres

Vultures breed slowly: they first breed at about five years of age, and a pair raises at most one chick a year. A population that crashes by 99% cannot recover quickly on its own. The breeding programme therefore aimed to hold at least 25 pairs of each species in each centre, enough to maintain genetic diversity.

- **Pinjore, Haryana**: a Vulture Care Centre set up in 2001 by BNHS and the Haryana Forest Department to study the dying birds was converted in 2004 into India's first **Vulture Conservation Breeding Centre (VCBC)**, now called the **Jatayu Conservation Breeding Centre**. With about 400 birds in 2023, it holds the world's largest captive collection of the three critically endangered *Gyps* species.
- Further BNHS-run centres were set up at **Rajabhatkhawa** (near Buxa Tiger Reserve, West Bengal), **Rani** (near Guwahati, Assam) and **Kerwa** (Bhopal, Madhya Pradesh). Zoos and state governments run other breeding facilities, for example in Odisha, Gujarat, Jharkhand and Telangana.

Centres use large **colony aviaries** in which birds can fly and choose partners, feed vultures meat from animals that were held for days without drugs, and monitor nests by CCTV to avoid disturbance. The technique of **double clutching** (removing the first egg for artificial incubation so that the female lays a second) has been used to increase output. By the mid-2020s the BNHS centres together held several hundred vultures, most of them bred in captivity.

Vulture Safe Zones and the 2020–25 Action Plan

A released vulture can fly 100 km or more in a day, so making a single forest safe is not enough. A **Vulture Safe Zone (VSZ)** is an area of about 100 km radius around a vulture colony or release site where pharmacies and vets are surveyed to confirm that diclofenac and other toxic NSAIDs are no longer used. The concept was developed by **SAVE (Saving Asia's Vultures from Extinction)**, a consortium of BNHS, the Royal Society for the Protection of Birds (RSPB, UK) and partners in Nepal, Pakistan and Bangladesh.

In November 2020 the Environment Ministry launched the **Action Plan for Vulture Conservation in India 2020–2025**. Its main measures were:

- new breeding centres in states such as Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu and Tripura;
- four regional **vulture rescue centres** (Pinjore, Bhopal, Guwahati and Hyderabad);
- at least one Vulture Safe Zone in every state;
- a system to test veterinary drugs for vulture toxicity and remove toxic ones from the market;
- regular nationwide vulture population surveys.

6.18.5 Achievements

- **The decline has slowed.** Surveys after the ban showed that the fall in White-rumped Vulture numbers had stopped or nearly stopped by the 2010s, and that diclofenac residues in cattle carcasses had dropped sharply.
- **Releases have begun.** Two Himalayan Griffons were released from Pinjore in 2016 as a trial. In October 2020 eight White-rumped Vultures, most of them captive-bred, were released near Pinjore; some later nested in the wild. Releases have followed in West Bengal (near Buxa), Madhya Pradesh (six birds near the Halali dam, Bhopal, in 2025) and Maharashtra, where captive-bred birds from Pinjore were moved to tiger reserves such as Melghat and Tadoba for acclimatisation and release. Released birds carry satellite tags and coloured leg rings so that their survival can be tracked.
- **State-level counts are rising** in some places. Madhya Pradesh, which runs regular statewide vulture counts, recorded about 12,900 vultures (all species) in 2025, up from about 8,400 in 2019.

> **Case Study: Nepal's vulture restaurants**
>
> Nepal faced the same crisis. Alongside its own ban on veterinary diclofenac (2006) and a breeding centre in Chitwan, Nepal pioneered **community-managed vulture restaurants**: feeding sites where villagers buy old cattle from farmers, keep them drug-free until they die naturally, and leave the carcasses for vultures. Visiting birdwatchers pay to watch the vultures feed, giving villagers an income. Nepal also campaigned to make whole districts diclofenac-free. White-rumped Vulture colonies in the lowlands have increased, and captive-bred birds released from Chitwan have bred in the wild. Nepal's experience shows that drug control, community income and breeding work best together.

6.18.6 Global context

Vultures are in trouble on every continent where they occur. In Africa, poachers poison carcasses so that circling vultures do not reveal their kills to rangers, and vulture body parts are used in traditional medicine; several African species are now Critically Endangered. In Europe, the Bearded Vulture was wiped out from the Alps by persecution and has been restored through releases of captive-bred birds since 1986. Europe also illustrates a policy risk: veterinary diclofenac was approved in Spain and Italy in 2013, despite warnings from Asia. The California Condor programme in the USA, in which every remaining wild bird was taken into captivity in 1987, is the classic example of breeding a bird back from the brink.

6.18.7 Challenges

- **Illegal and new drugs**: Diclofenac is still sometimes sold illegally for animals, and new NSAIDs can enter the market without vulture-safety testing.
- **Slow breeding**: Even with good breeding, numbers will take decades to recover.
- **Food and habitat**: Changing carcass disposal (burial, rendering plants), loss of tall nesting trees and cliffs, power-line electrocution, and kite-string (manja) injuries all harm released birds.

- **Funding**: Breeding centres are expensive to run for decades, and depend on a few organisations and individuals.

6.18.8 Lessons from the vulture programme

1. A single, invisible cause, here a common medicine, can wipe out abundant species in a few years. Monitoring common species, not only rare ones, is essential.
2. Science can identify a cause quickly, but policy must follow quickly too; India's ban came about three years after the cause was found, and gaps remained for years afterwards.
3. Ex-situ breeding works only alongside in-situ action: captive birds can be released only once the wild is made safe again.

6.19 Project Great Indian Bustard

The **Great Indian Bustard** (*Ardeotis nigriceps*), called **Godawan** in Rajasthan, is a tall, long-legged bird of dry grasslands. Standing about a metre high, with males weighing up to about 15 kg, it is one of the heaviest flying birds in the world. In the 1960s the ornithologist Salim Ali argued that it should be India's national bird. Today it is one of the rarest birds on Earth, listed as **Critically Endangered** by the IUCN since 2011 and placed in Schedule I of the Wildlife (Protection) Act. Fewer than 150 birds survive in the wild, nearly all in western Rajasthan.

6.19.1 Background: a grassland bird in a changing land

The bustard is a **flagship species of grasslands** and semi-arid scrub, ecosystems that India's land laws long classified as "wasteland". It eats insects (especially grasshoppers and beetles), small reptiles and rodents, seeds and berries. Females lay a single egg, directly on the ground, often once every year or two, so the species recovers very slowly from losses.

Once found across about 11 states of western and central India and in Pakistan, the bustard has lost about 90% of its range. Estimates suggest roughly 1,260 birds in 1969, about 745 in 1978, roughly 300 by 2008 and around 150 by 2018, when the Wildlife Institute of India (WII) carried out a range-wide survey.

Table 6.21: Great Indian Bustard population estimates in India

Year	Approximate number	Note
1969	1,260	Range across about 11 states
1978	745	Decline evident
2008	about 300	Several populations already lost
2018	about 150	Mostly in Jaisalmer and Barmer (WII survey)
2025–26	under 150 wild; about 70 captive	Captive flock at two breeding centres

Small remnant groups survive in Kachchh (Gujarat), where only a few, mostly female, birds remain; around Solapur (Maharashtra); and in Karnataka and Andhra Pradesh. Populations at the Ranebennur sanctuary in Karnataka and at Karera in Madhya Pradesh have disappeared.

6.19.2 Threats

- **Power-line collisions**: Bustards have wide side vision but poor forward vision, and being heavy they cannot swerve in time to avoid wires. A WII study published in 2018 estimated that overhead power lines in the Thar were killing about **18 bustards a year**, a loss no population of 150 slow-breeding birds can survive. The boom in wind and solar parks in western Rajasthan and Kachchh, each linked to the grid by new transmission lines, made the threat worse.
- **Loss of grasslands**: Grasslands have been ploughed for farming, irrigated by projects such as the Indira Gandhi Canal, planted with trees, or covered with solar panels. At Ranebennur, plantations of eucalyptus on open grassland contributed to the bird's local extinction.
- **Predation and disturbance**: Feral dogs, foxes and wild boar take eggs and chicks; free-ranging livestock trample nests.
- **Hunting**: Historically hunted for meat and sport; hunting continues in parts of Pakistan.

6.19.3 Objectives and strategies

Project Great Indian Bustard (Rajasthan, 2013)

The Rajasthan government launched **Project Great Indian Bustard** on World Environment Day, 5 June 2013. Its aims were to identify and fence breeding areas in and around **Desert National Park (DNP)**, protect them from disturbance and predators, improve habitat by planting native grasses such as *sewan* (*Lasiurus sindicus*), and involve local people. DNP covers about 3,162 km^2 of Jaisalmer and Barmer districts, but much of it is revenue or private land with villages inside, so predator-proof **enclosures** inside the park became the safest breeding sites.

Conservation breeding (Centre and state, from 2016)

In 2016 the Environment Ministry, WII and the Rajasthan Forest Department began a national programme, **Habitat Improvement and Conservation Breeding of the Great Indian Bustard: an Integrated Approach**, funded with about ₹33.85 crore from CAMPA (the Compensatory Afforestation Fund). It received technical help from the International Fund for Houbara Conservation in Abu Dhabi, which has decades of experience breeding a related bustard.

Because adult bustards are too sensitive to be caught, staff collect eggs from wild nests (wild females often lay again if the first egg is lost) and hatch them in incubators. Breeding centres were set up at **Sam** (Jaisalmer) and **Ramdevra** (near Pokhran). The first eggs were collected in 2019. By early 2026 the captive flock had grown to about 70 birds, up from 16 four years earlier, and captive-bred chicks were being hatched every season.

> **A bustard born by artificial insemination**
>
> On 16 October 2024, a Great Indian Bustard chick hatched at the Jaisalmer breeding facility from an egg produced by **artificial insemination**. A male named Suda at Ramdevra had been trained over several months for semen collection, and the semen was used to inseminate a female named Tony at the Sudasari centre. This was the first such success for the species anywhere. More chicks bred this way followed in 2025 and 2026. Artificial insemination lets managers pair the most genetically suitable birds even if they are housed far apart.

Power-line mitigation and the courts

The fight over power lines reached the **Supreme Court** in a petition filed by the conservationist M.K. Ranjitsinh in 2019.

- **April 2021**: The Court ordered that overhead power lines in the bustard's priority and potential habitat in Rajasthan and Gujarat (about 99,000 km^2) be laid **underground** where feasible, with **bird flight diverters** (reflective discs that make wires visible) on the remaining lines.
- **March 2024**: The Court accepted the Centre's argument that undergrounding high-voltage lines across such a large area was technically and financially very difficult and would slow India's renewable-energy goals. It modified its order and set up an **expert committee**. In this judgment the Court also recognised a **right against the adverse effects of climate change** as part of the rights to life and equality (Articles 21 and 14). The case thus became a clear example of a conflict between two green goals: protecting biodiversity and expanding clean energy.
- **December 2025**: Accepting the expert committee's report, the Court fixed **priority areas** of 14,013 km^2 in Rajasthan and 740 km^2 in Gujarat. Inside these areas, no new overhead lines are allowed except local distribution lines of 11 kV or less; no new wind turbines or solar parks above 2 MW may be built; and identified stretches of existing lines (including about 80 km of 33 kV lines in Rajasthan) must be undergrounded or re-routed by December 2027. The Court did not require bird diverters everywhere, noting that their effectiveness for bustards is unproven.

6.19.4 Achievements

- A captive **insurance population** of about 70 birds now exists, a remarkable result given that no one had bred this species in captivity before 2019.
- India has mastered egg collection, artificial incubation, hand-rearing and artificial insemination for a large bustard.
- Fenced enclosures in Desert National Park have improved breeding success in the wild.
- The Supreme Court's orders have, for the first time, placed binding limits on energy infrastructure to protect a single bird species.

6.19.5 Challenges and criticisms

- **Too little, too late**: Conservationists had warned of the decline since the 1980s, yet strong action began only around 2013–2019, when fewer than 200 birds remained.
- **Release is untested**: As of 2026 no captive-bred bustard had yet been released into the wild. Released birds will face the same power lines and dogs that killed wild birds.
- **Land ownership**: Much bustard habitat is private or community land, so protection depends on the goodwill of farmers and herders, who have sometimes opposed restrictions.
- **Energy versus wildlife**: Rajasthan and Gujarat are India's leading renewable-energy states. Every kilometre of line that is undergrounded adds cost, and developers have pushed back.

6.19.6 Lessons from the bustard programme

1. Grasslands deserve protection as ecosystems in their own right, not as "wastelands" for plantations or solar parks.

2. Even "green" development, such as solar and wind energy, has ecological costs that must be planned for.
3. Conservation breeding is a last resort; it buys time but cannot replace safe habitat.
4. Courts can force action, but lasting solutions need agreement among ministries, power companies and local communities.

6.20 Crocodile Conservation Project

India is home to three crocodilians, each adapted to a different kind of water:

- **Gharial** (*Gavialis gangeticus*): a fish-eater of large, fast-flowing rivers, with a long, thin snout. Males develop a pot-like knob (*ghara*) at the snout tip, which gives the species its name. It is **Critically Endangered**.
- **Mugger** or marsh crocodile (*Crocodylus palustris*): a broad-snouted generalist of lakes, ponds, reservoirs and slow rivers. It is **Vulnerable**.
- **Saltwater** or estuarine crocodile (*Crocodylus porosus*): the world's largest living reptile, found in mangroves and estuaries of Odisha, the Sundarbans and the Andaman and Nicobar Islands. It is listed as Least Concern globally but was close to disappearing from mainland India.

The **Crocodile Conservation Project**, launched on 1 April 1975, is one of the earliest and most successful examples of the **rear-and-release** method of conservation.

6.20.1 Background: crocodiles on the brink

By the early 1970s, all three species had been hunted almost out of existence for their skins, which were valuable in the leather trade, and for eggs and meat. Dams and barrages changed river flows, fishermen killed crocodiles caught in nets, and nesting sandbanks were farmed or mined. A survey in 1974 by **H.R. (Robert) Bustard**, a crocodile expert sent by the Food and Agriculture Organization (FAO), found only a few hundred gharials left in India, saltwater crocodiles confined to a few pockets, and muggers gone from much of their range. The Wildlife (Protection) Act, 1972 gave all three species legal protection, but protection alone could not rebuild populations of animals whose eggs and young suffer very high natural losses.

6.20.2 Objectives

- Protect the remaining populations through sanctuaries.
- Rebuild numbers quickly by collecting eggs, hatching and rearing the young in safety, and releasing them.
- Promote captive breeding where wild eggs are unavailable.
- Train Indian staff and build research capacity.

6.20.3 Strategies

Rear and release

In the wild, over 90% of crocodile eggs and hatchlings may be lost to predators (monitor lizards, jackals, birds, fish), floods and people. The project therefore:

1. collected eggs from wild nests soon after laying;

2. hatched them in protected hatcheries;
3. reared the young in pools for two to three years, until they were about 1.2 m long and safe from most predators; and
4. released them into protected stretches of river or estuary.

This multiplied the number of young that survived to join the breeding population.

Breeding and rearing centres

The project was a partnership of the Government of India and state forest departments, supported by the **UNDP** and **FAO**, with Bustard as chief technical adviser. Odisha took the lead, and it remains the only state with wild populations of all three species:

- **Tikarpada**, in the Satkosia gorge of the Mahanadi: gharial rearing.
- **Dangamal**, in **Bhitarkanika**: saltwater crocodile rearing.
- **Ramatirtha**, near Similipal: mugger rearing.
- **Nandankanan Zoological Park**, Bhubaneswar: the world's first captive breeding of gharials, in 1980.

Other centres included the **Kukrail Gharial Rehabilitation Centre** in Lucknow (Uttar Pradesh) and centres in Rajasthan, Madhya Pradesh, Andhra Pradesh, Tamil Nadu and West Bengal. A Central Crocodile Breeding and Management Training Institute was set up in Hyderabad in 1980 to train staff.

Protected areas

About a dozen crocodile sanctuaries were declared. The most important is the **National Chambal Sanctuary** (notified 1978–79), a tri-state protected area along roughly 400 km of the Chambal river shared by Madhya Pradesh, Rajasthan and Uttar Pradesh. The clean, fast-flowing Chambal now holds the world's largest gharial population, along with the Ganges river dolphin, rare turtles such as the red-crowned roofed turtle, and the Indian skimmer. Other key sites are **Bhitarkanika National Park** (Odisha), Katarniaghat (Uttar Pradesh), the Son and Ken gharial sanctuaries (Madhya Pradesh), and the Satkosia gorge (Odisha).

Case Study: Madras Crocodile Bank

In 1976 the herpetologist **Romulus Whitaker**, with Zai Whitaker, founded the **Madras Crocodile Bank Trust** near Mamallapuram, south of Chennai. It began as a "gene bank" to breed and hold India's three crocodilians at a time when they were vanishing, and became one of the world's largest collections of crocodiles, with thousands of animals of more than a dozen species. Many gharials and muggers bred there were sent for release in rivers and for other zoos.

The Croc Bank also became a centre for reptile research and education. It supported the Irula community's snake-catchers' cooperative, which collects venom for anti-venom production, and Whitaker later led field surveys and campaigns for gharial protection in the Chambal and elsewhere. The Croc Bank shows how a private, science-led organisation can complement government programmes in ex-situ conservation.

6.20.4 Achievements

- **Saltwater crocodiles in Bhitarkanika** grew from fewer than 100 in the mid-1970s to **1,826** in the January 2025 census, and 1,858 in 2026. Bhitarkanika now holds one of India's largest saltwater crocodile populations, including a few rare white (leucistic) individuals. Officials stopped rearing and releasing young in 2024, judging that the estuary had reached its natural carrying capacity.
- **Gharials**: India now holds about 80% of the world's wild gharials. Thousands of captive-reared gharials have been released into the Chambal, Girwa (Katarniaghat), Son, Ken, Mahanadi and other rivers. The Chambal population now numbers in the low thousands (counts of individuals of all ages), with hundreds of nests laid each year.
- **Muggers** have recovered widely and now live in reservoirs, rivers and even urban waterways, such as the Vishwamitri river in Vadodara (Gujarat).
- The project trained a generation of Indian crocodile biologists and gave India international leadership in crocodilian conservation.

> **The 2007–08 gharial deaths**
>
> Between December 2007 and March 2008, more than 100 gharials were found dead along the Chambal, many with damaged livers and kidneys. Investigators suspected toxins (possibly heavy metals entering via the Yamuna) and parasites, but no single cause was proved. The event was a warning that even a "safe" sanctuary depends on the health of the whole river system upstream.

6.20.5 Challenges

- **Gharials remain fragile**: The total number of breeding adults in the world is only a few hundred, mostly in one river, the Chambal. Any disaster there could be catastrophic.
- **Sand mining**: Illegal mining of sand from the Chambal's banks destroys nesting sites. It is organised, violent and difficult to control.
- **Dams, barrages and river linking**: Reduced flows, barrages and projects such as the Ken–Betwa river link threaten gharial habitat.
- **Fishing nets** drown gharials and turtles; river pollution reduces fish prey.
- **Human–crocodile conflict**: As saltwater crocodiles and muggers recover, attacks on people in the creeks around Bhitarkanika and in Gujarat and Maharashtra have increased, causing resentment among fishing communities.

In March 2025, at a meeting of the National Board for Wildlife, the Prime Minister announced a dedicated conservation project for the gharial, intended to strengthen its remaining populations and restore it to rivers in its former range.

6.20.6 Lessons from the crocodile project

1. For species whose eggs and young suffer heavy natural losses, rear-and-release can rebuild populations quickly.
2. Releasing animals is pointless unless their habitat (here, free-flowing, clean, unmined rivers) is protected.

3. Success brings new problems; recovered predators need conflict management and community support.

6.21 Silent Valley Movement

The **Silent Valley movement** (roughly 1973–1984) was a campaign by scientists, writers, teachers and ordinary citizens in Kerala to stop a hydroelectric dam that would have flooded part of one of India's last large tracts of undisturbed tropical evergreen forest. It ended with the dam being shelved and the forest declared a national park. It was the first major environmental campaign in India to win against a state-backed development project mainly on the grounds of biodiversity, and it shaped how India thinks about development and conservation.

6.21.1 Background: the valley and the dam

Silent Valley lies in the Palakkad district of Kerala, on the south-western slopes of the Nilgiri hills in the Western Ghats. Its forests are drained by the **Kunthipuzha**, a tributary of the Bharathapuzha river. Locally the forest is called *Sairandhrivanam*, the forest of Sairandhri (Draupadi in the Mahabharata). The English name is often said to come from the apparent absence of the noisy cicadas found in most forests.

The valley's remoteness had kept it free of logging and plantations. It holds tropical wet evergreen forest with a very high diversity of trees, orchids, ferns, frogs and insects, many of them **endemic** to the Western Ghats. Its best-known resident is the **lion-tailed macaque** (*Macaca silenus*), an endangered monkey with a silver-grey mane that lives only in the rainforests of the Western Ghats and depends on an unbroken tree canopy. Silent Valley held one of the largest remaining groups of this species, which became the symbol of the campaign.

British engineers had identified the Kunthipuzha as a good hydroelectric site in 1928. The **Kerala State Electricity Board (KSEB)** investigated it from 1958 and proposed a dam in 1970, and the Planning Commission approved the project in 1973 at a cost of about ₹25 crore. The **Silent Valley Hydroelectric Project** would generate about 120 MW of electricity and irrigate farmland in the dry districts of Palakkad and Malappuram, but its reservoir would submerge about **830 hectares** of primary forest along the river, the core habitat of the macaque.

6.21.2 The movement

Opposition grew slowly through the 1970s and came from several directions at once.

- **Scientists**: A 1976 task force of the National Committee on Environmental Planning and Coordination, led by Zafar Futehally, recommended that the project be abandoned or carried out only with strict safeguards. Studies of the lion-tailed macaque by primatologists, and a 1977 ecological study by the Kerala Forest Research Institute, showed the valley's value. In 1978 the **IUCN** passed a resolution calling for the protection of the lion-tailed macaque's habitat in Silent Valley. Salim Ali visited and appealed against the dam.
- **The Kerala Sastra Sahitya Parishad (KSSP)**: This people's science movement, which brings science to the public in Malayalam, published a **techno-economic and socio-political assessment** in 1979. It argued that Kerala's power needs could be met in other ways and that the valley's value as a store of genes and as a watershed outweighed the dam's benefits. KSSP

members carried the message to villages, schools and tea shops through talks, pamphlets and street plays.

- **Writers and poets**: The poet **Sugathakumari** wrote *Marathinu Stuthi* (*Ode to a Tree*), which became the anthem of the campaign, and helped form a Nature Protection Society (*Prakruthi Samrakshana Samithi*) that brought writers and artists into the movement. Poetry readings and articles turned a technical debate into a moral and cultural one.
- **The courts and the press**: Petitions in the Kerala High Court briefly halted tree-felling in the project area, and national newspapers carried the debate beyond Kerala.

Supporters of the dam, including the KSEB, political parties and many people in the Malabar region, argued that northern Kerala needed electricity and irrigation, that only a small area would be flooded, and that conservationists were placing monkeys above people. The movement therefore had to make a strong scientific case and to persuade people, not only officials.

> **A poem against a dam**
>
> Sugathakumari (1934–2020) was one of Malayalam's best-loved poets. Her poem *Marathinu Stuthi* was recited at meetings all over Kerala during the Silent Valley campaign. She went on to campaign for women's rights and mental-health care and received the Padma Shri in 2006. Her role shows that environmental movements need storytellers as well as scientists.

6.21.3 Government decisions and the outcome

In 1979 the Union Agriculture Secretary, **M.S. Swaminathan**, visited the valley and recommended that it be protected as part of a larger rainforest biosphere reserve. In 1980 Prime Minister **Indira Gandhi**, who had a strong personal interest in wildlife, asked Kerala to stop work, and a joint committee headed by the scientist **M.G.K. Menon** was set up to examine the issue. After reviewing its report, the Central Government decided in **November 1983** to **abandon the project**.

- On **15 November 1984**, about 89.5 km^2 of Silent Valley was notified as a **national park**.
- On **7 September 1985**, Prime Minister **Rajiv Gandhi** formally inaugurated **Silent Valley National Park**.
- In 1986 the park became part of the core area of the **Nilgiri Biosphere Reserve**, India's first biosphere reserve.
- In 2007 a buffer zone of about 148 km^2 was added around the park.

Later proposals for dams on the Kunthipuzha just downstream of the park, such as the Pathrakkadavu project in the 2000s, revived the campaign and were not carried out.

6.21.4 Significance and legacy

- **Biodiversity as a reason to say no**: For the first time in India, a major development project was cancelled mainly to protect a forest ecosystem and its species, rather than to protect people's livelihoods.
- **Science in public**: The KSSP showed that science could be explained to ordinary citizens and used to question official claims about costs and benefits. This model of a **people's science movement** has influenced later campaigns.

- **Institutions**: The campaign coincided with, and strengthened the case for, the creation of a central Department of Environment (1980), the Ministry of Environment and Forests (1985), and later the legal requirement for **environmental impact assessment** of large projects (made mandatory under the Environment (Protection) Act in 1994; see Chapter 5).
- **Flagship species in campaigns**: The lion-tailed macaque showed how one appealing species can carry public concern for a whole ecosystem.

Criticism has also been made. Some argued that the movement was led by an educated urban middle class and paid less attention to the electricity and irrigation needs of poorer districts. Others point out that its success depended heavily on the personal support of the Prime Minister, which later campaigns, such as those against the Narmada dams, did not enjoy (see Chapter 7).

6.21.5 Lessons from Silent Valley

1. Good science, clearly communicated to the public, can change government decisions.
2. Broad coalitions of scientists, writers, students and the press are stronger than any single group.
3. Development choices involve trade-offs; the question is not "development or environment" but which form of development, at what cost, and for whom.
4. Protection must be defended continuously, as later dam proposals near the park showed.

6.22 Save Western Ghats Movement

Silent Valley saved one valley. The **Save Western Ghats movement** asked a bigger question: how should India look after an entire mountain range? It began with a 100-day march along the Ghats in 1987–88 and lives on in the long, unresolved national debate over the **Gadgil** and **Kasturirangan** reports and the notification of an **ecologically sensitive area (ESA)** in the Western Ghats.

6.22.1 Background: why the Western Ghats matter

The **Western Ghats** (Sahyadri) run about 1,600 km along India's west coast, from the Tapi river near the Gujarat–Maharashtra border to near Kanyakumari, through six states: Gujarat, Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu. Together with Sri Lanka they form one of the world's **biodiversity hotspots** (see the section on hotspots earlier in this chapter), and in 2012 UNESCO inscribed 39 sites in the range as a **World Heritage Site**.

- **Biodiversity**: The Ghats hold a large share of India's plants and animals on less than 6% of its land, with very high endemism. Most of their amphibians, and many reptiles, freshwater fish and flowering plants, are found nowhere else. Endemic animals include the lion-tailed macaque, Nilgiri tahr, Malabar civet and the purple frog.
- **Water**: The Godavari, Krishna and Kaveri rise in the Ghats, along with dozens of short west-flowing rivers. Their forests and grasslands feed the water supply of several hundred million people in peninsular India.
- **Climate**: The range intercepts the south-west monsoon, producing heavy rain on the western slopes and a rain shadow to the east.

By the 1980s the Ghats were under heavy pressure: clearing for tea, coffee, rubber and eucalyptus plantations; dozens of dams; iron-ore and manganese mining in Goa and Karnataka; quarrying; roads;

and the loss of sacred groves and common lands. Each problem was fought locally, state by state. Activists realised that the range had to be seen as one ecological unit.

6.22.2 The Save Western Ghats March, 1987–88

In 1986, voluntary groups from the Ghats states met at a consultation called by the Goa-based organisation **Peaceful Society** (coordinator: Kumar Kalanand Mani) and decided on a foot march (**padayatra**) along the whole range. Ecologists such as **Madhav Gadgil** of the Indian Institute of Science advised the organisers.

The **Save the Western Ghats March** began on 1 November 1987 with two teams:

- a northern team starting from **Navapur** (Maharashtra, near the Gujarat border), flagged off by the Chipko leader **Chandi Prasad Bhatt**;
- a southern team starting from **Kanyakumari** (Tamil Nadu).

The two teams walked for about 100 days, covering the length of the Ghats, and met in **Goa** in February 1988 for a concluding conference. Participants from more than 150 organisations took part. Along the way they held hundreds of village meetings, surveyed villages about changes in rainfall, water, forests and crops, and documented mining, dams and deforestation. Villagers reported drying springs, shorter monsoons, loss of fodder and fuelwood, and invasion of weeds.

6.22.3 Objectives and demands

The march aimed to:

1. raise public awareness of the ecological crisis in the Ghats;
2. document the extent of degradation by direct observation;
3. link local struggles into a single regional network; and
4. promote locally led conservation, such as afforestation, protection of sacred groves and community control of forests.

The Goa conference supported local struggles against specific projects (for example, dams in Karnataka and Kerala and polluting industries in Goa), opposed the handing over of forest and common land to industries in the name of "social forestry", and called for the land to be used instead for the needs of local, especially landless, people.

6.22.4 Outcomes of the march

The march did not stop any single project on its own, and its organisers later admitted that the network was not sustained as a coordinated movement after 1988. Its lasting achievement was to establish the idea of the Western Ghats as a single ecological region requiring a common policy, and to build links between activists, scientists and villagers. These ideas resurfaced two decades later in official policy.

6.22.5 From movement to policy: Gadgil versus Kasturirangan

In 2010 the Environment Ministry set up the **Western Ghats Ecology Expert Panel (WGEEP)** chaired by Madhav Gadgil, a link with the 1987 march. Its report, submitted in August 2011, was made public only in 2012 after pressure through the Right to Information Act and the courts. The states

objected strongly, and in 2012 the Ministry appointed a **High Level Working Group (HLWG)** under the space scientist **K. Kasturirangan** to examine it. The HLWG reported in April 2013.

Table 6.22: Gadgil (WGEEP, 2011) and Kasturirangan (HLWG, 2013) reports compared

Feature	Gadgil report (WGEEP)	Kasturirangan report (HLWG)
Area treated as sensitive	Whole Western Ghats, about 1.29 lakh km^2	About 37% (about 60,000 km^2)
Zoning	Three Ecologically Sensitive Zones (ESZ 1, 2, 3) with graded rules	"Natural landscape" (ESA) and "cultural landscape" (outside ESA)
Mining and quarrying	No new mining in ESZ 1 and 2; phase-out in ESZ 1	Banned in the ESA
Dams	No new large dams in ESZ 1	Allowed in ESA with conditions
Governance	Bottom-up: gram sabhas plus a Western Ghats Ecology Authority	Top-down: existing laws and agencies

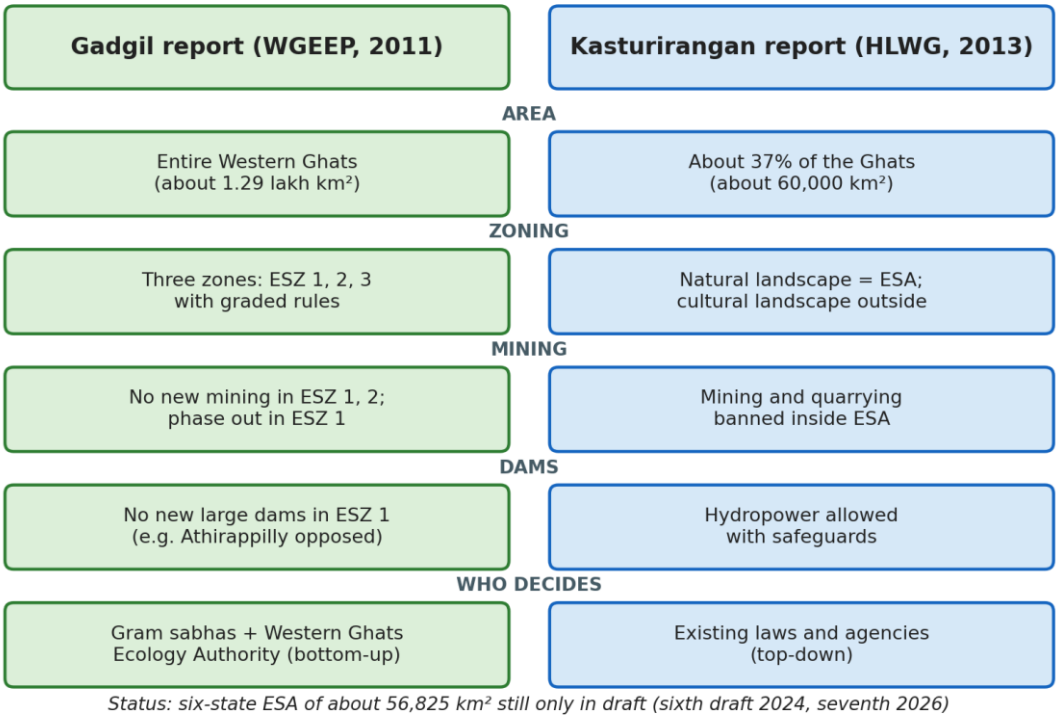


Figure 6.14: *Main differences between the Gadgil (2011) and Kasturirangan (2013) reports on the Western Ghats. Source: WGEEP (2011) and HLWG (2013) reports, MoEFCC.*

The **Gadgil report** treated the entire range as ecologically sensitive. It proposed three zones of decreasing protection, with ESZ 1 the strictest. It recommended no new mining in the two most sensitive zones, a ban on new large dams in ESZ 1 (it specifically opposed the Athirappilly and Gundia hydroelectric projects), a phase-out of chemical pesticides, and, above all, decision-making from below by gram sabhas, supervised by a statutory **Western Ghats Ecology Authority**.

The **Kasturirangan report** used satellite data to divide the Ghats into **cultural landscape** (about 58%, mostly farms, plantations and settlements) and **natural landscape** (about 41%). It proposed that about 90% of the natural landscape, about 37% of the Ghats, be declared an ESA. Inside the ESA

it banned mining, quarrying, sand mining, thermal power plants, the most polluting ("red category") industries and very large townships, but allowed hydropower and other activities with safeguards. Outside the ESA, ordinary laws would apply.

Supporters of Gadgil argued that the Kasturirangan approach protected mainly forests that were already protected and left the populated slopes, where most damage happens, without special rules. Critics of Gadgil, including many farmers, plantation owners, church groups in Kerala and state governments, feared that restrictions would hurt livelihoods and development; there were large protests in Kerala's hill districts in 2013–14.

6.22.6 The unfinished ESA notification

In 2013 the Environment Ministry accepted the Kasturirangan approach in principle. It has since issued a series of **draft notifications** under the Environment (Protection) Act declaring the ESA, in 2014, 2015, 2017, 2018, 2022, July 2024 (the sixth draft, covering about **56,825 km²** in six states, including about 20,668 km² in Karnataka and about 9,994 km² in Kerala) and again in July 2026 (a seventh draft repeating the 2024 proposal). None had been finalised by late 2026. States have repeatedly sought to shrink the area, especially by excluding villages and plantations, and each draft has lapsed before a final notification could be issued.

6.22.7 Floods, landslides and the debate

The issue returned to the headlines after disasters in the Ghats:

- **Kerala floods, August 2018**: Extremely heavy monsoon rain, combined with the release of water from full reservoirs, caused Kerala's worst floods in nearly a century; more than 400 people died. Hundreds of landslides struck the hill districts.
- **Landslides of 2019** at Kavalappara (Malappuram) and Puthumala (Wayanad).
- **Wayanad landslides, 30 July 2024**: After extremely heavy rain, debris flows swept through the villages of Mundakkai, Chooralmala and Punchirimattom, killing more than 200 people, with many more missing. It was one of the deadliest landslide disasters in Kerala's history.

Madhav Gadgil and others argued that unregulated quarrying, construction on steep slopes, road cutting and changes in land use had made the hills more vulnerable, and that these disasters showed the cost of ignoring the WGEEP. Others replied that the immediate trigger was extreme rainfall, which no zoning plan could prevent, and that the Wayanad slide began high on steep, forested slopes. A rapid study by the World Weather Attribution group estimated that climate change had made the extreme one-day rainfall in Wayanad about 10% heavier. Most scientists now agree that disasters in the Ghats result from a **combination** of heavier rainfall linked to climate change, natural slope instability and human changes to the land, which is why land-use planning in sensitive zones matters. The sixth ESA draft was issued within days of the Wayanad disaster.

Think and Do: Compare the two reports

Form two groups. One group represents a gram sabha in a hill village of Wayanad; the other represents the state government planning a highway and a stone quarry. Using the comparison table above, each group should list which recommendations of the Gadgil and Kasturirangan reports it would support and why. Then try to agree on three rules for development in the village's ward.

6.22.8 Significance and lessons

1. The Save Western Ghats march introduced **landscape-level thinking** to Indian environmental movements: ecosystems cross state boundaries, so policy must too.
2. The Gadgil–Kasturirangan debate shows a basic choice in conservation: protect nature through **local democratic control** or through **expert-drawn zones** enforced from above.
3. Delays have costs. More than a decade after the reports, the absence of a final ESA notification leaves the Ghats exposed to piecemeal decisions.
4. Disasters can open a window for policy change, but only if public attention lasts longer than the news cycle.

6.23 Lessons Across the Case Studies

Read together, the seven case studies show how conservation in India has changed over half a century. The programmes of the 1970s (tigers, crocodiles) relied on strict protection of reserves and on breeding and releasing animals. The movements of the late 1970s and 1980s (Silent Valley, Western Ghats) showed that citizens could challenge development projects and demand landscape-level planning. The programmes of the 1990s and later (elephants, vultures, bustards) had to deal with threats that lie far outside protected areas: corridors through farmland, veterinary drugs in village cattle, and power lines across private grasslands.

Several common threads stand out:

- **Science and monitoring**: Honest counts (tiger camera traps, vulture road transects, elephant DNA) revealed crises early and measured success.
- **Law and institutions**: Statutory backing (the NTCA, drug bans, Supreme Court orders) turned plans into obligations.
- **In-situ and ex-situ together**: Breeding centres for crocodiles, vultures and bustards worked, or will work, only where the wild was made safe again.
- **People**: Every programme ultimately depends on the communities who live with tigers, elephants, crocodiles and vultures, or who own the land that bustards and corridors need. Fair compensation, rights and benefits are conservation tools, not optional extras.
- **Development trade-offs**: Dams, mines, railways and even renewable-energy projects carry ecological costs that are best handled at the planning stage.

Key Points

- Project Tiger (1973) grew from 9 to 58 reserves; India had about 3,682 tigers in 2022, roughly 75% of the world's wild tigers.
- Project Elephant (1992) manages 33 elephant reserves and 150 corridors; human–elephant conflict is now its main challenge.
- Veterinary diclofenac caused a 97–99.9% collapse of *Gyps* vultures; the 2006 ban and breeding centres have halted the fall and releases have begun.
- The Great Indian Bustard survives with fewer than 150 wild birds; captive breeding and court-ordered limits on power lines are its last hope.
- The Crocodile Conservation Project (1975) used rear-and-release to rebuild all three Indian crocodilians.
- The Silent Valley movement stopped a dam (1983) and created a national park (1984); the Save Western Ghats movement introduced landscape-level conservation, but the Western Ghats ESA is still not finally notified.

Chapter 6 Review

Summary

- Biodiversity is the variability among living organisms at three levels: genetic, species and ecosystem (CBD, 1992). About 2.1–2.2 million species have been named, out of an estimated 8–9 million eukaryotic species.
- Species diversity combines richness and evenness, and is measured by indices such as the Shannon index. Alpha diversity is within a habitat, beta diversity is the turnover between habitats and gamma diversity is the total for a region.
- India is one of 17 megadiverse countries. It has about 2.4% of the world's land but about 7–8% of recorded species, including over 1,05,000 animal species and very high endemism among amphibians and plants. It is also a Vavilovian centre of origin of crops such as rice, pigeon pea, mango and sugarcane.
- Rodgers and Panwar divided India into 10 biogeographic zones, from the Trans-Himalaya to the Islands. The Deccan Peninsula is the largest (42% of the area) and the Islands the smallest (0.3%).
- A biodiversity hotspot has at least 1,500 endemic vascular plants and has lost at least 70% of its original vegetation. There are 36 hotspots, and India shares four: the Himalaya, Indo-Burma, the Western Ghats and Sri Lanka, and Sundaland (Nicobar Islands).
- India's endemic and threatened species include the lion-tailed macaque, Nilgiri tahr, purple frog, Malabar civet, pygmy hog, hangul, Namdapha flying squirrel, Gangetic dolphin, Great Indian bustard and gharial. The cheetah was declared extinct in India in 1952.
- The IUCN Red List uses nine categories, from Extinct to Not Evaluated. CR, EN and VU are the threatened categories, assigned using five criteria (A–E): population reduction, small range, small and declining population, very small population, and quantitative analysis.
- Biodiversity has ecological, economic (consumptive, productive and non-consumptive), social, ethical, aesthetic and informational value. The Jeevani case shows the value of traditional knowledge and the challenge of sharing its benefits fairly.
- Sacred groves, such as the kavu, devrai, sarna, oran and law kyntang, are community-protected forests. They conserve relict vegetation, springs and cultural traditions, but are threatened by weakening beliefs, encroachment and development.
- The main threats can be summed up as HIPPO. Habitat loss, degradation and fragmentation (including edge effects and linear infrastructure) are the most important. Poaching for trade targets tigers, rhinos, pangolins and star tortoises; India responds through the Wild Life (Protection) Act (amended 2022), the WCCB and CITES.
- Man–wildlife conflict is severe in India: elephants alone killed about 600 people a year in 2019–2024. Barriers, early warning, corridors, changes in land use and quick compensation (₹10 lakh for a death since 2023) reduce it.
- Invasive species such as lantana, Prosopis juliflora, water hyacinth, Parthenium, African catfish and tilapia degrade habitats and cost India billions of dollars.
- Five past mass extinctions each removed about three-quarters or more of species. IPBES (2019) estimates that about 1 million species are now threatened, and the Living Planet Index shows a 73% average decline in monitored vertebrate populations since 1970.

- Conservation can be in-situ (protected areas, biosphere reserves, sacred groves) or ex-situ (zoos, botanical gardens, seed and gene banks, cryopreservation). India has about 1,015 protected areas covering about 5.3% of its area, and 18 biosphere reserves, 13 of them in UNESCO's network.
- Keystone, flagship, umbrella and indicator species guide conservation. Under IUCN guidelines, reintroductions and translocations such as Indian Rhino Vision 2020 and Project Cheetah succeed only when habitat, prey, protection and local support are in place.
- India's conservation case studies show a shift from protecting single species inside reserves (1970s) to managing landscapes, corridors, drugs and infrastructure (2000s onwards).
- Project Tiger (1973) began with 9 reserves and had 58 by March 2025; the All India Tiger Estimation 2022 gave 3,682 tigers (range 3,167–3,925), about 75% of the world's wild tigers.
- The local extinction of tigers in Sariska (2004–05) led to the Tiger Task Force, the National Tiger Conservation Authority (2006), scientific camera-trap monitoring and India's first tiger reintroduction (2008).
- Tiger reserves are zoned into an inviolate core (Critical Tiger Habitat) and a buffer, linked by corridors; relocation of villages from cores remains controversial under the Forest Rights Act.
- In 2023 Project Tiger and Project Elephant were merged into one scheme run by a Project Tiger & Elephant division.
- Project Elephant (1992) works through 33 elephant reserves and 150 mapped corridors; human–elephant conflict, which kills about 500–600 people a year, and train collisions are its biggest challenges.
- India's elephants were estimated at 29,964 in 2017; a new DNA-based method gave 22,446 in 2021–25, a figure not directly comparable with earlier counts.
- The veterinary painkiller diclofenac caused 97–99.9% declines of three *Gyps* vultures in India between 1992 and 2007.
- Veterinary diclofenac was banned in 2006, aceclofenac and ketoprofen in 2023, and nimesulide in 2024–25; breeding centres begun at Pinjore (2004) now release captive-bred vultures into Vulture Safe Zones.
- The Great Indian Bustard has fewer than 150 wild birds; power-line collisions are the main threat; a captive flock of about 70 birds, including the first chick from artificial insemination (2024), offers hope.
- Supreme Court orders (2021, 2024, 2025) on the bustard show the conflict between biodiversity and renewable-energy expansion.
- The Crocodile Conservation Project (1975) used rear-and-release to rebuild gharial, mugger and saltwater crocodile populations; Bhitarkanika's saltwater crocodiles grew from under 100 to 1,826 (2025).
- The Silent Valley movement, led by the KSSP, scientists and the poet Sugathakumari, saved a rainforest from a hydroelectric dam (shelved 1983); the area became a national park in 1984.
- The Save Western Ghats march (1987–88) introduced landscape-level conservation; its ideas resurfaced in the Gadgil (2011) and Kasturirangan (2013) reports, but the Western Ghats ESA remains in draft.

Key Terms

Biodiversity: The variability among living organisms from all sources, including diversity within species, between species and of ecosystems.

Genetic diversity: The variety of genes and alleles within a species, between individuals and populations.

Species richness: The number of different species present in an area.

Species evenness: How equally individuals are distributed among the species in a community.

Shannon index: A diversity index, $H' = -\Sigma\, p_i \ln p_i$, that increases with both species richness and evenness.

Alpha diversity: The diversity of species within a single habitat or community.

Beta diversity: The change in species composition between habitats or along a gradient.

Biogeographic zone: A large region with a distinctive combination of climate, landform and plant and animal life; India has ten.

Biodiversity hotspot: A region with at least 1,500 endemic vascular plant species that has lost at least 70% of its original vegetation.

Endemic species: A species naturally found only in a particular area and nowhere else in the world.

IUCN Red List: The global inventory that assesses the extinction risk of species using standard categories and criteria.

Threatened species: A species listed as Critically Endangered, Endangered or Vulnerable on the IUCN Red List.

Consumptive use value: The value of natural products used directly by people without passing through a market, such as fuelwood and wild food.

Productive use value: The value of biological products sold in markets, such as timber, fish and medicines.

Sacred grove: A patch of natural vegetation protected by a local community because it is dedicated to a deity or ancestors.

Habitat fragmentation: The breaking up of a continuous habitat into smaller, isolated patches.

Edge effect: The changes in conditions and species at the boundary of a habitat patch compared with its interior.

Man–wildlife conflict: Interactions between people and wild animals that harm people, their property or the animals.

Invasive alien species: A species introduced outside its natural range that spreads and harms native biodiversity, the economy or health.

Mass extinction: A geologically short period in which at least about three-quarters of all species die out.

In-situ conservation: Conservation of species in their natural habitats.

Ex-situ conservation: Conservation of species or their genetic material outside their natural habitats.

National park: A protected area under the Wild Life (Protection) Act where all private rights are extinguished and grazing and extraction are not allowed.

Wildlife sanctuary: A protected area under the Wild Life (Protection) Act where some rights of local people may be allowed.

Biosphere reserve: A large area, zoned into core, buffer and transition zones, that combines conservation with sustainable development.

Keystone species: A species whose effect on its ecosystem is far greater than its abundance suggests.

Flagship species: A charismatic species used to raise public support and funds for conservation.

Umbrella species: A species with large area needs whose protection also protects many other species.

Indicator species: A species whose presence or condition reflects the health of its environment.

Reintroduction: Release of a species into part of its natural range from which it has disappeared.

Project Tiger: India's tiger conservation programme, launched in 1973, built around a network of tiger reserves.

Tiger reserve: A protected landscape notified for tiger conservation, with a core and a buffer zone.

Critical Tiger Habitat (core): The inviolate part of a tiger reserve kept free of settlements and resource use.

Buffer zone: The area around a core where people live and land use is regulated to support conservation.

National Tiger Conservation Authority (NTCA): Statutory body created in 2006 under the Wildlife (Protection) Act to oversee tiger conservation.

M-STrIPES: A GPS-based software and mobile app system for recording patrols and ecological data in tiger reserves.

Camera trapping: Use of motion-triggered cameras to photograph and identify individual animals for population estimates.

Elephant reserve: An administrative landscape, not a legal protected-area category, managed for elephant conservation under Project Elephant.

Elephant corridor: A narrow strip of habitat that connects two larger elephant habitats and allows movement between them.

Human–elephant conflict: Negative interactions such as crop raiding, property damage and deaths of people or elephants.

Diclofenac: A non-steroidal anti-inflammatory drug whose residues in cattle carcasses killed vultures across South Asia.

Visceral gout: Deposition of uric acid crystals on internal organs, a sign of kidney failure in poisoned vultures.

Meloxicam: A veterinary painkiller that is safe for vultures and is promoted as a substitute for diclofenac.

Conservation breeding: Breeding a threatened species in captivity to build an insurance population for later release.

Vulture Safe Zone: An area of about 100 km radius around a vulture colony where toxic veterinary drugs have been eliminated.

Double clutching: Removing a bird's first egg for artificial incubation so that it lays a second egg.

Artificial insemination: Placing semen collected from a male into a female to achieve fertilisation without natural mating.

Bird flight diverter: A reflective or coloured device fixed to power lines to make them visible to flying birds.

Rear and release: Collecting eggs, rearing young in safety until they are large enough to escape most predators, and releasing them.

Gharial: A critically endangered, fish-eating crocodilian of large Indian rivers with a long, slender snout.

Kerala Sastra Sahitya Parishad (KSSP): A people's science movement in Kerala that led the campaign to save Silent Valley.

Lion-tailed macaque: An endangered monkey endemic to the Western Ghats rainforests and the symbol of the Silent Valley campaign.

Padayatra: A long foot march used to spread awareness and mobilise people, as in the Save Western Ghats march.

Ecologically Sensitive Area (ESA): An area notified under the Environment (Protection) Act in which harmful activities are restricted.

WGEEP (Gadgil Committee): The Western Ghats Ecology Expert Panel (2011), which proposed treating the entire range as ecologically sensitive.

HLWG (Kasturirangan Committee): The High Level Working Group (2013), which proposed an ESA covering about 37% of the Western Ghats.

Fortress conservation: A model of conservation that excludes local people from protected areas.

Multiple-Choice Questions

1. The definition of biodiversity most widely used in law and policy comes from which agreement?

(a) Ramsar Convention, 1971

(b) Convention on Biological Diversity, 1992

(c) CITES, 1973

(d) Kyoto Protocol, 1997

2. Two ponds each contain four insect species. In Pond X each species makes up 25% of individuals; in Pond Y one species makes up 85%. Which statement is correct?

(a) Pond Y has higher species richness

(b) Both ponds have the same Shannon index

(c) Pond X has higher evenness and a higher Shannon index

(d) Pond X has lower richness but higher evenness

3. The change in species composition between a shola forest and the adjacent grassland is an example of:

(a) Beta diversity

(b) Alpha diversity

(c) Gamma diversity

(d) Genetic diversity

4. India occupies about 2.4% of the world's land area. Roughly what share of the world's recorded species does it hold?

(a) 1–2%
(b) 3–4%
(c) 15–20%
(d) 7–8%

5. Which is the largest biogeographic zone of India by area?

(a) Semi-arid
(b) Gangetic Plain
(c) Deccan Peninsula
(d) Western Ghats

6. Which pair of criteria must a region meet to be called a biodiversity hotspot?

(a) At least 1,500 endemic vascular plants and loss of at least 70% of original vegetation
(b) At least 1,000 endemic vertebrates and loss of at least 50% of forest
(c) At least 500 endemic plants and a protected-area cover below 10%
(d) At least 1,500 endemic animals and a human population above 300 per km^2

7. The Indian part of the Sundaland biodiversity hotspot is the:

(a) Andaman Islands
(b) Lakshadweep Islands
(c) Sundarbans
(d) Nicobar Islands

8. Which of the following is endemic to the Western Ghats?

(a) Pygmy hog
(b) Purple frog
(c) Hangul
(d) Namdapha flying squirrel

9. Under IUCN criterion D, a species with fewer than 50 mature individuals qualifies as:

(a) Critically Endangered
(b) Endangered
(c) Vulnerable
(d) Data Deficient

10. Which IUCN category is NOT a category of threat?

(a) Vulnerable
(b) Endangered
(c) Data Deficient
(d) Critically Endangered

11. Income earned by guides and hotels from tiger safaris is an example of which value of biodiversity?

(a) Consumptive use value
(b) Non-consumptive use value
(c) Existence value
(d) Option value

12. The Jeevani case of Kerala is best known as an example of:

(a) Access and benefit-sharing with a tribal community
(b) Biological control of an invasive weed
(c) Reintroduction of a carnivore
(d) A sacred grove declared a national park

13. "Sarna" sacred groves, usually of sal trees, are associated mainly with which state?

(a) Kerala
(b) Rajasthan
(c) Jharkhand
(d) Meghalaya

14. A small forest fragment surrounded by tea estates becomes hotter, drier and invaded by lantana near its boundary. This is called:

(a) Allelopathy
(b) Enemy release
(c) Habitat loss
(d) Edge effect

15. Which agency was set up by the Government of India in 2007 to combat organised wildlife crime?

(a) Wildlife Crime Control Bureau
(b) National Tiger Conservation Authority
(c) Central Zoo Authority
(d) National Biodiversity Authority

16. The beetle *Zygogramma bicolorata* has been used in India for the biological control of:

(a) Water hyacinth
(b) Lantana camara
(c) Parthenium hysterophorus
(d) Prosopis juliflora

17. According to the IPBES Global Assessment (2019), about how many species of plants and animals are threatened with extinction?

(a) 10,000
(b) 1 million
(c) 100,000
(d) 8 million

18. In a biosphere reserve, villages, farms and sustainable forestry are mainly found in the:

(a) Core zone
(b) Buffer zone
(c) Transition zone
(d) Corridor zone

19. Under the Wild Life (Protection) Act, in which type of protected area are all private rights extinguished and grazing prohibited?

(a) Wildlife sanctuary
(b) Community reserve
(c) Conservation reserve
(d) National park

20. Releasing African cheetahs at Kuno to fill the role of the extinct Asiatic cheetah is, in IUCN terms, an example of:

(a) Ecological replacement
(b) Reinforcement
(c) Head-starting
(d) Ex-situ conservation

21. In which year was Project Tiger launched?

(a) 1972
(b) 1973
(c) 1975
(d) 1992

22. According to the All India Tiger Estimation 2022, the estimated number of tigers in India was about:

(a) 2,226
(b) 2,967
(c) 3,682
(d) 4,500

23. The National Tiger Conservation Authority was created by an amendment to which Act?

(a) Environment (Protection) Act, 1986
(b) Forest (Conservation) Act, 1980
(c) Wildlife (Protection) Act, 1972
(d) Biological Diversity Act, 2002

24. The disappearance of all tigers from which reserve in 2004–05 led to the setting up of the Tiger Task Force?

(a) Sariska
(b) Corbett
(c) Kanha
(d) Sundarbans

25. Which part of a tiger reserve is meant to be kept inviolate?

(a) Buffer zone
(b) Corridor
(c) Eco-development zone
(d) Critical Tiger Habitat

26. Which technology is used in M-STrIPES?

(a) Satellite radar imaging of forests
(b) GPS-based recording of patrols and wildlife signs
(c) DNA barcoding of plants
(d) Acoustic monitoring of birds

27. The 2021–25 all-India elephant estimation differed from earlier counts mainly because it:

(a) counted only captive elephants
(b) used aerial photographs
(c) identified individual elephants from DNA in dung
(d) counted only tuskers

28. An elephant corridor is best described as:

(a) a fenced enclosure for problem elephants
(b) a strip of habitat linking two larger elephant habitats
(c) a railway line with speed limits
(d) a captive elephant camp

29. Diclofenac killed vultures mainly by causing:

(a) eggshell thinning
(b) lead poisoning
(c) kidney failure and visceral gout
(d) respiratory infection

30. India's first Vulture Conservation Breeding Centre was set up at:

(a) Pinjore, Haryana
(b) Bharatpur, Rajasthan
(c) Bhopal, Madhya Pradesh
(d) Rani, Assam

31. Which NSAID is considered safe for vultures and is promoted as an alternative to diclofenac?

(a) Ketoprofen
(b) Aceclofenac
(c) Nimesulide
(d) Meloxicam

32. The most immediate threat to the Great Indian Bustard in the Thar desert is:

(a) collision with overhead power lines
(b) diclofenac poisoning
(c) invasive fish
(d) forest fires

33. In 2024 the first Great Indian Bustard chick produced by artificial insemination hatched in which district?

(a) Kachchh
(b) Jaisalmer
(c) Solapur
(d) Kurnool

34. In the Great Indian Bustard case (March 2024), the Supreme Court recognised a right against:

(a) the adverse effects of climate change
(b) noise pollution
(c) displacement by dams
(d) genetically modified crops

35. The Crocodile Conservation Project relied mainly on which method?

(a) Translocating adults between countries
(b) Rear and release of young reared from wild-collected eggs
(c) Artificial insemination
(d) Culling of predators of adults

36. The world's largest population of gharials lives in:

(a) Bhitarkanika National Park
(b) Sundarbans Tiger Reserve
(c) National Chambal Sanctuary
(d) Silent Valley National Park

37. The Silent Valley hydroelectric project was proposed on which river?

(a) Periyar
(b) Kunthipuzha
(c) Chaliyar
(d) Kabini

38. Which species became the symbol of the Silent Valley campaign?

(a) Nilgiri tahr
(b) Great hornbill
(c) Lion-tailed macaque
(d) Malabar civet

39. Which statement correctly compares the two Western Ghats reports?

(a) Kasturirangan proposed the whole range as an ESA; Gadgil proposed 37%
(b) Gadgil proposed the whole range as ecologically sensitive with three zones; Kasturirangan proposed about 37% as ESA
(c) Both proposed exactly the same area
(d) Neither report dealt with mining

Short-Answer Questions

1. Define biodiversity according to the Convention on Biological Diversity and distinguish between its three levels, with one Indian example of each.

2. Explain species richness, species evenness and alpha, beta and gamma diversity. Calculate the Shannon index for a community of three species with 50, 30 and 20 individuals.
3. Why is India called a mega-biodiversity nation? Give four reasons for its high biodiversity.
4. List the ten biogeographic zones of India and name one characteristic species of any five of them.
5. What are the criteria for identifying a biodiversity hotspot? Name India's four hotspots and one endemic species from each.
6. Explain the IUCN Red List categories with the help of a labelled diagram, and give one Indian species in each threatened category.
7. Differentiate between consumptive, productive and non-consumptive use values of biodiversity with Indian examples.
8. What are sacred groves? Give their local names in any four Indian states and explain two threats they face.
9. What is habitat fragmentation? Explain edge effects and the impact of linear infrastructure on wildlife.
10. Why do invasive species succeed in new areas? Describe the impacts and control of *Lantana camara* in India.
11. Distinguish between a national park, a wildlife sanctuary and a biosphere reserve. Draw the zonation of a biosphere reserve.
12. Explain keystone, flagship, umbrella and indicator species with one Indian example of each.
13. Explain the difference between the core and buffer zones of a tiger reserve. Why are corridors also needed?
14. How did the Sariska crisis change the way Project Tiger was managed?
15. What is M-STrIPES, and how does it help in protecting tigers?
16. Why are the elephant population estimates of 2017 and 2021–25 not directly comparable?
17. List four methods used in India to reduce human–elephant conflict.
18. Describe how veterinary diclofenac entered the food chain of vultures and killed them.
19. What is a Vulture Safe Zone? Why is it necessary before captive-bred vultures are released?
20. Why are power lines especially dangerous for the Great Indian Bustard? What did the Supreme Court order in December 2025?
21. Explain the rear-and-release method with reference to the gharial.
22. What roles did scientists, the KSSP and writers play in the Silent Valley movement?
23. State the main objectives of the Save Western Ghats march of 1987–88.
24. Give three differences between the Gadgil and Kasturirangan reports.

Long-Answer Questions

1. "India's biodiversity is both exceptionally rich and exceptionally threatened." Discuss this statement with reference to India's megadiversity, biogeographic zones and biodiversity hotspots.

2. Explain the IUCN Red List categories and criteria (A–E) in detail. How can the Red List and the schedules of the Wild Life (Protection) Act together guide conservation of species such as the Great Indian bustard and the hangul?
3. Discuss the ecological, economic, social, ethical, aesthetic and informational values of biodiversity. Why is it difficult to put a monetary value on biodiversity? Illustrate with Indian examples, including sacred groves.
4. Critically examine the causes and consequences of man–wildlife conflict in India. Evaluate the measures adopted to reduce it, including compensation, and suggest improvements.
5. "Biological invasion is a silent threat to Indian biodiversity." Discuss with reference to at least four invasive species, their impacts and methods of control. Relate your answer to the current mass-extinction crisis.
6. Compare in-situ and ex-situ conservation. With reference to Indian Rhino Vision 2020, the Asiatic lion relocation debate and Project Cheetah, assess the role and limitations of species reintroduction and translocation.
7. "Project Tiger is a success story with unfinished business." Evaluate this statement using population data, institutional changes and criticisms of the programme.
8. Compare Project Tiger and Project Elephant in terms of objectives, strategies and challenges. Why is conflict with people a bigger problem for elephant conservation?
9. Trace the story of India's vulture crisis from the discovery of the decline to the release of captive-bred birds. What does it teach us about the links between ex-situ and in-situ conservation?
10. The Great Indian Bustard case has been described as a conflict between "green and green". Discuss, with reference to renewable energy, power lines and court orders.
11. Compare the Silent Valley and Save Western Ghats movements. Why did one achieve a clear victory while the other's main demand remains unfulfilled?
12. Using the Gadgil–Kasturirangan debate and the Kerala floods (2018) and Wayanad landslides (2024), discuss how land-use decisions in the Western Ghats affect both biodiversity and human safety.

Further Reading

- Primack, R. B. (2014). *Essentials of Conservation Biology* (6th ed.). Sinauer Associates.
- Singh, J. S., Singh, S. P. & Gupta, S. R. (2017). *Ecology, Environmental Science and Conservation*. S. Chand.
- Gadgil, M. & Guha, R. (1993). *This Fissured Land: An Ecological History of India*. Oxford University Press.
- Rodgers, W. A., Panwar, H. S. & Mathur, V. B. (2002). *Wildlife Protected Area Network in India: A Review*. Wildlife Institute of India, Dehradun.
- IPBES (2019). *Global Assessment Report on Biodiversity and Ecosystem Services*. IPBES Secretariat, Bonn. https://www.ipbes.net
- IUCN (2012). *IUCN Red List Categories and Criteria: Version 3.1* (2nd ed.) and the IUCN Red List website. https://www.iucnredlist.org
- IUCN/SSC (2013). *Guidelines for Reintroductions and Other Conservation Translocations*. IUCN, Gland.

- Wildlife Institute of India, ENVIS Centre on Wildlife and Protected Areas (protected-area database). https://wiienvis.nic.in
- National Tiger Conservation Authority and Wildlife Institute of India (2023). *Status of Tigers, Co-predators and Prey in India 2022*. NTCA, New Delhi, and WII, Dehradun.
- Ministry of Environment, Forest and Climate Change (2020). *Action Plan for Vulture Conservation in India 2020–2025*. Government of India.
- Western Ghats Ecology Expert Panel (2011). *Report of the Western Ghats Ecology Expert Panel* (Gadgil Report). MoEFCC, Government of India.
- Menon, V., Tiwari, S.K., Ramkumar, K., Kyarong, S., Ganguly, U. and Sukumar, R. (eds.) (2017). *Right of Passage: Elephant Corridors of India* (2nd ed.). Wildlife Trust of India.
- D'Monte, D. (1985). *Temples or Tombs? Industry versus Environment: Three Controversies*. Centre for Science and Environment.
- National Tiger Conservation Authority (official website): https://ntca.gov.in

Answers to Multiple-Choice Questions

1. (b)	2. (c)	3. (a)	4. (d)	5. (c)	6. (a)	7. (d)	8. (b)	9. (a)	10. (c)
11. (b)	12. (a)	13. (c)	14. (d)	15. (a)	16. (c)	17. (b)	18. (c)	19. (d)	20. (a)
21. (b)	22. (c)	23. (c)	24. (a)	25. (d)	26. (b)	27. (c)	28. (b)	29. (c)	30. (a)
31. (d)	32. (a)	33. (b)	34. (a)	35. (b)	36. (c)	37. (b)	38. (c)	39. (b)	

Chapter 7: Human Communities and the Environment

Every environmental problem studied in earlier chapters, from a polluted river to a warming climate, is ultimately about people: how many people there are, how much they consume, who gains from the use of nature and who pays the price. This chapter turns to the human side of environmental studies. It begins with the growth of human population and its effects on the environment, health and welfare, and introduces the carbon footprint as a way of measuring an individual's or a nation's contribution to climate change. It then examines the people displaced by dams, mines and other development projects, and the great environmental movements of India, from the Bishnois of Rajasthan to the Narmada Bachao Andolan, that grew out of such conflicts. The later sections cover environmental justice and the National Green Tribunal, the ethical and religious ideas that shape how people treat nature, and the ways in which communication and public campaigns such as the CNG transition in Delhi, Swachh Bharat and the eco-club programme can change behaviour.

Learning Objectives

- Describe trends in human population growth, explain the demographic transition, and analyse the impacts of population on environment, health and welfare using the I = PAT identity.
- Define carbon footprint, distinguish Scope 1, 2 and 3 emissions, and calculate a simple household footprint using Indian emission factors.
- Explain the risks faced by people displaced by development projects and summarise India's legal framework for resettlement and rehabilitation, with case studies.
- Narrate the history, leaders, methods and outcomes of the Bishnoi, Chipko, Appiko, Silent Valley and Narmada movements, and compare them.
- Explain the concept of environmental justice and the structure, jurisdiction and landmark decisions of the National Green Tribunal.
- Compare anthropocentric, biocentric and ecocentric ethics, and discuss the role of religions and cultural practices in conservation.
- Evaluate environmental communication and awareness programmes in India, including the CNG transition, Swachh Bharat Mission, NEAC and the National Green Corps eco-clubs.

7.1 Human Population Growth: Impacts on Environment, Health and Welfare

Demography is the scientific study of human populations: their size, composition, distribution and change over time. Population matters to environmental science because every person needs food, water, energy, shelter and space, and produces waste. Yet the number of people alone never tells the whole story. How much each person consumes, and with what technology, matters at least as much. This section looks at the trends first and then at the impacts.

7.1.1 Trends in human population growth

For most of human history the world's population grew very slowly, because high birth rates were cancelled out by equally high death rates from famine, disease and war. The world reached its first billion people only around 1800. After that, growth accelerated sharply as agriculture, sanitation, vaccines and modern medicine reduced death rates. The second billion came by about 1927, the third by 1960, and each later billion arrived in just 12–14 years. The United Nations marked the birth of the world's eight-billionth person on 15 November 2022.

According to the UN *World Population Prospects 2024* (WPP 2024), the world population was about 8.2 billion in 2025. The rate of growth, however, is falling. The global **total fertility rate (TFR)**, the average number of children a woman would have over her lifetime at current rates, has fallen from about 5 in 1950 to about 2.2 today. The UN's medium projection is that world population will peak at about 10.3 billion in the mid-2080s and then begin a slow decline. Most future growth will occur in sub-Saharan Africa and parts of South Asia, while populations in China, Japan, South Korea and much of Europe are already shrinking.

India's story is central to these trends. India's population rose from about 36 crore at the 1951 Census to 121 crore (1.21 billion) at the 2011 Census. In April 2023, according to UN estimates, India overtook China to become the world's most populous country. UN WPP 2024 estimates India's population at about 1.46 billion (146 crore) in 2025, roughly 18% of humanity, living on only about 2.4% of the world's land area and with about 4% of its fresh water. India's next Census, delayed from 2021, is scheduled with a reference date of 1 March 2027.

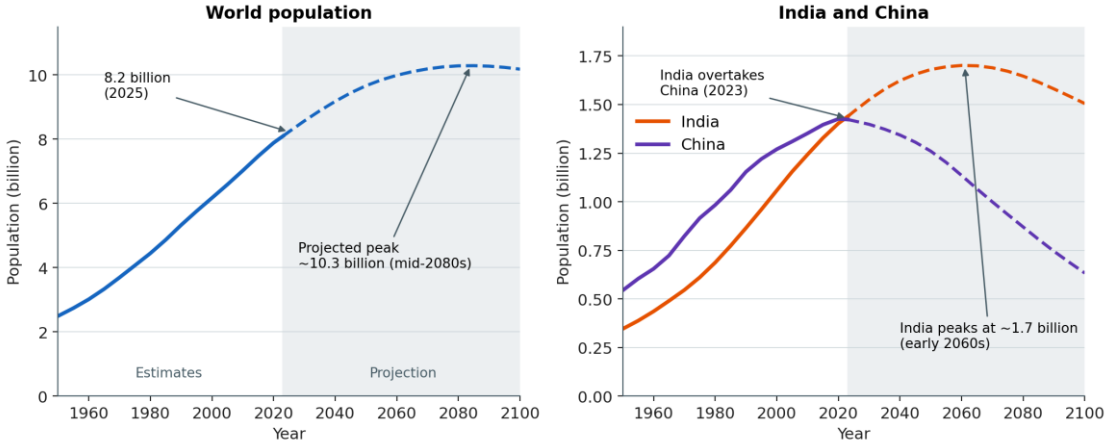


Figure 7.1: *World population (left) and the populations of India and China (right), 1950–2100. Solid lines are estimates; dashed lines are the UN medium-variant projection. Source: UN World Population Prospects 2024, via Our World in Data.*

The figure above shows that India's growth has slowed considerably. India's TFR fell from about 5.9 in 1950 to 2.0 in the National Family Health Survey (NFHS-5, 2019–21), which is below the **replacement level fertility** of about 2.1 children per woman, the level at which each generation exactly replaces itself. Why, then, is India's population still growing? The answer is **population momentum**: because of high fertility in the past, a very large share of Indians are young adults now entering their reproductive years. Even if each couple has only two children, the total number of births stays high for several decades. The UN projects that India's population will peak at about 1.7 billion in the early 2060s.

Fertility in India varies widely between states. Bihar (TFR about 3.0), Meghalaya and Uttar Pradesh remain above replacement level, while Kerala, Tamil Nadu, West Bengal, Punjab and Delhi have TFRs well below 2.0 (NFHS-5). This regional difference has major consequences for jobs, migration, ageing and even the future allocation of parliamentary seats.

Table 7.1: Key population terms

Term	Meaning
Crude birth rate (CBR)	Live births per 1,000 people per year
Crude death rate (CDR)	Deaths per 1,000 people per year
Natural growth rate	(CBR − CDR) ÷ 10, expressed as % per year
Total fertility rate (TFR)	Average children born per woman over her lifetime
Replacement level	TFR of about 2.1 at which a population stabilises in the long run
Doubling time	Years for a population to double; roughly 70 ÷ growth rate (%)
Dependency ratio	Children (0–14) plus elderly (65+) per 100 people of working age

> **The rule of 70**
>
> A population growing at 2% per year doubles in about 70 ÷ 2 = 35 years. India grew at about 2.2% per year in the 1970s (doubling time about 32 years); today it grows at under 1% a year, and will stop growing long before it doubles again.

7.1.2 The demographic transition

The **demographic transition model** describes how countries move from high birth and death rates to low birth and death rates as they develop. It was first developed by Warren Thompson (1929) and later by Frank Notestein (1945), based on the experience of European countries. The model has four classic stages, with a fifth sometimes added.

1. **Stage 1 (high stationary):** Both birth and death rates are high (around 35–40 per 1,000). Population grows very slowly. All countries were in this stage before about 1800; India was in it until about 1920.
2. **Stage 2 (early expanding):** Death rates fall rapidly because of better food supply, clean water, sanitation, vaccination and medicine, but birth rates stay high. The gap between the two causes a **population explosion**. India was in this stage from about 1921, the "great divide" year after which its population never declined again, until the 1970s.
3. **Stage 3 (late expanding):** Birth rates begin to fall as child mortality declines, education (especially of girls) spreads, people move to cities and contraception becomes available. Growth continues but slows.
4. **Stage 4 (low stationary):** Both rates are low (around 10–15 per 1,000). Population is large but stable, and the population ages. Most of Europe reached this stage in the late twentieth century.
5. **Stage 5 (declining):** Birth rates fall below death rates and the population shrinks, as in Japan, Italy and South Korea today.

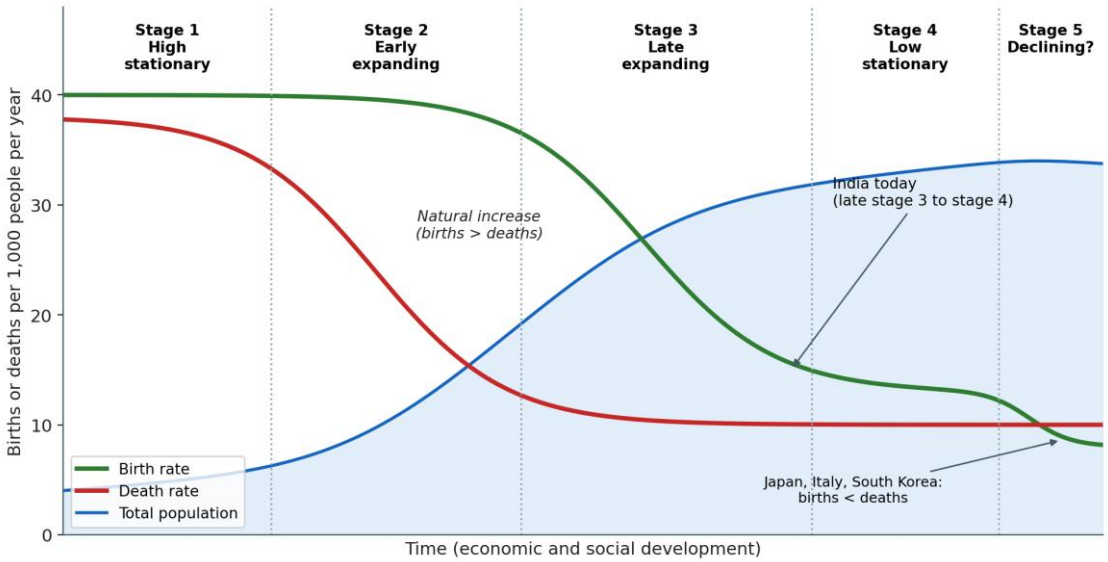


Figure 7.2: *The demographic transition model. The gap between birth and death rates (natural increase) is widest in stages 2 and 3; India today is moving from late stage 3 into stage 4. Source: schematic, after Thompson (1929) and Notestein (1945).*

India's crude birth rate was about 19 per 1,000 and its crude death rate about 7–8 per 1,000 in recent Sample Registration System (SRS) data, placing it in late stage 3 to stage 4. One consequence of passing through the transition is a **demographic dividend**: a period of several decades during which the working-age population (15–64 years) is large compared with dependent children and elderly people. About two-thirds of Indians are of working age today. The dividend can raise savings and economic growth, but only if young people are healthy, educated and employed. If they are not, the same bulge becomes a source of unemployment and social stress. Later, as in Kerala and Tamil Nadu already, the population will age, increasing the need for pensions, health care and elder care.

India was the first country to adopt an official family planning programme, in 1952. The forced sterilisation drive of the Emergency (1975–77) caused great suffering and a lasting backlash, showing that coercion is unethical and counterproductive. The **National Population Policy 2000** shifted the emphasis to voluntary choice, reproductive and child health, and women's empowerment. Kerala, which reached low fertility in the 1980s at modest income levels, shows that female literacy, low infant mortality and access to health care are the most powerful "contraceptives".

7.1.3 Impacts of population growth on the environment

The environmental impact of a population depends on three things at once: how many people there are, how much each person consumes, and how damaging the technology used to meet that consumption is. Ecologists Paul Ehrlich and John Holdren expressed this in 1971 as the **I = PAT identity**:

$$I = P \times A \times T$$

Here **I** is environmental impact (for example, tonnes of CO_2 emitted or hectares of forest cleared), **P** is population, **A** is affluence or consumption per person, and **T** is technology, the impact per unit of consumption.

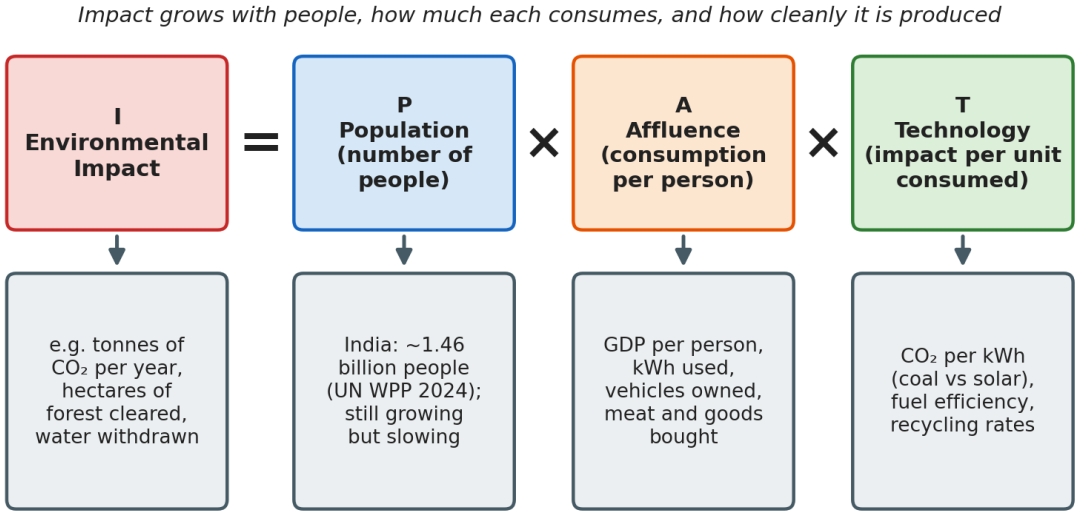


Figure 7.3: *The I = PAT identity. Environmental impact is the product of population, affluence (consumption per person) and technology (impact per unit consumed). Source: after Ehrlich and Holdren (1971).*

The identity shows why blaming population alone is misleading. The average person in the United States emits about seven times as much fossil CO_2 as the average Indian (see the carbon footprint section below), so a small increase in the US population can add as much to emissions as a much larger increase in India. It also shows where solutions lie: impact can fall even as population and affluence rise if technology improves enough. India's electricity sector illustrates this: the share of solar and wind power is rising quickly, so the CO_2 emitted per kilowatt-hour (T) is slowly falling while electricity use per person (A) is rising.

The main environmental impacts of a growing and more affluent population in India are summarised below. Each is treated in detail in earlier chapters.

- **Land and forests:** Demand for farmland, housing, roads and mines converts forests, grasslands and wetlands (see Chapter 3). Because the cultivated area has hardly grown while population has more than quadrupled, cropland per person in India has fallen by well over half since 1951, and the average farm holding has shrunk to just over one hectare (Agriculture Census 2015–16). Grazing and fuelwood collection degrade the forests that remain.
- **Water:** India's per-capita annual water availability fell from about 5,200 m^3 in 1951 to about 1,500 m^3 in 2021 (Ministry of Jal Shakti) and is expected to fall further, close to the "water stress" threshold of 1,700 m^3 and heading towards "scarcity" (below 1,000 m^3) in many basins. Groundwater is over-exploited in large parts of Punjab, Haryana, Rajasthan and Delhi (see Chapter 3).
- **Biodiversity:** Habitat loss, fragmentation and human–wildlife conflict increase as human settlements spread into forests and corridors (see Chapter 6).
- **Pollution and waste:** Indian cities generate roughly 1.5–1.6 lakh tonnes of municipal solid waste a day. Untreated sewage, vehicle exhaust and industrial emissions grow with urban population (see Chapter 4).
- **Energy and climate:** More people with rising incomes use more electricity, fuel and cement. India is now the world's third-largest emitter of CO_2 in total, though its emissions per person remain low (see Chapter 5).

- **Urbanisation:** About 36% of Indians lived in towns and cities in 2023 (World Bank estimate), and the share is rising. Rapid, unplanned urban growth produces slums, traffic congestion, heat islands and flooding of low-lying settlements.

Population growth does not act alone: poverty and insecure land rights push poor households to over-use common lands, while affluence drives high consumption of energy, meat, vehicles and goods.

> **Think and Do: Population and environment in your state**
>
> Using Census data (1991, 2001, 2011) and a state environment report, plot your state's population against one environmental indicator, such as forest cover, groundwater level or number of registered vehicles. Calculate the correlation coefficient. Does a strong correlation prove that population growth caused the change? What other variables (income, policy, technology) could explain it?

7.1.4 Impacts on human health

Population growth, crowding and environmental change interact to shape human health. Four sets of problems are most important in India.

Water, sanitation and hygiene (WASH)

Unsafe drinking water, poor sanitation and lack of handwashing cause diarrhoeal diseases, typhoid, cholera, hepatitis A and E, and intestinal worms. Repeated infections in early childhood contribute to **stunting** (low height for age), which affected about 35% of Indian children under five in NFHS-5. Crowded urban slums with shared toilets and irregular water supply are particularly vulnerable. The World Health Organization (WHO) estimates that unsafe WASH causes more than a million deaths worldwide every year. India's Swachh Bharat Mission and Jal Jeevan Mission (tap water to every rural household) are national responses, discussed later in this chapter.

Air pollution

Air pollution is the largest environmental health risk in India. The India State-Level Disease Burden Initiative estimated that about 1.67 million deaths in India in 2019, nearly 18% of all deaths, were attributable to air pollution, from both outdoor (ambient) air and household smoke from solid cooking fuels (*The Lancet Planetary Health*, 2020). Exposure to fine particulate matter ($PM_{2.5}$) causes heart disease, stroke, lung cancer, chronic obstructive pulmonary disease and lower respiratory infections, and harms children's lung development. Growing cities mean more vehicles, construction dust and waste burning. Household air pollution has fallen as the Pradhan Mantri Ujjwala Yojana (2016) provided LPG connections to poor households, although many still use firewood when refills are costly. Delhi's air pollution crisis is analysed in Chapter 4.

Vector-borne diseases

Diseases spread by mosquitoes and other **vectors** (organisms that carry a pathogen from one host to another) are closely linked to environmental conditions. Dengue and chikungunya, spread by the *Aedes aegypti* mosquito, which breeds in water stored in containers, coolers, tyres and construction sites, have spread from a few large cities to almost every state as urbanisation and warmer temperatures extend the mosquito's range. Malaria has declined sharply in India since 2015 but persists in forested tribal districts of Odisha, Chhattisgarh, Jharkhand and the North-East. Dams,

irrigation canals and urban water storage can create new breeding habitats, which is why health impact assessment should be part of project planning.

Heat stress

India is experiencing more frequent and intense heat waves as the climate warms (see Chapter 5), and the urban **heat island** effect, in which concrete, asphalt and the absence of trees make cities several degrees warmer than the countryside, adds to the danger. Heat stress kills through heat stroke and by worsening heart and kidney disease, and reduces the working capacity of outdoor workers such as construction labourers, street vendors and farm workers.

7.1.5 Impacts on human welfare

Human welfare refers to the overall well-being of people: income and livelihood, education, health, housing, security and the freedom to make choices. Rapid population growth combined with environmental degradation can reduce welfare in several ways:

- **Pressure on public services:** schools, hospitals, transport and housing in fast-growing cities struggle to keep up.
- **Livelihood loss:** degradation of soil, water, forests and fisheries hits small farmers, pastoralists, fishers and forest dwellers hardest, pushing many to migrate.
- **Food security:** India produces enough grain in total, but malnutrition persists because of poverty and unequal access, and climate change threatens future harvests.
- **Inequality and gender:** Women and girls carry most of the burden of fetching water and fuelwood, and lose time for school and paid work when these resources become scarce.
- **Conflict and displacement:** Competition for land, water and minerals can lead to conflict, and large development projects displace communities (see the section on resettlement below).

Investment in education, health and women's empowerment reduces fertility, improves welfare and lowers pressure on the environment at the same time, which is why the **Sustainable Development Goals** (see Chapter 1) treat these aims as connected.

Key Points

- The world population passed 8 billion in 2022 and is projected (UN WPP 2024) to peak at about 10.3 billion in the mid-2080s; India became the most populous country in 2023.
- India's TFR has fallen to about 2.0, but population momentum means growth continues until the early 2060s.
- The demographic transition explains how falling death rates followed by falling birth rates produce rapid growth and then stabilisation.
- I = PAT shows that environmental impact depends on consumption and technology as well as numbers of people.
- The major environmental health risks in India are unsafe WASH, air pollution, vector-borne diseases and heat stress.

7.2 Carbon Footprint

The I = PAT identity shows that consumption matters as much as population. The **carbon footprint** turns this idea into a number that individuals, households, companies, cities and countries can measure and reduce. It links everyday choices, such as how you travel to college, what you eat and how your home is cooled, to the global problem of climate change (see Chapter 5).

7.2.1 Concept of the carbon footprint

A **carbon footprint** is the total amount of greenhouse gases emitted directly and indirectly by a person, organisation, product, event or country over a given period, usually one year, expressed as **carbon dioxide equivalent (CO_2e)**. The term grew out of the broader idea of the **ecological footprint** developed by Mathis Wackernagel and William Rees in the 1990s, which measures the area of productive land and water needed to supply a population's resources and absorb its wastes. The carbon footprint focuses only on greenhouse gases and is measured in tonnes rather than hectares.

Gases other than CO_2 are converted to CO_2e by multiplying by their **global warming potential (GWP)**, a measure of how much heat one kilogram of the gas traps over 100 years compared with one kilogram of CO_2.

Table 7.2: Global warming potentials of major greenhouse gases (100-year, IPCC AR6)

Gas	Main human sources	GWP (CO_2 = 1)
Carbon dioxide (CO_2)	Burning coal, oil, gas; cement; deforestation	1
Methane (CH_4)	Paddy fields, livestock, landfills, gas leaks	about 27–30
Nitrous oxide (N_2O)	Nitrogen fertilisers, manure, some industry	273
Hydrofluorocarbons (HFCs)	Air conditioners, refrigerators	about 100 to over 10,000
Sulfur hexafluoride (SF_6)	Electrical switchgear	about 25,000

So 1 kg of methane released from a landfill counts as roughly 27–30 kg CO_2e. A product's footprint can be measured across its whole life cycle, "cradle to grave": extraction of raw materials, manufacture, transport, use and disposal. This method is called **life cycle assessment (LCA)**.

7.2.2 Scope 1, Scope 2 and Scope 3 emissions

The most widely used accounting framework, the **Greenhouse Gas Protocol** developed by the World Resources Institute and the World Business Council for Sustainable Development (first published 2001), divides emissions into three "scopes". The scheme was designed for organisations, but it works equally well for a household or a college.

- **Scope 1 (direct emissions):** greenhouse gases released from sources that you own or control. For a household: burning LPG or firewood for cooking, petrol or diesel in the family vehicle, a diesel generator. For a college: its diesel generator sets, college buses and kitchen fuel.
- **Scope 2 (indirect emissions from purchased energy):** emissions released at the power station that generated the electricity you buy. You do not burn the coal yourself, but your demand causes it to be burned. In India, where most electricity still comes from coal, Scope 2 is often the largest item for urban households and offices.

- **Scope 3 (all other indirect emissions):** emissions in the value chain, upstream and downstream. For a household these include the food you eat, the clothes and phones you buy, flights and train journeys, water supply and the treatment of your waste. For a company they include purchased materials, business travel, employee commuting and the use of its products by customers. Scope 3 is usually the largest but hardest category to measure.

In India, the Securities and Exchange Board of India (SEBI) requires the top 1,000 listed companies to publish a **Business Responsibility and Sustainability Report (BRSR)**, which includes disclosure of Scope 1 and Scope 2 emissions; Scope 3 reporting is encouraged. The government also notified a national **Carbon Credit Trading Scheme** in 2023 to create an Indian carbon market.

7.2.3 Calculating a carbon footprint

The basic method is simple. For each activity:

$$\text{Emissions (kg } CO_2\text{e)} = \text{Activity data} \times \text{Emission factor}$$

Activity data is the quantity of the activity: kilowatt-hours of electricity from the bill, litres of petrol from fuel receipts, kilograms of LPG, kilometres travelled. An **emission factor** is the mass of greenhouse gas released per unit of activity. Indian emission factors should be used where available, because India's electricity mix differs from that of other countries.

Table 7.3: Typical emission factors for Indian households

Activity	Unit	Emission factor (kg CO_2)	Source
Grid electricity	per kWh	about 0.72	CEA CO_2 Baseline Database (FY 2023–24)
LPG (cooking)	per kg	about 2.98	IPCC 2006 default values
Petrol	per litre	about 2.3	IPCC 2006 default values
Diesel	per litre	about 2.65	IPCC 2006 default values
CNG	per kg	about 2.7	IPCC 2006 default values

The **Central Electricity Authority (CEA)** publishes India's grid emission factor every year in its *CO_2 Baseline Database for the Indian Power Sector*. The weighted average for 2023–24 was about 0.72–0.73 tonne CO_2 per megawatt-hour, that is, about 0.72 kg per kWh. This is well above the world average of roughly 0.45–0.48 kg per kWh, because coal still produces about three-quarters of India's electricity. As solar and wind generation grow, the factor is slowly declining.

Case Study: Worked example: the carbon footprint of a Delhi household

Consider a family of four living in a flat in Delhi. In a typical month they use 250 kWh of electricity (fans, lights, a refrigerator, a television and an air conditioner for part of the year), one 14.2 kg LPG cylinder for cooking, 15 litres of petrol in a scooter and 30 litres of petrol in a small car.

Step 1, Scope 2 (electricity): 250 kWh × 0.72 kg/kWh = 180 kg CO_2.

Step 2, Scope 1 (cooking): 14.2 kg LPG × 2.98 kg/kg = about 42 kg CO_2.

Step 3, Scope 1 (transport): (15 + 30) litres of petrol × 2.3 kg/litre = about 104 kg CO_2.

Monthly total for direct energy use: 180 + 42 + 104 = 326 kg CO_2. Annual total: 326 × 12 = about 3.9 tonnes CO_2, or about 1 tonne per person.

Interpretation: electricity is the largest single item, so the biggest savings would come from energy-efficient appliances (a 5-star inverter air conditioner, LED lights, a star-rated refrigerator), setting the air conditioner at 24 °C, or rooftop solar panels. Shifting the car's 30 litres a month to the Delhi Metro would save about 70 kg of CO_2 a month. The calculation leaves out Scope 3: food, clothes, gadgets, flights and waste would add substantially to the total. A single return flight from Delhi to Mumbai, for example, adds a few hundred kilograms of CO_2 per passenger.

Think and Do: Calculate your own footprint

Collect your household's last three electricity bills, the number of LPG cylinders used in a year, and monthly fuel purchases. Use the emission factors in the table above to calculate your Scope 1 and Scope 2 emissions, then divide by the number of family members. Compare the result with India's per-capita figure in the chart below and identify your two largest sources. The same method, applied to a college's electricity, diesel and transport records, gives the college's footprint (see Appendix A).

7.2.4 Carbon footprints compared: India and the world

National carbon footprints are usually reported as **territorial emissions**, the greenhouse gases released within a country's borders. Dividing by population gives **per-capita emissions**, which vary enormously between countries.

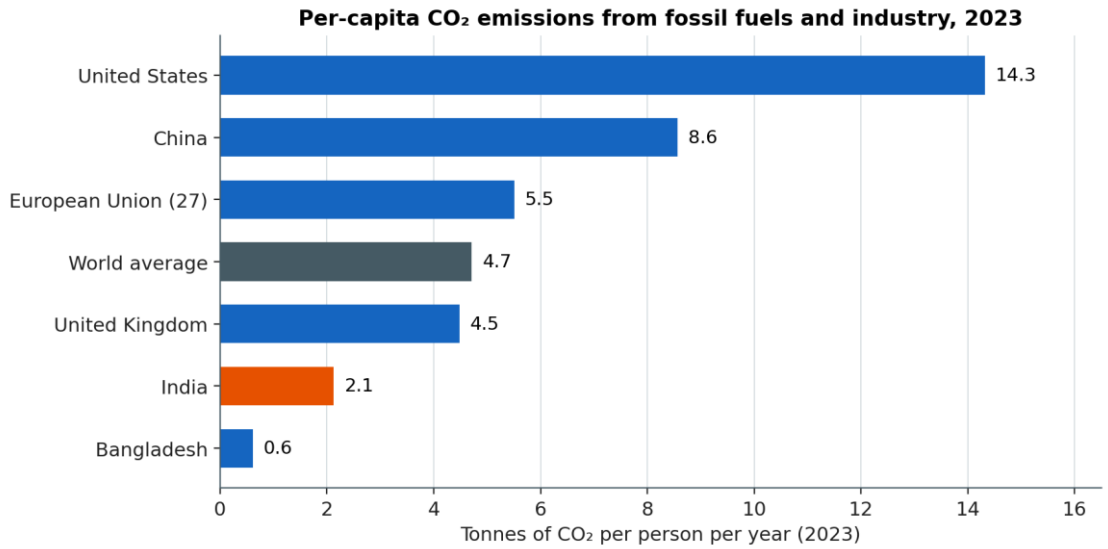


Figure 7.4: *Per-capita CO_2 emissions from fossil fuels and industry in 2023, in tonnes per person per year. Source: Global Carbon Budget 2024, via Our World in Data.*

In 2023 the average person in the United States emitted about 14.3 tonnes of fossil CO_2, the average person in China about 8.6 tonnes, the average European Union resident about 5.5 tonnes, and the world average was about 4.7 tonnes. The average Indian emitted about 2.1 tonnes, less than half the world average and about one-seventh of the US figure (Global Carbon Budget 2024). In total, however, India is the world's third-largest emitter, after China and the United States, because of its huge population.

Three further points are needed to read these numbers fairly:

- **Historical responsibility:** CO_2 stays in the atmosphere for centuries, so warming depends on cumulative emissions. The United States is responsible for about a quarter of all fossil CO_2 emitted since 1850, the European Union for about a fifth, and India for only about 3%. This is the basis of the principle of **common but differentiated responsibilities** in the climate treaties (see Chapter 5).
- **Consumption-based accounting:** Some emissions are "exported" in traded goods. Steel, cement and electronics made in China and India for consumers in Europe and North America are counted in the producing country's territorial emissions, so the consumption-based footprints of rich countries are higher than their territorial ones.
- **Inequality within countries:** Footprints differ greatly between rich and poor people in every country. Studies by Oxfam and the Stockholm Environment Institute estimate that the richest 10% of the world's people account for about half of consumption-related emissions, while the poorest half account for only about a tenth. In India too, the footprint of an affluent urban household with cars, air conditioners and frequent flights is many times that of a rural household.

Think and Do: Comparing countries with online data

Open the CO_2 emissions pages of Our World in Data or the Global Carbon Atlas. Download per-capita and total emissions for India, China, the United States, the United Kingdom, Brazil and Nigeria from 1990 to the latest year. Plot both measures and calculate each country's share of cumulative emissions since 1850. Which countries look most "responsible" on each measure? Write a 200-word note on what a fair basis for sharing emission cuts might be.

7.2.5 Reducing carbon footprints

Reductions are possible at every level, and individual and collective actions reinforce each other.

Individual and household actions

- **Energy at home:** Use LED lights, star-rated appliances and inverter air conditioners set at 24 °C or higher; switch devices off at the plug; use natural ventilation and shading; install rooftop solar where possible (the PM Surya Ghar: Muft Bijli Yojana, launched in 2024, subsidises rooftop solar for households).
- **Transport:** Walk, cycle, or use the metro, buses and shared transport; choose electric two-wheelers; combine trips; fly less and take trains for medium distances.
- **Food:** Eat seasonal, local and mainly plant-based food, and avoid waste, since food rotting in a landfill also produces methane.
- **Goods and waste:** Buy less and buy durable products; repair, reuse and share; segregate waste so that it can be recycled and composted.

Institutional and national actions

- Expanding renewable energy: India has a target of 500 GW of non-fossil electricity capacity by 2030 and has pledged net-zero emissions by 2070 (see Chapter 5).
- Energy-efficiency programmes such as UJALA (LED bulb distribution), the Perform, Achieve and Trade scheme for industry, and the Energy Conservation Building Code.
- Better public transport, electric mobility and compact city planning.
- Carbon markets, green buildings, afforestation, and carbon footprint reporting by companies and institutions.

Individual action is worthwhile, but most emissions are locked in by systems, such as how electricity is generated, how cities are built and how goods are made, that only policy and collective action can change. The most effective citizen reduces his or her own footprint *and* supports wider change.

7.2.6 Mission LiFE: Lifestyle for Environment

India has made sustainable lifestyles a theme of its climate diplomacy. Prime Minister Narendra Modi introduced the idea of **LiFE (Lifestyle for Environment)** at the climate summit COP26 in Glasgow in November 2021, calling for a shift from "mindless and destructive consumption" to "mindful and deliberate utilisation". **Mission LiFE** was formally launched on 20 October 2022 at Ekta Nagar (Kevadia), Gujarat, in the presence of UN Secretary-General António Guterres.

The mission aims to make environmental protection a mass movement by encouraging individuals to become "pro-planet people". It promotes simple, measurable LiFE actions grouped under seven themes:

1. Save energy
2. Save water
3. Say no to single-use plastics
4. Adopt sustainable food systems
5. Reduce waste
6. Adopt healthy lifestyles
7. Reduce e-waste

The mission aims to mobilise at least one billion Indians and other global citizens to take such actions in 2022–2028, and LiFE activities are now part of the Ministry of Environment's environment education programme and eco-clubs (see the section on the National Green Corps below). Supporters see LiFE as a culturally rooted approach that rightly asks the biggest consumers to change their habits; critics note that lifestyle change cannot cut emissions enough without rapid change in energy and industry, and that the campaign's impact has not been independently evaluated.

Key Points

- A carbon footprint is the total greenhouse gas emitted by an activity, person or organisation, measured in CO_2e using global warming potentials.
- Scope 1 covers direct emissions, Scope 2 purchased electricity and Scope 3 all other value-chain emissions.
- Emissions = activity data × emission factor; India's grid factor is about 0.72 kg CO_2/kWh (CEA, 2023–24).
- India's per-capita fossil CO_2 (about 2.1 t in 2023) is less than half the world average and about one-seventh that of the USA.
- Mission LiFE (2022) promotes sustainable lifestyles through simple actions under seven themes.

7.3 Resettlement and Rehabilitation of Project-Affected Persons and Communities

Dams, mines, power stations, highways, industrial corridors, airports and even wildlife reserves need land. When that land is already home to people, they must move. Such projects may benefit the nation as a whole, yet the people who lose their land, homes and livelihoods are often among the poorest and least powerful. How fairly they are treated is a central question of environmental and social justice.

7.3.1 Development-induced displacement

Development-induced displacement is the forced movement of people from their homes and lands to make way for development projects. Three related terms are used in policy:

- **Displacement:** the loss of home, land or livelihood because of a project. People may be *physically displaced* (forced to move) or *economically displaced* (they stay but lose fields, grazing land, fishing grounds or forests).
- **Resettlement:** the physical relocation of displaced people to a new site, with housing and basic services.
- **Rehabilitation:** the restoration, and ideally improvement, of displaced people's incomes, livelihoods, social networks and quality of life after resettlement.

The people affected are called **project-affected persons (PAPs)** or **project-affected families (PAFs)**. Resettlement without rehabilitation, moving people into new houses without land, jobs or community life, has been one of the most common failures of development policy.

Scale of displacement in India

India has no official national count of people displaced by development. The best-known estimates come from researchers such as Walter Fernandes, who compiled state-level records and concluded that roughly 60 million people (6 crore) were displaced or otherwise affected by development projects in India between 1947 and the early 2000s, with later estimates higher still. Some general patterns are clear from this and other studies:

- **Dams and irrigation projects** have been the largest single cause, followed by mines, industries, power plants, wildlife sanctuaries and national parks, and urban infrastructure.
- **Adivasis (Scheduled Tribes)** are hugely over-represented. They form about 8.6% of India's population (Census 2011) but an estimated 40% of those displaced, because many dams, mines and forests lie in tribal regions. Dalits and other landless groups are also heavily affected.
- **Only a minority were resettled.** Estimates suggest that perhaps only one in three displaced people received any formal resettlement; the rest received cash compensation, or nothing if they lacked legal title to the land they used.
- **Common property is ignored.** Under the old law, only owners with title deeds were compensated, so landless labourers, fishers, artisans, tenants and forest users who depended on common land often received nothing.

Until 2013, land was acquired under the colonial **Land Acquisition Act, 1894**, which allowed the government to take land for any "public purpose" (the principle of **eminent domain**), paid only cash for the market value of land, and said nothing about rehabilitation.

7.3.2 The Impoverishment Risks and Reconstruction model

The sociologist **Michael Cernea**, who worked for many years at the World Bank, studied resettlement projects across the world and found that forced displacement tends to impoverish people in predictable ways. His **Impoverishment Risks and Reconstruction (IRR) model** (1997) identifies eight major risks and, for each, a corresponding reconstruction strategy that planners must adopt.

Table 7.4: Cernea's Impoverishment Risks and Reconstruction model

Impoverishment risk	What happens	Reconstruction strategy
Landlessness	Loss of farmland, the main asset of rural families	Land-for-land resettlement

Impoverishment risk	What happens	Reconstruction strategy
Joblessness	Loss of wage work, crafts and small businesses	Re-employment, skills, credit
Homelessness	Loss of house and village; poor temporary shelter	Adequate housing at the new site
Marginalisation	Fall in income and social status; loss of confidence	Social inclusion, shared benefits
Food insecurity	Hunger and malnutrition after moving	Food support until incomes recover
Increased morbidity and mortality	Disease, stress, unsafe water at new sites	Health care, clean water, sanitation
Loss of common property resources	Loss of access to forests, grazing land, rivers, fish	Restore access to commons
Social disarticulation	Break-up of kinship, community and support networks	Resettle communities together; rebuild institutions

Later researchers added further risks, such as loss of access to education and the special vulnerability of women. The model's value is that it turns vague concern into a checklist: a resettlement plan must show how each risk will be prevented, not merely how much cash will be paid.

7.3.3 Policy and law on resettlement and rehabilitation in India

From the 1894 Act to national policies

For decades, resettlement in India depended on project-specific packages and state policies, such as Maharashtra's 1976 Act for project-affected persons and the special package under the Narmada Water Disputes Tribunal award (1979). Public pressure from movements such as the Narmada Bachao Andolan, and research on the failures of earlier projects, led the central government to adopt a **National Policy on Resettlement and Rehabilitation for Project Affected Families** in 2003. It was replaced by the **National Rehabilitation and Resettlement Policy, 2007 (NRRP 2007)**, whose main features were:

- Displacement should be minimised, and non-displacing or least-displacing alternatives should be considered.
- A **Social Impact Assessment (SIA)** was required for projects displacing 400 or more families in the plains, or 200 or more in hilly, tribal (Scheduled Area) and desert regions.
- Land for land was to be given where government land was available; otherwise preference in project jobs, training, and monetary benefits.
- Special provisions for Scheduled Tribes, including consultation with gram sabhas and resettlement as a group close to their original area.
- Grievance redress through an ombudsman and a National Rehabilitation Commission.

NRRP 2007 was a policy, not a law, so it could not be enforced in court, and the 1894 Act remained in force.

The Right to Fair Compensation Act, 2013

The **Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (RFCTLARR Act)** replaced the 1894 Act and came into force on 1 January 2014. For the first time, it combined land acquisition with legally binding resettlement and rehabilitation. Its key provisions are:

- **Higher compensation:** the market value of land is multiplied by a factor of 1 to 2 in rural areas (depending on distance from towns, as notified by the state) and 1 in urban areas, and then a **solatium** (an additional payment for the compulsory nature of acquisition) equal to 100% is added. Rural landowners can therefore receive up to about four times, and urban owners about twice, the market value.
- **Social Impact Assessment:** an SIA, reviewed by an independent expert group and discussed in public hearings, is required before most acquisitions.
- **Consent:** acquisition for private companies requires the consent of at least 80% of affected families, and for public–private partnership projects at least 70%.
- **Scheduled Areas:** acquisition in Fifth Schedule (tribal) areas requires the consent of the gram sabha, and should be a last resort.
- **Food security:** acquisition of irrigated, multi-cropped land is permitted only as a last resort and within limits set by the state, and an equivalent area of culturable wasteland must be developed for agriculture in its place.
- **R&R entitlements** (Second Schedule) for every affected family, including landless families who depended on the land: a house; a choice of a job in the project, a one-time payment of ₹5 lakh, or an annuity; a monthly subsistence allowance of ₹3,000 for one year; transport and resettlement allowances; and extra benefits for Scheduled Caste and Scheduled Tribe families.
- **Infrastructure at resettlement sites** (Third Schedule): schools, health centres, roads, drinking water, places of worship, burial grounds and other amenities.
- **Return of unused land:** land not used for five years can be returned to the original owners or to a land bank.
- **Retrospective application:** where compensation had not been paid or possession not taken under the old Act, the new Act's benefits could apply.

The Act's implementation has been contested. Industry argued that consent and SIA requirements made acquisition slow and costly. A central ordinance of 2014–15 to dilute these provisions lapsed in 2015, but several states, including Gujarat, Maharashtra, Telangana and Jharkhand, have used their own laws to exempt categories of projects from SIA and consent, which civil society groups say weakens the Act.

Other laws also protect affected communities. The **Panchayats (Extension to Scheduled Areas) Act, 1996 (PESA)** requires gram sabhas in Scheduled Areas to be consulted before land acquisition, and the **Forest Rights Act, 2006** (see Chapter 5) recognises the rights of forest dwellers. In the Niyamgiri case (2013), the Supreme Court directed that the gram sabhas of the Dongria Kondh and other forest dwellers decide whether bauxite mining in the Niyamgiri hills of Odisha would affect their religious and cultural rights; all twelve gram sabhas consulted rejected the project.

7.3.4 Case studies of displacement and rehabilitation

Case Study: Hirakud Dam, Odisha: the first of the big dams

The Hirakud Dam across the Mahanadi near Sambalpur, inaugurated by Jawaharlal Nehru in 1957, was one of independent India's first multipurpose river valley projects. With its earthen dykes it stretches about 25 km, among the longest dams in the world. Its reservoir submerged a large number of villages and displaced roughly 22,000 families (well over one lakh people affected, by most estimates), most of them farmers in the fertile Mahanadi valley.

There was no resettlement policy at the time. Compensation was paid only in cash and only to those with land records. The government's own estimate of compensation, about ₹12 crore, was later reduced to about ₹9.5 crore, and only about ₹3.3 crore was actually paid. Many families never received anything, and a few resettlement colonies were set up with poor land. Decades later, oustees and their children were still petitioning for compensation and land titles. Hirakud shows the long-term cost of treating displaced people as an afterthought: the benefits of irrigation and power went largely to downstream areas, while the oustees bore the losses.

Case Study: Tehri Dam, Uttarakhand

The Tehri Dam on the Bhagirathi river in the Garhwal Himalaya is, at 260.5 m, India's tallest dam. Construction began in 1978 and the first units were commissioned in 2006. The reservoir drowned the historic town of Old Tehri, the former capital of the Tehri Garhwal princely state, and dozens of villages, displacing more than one lakh people.

Urban residents were moved to a newly built hill town, New Tehri, and to colonies near Dehradun and Haridwar; many rural families were given land in the plains, far from their mountain villages, where climate, farming and neighbours were unfamiliar. Families on the reservoir rim, whose land suffered landslides and cracks after filling, struggled for years to be recognised as affected.

Sunderlal Bahuguna of the Chipko movement led opposition to the dam for nearly two decades, with long fasts in the 1990s demanding a review of seismic safety and rehabilitation, since the dam lies in an earthquake-prone zone. Review committees cleared the project and the Supreme Court dismissed petitions against it in 2003. Tehri illustrates how hard it is to resettle mountain communities whose way of life cannot be recreated in the plains.

Case Study: Singrauli: displaced again and again

Singrauli, on the border of Madhya Pradesh and Uttar Pradesh, is known as India's "energy capital" because of its coal mines and giant thermal power stations. The process of displacement began with the Rihand Dam (Govind Ballabh Pant Sagar), completed in the early 1960s, which submerged many villages; the oustees, many of them Adivasis, were given little or no rehabilitation and moved to nearby land.

From the late 1970s, the region's coal reserves attracted open-cast mines of Northern Coalfields Limited and thermal power stations of the National Thermal Power Corporation (NTPC) and private companies. Many of the same families were displaced a second, third or even fourth time as mines and ash ponds expanded onto the land where they had resettled. In the 1990s, residents complained to the World Bank's independent Inspection Panel about the resettlement record of a Bank-funded NTPC power project in the region, one of the first such complaints from India.

Singrauli is also one of India's most polluted industrial areas, with fly ash, mercury and coal dust contaminating air, water and soil. It shows how a series of projects, each assessed separately, can add up to a cumulative disaster for one community, which is why impacts should be assessed for a whole region.

Sardar Sarovar: resettlement under scrutiny

The Sardar Sarovar Project on the Narmada (its engineering and environmental impacts are described in Chapter 3) affected about 245 villages in Gujarat, Maharashtra and Madhya Pradesh, many of them Adivasi villages. The Narmada Water Disputes Tribunal award of 1979 promised land for land (at least two hectares of irrigable land per family) and resettlement of villages as units, which was progressive for its time. In practice, cultivable land was hard to find, records were incomplete and many Adivasi families without formal titles were initially not recognised. Gujarat's package is often cited as relatively generous, but oustees in Madhya Pradesh were still contesting their rehabilitation when the dam reached full height in 2017. The movement that grew around these issues is discussed in the next section.

Key Points

- Development projects have displaced or affected an estimated 6 crore or more people in India since 1947, a disproportionate share of them Adivasis.
- Cernea's IRR model lists eight impoverishment risks, from landlessness to social disarticulation, each with a reconstruction strategy.
- NRRP 2007 introduced social impact assessment and land-for-land principles; the RFCTLARR Act 2013 made higher compensation, SIA, consent and R&R entitlements legally binding.
- Hirakud, Tehri, Singrauli and Sardar Sarovar show the costs of cash-only compensation, distant resettlement and repeated displacement.

Think and Do: Visit a project-affected locality

With your teacher's help, visit a locality in or near Delhi whose residents were moved by a development project, such as a resettlement colony created for a road, metro line, riverfront or slum-clearance scheme. Prepare a short interview schedule based on Cernea's eight risks. Ask residents, respectfully and with consent, how their housing, work, travel time, schooling and community life changed after relocation. Present your findings as a short report that compares their experience with the entitlements in the RFCTLARR Act.

7.4 Environmental Movements in India

An **environmental movement** is a sustained, collective effort by people to protect the natural environment or to resist its damage, usually by challenging the decisions of governments or companies. India's environmental movements differ in an important way from the early conservation movements of Europe and North America, which focused mainly on protecting wilderness and wildlife. In India, most movements have been led by peasants, Adivasis, fishers and women whose survival depends directly on forests, rivers and land. The historians Ramachandra Guha and Joan Martínez-Alier called this the **"environmentalism of the poor"**: conflicts in which defending nature and defending livelihoods are the same struggle.

Indian movements have also drawn heavily on **Gandhian** methods: non-violent resistance (*satyagraha*), fasting, long marches on foot (*padyatras*) and constructive work such as tree planting. They have used courts, scientific evidence and the media, and they have influenced national laws and policies. The timeline below places the movements studied in this section in sequence.

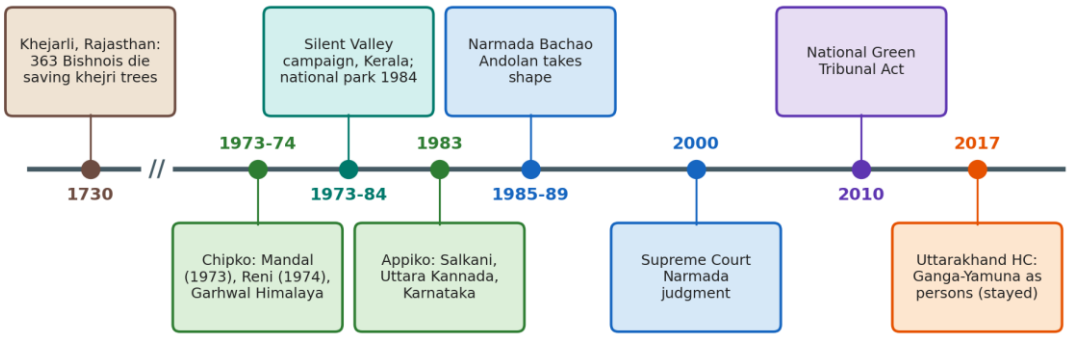


Figure 7.5: *Timeline of major Indian environmental movements and related legal milestones discussed in this chapter (positions are schematic, not to scale). Source: compiled from sources cited in the text.*

7.4.1 The Bishnois of Rajasthan

The **Bishnoi** community of western Rajasthan, with smaller populations in Haryana, Punjab and Madhya Pradesh, offers the oldest and perhaps most remarkable example of a religious community organised around the protection of trees and animals. For more than five centuries the Bishnois have protected the trees and wildlife of the Thar Desert as a matter of faith.

Guru Jambheshwar and the twenty-nine principles

The Bishnoi faith was founded in 1485 by **Guru Jambheshwar** (1451–1536), also known as Jambhoji, who was born in Pipasar village in the Nagaur region. According to tradition, he founded the sect at Samrathal Dhora, a sand dune near Bikaner, during a long drought in which people and animals were dying for want of water and fodder. Jambheshwar taught that people must live in harmony with the harsh desert environment rather than exploit it. His followers were asked to obey **29 principles** (*bis* means twenty and *noi* nine in the local language, giving the name Bishnoi).

The principles cover hygiene, honesty and worship, but several are directly ecological:

- Do not cut green trees (*rukh lilo nahi ghave*).
- Show compassion to all living beings (*jeev daya palni*).
- Do not kill animals and do not eat meat.
- Filter drinking water, milk and firewood to avoid harming small creatures.
- Do not castrate bullocks; provide shelter for abandoned animals.
- Avoid intoxicants, tobacco and alcohol, and avoid anger and theft.

Bishnoi villages traditionally use only dead wood for fuel, keep common grazing land and water bodies (*johads*) for animals, and leave grain and water out for wild animals and birds.

The khejri tree and the blackbuck

Two species are at the centre of Bishnoi conservation. The **khejri** (*Prosopis cineraria*), Rajasthan's state tree, is a hardy, deep-rooted, nitrogen-fixing tree whose leaves (*loong*) are excellent fodder, whose pods (*sangri*) are eaten as a vegetable, and which enriches the soil and shades crops in desert fields. For a desert community it is a tree of life.

The **blackbuck** (*Antilope cervicapra*) and the **chinkara** or Indian gazelle (*Gazella bennettii*) roam freely in and around Bishnoi villages, where hunting is forbidden; Bishnoi families rescue and raise injured or orphaned fawns. Blackbuck populations survive in good numbers around Bishnoi villages even where they have vanished elsewhere in Rajasthan, and Bishnoi volunteers still patrol against poachers, some at the cost of their lives.

Khejarli, 1730: Amrita Devi and the 363 martyrs

In 1730 (Samvat 1787), Maharaja **Abhay Singh** of Jodhpur (Marwar) needed wood to burn lime for the construction of a new palace. His minister **Giridhar Bhandari** sent soldiers to fell khejri trees in a Bishnoi village about 25 km south-east of Jodhpur, which was rich in trees despite the desert. When the axe-men arrived, a Bishnoi woman, **Amrita Devi**, refused to allow the sacred trees to be cut. She hugged a tree and declared, in words still repeated by Bishnois: *"Sar sānte rūkh rahe to bhī sasto jān"*, meaning "If a tree is saved even at the cost of one's head, it is a cheap bargain."

The soldiers beheaded her. Her three daughters, Asu, Ratni and Bhagu, followed her example and were also killed. News spread to the surrounding Bishnoi villages, and people came forward one by one to embrace the trees. According to Bishnoi tradition, **363 Bishnois** from dozens of villages, men and women, old and young, were killed before the Maharaja learned what was happening. Horrified, he stopped the felling, apologised to the community, and issued a royal decree, engraved on a copper plate, that prohibited the cutting of green trees and the hunting of animals in or near Bishnoi villages, with severe penalties for violators.

The village, named **Khejarli** after the khejri trees, became a place of pilgrimage, and a memorial to the martyrs stands there today. A fair is held every year on the anniversary, which falls in the month of Bhadrapad. In 2013 the Ministry of Environment and Forests declared **11 September** as **National Forest Martyrs Day**, in memory of the Khejarli sacrifice and of forest staff killed in the line of duty. The Khejarli story, of villagers hugging trees to stop axe-men, is widely regarded as a forerunner of the Chipko movement more than two centuries later.

The Bishnois and wildlife protection today

The Bishnois' commitment to wildlife became nationally known through the **blackbuck poaching case** against film actor Salman Khan. In October 1998, during a film shoot near Jodhpur, two blackbucks were killed at Kankani village. Bishnoi villagers raised the alarm and pursued the case for two decades; in April 2018 a Jodhpur court convicted the actor under the Wildlife (Protection) Act, 1972 and sentenced him to five years' imprisonment, though he was released on bail pending appeal. The blackbuck is a Schedule I species under the Act.

The Ministry of Environment, Forest and Climate Change honours this tradition through the **Amrita Devi Bishnoi Wildlife Protection Award** (₹1 lakh, a medal and a citation) for individuals or communities, especially rural ones, who show exceptional courage in protecting wildlife; an early recipient was Ganga Ram Bishnoi of Chirai village near Jodhpur, honoured posthumously after he was killed trying to stop poachers. In recent years Bishnoi groups have also protested against the felling of khejri trees for large solar power parks in western Rajasthan.

The Bishnoi example shows that conservation rules work best when they are part of a community's identity and daily life, and that protecting a few keystone species can sustain a whole desert ecosystem.

7.4.2 The Chipko movement

The **Chipko movement** (from the Hindi *chipakna*, to cling or to hug) arose in the early 1970s in the Garhwal Himalaya, in what was then Uttar Pradesh and is now Uttarakhand. It is India's best-known environmental movement and an international symbol of non-violent, community-based forest protection.

Causes and background

Several pressures came together in the hill districts of Garhwal in the 1960s and early 1970s:

- **Commercial forestry:** after the India–China war of 1962, new roads opened the inner Himalaya to logging, and the Forest Department auctioned large tracts of trees to contractors from the plains.
- **Denial of local needs:** hill villagers depended on forests for fuelwood, fodder, leaf manure, small timber and water, but colonial-era forest laws restricted their access while outside contractors felled trees.
- **Ecological disaster:** the Alaknanda flood of July 1970 swept away villages and bridges; many hill people linked it to deforestation of the catchment.
- **Local organisation:** The **Dasholi Gram Swarajya Sangh (DGSS)**, a Gandhian cooperative founded in 1964 in Gopeshwar (Chamoli district) by **Chandi Prasad Bhatt**, ran small forest-based industries, such as making agricultural tools, and had the experience to organise villagers.

Gandhian workers such as Sarala Behn and Mira Behn had earlier spread ideas of village self-reliance and forest protection in the hills.

Key events

1. **Mandal, 1973:** In early 1973 the Forest Department refused the DGSS's request for a few ash (*angu*) trees to make ploughs and yokes, but allotted ash trees in the nearby Mandal forest to the Symonds Company, a sports goods manufacturer from Allahabad, to make tennis rackets and other goods. In April 1973 the villagers of Mandal, led by Bhatt and the DGSS, declared that they would hug the trees rather than allow them to be felled. Faced with this resistance, the company's men withdrew. A similar confrontation followed at Phata, near Kedarnath, later that year.
2. **Reni, March 1974:** The most famous Chipko action took place at Reni village, near Joshimath in the upper Alaknanda valley, where nearly 2,500 trees had been auctioned. On 26 March 1974, the men of the village were away, having been called to the district headquarters to receive long-awaited compensation for land taken by the army. When the contractor's labourers arrived, **Gaura Devi**, head of the village women's group (*Mahila Mangal Dal*), gathered 27 women and girls and confronted them. She told the men that the forest was their *maika* (mother's home) and that they would have to shoot her before they could cut the trees. The women kept vigil through the night and the loggers left. A state government committee later recommended a ten-year ban on commercial felling in the upper Alaknanda catchment.
3. **Advani and the Henwal valley, 1977–78:** In Tehri Garhwal, activists including **Sunderlal Bahuguna**, Dhoom Singh Negi and Bachni Devi spread the movement. At Advani, women tied sacred threads (*raksha sutra*) around trees marked for felling, turning protection into a religious vow. Bahuguna undertook fasts to oppose forest auctions.
4. **National recognition, 1980–83:** In 1980 Prime Minister Indira Gandhi announced a **15-year ban on the commercial felling of green trees** in the Himalayan forests of Uttar Pradesh. Between 1981 and 1983 Bahuguna walked about 4,800 km on a padyatra from Kashmir to Kohima to spread the Chipko message across the Himalaya.

The movement's slogan captured its philosophy: *"Kya hain jangal ke upkar? Mitti, pani aur bayar. Mitti, pani aur bayar, zinda rehne ke adhar."* ("What do the forests bear? Soil, water and pure air. Soil, water and pure air are the basis of life.") This answered the official slogan that forests bear "resin, timber and profit".

Leaders and the role of women

Chipko had no single leader. **Chandi Prasad Bhatt** emphasised local control of forests, small forest-based industries and ecological restoration; the DGSS later organised large tree-planting camps and eco-development work. He received the Ramon Magsaysay Award (1982) and the Gandhi Peace Prize (2013). **Sunderlal Bahuguna** (1927–2021) became the movement's best-known voice, emphasising the spiritual and ecological value of the Himalaya and calling for a complete ban on commercial felling; he coined the slogan "Ecology is permanent economy" and later led the movement against the Tehri Dam. He received the Padma Vibhushan in 2009.

The real strength of Chipko lay with the hill **women**, who collected fuelwood, fodder and water and felt the effects of deforestation most directly; women like Gaura Devi sometimes acted against the wishes of village men tempted by contractors' wages. Vandana Shiva presented Chipko as a key example of **ecofeminism**, the idea that the domination of nature and of women are linked, while

Ramachandra Guha (*The Unquiet Woods*, 1989) placed it in a long tradition of peasant resistance to state forestry. Both agree that Chipko was about livelihoods and rights as much as trees.

Outcomes and legacy

- The 15-year felling ban in the UP Himalaya, and greater caution in commercial felling across the Himalaya.
- Influence on the **Forest (Conservation) Act, 1980**, which required central approval for diverting forest land, and on the **National Forest Policy, 1988**, which declared that forests should be managed primarily for ecological stability and local needs rather than revenue.
- Inspiration for other movements, notably Appiko in Karnataka, and worldwide recognition: the movement received the Right Livelihood Award in 1987.
- An enduring image of women as environmental leaders.

Chipko also had unintended consequences: strict felling bans sometimes prevented villagers themselves from using forests or building roads, and in the late 1990s some villages launched a "Jhapto Chheeno" (grab and snatch) protest demanding access to closed forests. The tension between conservation and local development remains a lesson for all movements.

7.4.3 The Appiko movement

The **Appiko movement** (*appiko* means "to hug" in Kannada) was the southern counterpart of Chipko, in the Western Ghats of Uttara Kannada (North Kanara) district, Karnataka.

Background

Uttara Kannada was once among the most densely forested districts of peninsular India. From the 1950s its natural forests were depleted to supply paper and plywood mills, submerged by hydroelectric projects on the Kali river, and clear-felled and replaced with monoculture plantations of teak and eucalyptus. Local farmers, especially arecanut and spice growers, depended on the forest for leaf manure, fodder, fuelwood and water; as forests disappeared, streams dried up and soil eroded.

The movement

In August 1983, the young people of **Salkani** village in Sirsi taluk asked forest officials to stop felling in the nearby Kalase forest, but were ignored. On **8 September 1983**, about 160 men, women and children of Salkani and neighbouring villages walked to the Kalase forest, hugged the trees and forced the fellers to stop. The protest continued for about six weeks, until the state government withdrew the felling orders. The movement soon spread to other forests in Sirsi, Siddapur and nearby taluks.

The leading figure was **Panduranga Hegde**, a young man from the region who had studied at the Delhi School of Social Work and had worked with the Chipko movement and Sunderlal Bahuguna in the Himalaya before returning home. The movement used foot marches, slide shows, folk dance (*Yakshagana*) and street plays to carry its message from village to village.

The three Rs of Appiko: save, grow, use rationally

Appiko summarised its programme in three Kannada words:

1. **Ulisu (save):** protect the remaining natural forest from commercial felling.
2. **Belesu (grow):** regenerate degraded land through planting and natural regeneration, using local species rather than monocultures.

3. **Balasu (use rationally):** reduce pressure on forests through sustainable use, for example through fuel-efficient cooking stoves (*chulhas*), biogas plants and careful harvesting of leaf manure.

This combination of protest with constructive work, including village nurseries and improved stoves, became the movement's distinctive feature.

Outcomes

Under pressure from Appiko, the Karnataka government restricted the felling of green trees in parts of the Western Ghats, and clear-felling of natural forests for plantations was reduced. The movement contributed to the wider campaigns to protect the Western Ghats (see Chapter 6) and showed that the Chipko model could be adapted to a different ecology, culture and economy. It also demonstrated that environmental movements gain lasting support when they offer practical alternatives, not just opposition.

7.4.4 The Silent Valley movement

The campaign to save **Silent Valley**, a tract of tropical evergreen rainforest in the Palakkad district of Kerala, ran from the early 1970s to the early 1980s against a proposed hydroelectric dam on the Kunthipuzha river. A coalition led by the Kerala Sastra Sahitya Parishad (a people's science movement), scientists, the poet Sugathakumari and other writers, and national and international conservation bodies argued that the dam would destroy one of India's last undisturbed rainforests and the habitat of the endangered lion-tailed macaque. After review committees and the intervention of Prime Minister Indira Gandhi, the project was abandoned, and Silent Valley was declared a national park in 1984 and formally inaugurated by Prime Minister Rajiv Gandhi in 1985. Unlike Chipko and Appiko, Silent Valley was mainly a science-led campaign for biodiversity rather than a livelihood struggle. Its history, ecology and significance for conservation are discussed fully in Chapter 6.

7.4.5 The Narmada Bachao Andolan

The **Narmada Bachao Andolan (NBA)**, or "Save the Narmada Movement", is the most prominent movement in India against large dams and the displacement they cause. It challenged not only a single project but the whole model of development that treats the losses of the poor as an acceptable price for national progress. The engineering, benefits and environmental impacts of the Sardar Sarovar Dam itself are covered in Chapter 3; this section focuses on the movement.

Origins

The Narmada Valley Development Project planned 30 large, 135 medium and about 3,000 small dams on the Narmada and its tributaries, with the Sardar Sarovar Dam in Gujarat and the Narmada Sagar Dam in Madhya Pradesh as the largest. The Narmada Water Disputes Tribunal (award of 1979) allocated the river's waters among Gujarat, Madhya Pradesh, Maharashtra and Rajasthan and laid down resettlement conditions. In 1985 the World Bank approved a loan of about US$450 million for Sardar Sarovar.

In the mid-1980s, social worker **Medha Patkar** began working with Adivasi villages in the submergence zone and found that many people had not been told they would be displaced and that no proper surveys or resettlement plans existed. Groups in the three states initially campaigned for fair resettlement, but by 1988–89, convinced that just resettlement of so many people was impossible,

they came together as the Narmada Bachao Andolan to oppose the dam itself. A large rally at Harsud in Madhya Pradesh in September 1989 brought the movement onto the national stage.

Leaders and methods

- **Medha Patkar** became the movement's best-known leader, undertaking long fasts and leading marches and satyagrahas.
- **Baba Amte** (1914–2008), the veteran social worker known for his work with leprosy patients, joined the movement, moved to live on the banks of the Narmada at Kasrawad in Madhya Pradesh in 1990, and wrote *Cry, O Beloved Narmada*.
- Adivasi leaders from the hill villages, such as Keshavbhau and Kevalsingh Vasave, and farmers from the Nimar plains formed the movement's base.

The NBA used a wide repertoire of non-violent methods: the **Jan Vikas Sangharsh Yatra** (December 1990–January 1991), a march of thousands towards the dam site that was stopped at the Gujarat border at Ferkuwa and followed by a 22-day fast by Patkar and others; the **jal samarpan** pledge to drown rather than leave as waters rose at villages such as Manibeli and Domkhedi; dharnas, research and international campaigns. Writers such as Arundhati Roy (*The Greater Common Good*, 1999) brought the issue to a wide audience. Patkar and Baba Amte received the Right Livelihood Award in 1991.

The World Bank and the Morse Report

Under pressure from the NBA and international campaigners, the World Bank in 1991 appointed its first-ever independent review, headed by **Bradford Morse**, a former head of the UN Development Programme. The **Morse Report (June 1992)** found that resettlement could not be carried out as planned, that environmental impacts had not been properly assessed, and that the Bank had violated its own policies; it recommended that the Bank "step back". The Government of India cancelled the remaining loan in **March 1993** and the Bank withdrew. The episode led the Bank to create a permanent **Inspection Panel** (1993) and contributed to the **World Commission on Dams** (1997–2000).

The Supreme Court judgment of 2000

The NBA petitioned the Supreme Court in 1994. In 1995 the Court halted construction at a height of about 80 m while it heard the case. On 18 October 2000, in *Narmada Bachao Andolan v. Union of India*, a three-judge bench by a majority of 2:1 allowed construction to resume, initially up to 90 m, on condition that the rehabilitation of affected families be completed *pari passu* (step by step, in advance of submergence) at each stage of raising the dam's height, under the supervision of a Narmada Control Authority and grievance redressal authorities. The majority held that the courts should not normally question policy decisions on large projects once they had been approved, while the dissenting judge held that construction should not proceed until environmental clearance conditions were fully met.

The dam's height was raised in stages over the next 17 years. Its gates were closed in 2017, raising it to the full height of 138.68 m, and the dam was dedicated to the nation in September 2017. Villages in the Nimar plains of Madhya Pradesh were submerged in 2017 and 2019 while many families still disputed their rehabilitation.

Legacy

The NBA did not stop the Sardar Sarovar Dam, which today supplies irrigation and drinking water to large parts of Gujarat and Rajasthan. Supporters of the dam accuse the movement of delaying vital water supplies to drought-prone regions such as Kutch and Saurashtra. Yet the movement had lasting effects:

- It forced resettlement and rehabilitation onto the national agenda, contributing to the R&R policies of 2003 and 2007 and the RFCTLARR Act of 2013.
- It changed how international lenders deal with affected people, through the Morse Report, the World Bank Inspection Panel and the World Commission on Dams.
- It made "development-induced displacement" a central issue of environmental justice in India and inspired many later struggles against mines, dams and industrial projects.

7.4.6 Comparing the movements

Table 7.5: Comparison of major Indian environmental movements

Movement	Place and period	Key leaders	Main issue	Main outcomes
Bishnoi (Khejarli)	Jodhpur, Rajasthan; 1730	Amrita Devi; teachings of Guru Jambheshwar	Felling of khejri trees; protection of wildlife	Royal decree banning felling and hunting; living conservation tradition
Chipko	Garhwal Himalaya; 1973 onwards	Chandi Prasad Bhatt, Gaura Devi, Sunderlal Bahuguna	Commercial logging; local forest rights	15-year felling ban (1980); influence on forest law and policy
Silent Valley	Palakkad, Kerala; 1973–84	KSSP, scientists, Sugathakumari	Hydroelectric dam in rainforest	Project dropped; national park 1984
Appiko	Uttara Kannada, Karnataka; 1983 onwards	Panduranga Hegde	Felling and monoculture plantations	Felling restrictions; save–grow–use rationally model
Narmada Bachao Andolan	Narmada valley; 1985 onwards	Medha Patkar, Baba Amte	Displacement by large dams	World Bank exit (1993); R&R reforms; dam completed 2017

All five movements were rooted in local communities and used non-violent methods, and women played leading roles in Khejarli, Chipko and Appiko. The forest movements combined protest with traditions of sustainable use, while Silent Valley relied mainly on scientific argument and the NBA on mass mobilisation, litigation and international pressure. Some stopped a project outright; others lost their immediate battle but changed laws, institutions and public thinking.

Key Points

- India's environmental movements are largely "environmentalism of the poor", linking nature protection with livelihoods and rights.
- The Bishnois, following Guru Jambheshwar's 29 principles, protected khejri trees and blackbuck; 363 people led by Amrita Devi died at Khejarli in 1730.
- Chipko (Mandal 1973, Reni 1974) led to a 15-year felling ban in the UP Himalaya in 1980; Gaura Devi, Chandi Prasad Bhatt and Sunderlal Bahuguna were key figures.
- Appiko (Salkani, 1983; Panduranga Hegde) adapted Chipko to the Western Ghats with its programme of Ulisu, Belesu and Balasu.
- The NBA, led by Medha Patkar and Baba Amte, led to the Morse Report (1992), the World Bank's exit (1993) and the Supreme Court's 2000 judgment requiring rehabilitation pari passu with dam construction.

7.5 Environmental Justice: The National Green Tribunal

The movements described above were, at heart, struggles for justice: about who bears the costs of pollution and resource extraction, and who has a say in decisions. This section explains the concept of environmental justice and then examines the National Green Tribunal, India's specialised court for environmental cases.

7.5.1 The concept of environmental justice

Environmental justice is the fair distribution of environmental benefits and burdens among all people, regardless of class, caste, race, gender or place, together with their meaningful participation in decisions that affect their environment. The term arose in the United States in the early 1980s, when African-American residents of Warren County, North Carolina, protested in 1982 against a toxic waste landfill sited in their community. Studies by sociologist Robert Bullard and others then showed that hazardous facilities were disproportionately located in poor and minority neighbourhoods.

Environmental justice has three dimensions:

- **Distributive justice:** who receives the benefits (electricity, irrigation, profits) and who bears the burdens (pollution, displacement, loss of forests).
- **Procedural justice:** whether affected people are informed, consulted and able to influence decisions, for example through public hearings in environmental impact assessment, gram sabha consent and access to courts.
- **Justice as recognition:** respect for the identities, cultures and knowledge of affected groups, such as Adivasis' relationship with their forests and sacred sites.

In India, injustice is visible in many forms: the victims of the Bhopal gas disaster (see Chapter 4), who waited decades for compensation; Adivasis displaced by dams and mines; waste pickers and residents living beside landfills such as Ghazipur and Bhalswa in Delhi; and the urban poor who breathe the most polluted air while contributing least to it. At the global level, **climate justice** points out that the countries and people least responsible for climate change suffer most from it.

Indian courts have developed strong environmental jurisprudence. The Supreme Court has interpreted the **right to life under Article 21** of the Constitution to include the right to a clean and healthy environment (for example, *Subhash Kumar v. State of Bihar*, 1991) and, in 2024, the right to be free from the adverse effects of climate change (*M.K. Ranjitsinh v. Union of India*). Through **public interest litigation (PIL)**, in which any citizen can approach the court on behalf of the public, the courts adopted principles such as **absolute liability** for hazardous industries (*M.C. Mehta v. Union of India*, Oleum gas leak case, 1986), the **polluter pays principle** and the **precautionary principle** (*Vellore Citizens' Welfare Forum v. Union of India*, 1996), and the **public trust doctrine**, under which the state holds natural resources such as rivers, forests and the seashore in trust for the public (*M.C. Mehta v. Kamal Nath*, 1997). However, the growing number of environmental cases in ordinary courts, and the need for scientific expertise, led to demands for a specialised tribunal.

7.5.2 The National Green Tribunal

Establishment

The Supreme Court and the Law Commission of India (186th Report, 2003) recommended specialised environmental courts with judicial and technical members, and Principle 10 of the 1992 Rio Declaration called for effective access to judicial remedies in environmental matters.

Parliament passed the **National Green Tribunal Act, 2010**, and the **National Green Tribunal (NGT)** was established on 18 October 2010. India thereby became one of the first countries, after Australia and New Zealand, to set up a specialised national environmental tribunal.

Structure and benches

- The NGT is headed by a **Chairperson**, who must be a retired judge of the Supreme Court or a retired Chief Justice of a High Court. Justice Prakash Shrivastava has been Chairperson since 2023.
- It has **judicial members** (retired High Court judges or persons qualified to be such judges) and **expert members** with professional qualifications and at least 15 years of experience in environmental science, forestry, engineering or related fields, including administrative experience. The Act provides for at least ten and up to twenty members of each kind.
- The **Principal Bench** sits in New Delhi, with four **zonal benches**: Central Zone at Bhopal, Western Zone at Pune, Eastern Zone at Kolkata and Southern Zone at Chennai. The tribunal can also hold circuit sittings elsewhere to improve access.

The combination of judges and scientists is the NGT's key innovation, allowing technical questions, such as how much pollution a river can absorb, to be examined by experts within the court.

Jurisdiction

The NGT has jurisdiction over civil cases involving a **substantial question relating to the environment** arising under seven laws listed in Schedule I of the Act:

1. Water (Prevention and Control of Pollution) Act, 1974
2. Water (Prevention and Control of Pollution) Cess Act, 1977
3. Forest (Conservation) Act, 1980
4. Air (Prevention and Control of Pollution) Act, 1981
5. Environment (Protection) Act, 1986

6. Public Liability Insurance Act, 1991
7. Biological Diversity Act, 2002

It has three kinds of powers: **original jurisdiction** over environmental disputes; **appellate jurisdiction** over orders such as environmental clearances, forest clearances and pollution control board decisions; and power to order **relief and compensation** for victims of pollution, **restitution** of damaged property and **restoration** of the environment. Two important laws, the Wildlife (Protection) Act, 1972 and the Indian Forest Act, 1927, are not in its schedule, so wildlife cases remain with ordinary courts.

The NGT follows the **principles of natural justice** rather than the detailed Code of Civil Procedure. The Act directs it to try to dispose of cases within **six months**, and appeals lie to the Supreme Court within 90 days. Non-compliance with its orders can be punished with imprisonment of up to three years, a fine of up to ₹10 crore (₹25 crore for companies), or both.

Principles applied

Section 20 of the Act requires the tribunal to apply three principles in its decisions:

- **Sustainable development:** development that meets present needs without compromising the ability of future generations to meet theirs.
- **Precautionary principle:** where there are threats of serious or irreversible damage, lack of full scientific certainty should not be used as a reason to postpone preventive measures; the burden of proof lies on the developer or polluter.
- **Polluter pays principle:** those who cause pollution must pay the cost of remedying the damage and compensating victims.

The NGT has also repeatedly used **environmental compensation**, ordering polluters and even government bodies to pay amounts calculated to cover restoration costs.

Suo motu powers

The NGT has often acted on its own initiative (**suo motu**) on the basis of news reports. When the central government questioned this power, the Supreme Court held in 2021 (*Municipal Corporation of Greater Mumbai v. Ankita Sinha*) that the NGT is a unique (*sui generis*) forum that can take suo motu cognisance of environmental issues.

7.5.3 Landmark NGT cases

Case Study: The Art of Living festival on the Yamuna floodplain, 2016

In March 2016 the Art of Living Foundation held a three-day "World Culture Festival" on about 1,000 acres of the Yamuna floodplain in Delhi, with a huge stage and lakhs of visitors. Environmentalists petitioned the NGT, arguing that levelling, debris dumping and compaction would destroy the floodplain's wetlands and vegetation. Just before the event, the tribunal allowed it to proceed but imposed an interim environmental compensation of ₹5 crore on the foundation. An expert committee later reported serious damage to the floodplain, and in December 2017 the NGT held the foundation responsible, directed that the amount deposited be used by the Delhi Development Authority for restoration, and allowed further costs to be recovered if restoration cost more. The case became a national example of the polluter pays principle being applied to a powerful organisation, although critics argued that the tribunal should have stopped the event altogether, in line with the precautionary principle.

Old vehicles in Delhi-NCR

In 2014 and 2015, while hearing cases on Delhi's air pollution, the NGT directed that diesel vehicles more than 10 years old and petrol vehicles more than 15 years old should not ply in Delhi, later extending this to the National Capital Region. The Supreme Court upheld the restriction in 2018. It led to the deregistration of lakhs of old vehicles, though critics noted that it burdened small owners while other pollution sources continued.

Illegal sand mining

In August 2013 the NGT restrained riverbed sand mining anywhere in India without prior environmental clearance. Unregulated mining had been lowering riverbeds, undermining bridges, reducing groundwater recharge and fuelling a violent "sand mafia". The order led to the Sustainable Sand Mining Management Guidelines (2016, 2020), although illegal mining remains widespread.

Other notable decisions

- **Sterlite copper smelter, Thoothukudi:** after police firing on protesters killed 13 people in 2018, Tamil Nadu closed the plant. The NGT set aside the closure, but the Supreme Court overturned the NGT's order in 2019 on grounds of jurisdiction, and the closure was finally upheld by the Madras High Court (2020) and the Supreme Court (2024).
- **LG Polymers styrene gas leak, Visakhapatnam (2020):** the NGT took suo motu cognisance of the leak that killed 12 people and ordered the company to deposit ₹50 crore as interim compensation.
- **River pollution and waste:** the NGT has issued continuing orders on the cleaning of the Ganga and Yamuna and imposed environmental compensation running into thousands of crores of rupees on state governments for failing to manage solid and liquid waste.

7.5.4 Assessment of the NGT

The NGT has made environmental justice faster, cheaper and more accessible, brought scientific expertise into the courtroom, and made the polluter pays and precautionary principles part of everyday decisions. Critics point to several problems:

- **Vacancies:** the tribunal has often worked with far fewer judicial and expert members than the minimum required, delaying cases and limiting the zonal benches.
- **Implementation:** many orders, especially those directed at government bodies on rivers and waste, are only partly complied with; imposing compensation on states does not automatically produce clean rivers.
- **Overreach and consistency:** some argue that the NGT has taken on administrative roles better left to the executive, issued sweeping orders without hearing all affected parties, or relied heavily on committees. Others criticise it for approving projects with conditions rather than stopping them.
- **Access:** benches are located in only five cities, which makes access difficult for rural and Adivasi communities, who are often the most affected.

Despite these limitations, the NGT is widely regarded as one of the most important environmental institutions created in India since the Environment (Protection) Act of 1986.

Think and Do: Mock Green Tribunal

Divide the class into a bench (one judicial and one expert member), petitioners, a project proponent and a state pollution control board. Choose a realistic local case, such as a proposed landfill near a residential colony or construction on a Yamuna floodplain. Each side presents evidence for 10 minutes. The bench must deliver a short reasoned order applying the principles of sustainable development, precaution and polluter pays.

7.6 Environmental Philosophy: Ethics, Religion and Culture

Laws and institutions such as the NGT rest on deeper beliefs about the value of nature and human duties towards it. **Environmental philosophy** is the branch of philosophy that examines the relationship between humans and the natural world. Its main sub-field, **environmental ethics**, asks moral questions: Does nature have value only because it is useful to humans? Do animals, plants, species or rivers have interests or rights? What do we owe to future generations?

7.6.1 Environmental ethics

Environmental ethicists distinguish between **instrumental value**, the value of something as a means to an end (a forest valued for timber or water), and **intrinsic value**, the value of something in itself, regardless of its usefulness. Different ethical positions draw the circle of moral concern at different points, as shown in the figure below.

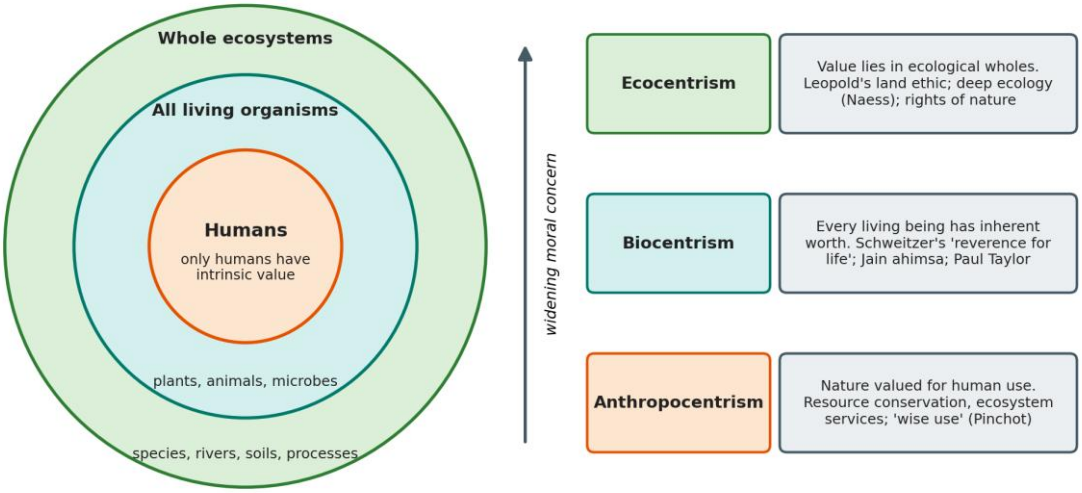


Figure 7.6: *The expanding circle of moral concern in environmental ethics, from anthropocentrism (humans only) through biocentrism (all living beings) to ecocentrism (whole ecosystems). Source: original diagram based on standard classifications.*

Anthropocentrism

Anthropocentrism (human-centred ethics) holds that only human beings have intrinsic value; nature matters because of the goods and services it provides to people. This view can still support strong conservation: forests should be protected because they supply water, prevent floods and store carbon. The American forester Gifford Pinchot's idea of "wise use" of resources for "the greatest good of the greatest number for the longest time", and today's language of **ecosystem services**, are anthropocentric. Critics argue that anthropocentrism allows the destruction of species that have no obvious use. The historian Lynn White Jr., in a famous essay "The Historical Roots of Our Ecologic Crisis" (1967), blamed a human-centred reading of the Bible for Western attitudes of domination over nature.

Biocentrism

Biocentrism (life-centred ethics) holds that every living organism has intrinsic value because it has its own good: it can grow, flourish or be harmed. The theologian and doctor Albert Schweitzer's principle of **"reverence for life"** and the philosopher Paul Taylor's *Respect for Nature* (1986) are biocentric. Peter Singer's *Animal Liberation* (1975) extended moral concern to all animals capable of suffering. Indian traditions such as Jain *ahimsa* (non-violence) are strongly biocentric.

Ecocentrism, the land ethic and deep ecology

Ecocentrism (ecosystem-centred ethics) gives moral value to whole ecological systems: species, communities, rivers, landscapes and ecological processes, not just individual organisms. Two influential versions are:

- **The land ethic** of the American forester and ecologist **Aldo Leopold**, set out in *A Sand County Almanac* (1949). Leopold argued that the moral community should be enlarged to include "soils, waters, plants, and animals, or collectively: the land". His summary rule was: "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong when it tends otherwise."
- **Deep ecology**, a term introduced by the Norwegian philosopher **Arne Naess** in 1973. Naess contrasted "shallow" environmentalism, which fights pollution and resource depletion mainly

for the health and affluence of people in rich countries, with a "deep" view that all forms of life have equal intrinsic value, that human interference in nature is excessive, and that fundamental changes in lifestyles and economies are needed.

Ramachandra Guha argued in 1989 that deep ecology's focus on wilderness, applied in India, could justify evicting poor forest dwellers to create parks; an Indian environmental ethic, he suggested, must combine respect for nature with justice for people who depend on it.

Intergenerational equity and sustainability

Intergenerational equity is the principle that each generation holds the Earth in trust for future generations and must pass it on in no worse condition than it received it. It underlies the definition of sustainable development in the Brundtland Report (1987) and is often illustrated by the saying that the Earth is not inherited from ancestors but borrowed from children. Indian courts have applied the principle in cases on mining and forests, holding that natural resources must be used in a way that preserves them for future generations. Mahatma Gandhi's remark that "the Earth provides enough to satisfy every man's needs, but not every man's greed" expresses a related ethic of restraint.

Rights of nature

The **rights of nature** movement goes further and argues that ecosystems themselves should be recognised as legal persons with rights that can be defended in court. Ecuador wrote the rights of nature into its Constitution in 2008, and New Zealand granted legal personhood to the Whanganui River in 2017 in recognition of the Māori people's relationship with it.

In India, on 20 March 2017, the **Uttarakhand High Court** declared the **Ganga and Yamuna rivers**, and their tributaries, to be legal persons, with officials appointed as their guardians (*Mohd. Salim v. State of Uttarakhand*). Days later the same court extended a similar status to glaciers, forests, lakes and other natural features of the state. The Uttarakhand government appealed, pointing out practical problems, for example who would be liable if a flood caused damage, and in July 2017 the **Supreme Court stayed** the High Court's order. Though not in force, the judgment sparked wide debate about whether legal personhood is a practical way to protect rivers or largely symbolic.

7.6.2 Role of religions and cultural practices in environmental conservation

Long before modern environmental laws, religions and cultural traditions shaped how people used nature through teachings of restraint, reverence and stewardship, although actual behaviour often falls short of these ideals, as the pollution of sacred rivers shows.

Table 7.6: Environmental teachings of major religious traditions

Tradition	Key concepts	Environmental teaching or practice
Hinduism	Prithvi as mother; *dharma*; divinity in rivers, trees and animals	Earth hymn of the Atharva Veda; worship of peepal, tulsi and banyan; sacred rivers and groves
Buddhism	Interdependence; compassion (*karuna*); middle way	Non-harm, moderation in consumption, protection of trees in monastic rules
Jainism	*Ahimsa*; *aparigraha* (non-possession)	Avoid harm to all life; minimal consumption; vegetarianism
Islam	Humans as *khalifa* (trustees); *mizan* (balance)	Forbids waste (*israf*); protected zones (*hima*); conserve water even when plentiful

Tradition	Key concepts	Environmental teaching or practice
Sikhism	Air as guru, water as father, earth as mother	Gurus' teaching in the Guru Granth Sahib; community kitchens; tree-planting drives
Christianity	Stewardship of creation	Pope Francis's *Laudato Si'* (2015) on "care for our common home"
Tribal traditions	Kinship with forest, hills and animals	Sacred groves, totemic species, seasonal hunting and harvesting rules

Hinduism

Hindu texts personify the Earth as a mother: the *Prithvi Sukta* of the Atharva Veda declares, "The Earth is my mother and I am her son" (*mātā bhūmiḥ putro ahaṃ pṛthivyāḥ*), and the *Isha Upanishad* teaches that one should enjoy only what is given, without coveting more. Rivers are worshipped as goddesses and trees such as the peepal, banyan, neem and tulsi are sacred. These beliefs have protected trees in temple compounds and village commons, though they have not prevented practices such as idol immersion from polluting rivers.

Buddhism and Jainism

Buddhism teaches that all phenomena arise in dependence on one another, and urges compassion for all sentient beings and moderation in desires; the Emperor Ashoka's edicts record the planting of shade trees along roads and restrictions on killing animals. Jainism carries *ahimsa* further than any other tradition, and its principle of *aparigraha* (limiting one's wants) is directly relevant to reducing consumption.

Islam, Sikhism and Christianity

In Islam, humans are *khalifa*, trustees of God's creation, who must maintain its balance (*mizan*) and avoid waste (*israf*), even of water at a flowing river; Islamic law also recognised protected zones (*hima*). Sikhism's Guru Granth Sahib states, "Air is the Guru, water is the father, and earth is the great mother of all"; this teaching inspired Sant Balbir Singh Seechewal's volunteer-led clean-up of the Kali Bein rivulet in Punjab from 2000 onwards. In Christianity, the idea of humans as stewards rather than masters of creation has grown in importance, and Pope Francis's *Laudato Si'* (2015) links care for the environment with justice for the poor.

Tribal and folk traditions

Many Adivasi communities regard forests, hills and rivers as sacred kin. The Dongria Kondh revere the Niyamgiri hills of Odisha as the seat of their god Niyam Raja, a belief central to their successful opposition to bauxite mining (see the resettlement section above). Many tribes follow rules on when and how much to harvest and protect **totemic** species associated with their clans.

Sacred groves are patches of forest dedicated to local deities or ancestors and protected by community taboos against cutting trees or hunting. India has an estimated 1 to 1.5 lakh sacred groves, known by local names such as *kavu* in Kerala, *devrai* in Maharashtra, *orans* in Rajasthan, *sarna* in Jharkhand and *law kyntang* in Meghalaya, where the grove at Mawphlang is famous. They often shelter rare and endemic plants, act as seed banks and protect springs. Their ecology and conservation value are discussed in Chapter 6.

Festivals and eco-friendly cultural practices

Festivals can either harm or help the environment. Idol immersion with toxic paints and plaster of Paris pollutes lakes and rivers; firecrackers worsen air quality and noise during Diwali; and large gatherings generate tonnes of waste. In response, many communities and governments now promote eco-friendly practices:

- **Clay idols** with natural colours, immersed in artificial ponds, as encouraged by CPCB guidelines.
- **Green crackers** developed by CSIR-NEERI after the Supreme Court restricted conventional firecrackers in 2018.
- **Holi with natural colours** from flowers such as *tesu* (*Butea monosperma*).
- **Tree festivals** such as Van Mahotsav (begun in 1950), Uttarakhand's Harela and Jharkhand's Sarhul.
- **Zero-waste weddings and community meals** using steel utensils or leaf plates instead of disposable plastic.

Key Points

- Anthropocentrism values nature for human use; biocentrism gives intrinsic value to all living beings; ecocentrism values whole ecosystems (Leopold's land ethic, Naess's deep ecology).
- Intergenerational equity and the rights of nature extend moral concern to future generations and to ecosystems; the Uttarakhand High Court's 2017 personhood order for the Ganga and Yamuna was stayed by the Supreme Court.
- All major religions contain ethics of restraint, reverence or stewardship; tribal traditions and sacred groves have conserved forests for centuries.
- Eco-friendly festivals show how cultural practices can adapt to reduce environmental harm.

7.7 Environmental Communication and Public Awareness

Laws, technologies and ethics achieve little unless people understand environmental problems and are willing to act. **Environmental communication** is the planned use of information, messages and media to help people understand environmental issues and to change attitudes, behaviour and policy. **Public awareness** is the knowledge and concern about environmental issues that exists among the general population. This section first sets out principles of effective communication and then examines four Indian case studies named in the syllabus.

7.7.1 Principles of effective environmental communication

Research shows a large **knowledge–action gap**: many people know that air pollution is harmful or that plastic chokes drains, yet do not change their behaviour because of cost, habit, convenience, social norms or distrust. Effective communication can be understood as a cycle in which a source sends a message through a channel to an audience, whose response feeds back into the next message.

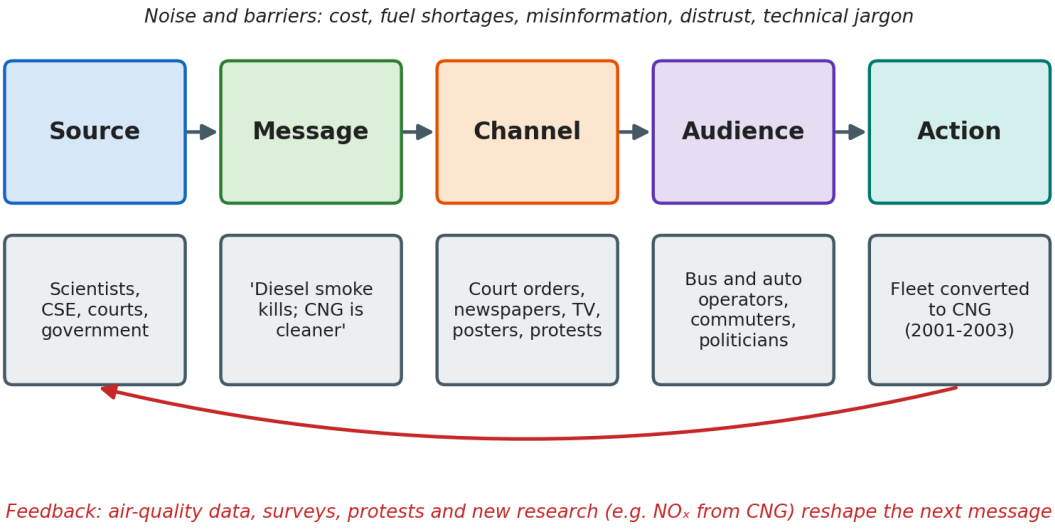


Figure 7.7: *Environmental communication as a cycle, illustrated with the Delhi CNG transition: source, message, channel, audience and action, with feedback shaping later messages. Source: original diagram based on the classic source–message–channel–receiver model.*

Key principles include:

- **Credible sources:** people trust messages from scientists, doctors, teachers and respected community or religious leaders more than from advertisers or politicians.
- **Know the audience:** a message for farmers about stubble burning must differ from one for urban car owners. Use local languages, examples and media.
- **Make the invisible visible:** air-quality index colours, photographs, maps and simple data such as a household carbon footprint make invisible problems concrete.
- **Frame for values:** linking an issue to health, children's future, savings, dignity or religious duty motivates more than abstract statistics.
- **Offer solutions and make action easy:** fear without a clear, affordable action leads to denial; the desired behaviour must be convenient and supported by infrastructure.
- **Use social norms and participation:** people copy neighbours and peers. Community leaders, competitions, pledges and public recognition spread new norms.
- **Listen and update:** two-way communication, honest admission of uncertainty and correction of earlier messages build trust.

7.7.2 Case study: CNG vehicles in Delhi

By the mid-1990s Delhi was among the most polluted cities in the world, and vehicles, especially diesel buses and two-stroke auto-rickshaws, were a major source of particulate matter. The analysis of Delhi's air pollution and its health effects is covered in Chapter 4; here the focus is on how a major change in public transport was achieved and communicated.

The legal path

In 1985 the environmental lawyer **M.C. Mehta** filed a public interest petition in the Supreme Court on air pollution in Delhi (*M.C. Mehta v. Union of India*). In 1996 the Centre for Science and Environment (CSE), a Delhi-based research and advocacy organisation, published *Slow Murder*, a study of vehicular

pollution, and launched a "Right to Clean Air" campaign. In January 1998 the central government, under the Supreme Court's direction, set up the **Environment Pollution (Prevention and Control) Authority (EPCA)**, chaired by Bhure Lal, to advise on measures. On **28 July 1998** the Supreme Court accepted EPCA's recommendations and ordered, among other steps, that old auto-rickshaws and taxis be replaced with vehicles using clean fuels and that the **entire city bus fleet be converted to compressed natural gas (CNG) by 31 March 2001**.

The deadline was not met. Transport operators and the government argued that CNG supply, filling stations and conversion kits were inadequate, and some claimed that low-sulfur diesel would be equally clean. In April 2001 the shortage of CNG buses caused chaos, with long queues at the few filling stations. The Court refused to retreat; in 2002 it imposed fines on operators running diesel buses. Gas supply and the network of filling stations were expanded, and by the end of 2002 Delhi's public transport, including buses, auto-rickshaws and many taxis, had largely switched to CNG. By 2003 Delhi had one of the largest CNG public-transport fleets in the world.

Results

Monitoring in the early 2000s showed falls in carbon monoxide, sulfur dioxide and, for a few years, particulate pollution, and the transition was celebrated internationally. The gains were later eroded by the rapid growth of private vehicles and other sources, and research has shown that CNG vehicles, though far cleaner than old diesel vehicles for particulates, still emit significant nitrogen oxides. Delhi has since moved to Bharat Stage VI standards (2020) and electric buses.

Lessons for environmental communication

- **Science-based advocacy:** CSE's studies, public meetings, newspaper advertisements and media campaigns turned technical data on particulate matter into a public demand. Making the health threat to children vivid built broad support.
- **The court as amplifier:** each hearing and order was widely reported, keeping the issue in public view and making clear that the change would not be reversed.
- **Preparing the audience:** the chaos of 2001 showed that communication must be matched by practical support: fuel supply, filling stations, subsidies, and clear information for bus and auto owners and commuters.
- **Honesty about limits:** presenting CNG as a complete "clean fuel" solution oversimplified the problem. Later evidence on NO_x and the rise in private vehicles showed the need to communicate trade-offs and to update messages, rather than claim permanent victory.

7.7.3 Case study: Swachh Bharat Abhiyan

The mission

Swachh Bharat Abhiyan (Clean India Mission) was launched by Prime Minister Narendra Modi on **2 October 2014**, Mahatma Gandhi's birth anniversary, with the goal of making India **open defecation free (ODF)** and improving solid waste management by 2 October 2019, Gandhi's 150th birth anniversary. It restructured the earlier Nirmal Bharat Abhiyan. When the mission began, about 60% of the world's people who defecated in the open lived in India, and the 2011 Census had found that only about a third of rural households had a toilet.

The mission has two components:

- **Swachh Bharat Mission–Gramin (SBM-G)**, for rural areas, implemented by the Department of Drinking Water and Sanitation (now under the Ministry of Jal Shakti). Rural households received an incentive of ₹12,000 to build a toilet.
- **Swachh Bharat Mission–Urban (SBM-U)**, implemented by the Ministry of Housing and Urban Affairs, covering household, community and public toilets and municipal solid waste management.

According to government data, more than 10 crore (100 million) household toilets were built in rural India, and on 2 October 2019 all villages were declared ODF. **SBM-G Phase II** (2020–21 to 2024–25) focuses on **ODF Plus**: sustaining toilet use and managing solid waste, grey water and faecal sludge in villages; by 2023 more than half of India's villages had been declared ODF Plus. **SBM-Urban 2.0**, launched on 1 October 2021 for 2021–26, aims at "garbage-free cities" through source segregation, scientific processing of waste and clearance of old dump sites. The annual **Swachh Survekshan** survey ranks cities on cleanliness; Indore was ranked the cleanest city for seven consecutive years from 2017.

Behaviour-change communication

The mission's planners recognised that building toilets would not end open defecation unless people chose to use them. Its **behaviour-change communication** strategy included:

- Framing sanitation as a national movement (*jan andolan*) linked to Gandhi, with celebrities as ambassadors and Gandhi's spectacles as the logo.
- The **"Darwaza Band"** (shut the door) mass-media campaign from 2017, featuring Amitabh Bachchan, targeting men who had toilets but did not use them.
- **Swachhagrahis**, trained local volunteers who used community-led total sanitation methods, and **Rani Mistris**, women trained as masons.
- Messages on the **dignity and safety of women**, and popular culture such as the film *Toilet: Ek Prem Katha* (2017).

Achievements and critiques

Independent surveys confirm real progress: the National Statistical Office found that about 71% of rural households had access to a toilet in 2018, up from about 40% in 2012, and the WHO estimated in 2018 that improved sanitation would avert hundreds of thousands of deaths from diarrhoea and malnutrition. Critics, however, questioned the ODF claims:

However, critics questioned the ODF claims:

- **Gap between toilets and use:** a 2018 survey by the Research Institute for Compassionate Economics (RICE) in four north Indian states found that about 44% of rural people still defecated in the open, many of them in households with a latrine. NFHS-5 (2019–21) found that about 19% of households used no toilet facility.
- **Verification:** ODF declarations were often based on toilet construction targets rather than on verified use, and some toilets existed only on paper.
- **Water and design:** toilets without water supply, or with single pits that fill quickly, were often abandoned.
- **Caste and manual scavenging:** purity beliefs and the caste-based practice of emptying pits by hand made people reluctant to use pit toilets.

- **Waste management:** in urban areas, segregation at source and processing of waste lag behind collection, and many legacy landfills remain.

Swachh Bharat shows both the power and the limits of a national campaign: political leadership, funding and mass communication changed norms rapidly, but lasting change needs continued attention to use, maintenance, water supply and social inequality.

7.7.4 Case study: National Environment Awareness Campaign (NEAC)

The **National Environment Awareness Campaign (NEAC)** was launched in **1986** by the then Ministry of Environment and Forests (now MoEFCC) to create environmental awareness among all sections of society and to make environmental protection a people's movement. It was one of the oldest components of the Ministry's **Environment Education, Awareness and Training (EEAT)** scheme.

How NEAC worked

- **Annual theme:** each year the Ministry chose a national theme, such as solid waste management, biodiversity conservation or climate change.
- **Regional Resource Agencies (RRAs):** about 30 organisations, including NGOs, universities and state bodies such as the Nagaland Pollution Control Board, managed the campaign in their regions, selected participants, released grants and monitored activities.
- **Participating organisations:** thousands of NGOs, schools, colleges, youth clubs, women's groups and panchayats received small one-time grants, typically from a few thousand to about ₹20,000.
- **Activities:** rallies, street plays and folk performances, workshops, tree-planting and cleanliness drives, exhibitions, competitions and demonstrations of composting or rainwater harvesting.

Strengths, weaknesses and current status

NEAC's strength was its reach through a decentralised network, often using local languages and folk media; its weaknesses were small one-off grants, symbolic events rather than sustained programmes, and little evaluation of behaviour change.

NEAC grants were last released in the late 2010s. In 2023 the MoEFCC issued a circular stating that NEAC and the other programmes of the old EEAT scheme were no longer being implemented. Environmental awareness activities are now funded through the **Environment Education Programme (EEP)** of the revamped central scheme **Environment Education, Awareness, Research and Skill Development (EEARSD)**, through government implementing agencies, and are linked to Mission LiFE.

7.7.5 Case study: National Green Corps "Eco-club" programme

Origins and structure

The **National Green Corps (NGC)** programme was launched by the Ministry of Environment and Forests in **2001–02** to promote **eco-clubs** in schools across the country. Its premise was that children, at an age when they are most receptive, can learn about the environment through action, and then influence their families and communities. An **eco-club** is a voluntary group of students, usually 30–50, guided by a teacher in-charge, that carries out environmental activities in the school and its neighbourhood.

The programme worked through a three-level structure:

- The **Ministry** framed guidelines, provided funds and organised national meetings and awards.
- **State Nodal Agencies** (for example, the Centre for Environmental Studies in Odisha and the GEER Foundation in Gujarat) coordinated the programme, trained teachers and produced material in local languages.
- **District Implementation and Monitoring Committees** selected schools and monitored activities.

Each eco-club received an annual grant, raised over time to ₹5,000, for its activities. In 2019–20 the programme was extended to **colleges**, with a ceiling of 500 school eco-clubs per district and 100 college eco-clubs per state for central funding. According to the Ministry's reply to Parliament in 2023, more than **one lakh eco-clubs** had been formed under NGC in schools and colleges. Many states, such as Tamil Nadu and Odisha, supported additional eco-clubs from their own funds.

Activities

Eco-clubs have carried out a wide range of activities:

- Tree planting and maintenance, school nurseries and herbal or butterfly gardens.
- Waste segregation, composting and vermicomposting, paper recycling and campaigns against single-use plastic.
- Water and energy audits of the school, rainwater harvesting and saving electricity.
- Surveys of local biodiversity, water bodies and pollution; nature camps.
- Rallies, street plays and competitions on environment days such as World Environment Day (5 June).

From NGC to the Environment Education Programme

The NGC ran as the Ministry's flagship programme for non-formal environment education until **2021–22**. It has since been subsumed in the **Environment Education Programme (EEP)** under the EEARSD scheme, which aims to build on the network of eco-clubs formed under NGC, together with youth clubs, NSS units and other groups, and to move from symbolic activities to focused workshops, projects, exhibitions, nature camps and campaigns on sustainable lifestyles linked to Mission LiFE. In parallel, the Central Board of Secondary Education directed in 2019 that every affiliated school must have an eco-club, the Ministry of Education supports eco-clubs in government schools under Samagra Shiksha, and the National Education Policy 2020 envisages eco-clubs as part of school and higher education.

Assessment

Eco-clubs have created a large network of environmentally active schools and trained teachers. Common problems are small and delayed grants, dependence on one enthusiastic teacher, examination pressure and activities limited to environment days; the most successful clubs run year-round projects that change how the institution manages its waste, water, energy and green spaces.

Starting an eco-club in a Delhi University college

Many Delhi University colleges already have eco-clubs or environment societies; others can start one. A practical sequence is:

1. **Form a core team:** a faculty convener and a student committee with clear roles (president, secretary, treasurer, outreach, documentation), and a short constitution stating the club's aims.
2. **Seek recognition:** register the club as a college society and link it with the NSS unit and University or government environment education programmes.
3. **Audit the campus:** measure waste generated, water and electricity use, the college's carbon footprint (using the method in this chapter), and trees and biodiversity.
4. **Plan year-round projects:** for example, segregated bins and composting of canteen waste, a plastic-free canteen, a campus tree census and energy-saving drives.
5. **Reach out:** visit the Yamuna or Aravalli Biodiversity Park or a sewage treatment plant, and run awareness sessions in a nearby school or resettlement colony.
6. **Measure and report:** record before-and-after data (kilograms composted, units of electricity saved) and publish an annual report.

Key Points

- Effective environmental communication closes the knowledge–action gap by using credible sources, audience-specific framing, visible data, easy actions, social norms and feedback.
- The Supreme Court's 1998 order in the M.C. Mehta case, backed by CSE's advocacy, converted Delhi's buses, autos and taxis to CNG by 2002–03; the case shows both the power of science-based campaigns and the need to communicate limits.
- Swachh Bharat (2014) built over 10 crore rural toilets using intensive behaviour-change communication, but surveys show gaps between toilet access and use.
- NEAC (1986) funded local awareness activities through Regional Resource Agencies; it is no longer implemented and has been replaced by the Environment Education Programme.
- The National Green Corps (2001–02) created more than one lakh school and college eco-clubs; since 2022 eco-clubs are supported through the Environment Education Programme and linked to Mission LiFE.

Chapter 7 Review

Summary

- The world population passed 8 billion in 2022 and, according to UN World Population Prospects 2024, will peak at about 10.3 billion in the mid-2080s; India overtook China as the most populous country in 2023.
- India's total fertility rate has fallen to about 2.0 (NFHS-5), below replacement level, but population momentum will keep its population growing until it peaks at about 1.7 billion in the early 2060s.
- The demographic transition model explains how falling death rates followed by falling birth rates produce rapid growth and then stabilisation; India is moving from late stage 3 into stage 4 and has a time-limited demographic dividend.
- The I = PAT identity shows that environmental impact depends on population, affluence and technology together, so consumption and clean technology matter as much as numbers.
- The main environment-related health risks in India are unsafe water, sanitation and hygiene, air pollution, vector-borne diseases and heat stress; environmental degradation also harms livelihoods, food security and welfare.
- A carbon footprint is the total greenhouse gas emitted by a person, organisation or product, expressed in CO_2e; it is calculated as activity data × emission factor and divided into Scope 1, 2 and 3 emissions.
- India's per-capita fossil CO_2 emissions (about 2.1 t in 2023) are less than half the world average and about one-seventh of the US figure, although India is the third-largest emitter in total; Mission LiFE (2022) promotes sustainable lifestyles.
- Development projects have displaced an estimated 6 crore or more people in India since 1947, disproportionately Adivasis; Cernea's IRR model identifies eight impoverishment risks of displacement.
- The RFCTLARR Act 2013 replaced the Land Acquisition Act 1894 with higher compensation, social impact assessment, consent requirements and legally binding R&R entitlements; Hirakud, Tehri, Singrauli and Sardar Sarovar show the costs of poor rehabilitation.
- The Bishnois (Khejarli, 1730), Chipko (1973–74), Appiko (1983), Silent Valley (1973–84) and Narmada Bachao Andolan (from 1985) show how communities, often led by women and Adivasis, have defended forests, rivers and livelihoods through non-violent action.
- Environmental justice concerns the fair distribution of environmental benefits and burdens and fair participation in decisions; the National Green Tribunal (2010) applies the principles of sustainable development, precaution and polluter pays.
- Environmental ethics ranges from anthropocentrism through biocentrism to ecocentrism; religions, tribal traditions, sacred groves and eco-friendly festivals show how culture can support conservation.
- Effective environmental communication must close the knowledge–action gap; the Delhi CNG transition, Swachh Bharat Abhiyan, NEAC and the National Green Corps eco-clubs illustrate both successes and limitations of awareness campaigns.

Key Terms

Demography: The scientific study of the size, composition, distribution and change of human populations.

Total fertility rate (TFR): The average number of children a woman would bear over her lifetime at current age-specific birth rates.

Replacement level fertility: A TFR of about 2.1, at which each generation exactly replaces itself in the long run.

Population momentum: Continued population growth after fertility falls to replacement level, because of a large number of young people entering reproductive age.

Demographic transition: The shift from high birth and death rates to low birth and death rates that accompanies economic and social development.

Demographic dividend: The potential economic gain when the working-age share of a population is large compared with dependants.

I = PAT: An identity stating that environmental impact equals population × affluence × technology.

Carbon footprint: The total greenhouse gas emissions caused directly and indirectly by a person, organisation, product or country, expressed in CO_2e.

Carbon dioxide equivalent (CO_2e): A unit that expresses the warming effect of any greenhouse gas as the amount of CO_2 that would cause the same warming.

Global warming potential (GWP): The heat trapped by one kilogram of a gas over a period (usually 100 years) relative to one kilogram of CO_2.

Emission factor: The mass of greenhouse gas released per unit of an activity, such as kg CO_2 per kWh of electricity.

Scope 1, 2 and 3 emissions: Direct emissions, indirect emissions from purchased energy, and all other value-chain emissions respectively.

Mission LiFE: India's Lifestyle for Environment mission (2022) that promotes individual pro-planet actions under seven themes.

Development-induced displacement: The forced movement of people from their homes or livelihoods to make way for development projects.

Rehabilitation: The restoration of displaced people's livelihoods, incomes, social networks and quality of life after resettlement.

Impoverishment Risks and Reconstruction (IRR) model: Michael Cernea's framework of eight risks faced by displaced people and the strategies needed to counter them.

Social Impact Assessment (SIA): A study of the effects of a project on affected people and communities, required before land acquisition under the 2013 Act.

Solatium: An additional payment made to landowners for the compulsory nature of land acquisition.

Environmentalism of the poor: Environmental struggles in which defending nature and defending the livelihoods of poor communities are the same cause.

Ecofeminism: The view that the domination of nature and the domination of women are linked.

Environmental justice: The fair distribution of environmental benefits and burdens and fair participation in environmental decisions for all people.

Precautionary principle: The rule that lack of full scientific certainty should not delay measures to prevent serious or irreversible environmental harm.

Polluter pays principle: The rule that those who cause pollution must bear the cost of remedying it and compensating victims.

Anthropocentrism: An ethical view that only humans have intrinsic value and nature matters for its usefulness to people.

Biocentrism: An ethical view that every living organism has intrinsic value.

Ecocentrism: An ethical view that gives moral value to whole ecosystems, species and ecological processes.

Land ethic: Aldo Leopold's principle that actions are right when they preserve the integrity, stability and beauty of the biotic community.

Intergenerational equity: The principle that each generation must pass on the Earth's resources to future generations in no worse condition.

Knowledge–action gap: The gap between knowing about an environmental problem and acting to address it.

Eco-club: A voluntary group of students guided by a teacher that carries out practical environmental activities in and around a school or college.

Multiple-Choice Questions

1. According to UN World Population Prospects 2024, the world population is projected to peak at about:

(a) 8.5 billion around 2040
(b) 9.2 billion around 2050
(c) 10.3 billion in the mid-2080s
(d) 12 billion around 2100

2. In which year did India overtake China as the world's most populous country, according to UN estimates?

(a) 2023
(b) 2011
(c) 2019
(d) 2030

3. India's population continues to grow even though its TFR has fallen to about 2.0 mainly because of:

(a) rising death rates
(b) immigration from neighbouring countries
(c) an increase in the replacement level
(d) population momentum from a large young population

4. In which stage of the demographic transition model do death rates fall sharply while birth rates remain high?

(a) Stage 1
(b) Stage 2
(c) Stage 4
(d) Stage 5

5. In the identity I = P × A × T, the term A stands for:

(a) Agriculture
(b) Area of land used
(c) Affluence, or consumption per person
(d) Air pollution

6. Emissions from the coal-fired power station that supplies electricity to a household are classified as:

(a) Scope 2
(b) Scope 1
(c) Scope 3
(d) Not part of the household's footprint

7. A household uses 300 kWh of grid electricity in a month. Using a grid emission factor of 0.72 kg CO_2/kWh, its monthly Scope 2 emissions are about:

(a) 72 kg CO_2
(b) 216 kg CO_2
(c) 300 kg CO_2
(d) 417 kg CO_2

8. Mission LiFE was formally launched in October 2022 at:

(a) Glasgow, United Kingdom
(b) New Delhi
(c) Paris, France
(d) Ekta Nagar (Kevadia), Gujarat

9. Which of the following is NOT one of the eight risks in Cernea's Impoverishment Risks and Reconstruction model?

(a) Landlessness
(b) Social disarticulation
(c) Increased literacy
(d) Loss of access to common property resources

10. Under the RFCTLARR Act 2013, land acquisition for a private company requires the consent of at least:

(a) 80% of affected families
(b) 50% of affected families
(c) 70% of affected families
(d) 100% of affected families

11. The Bishnoi sect, whose followers protect khejri trees and blackbuck, was founded by:

(a) Sant Kabir
(b) Guru Jambheshwar
(c) Guru Nanak
(d) Mirabai

12. The Khejarli sacrifice of 1730, in which 363 Bishnois died protecting trees, was led by:

(a) Gaura Devi
(b) Bachni Devi
(c) Sugathakumari
(d) Amrita Devi

13. The famous Chipko action of March 1974 in which Gaura Devi and other women guarded the forest took place at:

(a) Reni
(b) Mandal
(c) Salkani
(d) Khejarli

14. The Kannada words Ulisu, Belesu and Balasu, the guiding principles of the Appiko movement, mean:

(a) hug, march, fast
(b) protest, petition, prosecute
(c) save, grow, use rationally
(d) reduce, reuse, recycle

15. The independent review whose 1992 report led to the World Bank's withdrawal from the Sardar Sarovar Project was headed by:

(a) Baba Amte
(b) Bradford Morse
(c) Michael Cernea
(d) M.G.K. Menon

16. Which of the following Acts is NOT within the jurisdiction of the National Green Tribunal under Schedule I of the NGT Act 2010?

(a) Air (Prevention and Control of Pollution) Act, 1981
(b) Biological Diversity Act, 2002
(c) Forest (Conservation) Act, 1980
(d) Wildlife (Protection) Act, 1972

17. The NGT imposed environmental compensation on the Art of Living Foundation in 2016 for damage to the floodplain of which river?

(a) Yamuna
(b) Ganga
(c) Hindon
(d) Sabarmati

18. "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community." This statement expresses:

(a) Anthropocentrism
(b) Aldo Leopold's land ethic
(c) Utilitarianism
(d) The polluter pays principle

19. The Supreme Court order of July 1998 requiring Delhi's city bus fleet to be converted to CNG was passed in the case of:

(a) Vellore Citizens' Welfare Forum v. Union of India
(b) Narmada Bachao Andolan v. Union of India
(c) M.C. Mehta v. Union of India
(d) Subhash Kumar v. State of Bihar

20. The National Green Corps eco-club programme was launched by the Ministry of Environment and Forests in:

(a) 1986
(b) 1992
(c) 2014
(d) 2001–02

Short-Answer Questions

1. Distinguish between total fertility rate and replacement level fertility. Why does India's population continue to grow although its TFR is about 2.0?
2. Describe the five stages of the demographic transition model, and state where India stands today.
3. Explain the I = PAT identity with one Indian example for each term.
4. Explain how unsafe water and sanitation, air pollution and vector-borne diseases link population growth and environmental change to human health in India.
5. Define carbon footprint and distinguish between Scope 1, Scope 2 and Scope 3 emissions with household examples.
6. A college uses 20,000 kWh of grid electricity and 1,500 litres of diesel (for generators) in a year. Using emission factors of 0.72 kg CO_2/kWh and 2.65 kg CO_2/litre, calculate its Scope 1 and Scope 2 emissions in tonnes.
7. List Cernea's eight impoverishment risks of displacement and suggest a reconstruction strategy for any three.
8. State five key provisions of the RFCTLARR Act 2013.
9. Describe the Khejarli incident of 1730 and its significance for conservation.
10. Compare the Chipko and Appiko movements in terms of region, leaders, methods and outcomes.
11. What are the jurisdiction and guiding principles of the National Green Tribunal?
12. What lessons for environmental communication can be drawn from the CNG transition in Delhi?

Long-Answer Questions

1. "Population growth is only one of the drivers of environmental degradation." Critically examine this statement using the I = PAT identity, the demographic transition and data on per-capita carbon emissions for India and other countries.
2. Using Cernea's Impoverishment Risks and Reconstruction model, analyse the experience of people displaced by any two of the following: Hirakud Dam, Tehri Dam, Singrauli projects, Sardar Sarovar Dam. How far does the RFCTLARR Act 2013 address the problems you identify?
3. Trace the history, leadership, methods and outcomes of the Bishnoi, Chipko and Narmada Bachao Andolan movements. What do they reveal about the "environmentalism of the poor" in India?
4. Discuss the concept of environmental justice. Evaluate the role of the National Green Tribunal in delivering environmental justice in India, with reference to landmark cases and its limitations.
5. Compare anthropocentric, biocentric and ecocentric environmental ethics. How have Indian religious and cultural traditions, including tribal practices and sacred groves, contributed to conservation?
6. Evaluate Swachh Bharat Abhiyan and the National Green Corps eco-club programme as examples of environmental communication and public awareness. What made them succeed, and where did they fall short?

Further Reading

- Gadgil, M. and Guha, R. (1993). *This Fissured Land: An Ecological History of India*. Oxford University Press.
- Guha, R. (1989). *The Unquiet Woods: Ecological Change and Peasant Resistance in the Himalaya*. Oxford University Press.
- Divan, S. and Rosencranz, A. (2002). *Environmental Law and Policy in India: Cases, Materials and Statutes* (2nd ed.). Oxford University Press.
- McCully, P. (1996). *Silenced Rivers: The Ecology and Politics of Large Dams*. Zed Books.
- Raven, P.H., Hassenzahl, D.M., Hager, M.C., Gift, N.Y. and Berg, L.R. (2015). *Environment* (9th ed.). Wiley.
- United Nations, Department of Economic and Social Affairs (2024). *World Population Prospects 2024: Summary of Results*. United Nations. https://population.un.org/wpp/
- Central Electricity Authority (2024). *CO_2 Baseline Database for the Indian Power Sector, Version 20.0*. Government of India. https://cea.nic.in
- Ministry of Environment, Forest and Climate Change. *Mission LiFE* and *Environment Education Programme* pages. https://moef.gov.in

Answers to Multiple-Choice Questions

1. (c)	2. (a)	3. (d)	4. (b)	5. (c)	6. (a)	7. (b)	8. (d)	9. (c)	10. (a)
11. (b)	12. (d)	13. (a)	14. (c)	15. (b)	16. (d)	17. (a)	18. (b)	19. (c)	20. (d)

Appendix A: Practical and Outreach Activities

The University of Delhi course "Environmental Science: Theory into Practice" gives one of its two credits each year to practical work. This appendix turns every practical listed in the syllabus for Units 1–7 into a ready-to-run activity with an aim, short background, free data sources and materials, a numbered procedure, an observation table, analysis (with formulas and worked examples where needed) and viva questions. None needs a science laboratory: a smartphone, a spreadsheet, a measuring tape, a low-cost water-testing kit and internet access are enough for almost all of them.

Colleges choose which practicals to run, so the activities are independent. Where two share a method (questionnaire surveys, or carbon-footprint calculation, in Units 5 and 7), it is explained fully once and the later activity points back to it.

A.1 How Practical Work Is Organised and Assessed

A.1.1 Formats of practical work

The syllabus expects practical learning through **project work**, written **reports**, **field visits**, **presentations**, **posters**, **short videos**, **models** and **simulations** such as a model climate conference or a mock tribunal hearing. Most activities are designed for groups of four to six students, which mirrors real environmental assessment and lets students from different disciplines contribute their strengths.

A typical practical has three sessions: **preparation** (the teacher explains the aim, the group divides tasks and reads the relevant chapter), **data collection** (measuring, observing, surveying or downloading data) and **analysis and reporting** (computing results, drawing graphs, writing the report or preparing the presentation).

A.1.2 How practicals are assessed

Colleges follow the University's continuous-assessment scheme for the practical credit. Within that scheme, teachers usually assess four things:

- **Practical record or file**: the written report of each activity, checked for completeness, accuracy and honest presentation of data.
- **Project or field report**: a longer report on one chosen project, field visit or survey.
- **Presentation, poster, model or video**: communication of findings to an audience.
- **Viva voce**: a short oral examination in which students explain what they did, why, and what the results mean.

A rubric for marking practical reports appears at the end of this appendix. Students should read it before starting so that they know what is expected.

A.1.3 Safety in field and laboratory work

Fieldwork in Delhi carries modest but real risks (heat, smog, traffic, stray animals, open drains, contaminated water). Follow these rules on every visit:

- Go in groups, never alone. Tell the teacher-in-charge your route and return time, and carry the college identity card and a charged phone.
- Obtain written permission before entering a protected area, a treatment plant, a landfill area, a factory or private land.
- Wear closed shoes, full-length clothing, a cap and sunscreen; carry drinking water and a basic first-aid kit (antiseptic, bandages, oral rehydration salts).
- Wear disposable gloves when handling water samples, compost or waste, and wash hands with soap afterwards. Never taste a sample or smell it closely.
- Do not enter water bodies, climb on landfill slopes, lean over tanks at treatment plants, or touch electrical equipment at solar or biogas units.
- Do not disturb wildlife. Do not collect plants, eggs, nests or animals; photograph them instead. Collecting from a sanctuary or national park is an offence under the Wildlife (Protection) Act, 1972.
- Check the Air Quality Index before outdoor work in winter. If the AQI is "Very Poor" or "Severe", postpone strenuous outdoor work or wear a well-fitted N95 mask.
- Dispose of used gloves, test-kit reagents and strips as instructed on the kit; some reagents are corrosive or toxic.

A.1.4 Ethics in surveys and community work

Practicals that involve talking to people follow the ethical rules of social research.

- **Informed consent**: before asking any question, explain who you are, the purpose and duration of the study, that participation is voluntary and may be stopped at any time, and how the answers will be used. Record consent (a signature, a thumb impression, or a tick on the first page of an online form). For anyone under 18, obtain consent from a parent or guardian and the school, and assent from the child.
- **Anonymity and confidentiality**: do not record names, phone numbers or addresses unless they are essential. Use codes (R1, R2…) in the data sheet. Do not photograph faces or homes without explicit permission.
- **Do no harm**: avoid questions that could embarrass, frighten or endanger respondents, for example about legal status, caste, income details or disputes with authorities, unless they are essential and asked sensitively.
- **Respect and reciprocity**: respect people's time; visit at convenient hours; share results with the community (a summary sheet or a short talk) where possible.
- **Honesty**: never invent or "adjust" responses. Report the number of people who refused or did not answer.
- **Institutional approval**: some colleges require a teacher or an ethics committee to approve questionnaires. Keep a signed copy of the approved questionnaire in the practical file.

A.1.5 How to write a practical record

Every activity should be written up in the practical file in a standard format. Use the template below; adapt the headings slightly for surveys, simulations and models.

Table A.1: Standard format for a practical report

Section	What to write	Typical length
Title, date, group members	Activity title, date(s), names and roll numbers	3–4 lines
Aim	One sentence on what you set out to find or do	1–2 lines
Introduction	Background concept and why it matters, with one Indian example	100–200 words
Study area	Location, map sketch, GPS coordinates, date and time, weather	50–150 words
Materials and data sources	Equipment, kits, apps, websites with access dates	List
Method	Numbered steps actually followed, in past tense	100–250 words
Observations	Tables of raw data with units	As needed
Analysis and results	Calculations, graphs, indices, comparison with standards	As needed
Discussion	What the results mean, sources of error, comparison with published data	150–300 words
Conclusion and suggestions	Main finding and practical recommendations	50–150 words
References	Books, reports and websites actually used	List

Good reports share a few habits. Units are given for every number. Graphs have titles, labelled axes and a source line. Photographs carry a caption with place and date. Raw data are shown, not only averages. Limitations are stated plainly: a single sample, a short survey or a small quadrat area cannot support sweeping conclusions.

A.2 Unit 1: Introduction to Environmental Studies

A.2.1 Analysing a Country's Achievement of the Sustainable Development Goals

Aim

To assess how far a chosen country (or an Indian state) has progressed towards the 17 Sustainable Development Goals (SDGs) using official indicator data, and to identify its strongest and weakest goals.

Background

The 2030 Agenda for Sustainable Development, adopted by all UN member states in 2015, sets 17 goals, 169 targets and over 230 unique indicators (see Chapter 1). In India, NITI Aayog publishes the **SDG India Index**, which scores each state and Union Territory from 0 to 100 on each goal. India's composite score rose from 57 (2018) to 66 (2020–21) and 71 (2023–24), with Kerala and Uttarakhand leading at 79 (NITI Aayog, SDG India Index 2023–24).

Materials and Data Sources

- NITI Aayog SDG India Index dashboard: https://sdgindiaindex.niti.gov.in
- UN SDG Global Database: https://unstats.un.org/sdgs/dataportal
- Sustainable Development Report dashboards (SDSN): https://dashboards.sdgindex.org
- Our World in Data SDG Tracker: https://ourworldindata.org/sdgs
- Spreadsheet software (Excel, Google Sheets or LibreOffice Calc)

Procedure

1. Choose one country (or two Indian states) and, from the SDSN or SDG India Index dashboard, record the overall score and each goal's score or status for the latest year.
2. For four goals (at least one each from the social, economic and environmental groups), select one or two headline indicators, for example under-five mortality (SDG 3), renewable share of final energy (SDG 7), forest area as a share of land area (SDG 15) and CO_2 emissions per capita (SDG 13).
3. Download at least ten years of data for these indicators from the UN SDG database or Our World in Data, noting units and sources.
4. Draw a line graph for each indicator and a bar chart of goal scores for the latest year.
5. Write a two-page analysis of progress, gaps and probable reasons.

Observations

Table A.2: Goal-wise status of the selected country (example headings)

SDG	Headline indicator	Value (earliest year)	Value (latest year)	Trend
3 Good health	Under-five mortality per 1,000 live births			
7 Clean energy	Renewable share of final energy (%)			
13 Climate action	CO_2 emissions per capita (t)			

Analysis and Results

For each indicator compute the **compound annual growth rate** (CAGR) between the first year (V_0) and the last year (V_n), n years apart:

$$\text{CAGR} = (V_n / V_0)\text{^}(1/n) - 1$$

Example: if access to safely managed drinking water rose from 40% to 58% over 12 years, CAGR = (58/40)^(1/12) − 1 = 1.45^(0.0833) − 1 ≈ 0.031, or about 3.1% a year.

Extend each trend to 2030, compare it with the target, and classify each goal (achieved, on track, improving, stagnating, decreasing) in a colour-coded table. Discuss trade-offs: rising electricity access (SDG 7) may raise CO_2 emissions (SDG 13) unless supply comes from non-fossil sources.

Viva / Reflection Questions

1. Why does the SDG India Index exclude SDG 14 when calculating a state's composite score?
2. Which of your chosen goals shows the fastest progress, and what policies might explain it?

3. Give one example of a synergy and one of a trade-off between two SDGs in your country.
4. Why might two indices (SDSN and a national index) rank the same country differently?

A.2.2 Applying a Sustainability Framework to an Industrial Activity

Aim

To examine one industrial activity through a structured sustainability framework, using activity worksheets to identify its environmental, social and economic impacts and possible improvements.

Background

A **sustainability framework** is a structured checklist that makes an industry's flows of resources and wastes visible; examples are the **triple bottom line** (people, planet, profit), **life-cycle thinking** (cradle to grave) and reporting standards such as the Global Reporting Initiative (GRI). In India, SEBI requires the top 1,000 listed companies by market capitalisation to publish a **Business Responsibility and Sustainability Report (BRSR)** disclosing energy, water, emissions and waste data.

Materials and Data Sources

- Annual reports and BRSR disclosures of listed companies (company websites; BSE and NSE filings)
- CPCB categorisation of industries (red, orange, green, white): https://cpcb.nic.in
- Bureau of Energy Efficiency (BEE) PAT scheme information: https://beeindia.gov.in

Procedure

1. Choose an industrial activity: a cement plant, thermal power station, textile dyeing unit, sugar mill, brick kiln, dairy, or the college canteen as a small "industry".
2. Draw a flow diagram: raw materials, energy and water in; product, by-products, emissions, effluents and solid waste out.
3. Fill in Worksheet A using the company's BRSR or, for a small unit, an interview with the manager (with consent).
4. Fill in Worksheet B, scoring each dimension from 1 (poor) to 5 (excellent) with a justification.
5. Identify three improvements using the waste hierarchy (reduce, reuse, recycle, recover, dispose) and cleaner-production ideas, and present the analysis as a one-page poster.

Observations

Table A.3: Worksheet A, inputs and outputs of the chosen activity (example)

Flow	Item	Quantity per year	Unit	Source of data
Input	Electricity		MWh	BRSR / bills
Input	Fresh water		kL	BRSR / meter
Output	Solid waste		t	BRSR
Output	Greenhouse gas (Scope 1 + 2)		t CO_2e	BRSR

Table A.4: Worksheet B, sustainability scoring (example)

Dimension	Guiding question	Score (1–5)	Evidence
Resource efficiency	Is energy and water use per unit product falling?		
Pollution	Does it meet CPCB discharge and emission standards?		
Workers and community	Are health, safety and fair wages ensured?		
Circularity	Are wastes reused or sold as raw material?		

Analysis and Results

Calculate **intensity indicators**, which allow comparison between firms of different size: energy intensity (GJ per tonne of product), water intensity (kL per tonne) and emission intensity (t CO_2e per tonne). Compare two companies of the same sector where possible. Sum the Worksheet B scores, draw a spider chart of the dimensions, and identify the weakest dimension.

Viva / Reflection Questions

1. Why are intensity indicators more useful than total quantities when comparing two companies?
2. What do "Scope 1" and "Scope 2" emissions mean?
3. How does life-cycle thinking change the assessment of a plastic bottle versus a glass bottle?
4. Suggest one policy instrument that would push your chosen industry towards sustainability.

A.2.3 Worksheets on Interdependence of Environmental Components

Aim

To understand, through guided worksheets and observation, how the atmosphere, hydrosphere, lithosphere and biosphere interact in a familiar setting.

Background

The four components of the environment are not separate compartments. Rain (hydrosphere) falling through polluted air (atmosphere) can become acidic and leach nutrients from soil (lithosphere), affecting plants (biosphere). This activity traces such links in a real place (see Chapter 1).

Materials and Data Sources

- Printed worksheet, notebook, phone camera, thermometer
- CPCB AQI (see Unit 4) and IMD weather information for the day: https://mausam.imd.gov.in

Procedure

1. Choose a site of about 50 m × 50 m with vegetation, soil and paved surface, ideally with water (a pond, fountain or drain).
2. Observe for 30 minutes and list at least five features of each component.
3. Measure air temperature at about 1 m height over a paved surface in sun, over grass in sun, and in the shade of a large tree.

4. Fill the interaction matrix: in each cell, write one way in which the row component affects the column component at this site.
5. Draw a flow diagram linking at least six interactions and mark one **feedback loop** (for example, tree cover → shade → cooler soil → more soil moisture → healthier trees).

Observations

Table A.5: Interaction matrix of environmental components (example entries)

Effect of ↓ on →	Atmosphere	Hydrosphere	Lithosphere	Biosphere
Atmosphere	—	Rain fills pond	Wind erodes bare soil	Dust coats leaves
Hydrosphere	Evaporation cools air	—	Runoff carries silt	Water for plants
Lithosphere	Dust from bare ground	Soil filters rainwater	—	Soil supplies nutrients
Biosphere	Trees release O_2, shade	Roots aid infiltration	Leaf litter forms humus	—

Table A.6: Temperature at three surfaces

Location	Time	Air temperature (°C)	Remarks
Paved surface in sun			
Grass in sun			
Under tree canopy			

Analysis and Results

Compute the temperature difference between the paved surface and the tree shade. A difference of several degrees illustrates the **urban heat island** effect and the cooling service of trees. Discuss which human action at the site (paving, irrigation, tree felling, littering) affects the most interactions.

Viva / Reflection Questions

1. Give one example from your site of an interaction involving all four components.
2. What is a feedback loop? Was the loop you drew positive (amplifying) or negative (stabilising)?
3. Why is environmental studies called a multidisciplinary subject? Use your matrix to explain.

A.3 Unit 2: Ecosystems

A.3.1 Ecological Pyramids of the College Campus

Aim

To collect data systematically on producers and consumers in a part of the college campus and depict them as pyramids of numbers and of biomass.

Background

An **ecological pyramid** shows the number, biomass or energy at successive trophic levels of a food chain (see Chapter 2). In a grassland the pyramid of numbers is upright: many grass plants support fewer grasshoppers, which support still fewer spiders and birds. Under a single large tree the pyramid of numbers is **inverted**, because one tree supports thousands of insects, yet the pyramid of biomass remains upright.

Materials and Data Sources

- Measuring tape (30 m), pegs, string, four 1 m × 1 m quadrat frames (bamboo or PVC pipe)
- Sweep net (or a white cloth and stick for beating shrubs), plastic containers, hand lens
- iNaturalist (https://www.inaturalist.org) for identification; a random-number app

Procedure

1. Select two contrasting habitats: a grassy lawn and the area under a single large tree (neem, peepal or jamun).
2. In the lawn, lay out a 10 m × 10 m plot and place five 1 m × 1 m quadrats at random positions (see the figure below for a standard layout).
3. In each quadrat, count the plants (grass tufts and herbs); these are the **producers**.
4. Sweep the vegetation over each quadrat ten times. Count plant-eaters (grasshoppers, caterpillars, bugs, aphids) and predators (spiders, ladybird beetles, mantises) separately, then release them.
5. For 20 minutes, count birds and lizards feeding in the plot.
6. Under the tree, count the tree as one producer; beat three low branches over a white cloth, count insects and spiders, and multiply by the approximate number of similar branches.
7. For biomass, clip all above-ground plants from one quadrat, dry them (several days in sun or at 60 °C in an oven) and weigh; weigh 20 insects together to estimate average insect mass.

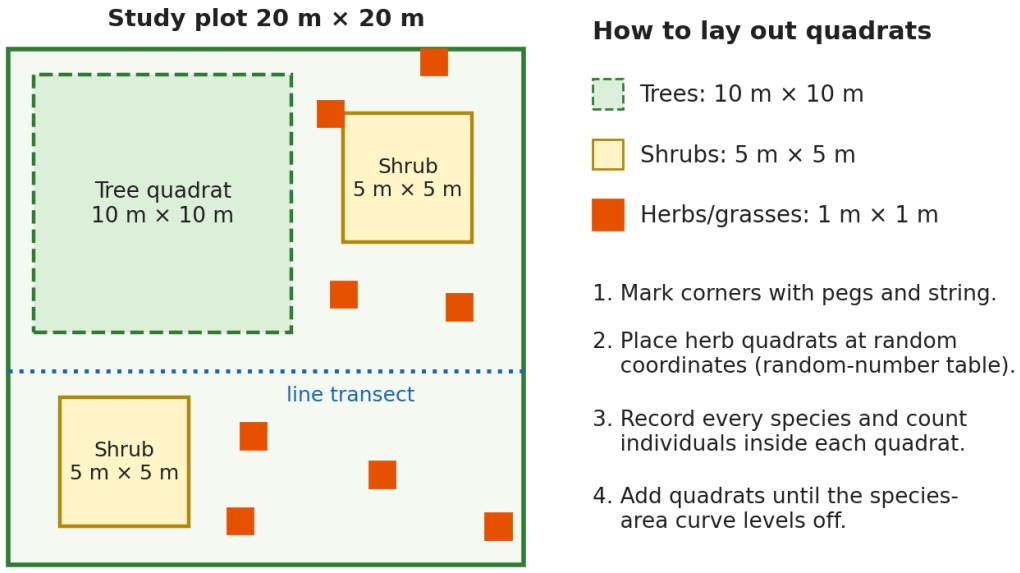


Figure A.1: *Layout of a 20 m × 20 m study plot with nested quadrats for trees, shrubs and herbs, randomly placed 1 m × 1 m quadrats and a line transect. Source: Author's schematic.*

Observations

Table A.7: Count of organisms by trophic level in five lawn quadrats (example headings)

Trophic level	Organisms	Q1	Q2	Q3
Producers	Grasses, herbs			
Primary consumers	Grasshoppers, aphids, caterpillars			
Secondary consumers	Spiders, ladybirds			
Tertiary consumers	Birds, lizards (plot total)			

Table A.8: Biomass estimate (example headings)

Trophic level	Number per m²	Average dry mass per individual (g)	Biomass (g/m²)
Producers			
Primary consumers			

Analysis and Results

Mean number per m^2 at each level = total count ÷ total area sampled; biomass = mean number per m^2 × average dry mass of one individual. Draw centred bars proportional to each level's value, producers at the bottom.

Example: five quadrats contained 1,150 grass tufts, 86 herbivorous insects and 9 spiders, giving densities of 230, 17.2 and 1.8 per m^2. Producers outnumber herbivores about 13:1 and herbivores

outnumber carnivores about 10:1, an upright pyramid. Under the tree, 1 tree : ~2,000 insects : ~60 spiders gives an inverted base. Relate the result to the **10% rule** of energy transfer.

Precautions

- Handle insects gently and release them at the same spot.
- Wear gloves; watch for wasps and thorny plants such as *Prosopis juliflora* (vilayati kikar).

Viva / Reflection Questions

1. Why is the pyramid of numbers under a tree inverted while its pyramid of biomass is upright?
2. Can a pyramid of energy ever be inverted? Explain.
3. What sources of error affected your counts, and how would you reduce them?

A.3.2 Distinguishing Natural and Managed Ecosystems Using Google Earth

Aim

To identify, map and compare natural and human-managed ecosystems in and around Delhi using satellite imagery from Google Earth or Google Maps.

Background

Natural ecosystems, such as the Ridge forest or the Yamuna floodplain, develop largely through natural processes and are diverse and self-regulating. **Managed ecosystems**, such as farms, lawns, plantations and fish ponds, are maintained by people through planting, irrigation, fertilisers and harvesting, and are usually simpler.

Materials and Data Sources

- Google Earth Pro (free desktop version with historical imagery): https://www.google.com/earth/about/versions
- Bhuvan, ISRO's geoportal, for land use/land cover layers: https://bhuvan.nrsc.gov.in
- Suggested sites: Central Ridge and Kamla Nehru Ridge; Asola Bhatti Wildlife Sanctuary; Yamuna floodplain near Wazirabad; farmland around Narela and Bawana; Lodhi Garden; Sunder Nursery; golf courses

Procedure

1. Open Google Earth Pro, locate your college and set the eye altitude to about 3 km.
2. Mark at least three natural and three managed ecosystems within about 20 km, outline each with the polygon tool (Add → Polygon) and note its area.
3. Describe colour, texture, shape, edges, structures and signs of irrigation.
4. Use the historical-imagery tool (clock icon) to view each site in two years at least ten years apart, and note changes.
5. Check your classification against Bhuvan's land use/land cover layer for Delhi, and save dated screenshots with a scale bar.

Observations

Table A.9: Comparison of natural and managed ecosystems from satellite imagery (example)

Site	Type (natural/managed)	Area (ha)	Visual features	Change over time
Central Ridge patch	Natural		Irregular edge, mixed dark green	
Farmland, Narela	Managed		Rectangular plots, seasonal colour	
Lodhi Garden	Managed		Lawns, paths, scattered trees	

Analysis and Results

Calculate the proportion of natural and managed area among your sites. Compare the two types for diversity, energy subsidy, stability and nutrient cycling (see Chapter 2). For sites viewed in two years, compute the percentage change in green cover.

Viva / Reflection Questions

1. List three visual clues that suggest an ecosystem is managed.
2. Why are managed ecosystems usually less stable than natural ones?
3. Is a biodiversity park a natural or a managed ecosystem? Justify your answer.
4. What are the limitations of identifying ecosystems only from satellite images?

A.3.3 Field Visit to Terrestrial and Aquatic Ecosystems

Aim

To observe the structure and functioning of a forest, grassland, wetland or biodiversity park near Delhi, and to record evidence of ecosystem services, threats and conservation measures.

Background

The **Delhi Ridge**, the northern tip of the Aravalli range, carries semi-arid thorn forest. The Delhi Development Authority (DDA) has restored degraded land into seven biodiversity parks: Yamuna, Aravalli, Northern Ridge (Kamla Nehru Ridge), Tilpath Valley, Neela Hauz, Tughlaqabad and Kalindi (also described as the South Delhi Biodiversity Park). Wetlands include the Najafgarh jheel, the Okhla Bird Sanctuary on the Yamuna (in Noida, Uttar Pradesh) and the wetlands of the Yamuna Biodiversity Park.

Table A.10: Suggested field-visit sites in and around Delhi

Site	Ecosystem type	What to observe
Yamuna Biodiversity Park, Wazirabad	Floodplain wetland, grassland, forest	Restored wetlands, native trees, migratory birds
Aravalli Biodiversity Park, Vasant Vihar	Ridge forest on former mining land	Restoration of mined land, native Aravalli species
Kamla Nehru Ridge (Northern Ridge)	Ridge thorn forest	Removal of vilayati kikar, native regeneration

Site	Ecosystem type	What to observe
Asola Bhatti Wildlife Sanctuary	Protected Ridge forest, old mining pits	Wildlife, water-filled pits, eco-restoration
Sanjay Van and Neela Hauz Biodiversity Park	Urban forest and restored lake	Forest structure, wetland restoration
Okhla Bird Sanctuary	Riverine wetland	Waterbirds, aquatic plants
Najafgarh jheel	Seasonal wetland	Waterbirds in winter, pollution, encroachment
Sunder Nursery	Managed heritage park with micro-habitats	Native plant collections, managed versus natural

Materials and Data Sources

- Permission letter from the college; entry rules on the DDA website (https://dda.gov.in/biodiversity-parks) or from the Delhi Forest Department
- Notebook, phone camera with GPS, binoculars
- eBird India (https://ebird.org/india) and Merlin Bird ID; iNaturalist for plants and insects

Procedure

1. Before the visit, prepare a checklist of expected species from eBird or iNaturalist records for the site.
2. Attend the guide's briefing, note the site's history, and record GPS location, time and weather.
3. Walk a fixed route of about 1 km. Every 200 m, stop for five minutes and record plants, animals, animal signs (droppings, nests, burrows) and human impacts (litter, fire scars, encroachment, invasive species).
4. At a water body, record colour, smell, floating plants (water hyacinth, duckweed), algae, fish, waterbirds and inflowing drains.
5. Identify abiotic and biotic components, at least two food chains, and examples of each category of ecosystem service; ask the guide (with consent) about threats and management.

Observations

Table A.11: Field observation sheet (example headings)

Stop	Habitat	Species or signs seen	Human impacts	Photo number
1				

Table A.12: Ecosystem services observed at the site (example)

Category	Example observed	Who benefits
Provisioning		
Regulating		
Cultural		

Category	Example observed	Who benefits
Supporting		

Analysis and Results

Count species recorded in each group (trees, shrubs, herbs, birds, insects, others) and compare with the checklist, suggesting why some were missed (season, time of day, observer skill). Draw two food chains and one simple food web, and rank the threats you saw by severity.

Precautions

- Stay on paths; do not feed animals, pluck plants or play loud music, and carry back all litter.
- In wetlands, stay well away from the water's edge and deep mud.

Viva / Reflection Questions

1. Which abiotic factor appeared to limit plant growth most at your site?
2. Name one invasive species you saw and explain its effect on native species.
3. How does a biodiversity park differ from an ordinary city park?

A.3.4 Developing a Working Model of an Ecosystem

Aim

To build a small, functioning model ecosystem (a bottle garden, aquarium or pond model) and observe interactions among its components over several weeks.

Background

A **closed or semi-closed model ecosystem** shows in miniature the processes of a real one: producers capture light energy, consumers and decomposers recycle nutrients, and water cycles by evaporation and condensation.

Materials and Data Sources

- Option A (terrarium): large transparent jar or bottle (5–20 L), gravel, charcoal pieces, garden soil, small moisture-loving plants (moss, small ferns, money plant cuttings), a few earthworms or woodlice
- Option B (aquarium): glass jar or tank (10–20 L), pond water and sediment, aquatic plants (*Hydrilla*, *Vallisneria*, duckweed), snails
- pH paper, notebook, phone camera

Procedure

1. For a terrarium, add 2–3 cm gravel, a thin layer of charcoal (prevents odour) and 5–8 cm soil; plant, moisten without waterlogging, add a few soil organisms and close the lid.
2. For an aquarium, place 3 cm of pond sediment, fill with pond water, add rooted and floating plants, and add snails after three days.
3. Place the model in indirect sunlight near a window; direct sun overheats it.

4. Observe weekly for six weeks: condensation, plant growth, water clarity and colour, algae, animal activity and deaths.
5. Keep a second, identical model in a dark cupboard to test the role of light.

Observations

Table A.13: Weekly observations of the model ecosystem (example headings)

Week	Condensation (yes/no)	Plant height (cm)	Water clarity / pH	Animals seen
1				
2				

Analysis and Results

Plot plant height against time for the light and dark models. Explain condensation as part of the water cycle and describe carbon and oxygen exchange between plants and animals. If the model failed (plants rotting, foul smell, animals dying), identify the likely cause, such as overcrowding, excess organic matter, lack of light or oxygen depletion, and relate it to **eutrophication** in real lakes.

Precautions

- Never release aquarium animals or plants into natural water bodies; exotic species can become invasive.
- Do not seal animals in a container without enough plants; check daily during the first week.

Viva / Reflection Questions

1. Why can a sealed bottle garden survive without being watered?
2. Which organisms in your model are decomposers, and what would happen without them?
3. In what ways is your model different from a real pond ecosystem?

A.3.5 Worksheets on the Structure and Function of Ecosystems

Aim

To identify and compare the structural components and functional processes of different ecosystem types using guided worksheets, pictures and field or video observations.

Background

Every ecosystem has **structure** (abiotic components, and biotic components arranged into trophic levels) and **function** (energy flow, nutrient cycling, productivity, decomposition, regulation). Comparing ecosystem types reveals the common pattern beneath the variety of organisms.

Materials and Data Sources

- Worksheet (table below) printed for each student
- Chapter 2 of this book; MoEFCC's EIACP (formerly ENVIS) resources on Indian ecosystems: https://www.envis.nic.in

Procedure

1. Assign one ecosystem type to each of six groups: forest (Western Ghats evergreen forest), grassland (Banni, Kachchh), desert (Thar), pond or lake (a local pond), river (Yamuna) and estuary or mangrove (Sundarbans).
2. Each group fills in the structure-and-function worksheet using the chapter, reliable websites and videos.
3. Each group draws a food web of at least eight organisms with labelled trophic levels, and identifies two threats and one conservation response specific to India.
4. Groups present for five minutes each while others fill in a summary comparison table.

Observations

Table A.14: Worksheet on ecosystem structure and function (example for a pond)

Feature	Example entries for a pond
Abiotic components	Water, dissolved O_2 and CO_2, sunlight, sediment, nutrients
Producers	Phytoplankton, *Hydrilla*, lotus
Consumers	Zooplankton, snails, fish, kingfisher
Decomposers	Bacteria and fungi in sediment
Main energy source	Sunlight captured by phytoplankton
Threats in India	Sewage inflow, encroachment, water hyacinth

Analysis and Results

Rank the six ecosystems by expected primary productivity (high in estuaries and evergreen forests, low in deserts) and explain the ranking using limiting factors such as water, temperature and nutrients. Identify one regulating and one cultural service of each ecosystem.

Viva / Reflection Questions

1. What limits productivity in a desert and in the open ocean?
2. Why are estuaries among the most productive ecosystems?
3. Name the producer and the top consumer in a grassland food chain from India.

A.4 Unit 3: Natural Resources

A.4.1 Visit to a Paper Recycling, Rainwater Harvesting, Solar or Biogas Unit on Campus

Aim

To study how a resource-saving facility on campus works, measure its output, and estimate the resources and emissions it saves.

Background

Many Delhi colleges now operate **rooftop solar power plants**, **rainwater harvesting (RWH)** structures (mandatory under Delhi's building bye-laws for plots of 100 m^2 and above), small **paper**

recycling units, and **biogas plants** or composters fed with canteen waste. Each converts a "waste" or an untapped flow (sunlight, rain, paper, food scraps) into a useful resource.

Materials and Data Sources

- Permission from the college administration; a staff member who operates the unit
- Solar: inverter or net-meter readings; MNRE rooftop solar information: https://solarrooftop.gov.in (PM Surya Ghar portal: https://pmsuryaghar.gov.in)
- RWH: rainfall data from IMD (https://mausam.imd.gov.in); CGWB guidance on rainwater harvesting: https://cgwb.gov.in
- Grid emission factor: CEA CO_2 Baseline Database for the Indian Power Sector: https://cea.nic.in

Procedure

1. Arrange the visit through the teacher and fix a time with the operator.
2. Draw a labelled flow diagram of the unit: inputs, process steps, outputs, wastes.
3. Record technical details. Solar: capacity (kWp), panels, tilt, daily and cumulative generation (kWh). RWH: catchment area (m^2), number of recharge pits or tanks, filter type. Paper unit: waste paper used (kg/day) and products made. Biogas: digester volume (m^3), daily feed (kg), hours of burner use.
4. Ask the operator (with consent) about costs, problems and benefits; photograph each component.

Observations

Table A.15: Technical data of the campus unit (example headings)

Parameter	Value	Unit	Source
Installed capacity / catchment area		kWp / m^2	Nameplate / measured
Daily output		kWh / kg / m^3	Display / records
Annual output (actual or estimated)			Records

Analysis and Results

Use the formula appropriate to your unit.

- **Solar**: annual generation (kWh) ≈ capacity (kWp) × average daily yield × 365. In Delhi a well-maintained system yields roughly 4 kWh per kWp per day, so a 100 kWp plant gives about 100 × 4 × 365 = 1,46,000 kWh (146 MWh) a year. With a grid factor of about 0.7 t CO_2/MWh (check the latest CEA value), avoided emissions ≈ 146 × 0.7 ≈ 102 t CO_2 a year.
- **Rainwater harvesting**: potential harvest (m^3) = catchment area (m^2) × annual rainfall (m) × runoff coefficient. For a 2,000 m^2 concrete roof (coefficient about 0.8) and about 0.75 m of rain, harvest ≈ 2,000 × 0.75 × 0.8 = 1,200 m^3 (12 lakh litres) a year.
- **Biogas**: food waste typically yields of the order of 0.05–0.1 m^3 of biogas per kg, so 100 kg/day gives roughly 5–10 m^3/day. Take 1 m^3 of biogas (about 60% methane) as roughly equivalent to 0.4–0.45 kg of LPG.

Calculate the simple **payback period** = capital cost ÷ annual saving (₹ per year).

Precautions

- Do not touch panels, cables or inverters; do not open recharge-pit covers or enter digesters.
- No naked flames near biogas pipes.

Viva / Reflection Questions

1. Why does a solar panel produce less electricity in December than in May in Delhi, even on clear days?
2. What is the purpose of the filter chamber in a recharge pit?
3. What happens to the slurry from a biogas plant?

A.4.2 Working Model of Renewable or Non-Renewable Energy Sources

Aim

To build a working model that demonstrates how a renewable or non-renewable energy source is converted into useful energy, and to explain its advantages and limitations.

Background

Energy resources are **non-renewable** (coal, petroleum, natural gas, nuclear fuels) or **renewable** (solar, wind, hydro, biomass, geothermal, tidal). Their energy content differs widely: Indian non-coking coal typically has a gross calorific value of about 15–20 MJ/kg, petroleum products about 42–46 MJ/kg, natural gas about 50 MJ/kg (roughly 35–40 MJ per cubic metre), and biogas about 20–24 MJ per cubic metre. A model makes the conversion chain (primary source → mechanical or thermal energy → electricity) visible.

Table A.16: Suggested working models and the conversions they show

Model	Energy conversion	Key materials
Solar-powered fan or charger	Light → electricity → motion	Small PV panel (6–12 V), DC motor, LED
Wind turbine	Wind (kinetic) → mechanical → electricity	PVC-pipe blades, DC motor as generator, table fan
Mini hydro	Potential energy of water → electricity	Spoon wheel, DC motor, bucket
Bottle biogas digester	Chemical (biomass) → methane	5 L bottle, cow dung, kitchen waste, balloon
Solar cooker	Light → heat	Cardboard box, foil, black vessel, glass sheet

Materials and Data Sources

- Materials listed in the table above; multimeter (borrowed from the physics department)
- MNRE information on renewable energy: https://mnre.gov.in
- CEA installed-capacity reports: https://cea.nic.in

Procedure

1. Choose a model and draw a design with dimensions; collect materials, preferably reused ones.
2. Assemble the model under the teacher's supervision.

3. Test it. Measure voltage (V) and current (A) with the multimeter and calculate power ($P = V \times I$). For the solar cooker, measure water temperature every 10 minutes; for the biogas digester, measure the balloon's circumference daily.
4. Vary one condition (light intensity, fan distance, water height, feed amount) and record the effect.
5. Prepare a chart explaining the conversion, India's installed capacity of the source (CEA) and its pros and cons.

Observations

Table A.17: Performance of the model under changing conditions (example for a wind model)

Fan distance (cm)	Voltage (V)	Current (mA)	Power (mW)
30			
60			

Analysis and Results

Plot output power against the variable changed. For a wind model, power falls sharply with distance because wind power varies with the cube of wind speed ($P \propto v^3$). For a solar cooker, heat gained by water = mass (kg) × 4.18 kJ/kg°C × temperature rise; compare it with the solar energy falling on the opening (about 0.6–1 kW/m^2 at noon in Delhi) to estimate efficiency.

Precautions

- Never ignite a bottle digester indoors, and do not connect models to mains electricity.

Viva / Reflection Questions

1. Trace the energy conversions in your model from the original source to the final use.
2. Why do solar and wind power need storage or backup?
3. Compare the energy content of coal and natural gas per kilogram. Which emits less CO_2 per unit energy, and why?

A.4.3 Mapping the Natural Resources of a Study Area Using Google Earth

Aim

To map land, water, forest and mineral resources of a chosen study area and detect changes in them over time using Google Earth's historical imagery.

Background

Land use/land cover (LULC) maps show how land is shared between forests, farms, water bodies, built-up areas and wasteland. Comparing images from different years reveals **land-use change**, such as shrinking farmland around Delhi, lost water bodies, or mining scars in the Aravallis. Google Earth Pro and ISRO's Bhuvan make such mapping possible without specialised software.

Materials and Data Sources

- Google Earth Pro with historical imagery (desktop version)

- Bhuvan LULC layers and thematic services: https://bhuvan.nrsc.gov.in
- India-WRIS for rivers, reservoirs and groundwater: https://indiawris.gov.in
- Global Forest Watch for tree-cover loss: https://www.globalforestwatch.org

Procedure

1. Define a study area of about 25–100 km^2 (for example, the Yamuna floodplain from Wazirabad to Okhla, or the Aravalli belt towards Gurugram) and outline it as a polygon.
2. Create folders for each resource class: water bodies, forest or tree cover, agricultural land, open scrub, mining or quarry sites, built-up area.
3. Digitise each feature as a polygon or line and record polygon areas from the Measurements tab.
4. Repeat the mapping for an image at least ten years older, from the same season and cloud-free.
5. Cross-check classes with Bhuvan's LULC layer and field knowledge.
6. Export the map as an image with legend, north arrow and scale, and the polygons as a KML file.

Observations

Table A.18: Area of resource classes in two years (example headings)

Resource class	Area in earlier year (ha)	Area in recent year (ha)	Change (ha)	Change (%)
Water bodies				
Forest / tree cover				
Agricultural land				
Built-up				

Analysis and Results

Percentage change = (recent area − earlier area) ÷ earlier area × 100. Present the two years as side-by-side maps and a grouped bar chart. Discuss the drivers of each change (urban expansion, mining, irrigation, restoration) and its consequences for groundwater recharge, biodiversity and flood risk.

Viva / Reflection Questions

1. Why should images from the same season be compared?
2. Which resource class changed most in your study area, and why?
3. How could a loss of water bodies affect groundwater in Delhi?

A.4.4 Time-Series Analysis of a Country's Natural-Resource Consumption

Aim

To analyse long-term trends in a country's consumption of one or more natural resources using public data, and to test whether resource use is being decoupled from economic growth.

Background

Growing economies use more energy, water, minerals and biomass; sustainability depends on **resource efficiency**, or less resource per unit of GDP. When GDP grows while resource use stays flat or falls, the economy shows **absolute decoupling**; when resource use grows more slowly than GDP, it shows **relative decoupling**. India's energy and material use have grown rapidly since 1990, but its energy use per unit of GDP has fallen.

Materials and Data Sources

- Our World in Data (energy, CO_2, land use, fertiliser): https://ourworldindata.org
- Energy Institute Statistical Review of World Energy: https://www.energyinst.org/statistical-review
- UNEP International Resource Panel Global Material Flows Database: https://www.resourcepanel.org/global-material-flows-database
- World Bank World Development Indicators (GDP, population): https://data.worldbank.org

Procedure

1. Choose a country (India by default) and one or two resources: primary energy, coal, crude oil, material footprint, freshwater withdrawal, or forest products.
2. Download annual data for at least 20 years, with population and GDP (constant prices) for the same years.
3. Calculate per-capita use (resource ÷ population) and resource intensity (resource ÷ GDP).
4. Plot total use, per-capita use and intensity, and all three as an index with the first year = 100.
5. Compare India with one developed and one developing country.

Observations

Table A.19: Resource-use data sheet (example headings)

Year	Resource use (unit)	Population (million)	GDP (constant US$ billion)	Per-capita use
2000				
2020				

Analysis and Results

An **index** expresses each year's value relative to a base year:

$$\text{Index (year t)} = \text{Value (year t)} \div \text{Value (base year)} \times 100$$

If GDP's index rises to 300 while the energy index rises to 200, energy intensity has fallen by a third (200 ÷ 300 = 0.67) and the economy shows relative decoupling. Also compute the CAGR of total use (formula in the Unit 1 SDG activity). Discuss whether India's per-capita use is converging towards that of richer countries.

Viva / Reflection Questions

1. Distinguish between absolute and relative decoupling with an example.

2. Why is per-capita consumption a fairer basis for international comparison than total consumption?
3. What might explain a sudden dip in your time series (for example, in 2020)?

A.4.5 Comparing a State's Energy Demand and Consumption over the Years

Aim

To compare the electricity demand and consumption of an Indian state over at least ten years using official data and graphs, and to explain the trends.

Background

Electricity demand in India has grown steadily with urbanisation, air-conditioning, irrigation pumping and industry,. Delhi's demand is dominated by households and commerce and peaks on summer afternoons; it reached 8,656 MW on 19 June 2024 (Delhi SLDC). Comparing **energy requirement** with **energy supplied** reveals shortages.

Materials and Data Sources

- Central Electricity Authority (CEA): Power Supply Position reports, Load Generation Balance Report, and All India Electricity Statistics (General Review): https://cea.nic.in
- NITI Aayog India Climate and Energy Dashboard: https://iced.niti.gov.in
- MoSPI Energy Statistics India: https://www.mospi.gov.in
- Delhi State Load Despatch Centre (daily peak demand): https://www.delhisldc.org

Procedure

1. Choose a state (Delhi or your home state) and a second state for contrast.
2. From CEA Power Supply Position reports, collect annual energy requirement (MU, million kWh), energy supplied (MU), peak demand (MW) and peak met (MW) for at least ten years.
3. From the General Review or ICED, collect sector-wise consumption for at least two years.
4. Collect state population to calculate per-capita consumption.
5. Plot energy requirement and supply over time, peak demand over time, sectoral consumption, and per-capita consumption of the two states.

Observations

Table A.20: Annual electricity data for the chosen state (example headings)

Year	Energy requirement (MU)	Energy supplied (MU)	Shortage (%)	Peak demand (MW)
2014–15				
2024–25				

Analysis and Results

Shortage (%) = (requirement − supplied) ÷ requirement × 100. Calculate the CAGR of requirement and of peak demand; a faster rise in peak than in total energy points to growing cooling loads. Per-

capita consumption (kWh) = consumption (MU) × 10^6 ÷ population. Explain differences in sectoral mix between the two states by economic structure and climate, and discuss how rooftop solar, efficient appliances (BEE star labels) and time-of-day tariffs could flatten the peak.

Viva / Reflection Questions

1. Why does Delhi's peak electricity demand occur in summer, whereas some hill states peak in winter?
2. What is the difference between energy (MU) and power (MW)?
3. Which sector's share grew most in your chosen state, and why?

A.4.6 Resource Consumption by a Dominant Local Industry and Status of the Supplying Area

Aim

To assess how much of a natural resource a dominant local industry consumes, trace where that resource comes from, and evaluate the condition of the supplying area.

Background

Every industry depends on a "resource shed", the area that supplies its raw materials. Around Delhi, **brick kilns** consume fertile topsoil and coal; the **construction industry** uses river sand and Aravalli stone; **packaged-water units** draw groundwater; and **thermal power stations** burn coal from Jharkhand, Odisha and Chhattisgarh. Linking consumption to its source shows that a city's environmental footprint extends far beyond its boundaries.

Materials and Data Sources

- Interviews (with consent) with industry managers, workers, traders or residents
- CGWB groundwater-level data and Dynamic Ground Water Resources assessment: https://cgwb.gov.in and India-WRIS: https://indiawris.gov.in
- Indian Bureau of Mines statistics: https://ibm.gov.in; Ministry of Coal statistics: https://coal.gov.in
- Google Earth Pro historical imagery; news reports and NGT orders (https://greentribunal.gov.in)

Procedure

1. Identify the dominant industry of your locality or a nearby area (for example, brick kilns around Narela or Jhajjar, stone crushing near Faridabad, a bottling plant, a thermal plant).
2. Estimate its annual consumption of the chosen resource. For a brick kiln, ask about bricks fired per season, soil per thousand bricks and coal per lakh bricks; for a bottling plant, use withdrawal permits or BRSR data.
3. Trace the supply chain to the village, district, mine or aquifer of origin.
4. Assess the supplying area using CGWB groundwater category, satellite imagery of pits and quarries, land-use change, and reports of conflict or court orders; visit if feasible.
5. Prepare a resource-flow diagram from source to industry to product to waste.

Observations

Table A.21: Resource-consumption profile of the industry (example for a brick kiln)

Item	Quantity per season	Unit	Source of resource	Data source
Bricks produced		lakh bricks	—	Owner interview
Clay / topsoil used		m^3 or t	Farmland in ___ village	Owner interview
Coal used		t	Coal from ___	Bills / interview

Analysis and Results

Calculate resource intensity (for example, tonnes of soil per lakh bricks) and scale it up using the number of similar units in the district. Express soil removal as farmland stripped: area (m^2) = volume (m^3) ÷ depth (m); 3,000 m^3 taken to a depth of 1 m strips 3,000 m^2, or 0.3 ha. Judge whether current use is sustainable and propose alternatives (fly-ash bricks, zig-zag kiln technology, manufactured sand, rainwater recharge).

Viva / Reflection Questions

1. What is a "resource shed"? Describe the resource shed of your chosen industry.
2. Why does removal of topsoil for bricks harm agriculture for many years?
3. What does "over-exploited" mean in the CGWB groundwater classification?
4. Who bears the environmental cost of the industry, and who gains the benefit?

A.5 Unit 4: Environmental Pollution and Control

A.5.1 Water Quality of a Location Using Rapid Pollution-Monitoring Kits

Aim

To measure key water-quality parameters at several points with a rapid field kit and judge the level of pollution.

Background

Rapid field kits, like those used under the Jal Jeevan Mission, give approximate values within minutes using colour strips, drop-count titrations and pocket meters. Sampling several points (for example, the Yamuna at Wazirabad and below the Najafgarh drain outfall) shows where pollution enters. **Dissolved oxygen (DO)** is the best single indicator of organic pollution: healthy rivers have more than 5 mg/L, while sewage-laden stretches may fall close to zero.

Materials and Data Sources

- Rapid water-testing kit (pH, total dissolved solids (TDS), DO, hardness, chloride, nitrate, fluoride, iron) and H_2S strip vials for faecal contamination
- Pocket TDS/EC meter, thermometer
- Clean 1 L bottles, a bottle on a rope for sampling from a ghat or bridge, gloves, labels

- CPCB water-quality data (National Water Quality Monitoring Programme) and DPCC Yamuna monitoring reports: https://cpcb.nic.in and https://www.dpcc.delhigovt.nic.in

Procedure

1. Select two to four points along a gradient, such as upstream and downstream of a drain, or a lake's inlet and centre.
2. At each point, record GPS location, time, weather, colour, odour (from a distance), floating matter and nearby activities.
3. Collect samples 20–30 cm below the surface. Fill the DO bottle without air bubbles and add fixing reagents on the spot; measure temperature, pH and TDS immediately.
4. For hardness, chloride and DO, add reagent drop by drop until the colour changes; convert drops to mg/L using the kit chart.
5. For nitrate, fluoride and iron, add reagent, wait the stated time, and match the colour against the comparator card.
6. Fill an H_2S strip vial to the mark, close it and keep it at 25–37 °C for 24–48 hours. Blackening indicates likely faecal contamination.
7. Repeat each test twice and record the mean; dispose of reagents as instructed and wash hands.

Observations

Table A.22: Water-quality results at sampling points (example headings)

Parameter (unit)	Point A	Point B	Point C	Standard / guideline
pH				6.5–8.5
TDS (mg/L)				500 (drinking, acceptable)
Dissolved oxygen (mg/L)				≥ 5 (bathing)
H_2S strip (black / no change)				No change

Analysis and Results

Plot each parameter by sampling point. Compare values with the CPCB **designated best-use classes** for surface water: Class A (drinking without conventional treatment but after disinfection: DO ≥ 6 mg/L, BOD ≤ 2 mg/L, pH 6.5–8.5), Class B (outdoor bathing: DO ≥ 5 mg/L, BOD ≤ 3 mg/L, pH 6.5–8.5), Class C (drinking after conventional treatment: DO ≥ 4 mg/L), Class D (wildlife and fisheries: DO ≥ 4 mg/L) and Class E (irrigation and industrial cooling). Assign each point to the highest class it satisfies, and explain low DO below drains by sewage decomposition (see Chapter 4). Kits cannot measure BOD.

Precautions

- Never sample from a slippery bank or by entering the water; use a rope-and-bottle from a safe platform.
- Wear gloves; treat all river and drain water as infectious; analyse samples within six hours.

Viva / Reflection Questions

1. Why does dissolved oxygen fall downstream of a sewage drain?

2. What does blackening in an H_2S strip test indicate, and why is it not a confirmed coliform count?
3. Why must there be no air bubble in a DO sample bottle?

A.5.2 Air Quality Index of a Location from Real-Time Data

Aim

To compute the Air Quality Index (AQI) of a location from real-time data and analyse its daily and seasonal variation.

Background

The **National Air Quality Index**, launched by CPCB in 2014, converts concentrations of up to eight pollutants (PM_{10}, $PM_{2.5}$, NO_2, SO_2, CO, O_3, NH_3 and Pb) into a number from 0 to 500 in six categories: Good (0–50), Satisfactory (51–100), Moderate (101–200), Poor (201–300), Very Poor (301–400) and Severe (401–500). A **sub-index** is calculated for each pollutant, and the AQI is the highest sub-index; that pollutant is the **prominent pollutant**. At least three pollutants, including $PM_{2.5}$ or PM_{10}, are required.

Materials and Data Sources

- CPCB Central Control Room for Air Quality Management (real-time station data): https://airquality.cpcb.gov.in/ccr
- CPCB National AQI bulletin and SAMEER mobile app
- SAFAR (System of Air Quality and Weather Forecasting and Research), IITM Pune: https://safar.tropmet.res.in

Table A.23: CPCB breakpoints for selected pollutants (concentrations in µg/m³; 24-hour averages for PM and NO_2, 8-hour for O_3)

AQI category (range)	PM_{10}	$PM_{2.5}$	NO_2	O_3
Good (0–50)	0–50	0–30	0–40	0–50
Satisfactory (51–100)	51–100	31–60	41–80	51–100
Moderate (101–200)	101–250	61–90	81–180	101–168
Poor (201–300)	251–350	91–120	181–280	169–208
Very Poor (301–400)	351–430	121–250	281–400	209–748
Severe (401–500)	430+	250+	400+	748+

Procedure

1. Choose the station nearest your college (for example, North Campus DU, ITO or Anand Vihar).
2. On the CCR portal, download hourly data for $PM_{2.5}$, PM_{10}, NO_2, O_3, SO_2 and CO for one week in October–November and one week in July–August.
3. Calculate 24-hour averages for $PM_{2.5}$, PM_{10}, NO_2 and SO_2, and the maximum 8-hour average for O_3 and CO.
4. Calculate each pollutant's sub-index with the formula below, and the AQI for each day.

5. Check your AQI against the CPCB daily bulletin, and plot the hourly $PM_{2.5}$ curve for one day and the daily AQI for both weeks.

Observations

Table A.24: Daily average concentrations and sub-indices (example headings)

Date	$PM_{2.5}$ (µg/m³)	PM_{10} (µg/m³)	NO_2 (µg/m³)	AQI and prominent pollutant

Analysis and Results

The sub-index I_p for a pollutant with concentration C_p is found by linear interpolation within the breakpoint range that contains it:

$$I_p = [(I_HI - I_LO) \div (B_HI - B_LO)] \times (C_p - B_LO) + I_LO$$

where B_LO and B_HI are the concentration breakpoints of the range and I_LO and I_HI the corresponding AQI values.

Worked example (a November day): $PM_{2.5}$ = 180 µg/m³, PM_{10} = 290 µg/m³, NO_2 = 85 µg/m³.

- $PM_{2.5}$ lies in 121–250 (AQI 301–400): I = (99 ÷ 129) × (180 − 121) + 301 = 0.767 × 59 + 301 ≈ 346.
- PM_{10} lies in 251–350 (AQI 201–300): I = (99 ÷ 99) × (290 − 251) + 201 = 240.
- NO_2 lies in 81–180 (AQI 101–200): I = (99 ÷ 99) × (85 − 81) + 101 = 105.

AQI = maximum sub-index = 346, category Very Poor, prominent pollutant $PM_{2.5}$.

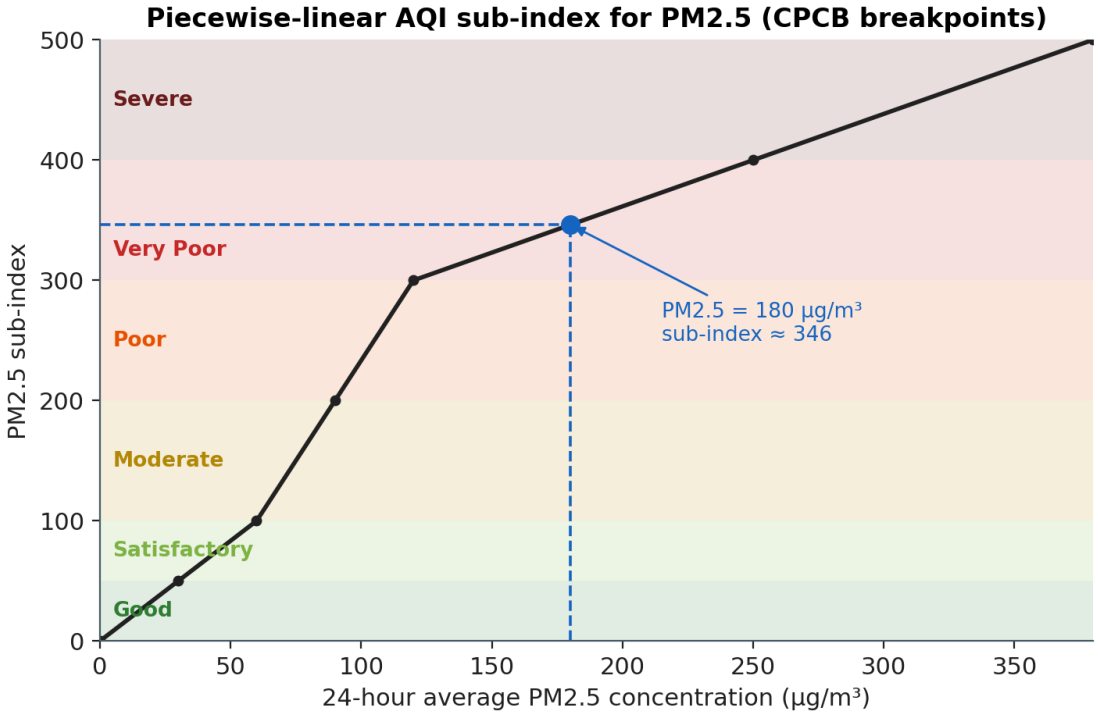


Figure A.2: *Piecewise-linear relationship between 24-hour PM2.5 concentration and its AQI sub-index, with the worked example of 180 µg/m³ giving a sub-index of about 346. Source: Plotted from CPCB National Air Quality Index breakpoints (2014).*

Explain differences between the two weeks by meteorology (low mixing height and calm winds in winter; rain washout in monsoon) and sources (stubble burning, vehicles, dust, firecrackers), and note which Graded Response Action Plan (GRAP) stage would have applied.

Viva / Reflection Questions

1. Why is the AQI the maximum sub-index rather than the average of sub-indices?
2. Why are $PM_{2.5}$ levels in Delhi much higher in November than in August?
3. Why is ozone often higher on sunny afternoons?

A.5.3 Monthly Solid Waste Generated in a Home or College

Aim

To measure the quantity and composition of solid waste from a household or college, calculate per-capita generation, and suggest reductions.

Background

Indian cities generate roughly 0.3–0.6 kg of municipal solid waste per person per day, about half of it wet, biodegradable waste. The **Solid Waste Management Rules, 2026**, effective from 1 April 2026, replace the 2016 Rules and require segregation at source into four streams: wet, dry, sanitary and special-care (domestic hazardous) waste. Institutions generating more than 100 kg/day are **bulk waste generators**, with extra duties to process wet waste on site.

Materials and Data Sources

- Digital luggage scale (0–50 kg), labelled bins or bags, gloves, plastic sheet for sorting
- MoHUA Swachh Bharat Mission (Urban): https://sbmurban.org; CPCB annual reports on implementation of SWM Rules: https://cpcb.nic.in

Procedure

1. Choose the unit: your household or one college building, canteen or hostel.
2. Prepare labelled containers: wet, dry recyclable (paper, plastics, metal, glass), dry non-recyclable (multilayer packets, thermocol), sanitary, special-care (batteries, bulbs, medicines, e-waste) and inert. Wear gloves when sorting.
3. Every day at the same time, weigh each container (minus its empty weight).
4. Continue for at least seven days (ideally 30).

Observations

Table A.25: Daily waste generation by category (kg) (example headings)

Day	Wet	Dry recyclable	Dry non-recyclable	Sanitary + special-care
1				
…				
Total				

Analysis and Results

$$\text{Per-capita generation (kg/person/day)} = \text{Total waste (kg)} \div (\text{Number of persons} \times \text{Number of days})$$

Example: a household of five generates 21.7 kg in 7 days. Per-capita generation = 21.7 ÷ (5 × 7) = 0.62 kg/person/day; monthly total ≈ 21.7 × 30 ÷ 7 ≈ 93 kg. If 12.4 kg was wet waste, the wet fraction is 57%. For a college, divide by the number of people present daily.

Draw a pie chart of composition and estimate how much could be diverted from landfill by composting and recycling.

Viva / Reflection Questions

1. What fraction of your waste was biodegradable, and how could it be managed at source?
2. Which items in your dry waste cannot be recycled, and why?
3. What are the four segregation streams under the Solid Waste Management Rules, 2026?

A.5.4 Making Compost or Vermicompost from Biodegradable Waste in College

Aim

To set up, maintain and monitor a compost pit or vermicompost bed on campus using canteen and garden waste.

Background

Composting is controlled aerobic decomposition of organic matter into stable, humus-like material; **vermicomposting** uses earthworms, usually *Eisenia fetida* or *Eudrilus eugeniae*, whose castings are rich in nutrients. A good mix balances "greens" (nitrogen-rich: food scraps, fresh grass) and "browns" (carbon-rich: dry leaves, straw, cardboard) at a C:N ratio of about 25–30:1, with 40–60% moisture. Composting campus leaf litter, often burnt in spring, turns a pollution source into a soil improver.

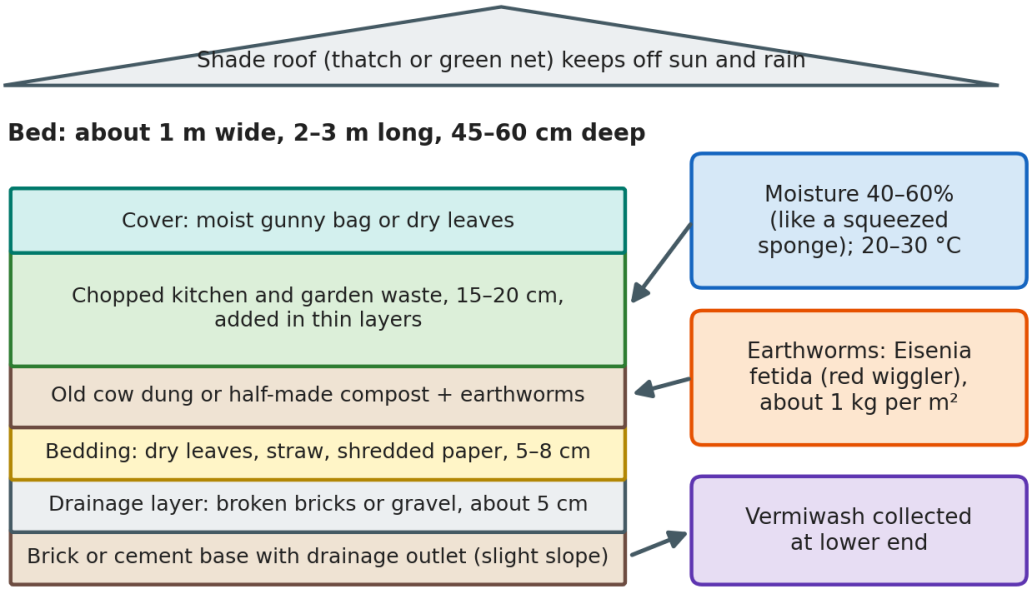


Figure A.3: *Cross-section of a vermicompost bed showing the drainage layer, bedding, worm inoculum, waste layers and cover, with key operating conditions. Source: Author's schematic.*

Materials and Data Sources

- A shaded site of about 3 m × 2 m, with permission from the estate office
- For vermicompost: bricks and cement or a ready-made HDPE vermibed (about 3.6 m × 1.2 m × 0.6 m); 1–2 kg of *Eisenia fetida* (from a nursery, agricultural university or Krishi Vigyan Kendra); old cow dung
- For aerobic compost: a pit (1 m × 1 m × 1 m), a bin with holes, or a pile
- Dry leaves, canteen waste, a 2–4 mm sieve, garden fork, watering can, thermometer, pH strips, spring balance

Procedure

1. **Set up the bed.** Build the vermibed on a slightly raised, sloping floor with an outlet at the lower end for vermiwash. Add a 5 cm drainage layer of broken bricks or gravel, then 5–8 cm of moistened bedding (dry leaves, straw, shredded paper).
2. **Pre-decompose the waste.** Mix chopped canteen and garden waste with dry leaves (about one part green to two parts brown by volume) and some cow dung; heap it for 10–15 days, turning twice. This stops the heat of fresh decomposition from killing worms.
3. **Introduce worms.** Spread a layer of old cow dung or half-made compost and release worms at about 1 kg per square metre of bed.
4. **Feed regularly.** Add pre-decomposed waste in thin layers (5–10 cm) two or three times a week. Do not add meat, fish, dairy, oil, large amounts of citrus or onion peel, plastics or glass.
5. **Maintain conditions.** Cover with a moist gunny bag, keep the material damp like a squeezed sponge, keep it shaded, and use wire mesh against rats and birds. Turn only the top layer weekly.
6. **Monitor weekly.** Record temperature, moisture (squeeze test), pH, odour, worm activity and waste added (kg).

7. **Harvest.** After 45–60 days the material is dark, crumbly and earthy-smelling. Stop feeding for a week, heap the bed under light (worms move down), scrape off the top compost and sieve it. Return worms to the bed.
8. **Collect vermiwash** from the outlet, dilute about 1:10 with water, and use as a foliar spray.
9. For a simple aerobic **compost pit**, layer greens and browns, keep moist, turn every 7–10 days, and harvest after 2–3 months.

Observations

Table A.26: Weekly monitoring log of the vermicompost bed (example headings)

Week	Waste added (kg)	Temperature (°C)	Moisture / odour	Worm activity
1				
…				

Analysis and Results

Calculate the **conversion ratio** = compost harvested (kg) ÷ total waste added (kg); 0.3–0.5 is typical, as much mass is lost as water vapour and CO_2. Estimate landfill diversion: if the bed handles 10 kg of canteen waste a day, it diverts 10 × 30 = 300 kg a month. Compare the compost's colour, smell and pH with the raw waste. Report problems and fixes: a foul smell signals waterlogging and anaerobic conditions; worms escaping signals heat, acidity or excess moisture.

Precautions

- Wear gloves; keep the bed covered to prevent flies and mosquito breeding.
- Protect the bed from direct sun and heavy rain; worms die above about 35 °C.

Viva / Reflection Questions

1. Why is waste pre-decomposed before being fed to earthworms?
2. What is the role of "browns" in a compost mix?
3. Why does a badly managed compost pit smell, and how would you fix it?
4. How does vermicomposting help a bulk waste generator meet its duties?

A.5.5 Suitability of Water Samples for Various Purposes Using Kits

Aim

To test water from different sources with a rapid kit and judge its suitability for drinking, bathing or irrigation against Indian standards.

Background

Drinking water must meet **IS 10500:2012** of the Bureau of Indian Standards, which gives an "acceptable limit" and a higher "permissible limit in the absence of an alternate source". Irrigation water is judged mainly by salinity (electrical conductivity, EC). Parts of south-west and north-west Delhi have saline or fluoride-rich groundwater.

Table A.27: Selected drinking-water limits under IS 10500:2012 (mg/L unless stated)

Parameter	Acceptable limit	Permissible limit (no alternate source)
pH	6.5–8.5	No relaxation
Total dissolved solids	500	2,000
Total hardness (as $CaCO_3$)	200	600
Chloride	250	1,000
Nitrate (as NO_3)	45	No relaxation
Fluoride	1.0	1.5
Iron	0.3	No relaxation
Total and faecal coliforms	Absent in 100 mL	Absent in 100 mL

Materials and Data Sources

- Rapid kit, H_2S strips and TDS/EC meter (as in the first activity of this unit)
- IS 10500:2012 (Bureau of Indian Standards; Jal Jeevan Mission portal: https://jaljeevanmission.gov.in)
- CGWB groundwater-quality reports for Delhi: https://cgwb.gov.in

Procedure

1. Collect five samples: municipal tap, groundwater (hand pump or borewell), RO-filtered water, harvested rainwater or packaged water, and a pond or lake. Label each with source, location, date and time.
2. Test each for pH, TDS, hardness, chloride, nitrate, fluoride and iron, and set up an H_2S strip test, following the kit instructions; repeat and average.
3. Measure EC with the meter, or estimate it from TDS: EC (µS/cm) ≈ TDS (mg/L) ÷ 0.64.
4. For an RO unit, measure the volume and TDS of reject water per litre purified.

Observations

Table A.28: Results for water sources (example headings)

Parameter	Tap	Groundwater	RO	Pond
pH				
TDS (mg/L)				
H_2S strip				

Analysis and Results

Mark each parameter as within the acceptable limit, within the permissible limit, or exceeding it; a sample is fit for drinking only if every parameter is within limits and the H_2S test is negative. For irrigation, classify salinity hazard by EC: below 750 µS/cm low to medium, 750–2,250 µS/cm high, above 2,250 µS/cm very high (the CPCB Class E limit is 2,250 µS/cm). Prepare a "suitability matrix" of sources against uses (drinking, bathing, washing, gardening, flushing).

Example: groundwater with TDS 1,450 mg/L and fluoride 1.3 mg/L exceeds the acceptable limits but lies within permissible limits; it may be drunk only if no alternative exists, and is better used for washing and flushing. RO water with TDS 40 mg/L is safe but very low in minerals, and the RO process may waste two to three litres for every litre purified.

Precautions

- Do not drink any sample; follow kit instructions on reagent disposal, as some reagents are corrosive.

Viva / Reflection Questions

1. Why does IS 10500 allow a permissible limit for TDS but no relaxation for nitrate?
2. What health problem is caused by long-term intake of water with fluoride above 1.5 mg/L?
3. Why is RO purification not recommended where TDS is already below 500 mg/L?

A.5.6 Water Audit of a College, House or Locality

Aim

To account for all water entering, used in and leaving a college, house or locality, identify losses, and recommend savings.

Background

A **water audit** is a systematic accounting of water based on a **water balance**: total input equals total use plus losses plus change in storage. Indian urban norms (CPHEEO) are 135 litres per capita per day (lpcd) for sewered cities, while the National Building Code suggests about 45 lpcd for day institutions such as colleges.

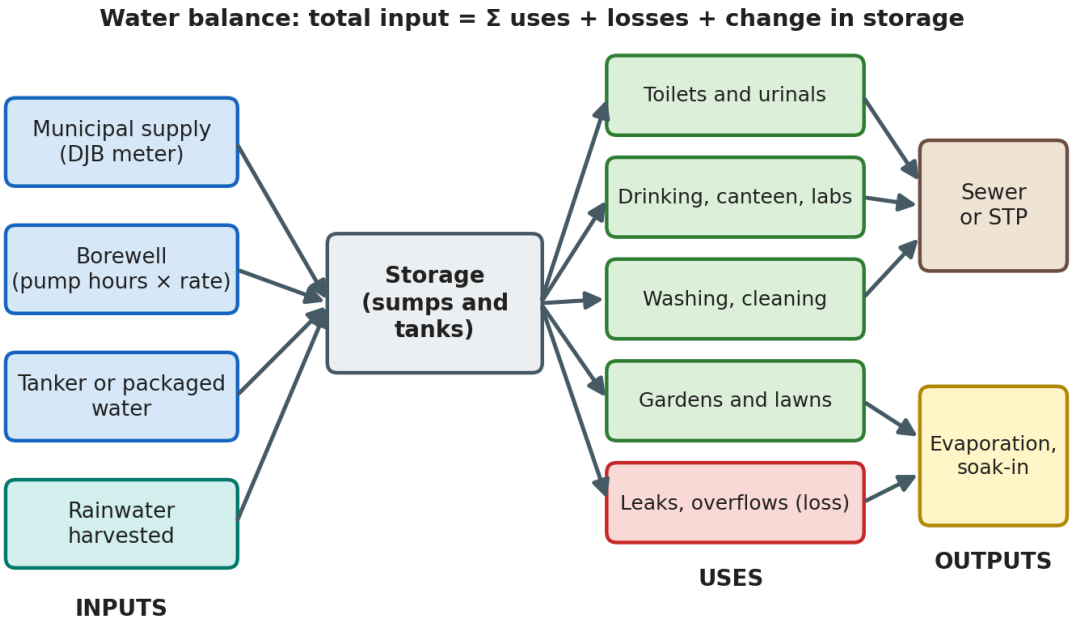


Figure A.4: *Flow diagram of a water audit showing inputs (municipal supply, borewell, tanker, harvested rainwater), storage, end uses including leaks, and outputs to sewer and evaporation. Source: Author's schematic.*

Materials and Data Sources

- Water bills and meter readings (Delhi Jal Board, https://delhijalboard.delhi.gov.in)
- Bucket of known volume (10–20 L), 1 L measuring jug, stopwatch (phone), measuring tape
- CPHEEO Manual on Water Supply and Treatment (MoHUA) for norms

Procedure

1. **Inventory.** List all sources and points of use (taps, toilets and cistern sizes, urinals, showers, coolers, RO units, kitchen, labs, garden taps). Sketch the supply network.
2. **Measure inputs.** Read the water meter at the same time on consecutive days (ideally for a week). For a borewell, multiply pump hours by discharge (litres per hour). Count tankers and their capacity.
3. **Measure flow rates.** Time how long sample taps and showers take to fill the 10 L bucket; flow rate (L/min) = volume ÷ time. Measure cistern volume (length × breadth × water depth).
4. **Count uses.** For a house, record daily uses per person (flushes, buckets per bath, washing-machine loads). For a college, count flushes and hand-washes at a toilet block for one busy hour and scale up using attendance.
5. **Detect leaks.** Collect drips from leaking taps in the jug for 10 minutes. Put food colour in a cistern; colour in the pan without flushing means a leak. For the night-flow test, read the meter after the last user leaves and before the first arrives; any flow indicates a leak.
6. **Check overflows** from overhead tanks during pumping.
7. **Calculate** the water balance and identify the largest uses and losses.
8. **Recommend** measures and estimate their savings.

Observations

Table A.29: Water-use estimate for a house of five (example)

End use	Unit volume	Uses per day	Daily volume (L)
Toilet flushing	9 L per flush	20 flushes	180
Bathing (bucket)	20 L per bath	5 baths	100
Washing machine	70 L per load	1 load	70
Kitchen and dishes	—	—	80
Drinking and cooking	4 L per person	5 persons	20
Mopping, cleaning	—	—	30
RO reject water	2 L per litre purified	15 L purified	30
Leaking tap (measured)	0.4 L per hour	24 hours	10
Total			520

Analysis and Results

Per-capita use = total daily use ÷ number of persons; in the example, 520 ÷ 5 = 104 lpcd, below the 135 lpcd norm. Compare the metered input with the sum of estimated uses:

$$\text{Unaccounted water (\%)} = (\text{Metered input} - \text{Sum of estimated uses}) \div \text{Metered input} \times 100$$

If the meter shows 580 L/day, unaccounted water = (580 − 520) ÷ 580 × 100 ≈ 10%, suggesting unmeasured leaks or overflow.

Rank end uses in a chart. Estimate savings from dual-flush cisterns (9 L to 3/6 L), tap aerators (often halving flow), fixing leaks, reusing RO reject water and rainwater harvesting, with cost and payback period for each.

Precautions

- Do not climb on overhead tanks or open sumps without staff supervision; take permission before entering toilets, kitchens or labs.

Viva / Reflection Questions

1. State the water-balance equation used in a water audit.
2. What is the night-flow test and what does it reveal?
3. Which end use in your audit offered the largest saving for the least cost?

A.5.7 Mapping Solid and Liquid Discharge of a College or Colony and Developing a Management Plan

Aim

To map where solid waste and wastewater are generated and discharged in a college or colony, and to prepare a management plan with a schematic map and photographs.

Background

Many Delhi colonies have overflowing community bins (dhalaos), open drains carrying sewage and garbage, and vacant plots used as dumps; wastewater reaches the Yamuna through drains such as Najafgarh and Shahdara. **Waste mapping** combines observation, simple measurement and community input to locate "hotspots" and design solutions.

Materials and Data Sources

- Printed Google Maps or Google Earth base map; phone camera with GPS; measuring tape
- Data from the waste-generation and water-audit activities of this unit

Procedure

1. Divide the area into zones (academic blocks, canteen, hostel, lawns; or colony lanes).
2. Walk each zone and mark on the base map: waste generation points, bins (segregated or mixed), dhalaos, illegal dumping and burning spots, drains and their direction of flow, manholes, septic tanks, outfalls, stagnant water, and existing composters or recycling points.
3. Photograph each point, note GPS coordinates, and give it a code (S1, S2… for solid; L1, L2… for liquid).
4. Estimate quantities: bin waste = bin volume × fill fraction × density (about 0.2–0.4 kg/L for mixed waste); drain flow = velocity (time a floating object over a measured distance) × wetted cross-section area.
5. Talk to sanitation workers or the estate office (with consent) about collection and problems.

6. Draw a clean schematic map with symbols and a legend, and draft the management plan.

Observations

Table A.30: Inventory of waste and discharge points (example headings)

Code	Type	Location / description	Estimated quantity	Problem observed
S1	Canteen bins	Behind canteen	3 bins/day	Mixed waste, stray dogs
S2	Open dumping	Corner of parking	—	Burning of leaves
L1	Drain	Along boundary wall	2 L/s	Kitchen greywater, plastic

Analysis and Results

Rank hotspots by severity using volume, health risk and frequency of complaints. Prepare the management plan as a table with five columns: problem, proposed action, responsibility, cost estimate and time frame. Typical actions: four-stream bins, a composting unit, a tie-up with an authorised recycler, e-waste drop boxes, drain screens, a canteen grease trap and greywater reuse. Present the plan with the map and "before" (and later "after") photographs.

Viva / Reflection Questions

1. What is a waste "hotspot" and how did you identify one?
2. How did you estimate the flow in a drain?
3. Why does mixed waste in drains increase flooding during the monsoon?

A.5.8 Repurposing Waste for Economic and Environmental Benefit: Short Video

Aim

To repurpose a waste item into a useful product, assess its benefits, and communicate it through a short video.

Background

Upcycling turns waste into products of equal or higher value, while **recycling** breaks materials down into raw material; both move towards a **circular economy**. Indian examples include products from temple flowers, artwork from scrap at Delhi's Waste to Wonder Park, and the informal economy of waste pickers and kabadiwalas.

Materials and Data Sources

- Waste items (PET bottles, old clothes, sacks, glass jars, newspapers), basic tools, phone camera and a free video editor
- Plastic Waste Management Rules and EPR information (CPCB): https://cpcb.nic.in

Procedure

1. Choose a waste stream common on campus or at home and estimate how much is generated (from your waste audit).

2. Select a safe, practical upcycling idea (for example, cloth bags from old clothes, planters from bottles, seed paper from waste paper, eco-bricks of clean dry plastic packed into PET bottles).
3. Make a prototype and record materials, time and cost.
4. Estimate the economic benefit (market price of a similar new product minus cost of making yours) and the mass of waste diverted per product.
5. Shoot a 2–3 minute video showing the problem (with data), the process, the product and its benefit, with English and Hindi subtitles.

Observations

Table A.31: Cost-benefit sheet for an upcycled product (example)

Item	Quantity	Value (₹)	Remarks
Waste used	2 old shirts	0	Diverted 0.4 kg textile
Other inputs	Thread, buttons	15	
Market price of similar bag		150	Local shop

Analysis and Results

Net benefit per unit = market value − cost of inputs (and labour, if valued). Scale up: if 200 students each make one bag and stop using 50 plastic carry bags a year, 10,000 bags are avoided annually. Discuss limitations, such as durability, and evaluate videos with the rubric at the end of this appendix.

Viva / Reflection Questions

1. Distinguish between upcycling, recycling and downcycling.
2. What is Extended Producer Responsibility (EPR) under the Plastic Waste Management Rules?
3. Why is the informal recycling sector important in Indian cities?

A.5.9 River–Society–Economy Nexus: Poster from Primary and Secondary Data

Aim

To study the links between a river, its dependent communities and the economy using primary and secondary data, and to present them as a poster.

Background

The Yamuna flows about 48 km through Delhi, from Palla to Okhla; the roughly 22 km stretch from Wazirabad to Okhla receives most of the river's pollution load. Farmers, fishers, washer communities, boatmen and the city's water supply all depend on it. The **nexus** view studies how river health, livelihoods and the economy shape one another.

Materials and Data Sources

- Primary data: observation and short interviews (with consent) at a ghat or floodplain site (for example, Wazirabad, Nigambodh Ghat or Kalindi Kunj), photographs
- CPCB and DPCC Yamuna water-quality data: https://cpcb.nic.in and https://www.dpcc.delhigovt.nic.in

- National Mission for Clean Ganga (Namami Gange): https://nmcg.nic.in

Procedure

1. Choose a river stretch (the Yamuna in Delhi or a river in your home district).
2. From secondary sources, compile upstream and downstream water quality (DO, BOD, faecal coliform), major drains, barrages (Wazirabad, Okhla) and government programmes.
3. Visit one site with the teacher. Record activities (farming, fishing, washing, bathing, rituals, boating, sand mining, dumping) and interview five to ten people about how they use the river and how it has changed.
4. Build a nexus diagram linking the river (quantity, quality, floodplain) to society (health, culture, livelihoods) and the economy (agriculture, industry, water supply, tourism).
5. Design an A1 or A2 poster with a title, a sketch map, three or four graphs, photographs, anonymised quotes, the nexus diagram and recommendations.

Observations

Table A.32: Activities observed and dependence on the river (example headings)

Activity	Number of people seen	Dependence (high/medium/low)	Change reported by respondents
Floodplain farming			
Fishing			
Washing clothes			

Analysis and Results

Graph water-quality parameters from upstream to downstream and show where the river fails Class B bathing criteria. Identify at least two **feedback loops**, for example: untreated sewage → low DO → fish kills → loss of fishers' income. Discuss trade-offs between floodplain development and flood protection.

Precautions

- Visit ghats in daylight with the teacher; stay away from the water's edge and sand-mining pits; ask before photographing people.

Viva / Reflection Questions

1. What is meant by the river–society–economy nexus? Give one example from your study.
2. Why is the Delhi stretch of the Yamuna more polluted than the stretch upstream of Wazirabad?
3. Who gains and who loses when a river's floodplain is developed?

A.6 Unit 5: Global Environmental Issues and Policies

A.6.1 Temperature and Precipitation Trends of a Study Area Using Online Data

Aim

To download long-term temperature and rainfall data for a study area, detect trends, and interpret them in the context of climate change.

Background

IMD reports that India's annual mean temperature has risen by about 0.7 °C since 1901, with 2024 the warmest year on record for the country. A simple **linear trend** fitted to 30–40 years of data shows whether a local area is warming and by how much.

Materials and Data Sources

- NASA POWER (daily and monthly gridded weather data from 1981 onwards, any location): https://power.larc.nasa.gov/data-access-viewer
- IMD Pune data portal (gridded rainfall and temperature): https://www.imdpune.gov.in
- India-WRIS rainfall data: https://indiawris.gov.in
- Spreadsheet software with charting and trend-line functions

Procedure

1. Choose a study area (your college or home district) and note its latitude and longitude.
2. In NASA POWER, select "Single Point", community "Agroclimatology", temporal average "Monthly and Annual", and parameters "Temperature at 2 m (T2M)", "Maximum temperature (T2M_MAX)" and "Precipitation corrected (PRECTOTCORR)". Download data from 1981 to the latest full year as a CSV file.
3. Calculate annual mean temperature, mean daily maximum temperature and total rainfall for each year.
4. Plot each series against year with a linear trend line, its equation and R^2.

Observations

Table A.33: Annual climate data for the study area (example headings)

Year	Mean temperature (°C)	Mean maximum temperature (°C)	Total rainfall (mm)
1981			
…			

Analysis and Results

The slope b of the least-squares trend line y = a + bx (x = year, y = temperature) is:

$$b = \Sigma(x - \bar{x})(y - \bar{y}) \div \Sigma(x - \bar{x})^2$$

A slope of 0.025 °C per year means warming of 0.25 °C per decade, or about 1.1 °C over 44 years. Report the decadal rate and the difference between the 1981–1990 and latest ten-year means. For

rainfall, compute the **coefficient of variation** (standard deviation ÷ mean × 100) and count deficit years (rainfall more than 20% below the mean). A low R^2 signals large natural variability.

Viva / Reflection Questions

1. Why is a 30-year period used to describe climate?
2. What does the slope of your trend line mean in plain words?
3. Why might a city station show more warming than the surrounding countryside?

A.6.2 Questionnaire Survey on the Impact of Climate Change on People

Aim

To design and conduct a questionnaire or online survey on how people perceive and experience the impacts of climate change, and to analyse the responses.

Background

Perceptions of climate impacts (heat, erratic rain, floods, water shortage, disease) shape public support for adaptation. A well-designed survey follows the steps in the figure below: a clear question, a defined population, carefully worded items, ethical safeguards, a pilot test and systematic analysis. The same method is used for the environmental-education survey in Unit 7.

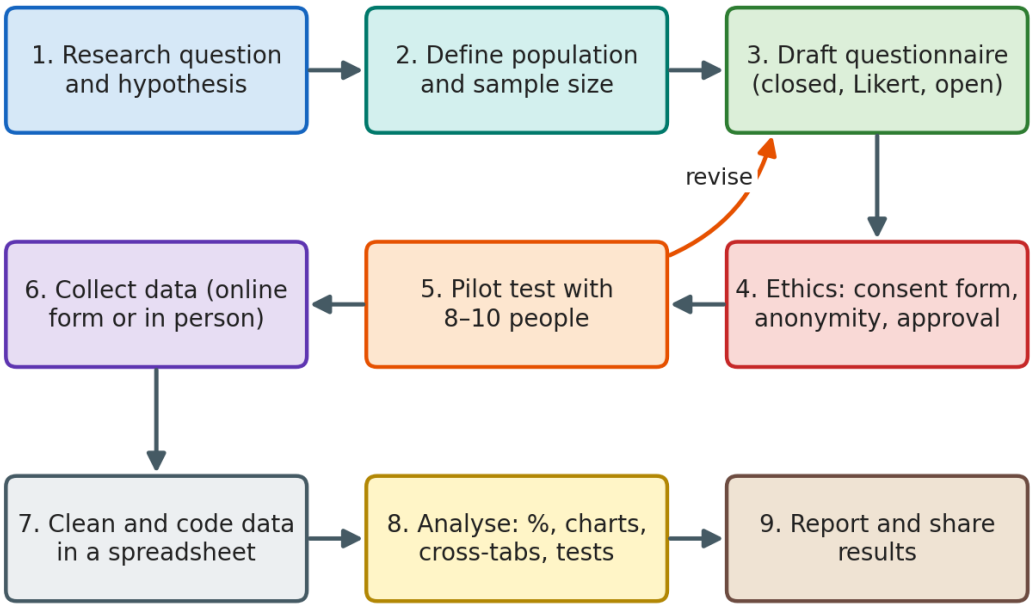


Figure A.5: *Steps in designing and conducting a questionnaire survey, including ethics approval and a pilot test that feeds back into revising the questionnaire. Source: Author's schematic.*

Materials and Data Sources

- Google Forms or Microsoft Forms (free), or printed questionnaires
- Consent statement (see the ethics guidance at the start of this appendix)
- Spreadsheet software; optional free statistics tools (PSPP, JASP)

Procedure

1. Frame a research question, for example: "How do residents of [locality] experience changes in summer heat, and how do they cope?"
2. Define the population (for example, households in a colony or street vendors) and a sampling method: random, systematic (every fifth house) or stratified.
3. Decide the sample size. For large populations, Cochran's formula gives $n = z^2p(1 - p) \div e^2$; with $z = 1.96$ (95% confidence), $p = 0.5$ and $e = 0.05$, $n \approx 384$. A class project may use 60–150 respondents and must state the resulting margin of error.
4. Draft 15–20 questions: profile, experience of impacts, perception (5-point Likert items from "strongly disagree" to "strongly agree") and coping. Avoid leading and double-barrelled questions; translate into the local language; add a consent section and collect no names.
5. Pilot with 8–10 people, revise, collect data, and code answers numerically in a spreadsheet.

Observations

Table A.34: Sample coding sheet for Likert items (example)

Respondent	Age group	Occupation	"Summers are hotter than 10 years ago" (1–5)	"Heat has reduced my income" (1–5)
R1	18–30	Vendor	5	4
R2	31–50	Office worker	4	2

Analysis and Results

Report percentages and mean Likert scores as bar charts. Use **cross-tabulation** to compare groups, for example outdoor versus indoor workers reporting income loss. The chi-square test ($\chi^2 = \Sigma(O - E)^2 \div E$, where O is observed and E expected frequency) shows whether a difference is statistically significant. Note that online surveys miss people without smartphones.

Viva / Reflection Questions

1. Give an example of a leading question and rewrite it neutrally.
2. Why is a pilot test necessary?
3. What is a Likert scale and how is it analysed?

A.6.3 Assessing Nationally Determined Contributions of Developed and Developing Countries

Aim

To compare the climate pledges (Nationally Determined Contributions, NDCs) of selected developed and developing countries under the Paris Agreement and evaluate their ambition and fairness.

Background

Under the Paris Agreement (2015), each country submits an **NDC**, updated every five years; targets may be absolute cuts, cuts in **emissions intensity** of GDP, cuts below "business as usual", or peaking dates. India's updated NDC (August 2022) commits to reducing the emissions intensity of GDP by 45%

from the 2005 level by 2030, achieving about 50% of installed electric power capacity from non-fossil sources by 2030, and creating an additional carbon sink of 2.5–3 billion tonnes of CO_2 equivalent through forest and tree cover; its long-term goal is net zero by 2070. A new round of NDCs with 2035 targets began in 2025.

Materials and Data Sources

- UNFCCC NDC Registry: https://unfccc.int/NDCREG
- Climate Watch (NDC explorer, emissions data): https://www.climatewatchdata.org
- Climate Action Tracker (independent ratings): https://climateactiontracker.org
- Our World in Data CO_2 and GHG data: https://ourworldindata.org/co2-and-greenhouse-gas-emissions

Procedure

1. Choose at least five parties, including India, China, the USA, the EU and a least-developed or small island state.
2. Record each party's latest NDC: target type, base and target year, coverage, and conditional components.
3. Add total, per-capita and cumulative emissions (Our World in Data) and Climate Action Tracker ratings.

Observations

Table A.35: Comparison of NDCs (example headings)

Party	Target type and value	Base / target year	Per-capita emissions (t CO_2e)	Conditional on finance?
India	Intensity −45%; 50% non-fossil capacity	2005 / 2030		Partly
United States				
European Union				
Bangladesh				

Analysis and Results

Rank parties by (a) ambition (percentage change in absolute emissions), (b) per-capita emissions in the target year, and (c) historical responsibility (cumulative emissions). Discuss **common but differentiated responsibilities and respective capabilities** (CBDR-RC), why intensity targets suit growing economies.

Viva / Reflection Questions

1. Why does India use an emissions-intensity target rather than an absolute cap?
2. What is the difference between conditional and unconditional NDC targets?
3. Which party in your table carries the greatest historical responsibility, and why does that matter?

A.6.4 Model UNFCCC Simulation for Negotiation Skills

Aim

To simulate a session of the Conference of the Parties (COP) to the UNFCCC, with students representing countries and negotiating groups, in order to understand climate diplomacy and build negotiation, research and public-speaking skills.

Background

COP decisions are taken by **consensus**: a decision passes only when no party formally objects. Countries negotiate through blocs such as the G77 and China, BASIC (Brazil, South Africa, India, China), the Alliance of Small Island States (AOSIS), the Least Developed Countries group, the European Union and the Umbrella Group (including the USA, Japan and Australia). At COP29 in Baku (2024), parties agreed a New Collective Quantified Goal of at least US$ 300 billion a year by 2035 for developing countries, with a call to scale up total finance to US$ 1.3 trillion a year.

Materials and Data Sources

- UNFCCC website (COP decisions, negotiating groups, submissions): https://unfccc.int
- Climate Watch country profiles: https://www.climatewatchdata.org
- Carbon Brief and Earth Negotiations Bulletin (IISD) summaries of COP outcomes: https://www.carbonbrief.org and https://enb.iisd.org
- Country placards, a gavel or bell, a projector for drafting text, timer

Table A.36: Roles in the Model UNFCCC

Role	Number	Responsibilities
COP President	1	Chairs plenary, rules on procedure, seeks consensus
Secretariat	2	Keeps speakers' list and time, records decisions, projects draft text
Co-facilitators	2	Run informal consultations on the draft decision
Country delegations	15–25 (1–2 students each)	Represent national positions; negotiate in blocs
Observers (NGOs, youth)	3–5	Make short interventions; lobby delegates

Procedure

1. **Allocation and research (two weeks before).** Assign countries so that all major blocs are represented (for example India, China, Brazil, Saudi Arabia, the EU, the USA, Japan, Tuvalu, Bangladesh, Kenya). Each delegation writes a one-page **position paper**: emissions profile, NDC, vulnerability, bloc, and its position on the agenda item with "red lines" and possible compromises.
2. **Opening plenary (30 min).** The President adopts the agenda; bloc spokespersons make 2-minute statements.
3. **Informal consultations (45–60 min).** Co-facilitators present a draft decision with bracketed options, for example "[developed countries shall] [all parties in a position to do so should]

provide finance". Delegates negotiate in blocs and then together, proposing amendments, followed by a moderated debate on contentious paragraphs.

4. **Closing plenary (30 min).** The President presents the final text and asks whether any party objects. If none, the decision is gavelled through; if a party objects, the President may suspend for a "huddle" and try once more.
5. **Debriefing (20 min).** Discuss who compromised and how the outcome compares with real COPs.

Rules of procedure (simplified from the UNFCCC draft rules applied in practice):

- Delegates speak only when recognised by the President and refer to other parties by country name.
- Speaking time is 2 minutes in plenary and 1 minute in debate.
- A **point of order** (procedural objection) may interrupt a speaker; a **point of information** (question to a speaker) is allowed only if the speaker accepts it.
- Decisions require consensus; the President judges whether consensus exists.
- Written amendments need the support of at least two parties.

Sample agenda item and draft decision text:

- Agenda item 4: Mobilising climate finance for adaptation and loss and damage in developing countries.
- Draft operative paragraphs: (1) "Urges developed country Parties to [double] [triple] adaptation finance from 2019 levels by [2027] [2030]"; (2) "Decides that contributions to the Fund for responding to Loss and Damage [shall] [may] come from [developed countries] [all Parties in a position to do so] [innovative sources including levies on fossil-fuel profits]"; (3) "Requests the Secretariat to prepare a report on progress towards the New Collective Quantified Goal for consideration at the next session."

Observations

Table A.37: Negotiation record sheet kept by each delegation (example headings)

Paragraph	Our position	Allies	Opponents	Final outcome
Op. 1 (adaptation finance)				
Op. 2 (loss and damage)				

Analysis and Results

Each delegation writes a one-page reflection on how far the outcome matched its position paper, which coalitions formed and which bracketed options survived, comparing it with the actual COP decision. Assess performance with the rubric below.

Table A.38: Assessment rubric for the Model UNFCCC (total 25 marks)

Criterion	Excellent (5)	Adequate (3)	Weak (1)
Research and position paper	Accurate data, clear red lines	Mostly accurate, general	Inaccurate or missing

Criterion	Excellent (5)	Adequate (3)	Weak (1)
Fidelity to country role	Consistently in character	Occasionally drifts	Personal views dominate
Speeches and interventions	Clear, timed, evidence-based	Clear but little evidence	Unclear or off-topic
Negotiation and coalition-building	Drafts amendments, builds consensus	Supports others' proposals	Passive
Reflection report	Insightful comparison with real COP	Descriptive	Missing

Viva / Reflection Questions

1. Why does consensus decision-making make climate agreements slow but durable?
2. Which bloc did your country belong to, and why?
3. What is "loss and damage", and why is it contentious?

A.6.5 Moot Court or Mock Trial in the National Green Tribunal

Aim

To conduct a simulated hearing before the National Green Tribunal (NGT) on an environmental dispute, in order to understand environmental law, legal reasoning and environmental justice.

Background

The **National Green Tribunal**, set up under the National Green Tribunal Act, 2010, hears cases involving a substantial question relating to the environment under the laws in Schedule I of the Act, such as the Water, Air and Environment (Protection) Acts (see Chapters 5 and 7). Its principal bench sits in New Delhi, and each bench has at least one **judicial member** and one **expert member**. Section 20 of the Act directs the Tribunal to apply the principles of **sustainable development**, **precaution** and **polluter pays**.

Materials and Data Sources

- National Green Tribunal Act, 2010 (India Code): https://www.indiacode.nic.in
- NGT website (judgments and cause lists): https://greentribunal.gov.in
- Divan and Rosencranz, *Environmental Law and Policy in India* (syllabus reading)

Table A.39: Roles in the NGT moot court

Role	Number	Responsibilities
Bench: Judicial Member (presiding) and Expert Member	2–3	Conduct hearing, question counsel, deliver reasoned order
Counsel for Applicants	2–3	Present the case for affected residents or an NGO
Counsel for Respondents	3–4	Defend the project proponent, the State Pollution Control Board and the government department
Expert witnesses	2	Give technical evidence (for example, a hydrologist and an air-quality scientist)

Role	Number	Responsibilities
Court master / registrar	1	Calls the case, keeps time and records
Observers	rest of class	Evaluate arguments using the rubric

Procedure

1. **Case brief (two weeks before).** The teacher distributes a fictional but realistic problem. Sample: *Residents' Welfare Association of Green Valley v. State of Delhi and others*. A developer has begun building a 20-storey housing complex on the edge of a seasonal wetland in the Yamuna floodplain. Residents allege that it lacks environmental clearance, that its dust violates air-quality norms, and that debris is dumped into the wetland. The developer claims that the land is not a notified wetland, that the project is below the clearance threshold, and that it provides affordable housing.
2. **Framing issues.** Is the site protected under the Wetlands (Conservation and Management) Rules 2017? Did it need prior clearance under the EIA Notification 2006? Have the Air and Water Acts been violated? Should precaution and polluter pays apply?
3. **Written memorials.** Each side submits a 4–6 page memorial: facts, issues, arguments citing statutes and landmark cases (for example, *Vellore Citizens' Welfare Forum v. Union of India* (1996) on the precautionary and polluter pays principles; the *M.C. Mehta v. Union of India* cases on Delhi's environment), and the relief sought.
4. **Hearing (60–75 min).** Applicants open (15 min), then respondents (15 min each for the developer and the State); the bench may question counsel at any time. Experts are examined and cross-examined (5 min each), and each side gives a 5-minute rebuttal.
5. **Order and debriefing.** The bench delivers a reasoned oral order with findings and directions (for example, a stay, a joint inspection committee, restoration, environmental compensation) referring to Section 20; the class compares it with real NGT orders on floodplain construction.

Rules of procedure:

- Address the bench as "the Hon'ble Tribunal"; stand while speaking.
- Only facts in the case brief may be used, though reasonable inferences are allowed.
- Every legal claim must be supported by a statute, rule or judgment.
- Counsel may object to misleading questions; the presiding member rules on objections and the court master enforces time limits.

Observations

Table A.40: Issue-wise record kept by observers (example headings)

Issue	Main argument of applicants	Main argument of respondents	Bench's finding
Wetland status			
Environmental clearance			
Air and water violations			

Analysis and Results

Each student writes a one-page case comment on the order, stating whether they agree, with reasons drawn from law and environmental science, and how environmental principles were balanced against housing needs. Assess performance with the rubric below.

Table A.41: Assessment rubric for the moot court (total 25 marks)

Criterion	Excellent (5)	Adequate (3)	Weak (1)
Memorial (research and structure)	All issues, correct citations	Most issues, some citations	Incomplete, no citations
Knowledge of law and principles	Accurate and applied to facts	Accurate but general	Errors
Use of scientific evidence	Data and expert evidence used well	Some evidence	None
Oral advocacy and court manners	Clear, confident, responsive to bench	Clear but scripted	Unclear, disrespectful
Rebuttal and reasoning	Addresses opponents' strongest points	Partial	Repetitive

Viva / Reflection Questions

1. Under what kinds of laws can a case be brought to the NGT?
2. Explain the precautionary principle with reference to your case.
3. What is environmental compensation, and how is it linked to the polluter pays principle?

A.6.6 Carbon Footprint of a College, Home or Locality

Aim

To estimate the annual greenhouse-gas emissions (carbon footprint) of a household, college or locality from activity data and emission factors, identify the largest sources, and propose reductions.

Background

A **carbon footprint** is the total greenhouse-gas emission caused directly and indirectly by a person, organisation or activity, expressed in tonnes of CO_2 equivalent (t CO_2e) (see Chapter 7). It is calculated as activity data × emission factor. Organisations group emissions into **Scope 1** (direct fuel burning on site, such as LPG and diesel generators), **Scope 2** (purchased electricity) and **Scope 3** (other indirect emissions, such as commuting, travel, purchased goods and waste).

Table A.42: Approximate emission factors for common activities in India

Activity	Unit	Emission factor (kg CO_2 per unit)	Source
Grid electricity	kWh	about 0.7 (use latest CEA value)	CEA CO_2 Baseline Database
LPG	kg	2.98	IPCC 2006 default
Petrol	litre	2.3	IPCC 2006 default
Diesel	litre	2.7	IPCC 2006 default
CNG	kg	about 2.7	IPCC 2006 default

Materials and Data Sources

- Electricity bills (12 months), LPG records, fuel bills; commuting survey (for a college)
- CEA CO_2 Baseline Database: https://cea.nic.in; IPCC Emission Factor Database: https://www.ipcc-nggip.iges.or.jp/EFDB
- GHG Protocol guidance: https://ghgprotocol.org

Procedure

1. Define the boundary (household, college building or colony block) and collect one year of activity data: electricity (kWh), LPG (cylinders × 14.2 kg for domestic or 19 kg for commercial), generator diesel (litres), vehicle fuel (litres), and, for a college, commuting mode and daily distance for a sample of students and staff.
2. Multiply each activity by its emission factor, add up emissions by scope and calculate per-capita values.
3. Identify the three largest sources and propose measures, estimating the emissions each would save.

Observations

Table A.43: Annual carbon footprint of a household of four (worked example)

Source	Annual activity	Emission factor	Emissions (t CO_2)
Electricity	3,000 kWh	0.7 kg/kWh	2.10
LPG	12 cylinders × 14.2 kg = 170 kg	2.98 kg/kg	0.51
Petrol (car and scooter)	480 L	2.3 kg/L	1.10
Total			3.71

Analysis and Results

In the example, the household's direct energy footprint is 3.71 t CO_2 a year, or about 0.93 t per person, of which electricity contributes 57%. A 3 kWp rooftop solar system (about 4,400 kWh a year in Delhi) could offset all grid electricity. Present a pie chart by source and compare per-capita results with India's national average (see Unit 7). Note that this estimate omits food, goods and waste (Scope 3), which can be large.

Viva / Reflection Questions

1. What are Scope 1, 2 and 3 emissions? Give a college example of each.
2. Why is the emission factor for electricity in India relatively high?
3. Which single measure would reduce your footprint most, and at what cost?

A.6.7 Status of SDGs in Your Neighbourhood and a Project Proposal

Aim

To assess the local status of at least three SDGs in a neighbourhood through observation and data, and to prepare and present a costed project proposal that addresses one identified problem.

Background

The SDGs are achieved locally: in a ward's water supply, a school's toilets, a market's waste. A neighbourhood checklist reveals concrete gaps, and a project proposal is the standard tool used by governments, NGOs and funders to plan action.

Materials and Data Sources

- NITI Aayog SDG India Index dashboard: https://sdgindiaindex.niti.gov.in
- Delhi Directorate of Economics and Statistics (Delhi Statistical Hand Book, Economic Survey of Delhi): https://des.delhi.gov.in
- National Family Health Survey (NFHS-5) district fact sheets: http://rchiips.org/nfhs
- Observation checklist, camera, short interviews with residents (with consent)

Procedure

1. Choose a neighbourhood of manageable size and at least three relevant SDGs, for example SDG 6 (water and sanitation), SDG 11 (sustainable cities) or SDG 12 (responsible consumption).
2. For each goal, choose two or three local indicators that can be observed or obtained: households with piped water, functional public toilets, segregated waste collection, green space, open drains, footpath condition, access to a dispensary.
3. Collect data through a transect walk, photographs, 10–20 short interviews and secondary data; rate each indicator good, fair or poor.
4. Select one problem and write a project proposal using the template below.
5. Present the assessment and proposal in 10 minutes, followed by 5 minutes of questions.

Observations

Table A.44: Neighbourhood SDG status sheet (example headings)

SDG	Local indicator	Data / observation	Status (good/fair/poor)
6	Functional public toilets		
11	Footpaths usable by pedestrians		

Analysis and Results

Summarise the status with a traffic-light chart and justify each rating with evidence. The project proposal should follow this structure:

1. **Title** and name of the proposing group.
2. **Problem statement**: the problem, whom it affects, how severe it is (with data and photographs), and the SDG target it relates to.

3. **Objectives**: two to four specific, measurable, achievable, relevant and time-bound (SMART) objectives, for example "to raise the share of households in Block C segregating wet waste from 20% to 60% within six months".
4. **Methodology**: activities, stakeholders (residents' welfare association, MCD, college NSS or eco-club), a month-by-month timeline, and how progress will be measured.
5. **Budget**: an itemised table (see example).
6. **Expected outcomes**: measurable results, monitoring indicators, risks and their management, and plans for continuing after the project ends.

Table A.45: Example budget for a waste-segregation project in one colony block (100 households)

Item	Quantity	Unit cost (₹)	Total (₹)
Pairs of household bins (green and blue)	100	250	25,000
Composting units (community, 100 L)	4	3,500	14,000
Printed awareness material	300	10	3,000
Volunteer travel and refreshments	6 sessions	1,000	6,000
Contingency (10%)	—	—	4,800
Total			52,800

Viva / Reflection Questions

1. Why are SMART objectives better than general aims?
2. Which SDG in your neighbourhood performed worst, and what evidence supports your judgement?
3. Who are the key stakeholders for your project, and what role will each play?

A.7 Unit 6: Biodiversity and Conservation

A.7.1 Exploring Open-Source Biodiversity Databases

Aim

To learn how to search, filter and download species records from open-source biodiversity databases, and to judge the quality of the data.

Background

Millions of species records from museum and herbarium collections, surveys and **citizen science** (members of the public uploading photographs or checklists) are freely available online. They underpin distribution studies, conservation planning and Red List assessments, but vary in reliability and reflect where people look as much as where species live.

Materials and Data Sources

- Global Biodiversity Information Facility (GBIF): https://www.gbif.org
- iNaturalist: https://www.inaturalist.org

- India Biodiversity Portal: https://indiabiodiversity.org
- eBird (India portal): https://ebird.org/india
- IUCN Red List of Threatened Species: https://www.iucnredlist.org

Procedure

1. Search GBIF, iNaturalist and eBird for the same species, for example the Indian peafowl (*Pavo cristatus*), the neem (*Azadirachta indica*) or the Indian flying fox (*Pteropus medius*).
2. Record the number of records, date range and main data sources (museum, citizen science).
3. On GBIF, filter records to India, then Delhi, and download an occurrence file (CSV); note the citation (DOI) GBIF generates.
4. In a spreadsheet, check quality: missing coordinates, duplicates, records far outside the known range, and the "basis of record" field.
5. Look up the species' IUCN Red List category, population trend and threats.

Observations

Table A.46: Comparison of databases for one species (example headings)

Database	Number of records (India)	Date range	Main source of records	Quality issues noted
GBIF				
eBird				

Analysis and Results

Calculate the percentage of records with usable coordinates and the percentage from citizen science. A sharp rise in records after 2010 usually reflects smartphone apps, not a real population increase. Discuss **sampling bias**: records cluster near cities, roads and protected areas.

Viva / Reflection Questions

1. What is citizen science? Give two Indian examples.
2. Why might a rise in records over time not mean a species is increasing?
3. What is a "research grade" observation on iNaturalist?

A.7.2 Locating Species in a Study Area

Aim

To record and map the locations of selected species in a study area, combining field observations with database records.

Background

Knowing where species occur shows which habitats matter and where threats overlap with populations. Field records with GPS coordinates, combined with database records, produce **occurrence maps** that reveal patterns, such as a bird found only near water or a tree confined to rocky Ridge slopes.

Materials and Data Sources

- Phone with GPS and iNaturalist or eBird app; Google Earth Pro; notebook
- GBIF and India Biodiversity Portal occurrence data (see the previous activity)
- Study area: the college campus, a biodiversity park or a Ridge forest patch

Procedure

1. Select five to ten target species that are easy to recognise (for example, neem, peepal, amaltas, babool, vilayati kikar; or common myna, house crow, rose-ringed parakeet, Indian grey hornbill, Oriental magpie-robin).
2. Walk a fixed route; for each sighting record GPS location, habitat (tree cover, lawn, built-up, water edge) and number of individuals. Upload observations to iNaturalist or eBird.
3. Download GBIF records for the same species within the study area.
4. Import your points and the GBIF points into Google Earth Pro (CSV with latitude and longitude columns), one colour per species.

Observations

Table A.47: Species location record (example headings)

Species	Latitude	Longitude	Habitat	Number seen
Indian grey hornbill (*Ocyceros birostris*)			Tall trees	

Analysis and Results

Count records of each species by habitat type and present a table of percentages. Identify species that are **habitat specialists** (found mostly in one habitat) and **generalists**. Note species you found that databases lacked, or vice versa.

Viva / Reflection Questions

1. Which species showed the strongest habitat preference, and why?
2. How accurate is a phone's GPS, and does that matter for your map?
3. How could your map help manage the study area?

A.7.3 Biodiversity Distribution Across Latitude and Altitude

Aim

To examine how species richness changes with latitude and with altitude, using open data from India and the world.

Background

The **latitudinal diversity gradient**, with species richness highest in the tropics and declining towards the poles, is explained by greater solar energy and productivity, long-term climatic stability, and more time for species to evolve. On mountains an **altitudinal gradient** often appears, though richness frequently peaks at middle elevations. India spans roughly 8° N to 37° N and from sea level to over 8,000 m, making it ideal for studying both.

Materials and Data Sources

- eBird regional and hotspot bar charts and species counts: https://ebird.org/india
- GBIF occurrence search with latitude and elevation filters: https://www.gbif.org
- Map of Life (species ranges): https://mol.org

Procedure

1. **Latitude.** In eBird, record the number of bird species and checklists for one state in each latitude band, from Kerala (about 8–13° N) through Maharashtra, Madhya Pradesh and Delhi to Ladakh (about 32–36° N).
2. **Altitude.** Choose a mountain transect, for example in Uttarakhand from Haridwar (about 300 m) through Dehradun (about 650 m), Mussoorie (about 2,000 m), Chopta (about 2,700 m) to Tungnath (about 3,600 m). Record bird species at an eBird hotspot at each point.
3. Tabulate the data and plot species richness against latitude and against altitude.

Observations

Table A.48: Species richness along an altitudinal transect (example headings)

Site	Elevation (m)	Number of species	Number of checklists
Haridwar	about 300		
Tungnath	about 3,600		

Analysis and Results

Plot richness against latitude and elevation and describe the trend. Because richness rises with sampling effort, calculate species per 100 checklists or compare only sites with similar effort. Discuss confounding factors: area, habitat diversity, rainfall (the Western Ghats and North-East are richer than their latitude alone predicts) and the mid-elevation peak.

Viva / Reflection Questions

1. State two hypotheses that explain the latitudinal diversity gradient.
2. Why might species richness peak at middle elevations?
3. How did you correct for differences in sampling effort?

A.7.4 Species Distribution Across Space and Time

Aim

To study how the geographical distribution of a species has changed over time using occurrence records, and to interpret the causes.

Background

Species ranges change: invasive plants expand, some birds move uphill as climates warm, and others contract as habitats are lost. The Great Indian Bustard once occurred across much of western and central India but is now confined mainly to parts of Rajasthan and Gujarat. Comparing occurrence

records from different decades can reveal such shifts, provided the bias in sampling effort is kept in mind.

Materials and Data Sources

- GBIF occurrence data with year filters: https://www.gbif.org
- eBird species maps and State of India's Birds reports: https://stateofindiasbirds.in
- IUCN Red List range maps: https://www.iucnredlist.org

Procedure

1. Choose a species with known change: an expanding invasive plant (*Lantana camara*, *Parthenium hysterophorus*), a declining bird (Great Indian Bustard, house sparrow in some cities), or a range-shifting bird.
2. Download GBIF records for India in two or three periods, for example before 2000, 2000–2014, and 2015 onwards.
3. Map each period in a different colour and count the 1° grid cells occupied in each period.
4. Read the species' Red List account or State of India's Birds assessment and note the reported trend.

Observations

Table A.49: Occupied grid cells by period (example headings)

Period	Number of records	Number of 1° grid cells occupied	Northernmost or highest record
Before 2000			
2015 onwards			

Analysis and Results

Compute the change in the number of occupied cells, and the shift in the mean latitude or elevation of records. Compare it with the published trend and explain any disagreement. Link the change to drivers such as habitat loss, power lines (a major cause of bustard deaths), climate warming or the spread of invasive species.

Viva / Reflection Questions

1. Why must records per period be interpreted with caution?
2. What drove the change in your chosen species' distribution?
3. How can distribution maps guide conservation action?

A.7.5 Quantifying Species Loss Across Time Periods

Aim

To quantify changes in species status and population over time using Red List assessments and population indices, and to relate them to the current extinction crisis.

Background

Biodiversity loss is measured by recorded extinctions, movement of species into higher threat categories on the **IUCN Red List** (Extinct, Extinct in the Wild, Critically Endangered, Endangered, Vulnerable, Near Threatened, Least Concern, plus Data Deficient), the **Red List Index**, and population indices such as the **Living Planet Index**, which showed an average 73% decline in monitored wildlife populations between 1970 and 2020 (WWF Living Planet Report 2024). In India, the State of India's Birds 2023 report found that of 338 species with long-term trend data, 204 (60%) had declined. India's three resident *Gyps* vulture species fell by more than 95% from the 1990s, mainly because of the veterinary drug diclofenac.

Materials and Data Sources

- IUCN Red List (advanced search by country and taxon; assessment history for each species): https://www.iucnredlist.org
- Living Planet Index: https://www.livingplanetindex.org
- State of India's Birds: https://stateofindiasbirds.in

Procedure

1. In the Red List, record the number of Indian species of one group (for example, mammals or birds) in each category.
2. For ten Indian species (for example, Great Indian Bustard, Indian vulture, Bengal florican, tiger, gharial), record the category in each assessment year.
3. Record trends by habitat group from the State of India's Birds report, and global and Asia-Pacific trends from the Living Planet Index.

Observations

Table A.50: Red List category history of selected Indian species (example headings)

Species	Earlier category (year)	Recent category (year)	Direction	Main threat
Great Indian Bustard				
Tiger				

Analysis and Results

Calculate the share of assessed species that are threatened (CR + EN + VU) ÷ (all assessed − Data Deficient) × 100. Count species that moved to a higher threat category (deteriorated) and those that moved to a lower one (improved), and present them as a bar chart. Distinguish genuine changes in status from changes caused by better knowledge, relate declines to the main drivers (habitat loss, overexploitation, pollution, invasive species, climate change) and note conservation successes such as the rise in India's tiger population.

Viva / Reflection Questions

1. List the IUCN Red List categories from highest to lowest risk.
2. What does the Living Planet Index measure, and what does it not measure?
3. Why did vulture populations collapse in India, and what has been done?

A.7.6 Sampling Plant and Animal Biodiversity of the Campus

Aim

To sample plants and selected animal groups on the college campus using standard methods and to calculate diversity indices.

Background

Ecologists **sample** rather than count every organism. Plants are sampled with **quadrats** of a size suited to the life form (see the quadrat layout figure in Unit 2); birds with **point counts** or line transects; insects with sweep nets or pitfall traps. **Diversity indices** combine **species richness** (number of species) and **evenness** (how equally individuals are spread among species) into a single number for comparing sites.

Materials and Data Sources

- Quadrat frames, measuring tape, pegs, string, sweep net, binoculars, notebook
- Identification aids: iNaturalist, Merlin Bird ID, eBird; floras of Delhi

Procedure

1. Choose two contrasting campus habitats (for example, a manicured lawn with ornamental plants, and an unmanaged wooded corner).
2. **Plants.** Lay out ten 1 m × 1 m quadrats for herbs in each habitat at random positions, and one 10 m × 10 m quadrat for trees. Record each species and its count in every quadrat.
3. **Birds.** At three points per habitat, record for 10 minutes in the early morning every bird seen or heard within 25 m.
4. **Insects.** Make 20 standard sweeps in each habitat and count individuals by group (or morphospecies).

Observations

Table A.51: Plant species data from herb quadrats (example headings)

Species	Total individuals (N)	Quadrats in which present	Frequency (%)	Density (per m^2)

Analysis and Results

For each species: **density** = total individuals ÷ total area sampled; **frequency** = (quadrats in which present ÷ total quadrats) × 100; **abundance** = total individuals ÷ quadrats in which present.

Shannon–Wiener index, where p_i is the proportion of individuals belonging to species i:

$$H' = -\Sigma\, p_i \ln p_i$$

Simpson's index of diversity, where n is the number of individuals of each species and N the total:

$$D = 1 - \Sigma\, n(n-1) \div [N(N-1)]$$

Worked example: a habitat has four species with 20, 15, 10 and 5 individuals (N = 50). The proportions are 0.4, 0.3, 0.2 and 0.1. The terms $p_i \ln p_i$ are −0.367, −0.361, −0.322 and −0.230, so H′ =

1.28. **Evenness** J = H′ ÷ ln S = 1.28 ÷ ln 4 = 1.28 ÷ 1.386 = 0.92. Simpson's D = 1 − (20×19 + 15×14 + 10×9 + 5×4) ÷ (50 × 49) = 1 − 700 ÷ 2,450 = 0.71.

Compare the two habitats using H′, D and species richness, and draw a **species–area curve** (cumulative species against number of quadrats) to judge whether sampling was adequate.

Precautions

- Do not uproot plants or harm animals; release insects immediately and do not use bird playback calls.

Viva / Reflection Questions

1. Why are quadrats of different sizes used for herbs, shrubs and trees?
2. What does a higher Shannon index indicate?
3. Two sites have the same number of species. How could their diversity indices differ?

A.7.7 Floral Diversity of Delhi and Other States

Aim

To document plant species of Delhi and another state (scientific and local names, IUCN status, medicinal properties, water requirement, socio-economic-environmental importance) in a short illustrated report.

Background

Delhi's native vegetation is semi-arid thorn forest on the Ridge and riverine vegetation along the Yamuna, with trees such as dhau (*Anogeissus pendula*), kair (*Capparis decidua*), hingot (*Balanites aegyptiaca*), khejri (*Prosopis cineraria*), dhak or palash (*Butea monosperma*), amaltas (*Cassia fistula*), neem (*Azadirachta indica*) and peepal (*Ficus religiosa*). The Mexican mesquite or vilayati kikar (*Prosopis juliflora*), planted widely in the twentieth century, now dominates much of the Ridge and suppresses native species; biodiversity-park restoration is replacing it.

Materials and Data Sources

- Phone camera; iNaturalist, Pl@ntNet or Google Lens for preliminary identification (confirm with a flora or expert)
- eFlora of India: https://efloraofindia.com; India Biodiversity Portal: https://indiabiodiversity.org
- Botanical Survey of India: https://bsi.gov.in; IUCN Red List: https://www.iucnredlist.org
- National Medicinal Plants Board: https://nmpb.nic.in

Procedure

1. Photograph at least 20 plant species (leaves, flowers, fruits, bark) on campus, in a biodiversity park or at Sunder Nursery, which has labelled native collections.
2. Identify each with apps, then confirm with a flora, eFlora of India or a botany teacher.
3. For each species, record the scientific name, family, local (Hindi or regional) name, native or introduced status, IUCN category (many common species are "Least Concern" or "Not Evaluated"), medicinal uses from authentic sources, water requirement (low, medium, high)

and socio-economic-environmental importance (shade, fodder, fruit, religious value, pollinators, invasiveness).

4. Collect similar information for ten species from another state using databases.
5. Compile a short report (5–8 pages) with a species table, photographs and a conclusion recommending species for planting in Delhi.

Observations

Table A.52: Floral documentation sheet (example entries)

Scientific name (family)	Local name	Native / introduced; IUCN	Water need	Importance
Cassia fistula (Fabaceae)	Amaltas	Native; Least Concern	Low	Ornamental, medicinal bark and pods
Prosopis juliflora (Fabaceae)	Vilayati kikar	Introduced, invasive	Very low	Fuelwood, but suppresses native flora

Analysis and Results

Calculate the percentage of native versus introduced species and the share with low water needs. Recommend five native, drought-tolerant species suitable for campus planting, giving reasons. Note that medicinal information is for knowledge only, not for self-medication.

Viva / Reflection Questions

1. Why are native species preferred over exotic ones in urban greening?
2. Why has vilayati kikar been removed from parts of the Delhi Ridge?
3. Why is an app-based identification not sufficient on its own?

A.7.8 Socio-Economic-Environmental Impact of Conserving a Global Animal Species

Aim

To report on the conservation of one globally important animal species: its relevance, range, threats, conflicts and poaching, IUCN status, protected areas and programmes, and suggestions.

Background

Flagship species such as the tiger, snow leopard, African elephant, giant panda or orangutan attract funding and public support; **umbrella species** need such large habitats that protecting them conserves many other species. Conservation brings people benefits (tourism income, ecosystem services) and costs (crop damage, livestock loss, restrictions on resource use, sometimes displacement).

Materials and Data Sources

- IUCN Red List species accounts (status, range map, population trend, threats, conservation actions): https://www.iucnredlist.org
- CITES species database (trade controls): https://checklist.cites.org
- Protected Planet (World Database on Protected Areas): https://www.protectedplanet.net

- National Tiger Conservation Authority (for comparison with Indian programmes): https://ntca.gov.in

Procedure

1. Each group chooses one species from outside or across India's borders, for example the snow leopard (*Panthera uncia*), African savanna elephant (*Loxodonta africana*), giant panda (*Ailuropoda melanoleuca*), Bornean orangutan (*Pongo pygmaeus*), polar bear (*Ursus maritimus*) or a pangolin.
2. Collect information under the headings in the observation table, identifying at least two stakeholder groups who gain from conservation and two who bear its costs.
3. Write a 4–6 page report with a range sketch, credited photographs and suggestions.

Observations

Table A.53: Species profile sheet (example headings)

Heading	Information	Source
Ecological relevance (keystone, flagship, umbrella role)		
Range and habitat		
IUCN status and population trend		
Threats, conflicts and poaching		
Protected areas and programmes		
Socio-economic impacts (benefits and costs)		
Suggestions		

Analysis and Results

Summarise costs and benefits in a stakeholder matrix (stakeholder, benefit, cost, suggested measure). Evaluate whether current programmes balance conservation and human welfare, and compare with an Indian programme such as Project Tiger or Project Elephant (see Chapter 6). Recommend measures such as compensation, community ecotourism, corridors and anti-poaching intelligence.

Viva / Reflection Questions

1. Is your species a flagship, umbrella or keystone species? Justify.
2. How does CITES help protect it?
3. Who bears the cost of conserving it, and how can they be compensated?

A.8 Unit 7: Human Communities and the Environment

A.8.1 Carbon Footprint of Different Countries Using Online Databases

Aim

To compare the carbon emissions of selected countries in total, per person, per unit of GDP and cumulatively over history, using online databases and simple mathematical tools.

Background

China is now the largest annual emitter, the United States has the largest cumulative emissions since 1750, and India is the third-largest annual emitter but has per-capita emissions of roughly 2 tonnes of CO_2 a year, well below the world average of roughly 4.7 tonnes (Global Carbon Project, 2023 data). **Consumption-based** accounts assign emissions to the country where goods are consumed rather than produced.

Materials and Data Sources

- Our World in Data CO_2 and greenhouse-gas explorer (territorial, consumption-based, per capita, cumulative): https://ourworldindata.org/co2-and-greenhouse-gas-emissions
- Global Carbon Atlas: https://globalcarbonatlas.org
- World Bank population and GDP data: https://data.worldbank.org
- Spreadsheet software

Procedure

1. Choose about eight countries of different income levels, for example India, China, the USA, Germany, Brazil, Nigeria, Bangladesh and Qatar.
2. Download, for the latest year, total, per-capita, per-GDP, consumption-based and cumulative (since 1850) CO_2 emissions, plus per-capita emissions for 1990.
3. Calculate each country's share of world emissions (annual and cumulative) and of world population.
4. Prepare bar charts for each measure and a scatter plot of per-capita emissions against GDP per capita.

Observations

Table A.54: Emission indicators by country (example headings)

Country	Total CO_2 (Mt)	Per capita (t)	Share of world population (%)	Share of cumulative CO_2 (%)
India				
USA				

Analysis and Results

Compute the ratio of each country's per-capita emissions to the world average, and the **responsibility ratio** = share of cumulative emissions ÷ share of world population; values above 1

indicate a larger-than-proportionate historical contribution. Calculate the CAGR of per-capita emissions between 1990 and the latest year (formula in Unit 1). On the scatter plot, describe the income–emissions relationship and identify outliers (for example, rich countries with low emissions due to hydro or nuclear power), and discuss equity in climate negotiations.

Viva / Reflection Questions

1. Why is per-capita emission a key argument in India's climate diplomacy?
2. What is the difference between territorial and consumption-based emissions?
3. Which country in your sample has the highest responsibility ratio?

A.8.2 Environmental Education and Awareness in Marginalised Localities

Aim

To plan and deliver an environmental-awareness session for residents or school students in a marginalised locality, and to learn from the community's own environmental knowledge and problems.

Background

Marginalised communities, such as residents of slums and resettlement colonies, waste pickers and migrant workers, often face the worst pollution, flooding, heat and water scarcity while contributing least to them. Outreach must be practical and two-way: it should build on what people already do (many reuse and recycle far more than affluent households) and focus on actions within their control, such as safe water storage, protection from heat and smog, and access to government schemes. The **National Green Corps (NGC)** eco-clubs and the **National Environment Awareness Campaign (NEAC)** support such outreach (see Chapter 7).

Materials and Data Sources

- A partner (local NGO, college NSS unit or government school) to arrange the visit
- Pictures, short videos, simple demonstrations (for example, the H_2S strip test for drinking water), handouts in the local language
- Information on schemes and advisories: Swachh Bharat Mission, Jal Jeevan Mission, heat action plans, CPCB AQI health advice

Procedure

1. With the partner, choose a locality and a topic that matters to residents (for example, safe water, waste, mosquitoes, heat or smog).
2. Make a preliminary visit to understand the community and convenient timings, and obtain permission from community leaders or the school.
3. Prepare a 30–45 minute session: an icebreaker, a question that draws out residents' experiences, a short demonstration, two or three simple actions, and questions and answers.
4. Deliver it in the local language using pictures rather than text; note questions, suggestions and local practices shared.
5. Leave a simple handout and write a reflective report, including what you learnt from the community.

Observations

Table A.55: Session record (example headings)

Item	Details
Locality, date, partner organisation	
Participants (number, age groups, gender)	
Key questions and concerns raised	
Local practices and knowledge shared	

Analysis and Results

Summarise participation and the main concerns raised. Classify concerns as within individual control, needing community action, or needing government action. Reflect on what worked and how the session could be improved; on a repeat visit, measure change with the survey method in the next activity.

Precautions

- Visit only with a teacher and a partner organisation; never go alone.
- Respect privacy; photograph only with permission and avoid photographing children's faces.

Viva / Reflection Questions

1. Why is environmental injustice a concern in Indian cities?
2. What did you learn from the community that you did not know before?
3. Why should outreach be two-way rather than a lecture?

A.8.3 Survey on the Impact of Environmental Education

Aim

To measure, through a questionnaire or online survey, whether environmental education has changed the knowledge, attitudes and practices of a target group.

Background

The real goal of education programmes is change in **knowledge, attitudes and practices (KAP)**. A **pre-test and post-test design**, in which the same people answer the same questions before and some weeks after an educational intervention, measures this change. The method follows the survey steps described under the climate-change survey in Unit 5.

Materials and Data Sources

- Questionnaire with knowledge items (true/false, multiple choice), attitude items (Likert scale) and practice items (frequency of actions such as segregating waste)
- Google Forms or printed sheets; consent forms (parental consent for school students); spreadsheet or JASP/PSPP

Procedure

1. Choose a target group and an intervention: first-year students before and after the EVS course, school students in an eco-club, or residents reached by the outreach activity above.
2. Design and pilot 15–20 items covering knowledge, attitudes and practices.
3. Administer the pre-test, giving each respondent a code so that answers can be matched without names.
4. Four to six weeks after the intervention, administer the post-test to the same people (and, if possible, to a comparison group).
5. Score knowledge items (1 for correct, 0 for incorrect), attitude items (1–5) and practice items (for example, 0 = never, 1 = sometimes, 2 = always).

Observations

Table A.56: Pre- and post-test scores (example headings)

Respondent code	Knowledge pre (out of 10)	Knowledge post (out of 10)	Practice pre (out of 10)	Practice post (out of 10)
S01				
S02				

Analysis and Results

Calculate mean scores before and after, and the mean change. Percentage change = (post-mean − pre-mean) ÷ pre-mean × 100. To test significance, use the **paired t-test**: compute the difference d for each respondent, then $t = \bar{d} \div (s_d \div \sqrt{n})$, where $\bar{d}$ is the mean difference, s_d its standard deviation and n the number of pairs. Example: for 25 students, mean knowledge rose from 5.2 to 7.0 ($\bar{d} = 1.8$) with $s_d = 1.5$; $t = 1.8 \div (1.5 \div 5) = 6.0$, far above the critical value of about 2.06 at 24 degrees of freedom, so the gain is significant. Discuss why knowledge often rises more than practice.

Viva / Reflection Questions

1. Why is a pre-test and post-test design better than a single survey?
2. What is the purpose of a comparison group?
3. Why might practice change less than knowledge?

A.8.4 Visit to a Development-Project-Affected Locality

Aim

To visit a locality affected by a development project (displacement, resettlement or environmental change) and assess the impacts on people's lives and livelihoods through observation and interviews.

Background

Dams, mines, highways and urban redevelopment bring benefits but often displace people. India's Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 requires social impact assessment and rehabilitation and resettlement (R&R) for affected families (see Chapter 7). In Delhi, resettlement colonies such as Bawana and Savda Ghevra house

families moved from riverbank and inner-city settlements in the 2000s; families displaced by the Tehri and Sardar Sarovar dams offer longer-term lessons.

Materials and Data Sources

- A partner NGO, researcher or local leader who can introduce the group
- Semi-structured interview guide, consent forms, notebook, camera (with permission)
- Secondary sources: social impact assessment reports, news reports, academic studies, court or NGT orders

Procedure

1. Read about the project: when and why people were moved, the R&R package promised, and documented outcomes.
2. Prepare an interview guide covering housing, livelihood, distance to work, water, sanitation, health, schooling, social networks and compensation received versus promised.
3. Visit with the teacher and partner; observe infrastructure (water points, toilets, drains, schools, transport).
4. Conduct 8–15 interviews with informed consent, including women, elders and young people, using a **before-and-after** recall: respondents rate their present situation against the pre-project one as worse, same or better.
5. Write a report combining observations, interviews and secondary data.

Observations

Table A.57: Before-and-after comparison reported by respondents (example headings)

Aspect	Worse (number)	Same (number)	Better (number)	Typical comment
Housing quality				
Livelihood and income				
Access to water				
Travel time to work				

Analysis and Results

Calculate the percentage of respondents reporting improvement or deterioration for each aspect, and present a stacked bar chart. Compare outcomes with the R&R package promised. Identify the groups most affected (for example, women who lost nearby work), discuss whether benefits and costs were fairly shared, recommend improvements to R&R policy, and acknowledge the limits of recall data.

Precautions

- Be sensitive: displacement can be painful. Do not press for details people do not wish to share.
- Do not raise expectations of help; keep identities confidential.

Viva / Reflection Questions

1. What does R&R mean, and what does the 2013 Act require?

2. Which impact of displacement did respondents describe as most serious?
3. Why do livelihood losses often outlast housing improvements?

A.8.5 Correlation Analysis of Population Growth and Its Impacts on Environment and Health

Aim

To test, using public data and Pearson's correlation coefficient, whether population size or growth is associated with indicators of environmental pressure and health.

Background

Human impact on the environment is often summarised by the **IPAT identity**: Impact = Population × Affluence × Technology. India's population, about 1.45 billion (145 crore) in 2024 according to UN World Population Prospects, is now the world's largest. A **correlation** measures how closely two variables move together, but correlation does not prove causation.

Materials and Data Sources

- UN World Population Prospects: https://population.un.org/wpp
- Our World in Data (population, CO_2, forest area, water use, health indicators): https://ourworldindata.org
- World Bank World Development Indicators: https://data.worldbank.org
- National Family Health Survey state fact sheets: http://rchiips.org/nfhs; Census of India: https://censusindia.gov.in

Procedure

1. Frame a question, for example: "Do states with higher population density generate more waste per person?"
2. Choose a **time series** (one country, many years) or a **cross-section** (many states or countries, one year) design.
3. Download population (or density) and two or three impact indicators, such as CO_2 emissions, forest area, water withdrawal, $PM_{2.5}$ exposure or under-five mortality, and draw scatter plots of each pair.
4. Calculate Pearson's r with the formula below (or the spreadsheet function CORREL) and test its significance.

Observations

Table A.58: Data sheet for correlation analysis (example headings)

Unit (year or state)	Population (million)	Indicator 1	Indicator 2	Indicator 3

Analysis and Results

Pearson's correlation coefficient between variables x and y is:

$$r = \Sigma(x - \bar{x})(y - \bar{y}) \div \sqrt{[\Sigma(x - \bar{x})^2 \times \Sigma(y - \bar{y})^2]}$$

r ranges from −1 (perfect negative) through 0 to +1 (perfect positive).

Worked example with illustrative numbers for five districts: x (population density, thousand per km^2) = 1, 2, 3, 4, 5; y (respiratory cases per 1,000 people) = 2, 4, 5, 4, 5. Here $\bar{x} = 3$ and $\bar{y} = 4$. The deviations are x: −2, −1, 0, 1, 2 and y: −2, 0, 1, 0, 1. $\Sigma(x - \bar{x})(y - \bar{y}) = 4 + 0 + 0 + 0 + 2 = 6$; $\Sigma(x - \bar{x})^2 = 10$; $\Sigma(y - \bar{y})^2 = 6$. So $r = 6 \div \sqrt{(10 \times 6)} = 6 \div 7.75 = 0.77$, a fairly strong positive correlation.

Test significance with $t = r\sqrt{(n - 2)} \div \sqrt{(1 - r^2)} = 0.77 \times 1.73 \div 0.64 \approx 2.1$. The critical value at 3 degrees of freedom (5% level, two-tailed) is 3.18, so this strong-looking correlation is not significant with only five data points: use as many units as possible. Report r^2 and discuss **confounding variables** (for example, vehicle density may drive both density and respiratory illness). In time series, both variables may rise simply with time; correlating year-to-year changes reduces this problem.

Viva / Reflection Questions

1. Why does correlation not imply causation? Give an example from your data.
2. What does the IPAT identity add to a simple population–impact correlation?
3. Why was the example correlation not statistically significant?

A.9 Assessment Rubric for Practical Reports

The rubric below can be used for any practical report in this appendix. Each criterion is scored from 1 to 4 and the total (out of 32) can be scaled to the marks allotted by the college. For posters, videos and presentations, replace "report structure" with "clarity of communication"; simulations use their own rubrics in Unit 5.

Table A.59: Assessment rubric for practical reports (score each criterion 1–4)

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs improvement (1)
Aim and background	Clear aim; concise, accurate background with Indian example	Clear aim; adequate background	Aim vague; thin background	Aim missing or wrong
Method and data collection	Systematic, replicable, ethical and safe; consent recorded	Mostly systematic	Gaps in method	Unclear or unsafe
Observations and data quality	Complete raw data with units, dates and sources	Minor omissions	Several omissions	Data missing or doubtful
Analysis and calculations	Correct formulas, worked steps, appropriate graphs	Minor errors	Several errors or weak graphs	Little or no analysis
Interpretation and discussion	Links results to concepts; compares with standards or published data; states limitations	Sound interpretation	Mostly descriptive	No interpretation
Conclusions and recommendations	Specific, feasible, evidence-based	Reasonable	General	Absent
Report structure and presentation	Standard format; clear figures and tables; well written	Minor lapses	Disorganised in places	Poorly organised

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs improvement (1)
Viva / reflection and teamwork	Explains every step confidently; fair share of group work	Explains most steps	Partial understanding	Cannot explain work

Appendix B: Glossary

Key terms from every chapter, in alphabetical order. The number in brackets is the chapter where the term is defined.

A

Abiotic component: A non-living physical or chemical factor of the environment, such as sunlight, temperature, water or soil. (1)

Absolute liability: The rule that an enterprise engaged in a hazardous activity is fully liable for any harm it causes, without exceptions. (5)

Access and benefit-sharing (ABS): Rules that require permission for using genetic resources and fair sharing of the resulting benefits with providers. (5)

Acid deposition: The fall of acidic rain, snow, fog, gases and particles formed from SO_2 and NO_x. (5)

Acid mine drainage: acidic, metal-rich water formed when sulfide minerals exposed by mining react with air and water. (3)

Activated sludge process: Biological wastewater treatment in which aerated microbial flocs consume organic matter and are then settled and partly recycled. (4)

Air Quality Index (AQI): a single number and colour category that summarises air pollution levels for the public. (4)

ALARA: the principle that radiation doses should be kept as low as reasonably achievable. (4)

Alpha diversity: The diversity of species within a single habitat or community. (6)

Anaerobic digestion: The breakdown of organic matter by micro-organisms in the absence of oxygen, producing biogas. (3)

Anthropocene: An informal name for the present era in which human activity has become the dominant influence on the Earth's environment. (1)

Anthropocentrism: An ethical view that only humans have intrinsic value and nature matters for its usefulness to people. (7)

Aquifer: A layer of rock or sediment that stores and transmits enough groundwater to supply wells. (3)

Artesian well: A well tapping a confined aquifer in which water pressure pushes water up without pumping. (3)

Artificial insemination: Placing semen collected from a male into a female to achieve fertilisation without natural mating. (6)

Assimilative capacity: the ability of an environment to absorb, dilute or break down a pollutant without harm. (4)

Atmosphere: The layer of gases surrounding the Earth, held by gravity. (1)

B

Beta diversity: The change in species composition between habitats or along a gradient. (6)

Biocapacity: The capacity of an area's ecosystems to produce useful biological materials and absorb wastes. (1)

Biocentrism: An ethical view that every living organism has intrinsic value. (7)

Biochemical oxygen demand (BOD): the oxygen consumed by micro-organisms decomposing organic matter in a water sample. (4)

Biodegradable pollutant: a pollutant that decomposers break down into harmless substances. (4)

Biodiversity: The variability among living organisms from all sources, including diversity within species, between species and of ecosystems. (6)

Biodiversity hotspot: A region with at least 1,500 endemic vascular plant species that has lost at least 70% of its original vegetation. (6)

Biogas: A fuel gas of about 50–70% methane and 30–45% carbon dioxide, produced by anaerobic digestion. (3)

Biogeochemical cycle: The movement of a chemical element between living organisms and the non-living environment. (2)

Biogeographic zone: A large region with a distinctive combination of climate, landform and plant and animal life; India has ten. (6)

Biomagnification: The increase in concentration of a persistent substance at each successive trophic level of a food chain. (2)

Biome: A large terrestrial ecosystem type defined by its climate and dominant vegetation, such as tundra, forest, grassland or desert. (2)

Biomethanation: Anaerobic digestion of organic waste in a sealed digester to produce biogas and digestate. (4)

Biomining: Excavation, stabilisation and sieving of legacy waste to recover soil, fuel and inert fractions and reclaim land. (4)

Bioremediation: The use of microorganisms to remove or detoxify pollutants in soil or water. (4)

Biosphere: The part of the Earth where life exists, together with all living organisms. (1)

Biosphere reserve: A large area, zoned into core, buffer and transition zones, that combines conservation with sustainable development. (6)

Biotic component: A living part of the environment, such as a plant, animal or micro-organism. (1)

Bird flight diverter: A reflective or coloured device fixed to power lines to make them visible to flying birds. (6)

Buffer zone: The area around a core where people live and land use is regulated to support conservation. (6)

Bulk waste generator: An entity generating large amounts of waste (100 kg or more a day under the SWM Rules) with special duties to process its own waste. (4)

C

Calorific value: The amount of heat released by complete combustion of a unit mass or volume of a fuel. (3)

Camera trapping: Use of motion-triggered cameras to photograph and identify individual animals for population estimates. (6)

Capacity utilisation factor: The ratio of the actual energy produced by a plant over a period to the energy it would produce at full capacity throughout. (3)

Carbon budget: The total amount of CO_2 that can still be emitted while keeping warming below a given limit. (5)

Carbon dioxide equivalent (CO_2e): A unit that expresses the warming effect of any greenhouse gas as the amount of CO_2 that would cause the same warming. (7)

Carbon footprint: The total greenhouse gas emissions caused directly and indirectly by a person, organisation, product or country, expressed in CO_2e. (7)

Carrying capacity: The maximum population of a species that an environment can support indefinitely without being degraded. (1)

Catalytic converter: an exhaust device that uses a catalyst to convert CO, hydrocarbons and NO_x into less harmful gases. (4)

Chapman cycle: The natural set of reactions by which UV light forms and destroys ozone in the stratosphere. (5)

Chemical oxygen demand (COD): the oxygen needed to chemically oxidise all organic matter in a water sample. (4)

Circular economy: An economic system that keeps materials in use through durable design, reuse, repair and recycling instead of take-make-dispose. (4)

CITES Appendix I: The list of species threatened with extinction, in which commercial international trade is banned. (5)

Clean Development Mechanism (CDM): A Kyoto Protocol mechanism that let developed countries earn credits by funding emission-reducing projects in developing countries. (5)

Cleaner production: A preventive strategy that reduces raw material, energy and water use and waste generation at source. (4)

Climate feedback: A process that amplifies or dampens an initial climate change, such as the water vapour or ice–albedo feedback. (5)

Climax community: The relatively stable, self-maintaining community at the end of succession. (2)

Co-firing: Burning biomass pellets together with coal in a thermal power station. (3)

Coagulation: Addition of chemicals such as alum to neutralise charges on fine particles so that they clump into settleable flocs. (4)

Coliform bacteria: intestinal bacteria whose presence in water indicates faecal contamination. (4)

Common but differentiated responsibilities and respective capabilities (CBDR-RC): The principle that all countries must act on climate change but developed countries should take the lead. (5)

Common effluent treatment plant (CETP): A shared plant that treats the combined wastewater of a cluster of small industries. (4)

Community Forest Resource (CFR) rights: Rights of a forest-dwelling community under the Forest Rights Act to protect, regenerate and manage its traditional forest. (5)

Community Forest Resource rights: rights under the Forest Rights Act, 2006 that allow a gram sabha to protect, regenerate and manage its customary forest. (3)

Compensation level: The depth at which photosynthesis just balances respiration. (2)

Compensatory afforestation: Tree planting on non-forest or degraded land required when forest land is diverted for other uses. (5)

Composting: Controlled aerobic decomposition of organic waste by microorganisms into humus-like compost. (4)

Compressed biogas (CBG): Biogas purified to at least 90% methane and compressed for use like CNG. (3)

Consent to Operate: Permission from a State Pollution Control Board, with conditions, required before an industry starts production. (4)

Conservation breeding: Breeding a threatened species in captivity to build an insurance population for later release. (6)

Consumer (heterotroph): An organism that obtains energy by eating other organisms. (2)

Consumptive use value: The value of natural products used directly by people without passing through a market, such as fuelwood and wild food. (6)

Coral bleaching: The whitening of corals when heat or other stress causes them to expel their symbiotic algae (zooxanthellae). (2)

Critical Tiger Habitat (core): The inviolate part of a tiger reserve kept free of settlements and resource use. (6)

Cropping intensity: the ratio of gross cropped area to net sown area, expressed as a percentage. (3)

D

Decibel (dB): a logarithmic unit of sound level in which a 10 dB rise is a tenfold increase in intensity. (4)

Decomposer: A microorganism, mainly a bacterium or fungus, that breaks down dead organic matter into simple inorganic substances. (2)

Deforestation: permanent clearing of forest and conversion of the land to a non-forest use. (3)

Demographic dividend: The potential economic gain when the working-age share of a population is large compared with dependants. (7)

Demographic transition: The shift from high birth and death rates to low birth and death rates that accompanies economic and social development. (7)

Demography: The scientific study of the size, composition, distribution and change of human populations. (7)

Denitrification: The conversion of nitrate back into nitrogen gas by bacteria in oxygen-poor soils and sediments. (2)

Desertification: Land degradation in arid, semi-arid and dry sub-humid areas caused by climatic variations and human activities. (2)

Designated best use: the most demanding use assigned to a water body, which determines its water quality criteria. (4)

Detritivore: An animal, such as an earthworm or termite, that feeds on dead organic matter and breaks it into smaller pieces. (2)

Development-induced displacement: The forced movement of people from their homes or livelihoods to make way for development projects. (7)

Diclofenac: A non-steroidal anti-inflammatory drug whose residues in cattle carcasses killed vultures across South Asia. (6)

Dissolved oxygen (DO): the amount of oxygen dissolved in water, needed by aquatic animals. (4)

District Mineral Foundation: a trust in each mining-affected district, funded by a share of mining royalty, that works for the benefit of people and areas affected by mining. (3)

Double clutching: Removing a bird's first egg for artificial incubation so that it lays a second egg. (6)

E

E-waste: Discarded electrical and electronic equipment, whole or in parts. (4)

Eco-club: A voluntary group of students guided by a teacher that carries out practical environmental activities in and around a school or college. (7)

Ecocentrism: An ethical view that gives moral value to whole ecosystems, species and ecological processes. (7)

Ecofeminism: The view that the domination of nature and the domination of women are linked. (7)

Ecological footprint: The area of biologically productive land and water needed to supply the resources a population consumes and absorb its wastes, measured in global hectares. (1)

Ecological pyramid: A diagram showing the number, biomass or energy at successive trophic levels. (2)

Ecological restoration: The process of assisting the recovery of an ecosystem that has been degraded, damaged or destroyed. (2)

Ecological succession: The orderly, directional change in the species composition of a community over time. (2)

Ecologically Sensitive Area (ESA): An area notified under the Environment (Protection) Act in which harmful activities are restricted. (6)

Ecosystem: A community of living organisms together with its physical environment, functioning as a unit through energy flow and nutrient cycling. (2)

Ecosystem services: The benefits people obtain from ecosystems, grouped as provisioning, regulating, cultural and supporting services. (2)

Ecotone: A transition zone between two adjacent ecosystems, such as a mangrove belt or the reed fringe of a lake. (2)

Edge effect: The changes in conditions and species at the boundary of a habitat patch compared with its interior. (6)

Electrostatic precipitator: a device that charges dust particles electrically and collects them on earthed plates. (4)

Elephant corridor: A narrow strip of habitat that connects two larger elephant habitats and allows movement between them. (6)

Elephant reserve: An administrative landscape, not a legal protected-area category, managed for elephant conservation under Project Elephant. (6)

Emission factor: The mass of greenhouse gas released per unit of an activity, such as kg CO_2 per kWh of electricity. (7)

Emissions intensity: Greenhouse gas emissions per unit of GDP. (5)

Endemic species: A species naturally found only in a particular area and nowhere else in the world. (6)

Enhanced greenhouse effect: The additional warming caused by human-added greenhouse gases. (5)

Environment: The sum of all living and non-living things surrounding an organism, together with the relationships among them. (1)

Environmental flow: The quantity, timing and quality of water flow required to sustain a river ecosystem and the people who depend on it. (2)

Environmental Impact Assessment (EIA): A process for predicting and evaluating the environmental effects of a proposed project before it is approved. (4)

Environmental justice: The fair distribution of environmental benefits and burdens and fair participation in environmental decisions for all people. (7)

Environmental science: The scientific study of natural processes and of how human activities affect them. (1)

Environmental studies: A multidisciplinary field that combines the natural sciences, social sciences and humanities to understand and solve environmental problems. (1)

Environmentalism of the poor: Environmental struggles in which defending nature and defending the livelihoods of poor communities are the same cause. (7)

Ephemeral plant: A plant that completes its life cycle in the short period after rain and survives drought as seed. (2)

Epiphyte: A plant that grows on another plant for support without taking nutrients from it, such as many orchids. (2)

Equitable and reasonable utilisation: The principle of international water law that shared rivers should be used fairly by all riparian states. (3)

Estuary: A partly enclosed coastal water body where fresh river water mixes with seawater. (2)

Eutrophication: The excessive growth of algae and plants in a water body caused by added nutrients, often leading to oxygen depletion. (2)

Ex-situ conservation: Conservation of species or their genetic material outside their natural habitats. (6)

Extended producer responsibility (EPR): A policy making producers responsible for the collection and recycling of their products after use. (4)

F

Flagship species: A charismatic species used to raise public support and funds for conservation. (6)

Floodplain zoning: Regulating land use on floodplains according to flood risk to reduce damage. (3)

Flue gas desulfurisation: removal of SO_2 from flue gas, usually by reaction with limestone to form gypsum. (4)

Flue gas desulfurisation (FGD): A technology that removes SO_2 from power plant exhaust, usually using limestone. (5)

Fly ash: Fine ash produced by burning pulverised coal in thermal power plants, usable in cement, bricks and roads. (4)

Food chain: A linear sequence of organisms through which energy and nutrients pass as one eats another. (2)

Food web: A network of interconnected food chains in a community. (2)

Forest cover: All land of more than one hectare with a tree canopy density above 10 per cent, irrespective of ownership or legal status (FSI definition). (2)

Fortress conservation: A model of conservation that excludes local people from protected areas. (6)

G

Genetic diversity: The variety of genes and alleles within a species, between individuals and populations. (6)

Gharial: A critically endangered, fish-eating crocodilian of large Indian rivers with a long, slender snout. (6)

Glacial lake outburst flood (GLOF): A sudden flood caused by the failure of a natural dam holding back a glacial lake. (3)

Global warming potential (GWP): The warming caused by 1 kg of a gas over a set period, usually 100 years, relative to 1 kg of CO_2. (5)

Green hydrogen: Hydrogen produced by electrolysis of water using electricity from renewable sources. (3)

Green Revolution: the rapid rise in Indian food production from the mid-1960s based on high-yielding seeds, irrigation, fertilisers and price support. (3)

Greenhouse effect: The warming of Earth's surface caused by atmospheric gases that absorb outgoing infrared radiation and re-emit part of it downwards. (5)

Greenwashing: Giving a misleading impression that a product, company or policy is environmentally friendly. (1)

Gross calorific value: The calorific value that includes the heat recovered from condensing water vapour formed in combustion. (3)

Gross primary productivity (GPP): The total rate at which producers fix energy or carbon in photosynthesis. (2)

Gully erosion: erosion in which runoff cuts deep channels that cannot be removed by ploughing. (3)

H

Habitat fragmentation: The breaking up of a continuous habitat into smaller, isolated patches. (6)

Half-life: the time taken for half of the atoms of a radioactive substance to decay. (4)

Hazardous waste: Waste that is ignitable, corrosive, reactive, toxic or otherwise likely to harm health or the environment. (4)

HLWG (Kasturirangan Committee): The High Level Working Group (2013), which proposed an ESA covering about 37% of the Western Ghats. (6)

Homeostasis: The ability of an ecosystem to maintain a relatively stable state through self-regulating feedback. (2)

Human–elephant conflict: Negative interactions such as crop raiding, property damage and deaths of people or elephants. (6)

Humus: dark, stable organic matter in soil formed by the decomposition of plant and animal remains. (3)

Hydrarch succession: Succession that begins in water, such as a pond, and ends in a land community. (2)

Hydrosphere: All the water on Earth in liquid, solid and gaseous forms. (1)

I

I = PAT: An identity stating that environmental impact equals population × affluence × technology. (7)

Impoverishment Risks and Reconstruction (IRR) model: Michael Cernea's framework of eight risks faced by displaced people and the strategies needed to counter them. (7)

In-situ conservation: Conservation of species in their natural habitats. (6)

Indicator species: A species whose presence or condition reflects the health of its environment. (6)

Interdisciplinary approach: Study of a problem in which knowledge and methods from different disciplines are integrated into a combined understanding. (1)

Intergenerational equity: The principle that present generations must pass on a healthy and productive environment to future generations. (1)

Invasive alien species: A species introduced outside its natural range that spreads and harms native biodiversity, the economy or health. (6)

IUCN Red List: The global inventory that assesses the extinction risk of species using standard categories and criteria. (6)

J

Johad: A traditional crescent-shaped earthen check dam in Rajasthan that stores runoff and recharges groundwater. (3)

Joint Forest Management: an arrangement in which village committees protect state forests in return for a share of forest produce. (3)

K

Kerala Sastra Sahitya Parishad (KSSP): A people's science movement in Kerala that led the campaign to save Silent Valley. (6)

Keystone species: A species whose effect on its ecosystem is far greater than its abundance suggests. (6)

Knowledge–action gap: The gap between knowing about an environmental problem and acting to address it. (7)

L

Land cover: the physical and biological material covering the land surface, such as forest, crops, water or buildings. (3)

Land degradation: a reduction or loss of the biological or economic productivity of land. (3)

Land Degradation Neutrality: a state in which the amount and quality of land resources remains stable or increases (SDG Target 15.3). (3)

Land ethic: Aldo Leopold's principle that actions are right when they preserve the integrity, stability and beauty of the biotic community. (7)

Land use: the purpose for which people use land, such as agriculture, settlement or conservation. (3)

Law of tolerance: Shelford's principle that each organism can survive only between a minimum and a maximum level of each environmental factor. (2)

Leachate: The contaminated liquid formed when water percolates through waste, carrying dissolved and suspended pollutants. (4)

Lentic ecosystem: A standing-water ecosystem such as a pond, lake or reservoir. (2)

Limiting factor: The environmental factor in shortest supply, or furthest from the optimum, that restricts the growth or distribution of an organism. (2)

Lion-tailed macaque: An endangered monkey endemic to the Western Ghats rainforests and the symbol of the Silent Valley campaign. (6)

Lithosphere: The rigid outer shell of the Earth, made up of the crust and the uppermost mantle. (1)

Littoral zone: The shallow margin of a lake where light reaches the bottom and rooted plants grow. (2)

Living modified organism (LMO): A living organism with a new combination of genetic material obtained through modern biotechnology. (5)

Loss and damage: Harm from climate change that cannot be avoided through mitigation or adaptation. (5)

Lotic ecosystem: A flowing-water ecosystem such as a spring, stream or river. (2)

M

M-STrIPES: A GPS-based software and mobile app system for recording patrols and ecological data in tiger reserves. (6)

Man–wildlife conflict: Interactions between people and wild animals that harm people, their property or the animals. (6)

Mass extinction: A geologically short period in which at least about three-quarters of all species die out. (6)

Meloxicam: A veterinary painkiller that is safe for vultures and is promoted as a substitute for diclofenac. (6)

Meteorological drought: A significant shortfall of rainfall compared with the long-term normal. (3)

Mineral: a naturally occurring inorganic solid with a definite chemical composition and ordered crystalline structure. (3)

Minor forest produce (MFP): non-timber products of plant origin collected from forests, such as tendu leaves, mahua, bamboo, lac and honey. (3)

Mission LiFE: India's campaign for Lifestyle for Environment, promoting mindful consumption by individuals and communities. (5)

Montreux Record: A register of Ramsar sites whose ecological character is changing and that need priority attention. (5)

Multidisciplinary approach: Study of a problem by several disciplines working side by side, each using its own methods. (1)

N

National park: A protected area under the Wild Life (Protection) Act where all private rights are extinguished and grazing and extraction are not allowed. (6)

National Tiger Conservation Authority (NTCA): Statutory body created in 2006 under the Wildlife (Protection) Act to oversee tiger conservation. (6)

Nationally Determined Contribution (NDC): A country's own climate action plan submitted every five years under the Paris Agreement. (5)

Natural capital: The stock of natural resources and ecosystems that provides goods and services to people. (1)

Natural farming: chemical-free farming that relies on on-farm inputs, mulching and mixed cropping. (3)

Natural resource: any material, energy source or service from nature that people use to meet their needs. (3)

Neritic zone: The relatively shallow marine zone above the continental shelf. (2)

Net primary productivity (NPP): Gross primary productivity minus the respiration of the producers themselves. (2)

Net zero: A state in which human greenhouse gas emissions are balanced by removals from the atmosphere. (5)

Nitrification: The conversion of ammonium to nitrite and then nitrate by bacteria such as *Nitrosomonas* and *Nitrobacter*. (2)

Nitrogen fixation: The conversion of atmospheric nitrogen gas into ammonia or nitrate by bacteria, lightning or industry. (2)

Non-point source: pollution released over a wide area with no single outlet, such as farm runoff. (4)

Non-renewable resource: a resource formed over geological time whose stock is effectively fixed, such as coal or metallic ores. (3)

O

OECM (Other effective area-based conservation measure): An area outside the formal protected-area system that achieves effective long-term conservation of biodiversity. (2)

Open natural ecosystem: A naturally treeless or sparsely wooded ecosystem, such as a grassland, savanna, scrubland or rocky outcrop. (2)

Ore: a rock containing a valuable mineral in high enough concentration to be mined at a profit. (3)

Over-exploited unit: An assessment unit where annual groundwater extraction exceeds annual extractable recharge (stage above 100%). (3)

Overburden: soil and rock lying above a mineral deposit that must be removed in surface mining. (3)

Oxygen sag curve: the fall and recovery of dissolved oxygen in a river downstream of an organic waste discharge. (4)

Ozone hole: The region over Antarctica where total ozone falls below 220 Dobson units each spring. (5)

Ozone layer: The region of the stratosphere, mainly between 15 and 35 km, where ozone is concentrated and absorbs harmful ultraviolet radiation. (1)

Ozone-depleting substance (ODS): A chemical, such as a CFC or halon, that releases chlorine or bromine that destroys stratospheric ozone. (5)

P

Padayatra: A long foot march used to spread awareness and mobilise people, as in the Save Western Ghats march. (6)

Particulate matter (PM2.5): airborne particles 2.5 micrometres or smaller that penetrate deep into the lungs. (4)

Pedogenesis: the process of soil formation from weathered rock and organic matter. (3)

Per-capita water availability: The average annual water resource of a region divided by its population. (3)

Permafrost: Ground that remains frozen for at least two consecutive years, found beneath Arctic tundra and parts of high mountains. (2)

Photochemical smog: an oxidising haze of ozone, PAN and particles formed when NO_x and VOCs react in sunlight. (4)

Phytoremediation: the use of plants to remove, stabilise or degrade pollutants in soil or water. (4)

Pioneer species: The first organisms to colonise a bare or disturbed area. (2)

Planetary boundaries: Limits for nine Earth-system processes that together define a safe operating space for humanity. (1)

Pneumatophore: An upward-growing breathing root of mangroves that takes in air from above the waterlogged mud. (2)

Point source: a single, identifiable outlet of pollution, such as a chimney or effluent pipe. (4)

Pollutant: a substance or form of energy that causes an undesirable change in the environment. (4)

Polluter pays principle: The principle that those who cause pollution should bear the cost of preventing and repairing it. (1)

Polluter-pays principle: The principle that the polluter must bear the cost of preventing and remedying the pollution it causes. (4)

Population momentum: Continued population growth after fertility falls to replacement level, because of a large number of young people entering reproductive age. (7)

Precautionary principle: The principle that lack of full scientific certainty should not delay action to prevent serious or irreversible environmental harm. (1)

Primary air pollutant: a pollutant emitted directly into the air from a source. (4)

Producer (autotroph): An organism that makes its own food from inorganic substances, using light (photosynthesis) or chemical energy (chemosynthesis). (2)

Productive use value: The value of biological products sold in markets, such as timber, fish and medicines. (6)

Profundal zone: The deep, dark water of a lake below the compensation level, where photosynthesis is not possible. (2)

Project Tiger: India's tiger conservation programme, launched in 1973, built around a network of tiger reserves. (6)

Public interest litigation (PIL): A legal action filed in court on behalf of the public or of people unable to approach the court themselves. (1)

R

Radiative forcing: The change in Earth's energy balance, in W/m^2, caused by a change in a climate driver. (5)

Rainwater harvesting: Collecting and storing rain where it falls, for direct use or to recharge groundwater. (3)

Rear and release: Collecting eggs, rearing young in safety until they are large enough to escape most predators, and releasing them. (6)

Red mud: Highly alkaline residue left after extracting alumina from bauxite. (4)

Refuse-derived fuel (RDF): Fuel made from the dry, combustible, non-recyclable fraction of waste, often burned in cement kilns. (4)

Rehabilitation: The restoration of displaced people's livelihoods, incomes, social networks and quality of life after resettlement. (7)

Reintroduction: Release of a species into part of its natural range from which it has disappeared. (6)

Renewable energy: Energy from sources that are naturally replenished as fast as they are used, such as sunlight, wind and flowing water. (3)

Renewable resource: a resource replenished by natural processes within a human lifetime, provided it is not used faster than it regenerates. (3)

Replacement level fertility: A TFR of about 2.1, at which each generation exactly replaces itself in the long run. (7)

Resilience: The ability of an ecosystem to recover its structure and function after a disturbance. (2)

Riparian state: A state or country through which a river flows or along whose banks it lies. (3)

River continuum concept: The idea that physical conditions, food sources and communities change in a predictable way from the headwaters of a river to its mouth. (2)

S

Sacred grove: A patch of natural vegetation protected by a local community because it is dedicated to a deity or ancestors. (6)

Sanitary landfill: An engineered disposal site with liners, leachate and gas collection, daily cover and monitoring. (4)

Savanna: A grassland with scattered trees, maintained by seasonal drought, fire and grazing. (2)

Scope 1, 2 and 3 emissions: Direct emissions, indirect emissions from purchased energy, and all other value-chain emissions respectively. (7)

Seawater intrusion: The movement of salt water into coastal freshwater aquifers caused by over-pumping. (3)

Second-generation (2G) ethanol: Ethanol made from cellulose in inedible plant material such as straw, rather than from sugar or grain. (3)

Secondary air pollutant: a pollutant formed in the atmosphere by reactions among primary pollutants and natural air components. (4)

Segregation at source: Separation of waste into different streams at the point where it is generated. (4)

Shannon index: A diversity index, $H' = -\Sigma\, p_i \ln p_i$, that increases with both species richness and evenness. (6)

Shelterbelt: rows of trees or shrubs planted across the prevailing wind to reduce wind speed and erosion. (3)

Shola: A patch of stunted montane evergreen forest in the sheltered valleys of the high Western Ghats, set within rolling grassland. (2)

Sievert (Sv): the unit of radiation dose that accounts for the biological harm caused by different radiation types. (4)

Silence zone: an area of at least 100 m around hospitals, educational institutions and courts with the strictest noise limits. (4)

Social Impact Assessment (SIA): A study of the effects of a project on affected people and communities, required before land acquisition under the 2013 Act. (7)

Soil erosion: the detachment and removal of soil by water, wind, ice or gravity. (3)

Soil Health Card: a card given to farmers that reports the nutrient status of their soil and recommends fertiliser doses. (3)

Soil profile: a vertical section through the soil showing its horizons from the surface to bedrock. (3)

Solatium: An additional payment made to landowners for the compulsory nature of land acquisition. (7)

Solid waste management: The planned control of waste generation, storage, collection, transport, processing and disposal to protect health and the environment. (4)

Species evenness: How equally individuals are distributed among the species in a community. (6)

Species richness: The number of different species present in an area. (6)

Stage of groundwater extraction: Annual groundwater extraction expressed as a percentage of the annual extractable groundwater resource. (3)

Stepwell (baoli): A traditional well with steps leading down to the water, built to store and give access to groundwater. (3)

Stochastic effect: a radiation effect, such as cancer, whose probability rather than severity increases with dose. (4)

Stratification (forest): The vertical layering of a forest into emergent, canopy, understorey, shrub, herb and floor layers. (2)

Stratosphere: The atmospheric layer from the tropopause to about 50 km, containing the ozone layer, where temperature rises with height. (1)

Strong sustainability: The view that critical natural capital cannot be replaced and must be maintained in its own right. (1)

Sustainability: The ability of a system to continue functioning over the long term without exhausting the resources and conditions it depends on. (1)

Sustainable development: Development that meets the needs of the present without compromising the ability of future generations to meet their own needs. (1)

Sustainable Development Goals (SDGs): The 17 global goals with 169 targets adopted by the United Nations in 2015 for achievement by 2030. (1)

T

Tailings: finely ground waste left after the valuable mineral has been separated from ore. (3)

Temperature inversion: a layer of warm air above cooler surface air that traps pollutants near the ground. (4)

Ten per cent law: The generalisation that, on average, only about 10% of the energy at one trophic level is transferred to the next. (2)

Thermal pollution: degradation of water quality by a change in its natural temperature, usually by heated discharges. (4)

Thermocline: The layer in a stratified water body in which temperature drops sharply with depth, separating the epilimnion from the hypolimnion. (2)

Threatened species: A species listed as Critically Endangered, Endangered or Vulnerable on the IUCN Red List. (6)

Tiger reserve: A protected landscape notified for tiger conservation, with a core and a buffer zone. (6)

Tipping point: A threshold beyond which an ecosystem shifts suddenly to a different, self-reinforcing state. (2)

Total Economic Value: The sum of use values (direct, indirect, option) and non-use values (existence, bequest) of an ecosystem. (2)

Total fertility rate (TFR): The average number of children a woman would bear over her lifetime at current age-specific birth rates. (7)

Treatment, storage and disposal facility (TSDF): A common facility for the safe treatment and secure disposal of hazardous waste. (4)

Treeline: The altitude or latitude beyond which trees cannot grow because of cold and wind. (2)

Trophic level: The position of an organism in the sequence of energy transfer, such as producer, herbivore or carnivore. (2)

Troposphere: The lowest layer of the atmosphere, up to about 8–18 km, where weather occurs and temperature falls with height. (1)

U

Umbrella species: A species with large area needs whose protection also protects many other species. (6)

Upwelling: The rising of cold, nutrient-rich deep water to the ocean surface, which raises productivity. (2)

V

Van Dhan Vikas Kendra: a cluster of tribal self-help groups that processes and markets forest produce under the Pradhan Mantri Van Dhan Yojana. (3)

Vermicomposting: Conversion of organic waste into nutrient-rich manure (vermicast) using earthworms. (4)

Virtual water: The water used to produce a good or service, which is hidden in the product and in trade. (3)

Visceral gout: Deposition of uric acid crystals on internal organs, a sign of kidney failure in poisoned vultures. (6)

Vivipary: Germination of a seed while it is still attached to the parent plant, as in many mangroves. (2)

Vulture Safe Zone: An area of about 100 km radius around a vulture colony where toxic veterinary drugs have been eliminated. (6)

W

Waste hierarchy: A ranking of waste-management options from prevention (most preferred) to disposal (least preferred). (4)

Water disputes tribunal: A quasi-judicial body set up under the Inter-State River Water Disputes Act 1956 to adjudicate disputes between states. (3)

Water footprint: The total volume of fresh water used to produce the goods and services consumed by a person, product or country. (3)

Water stress: A condition in which annual water availability falls below 1,700 m^3 per person. (3)

Water table: The upper surface of the zone in which all pores in soil or rock are filled with water. (3)

Weak sustainability: The view that natural capital can be replaced by human-made capital as long as the total capital stock does not fall. (1)

WGEEP (Gadgil Committee): The Western Ghats Ecology Expert Panel (2011), which proposed treating the entire range as ecologically sensitive. (6)

Wildlife sanctuary: A protected area under the Wild Life (Protection) Act where some rights of local people may be allowed. (6)

Wise use (Ramsar): Maintaining the ecological character of wetlands within the context of sustainable development. (5)

X

Xerarch succession: Succession that begins in a dry habitat, such as bare rock. (2)

Z

Zero liquid discharge (ZLD): A system in which all wastewater is treated and recycled and dissolved salts are recovered as solids, so no effluent is released. (4)